

z/OS
3.1

*IBM z/OS Management Facility
Configuration Guide*



Note

Before using this information and the product it supports, read the information in [“Notices” on page 431.](#)

This edition applies to IBM® z/OS® 3.1 (5655-ZOS) and to all subsequent releases and modifications until otherwise indicated in new editions.

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About this document

This document provides information for configuring IBM z/OS Management Facility (z/OSMF). This document also provides information for troubleshooting problems related to the use of z/OSMF.

Who should use this document

This document provides information for the person who is responsible for setting up z/OSMF on a z/OS system and for diagnosing problems with the product. This document assumes that the user is familiar with the z/OS operating system and its accompanying products.

Where to find more information

For an overview of the information associated with z/OS, see [z/OS Information Roadmap](#).

IBM Z and LinuxOne Community

To access a rich community of business and technical expert blogs and forums for z/OSMF, visit the IBM Z and LinuxOne Community at <https://ibm.github.io/zOSMF/>.

z/OSMF One Stop Hub

For the latest developer news about z/OSMF, visit the z/OSMF One Stop Hub at [z/OSMF One Stop Hub \(ibm.github.io/zOSMF/\)](https://ibm.github.io/zOSMF/).

z/OSMF Community Guild

To join z/OSMF subject matter experts in deep dives of technical demos and learn about improvements to the platform, visit the z/OSMF Community Guild at <https://ibm.biz/zosmfguildhome>.

z/OS Basic Skills Information Center

The z/OS Basic Skills Information Center is a web-based information resource intended to help users learn the basic concepts of z/OS, the operating system that runs most of the IBM mainframe computers in use today. The Information Center is designed to introduce a new generation of Information Technology professionals to z/OS concepts and help them prepare for a career as a z/OS professional, such as a z/OS system programmer.

Specifically, the z/OS Basic Skills Information Center is intended to achieve the following objectives:

- Provide basic education and information about z/OS without charge
- Shorten the time that it takes for people to become productive on the mainframe
- Make it easier for new people to learn z/OS.

To access the z/OS Basic Skills Information Center, open your web browser to the following website, which is available to all users (no login required): <https://www.ibm.com/docs/en/zos-basic-skills?topic=zosbasics/com.ibm.zos.zbasics/homepage.htm>.

How to send your comments to IBM

We invite you to submit comments about the z/OS product documentation. Your valuable feedback helps to ensure accurate and high-quality information.

Important: If your comment regards a technical question or problem, see instead [“If you have a technical problem”](#) on page xix.

Submit your feedback by using the appropriate method for your type of comment or question:

Feedback on z/OS function

If your comment or question is about z/OS itself, submit a request through the <https://ibm-z-hardware-and-operating-systems.ideas.ibm.com/>.

Feedback on IBM Documentation function

If your comment or question is about the IBM Documentation functionality, for example search capabilities or how to arrange the browser view, send a detailed email to IBM Documentation Support at ibmdocs@us.ibm.com.

Feedback on the z/OS product documentation and content

If your comment is about the information that is provided in the z/OS product documentation library, send a detailed email to mhvrcfs@us.ibm.com. We welcome any feedback that you have, including comments on the clarity, accuracy, or completeness of the information.

To help us better process your submission, include the following information:

- Your name, company/university/institution name, and email address
- The following deliverable title and order number: IBM z/OSMF Configuration Guide, SC27-8419-50
- The section title of the specific information to which your comment relates
- The text of your comment.

When you send comments to IBM, you grant IBM a nonexclusive authority to use or distribute the comments in any way appropriate without incurring any obligation to you.

IBM or any other organizations use the personal information that you supply to contact you only about the issues that you submit.

If you have a technical problem

If you have a technical problem or question, do not use the feedback methods that are provided for sending documentation comments. Instead, take one or more of the following actions:

- Go to the [IBM Support \(www.ibm.com/mysupport/s\)](http://www.ibm.com/mysupport/s).
- Contact your IBM service representative.
- Call IBM technical support.

Summary of changes

This information includes terminology, maintenance, and editorial changes. Technical changes or additions to the text and illustrations for the current edition are indicated by a vertical line to the left of the change.

Note: IBM z/OS policy for the integration of service information into the z/OS product documentation library is documented on the z/OS Internet Library under [IBM z/OS Product Documentation Update Policy](http://www.ibm.com/docs/en/zos/latest?topic=zos-product-documentation-update-policy) (www.ibm.com/docs/en/zos/latest?topic=zos-product-documentation-update-policy).

Summary of changes for z/OSMF Configuration Guide for 3.1

The following content is new, changed, or no longer included in z/OS 3.1.

New

The following content is new.

October 2024 release

Added configuration information to setup Monitor III REST Services for z/OS Data Gatherer. For more information, see [Chapter 31, “Configure z/OS Data Gatherer: Monitor III REST Services,”](#) on page 199.

August 2024 release

Added two more resources to the Resource authorizations for the z/OSMF core functions. For more information, see [Appendix A, “Security structures for z/OSMF,”](#) on page 371.

Added whitepaper for additional guidance on requirements and getting started with z/OSMF. For more information, see [“What setup is needed for z/OSMF?”](#) on page 8.

March 2024 release

The Parmlib Management component provides a solution to simplify and modernize the user experience of managing Parmlib. For more information, see [Chapter 34, “Configure the Parmlib Management service,”](#) on page 205.

February 2024 release

You can now configure your z/OSMF server to automatically send feedback monthly to IBM. For more information, see [Chapter 9, “Configure the z/OSMF settings service,”](#) on page 67.

September 2023 release

- The AI Control Interface for IBM z/OS automates and enhances z/OS system management tasks with prebuilt AI models. For more information, see [Chapter 33, “Configure the AI Control Interface,”](#) on page 203.
- The z/OSMF CFRM Policy Editor has received a major enhancement that enables performing CF structure sizing. For more information, see [Chapter 24, “Configure the Sysplex Management service,”](#) on page 121.

Changed

The following content is changed.

April 2025 release

- With APAR PH63501, the default Java level requirement was changed to IBM® Semeru Runtime Certified Edition for z/OS® Version 21. For more information, see [“Software prerequisites for z/OSMF”](#) on page 7.

March 2024 release

- With APAR PH59439, the default Java level requirement was changed to IBM® Semeru Runtime Certified Edition for z/OS® Version 17. For more information, see [“Software prerequisites for z/OSMF”](#) on page 7.
- Product names for CA Top Secret® and ACF2™ were updated to reflect an ownership change from CA Technologies to Broadcom. For more information, see [“Step 1: Run job IZUNUSEC to create the z/OSMF security authorizations”](#) on page 24.

September 2023 release

- IBM zERT Network Analyzer Enhanced Upgrade Support, see
 - [“Db2 for z/OS customization for the IBM zERT Network Analyzer task”](#) on page 131
 - [“Connect the IBM zERT Network Analyzer task with the Db2 for z/OS database”](#) on page 138
 - [“Deployment guidelines for IBM zERT Network Analyzer”](#) on page 139
- The Workload Management task no longer requires a Common Information Model (CIM) server as a prerequisite to be configured. For more information, see [Chapter 23, “Configure the Workload Management service,”](#) on page 117.
- The asynchronous job notifications of z/OS jobs REST services no longer require a Common Information Model (CIM) server as a prerequisite to be configured. For more information, see [Chapter 48, “Configuring your system for asynchronous job notifications,”](#) on page 255
- The web browsers that are supported by z/OSMF have changed. For more information, see [“Software prerequisites for z/OSMF”](#) on page 7.

Deleted

The following content was deleted.

September 2023 release

- None.

Information applicable to all releases

This document contains terminology, maintenance, and editorial changes. Technical changes or additions to the text and illustrations are indicated by a vertical line in the margin by the change.

The *Readers' Comments - We'd Like to Hear from You* section at the end of this publication has been replaced with a new section [“How to send your comments to IBM”](#) on page xix. The hardcopy mail-in form has been replaced with a page that provides information appropriate for submitting comments to IBM.

Part 1. Introduction to z/OSMF

An introduction to z/OSMF includes the following topics:

- [Chapter 1, “Overview of z/OSMF,” on page 3](#)
- [Chapter 2, “Selecting which z/OSMF services to add,” on page 11](#)
- [Chapter 3, “Ensure that the required z/OS elements are enabled,” on page 17](#)

Chapter 1. Overview of z/OSMF

IBM z/OS Management Facility (z/OSMF) provides system management functions in a task-oriented, web browser-based user interface with integrated user assistance so that you can more easily manage the day-to-day operations and administration of your mainframe z/OS systems. By streamlining some traditional tasks and automating others, z/OSMF can help to simplify some areas of z/OS system management.



Figure 1. z/OSMF desktop user interface (UI)

z/OSMF provides a framework for managing various aspects of a z/OS system through a web browser interface.

z/OSMF provides you with a single point of control for:

- Viewing, defining, and updating policies that affect system behavior
- Monitoring the performance of the systems in your enterprise
- Managing software that runs on z/OS
- Performing problem data management tasks
- Consolidating your z/OS management tools.

z/OSMF allows you to communicate with the z/OS system through a web browser, so you can access and manage your z/OS system from anywhere. Multiple users can log in to z/OSMF using different computers, different browsers, or multiple instances of the same browser.

This chapter introduces you to the major functions, architecture, and facilities of z/OSMF. Later chapters provide more details about configuration, administration, and troubleshooting. Usage information is provided in the z/OSMF online help.

z/OSMF and related system components

z/OSMF is shipped as part of z/OS.

z/OSMF can run on a parallel sysplex, monoplex, or XCF local mode environment.

Structurally, z/OSMF is a set of web applications that are hosted on your z/OS system. Depending on the task to be performed, z/OSMF interfaces with other z/OS components to offer you a simplified interface for performing those tasks.

The z/OS components make up the environment necessary for using the z/OSMF functions. z/OSMF neither requires a client workstation component nor does it provide one. All that is needed is a compatible web browser for accessing z/OSMF from your workstation.

z/OSMF includes the following software:

- z/OSMF server.
- WebSphere® Liberty profile, which provides an application server runtime environment for z/OSMF.
- Set of optional, system management functions or *services*, which you can enable in z/OSMF.
- Technologies for serving the web browser interface, such as JavaScript, Dojo, and Angular.

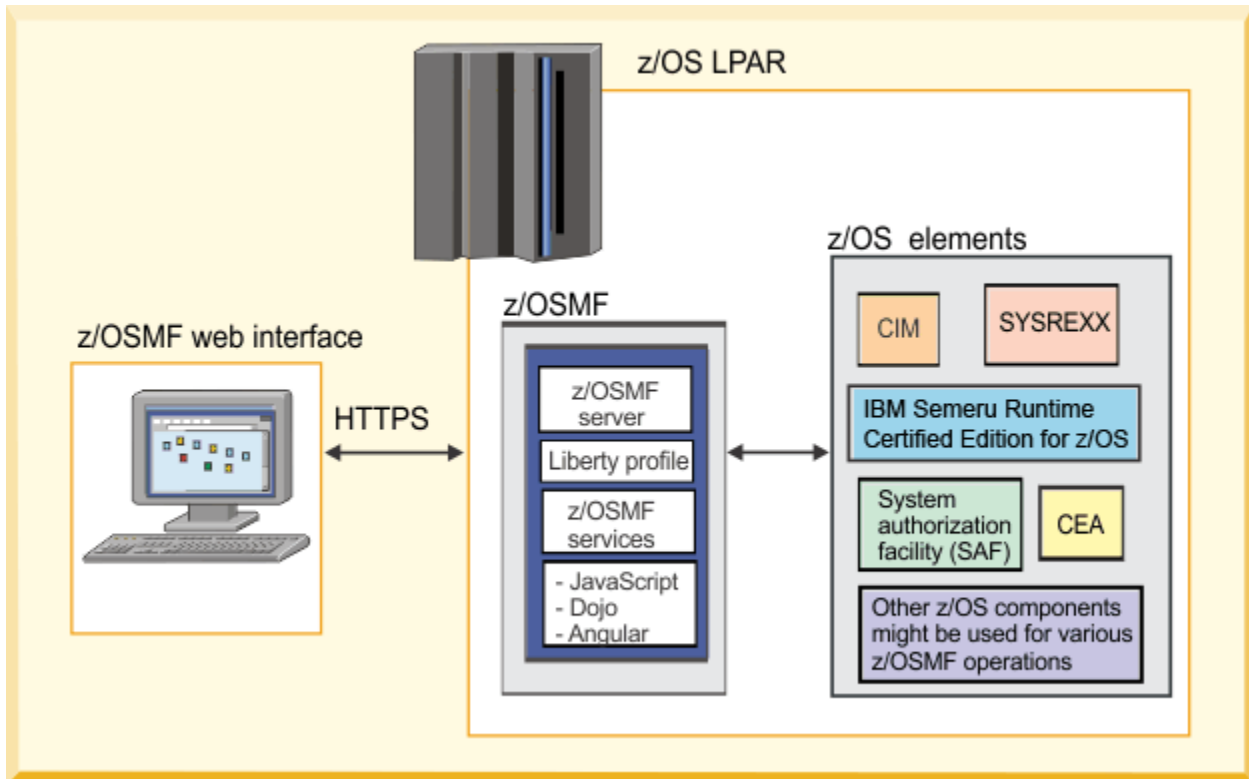


Figure 2. z/OSMF and related system components

The goal of this architecture is to provide simplified systems management function through a common, easy-to-use, graphical user interface. Figure 2 on page 4 shows a typical architecture and flow, starting with the user's browser session and continuing through z/OSMF, with information passed to various z/OS system components as needed.

Depending on the particular task that is being performed, z/OSMF uses various enabling technologies on z/OS, such as the following:

- IBM Semeru Runtime Certified Edition for z/OS. This IBM software development kit (SDK) contains the Java™ Runtime Environment (JRE) and other tools that support Java applications.
- Common Information Model (CIM) server on the host z/OS system. This component provides the z/OS data and administrative capability.
- Common event adapter (CEA). This component enables CIM providers to identify, receive, and process the selected z/OS events.
- System authorization facility (SAF). This component enables programs to use system authorization services to control access to resources, such as data sets and MVS commands. SAF either processes security authorization requests directly or works with RACF, or other external security manager, to process them.

- System REXX (SYSREXX). This component provides an infrastructure through which programs that are written in the REXX language can be run outside the normal TSO/E or batch environments, using a programming interface.

Security concepts in z/OSMF

As with other z/OS elements, security in z/OSMF is based on the concepts of *user authentication* and *user authorization*. User authentication occurs when a user attempts to log in to a system and the system's security management function examines the user's permission to do so. For z/OSMF, authentication occurs when the user attempts to log in to z/OSMF through a web browser. At the z/OSMF log-in page, the user enters a z/OS user ID and password in the appropriate input fields. The login request is verified by the z/OS host system's security management product (for example, RACF) through the SAF interface. This processing ensures that the user ID is known to the z/OS system, and the password is valid.

Besides the ability to authenticate, a would-be z/OSMF user requires authorization to one or more z/OSMF resources (tasks and links), which is necessary before the user can do useful work in z/OSMF (Figure 3 on page 5).

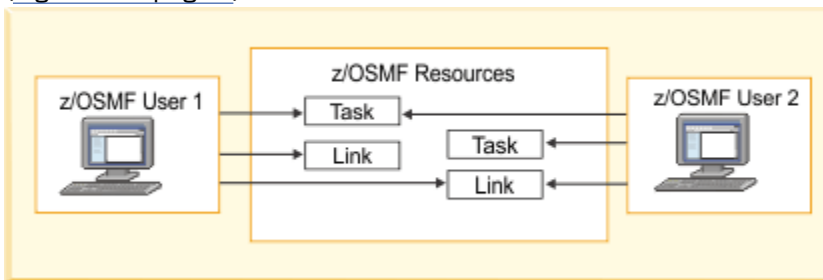


Figure 3. User authorizations in z/OSMF

Establishing security in z/OSMF requires the help of your security administrator. This person is responsible for ensuring that users and resources are defined in accordance with the security policies in use at your installation. For example, this work includes running security commands to allow z/OSMF to use various system functions to protect z/OSMF resources (tasks), and to authorize users to these resources.

z/OSMF also includes options for managing the access of *guest users*, that is, users who enter z/OSMF without authorization to tasks. Depending on how a guest user enters z/OSMF, the user is considered either authenticated or non-authenticated. A non-authenticated guest is a user who has displayed the **z/OSMF log-in** page, but has not logged in. An authenticated guest has logged in, but has not been granted authority to z/OSMF tasks.

Help with setting up security

IBM provides a set of security jobs in SYS1.SAMPLIB with sample RACF commands to help with your z/OSMF configuration and its prerequisites. Your security administrator can edit and run these jobs to secure various resources on the z/OS system:

- Job IZUNUSEC represents the authorizations that are needed to set up z/OSMF in a minimal configuration called the *nucleus*.
- Job IZUSEC represents the authorizations that are needed to set up z/OSMF in a full configuration: Nucleus, plus the core services.
- Each of the other IZUxxSEC jobs is associated with a particular z/OSMF service or an advanced configuration setup.

To create user authorizations for the services, your security administrator can use the IZUAUTH job in SYS1.SAMPLIB. It is assumed that your security administrator has a user ID with the RACF SPECIAL attribute. If your installation uses a security management product other than RACF, your security administrator can refer to the SAMPLIB jobs for examples when creating equivalent commands for the security management product on your system.

Table 1 on page 6 lists the z/OSMF security jobs that IBM provides in SYS1.SAMPLIB.

<i>Table 1. z/OSMF security jobs in SYS1.SAMPLIB</i>		
z/OSMF area to be configured	Description • Nucleus • Core service • Optional service • Advanced configuration	Security job in SYS1.SAMPLIB
Nucleus	Nucleus	IZUNUSEC
Notifications task	Core service	IZUNFSEC
z/OS data set and file REST services	Core service	IZURFSEC
z/OS jobs REST services	Core service	IZURJSEC
Swagger service	Core service	IZUSWSEC
TSO/E address space services	Core service	IZUTSSEC
z/OSMF administrative tasks	Core service	IZUATSEC
z/OSMF settings service	Core service	IZUSTSEC
z/OSMF Workflows task	Core service	IZUWFSEC
All of the above	Nucleus, plus all core services	IZUSEC
Capacity Provisioning service	Optional service	IZUCPSEC
Cloud Provisioning services	Optional service	IZUPRSEC
Console services	Optional service	IZUGCSEC
Incident Log service	Optional service	IZUILSEC
ISPF service	Optional service	IZUISSEC
Network Configuration Assistant service	Optional service	IZUCASEC
Resource Monitoring service	Optional service	IZURMSEC
Security Configuration Assistant	Optional service	IZUSASEC
Software Management service	Optional service	IZUDMSEC
Sysplex Management service	Optional service	• IZUSPSEC • IZUDCSEC
Workload Management service	Optional service	IZUWMSEC
Storage Management service	Optional service	IZUSGSEC
z/OS Encryption Readiness Technology (zERT) Network Analyzer	Optional service	IZUNASEC
Parmlib Management service	Optional service	IZUPMSEC
Use the autostart capability	Advanced configuration	IZUASSEC
Use ICSF services	Advanced configuration	IZUICSEC
Use AT-TLS connections	Advanced configuration	IZUTLSEC
Create the z/OSMF key ring and certificate	Advanced configuration	IZUSKSEC

z/OSMF does not support multilevel security

If the z/OSMF server is running in a *multilevel secure (MLS)* z/OS system, some z/OSMF functions might fail to work properly. The failures can occur because z/OSMF does not assign a SECLABEL to its started task address space. As a result, the functions that use inter-address space communication might fail because of a SECLABEL mismatch. For example, a failure can occur in the ISPF task because it starts a TSO address space with the SECLABEL of the current z/OSMF user. Other z/OSMF functions that might fail include the z/OS data set and file REST interface and the TSO/E address space services.

Software delivery options for z/OSMF

z/OSMF is a base element of the z/OS operating system; it is installed when you install z/OS. As part of z/OS, z/OSMF is available for installation through the ServerPac order delivery process or as a Custom-Built Product Delivery Option (CBPDO) software delivery package.

How your installation sets up z/OSMF — the procedures you use and the instructions that you follow — depends in part on the software delivery option that you use. These differences are explained as follows:

ServerPac users:

Use the jobs and documentation that is supplied with your ServerPac order to create an initial instance of z/OSMF. For customization guidance, see the copy of *ServerPac: Installing Your Order*, which is supplied with your order.

When you install your z/OS ServerPac order, a z/OSMF nucleus configuration is created on the target system through a ServerPac postinstallation job that uses IBM-supplied defaults. Another ServerPac job, RACFTGT, includes RACF commands for creating z/OSMF security definitions on the target system.

The default instance of z/OSMF does not include any of the optional services, such as Network Configuration Assistant, Incident Log, and so on. Refer to this document for information about performing various post-configuration actions, such as configuring the optional services.

CBPDO users:

If you receive z/OSMF in a Custom-Built Product Delivery Option (CBPDO) software delivery package, you require the planning and configuration information in this document. Your installation's system programmer can create a customized IZUPRMxx member to define an instance of z/OSMF on your system.

Hardware prerequisites for z/OSMF

For z/OS V3R1 with the IBM z14[®], IBM z15, or IBM z14 (z14) server, the z/OS operating system requires a minimum of 8 GB of processor storage to IPL. This minimum requirement helps to ensure that sufficient memory is available for the z/OS elements and components, and satisfies the minimum storage requirements for z/OSMF.

For z/OS running on an earlier processor, it is recommended that the target system for z/OSMF have at least 4 gigabytes (4G) of processor storage.

For a system that is running in an LPAR, ensure that the LPAR configuration has a Processor Capacity Index (PCI) of at least 50, though a smaller value might be sufficient on a system with an IBM z Integrated Information Processor (zIIP).

Software prerequisites for z/OSMF

It is recommended that you complete the planning for z/OSMF before you configure it.

Be sure to obtain the latest PTFs; see [“Receiving service updates for z/OSMF” on page 10](#).

The core services and optional services in z/OSMF have additional system set-up requirements. Enabling these services requires some customization of the z/OS host system, as described in [Part 3, “z/OSMF core services,” on page 59](#) and [Part 4, “z/OSMF optional services,” on page 81](#).

Minimum Java level

Java must be installed and operational on your z/OS system, z/OSMF V3R1 requires IBM Semeru Runtime Certified Edition for z/OS Version 11 at the required minimum level:

- IBM Semeru Runtime Certified Edition for z/OS, (formerly known as IBM 64-bit SDK for z/OS, Java Technology Edition), Version 11.0.19.1 (5655-DGJ).

With APAR PH59439, which is identified with SMP/E FIXCAT IBM.TargetSystem-RequiredService.Semeru.17, the default Java level for z/OSMF was changed to IBM® Semeru Runtime Certified Edition for z/OS® Version 17, the required IBM® Semeru Runtime Certified Edition for z/OS® Version 17 at the minimum service level:

- IBM Semeru Runtime Certified Edition for z/OS, Version 17.0.10.0 (5655-UA1).

With APAR PH63501, which is identified with SMP/E FIXCAT IBM.TargetSystem-RequiredService.Semeru.21, the default Java level for z/OSMF was changed to IBM® Semeru Runtime Certified Edition for z/OS® Version 21, the required IBM® Semeru Runtime Certified Edition for z/OS® Version 21 at the minimum service level:

- IBM Semeru Runtime Certified Edition for z/OS, Version 21.0.6.0 (5655-JB1)

This set-up must be done before you configure z/OSMF. By default, the SDK resides in the directory `/usr/lpp/java/J21.0_64` on your system. If you installed it in another location, be sure to include the `JAVA_HOME` statement in your `IZUPRMxx` parmlib member, as shown in [“IZUPRMxx reference information” on page 35](#).

Note: The default Java level for z/OSMF was changed to IBM® Semeru Runtime Certified Edition for z/OS® Version 21 with APAR PH63501 because IBM® Semeru Runtime Certified Edition for z/OS® Version 17 is planned to be end of marketing on October 6, 2025. IBM® Semeru Runtime Certified Edition for z/OS® Version 17 is planned to be end of support on September 30, 2026. If you upgrade to z/OS 3.1 after October 6, 2025, you can use and install IBM® Semeru Runtime Certified Edition for z/OS® Version 21 without needing IBM® Semeru Runtime Certified Edition for z/OS® Version 17 to satisfy this z/OSMF dependency.

WebSphere® Liberty profile

z/OSMF uses the Liberty Profile that is supplied with z/OS, rather than its own copy of Liberty. The WebSphere Liberty profile must be mounted on your z/OS system. The default mount point is: `/usr/lpp/liberty_zos`. To determine whether WebSphere® Liberty profile is mounted, check for the existence of the mount point directory on your z/OS system.

If WebSphere® Liberty profile is mounted at a non-default location, you must specify the location in the `IZUSVR1` started procedure on the keyword `WLPDIR=`. For more information, see [“IZUSVR reference information” on page 44](#).

Note: Whenever you apply PTFs for z/OSMF, you might be prompted to install outstanding WebSphere Liberty service. It is recommended that you do so to maintain z/OSMF functionality.

Web browser

The following web browsers are supported by z/OSMF, and are recommended for best results:

- Google Chrome V111 or later (Windows 11, macOS 12.6 and later version)
- Mozilla Firefox 102.9 or later (Windows 11, macOS 12.6 and later version)
- Microsoft Edge V111 (Windows 11)

What setup is needed for z/OSMF?

As a base element of the operating system, z/OSMF is installed when you install z/OS. By default, z/OSMF is installed into the z/OS root file system, in the directory `/usr/lpp/zosmf`.

Enabling z/OSMF on your system involves the following phases:

- Planning for z/OSMF. The z/OSMF server requires a minimum of 4 GB of system memory to be configured.
- Configuring the z/OSMF nucleus on your system and then adding core services and optional services. This phase requires certain z/OS resources to be set up, commands to be run, and security setup to be performed for RACF (or the equivalent). Information for these activities is provided in this document. A suggested approach for getting started is described in [“Faster set-up with a z/OSMF Lite configuration” on page 9](#).

For a full list of requirements, please refer to the whitepaper found here: <https://www.ibm.com/support/pages/node/6355781>.

Using z/OSMF requires sufficient authority in z/OS. Specifically, on the z/OS system to be managed, the resources to be accessed on behalf of users (data sets, operator commands, and so on) are secured through the external security manager at your installation, such as RACF. Your installation's security administrator must create the authorizations in your external security manager. To assist your security administrator, IBM provides sample jobs for z/OSMF in SYS1.SAMPLIB. More information about security setup is provided in [“Security concepts in z/OSMF” on page 5](#).

Unless you choose to manage the start-up and shutdown of the z/OSMF server through an automation product, z/OSMF is started automatically when you IPL your z/OS system. This behavior, which is referred to as *z/OSMF autostart*, means that z/OSMF is available for use as soon as the system is up. To make the best use of the z/OSMF autostart capability, you must plan how to deploy one or more z/OSMF servers in your environment. Generally, having one z/OSMF server active in a sysplex or monoplex is sufficient, but you might choose to have more, based on your workload requirements. The goal is to ensure that at least one z/OSMF server is always active in your environment.

For more information, see [Chapter 36, “Autostart concepts in z/OSMF,” on page 213](#).

Faster set-up with a z/OSMF Lite configuration

For a quick start with z/OSMF, you can set up z/OSMF in a *z/OSMF Lite* configuration. This approach requires only a minimal amount of z/OS customization, but provides the key z/OSMF functions that are required by many z/OS installations.

In short, *z/OSMF Lite* means configuring the z/OSMF nucleus on your system and then adding only the core services that you require. By following the steps in [Part 2, “The z/OSMF nucleus,” on page 21](#) and [Part 3, “z/OSMF core services,” on page 59](#), you can quickly enable z/OSMF on your z/OS system. Later, you can easily expand z/OSMF Lite into a full function z/OSMF configuration by adding optional services.

It takes approximately 2 - 3 hours to set up a z/OSMF Lite configuration. This work requires certain z/OS resources to be set up, commands to be run, and security setup to be performed. Some steps might require the assistance of your security administrator. IBM provides a program, IZUNUSEC, to help the security administrator set up basic security for a z/OSMF Lite configuration.

A z/OSMF Lite configuration is applicable to any future expansions you make to z/OSMF, such as adding the optional services.

Assumptions

A z/OSMF Lite configuration is intended for a first-time z/OSMF setup. If z/OSMF is already configured on your system, you do not need to create a z/OSMF Lite configuration.

System defaults are used for the z/OSMF environmental settings. Wherever possible, it is recommended that you use the default values. However, if necessary, you can override the defaults by supplying an IZUPRMxx parmlib member, as described in [“IZUPRMxx reference information” on page 35](#).

It is recommended that you use the following procedures, which are provided by IBM:

- Started procedures IZUSVR1 and IZUANG1
- Logon procedure IZUFPROC.

Information about installing these procedures is provided in [“Step 3: Copy the z/OSMF procedures into JES PROCLIB”](#) on page 27.

Preparing your workstation for z/OSMF

In preparing your workstation for use with z/OSMF, observe the considerations listed in this section.

- Your workstation requires a compatible operating system and web browser. For more information, see [“Software prerequisites for z/OSMF”](#) on page 7.
- z/OSMF requires a minimum screen resolution of 1024 by 768 pixels. If your workstation is set to a lower resolution, some content might not be displayed.
- Ensure that your browser is enabled for JavaScript. For instructions, see [Table 46 on page 278](#) or [Table 47 on page 280](#), as appropriate.
- z/OSMF uses session cookies to track which users are logged in from a specific browser. If you want to use multiple z/OSMF servers from the same workstation, you might need to either launch another browser instance (as with Microsoft Edge), or, configure another browser profile (as with Firefox). For information about creating Firefox profiles, see the Mozilla web site: <http://www.mozilla.com>.

After you have configured z/OSMF, you can use the included environment checker tool to verify your browser and workstation settings at any time. For more information, see [“Verifying your workstation with the environment checker”](#) on page 276.

Migrating to a new release

When you migrate to a new release of z/OSMF, you can reuse much of the customization from your current configuration.

The migration actions for z/OSMF are described in [z/OS Upgrade Workflow](#).

Receiving service updates for z/OSMF

IBM ships service for z/OSMF in the form of program temporary fixes (PTFs). As a z/OS element, z/OSMF should be updated on the same schedule that you use to update the rest of z/OS.

Here are some useful links:

- IBMLink web site at [IBM ServiceLink \(www.ibm.com/ibmlink/servicelink\)](http://www.ibm.com/ibmlink/servicelink).
- Information about IBM Recommended Service Upgrade (RSU) and service testing at [Consolidated Service Test and the RSU \(www.ibm.com/support/pages/ibm-zos-consolidated-service-test-and-rsu\)](http://www.ibm.com/support/pages/ibm-zos-consolidated-service-test-and-rsu).

Chapter 2. Selecting which z/OSMF services to add

To enable z/OSMF for useful work, you must add function to the z/OSMF nucleus through the addition of core (recommended) services and optional (suggested) services. Which services you add depends on your goals. To help you decide, this topic provides a functional overview of each service. After you choose the services you require, you then configure those services.

The z/OSMF nucleus includes only the following functions, which are enabled when the z/OSMF server is started.

- WebSphere Liberty profile runtime
- z/OSMF desktop user interface (UI)
- z/OSMF online help system.

After creating the nucleus, most installations will choose to configure additional, core services, which include the Workflows task, and a number of REST API interfaces. These services are described in [Part 3, “z/OSMF core services,”](#) on page 59.

Beyond the core services, you can further extend the functions of z/OSMF by adding optional services, which include a number of z/OS system management tasks, such as Network Configuration Assistant, ISPF, and Workload Management. These services are described in [Part 4, “z/OSMF optional services,”](#) on page 81.

Your decision of which services to configure will depend in part on your installation's readiness to perform the various z/OS system customizations associated with each service. When planning for the services, review the dependencies and system setup requirements for each service, as described in this book. You might find it easier to start with services that require little or no system customization, such as Configuration Assistant or ISPF, and then progress to plug-ins with more extensive requirements, such as Incident Log.

In general, enabling a service involves the following activities:

- Configuring any prerequisite services that might be required by the service.
- Creating security profiles for the z/OSMF tasks and REST services that are associated with the service. IBM provides a set of IZUnnSEC jobs in SYS1.SAMPLIB with RACF commands to help with performing these changes. Each IZUnnSEC job is associated with a service, as described in [“Security concepts in z/OSMF”](#) on page 5.
- Performing the various z/OS system customization updates, if any, that are associated with each service.

Table 2 on page 11 shows the z/OSMF services that are available in the current release of z/OS. The table provides a brief description of each task; indicates whether the service includes programming interfaces (REST APIs); and indicates the relative complexity of configuring the service (Low, Medium, or High). Notice that some services are footnoted; in earlier releases of z/OSMF, these services were enabled by default. Therefore, these services might be enabled already if you are upgrading from an earlier release of z/OSMF.

Table 2. z/OSMF core services and optional services				
Service Name	Description	Core or Optional	REST APIs (Y/N)	Complexity of Set Up: Low, Medium, or High
z/OSMF Administration tasks¹	Allows the user to perform administrative work on behalf of z/OSMF users.	Core	Y	Low

Table 2. z/OSMF core services and optional services (continued)

Service Name	Description	Core or Optional	REST APIs (Y/N)	Complexity of Set Up: Low, Medium, or High
z/OSMF Workflows (including the Workflow Editor)¹	Simplifies tasks through guided step-based workflows, and provides administrative functions for assigning workflow responsibilities and following progress.	Core	Y	Low
Notifications ¹	Allows you to view and work with the notifications that are assigned to you.	Core	Y	Low
z/OSMF Settings¹	Define global settings for z/OSMF, such as FTP servers for use by other z/OSMF services, Notification Settings.	Core	Y	Low
Swagger (API Discovery) ¹	You can use the Swagger interface to display information about REST APIs.	Core	N	Low
z/OS jobs REST services¹	A set of REST services for working with batch jobs on a z/OS system	Core	Y	Low
z/OS data set and file REST services¹	A set of REST services for working with data sets and UNIX files on a z/OS system.	Core	Y	Low
TSO/E address space services ¹	A set of REST services for working with TSO/E address spaces on a z/OS system.	Core	Y	Low
z/OSMF Security Configuration Assistant	Provides a visual framework for examining the different elements of z/OSMF security.	Optional	N	Low
Network Configuration Assistant	Configure TCP/IP policy-based networking functions.	Optional	N	Low
Software Management	Manage your z/OS software inventory, deploy SMP/E packaged and installed software, and generate reports about your software.	Optional	Y	Low
ISPF	Access traditional ISPF applications through a web browser UI.	Optional	N	Low
Resource Monitoring	- Monitor the performance of the z/OS, AIX®, Linux, and Windows systems in your enterprise. - Quickly assess the workload performance on the systems in your enterprise, and define the systems to be monitored.	Optional	N	Medium
Console services ¹	Provides functions for working with z/OS consoles, such as viewing system messages and entering system commands.	Optional	Y	Medium

Table 2. z/OSMF core services and optional services (continued)				
Service Name	Description	Core or Optional	REST APIs (Y/N)	Complexity of Set Up: Low, Medium, or High
Workload Management	Administer and operate WLM, and manage WLM service definitions and policies.	Optional	N	Medium
Sysplex Management	Manage the sysplex resources in your enterprise.	Optional	Y	Medium
Capacity Provisioning	Monitor your systems for capacity bottlenecks, and manage the physical capacity of your servers and the defined capacity and group capacity limits in use.	Optional	N	Medium
IBM zERT Network Analyzer	Analyze SMF records to identify the cryptographic protection attributes of TCP and Enterprise Extender (EE) traffic in your enterprise.	Optional	N	Medium
Cloud Provisioning services	Perform software provisioning, such as creating instances of CICS®, IBM Db2, IMS, IBM MQ, and IBM WebSphere Application Server, and creating middleware resources, such as IBM MQ queues, CICS regions, and Db2 databases.	Optional	Y	High
Incident Log	Diagnose system problems, and send diagnostic data to IBM or other vendors for further diagnostics.	Optional	N	High
Storage management services	A set of REST services for working with system storage elements.	Optional	Y	Low
Parmlib Management	Manage the Parmlib resources in your enterprise.	Optional	Y	Low

Table note: ¹ Was enabled by default in previous releases of z/OSMF. This service might be already enabled on your system if you are upgrading from an earlier release of z/OSMF.

z/OSMF dependencies matrix

Some z/OSMF services require other z/OSMF services to be enabled. Therefore, you might need to configure more services than just the ones you plan to use. [Table 3 on page 14](#) shows which services require other z/OSMF services to be enabled. Where applicable, the description of each service notes the dependencies that it might have for other z/OSMF services.

Table 3. z/OSMF dependencies matrix

To use this z/OSMF service...	... Configure these required services	You might also need to configure these optional services, depending on your intended use.
Capacity Provisioning	Common Information Model (CIM) server	None.
Cloud Provisioning	- Console services (UI and API) - Network Configuration Assistant - Notifications - Swagger (API Discovery) - z/OSMF Settings - z/OSMF Workflows	- Common Information Model (CIM) server, which is used by Resource Monitoring and Workload Management. - z/OS data set and file REST services because these services are used by z/OSMF Workflows. - TSO/E address space services because these services are used by the console services. - Resource Monitoring because it is used by Cloud Provisioning to obtain CPU and memory metering data. - Workload Management because it is used by Cloud Provisioning to set CPU and memory capping.
Console services	- Common event adapter (CEA) - TSO/E address space services - z/OSMF Settings	None.
Incident Log	- Common event adapter (CEA) - z/OSMF Settings - Common Information Model (CIM) server	None.
ISPF	- Common event adapter (CEA) - TSO/E address space services	None.
Network Configuration Assistant	None.	- Notifications, if your installation is using IBM Cloud Provisioning and Management for z/OS. - z/OSMF Settings, if your installation is using IBM Cloud Provisioning and Management for z/OS. - z/OSMF Workflows to assist with the initial setup of Network Configuration Assistant, or to assist with the initial setup or particular configuration steps for IBM Cloud Provisioning and Management for z/OS networking.
Notifications	None.	None.
Parmlib Management	TSO/E address space services	None.

<i>Table 3. z/OSMF dependencies matrix (continued)</i>		
To use this z/OSMF service...	... Configure these required services	You might also need to configure these optional services, depending on your intended use.
Resource Monitoring	• None.	• Workload Management because the Resource Monitoring task can open to the Workload Management task. • z/OSMF administration tasks because the Resource Monitoring task uses the Application Linking task to launch the Workload Management task.
Software Management	• TSO/E address space services • z/OS data set and file REST services • z/OS jobs REST services	• If you want to open a workflow from a software instance in Software Management, you require the z/OSMF Workflows service.
Storage Management service	• None.	Console service when the user activates an SCDS or gets an activation result.
Swagger (API Discovery)	• z/OS jobs REST services • z/OS data set and file REST services	• None.
Sysplex Management	• Common event adapter (CEA) • z/OSMF Settings • z/OS data set and file REST services • TSO/E address space services • Console services (UI and API)	• Discover CPC function, if you want the ability to query the topology of interconnected CPCs and LPARs in the sysplex.
z/OS jobs REST services	• None.	• Common Information Model (CIM) server, if your installation uses JES3.
z/OS data set and file REST services	• Common event adapter (CEA)	• None.
TSO/E address space services	• Common event adapter (CEA)	• None.
Workload Management	• None.	• Resource Monitoring because the Workload Management task can open to the Resource Monitoring task. • z/OSMF administration tasks because the Workload Management task uses the Application Linking task to launch the Resource Management task.
z/OSMF Administrator tasks	• TSO/E address space services	• None.
z/OSMF Settings	• None.	• None.

Table 3. z/OSMF dependencies matrix (continued)

To use this z/OSMF service...	... Configure these required services	You might also need to configure these optional services, depending on your intended use.
z/OSMF Workflows	• Common event adapter (CEA) • Notifications • z/OSMF Settings • TSO/E address space services • z/OS data set and file REST services • z/OS jobs REST services	• None.

Chapter 3. Ensure that the required z/OS elements are enabled

For most installations, it is recommended that you ensure that the following z/OS element is enabled:

- Common event adapter (CEA) must be enabled in full function mode, as described in [“Ensure that common event adapter \(CEA\) is configured and active”](#) on page 17.

Ensure that common event adapter (CEA) is configured and active

A number of z/OSMF services require the common event adapter (CEA) component to be enabled and running on your system in **full function mode**.

Specifically, the following z/OSMF services use CEA:

- z/OSMF Workflows
- z/OS data set and file REST services
- TSO/E address space services
- Software Management
- ISPF
- Console services
- Sysplex Management
- Incident Log

Usually, the CEA address space is started automatically during z/OS initialization. However, if your installation has stopped CEA, it is recommended that you restart it. For information about how to configure CEA, see [z/OS Planning for Installation](#).

For more information about using CEA with z/OSMF, see the following topics:

- [“Create the security authorizations for CEA”](#) on page 17
- [“How to check whether CEA is active”](#) on page 18
- [“Starting the CEA address space”](#) on page 18

Create the security authorizations for CEA

If your z/OSMF configuration includes services that use CEA on the z/OS host system, users of the services require the proper level of access to CEA resources. In SYS1.SAMPLIB, IBM provides the CEASEC job and the z/OSMF security sample jobs to help you create these authorizations. For the SAMPLIB jobs to use for each z/OSMF service, see [Table 4 on page 17](#).

Also, if your installation plans to use the ISPF task, you must ensure that the TRUSTED attribute is assigned to the CEA started task, as described in [“Updating z/OS for the ISPF service”](#) on page 101.

Table 4. z/OSMF services that depend on CEA being enabled on your z/OS system		
Service Name	Description	Security job in SYS1.SAMPLIB
Common event adapter (CEA)	CEA provides the ability to deliver z/OS events to clients, such as the CIM server, and create or manage TSO user address spaces under the ISPF task.	CEASEC

Table 4. z/OSMF services that depend on CEA being enabled on your z/OS system (continued)		
Service Name	Description	Security job in SYS1.SAMPLIB
z/OSMF Workflows	Simplifies tasks through guided step-based workflows, and provides administrative functions for assigning workflow responsibilities and following progress.	IZUWFSEC
z/OS data set and file REST services	A set of REST services for working with data sets and UNIX files on a z/OS system.	IZURFSEC
TSO/E address space services	A set of REST services for working with TSO/E address spaces on a z/OS system.	IZUTSSEC
Software Management	Manage your z/OS software inventory, deploy SMP/E packaged and installed software, and generate reports about your software.	IZUDMSEC
ISPF	Access traditional ISPF applications through a web browser UI.	IZUISSEC
Console services	Provides functions for working with z/OS consoles, such as viewing system messages and entering system commands.	IZUGCSEC
Sysplex Management	Manage the sysplex resources in your enterprise.	• IZUSPSEC • IZUDCSEC
Incident Log	Diagnose system problems, and send diagnostic data to IBM or other vendors for further diagnostics.	IZUILSEC

How to check whether CEA is active

To determine whether the CEA address space is active, enter the following command:

```
D A,CEA
```

Figure 4 on page 18 shows the expected results:

```

IEE115I 15.32.17 2010.132 ACTIVITY 109
  JOBS      M/S      TS USERS      SYSAS      INITS      ACTIVE/MAX VTAM      OAS
00018      00040      00002      00043      00246      00002/03500      00043
  CEA       CEA       IEFPROC      NSWPR*0      A=001A      PER=YES      SMC=000
                        PGN=N/A      DMN=N/A      AFF=NONE
                        CT=000.425S      ET=45.32.29
                        WKL=SYSTEM      SCL=SYSTEM      P=1
                        RGP=N/A      SRVR=NO      QSC=NO
                        ADDR SPACE ASTE=05A34680
                        DSPNAME=CEACTDSP ASTE=1002D600
                        DSPNAME=CEAPDWB  ASTE=1002D580
                        DSPNAME=CEACADS  ASTE=7EF42700
                        DSPNAME=CEACOMP  ASTE=1002D480

```

Figure 4. Expected results from the **D A,CEA** command

Starting the CEA address space

To start the CEA address space, enter the following command from the operator console: **START CEA**

It is recommended that you edit your active IEASYSxx parmlib member to identify the CEAPRMxx parmlib member to be used for the next IPL of the system. Specify the CEAPRMxx member suffix on the CEA=xx statement of IEASYSxx. The member that is specified in IEASYSxx will be in effect after the next system IPL.

To dynamically change the active CEA configuration, enter the MODIFY command, as follows: F CEA,CEA=xx, where xx is the suffix of the CEAPRMxx member to be used.

You can specify multiple CEAPRMxx members, for example:

```
F CEA,CEA=(01,02,03)
```

To check the resulting CEA configuration, enter the following command:

```
F CEA,D,PARMS
```

Identifying the CEAPRMxx member to use at IPL time

To ensure that common event adapter (CEA) is always active and using the correct settings, it is recommended that you edit your active IEASYSxx parmlib member to identify the CEAPRMxx parmlib member to use for the next IPL of the system. Specify the CEAPRMxx member suffix on the CEA=xx statement of IEASYSxx.

Modifying the common event adapter (CEA) settings

At any time during z/OSMF operations, you can modify CEA settings by selecting a new CEAPRMxx member. You can do so dynamically, that is, without having to restart CEA.

You might want to update the CEA settings to do the following:

- Add an eighth volume to CEA. Earlier, during the configuration prompts, if you provided VOLSER values to be used in the target CEAPRMxx member, you specified up to seven volumes as input. If you want to add an eighth volume, for example, to allow more space for diagnostic snapshots, you can update the CEAPRMxx member manually.
- Adjust the duration of OPERLOG or logrec that the system should capture for all future incidents.

If needed, you can restart CEA and specify a new CEAPRMxx member dynamically. To do so, enter the START command, as follows: START CEA. Then, enter the MODIFY command, as follows:

```
F CEA,CEA=xx
```

where xx represents the CEAPRMxx member suffix. You can specify multiple CEAPRMxx members, for example: F CEA,CEA=(01,02,03)

To check the results of these commands, enter the MODIFY command, as follows:

```
F CEA,D,PARMS
```

For information about how to configure CEA, see [z/OS Planning for Installation](#).

Part 2. The z/OSMF nucleus

In this part, you create the z/OSMF nucleus.

You can skip this part of the configuration process if you:

- Already have z/OSMF configured and running on at least one system in your sysplex.
- Are migrating from an earlier release of z/OS that includes an already-configured z/OSMF.
- Are receiving z/OSMF as part of a z/OS ServerPac order and you plan to run the RACFTGT job (or its equivalent) to create the security authorizations for your order.

If any of these conditions are true, you can skip this part. Instead, continue with the instructions for adding z/OSMF functions in [Part 3, “z/OSMF core services,” on page 59](#) and [Part 4, “z/OSMF optional services,” on page 81](#).

Chapter 4. Create a z/OSMF nucleus on your system

This information is intended for a first-time z/OSMF setup. If z/OSMF is already configured on your system, you do not need to create a z/OSMF nucleus on your system.

IBM provides a sample job, IZUNUSEC, to help you set up basic security for a z/OSMF nucleus configuration.

System defaults are used for the z/OSMF environmental settings. Wherever possible, it is recommended that you use the default values. However, if necessary you can override the defaults by supplying an IZUPRMxx member, as described in [“IZUPRMxx reference information” on page 35](#).

Creating a z/OSMF nucleus on your system can be accomplished in a short time by following these steps:

- [“Step 1: Run job IZUNUSEC to create the z/OSMF security authorizations” on page 24](#)
- [“Step 2: Run job IZUMKFS to create the z/OSMF user file system” on page 26](#)
- [“Step 3: Copy the z/OSMF procedures into JES PROCLIB” on page 27](#)
- [“Step 4: Start the z/OSMF server” on page 28](#)
- [“Step 5: Log in to z/OSMF” on page 29](#)
- [“Step 6: Mount the z/OSMF user file system at IPL time” on page 30](#)

The sample jobs that you might use are available from SYS1.SAMPLIB.

Additional usage and reference information is provided in the following topics:

- [“Stopping and starting z/OSMF manually” on page 31](#)
- [“Displaying the z/OSMF server settings” on page 32](#)
- [“IZUPRMxx reference information” on page 35](#)
- [“IZUSVR reference information” on page 44](#)

System setup requirements

This document assumes that the following is true of the z/OS host system:

- Port 443 is available for use. To check, enter either of the following TSO/E commands to determine whether the port is being used: NETSTAT SOCKET or NETSTAT BYTE
- The system host name is unique and maps to the system on which z/OSMF is being installed. To retrieve this value, enter either "hostname" z/OS UNIX command or TSO/E command "HOMETEST". If your system uses another method of assigning the system name, such as a multi-home stack, dynamic VIPA, or System Director, see [Chapter 40, “Configuring z/OSMF to provide redundancy for subset of z/OSMF function,” on page 231](#).
- The global mount point exists. On a z/OS 3.1 system, the system includes this directory by default at the following location: /global/zosmf/.

If you find that a different value is used on your z/OS system, you can edit the IZUPRMxx parmlib member to specify the correct setting. For details, see [“IZUPRMxx reference information” on page 35](#).

Dependencies on other z/OSMF services

Some functions in the z/OSMF desktop have dependencies on z/OSMF services. To use these desktop functions, you must enable the required service, as follows:

Data set search function

Requires the z/OS data set and file REST services, which are enabled by running the IZURFSEC security job.

Change password function

Requires the TSO/E address space services, which are enabled by running the IZUTSSEC security job.

You can perform this setup after you create a z/OSMF nucleus on your system.

Step 1: Run job IZUNUSEC to create the z/OSMF security authorizations

The security job IZUNUSEC contains a minimal set of RACF® commands for creating security profiles for the z/OSMF nucleus. The profiles are used to protect the resources that are used by the z/OSMF server, and to grant users access to the z/OSMF core functions. IZUNUSEC is a simplified version of the sample job IZUSEC, which is intended for a full installation of z/OSMF (the nucleus and core services).

Before you begin

It is strongly recommended that your security administrator review the contents of the job before running it, and make any changes that are needed to be consistent with your installation's security policies.

In a z/OS ServerPac order, the z/OSMF security profiles are created by the RACFTGT job. Thus, if you received z/OSMF as part of a ServerPac order, and you already ran RACFTGT (or its equivalent), you can skip this step.

If your installation uses an external security manager other than RACF, you must provide equivalent commands for your environment. For more information, see the following Broadcom product documentation:

- [Configure z/OS Management Facility for Top Secret for z/OS.](#)
- [Configure z/OS Management Facility for ACF2.](#)

About this step

The security job IZUNUSEC assigns the user ID IZUSVR to the z/OSMF started tasks, IZUANG1 and IZUSVR1. You can assign different user IDs by using profiles in the RACF STARTED class or by using the RACF Started Procedures Table. For more information about assigning user IDs to started tasks, see [z/OS Security Server RACF Security Administrator's Guide](#).

The security job IZUNUSEC also contains sample RACF commands for defining the z/OSMF started procedures to the STARTED class. [Figure 5 on page 24](#) shows the commands that are provided in the job.

```
/* Define the STARTED profiles for the z/OSMF server *  
RDEFINE STARTED IZUSVR1.* UACC(NONE) STDATA(USER(IZUSVR) +  
GROUP(IZUADMIN) PRIVILEGED(NO) TRUSTED(NO) TRACE(YES))  
RDEFINE STARTED IZUANG1.* UACC(NONE) STDATA(USER(IZUSVR) +  
GROUP(IZUADMIN) PRIVILEGED(NO) TRUSTED(NO) TRACE(YES))
```

Figure 5. RACF commands for defining the started procedures to the STARTED class

Notes:

1. When you use the STARTED class, you can modify the security definitions for started procedures dynamically. For more information, see the topic on using started procedures in [z/OS Security Server RACF Security Administrator's Guide](#).
2. If your installation uses hardware cryptography with z/OS Integrated Cryptographic Service Facility (ICSF), the z/OSMF server uses a number of ICSF callable services, such as CSFRNGL, CSFDSV, CSFOWH, CSFIQF, and others. These services might be protected through profiles that are established in your external security manager. Because z/OSMF uses these services, you must permit the z/OSMF started task user ID to these profiles. For more information, see [Chapter 44, "Considerations for using ICSF services," on page 247](#).
3. Before you submit the job, review the certificate expiration date in the RACDCERT/GENCERT command to ensure that the certificate is not approaching expiration. If the expiration date is less than a year from the current date, set this value to a date that is at least one year after the current date.

Before running the job

In most cases, you can run the IZUNUSEC security job without modification. To verify that the job is okay to run as is, ask your security administrator to review the job and modify it as necessary for your security environment. If security is not a concern for the host system, you can run the job without modification.

Running the job

1. Make a copy of job IZUNUSEC from SYS1.SAMPLIB.
2. Review and edit the job, if necessary.
3. Submit IZUNUSEC as a batch job on your z/OS system.
4. Connect your user ID to IZUADMIN group. For a system with RACF as the external security manager, you can use the following command:

```
CONNECT userid GROUP(IZUADMIN)
```

Where *userid* is your user ID.

Tip: Job IZUAUTH in SYS1.SAMPLIB includes sample RACF commands for connecting users to the various groups that you might use for z/OSMF: Capacity Provisioning groups (CPOCTRL and CPOQUERY), Workload Management group (WLMGRP), and the z/OSMF user group (IZUUSER) and administrator group (IZUADMIN). To use this job, make a copy of it, then uncomment the commands that you want to use and submit the job.

Results

Ensure that the IZUNUSEC job completes with return code 0000.

If the job is run more than once, message IKJ56702I *INVALID data* is issued for any user IDs or groups that were defined previously. You can ignore this message.

Common errors

Table 5 on page 25 shows the most common errors for this step and suggests resolutions.

Table 5. Common errors when you run job IZUNUSEC to create security		
Symptom	Cause	Resolution
Job does not run.	The job issues the following TSO RACF commands: • RACDCERT • RACMAP • RACPRIV • RACLINK However, these commands are not defined to the TSO/E parmlib member IKJTSOxx on your system.	The TSO/E parmlib member IKJTSOxx is used to identify the TSO/E commands and programs the system is to use. Ask the system programmer to check the active IKJTSOxx member for your system to ensure that it includes these RACF command names. To list the values in the active IKJTSOxx parmlib member, you can enter either of the following commands: DISPLAY IKJTSO or PARMLIB LIST.
Message IKJ56702I: INVALID data is issued	The job is submitted more than once.	You can ignore this message.

Table 5. Common errors when you run job IZUNUSEC to create security (continued)

Symptom	Cause	Resolution
Job fails with an authorization error.	Your user ID lacks superuser authority.	Contact your security admin to run IZUNUSEC. If you are using RACF®, select a user ID with SPECIAL attribute, which can issue all RACF® commands.
Job fails with an authorization error.	Your installation uses the RACF PROTECT-ALL option.	If your installation uses PROTECT-ALL, you must define a CEA.* data set profile to RACF and permit CEA and the z/OSMF installer user ID. For example: <pre>ADDSD 'CEA.*' UACC(NONE) PERMIT 'CEA.*' ID(CEA) ACCESS(ALTER) PERMIT 'CEA.*' ID(USER-ID) ACCESS(ALTER)</pre>
ADDGROUP and ADDUSER commands fail.	The automatic GID and UID assignment is required.	Define SHARED.IDS and BPX.NEXT.USER profiles to enable the use of AUTOUID and AUTOUID.

Step 2: Run job IZUMKFS to create the z/OSMF user file system

In this step, you run job IZUMKFS, which initializes the z/OSMF data directory (sometimes called the *user directory*). This file system contains the configuration settings and the persistence information for z/OSMF.

Before you begin

Run this job only if you did not receive z/OSMF as part of a z/OS ServerPac order. ServerPac includes a job to create the z/OSMF data directory.

To perform this step, you need a user ID with “superuser” authority on the z/OS host system. For more information about how to define a user with superuser authority, see the publication [z/OS UNIX System Services](#).

About this step

The job IZUMKFS mounts the z/OSMF user file system at the following mount point `/global/zosmf`.

Running the job

1. In the system library SYS1.SAMPLIB, locate job IZUMKFS.
2. Copy the job.
3. Review and edit the job:
 - Modify the job information so that the job can run on your system.
 - Specify a volume serial (VOLSER) to be used for allocating a data set for the z/OSMF data directory.
4. Submit IZUMKFS as a batch job on your z/OS system.

Results

The z/OSMF file system is allocated, formatted, and mounted, and the necessary directories are created.

To verify that this work is done, locate the following messages in IZUMKFS job output.

```
IDC0002I IDCAMS PROCESSING COMPLETE. MAX CONDITION CODE WAS 0.
IOEZ00077I HFS-compatibility aggregate izu.sizusrd has been successfully created.
```


Common errors

Table 6 on page 27 shows the most common errors for this step and suggests resolutions.

Table 6. Common errors when you run job IZUMKFS to create the z/OSMF user file system		
Symptom	Cause	Resolution
Job fails with an FSM error.	Your user ID lacks superuser authority.	For more information about how to define a user with superuser authority, see the publication z/OS UNIX System Services .
Job fails with an authorization error.	Job statement errors.	Correct the job statements.

Step 3: Copy the z/OSMF procedures into JES PROCLIB

In this step, you copy the z/OSMF started procedures and logon procedure from SYS1.PROCLIB into your JES concatenation.

About this step

You can use the \$D PROCLIB command to display your JES2 PROCLIB definitions.

Before you begin

Locate the IBM procedures. IBM supplies procedures for z/OSMF in your z/OS order:

- ServerPac orders: IBM supplies the z/OSMF procedures in the SMP/E managed proclib data set. The default name for the data set is SYS1.IBM.PROCLIB.
- CBPDO orders: For a CBPDO order, the SMP/E-managed proclib data set is named as SYS1.PROCLIB.
- Application Development CD.

Procedure

1. Use ISPF option 3.3 or 3.4 to copy the following procedures from SYS1.PROCLIB into your JES concatenation:
 - IZUSVR1
 - IZUANG1
 - IZUFPROC

Results

The procedures now reside in your JES PROCLIB.

Common errors

Table 7 on page 27 shows the most common errors for this step and suggests resolutions.

Table 7. Common errors when you copy the IBM procedures into JES PROCLIB		
Symptom	Cause	Resolution
Not authorized to copy into PROCLIB.	Your user ID does not have the permission to modify PROCLIB.	Contact your security administrator.

Table 7. Common errors when you copy the IBM procedures into JES PROCLIB (continued)		
Symptom	Cause	Resolution
Abend code B37 or E37.	The data set runs out of space.	Use IEBCOPY utility to compress PROCLIB data set before you copy it.

Step 4: Start the z/OSMF server

In this step, you start the z/OSMF server on your system.

About this step

z/OSMF processing is managed through the z/OSMF server, which runs as the started tasks IZUANG1 and IZUSVR1. z/OSMF is started with the START command.

Before you begin

Ensure that you have access to the operations console and can enter the START command.

Procedure

In the operations console, enter the START commands sequentially:

```
S IZUANG1
S IZUSVR1
```

Note: The z/OSMF angel (IZUANG1) must be started before the z/OSMF server (IZUSVR1).

You must enter these commands manually at subsequent IPLs. If necessary, you can stop z/OSMF processing by entering the STOP command for each of the started tasks IZUANG1 and IZUSVR1.

Note: z/OSMF offers an autostart function, which you can configure to have the z/OSMF server started automatically. For more information about the autostart capability, see [Chapter 36, “Autostart concepts in z/OSMF,”](#) on page 213

Results

When the z/OSMF server is initialized, you should see the following messages in the operations console:

```
CWWKB0069I: INITIALIZATION IS COMPLETE FOR THE IZUANG1 ANGEL PROCESS.
IZUG400I: The z/OSMF Web application services are initialized.
CWWKF0011I: The server zosmfServer is ready to run a smarter planet.
```

Common errors

Table 8 on page 28 shows the most common errors for this step and suggests resolutions.

Table 8. Common errors when you start the z/OSMF server		
Symptom	Cause	Resolution
The following message is written to the server job log: <pre> ICH420I PROGRAM CELQLIB FROM LIBRARY CEE.SCEERUN2 CAUSED THE ENVIRONMENT TO BECOME UNCONTROLLED. BPXP014I ENVIRONMENT MUST BE CONTROLLED FOR DAEMON (BPX.DAEMON) PROCESSING. </pre>	The IZUANG1 procedure is down-level.	Copy the latest procedure from SYS1.PROCLIB into your JES concatenation:

Table 8. Common errors when you start the z/OSMF server (continued)

Symptom	Cause	Resolution
One or more occurrences of message CWWKS2911E are written to the server job log. For example: <pre>CWWKS2911E: SAF Service RACROUTE_AUTH did not succeed because the resource profile IZUDFLT.ZOSMF.PROVISIONING.RESOURCE_MANAGEMENT in class ZMFAPLA does not exist.</pre>	These messages occur during the initial start-up of the z/OSMF server. They should not appear during subsequent restarts of the server.	You can ignore these messages.

Step 5: Log in to z/OSMF

In this step, you verify the results of your work by opening a web browser to z/OSMF and logging in with a z/OS user ID. By default, the z/OSMF configuration process creates security groups for administrator and users. You can use a user ID connected to either group to log in to z/OSMF.

About this step

The user ID that you use to log in to z/OSMF must have been used previously to authenticate on the z/OS system in another way before you can use it to log on to z/OSMF. For example, it can have been used to log on to TSO/E, or to run a batch job.

Before you begin

To find the URL of the z/OSMF log-in page, look for message IZUG349I in the z/OSMF server job log.

Procedure

1. Open a web browser to the z/OSMF log-in page. The URL for the log-in page has the following format: `https://hostname:port/zosmf/` Where:
 - *hostname* is the host name or IP address of the system in which z/OSMF is installed.
 - *port* is the secure port for the z/OSMF configuration. If you specified a secure port for SSL encrypted traffic during the configuration process through parmlib statement HTTP_SSL_PORT, the port is required to log in. Otherwise, it is assumed that you use the default port 443.
2. In the **z/OS USER ID** field on the log-in page, enter the z/OS user ID that you used to configure z/OSMF (the installer user ID).
3. In the **z/OS PASSWORD** field, enter the password or pass phrase that is associated with the z/OS user ID.
4. Click **Log In**.

When you log in to z/OSMF, the desktop interface is displayed. Only the options you are authorized to use are displayed. For more information about the z/OSMF user interface, see the topic *"Getting started with z/OSMF"* in the online help.

Results

If the user ID and password or pass phrase are valid, you are authenticated to z/OSMF.

You have successfully configured the z/OSMF nucleus.

Common errors

"Step 5: Log in to z/OSMF" on page 29 shows the most common errors for this step and suggests resolutions.

Table 9. Common errors when you access the z/OSMF log-in page

Symptom	Cause	Resolution
z/OSMF log-in page does not load in your web browser.	The SSL handshake was not successful. This problem can be related to the browser certificate.	See “Browser cannot connect to z/OSMF” on page 292 “Certificate error in the Mozilla Firefox browser” on page 294
Your log-in attempt fails.	The user ID is not connected to the IZUADMIN group.	Connect your user ID to the IZUADMIN group.
Your log-in attempt fails.	The password is expired.	Log on to TSO/E with your z/OS user ID and password. You are asked to change your password if it is expired.
Your log-in attempt fails.	Your log-in attempt fails.	Ask your administrator to reset your z/OS user ID and password.

Step 6: Mount the z/OSMF user file system at IPL time

In this step, you ensure that the user file system is mounted at IPL time.

About this step

Previously, in “[Step 2: Run job IZUMKFS to create the z/OSMF user file system](#)” on page 26 you created and mounted the z/OSMF user file system. Now, you can ensure that the z/OSMF user file system is mounted automatically for subsequent IPLs. Doing this work will involve updating the active BPXPRMxx parmlib member on your z/OS system.

Before you begin

By default, the z/OSMF file system uses the name IZU.SIZUUSRD, and is mounted in read/write mode. It is recommended that this file system is mounted automatically at IPL time. It is also highly recommended that the RWSHARE option is used when mounting the file system if you are running z/OSMF in a SYSPLEX, particularly when you run z/OSMF on a non-owning system of the file system.

If you do not know which BPXPRMxx member is active, follow these steps to find out:

1. In the operations console, enter the following command to see which parmlib members are included in the parmlib concatenation on your system:

```
D PARMLIB
```

2. Make a note of the BPXPRMxx member suffixes that you see.
3. To determine which BPXPRMxx member takes precedence, enter the following command:

```
D OMVS
```

The output of this command should be similar to the following messages:

```
BPX0042I 04.01.03 DISPLAY OMVS 391
OMVS 000F ACTIVE OMVS=(ST,3T)
```

In this example, the member BPXPRMST takes precedence. If BPXPRMST is not present in the concatenation, member BPXPRM3T is used.

Procedure

Add a MOUNT command for the z/OSMF user file system to your currently active BPXPRMxx parmlib member. For example:

```
MOUNT FILESYSTEM('IZU.SIZUUSR') TYPE(ZFS) MODE(RDWR)
MOUNTPOINT('/global/zosmf') PARM('AGGRGROW, RWSHARE') UNMOUNT
```

Results

The BPXPRMxx member is updated. At the next system IPL, the following message is issued to indicate that the z/OSMF file system is mounted automatically.

```
BPXF013I FILE SYSTEM IZU.SIZUUSR WAS SUCCESSFULLY MOUNTED.
```

Stopping and starting z/OSMF manually

Although z/OSMF starts automatically when you start z/OS, you can also stop and start z/OSMF manually.

To stop the z/OSMF server, you can use the **STOP** command from the operator console. Enter **STOP** for each started task in the following sequence:

```
STOP IZUSVR1
STOP IZUANG1
```

Figure 6 on page 31 shows an example of the expected results on a system that is running JES2:

```
stop izusvr1
+CWWKB0001I: Stop command received for server zosmfServer.
$HASP395 IZUSVR1 ENDED

stop izuang1
CWWKB0073I: THE IZUANG1 ANGEL PROCESS ENDED NORMALLY.
$HASP395 IZUANG1 ENDED
```

Figure 6. Example result from a **STOP** command

To start the z/OSMF server, you can use the **START** command from the operator console. Enter **START** for the two started tasks in the following sequence:

```
START IZUANG1
START IZUSVR1, IZUPRM=PREV
```

Specifying IZUPRM=PREV, which is the default, ensures that you use the same set of IZUPRMxx parmlib values that were in effect in the previous instance of z/OSMF.

- The z/OSMF server is available when the following message is displayed: CWWKF0011I: The server zosmfServer is ready to run a smarter planet.
- Generally, you should not cancel the z/OSMF angel process because the z/OSMF applications in your system might depend on it. However, in some rare cases, you might find it necessary to cancel the angel process to avoid a system shutdown, such as a re-IPL. If you ever need to cancel the angel process or if the angel process abends, your z/OSMF administrator should cancel the z/OSMF servers with applications dependent on the successful running of the angel process. Otherwise, leaving the servers and applications running can result in a server hang condition.

Displaying the z/OSMF server settings

Use the MODIFY command with the option 'DISPLAY IZU' to display the IZUPRMxx parmlib settings from the most recent z/OSMF server initialization. The display output is displayed in message IZUG013I.

The display output includes:

- z/OSMF home page URI.
- AUTOSTART group name, if the command is entered for an autostarted z/OSMF server.
- One of the following enablement status values for each of the optional z/OSMF services:

STARTED

Service is enabled.

STOPPED

Service is disabled.

UNSPECIFIED

Service name is not specified in a currently active IZUPRMxx parmlib member and is therefore not enabled.

The display output also includes the server start time and an indication of how long the server has been running.

The display output does not reflect any configuration changes that you might have applied to the IZUPRMxx member after server start-up, such as through the SET IZU or SETIZU command.

For descriptions of the IZUPRMxx parmlib member settings, see [“IZUPRMxx reference information” on page 35](#).

The syntax of the MODIFY, DISPLAY IZU command is as follows:

```
F <server-name>,DISPLAY IZU
```

Where:

server-name

Is the job name of the currently active z/OSMF server started task. By default, the job name is IZUSVR1, but your installation might use a different job name.

DISPLAY IZU

Displays the settings for the currently active IZUPRMxx parmlib members. This cumulative set of values is derived from the currently active parmlib members.

The display output does not reflect any configuration changes that you might have applied to the IZUPRMxx member after server start-up, such as through the SET IZU or SETIZU command.

If the command is entered incorrectly, an error message from the WebSphere Liberty run time is displayed to describe the error. The WebSphere Liberty messages are prefixed by CW and are described online at the following link: [WebSphere Liberty message descriptions \(www.ibm.com/docs/en/was-liberty/zos?topic=liberty-messages\)](http://www.ibm.com/docs/en/was-liberty/zos?topic=liberty-messages).

Example 1

[Figure 7 on page 33](#) shows the display output for an autostarted z/OSMF server named IZUSVR1.

```

F IZUSVR1,DISPLAY IZU

+CWWKB0004I: z/OSMF PARMLIBs DISPLAY
IZUG013I The home page of z/OSMF server in SYSTEM(SY1)
in AUTOSTART_GROUP(IZUDFLT) can be accessed at :
https://XXXXXX.XXX.XXX:443/zosmf
IZUG041I The server started at
and has been running for 0004(hhhh):424681(mm):02(ss)

Current z/OSMF settings
Source
  HOSTNAME(XXXXXX.XXX.XXX) IZUPRM3S
  HTTP_SSL_PORT(443) IZUPRM3S
  LOGGING('*=warning:com.ibm.zosmf.*=info:com.ibm.zosmf.environment.ui=finer') IZUPRM3S
  UNAUTH_USER(IZUGUEST) IZUPRM3S
  SEC_GROUPS
    ADMIN(IZUADMIN) IZUPRM3S
    USER(IZUUSER) IZUPRM3S
    SECADMIN(IZUSECAD) IZUPRM3S
  SAF_PREFIX(IZUDFLT) IZUPRM3S
  CLOUD_SAF_PREFIX(IYU) IZUPRM3S
  KEYRING_NAME(IZUKeyring.IZUDFLT) IZUPRM3S
  SESSION_EXPIRE(495) IZUPRM3S
  WLM_CLASSES
    LONG_WORK(IZUGWORK) IZUPRM3S
    DEFAULT(IZUGHTTP) IZUPRM3S
  JAVA_HOME(usr/lpp/java/J17.0_64) IZUPRM3S
  TEMP_DIR(/tmp) IZUPRM3S
  INCIDENT_LOG_UNIT(SYSALLDA) IZUPRM3S
  RESTAPI_FILE
    ACCT(IZUACCT) IZUPRM3S
    PROC(IZUFPROC) IZUPRM3S
    REGION(65536) IZUPRM3S
  COMMON_TSO
    ACCT(IZUACCT) IZUPRM3S
    PROC(IZUFPROC) IZUPRM3S
    REGION(50000) IZUPRM3S
  AUTOSTART_GROUP(IZUDFLT) IZUPRM3S
  AUTOSTART(LOCAL) IZUPRM3S
  SERVER_PROC(IZUSVR1) IZUPRM3S
  ANGEL_PROC(IZUANG1) IZUPRM3S
  USER_DIR(/var/zosmf) IZUPRM3S
  CSRF_SWITCH(ON) IZUPRM3S

Status of z/OSMF plugins
  Configuration Assistant(STARTED) IZUPRM3S
  Capacity Provisioning(STARTED) IZUPRM3S
  Workload Management(STARTED) IZUPRM3S
  Resource Monitoring(STARTED) IZUPRM3S
  Incident Log(STARTED) IZUPRM3S
  Software Management(STARTED) IZUPRM3S
  WebISPF(STARTED) IZUPRM3S
  Sysplex Management(STARTED) IZUPRM3S
+CWWKB0005I: COMMAND RESPONSES COMPLETED SUCCESSFULLY FROM display izu
348
Command Handler.
+CWWKB0002I: MODIFY COMMAND DISPLAY IZU COMPLETED SUCCESSFULLY.

```

Figure 7. MODIFY command output for an autostarted z/OSMF server

Example 2

Figure 8 on page 34 shows the display output for a stand-alone z/OSMF server named IZUSVR2.

```

F IZUSVR2,DISPLAY IZU

+CWWKB0004I: z/OSMF PARMLIBs DISPLAY 659
IZUG013I The home page of z/OSMF server in SYSTEM(SY1)
https://PEV051.POK.IBM.COM:443/zosmf
IZUG041I The server started at
and has been running for 0004(hhhh):424682(mm):02(ss)

Current z/OSMF settings                                         Source

HOSTNAME(PEV051.POK.IBM.COM)                                   IZUPRM3S
HTTP_SSL_PORT(443)                                             IZUPRM3S
+CWWKB00061I CONTINUATION 1 FOR MESSAGE IDENTIFIER 1862284635 660
LOGGING('*=warning:com.ibm.zosmf.*=info:com.ibm.zosmf
f.environment.ui=finer')                                       IZUPRM3S
UNAUTH_USER(IZUGUEST)                                          IZUPRM3S
SEC_GROUPS
  ADMIN(IZUADMIN)                                              IZUPRM3S
  USER(IZUUSER)                                                IZUPRM3S
  SECADMIN(IZUSECAD)                                           IZUPRM3S
  SAF_PREFIX(IZUDFLT)                                          IZUPRM3S
  CLOUD_SAF_PREFIX(IYU)                                         IZUPRM3S
+CWWKB00061I CONTINUATION 2 FOR MESSAGE IDENTIFIER 1862284635 661
KEYRING_NAME(IZUKeyring.IZUDFLT)                               IZUPRM3S
SESSION_EXPIRE(495)                                           IZUPRM3S
WLM_CLASSES
  LONG_WORK(IZUGWORK)                                          IZUPRM3S
  DEFAULT(IZUGHTTP)                                           IZUPRM3S
  JAVA_HOME/usr/lpp/java/J17.0_64                             IZUPRM3S
  TEMP_DIR(/tmp)                                               IZUPRM3S
  INCIDENT_LOG_UNIT(SYSALLDA)                                   IZUPRM3S
  RESTAPI_FILE
+CWWKB00061I CONTINUATION 3 FOR MESSAGE IDENTIFIER 1862284635 662
  ACCT(IZUACCT)                                                IZUPRM3S
  PROC(IZUFPROC)                                               IZUPRM3S
  REGION(65536)                                                 IZUPRM3S
  COMMON_TSO
    ACCT(IZUACCT)                                              IZUPRM3S
    PROC(IZUFPROC)                                             IZUPRM3S
    REGION(50000)                                              IZUPRM3S
  AUTOSTART_GROUP(IZUDFLT)                                     IZUPRM3S
  AUTOSTART(CONNECT)                                           IZUPRM3S
+CWWKB00061I CONTINUATION 4 FOR MESSAGE IDENTIFIER 1862284635 663
  SERVER_PROC(IZUSVR2)                                          IZUPRM3S
  ANGEL_PROC(IZUANG2)                                           IZUPRM3S
  USER_DIR(/var/zosmf)                                         IZUPRM3S
  CSRF_SWITCH(ON)                                              IZUPRM3S

Status of z/OSMF plugins

Configuration Assistant(STARTED)                               IZUPRM3S
Capacity Provisioning(STARTED)                                  IZUPRM3S
+CWWKB00061I CONTINUATION 5 FOR MESSAGE IDENTIFIER 1862284635 664
Workload Management(STARTED)                                   IZUPRM3S
Resource Monitoring(STARTED)                                    IZUPRM3S
Incident Log(STARTED)                                          IZUPRM3S
Software Management(STARTED)                                   IZUPRM3S
WebISPF(STARTED)                                              IZUPRM3S
Sysplex Management(STARTED)                                    IZUPRM3S
+CWWKB0005I: COMMAND RESPONSES COMPLETED SUCCESSFULLY FROM display izu
665
Command Handler.
+CWWKB0002I: MODIFY COMMAND DISPLAY IZU COMPLETED SUCCESSFULLY.

```

Figure 8. MODIFY command output for a stand-alone z/OSMF server

IZUPRMxx reference information

The IZUPRMxx parmlib member specifies options for z/OSMF. SYS1.SAMPLIB contains a copy of the IZUPRMxx member that you can copy to SYS1.PARMLIB and modify.

Operator commands

You can use the operator commands SET IZU and SETIZU to change z/OSMF parmlib options dynamically:

- Use the SET IZU to select one or more IZUPRMxx parmlib members for the next restart of the z/OSMF server. For example, the following specification indicates that IZUPRM01 and IZUPRM02 are to be used on the next server restart:

```
SET IZU=(01,02)
```

- Use the SETIZU command to modify one or more options in the currently active IZUPRMxx member. For example, the following specification indicates that SYSDA is to be used for storing output from the Incident Log FTP jobs:

```
SETIZU ILUNIT=SYSDA
```

For more information, see [z/OS MVS System Commands](#).

Syntax rules for IZUPRMxx

For general rules of parmlib member syntax, see [z/OS MVS Initialization and Tuning Reference](#).

Additionally, the following rules apply to the creation of IZUPRMxx parmlib members:

- Use columns 1-71 for data; columns 72-80 are ignored.
- If a statement is omitted, the default is used.
- You can enter one or more statements on a line, or use several lines for one statement.
- Blanks are treated as delimiters. The system interprets multiple blanks as a single blank. You can use blanks between parameters and values. For example, all of the following parameter specifications are equally valid:

```
SESSION_EXPIRE(495)  
SESSION_EXPIRE (495)  
SESSION_EXPIRE ( 495 )
```

- Comments can appear in columns 1-71 and must begin with "/" and end with "/". Any number of blank lines can appear between statements to improve readability.
- Enter values in uppercase, lowercase, or mixed case. The system converts input to uppercase, unless the values are enclosed in single quotation marks, which are processed without altering the case.

These values that you set for these parameters might require mixed casing, and therefore should be enclosed in single quotation marks:

- HOSTNAME
- INCIDENT_LOG UNIT
- JAVA_HOME
- KEYRING_NAME
- LOGGING
- SAF_PREFIX
- CLOUD_SAF_PREFIX
- CLOUD_SEC_ADMIN
- TEMP_DIR

- AUTOSTART_GROUP
- USER_DIR
- Enclose any value that contains special characters in single quotation marks.
- You can use system symbols in IZUPRMxx. Suppose, for example, that your installation defines a symbol in IEASYMxx for the Java directory, such as /usr/lpp/java/J21.0_64. To reference this symbol on the JAVA_HOME parameter in IZUPRMxx, specify the symbol as follows: JAVA_HOME(&JAVA21). The example in [“Syntax format of IZUPRMxx” on page 36](#) shows the use of a system symbol in IZUPRMxx.
- Enclose any value that is the same as a keyword in single quotation marks so that the system interprets the value as a value and not as a keyword.
- Enclose values in single quotation marks, according to the following rules:
 - Two single quotations next to each other on the same line are processed as a single quotation mark. For example, the system interprets Jane ' 's file as Jane's file.
 - If the length of a parameter and its value exceeds 71 characters, it requires multiple lines. Specify the first part of such a value in columns 1-71 and use as many subsequent lines as necessary to complete it. When a value spans multiple lines, place one quotation mark at the beginning of the value, stop the value in column 71 of the line, continue the value in column 1 of the next line, and complete the value with one quotation mark.
- You can specify multiple IZUPRMxx parmlib members on the IZU= parameter of IEASYSxx. If the same statement is used more than once, either in the same member or in multiple members, the value from the last occurrence is used. For example, suppose that your installation uses two members, IZUPRM01 and IZUPRM02. If the HOSTNAME parameter is specified in both IZUPRM01 and IZUPRM02, the system uses the HOSTNAME value from IZUPRM02.

Syntax format of IZUPRMxx

```

HOSTNAME('*')
HTTP_SSL_PORT(443)
INCIDENT_LOG UNIT('SYSALLDA')

JAVA_HOME('&JAVA21')      /* System symbol used to define Java home directory */
KEYRING_NAME('IZUKeyring.IZUDFLT')
LOGGING('*=warning:com.ibm.zosmf.*=info:com.ibm.zosmf.environment.ui=finer')
RESTAPI_FILE ACCT(IZUACCT) REGION(65536) PROC(IZUFPROC)
/* Common TSO logon proc, account, and region size, used by all services by default. */
COMMON_TSO ACCT(IZUACCT) REGION(50000) PROC(IZUFPROC)
SAF_PREFIX('IZUDFLT')
CLOUD_SAF_PREFIX('IYU')
CLOUD_SEC_ADMIN(userid)
SEC_GROUPS USER(IZUUSER),ADMIN(IZUADMIN),SECADMIN(IZUSECAD)
SESSION_EXPIRE(495)
TEMP_DIR('/tmp')
CSRF_SWITCH(ON)
SERVER_PROC(IZUSVR1)
ANGEL_PROC(IZUANG1)
AUTOSTART(LOCAL)
AUTOSTART_GROUP('IZUDFLT')
USER_DIR('/global/zosmf/')
UNAUTH_USER(IZUGUEST)
WLM_CLASSES DEFAULT(IZUGHHTTP)
LONG_WORK(IZUGWORK)

/* Uncomment the following statement and any plugins that are desired */
/* PLUGINS( INCIDENT_LOG,COMMSERVER_CFG,WORKLOAD_MGMT,RESOURCE_MON,
CAPACITY_PROV,SOFTWARE_MGMT, SYSPLEX_MGMT, ISPF ) */

```

IBM-supplied defaults for IZUPRMxx

There is no default IZUPRMxx parmlib member. IBM provides a sample IZUPRM00 parmlib member in the SAMPLIB data set.

“Syntax format of IZUPRMxx” on page 36 shows the IBM-supplied IZUPRM00 member. Notice that the PLUGINS statement is commented out; to use it, you must remove the comment characters.

Statements and parameters for IZUPRMxx

HOSTNAME('hostname')

Specifies the hostname, as defined by DNS, where the z/OSMF server is located. Specify the IP address for your system. If you are using z/OSMF in a multisystem sysplex, IBM recommends that you use a dynamic virtual IP address (DVIPA), which resolves to the correct IP address if the z/OSMF server is moved to a different system.

Note: HOSTNAME="*" means listen on all adapters. By default, the server is listening only on address 127.0.0.1/localhost. You can also use the HOSTNAME parameter to specify a single IP address to have the system listen only on the specified IP address.

Rules: Must be a valid TCP/IP HOSTNAME or an asterisk (*).

Default: *

HTTP_SSL_PORT(nnn)

Identifies the port number that is associated with the z/OSMF server. This port is used for SSL encrypted traffic from your z/OSMF configuration. The default value, 443, follows the Internet Engineering Task Force (IETF) standard.

By default, the z/OSMF server uses the SSL protocol TLS for secure TCP/IP communications. As a result, the server can accept incoming connections that use SSL V3.0 and the TLS 1.0, 1.1, 1.2 and 1.3 protocols.

The z/OSMF server port uses Java SSL encryption to protect its outbound HTTPS connections. Therefore, it is not necessary (or possible) to configure AT-TLS on the z/OSMF server port. If you attempt to do so, the z/OSMF server encounters HTTP connection failures and errors, such as the following, in the server logs directory:

- IZUG476E: The HTTP request to the secondary z/OSMF instance "209" failed with error type "CertificateError" and response code "0"
- javax.net.ssl.SSLException: Unrecognized SSL message, plaintext connection?

Rules: Must be a valid TCP/IP port number.

Value range: 1 - 65535 (up to 5 digits)

Default: 443

INCIDENT_LOG UNIT('device-name')

Specifies the device to be used for storing data sets and z/OS UNIX files for the FTP jobs that are used for the Incident Log service.

Rules: You must specify a generic name (such as "3390") or an esoteric name (such as "DISK"). The esoteric name SYSALLDA, which is used by default, is automatically defined by the system to include all direct-access disk devices.

Default: SYSALLDA

JAVA_HOME('directory-name')

Specifies the fully qualified path name for IBM Semeru Runtime Certified Edition on your system.

Rules:

- Must be a valid z/OS UNIX System Services path name.
- Must begin with a forward slash (/).
- Must specify a full or absolute path name.

Default: /usr/lpp/java/J21.0_64

KEYRING_NAME('keyring-name')

Specifies the key ring name for the z/OSMF server. The format is IZUKeyring.<SAF_PREFIX>.

Rules: Must be the name of a valid RACF profile in the DIGTRING class.

Note: The IZUSEC job contains statements that include the generation of digital certificates and the key ring. The value that is specified here must match the key ring name that you defined for z/OSMF in the IZUSEC job or by entering equivalent commands.

Default: IZUKeyring.IZUDFLT

LOGGING('trace_specification')

Initial trace state for the z/OSMF server. These settings are read when the server is started. Changes to this value are provided, when necessary, by IBM Support.

Rules:

- 1 - 2048 characters
- Case sensitive.

Default: *=warning:com.ibm.zosmf.*=info:com.ibm.zosmf.environment.ui=finer

RESTAPI_FILE ACCT(account-number) REGION(region-size) PROC(proc-name)

Specifies values for the TSO logon procedure that is used internally by the z/OS data set and file REST interface services. Except for the account number, it is recommended that you use the defaults, which should be adequate for most z/OS installations. If you specify alternative values, you must ensure that the z/OSMF user and z/OSMF administrator security groups are authorized to use the logon procedure name and account number that you specify, and that the region size is at least 65536 kilobytes (KB).

All z/OSMF users must have TSO segments that are defined in the external security manager, such as RACF. Failure to have a TSO segment for each user ID prevents some z/OSMF functions from working.

ACCT(account-number)

Account number to be used for the TSO/E logon procedure that is used for the z/OS data set and file REST interface services.

Rules: A valid accounting number for your installation.

Default: IZUACCT

REGION(region-size)

Region size (in kilobytes) to be used for the TSO/E logon procedure for the z/OS data set and file REST interface services.

Value range: 65536 – 2096128

Default: 65536

In z/OS 2.5 and earlier releases, or after applying PH61494 in z/OS 3.1, a value between 32768 to 65535 is accepted as a valid *REGION* value. If a value lower than 65536 is provided, the File and Dataset API overrides the *REGION* value to 65536.

PROC(proc-name)

TSO/E logon procedure to be used for operations with the z/OS data set and file REST interface services. It is recommended that you accept the default procedure, IZUFPROC, which is supplied by IBM as a cataloged procedure in SYS1.PROCLIB.

Rules: Must be a valid partitioned data set member name.

Default: IZUFPROC

COMMON_TSO ACCT(account-number) REGION(region-size) PROC(proc-name)

Specifies values for the TSO/E logon procedure that is used internally for various z/OSMF activities. This setting is applicable if your z/OSMF configuration uses:

- z/OS console REST interface services
- Software Management task
- Workflows task

Except for the account number, it is recommended that you use the default values, which should be adequate for most z/OS installations. If you specify alternative values, you must ensure that the

z/OSMF user and z/OSMF administrator security groups are authorized to use the logon procedure name and account number that you specify, and that the region size is at least 50 MB.

All z/OSMF users must have a TSO segment that is defined in the USER profiles that are used by the external security manager, such as RACF. Failure to have a TSO segment for each user ID prevents some z/OSMF functions from working.

ACCT(account-number)

Account number to be used for the common TSO/E logon procedure for z/OSMF.

Rules: A valid accounting number for your installation.

Default: IZUACCT

REGION(region-size)

Region size (in kilobytes) to be used for the common logon procedure for z/OSMF.

Value range: 50000 – 2096128

Default: 50000

PROC(proc-name)

TSO/E logon procedure to be used for z/OSMF. It is recommended that you accept the default procedure, IZUFPROC, which is supplied by IBM as a cataloged procedure in SYS1.PROCLIB.

Rules: Must be a valid partitioned data set member name.

Default: IZUFPROC

SAF_PREFIX('IZUDFLT')

SAF profile prefix that is prepended to the names of any resource profile names to be used for the z/OSMF core functions and optional services.

Note: The IZUxxSEC sample jobs contain commands that include the SAF profile prefix for creating resource profile names. The value that is specified here must match the prefix name that you define for z/OSMF in the IZUxxSEC jobs or by entering equivalent commands.

Rules:

- Must follow the rules for RACF profile names.
- 1 – 3 characters.

Default: IZUDFLT

CLOUD_SAF_PREFIX('IYU')

SAF profile prefix that is prepended to the names of any groups to be used for authorizing users to IBM Cloud Provisioning and Management for z/OS task activities.

Note: The IZUPRSEC sample job contains commands that include the group name for creating authorizations for IBM Cloud Provisioning and Management for z/OS. The value that is specified here must match the prefix name that you define for Cloud Provisioning authorizations in the IZUPRSEC job or by entering equivalent commands.

Rules:

- Must follow the rules for RACF profile names.
- 1 – 3 characters.

Default: IYU

CLOUD_SEC_ADMIN('user-id')

Specifies the security administrator user ID to be used for automatic security management in Cloud Provisioning. When specified, automatic security updates are performed under this user ID. Otherwise, if this value is omitted, security updates for Cloud Provisioning must be performed manually by your security administrator.

The user ID that is specified here must be connected to the z/OSMF security administrator group, which is named IZUSECAD by default. The IZUPRSEC job in SYS1.SAMPLIB contains a commented RACF command for creating this authorization. Minimally, this user ID requires:

- READ access to the ZMFCLOUD class resource profile IZUDFLT.ZOSMF.SECURITY.ADMIN.
- Authorization to manage resource profiles in the ZMFAPLA and ZMFCLOUD resource classes.
- Authorization to manage security groups.

During regular operations with Cloud Provisioning, your installation might periodically update Resource Management domains and tenants to add or remove users. Such changes require updates to your security setup. By specifying a user ID for the CLOUD_SEC_ADMIN keyword, you indicate that *automatic security* is to be used for performing user authorizations. If so, the authorizations are performed automatically by the Resource Management task, by using a security REXX exec that is provided by the external security manager. For example, IBM supplies the REXX exec **izu.provisioning.security.config.rexx** for use with RACF. For more information, see [“Automatic security management for Cloud Provisioning” on page 155.](#)

If the CLOUD_SEC_ADMIN value is changed, the new setting applies only to domains that are created after the change. Any existing domains continue to operate with manual or automated security, based on the value that was in effect when these domains were created.

Note: With the installation of the PTF for APAR PH29813, the default domain now supports manual security mode for creating templates and tenants. This option is intended for provisioning environments that cannot use automatic security mode. Previously, the default domain was required to run in automatic security mode. Now, when the default domain is created at z/OSMF startup time, it is placed in manual security mode if no security administrator is specified on the CLOUD_SEC_ADMIN statement in the IZUPRMxx parmlib member.

If you have incorrectly configured the security mode for Cloud Provisioning and Management, it is possible to change it. Doing so requires only that you edit the CLOUD_SEC_ADMIN statement in the IZUPRMxx parmlib member and restart the z/OSMF server. You can switch a domain from automatic security to manual security, and vice versa. Your changes to the CLOUD_SEC_ADMIN statement affect the security mode of all existing domains. The suggested practice is that you run Cloud Provisioning and Management in automatic security mode.

Rules:

- Must follow the rules for z/OS user IDs.
- 1 – 8 characters.

Default: None. If you do not provide a valid z/OS user ID, the Resource Management task does not perform automatic security updates.

SEC_GROUPS USER(group-name),ADMIN(group-name),SECADMIN(group-name)

Specifies group names for the base set of z/OSMF security groups: user, administrator, and z/OS security administrator.

USER(group-name)

Security group to be used for the z/OSMF user role. The user IDs that are connected to this group are considered to be z/OSMF users.

Default: IZUUSER

ADMIN(group-name)

Security group to be used for the z/OSMF administrator role. The user IDs that are connected to this group are considered to be z/OSMF administrators.

Default: IZUADMIN

SECADMIN(group-name)

Group name to be used for the z/OS Security Administrator role. This group is permitted to the Workflows task.

Default: IZUSECAD

SESSION_EXPIRE(*nnn*)

Amount of time (in minutes) for the session timeout. z/OSMF user sessions expire when this period elapses. For more information, see [“Re-authenticating in z/OSMF” on page 297](#).

Value range: 15-999999

Default: 495

TEMP_DIR(*'path-name'*)

Temporary directory for various z/OSMF activities. This setting is applicable if your z/OSMF configuration uses:

- Incident Log task
- Workflows task
- z/OSMF Diagnostic Assistant task

The temporary directory is used, as follows:

- Incident Log task uses this directory for sending z/OS UNIX file attachments through FTP.
- Workflows task uses this directory for storing temporary files.
- z/OSMF Diagnostic Assistant task uses this directory for storing temporary files.

Users of these z/OSMF tasks require write access to the temporary directory. Otherwise, the task might fail with an authorization error (the user encounters message ICH408I).

Notes:

- As part of its data collection, the z/OSMF Diagnostic Assistant task copies the z/OSMF log files and configuration files into a compressed (.zip) file and saves the file in the TEMP_DIR directory. The amount of storage needed to contain the compressed file varies, depending on your installation's use of z/OSMF. If the size of the compressed file exceeds the TEMP_DIR space, an error message is issued to the user of the z/OSMF Diagnostic Assistant task. If this problem occurs, increase the storage amount for the TEMP_DIR directory.
- In IBM Cloud Provisioning and Management for z/OS provisioning, a number of functions are performed by using workflows. For example, a software template is composed of one or more workflows. Therefore, any user who is involved in IBM Cloud Provisioning and Management for z/OS provisioning is also a potential user of the Workflows task. You must ensure that these users have write access to the TEMP_DIR location.

Rules:

- Must be a valid z/OS UNIX path name.
- Must specify the full or absolute path name, and a maximum of 255 characters between slashes.

Default: /tmp

CSRF_SWITCH(*ON/OFF*)

Indicates whether Cross Site Request Forgery (CSRF) custom header checking is enabled for REST API requests. By default, CSRF_SWITCH is set to ON to ensure that your installation is protected against CSRF attacks. However, in some limited cases, such as for testing, you might choose to temporarily disable CSRF checking by setting CSRF_SWITCH=OFF. However, it is recommended that you leave this setting enabled to prevent CSRF attacks. For more information, see [IBM z/OS Management Facility Programming Guide](#).

Default: ON

SERVER_PROC(*proc-name*)

Specifies the name of the started procedure that is used to start the z/OSMF server on this system. It is recommended that you use the default started procedure, which should be adequate for most z/OS installations. If you specify an alternative procedure name, ensure that the z/OSMF user and z/OSMF administrator security groups are authorized to the started procedure name.

Rules: Must specify a valid partitioned data set member name.

Default: IZUSVR1

ANGEL_PROC(*proc-name*)

Specifies the started procedure that is used to start the z/OSMF angel process on this system. It is recommended that you use the default started procedure, which should be adequate for most z/OS installations. If you specify an alternative procedure name, ensure that the z/OSMF user and z/OSMF administrator security groups are authorized to the started procedure name.

The ANGEL_PROC statement specifies both the name of the angel process and its started procedure name.

Example: ANGEL_PROC(IZUANG1) indicates that both the angel procedure member name and angel process name are IZUANG1.

Rules: Must specify a valid partitioned data set member name.

Default: IZUANG1

AUTOSTART(LOCAL|CONNECT)

Specifies whether the z/OSMF server is to be started automatically on this system.

The valid settings for AUTOSTART are, as follows:

LOCAL

Indicates that the system is to automatically start the z/OSMF server.

CONNECT

Indicates that the z/OSMF server is not to be autostarted on the local z/OS system.

1. IBM recommends that you specify LOCAL for all systems in a sysplex, or let it default to LOCAL, if you are using shared file systems for the z/OSMF data directory for each AUTOSTART group. If all systems in a sysplex are part of the same AUTOSTART group, the default, /global/zosmf, allows this. z/OSMF starts only on one system in the sysplex, if the sysplex has only one AUTOSTART group.
2. If you have more than one AUTOSTART group in a sysplex, you should use a shared file system for each one, with a unique mount point. For example, if you have AUTOSTART groups that are named ZOSMFA and ZOSMFB, you might use /global/zosmf/zosmfa for the first and /global/zosmf/zosmfb for the second. In this case, specifying LOCAL on all systems results in z/OSMF starting on one system per AUTOSTART group.
3. When the z/OSMF server has been started automatically on another system in the same AUTOSTART group in the same sysplex, requests for z/OSMF services that originate on the local system are routed to the remote server.
4. When AUTOSTART(CONNECT) is specified for every system in a sysplex, the z/OSMF server is not autostarted on any system in the sysplex. The z/OSMF server can be started with the START command or through automation when no other z/OSMF server is active in the system's AUTOSTART group.

If a z/OSMF server fails, it must be restarted to restore z/OSMF operations for the autostart group. The server can be restarted on this system or another system, regardless of whether the system is specified as AUTOSTART(LOCAL) or AUTOSTART(CONNECT), using the START command or through automation.

Default: LOCAL

AUTOSTART_GROUP(IZUDFLT|nnnnnnnn)

Associates the local system with other systems that can share an autostarted z/OSMF server. AUTOSTART_GROUP defines a domain for z/OSMF work and associated persistent data within a sysplex. By default, one autostart group that is called IZUDFLT exists per sysplex. To associate the z/OSMF server on this system with a different autostart group, specify the desired name here.

Rules:

- Must consist of 1-32 alphanumeric characters (A-Z, a-z, 0-9) or special characters (#, \$, or @).
- Alphabetic characters are case insensitive.

Default: IZUDFLT

USER_DIR

z/OSMF data directory path. By default, the z/OSMF data directory is located in /global/zosmf. If you want to use a different path for the z/OSMF data directory, specify that value here, for example: USER_DIR= '/the/new/config/dir'.

Every autostart group within a sysplex must have a unique specification for USER_DIR. If you plan to use an autostarted z/OSMF server, this file system must be mounted when you IPL the system. Otherwise, the z/OSMF server cannot be autostarted.

If you specify both USER_DIR= in IZUPRMxx and USERDIR= on the PRC statement of the started procedure, the system uses the path that is specified by USERDIR= in the started procedure.

Rules: Must be a valid z/OS UNIX path name.

Default: /global/zosmf/

UNAUTH_USER(user-id)

Represents an unauthenticated user. Provides an unknown user with basic privileges to access the z/OSMF log-in page, but nothing more.

Rules:

- Must follow the rules for z/OS user IDs.
- 1 – 8 characters.

Default: IZUGUEST

WLM_CLASSES DEFAULT(class-name)

Specifies the WLM transaction classes for managing z/OSMF work.

DEFAULT(class-name)

WLM transaction class to be used for managing z/OSMF work, except for long-running work. See the description of the LONG_WORK(class-name) statement.

Rules: Must specify a valid WLM transaction class name.

Default: IZUGHTTP

LONG_WORK(class-name)

WLM transaction class to be used for managing the execution of long-running work.

Rules: Must specify a valid WLM transaction class name.

Default: IZUGWORK

PLUGINS(plugin-id,plugin-id,plugin-id,...)

Specifies the optional services to be made available in your configuration. Enter one or more of the service identifiers that are shown in [Table 10 on page 43](#).

Table 10. z/OSMF optional services and associated service IDs	
Service ID	Service name
CAPACITY_PROV	Capacity Provisioning
COMMSERVER_CFG	Network Configuration Assistant
INCIDENT_LOG	Incident Log
ISPF	ISPF
RESOURCE_MON	Resource Monitoring
SOFTWARE_MGMT	Software Deployment
SYSPLEX_MGMT	Sysplex Management
WORKLOAD_MGMT	Workload Management
ZERT_ANALYZER	IBM zERT Network Analyzer

After a service is enabled, you might later decide to remove it. To do so, edit the IZUPRMxx parmlib member and remove the service identifier from the PLUGINS statement. Then, restart the z/OSMF server. This action removes the services from the z/OSMF desktop interface. Any residual data that is associated with the service is saved in z/OSMF, in case you decide to enable it again later.

Default: No optional services are enabled by default.

Example of IZUPRMxx parmlib member

In the example that follows, an IZUPRMxx parmlib member is used to set these values:

- Port 30443.
- System symbol for the Java home directory. The symbol must also be defined in your IEASYMxx member.
- On startup, the system autostarts a z/OSMF server. The autostarted z/OSMF server processes requests from all systems that are members of the z/OSMF autostart group IZUDFLT.
- These optional services are selected: Network Configuration Assistant, Software Deployment, and Sysplex Management. The services are enabled for use when your installation completes the required host system customization. See [Part 4, “z/OSMF optional services,” on page 81](#).

```
HTTP_SSL_PORT(30443)
JAVA_HOME('&JAVA17')      /* System symbol used to define Java home directory */
AUTOSTART(LOCAL)
AUTOSTART_GROUP(IZUDFLT)
PLUGINS(COMMSERVER_CFG,SOFTWARE_MGMT,SYSPLEX_MGMT)
```

IZUSVR reference information

The following parameters are supported for use in the IZUSVR1 procedure.

ROOT='directory-path'

z/OSMF root code directory path. This value cannot be changed.

Default: /usr/lpp/zosmf

WLPDIR='directory-path'

WebSphere Liberty server code path.

The directory path must:

- Be a valid z/OS UNIX path name
- Be a full or absolute path name
- Be enclosed in quotation marks
- Begin with a forward slash ('/').

Default: /usr/lpp/zosmf/liberty

OUTCLS='output-class'

Suitable output class for writing system output. By default, the z/OSMF procedures use output class *.

The value must be in quotation marks.

Default: *

USERDIR='directory-path'

z/OSMF data directory path. By default, the IZUSVR1 procedure uses the directory /global/zosmf. If your installation configured z/OSMF to use another path for the data directory, specify that value here, for example: USERDIR= '/the/new/config/dir'.

The directory path must:

- Be a valid z/OS UNIX path name

- Be a full or absolute path name
- Be enclosed in quotation marks
- Begin with a forward slash (/).

Default: /global/zosmf

TRACE='Y | N'

Enables tracing for configuration-time errors, such as parmlib parsing errors. The error data is written to the server job log. Use this option only at the direction of IBM Support.

Default: N

IZUPRM=(PREV|SYSPARM|NONE|xx|(xx,...,zz))

The following values are valid:

PREV

Use the IZUPRMxx suffixes, if any were used by the previous instance of z/OSMF within the current IPL. IZUPRM='PREV' is used as the default in the standard IZUSVR1 procedure. IZUPRM='PREV' behaves like IZUPRM='SYSPARM' when the system encounters it during the initial IPL time (the first use of the IZUSVR1 procedure) because there is no previous instance of z/OSMF to use.

This setting is not valid if the SERVER parameter is set to STANDALONE.

SYSPARM

Use the IZUPRMxx suffixes that are specified on the IZU system parameter in IEASYSxx.

This setting is not valid if the SERVER parameter is set to STANDALONE.

NONE

No parmlib members are specified and the z/OSMF defaults are used. For the default values, refer to the parameter descriptions in [“IZUPRMxx reference information” on page 35](#).

xx|(xx,...,zz)

Specify the specific suffixes for the IZUPRMxx parmlib member or members that you want the procedure to use. If you specify a suffix, the member must exist in your parmlib concatenation. Otherwise, the procedure cannot be started. Multiple suffixes must be enclosed in parentheses.

The following syntax forms are valid:

```
IZUPRM=PREV
IZUPRM=SYSPARM
IZUPRM=(xx,yy,zz)
IZUPRM=xx
IZUPRM=NONE
```

Note: The IZUPRMxx suffixes you specify, explicitly or implicitly, in the IZUPRM parameter of the procedure override any suffixes that are specified by using the IZU system parameter in IEASYSxx.

Default: PREV

SERVER=(AUTOSTART|STANDALONE)

Indicates the start-up behavior of the z/OSMF server, as follows:

AUTOSTART

The z/OSMF server can be started automatically during system IPL. To do so, the system uses the started procedures that you identify on the SERVER_PROC and ANGEL_PROC statements in the active IZUPRMxx parmlib member for this system or uses their defaults. It is also possible to start the server manually, by using the START command, if no other z/OSMF server is active in the system's AUTOSTART group.

STANDALONE

The system does not start the z/OSMF server automatically. It can be started manually by using the START operator command or by automation. The system uses the started procedures that

you identify on the SERVER_PROC and ANGEL_PROC statements in the active IZUPRMxx parmlib member for this system or uses their defaults.

- If you choose not to autostart the z/OSMF server, you can use a started procedure similar to IZUSVR2 in SYS1.SAMPLIB for starting the z/OSMF server manually.

Default: AUTOSTART

IZUMEM='maxmemlimit | NOLIMIT'

Maximum amount (*maxmemlimit*) of usable, above-the-bar, virtual storage for the z/OSMF server address space. This value can be expressed in megabytes (M), gigabytes (G), terabytes (T), or petabytes (P). *nnnnn* can be a value 0 - 99999, with a maximum value of 16384P. By default, the limit is 4 gigabytes (4G).

Observe the following considerations:

- The amount of virtual memory that you request above 4G by using IZUMEM can be reduced by an SMFLIMxx or SMFPRMxx member of parmlib, or by an IEFUSI installation exit that lowers the memory limit for the IZUSVR1 started task.
- To indicate no limit to the amount of above-the-bar virtual storage, specify NOLIMIT.

Default: 4G

Chapter 5. Security validation for z/OSMF

Security validation is designed to validate if the necessary security profiles are set up properly after running the job IZUNUSEC or before starting the z/OSMF server.

For a first time z/OSMF setup, it is recommended to create a z/OSMF nucleus system. For more information, see [Chapter 4, “Create a z/OSMF nucleus on your system,” on page 23](#).

In the procedure, running job IZUNUSEC to create the z/OSMF security authorization is critical to properly set up security profiles. This modification is error-prone and can result in the necessary security profiles either missing or being incorrect.

Security validation is designed to validate if the necessary security profiles are set up properly after running job IZUNUSEC or before starting the z/OSMF server. Through security validation, you are able to validate if the z/OSMF started task ID has the needed authorization to the security profiles for the z/OSMF nucleus. You are also able to validate if a specified user has the needed authorization to the security profiles for z/OSMF Security Configuration Assistant.

You can choose between two available methods to run security validation on your system:

- Run the job IZUSECJL to start the procedure IZUSECSV to run security validation. Job IZUSECJL is available from SYS1.SAMPLIB, and a sample procedure IZUSECSV is available in SYS1.PROCLIB. You can modify the job IZUSECJL based on your needs and submit it. For more information, see [“Run job IZUSECJL to perform security validation” on page 48](#).
- Run the procedure IZUSECSV as a started task to run the security validation. This method requires extra security setup to specify the procedure IZUSECSV as a started task. For more information, see [“Run procedure IZUSECSV to perform security validation” on page 49](#).

Additional usage and reference information is provided in the following topics:

- [“IZUSECSV reference information” on page 50](#)
- [“Examples of security validation” on page 52](#)

System setup requirements

- Security validation uses Java Technology. It is recommended to use the same JDK version as z/OSMF requires. For more information, see [“Software prerequisites for z/OSMF” on page 7](#).

Note: If the SDK is not installed in the default location `/usr/lpp/java/J17.0_64` on your system, be sure to include JAVA_HOME in your IZUPRMxx parmlib members. Specify the parmlib members in either job IZUSECJL or PROC IZUSECSV.

- Locate the IBM procedures. IBM supplies procedures for z/OSMF in your z/OS order:
 - ServerPac orders: IBM supplies the z/OSMF procedures in the SMP/E managed proclib data set. The default name for the data set is SYS1.IBM.PROCLIB.
 - CBPDO orders: For a CBPDO order, the SMP/E-managed proclib data set is named as SYS1.PROCLIB.

Software requirements

Security validation relies on z/OSMF Security Configuration Assistant as it uses the security descriptor template file that is shipped by SCA. For more information, see [Appendix B, “Creating security descriptor files for the Security Configuration Assistant task,” on page 411](#).

Prerequisite requirement

The owner of either job IZUSECJL or PROC IZUSECSV must have READ access to profile BPX.SERVER in class FACILITY and also have `_BPX_SHAREAS=NO` set in their .profile.

For example, in a system with RACF as the security management product, your security administrator can enter RACF commands like those shown in [Figure 9 on page 48](#). If your installation uses an external security manager other than RACF, you can refer to this example when creating equivalent commands for your environment.

```
PERMIT BPX.SERVER CLASS(FACILITY) ID(OWNER_ID) ACCESS(READ) .  
SETR RACLIST(FACILITY) REFRESH
```

Figure 9. RACF commands to enable READ access to profile BPX.SERVER in class FACILITY

Run job IZUSECJL to perform security validation

The job IZUSECJL is used to invoke the procedure IZUSECSV to run security validation.

About the job

The job IZUSECJL is used to invoke the procedure IZUSECSV to run security validation without needing any additional requirements. You can instead choose to run the procedure IZUSECSV as a started task, which requires extra security setup. For more information, see [“Run procedure IZUSECSV to perform security validation” on page 49](#).

Running the job

1. Make a copy of job IZUSECJL from SYS1.SAMPLIB.
2. Modify the parameters USER and NOTIFY based on your system environment.
3. Modify the parameters ROOT/USERDIR/TRACE/OUTCLS/IZUPRM/USERID/LOGLEVEL/SVRID as needed. The parameters are passed to procedure IZUSECSV to run the security validation.
4. The job IZUSECJL must be submitted on the system in which it resides. If the job is submitted in the JESplex environment, it could be routed to other systems in the JESplex and cause an unexpected result. It is recommended to specify the expected system to parameter SYSAFF.

```
/*JOBPARM SYSAFF=P00  
//DO      EXEC PROC=IZUSECSV
```

Important: The user performing the job has to be configured with the OMVS segment.

Example of IZUSECJL

```
//IZUSECJL JOB MSGCLASS=C,MSGLEVEL=(1,1),USER=XXXXXXX,NOTIFY=XXXXXXX  
//DO      EXEC PROC=IZUSECSV,  
//          ROOT='/usr/lpp/zosmf', /* zOSMF install root */  
//          USERDIR='/tmp', /* tmp dir */  
//          TRACE=N, /* Trace option */  
//          OUTCLS='*', /* Sysout class */  
//          IZUPRM='NONE', /* Parmlib suffixes  
*/  
//          USERID='NOT_SPECIFIED', /* User ID validated  
*/  
//          LOGLEVEL='WARNING', /* LOG LEVEL  
*/  
//          SVRID='NOT_SPECIFIED' /* Server ID validated */
```

Run procedure IZUSECSV to perform security validation

Run the procedure IZUSECSV for security validation as a started task.

About this procedure

Through security validation, you are able to validate if a z/OSMF started task ID or user ID has the needed authorization to the security profiles for z/OSMF nucleus. Running the procedure IZUSECSV as a started task requires extra security setup. You can instead choose to run the job IZUSECJL is used to start the procedure IZUSECSV to run security validation without needing any additional requirements. For more information, see [“Run job IZUSECJL to perform security validation”](#) on page 48.

Running the procedure

1. In the system library SYS1.PROCLIB, locate job IZUSECSV.
2. Set up the procedure IZUSECSV to be a started task.

If you use RACF to manage security, you can use this command to define the procedure IZUSECSV as started task. Otherwise, refer to this example to create an equivalent command.

```
RDEFINE STARTED IZUSVRSEV.* UACC(NONE) STDATA(USER(IZUSVR) GROUP(IZUADMIN) PRIVILEGED(NO)
TRUSTED(NO) TRACE(YES))
SETR RACLIST(STARTED) REFRESH
```

The owner of the started task should match the USERID that is used to start the z/OSMF server. The owner USERID is validated if it has the needed authorization to the security profiles for the z/OSMF nucleus.

3. Run the procedure IZUSECSV via START command with the corresponding parameter provided.

For example to validate the owner of the procedure:

```
S IZUSECSV
```

Note: The user performing the started task must be configured with OMVS segment.

Example

The z/OSMF file system is allocated, formatted, and mounted, and the necessary directories are created.

```
//IZUSECSV PROC ROOT='/usr/lpp/zosmf', /* zOSMF install root */
//          USERDIR='/tmp',          /* tmp dir */
//          TRACE='N',                /* Trace option */
//          OUTCLS='*',               /* Sysout class */
//          IZUPRM='NONE',            /* Parmlib suffixes */
//          USERID='NOT_SPECIFIED',  /* User ID validated */
//          LOGLEVEL='WARNING',       /* LOG LEVEL */
//          SVRID='NOT_SPECIFIED'     /* USER ID TYPE */
//*
//*-----*
//* z/OSMF security validation utility                                */
//* procedure                                                         */
//* PROPRIETARY STATEMENT:                                           */
//*                                                                    */
//*    LICENSED MATERIALS - PROPERTY OF IBM                          */
//*    5650-ZOS                                                        */
//*    COPYRIGHT IBM CORP. 2022                                       */
//*    STATUS = HSMA250                                               */
//*-----*
//* Parse z/OSMF PARMLIB member
//*-----*
//ZPARM EXEC PGM=IZUSECCN,REGION=0M,
// PARM='/IZUPRM=&IZUPRM,TRACE=&TRACE,USERDIR=&USERDIR'
//DFLTCTFG DD PATH='&ROOT./defaults/configuration.defaults'
//STDOUT DD SYSOUT=&OUTCLS
//STDERR DD SYSOUT=&OUTCLS
//CEEDUMP DD SYSOUT=&OUTCLS
//*
```

```

// *-----
// * Security Validation
// *-----
// SECVAL EXEC PGM=BPXBATCH,REGION=0M,COND=(0,LT),
// PARM='SH &ROOT./bin/izusecval.sh &ROOT &USERDIR &TRACE &LOGLEVEL
// &USERID &SVRID'
// *
// SYSPRINT DD SYSOUT=&OUTCLS
// SYSOUT DD SYSOUT=&OUTCLS
// STDERR DD SYSOUT=&OUTCLS
// STDOUT DD SYSOUT=&OUTCLS
// *-----
// * Copy report of security validation into DD:REPORT
// *-----
// COPYRPT EXEC PGM=IZUSECRP,REGION=0M,
// PARM=' /TRACE=&TRACE,USERDIR=&USERDIR '
// STDOUT DD SYSOUT=&OUTCLS
// STDERR DD SYSOUT=&OUTCLS
// CEEDUMP DD SYSOUT=&OUTCLS
// REPORT DD SYSOUT=&OUTCLS
// SECCFG DD SYSOUT=&OUTCLS
// SECLOG DD SYSOUT=&OUTCLS

```

IZUSECSV reference information

The following parameters are supported for use in the IZUSECSV procedure.

ROOT='directory-path'

z/OSMF root code directory path. If your installation was configured z/OSMF to use another path for the root code directory, specify that path instead. The path must be enclosed in quotation marks, begin with a forward slash ('/'), and be fully qualified (it cannot be relative). Mixed-case file system names are allowed.

Default: /usr/lpp/zosmf

USERDIR='temporary-path'

Temporary path is a directory that is used to temporarily store intermediate files that such as the report file, log file, and configuration file. If you specify a location other than '/tmp', you need to ensure the owner that runs the security validation has both READ and WRITE permission on the directory. If the security validation is submitted by job IZUSECJL, the owner represents the submitter. If security validation is performed by started task IZUSECSV, the owner represents started task USERID.

The directory path must:

- Be a valid z/OS UNIX path name.
- Be a full or absolute path name.
- Be enclosed in quotation marks.
- Begin with a forward slash ('/').

Default: /tmp

IZUPRM='(NONE/xx/(xx,...,zz))'

z/OSMF parmlib members contain the parameters that are used to define the z/OSMF server. Security validation uses some of the parameters in the parmlib members. It is recommended to use the same set of z/OSMF parmlib members that are specified for the z/OSMF server to run security validation. The parameters JAVA_HOME, SAF_PREFIX, UNAUTH_USER should be the same as the ones specified for the z/OSMF server.

NONE

No parmlib members are specified and the z/OSMF defaults are used. For the default values, refer to the parameter descriptions in [“IZUPRMxx reference information” on page 35](#).

xx|(xx,...,zz)

Specify the specific suffixes for the IZUPRMxx parmlib member or members that you want the procedure to use. If you specify a suffix, the member must exist in your parmlib concatenation. Otherwise, the procedure cannot be started. Multiple suffixes must be enclosed in parentheses.

The following syntax forms are valid:

```
IZUPRM=(xx,yy,zz)
IZUPRM=xx
IZUPRM=NONE
```

Default: NONE

SVRID='server-id'

SVRID represents the z/OSMF started task USERID called server ID. The server ID is validated for the required authorizations to the security profiles used for the z/OSMF nucleus.

If you choose to run the security validation for the server ID with job IZUSECJL, you need to specify the server ID to be validated in the parameter.

If procedure IZUSECSV is chosen, the owner of the procedure IZUSECSV is to be validated as server ID. It is recommended to use the same server ID between procedure IZUSECSV and z/OSMF server.

The server ID must:

- Follow the rules for z/OS user IDs. Partial implementation of security validation is written with unix script. In unix script, the character '\$' is a keyword that is used for representing a variable. If the userid contains the character '\$', backslash ('\') needs to be specified before '\$'. Otherwise, the character '\$' cannot be kept in unix script because it is treated as a keyword.
- Be 1 – 8 characters long.

Default: NOT_SPECIFIED

Note: If the z/OSMF server ID does not have the authorization to the specific security profile required to validate, the message ICH408I appears on SYSLOG.

This example shows when server ID *IZUSVR* has not READ access to profile *BBG.ANGEL.IZUANG1* in the server class. This message cannot be suppressed.

```
ICH408I USER(IZUSVR ) GROUP(IZUADMIN) NAME(ZOSMF STARTED TASK 0) 690
BBG.ANGEL.IZUANG1 CL(SERVER )
INSUFFICIENT ACCESS AUTHORITY
ACCESS INTENT(READ ) ACCESS ALLOWED(NONE )
```

USERID='user-id'

USERID represents a z/OSMF user to be validated for the required authorizations to the security profiles used for z/OSMF Security Configuration Assistant.

The user id must:

- Follow the rules for z/OS user IDs. Partial implementation of security validation is written with unix script. In unix script, the character '\$' is a keyword that is used for representing a variable. If the userid contains the character '\$', backslash ('\') needs to be specified before '\$'. Otherwise, the character '\$' cannot be kept in unix script because it is treated as a keyword.
- Be 1 – 8 characters long.

Default: NOT_SPECIFIED

Note: Security validation supports having both the SVRID and USERID specified.

LOGLEVEL='WARNING'

LOGLEVEL represents log level of Java. Security validation uses Java Technology. Use this option only at the direction of IBM Support.

The following values are valid:

```
OFF
INFO
WARNING
SEVERE
FINE
```

FINER
FINEST
ALL

Default: WARNING

TRACE='Y/N'

Enables tracing for configuration-time errors, such as parmlib parsing errors. The error data is written to the server job log. Use this option only at the direction of IBM Support.

Default: N

OUTCLS='output-class'

Suitable output class for writing system output. By default, the z/OSMF procedures use output class *. The value must be in quotation marks.

Default: *

Examples of security validation

You can refer to these examples to run security validation and review the generated report from output of the submitted job.

Example of validating a server ID by using job IZUSECJL

This example shows how to customize job IZUSECJL to run security validation on a specified server ID.

- [Figure 10 on page 53](#)
- [Figure 11 on page 53](#)
- [Figure 12 on page 53](#)
- [Figure 13 on page 54](#)
- [Figure 14 on page 54](#)

```
//IZUSECJL JOB CLASS=A,MSGCLASS=H,NOTIFY=&SYSUID,MSGLEVEL=(1,1)
//DO EXEC PROC=IZUSECSV,
// SVRID=IZUSVR /* SERVER ID VALIDATED */
```

Figure 10. Example of customized job IZUSECJL to specify the server ID to be validated (Part 1 of 5)

Submit the job IZUSECJL. If the job completion code is 0, a similar output of job IZUSECJL is shown.

NP	DDNAME	StepName	ProcStep	DSID	Owner	C	Dest	Rec-Cnt	Page-Cnt	Byte-Cnt	CC	Rmt	Node	O-Grp-N	SecLabel	PrMod
	JESMSGLG	JES2		2	IBMUSER	H	LOCAL	14		532	1		1	1		LINE
	JESJCL	JES2		3	IBMUSER	H	LOCAL	76		4,673	1		1	1		LINE
	JESYSMSG	JES2		4	IBMUSER	H	LOCAL	61		3,444	1		1	1		LINE
	STDOUT	DO	SECVAL	107	IBMUSER	H	LOCAL	11		565	1		1	1		LINE
	REPORT	DO	COPYRPT	111	IBMUSER	H	LOCAL	40		3,017	1		1	1		LINE
	SECCFG	DO	COPYRPT	112	IBMUSER	H	LOCAL	131		4,242	1		1	1		LINE

Figure 11. Example of a job IZUSECJL output with job completion code of 0 (Part 2 of 5)

Go to DD:REPORT to review the validation results.

```
SUCC: Class APPL is activated.
SUCC: Class SERVER is activated.
SUCC: Class FACILITY is activated.
SUCC: Class SERVAUTH is activated.
SUCC: Class ACCTNUM is activated.
SUCC: Class TSOPROC is activated.
SUCC: Class TSOAUTH is activated.
SUCC: Class OPERCMDS is activated.
SUCC: Class EJBROLE is activated.
SUCC: Class ZMFAPLA is activated.
SUCC: User IZUSVR has READ access to resource BBG.ANGEL.IZUANG1 in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM.SAFCD in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM.ZOSWLM in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM.TXRRS in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM.ZOSDUMP in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECPF.XIZUDFLT in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.ZMFAPLA in SERVER class.
SUCC: User IZUSVR has CONTROL access to resource BBG.SYNC.IZUDFLT in FACILITY class.
SUCC: User IZUSVR has READ access to resource BPX.WLMSEVER in FACILITY class.
SUCC: User IZUSVR has READ access to resource BPX.CONSOLE in FACILITY class.
SUCC: User IZUGUEST has READ access to resource IZUDFLT in APPL class.
SUCC: User IZUSVR has READ access to resource IRR.DIGTCERT.LISTRING in FACILITY class.
SUCC: User IZUSVR has READ access to resource CEA.SIGNAL.ENF83 in SERVAUTH class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.SERVER in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.APPL in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.FACILITY in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.EJBROLE in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.SERVAUTH in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.STARTED in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.ZMFACLOUD in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.ACCTNUM in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.TSOPROC in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.TSOAUTH in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.OPERCMDS in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.CSFSEVER in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.JESSPOOL in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.LOGSTRM in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.UNIXPRIV in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.RDATALIB in SERVER class.
```

Figure 12. Example of successful validation results from job IZUSECJL (Part 3 of 5)

If the specified server ID does not exist, the job completion code is 512. The message IZUG116E appears in DD:STDOUT to indicate that the server ID does not exist.

```
//IZUSECJL JOB CLASS=A,MSGCLASS=H,NOTIFY=&SYSUID,MSGLEVEL=(1,1)
//DO EXEC PROC=IZUSECSV,
// SVRID=JZUSVR /* SERVER ID VALIDATED */
IZUG116E: User JZUSVR to be validated doesn't exist.
```

Figure 13. Example of validation results from job IZUSECJL when the server ID does not exist (Part 4 of 5)

If the server ID does not have the authorizations to the security profiles required for the z/OSMF nucleus, the job completion code is 768. In this example, the server ID does not have the needed authorization to BBG . ANGEL . IZUANG1. The failed validation for the server ID to the security profile BBG . ANGEL . IZUANG1 appears in the first line. Grant the necessary authorization for the security profile to the server ID and rerun the validation to ensure that the validation passes.

```
FAIL: User IZUSVR doesn't have READ access to resource BBG.ANGEL.IZUANG1 in SERVER class.
SUCC: Class APPL is activated.
SUCC: Class SERVER is activated.
SUCC: Class FACILITY is activated.
SUCC: Class SERVAUTH is activated.
SUCC: Class ACCTNUM is activated.
SUCC: Class TSOPROC is activated.
SUCC: Class TSOAUTH is activated.
SUCC: Class OPERCMDS is activated.
SUCC: Class EJBROLE is activated.
SUCC: Class ZMFAPLA is activated.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM.SAFCRED in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM.ZOSWLM in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM.TXRRS in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.AUTHMOD.BBGZSAFM.ZOSDUMP in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECPF.XIZUDFLT in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.ZMFAPLA in SERVER class.
SUCC: User IZUSVR has CONTROL access to resource BBG.SYNC.IZUDFLT in FACILITY class.
SUCC: User IZUSVR has READ access to resource BPX.WLMSEVER in FACILITY class.
SUCC: User IZUSVR has READ access to resource BPX.CONSOLE in FACILITY class.
SUCC: User IZUGUEST has READ access to resource IZUDFLT in APPL class.
SUCC: User IZUSVR has READ access to resource IRR.DIGTCERT.LISTRING in FACILITY class.
SUCC: User IZUSVR has READ access to resource CEA.SIGNAL.ENF83 in SERVAUTH class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.SERVER in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.APPL in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.FACILITY in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.EJBROLE in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.SERVAUTH in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.STARTED in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.ZMFACLOUD in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.ACCTNUM in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.TSOPROC in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.TSOAUTH in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.OPERCMDS in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.CSFSERV in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.JESSPOOL in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.LOGSTRM in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.UNIXPRIV in SERVER class.
SUCC: User IZUSVR has READ access to resource BBG.SECCLASS.RDATALIB in SERVER class.
```

Figure 14. Example of validation results from job IZUSECJL when the server ID does not have necessary authorizations (Part 5 of 5)

Example of validating a user ID by using job IZUSECJL

Figure 10 on page 53 shows how to customize job IZUSECJL to run security validation on a specified user ID. Figure 11 on page 53

- Figure 15 on page 55
- Figure 16 on page 55
- Figure 17 on page 55
- Figure 18 on page 55
- Figure 19 on page 56

```
//IZUSECJL JOB CLASS=A,MSGCLASS=H,NOTIFY=&SYSUID,MSGLEVEL=(1,1)
//DD EXEC PROC=IZUSECSV,
// USERID=ZOSMFAD /* USER ID VALIDATED */
```

Figure 15. Example of customized job IZUSECJL to specify the user ID to be validated (Part 1 of 5)

Submit the job IZUSECJL. If the job completion code is 0, a similar output of job IZUSECJL is shown.

NP	DDNAME	StepName	ProcStep	DSID	Owner	C	Dest	Rec-Cnt	Page-Cnt	Byte-Cnt	CC	Rmt	Node	O-Grp-N	SecLabel	PrMod
	JESMSG LG	JES2		2	IBMUSER	H	LOCAL	14		532	1		1	1		LINE
	JESJCL	JES2		3	IBMUSER	H	LOCAL	76		4,673	1		1	1		LINE
	JESYSMSG	JES2		4	IBMUSER	H	LOCAL	61		3,444	1		1	1		LINE
	STDOUT	DD	SECVAL	107	IBMUSER	H	LOCAL	11		565	1		1	1		LINE
	REPORT	DD	COPYRPT	111	IBMUSER	H	LOCAL	40		3,017	1		1	1		LINE
	SECCFG	DD	COPYRPT	112	IBMUSER	H	LOCAL	131		4,242	1		1	1		LINE

Figure 16. Example of job IZUSECJL output with completion code of 0 (Part 2 of 5)

Go to DD:REPORT to review the validation results.

```
SUCC: Class APPL is activated.
SUCC: Class SERVER is activated.
SUCC: Class FACILITY is activated.
SUCC: Class SERVAUTH is activated.
SUCC: Class ACCTNUM is activated.
SUCC: Class TSOPROC is activated.
SUCC: Class TSOAUTH is activated.
SUCC: Class OPERCMDS is activated.
SUCC: Class EJBROLE is activated.
SUCC: Class ZMFAPLA is activated.
SUCC: User ZOSMFAD has READ access to resource IZUDFLT in APPL class.
SUCC: User ZOSMFAD has READ access to resource IZUDFLT.IzuManagementFacility.IzuUsers in EJBROLE class.
SUCC: User ZOSMFAD has READ access to resource IZUDFLT.IzuManagementFacilityHelpApp.IzuUsers in EJBROLE class.
SUCC: User ZOSMFAD has READ access to resource IZUDFLT.IzuManagementFacilityImportUtility.IzuUsers in EJBROLE class.
SUCC: User ZOSMFAD has READ access to resource IZUDFLT.ZOSMF in ZMFAPLA class.
SUCC: User ZOSMFAD has READ access to resource IZUDFLT.ZOSMF.CONFIGURATION.SECURITY_ASSISTANT in ZMFAPLA class.
SUCC: User ZOSMFAD has READ access to resource IZUDFLT.IzuManagementFacilitySecurityConfigurationAssistant.IzuUsers in EJBROLE class
```

Figure 17. Example of successful user ID validation results from job IZUSECJL (Part 3 of 5)

If the specified user ID does not exist, the job completion code is 512. The message IZUG116E appears in DD:STDOUT to indicate that the user ID does not exist.

```
//IZUSECJL JOB CLASS=A,MSGCLASS=H,NOTIFY=&SYSUID,MSGLEVEL=(1,1)
//DD EXEC PROC=IZUSECSV,
// SVRID=JOSMFAD /* USER ID VALIDATED */
IZUG116E: User JOSMFAD to be validated doesn't exist.
```

Figure 18. Example of validation results from job IZUSECJL when the server ID does not exist (Part 4 of 5)

If the user ID does not have the necessary authorizations to the security profiles required for z/OSMF Security Configuration Assistant, the job completion code is 768. In this example, user ID ZOSMFAD does not have the needed authorization to IZUDFLT.ZOSMF.CONFIGURATION.SECURITY_ASSISTANT. The failure validation for the user ID to security profile BBG.ANGEL.IZUANG1 appears in the first line. Grant

the needed authorization that is required for the security profile to the user ID and rerun the validation to ensure the validation passes.

```
//IZUSECJL JOB CLASS=A,MSGCLASS=H,NOTIFY=&SYSUID,MSGLEVEL=(1,1)
//DO EXEC PROC=IZUSECSV,
// SVRID=JOSMFAD /* USER ID VALIDATED */
IZUG116E: User JOSMFAD to be validated doesn't exist.
```

Figure 19. Example of validation results from job IZUSECJL when the user ID does not have necessary authorizations (Part 5 of 5)

Example of validating a user ID by using procedure IZUSECSV

Figure 20 on page 57 shows by using procedure IZUSECSV to validate a user ID. Start the started task IZUSECSV with the START command. The result is same as running job IZUSECJL.

```
S IZUSECSV,USERID=ZOSMFAD
```

Figure 20. Example of using procedure IZUSECSV to validate a user ID

Part 3. z/OSMF core services

To create a viable z/OSMF system, you must add z/OSMF core services to the z/OSMF nucleus. In this part, you choose the z/OSMF core services that you plan to use, then configure those services.

You can choose to install only a subset of the core services, or you can configure all of them. Because some core services require other services to be enabled, you might need to configure more services than the ones you plan to use. If so, these dependencies are noted in each service description.

For most installations, it is recommended that you configure all of the core services.

Table 11 on page 59 lists the z/OSMF core services.

<i>Table 11. z/OSMF core services</i>
Instructions for configuring the z/OSMF service
Chapter 6, “Configure the z/OSMF administration tasks,” on page 61
Chapter 7, “Configure the z/OSMF Workflows task,” on page 63
Chapter 8, “Configure the Notifications task,” on page 65
Chapter 9, “Configure the z/OSMF settings service,” on page 67
Chapter 10, “Configure the Swagger service,” on page 69
Chapter 11, “Configure the z/OS jobs REST services,” on page 71
Chapter 12, “Configure the z/OS data set and file REST services,” on page 73
Chapter 13, “Configure the TSO/E address space services,” on page 77

Chapter 6. Configure the z/OSMF administration tasks

To use the z/OSMF administration tasks, you must configure it as described in this topic.

The z/OSMF administration tasks include the following functions:

- Application Linking Manager
- Import Manager
- Usage Statistics
- Links
- z/OSMF Diagnostic Assistant

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides job IZUATSEC in SYS1.SAMPLIB. The job contains RACF commands for creating the required security authorizations.

The IZUATSEC job includes commands for:

- Authorizing z/OSMF administrators to the z/OSMF administration tasks
- Authorizing z/OSMF administrators and z/OSMF users to view the web links that are defined in the z/OSMF Links task.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary. To authorize users, grant the required users or groups READ access to the resource, as shown in the IZUATSEC job.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUATSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

Depending on your installation's security procedures, a web link might require more protection through a discrete profile. It is recommended that the z/OSMF administrator work with the security administrator to determine whether a new link requires protection through a discrete profile.

In the Links task, the z/OSMF administrator defines a link by specifying a name for the link and its URL. The Links task also includes a text entry window that requires the z/OSMF administrator to further qualify the link resource name with a SAF resource name, if a discrete profile is required for the link. If so, the z/OSMF administrator can provide this fully qualified resource name to the security administrator to use to create the user authorizations for the link.

For example, the following RACF commands can be used to define a discrete profile for a new link (the z/OS Basics Information Center website) and permit a group (IZUUSER) to that link:

```
RDEFINE ZMFAPLA (IZUDFLT.ZOSMF.LINK.Z_OS_BASICS_INFORMATION_CENTER) UACC(NONE)
PERMIT IZUDFLT.ZOSMF.LINK.Z_OS_BASICS_INFORMATION_CENTER CLASS(ZMFAPLA) ID(IZUUSER) ACC(READ)
```

If you change a link SAF resource name through the Links task, ensure that the new link resource name is adequately protected through a ZMFAPLA resource profile definition. You might need to create a new profile to properly secure the link. Deleting an existing link might require your security administrator to delete the discrete profile, if one is used to secure access to the link.

Note: If your system uses JES3 as its primary subsystem, and you find that jobs are not running, verify that JES3 is configured to allow multiple jobs with the same name. The JES3 option DUPJOBNM option must be set to YES. For more information, see the note regarding JES3 in [“Host system customization” on page 77](#).

Optional extensions to this service

None.

Chapter 7. Configure the z/OSMF Workflows task

To use the z/OSMF Workflows task, you must configure it as described in this topic.

Dependencies on other z/OSMF services

The z/OSMF Workflows task requires the following z/OSMF services to be configured:

- Common event adapter (CEA); see [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#).
- Notifications service; see [Chapter 8, “Configure the Notifications task,” on page 65](#).
- z/OSMF Settings service; see [Chapter 9, “Configure the z/OSMF settings service,” on page 67](#).
- z/OS jobs REST services; see [Chapter 11, “Configure the z/OS jobs REST services,” on page 71](#).
- z/OS data set and file REST services; see [Chapter 12, “Configure the z/OS data set and file REST services,” on page 73](#).
- TSO/E address space services; see [Chapter 13, “Configure the TSO/E address space services,” on page 77](#).

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUWFSEC in SYS1.SAMPLIB. This security profile defines the required profiles for z/OSMF Workflows and the Workflow Editor, including the creation of the workflow administrator role and the refreshing of several resource classes.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUWFSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

If you want to use the workflow signing function, refer to the chapter [Configuring the z/OSMF workflow signing certificate](#) for additional configuration information.

Maximum number of active and archived workflows

It is recommended to use Workflow Settings to set the maximum number of active and archived workflows. In the main Workflows panel, click **Settings** to open Workflows Settings and input the maximum number of active or archived workflows. Click **OK** to save the new value. The default maximum number of active and archived workflows is 200.

For the maximum number of active workflows, if your system increased the maximum workflows previously using IZU_WORKFLOWS_MAXIMUM in `local_override.cfg`, and you want to remove IZU_WORKFLOWS_MAXIMUM from `local_override.cfg`, you need to perform the following steps:

1. Open Workflows Settings and set your new maximum number of active workflows.
2. Click **OK** to save the new maximum value.
3. Remove the IZU_WORKFLOWS_MAXIMUM entry from `local_override.cfg`. If IZU_WORKFLOWS_MAXIMUM is the only parameter set in `local_override.cfg`, you can delete `local_override.cfg` directly.
4. Recycle the z/OSMF server.

Host system customization

None.

Optional extensions to this service

None.

Chapter 8. Configure the Notifications task

To use the Notifications service, you must configure it as described in this topic.

Dependencies on other z/OSMF services

None

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUNFSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUNFSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Optional extensions to this service

None.

Chapter 9. Configure the z/OSMF settings service

To use the z/OSMF settings service, you must configure it as described in this topic.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUSTSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUSTSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Optional extensions to this service

You can further extend the function of this service by enabling other, corequisite services in z/OSMF. [Table 12 on page 67](#) describes these dependencies.

Table 12. Optional extensions to the z/OSMF Settings service

Functional extensions to this service	Co-requisite z/OSMF services	Reason for enabling
Automatic feedback collection in the z/OSMF General Settings task. For set up instructions, see Chapter 8, “Configure the Notifications task,” on page 65, and Chapter 30, “Configure z/OS Data Gatherer: Monitor II REST Services,” on page 197.	• Notifications task • z/OS Data Gatherer: Monitor II REST Services	Allow IBM to collect the following information monthly to help us improve your experience. • z/OSMF Services enabled • z/OSMF tasks and REST services usage • System Operational Information

Chapter 10. Configure the Swagger service

To use the Swagger service, you must configure it as described in this topic.

Dependencies on other z/OSMF services

You can use the Swagger service to display information about the following z/OSMF Representational State Transfer (REST) APIs:

- z/OSMF Security Configuration Assistant
- Cloud provisioning services
- z/OS data set and file REST services
- z/OS jobs REST services
- z/OS console services
- TSO/E address space services
- Notification services
- z/OSMF information retrieval service
- z/OSMF workflow services
- z/OSMF Sysplex Management
- z/OSMF Storage Management
- z/OSMF Software Management
- z/OS Management Services Catalog
- z/OSMF Parmlib Management

Before you can display information about these z/OSMF services, you must configure them.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUSWSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUSWSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Optional extensions to this service

None.

Chapter 11. Configure the z/OS jobs REST services

To use the z/OS jobs REST services, you must configure it as described in this topic.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZURJSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary. For example, update the job card information.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZURJSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Optional extensions to this service

You can further extend the function of this service by enabling other, corequisite services in z/OSMF. [Table 13 on page 71](#) describes these dependencies.

Table 13. Optional extensions to the z/OS jobs REST services		
Functional extensions to this service	Co-requisite z/OSMF services	Reason for enabling
Asynchronous processing for subset of REST Jobs API	• Common Information Model (CIM) server is active.	For a JES3 environment, the z/OS Common Information Model (CIM) server must be configured, as described in Chapter 47 , “ Configuring the CIM server for your system ,” on page 253 .

Chapter 12. Configure the z/OS data set and file REST services

To use the z/OS data set and file REST services, you must configure it as described in this topic.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZURFSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZURFSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

Host system customization consists of the following tasks:

- [“Ensure that CEA is active in full-function mode” on page 73](#)
- [“Logon procedure for the z/OSMF REST interfaces” on page 73](#)
- [“Output messages file” on page 74](#)
- [“Reviewing the IPCMSGQBYTES option of BPXPRMxx” on page 75](#)

Ensure that CEA is active in full-function mode

The z/OS common event adapter (CEA) service must be configured and started in full-function mode to run this service, as described in [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#).

Logon procedure for the z/OSMF REST interfaces

IBM supplies a default logon procedure, IZUFPROC, in the PROCLIB data set. The procedure is used internally by the z/OSMF REST interfaces. Review the procedure to ensure that it is suitable for use in your environment.

Ensure that the SMP/E-managed PROCLIB data set resides in the JES PROCLIB concatenation that is used for TSO/E logon procedures. Or, copy IZUFPROC to a data set that is in the JES PROCLIB concatenation. For more information, see [z/OS TSO/E Customization](#).

Note: z/OSMF uses the ISPEXEC load module in the ISPF library SISPLoad. If your installation does not include the SISPLoad data set in the link list, you must add SISPLoad to the ISPLLIB DD concatenation in the logon procedure.

If you prefer, you can use a different logon procedure, if it provides the same function as the shipped IZUFPROC procedure. Specifically, the logon procedure must contain, at a minimum:

- All of the DD statements from IZUFPROC; these must reference the system data sets that contain the z/OS UNIX REXX exec programs and ISPF libraries.

- The PROC statement must specify the z/OSMF root code directory path on the ROOT variable, for example: `ROOT= '/usr/lpp/zosmf'`

If your installation configured z/OSMF to use another path for the root code directory, specify that path instead. The path must be enclosed in quotation marks, begin with a forward slash ('/'), and be fully qualified (it cannot be relative). Mixed-case file system names are allowed.

- If your installation uses permanent (non-temporary) data set for ISPFPROF, the logon procedure must be configured to allow profile sharing.
- The PROC statement must specify the Language Environment message file (MSGFILE), which is used for storing runtime messages from the z/OS data set and file REST services. For details, see [“Output messages file”](#) on page 74.

The topic [“IZUPRMxx reference information”](#) on page 35 describes options for the TSO/E logon procedure that can be specified on the COMMON_TSO statement. You can specify a different TSO/E logon procedure name, account number, and address space region size, or use the default specifications.

The defaults should be adequate for most z/OS installations. If you specify alternative values, you must ensure that the z/OSMF user and z/OSMF administrator security groups are authorized to use the logon procedure name and account number that you specify. Also, ensure that the address space region size is at least 50 MB, and that your SMFLIMxx parmlib member and IEFUSI exit allow TSO/E users to use this amount of memory.

All z/OSMF users must have TSO segments that are defined in your installation's security database. Failure to have a TSO segment causes some z/OSMF functions not to work.

Output messages file

The z/OS data set and file REST services write runtime messages to a common output messages file. The messages describe error conditions and suggest possible solutions to the errors. By default, these messages are written to SYSOUT.

Your installation can select another destination for message output by using the Language Environment MSGFILE runtime option. In the logon procedure that is used by the z/OS data set and file REST services, ensure that the message file ddname is specified, as follows:

- If your installation does not specify a message file ddname, you must ensure that the SYSOUT DD statement is specified in the logon procedure. For example:

```
//SYSOUT DD SYSOUT=H
```

- If your installation uses the Language Environment MSGFILE runtime option, you must ensure that the logon procedure is changed accordingly.

To view the current MSGFILE definition on your system, you can use the following command:

```
D CEE,ALL
```

Which displays output, such as the following:

```
PARMLIB(CEEPRM60) MSGFILE(LEMSG,FBA,121,0,NOENQ)
```

In this example, the message file ddname is LEMSG (not SYSOUT). Thus, you would modify the logon procedure, as follows:

```
//LEMSG DD SYSOUT=H
```

Or, to write the data to a specific data set, you can modify the logon procedure, as follows:

```
//LEMSG DD DSN=YOUR.CREATED.DATASET,DISP=OLD
```

In this example, the diagnostic logs are written to the data set that you specify. Ensure that the data set is created with the format FBA, 121, 0, NOENQ.

Reviewing the IPCMSGQBYTES option of BPXPRMxx

The z/OS data set and file REST interface uses the z/OS UNIX System Services interprocess communications (IPC) message queue for communications between TSO/E and z/OSMF. The maximum message size is controlled by the size of the queue that is defined by the IPCMSGQBYTES option of parmlib member BPXPRMxx.

It is required that you specify an IPCMSGQBYTES value of at least 20971520 (20 M) in BPXPRMxx. To set this value dynamically, you can enter the following operator command:

```
SETOMVS IPCMSGQBYTES=20971520
```

Optional extensions to this service

None.

Chapter 13. Configure the TSO/E address space services

To use the TSO/E address space services, you must configure it as described in this topic.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUTSSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUTSSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

The z/OS common event adapter (CEA) service must be configured and started in full-function mode to run this service. See [“Ensure that common event adapter \(CEA\) is configured and active”](#) on page 17.

Note: If your system uses JES3 as its primary subsystem, ensure that JES3 is configured to allow multiple jobs with the same name; the JES3 option DUPJOBNM option must be set to YES. If DUPJOBNM is set to NO, you can change it to YES, for example, by entering the following command at the system console:

```
*MODIFY Q,DUPJOBNM=YES
```

Optional extensions to this service

None.

Chapter 14. Configure the z/OS System Variable services

System Variable services have two parts: z/OSMF variable services and System Symbol services. If you only need to use z/OSMF variable services, no configuration is needed. However, to use System Symbol services you must configure it as described in this topic.

Dependencies on other z/OSMF services

The System Symbols services require the following services to be configured:

Common event adapter (CEA)

For more information, see [“Ensure that common event adapter \(CEA\) is configured and active” on page 17.](#)

z/OSMF Settings

For more information, see [Chapter 9, “Configure the z/OSMF settings service,” on page 67.](#)

TSO/E address space services

For more information, see [Chapter 13, “Configure the TSO/E address space services,” on page 77.](#)

z/OS Console service

For more information, see [Chapter 22, “Configure the Operator Console task,” on page 113.](#)

Security setup

To assist you with the security setup, IBM provides the sample security job IZUSVSEC in SYS1.SAMPLIB.

Follow these steps:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUSVSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Optional extensions to this service

None.

Part 4. z/OSMF optional services

You can add significant function to z/OSMF by adding one or more of the optional services. Which services you select depends on your goals for z/OSMF.

In this part, you choose the z/OSMF optional services that you require, then configure those services. Because some z/OSMF services require other z/OSMF services to be enabled, you might need to configure more services than only the ones you plan to use. These dependencies are noted in each service description.

When you configure the optional services, you might find it easier to start with services that require little or no system customization, such as ISPF and Network Configuration Assistant, and then progress to services with more extensive requirements, such as Cloud Provisioning and Incident Log. In this document, the optional services are presented in ascending order, based on their complexity of setup. Table 14 on page 81 lists the services in this order.

Table 14. z/OSMF optional services: Listed in order of setup complexity		
Complexity of setup	Instructions for configuring the z/OSMF service	Notes:
Low	Chapter 15, “Configure the Security Configuration Assistant service,” on page 83	None.
Low	Chapter 16, “Configure the Network Configuration Assistant service,” on page 85	None.
Low	Chapter 17, “Configure the Software Management service,” on page 87	Requires common event adapter (CEA) to be active on your system in full function mode.
Low	Chapter 19, “Configure the ISPF service,” on page 101	Requires common event adapter (CEA) to be active on your system in full function mode.
Low	Chapter 20, “Configure the z/OS Management Services Catalog task,” on page 103	None.
Low	Chapter 18, “Configure the Storage Management service,” on page 99	None.
Low	Chapter 34, “Configure the Parmlib Management service,” on page 205	None.
Moderate	Chapter 21, “Configure the Resource Monitoring service,” on page 105	Requires the RMF Distributed Data Server (DDS) to be active. Might require the use of a PassTicket for authentication.
Moderate	Chapter 22, “Configure the Operator Console task,” on page 113	Requires common event adapter (CEA) to be active on your system in full function mode.
Moderate	Chapter 23, “Configure the Workload Management service,” on page 117	None.
Moderate	Chapter 24, “Configure the Sysplex Management service,” on page 121	Intended for sysplex configurations.
Moderate	Chapter 25, “Configure the Capacity Provisioning service,” on page 125	Requires the CIM server to be active.

Table 14. z/OSMF optional services: Listed in order of setup complexity (continued)		
Complexity of setup	Instructions for configuring the z/OSMF service	Notes:
Moderate	Chapter 26, “Configure IBM z/OS Encryption Readiness Technology (zERT) Network Analyzer,” on page 129	Requires Db2
High	Chapter 27, “Configure the Cloud Provisioning services,” on page 141	Requires a number of other z/OSMF services to be configured.
High	Chapter 28, “Configure the Incident Log service,” on page 177	Requires a number of z/OS components to be enabled, such as CEA, CIM, DAE, EREP, IPCS dump services, SDUMP, and System Logger.

In general, enabling an optional service involves the following activities:

- In your active IZUPRMxx member, ensure that the PLUGINS statement is uncommented and includes the service ID for the required service. See “Statements and parameters for IZUPRMxx” on page 37.
- Create security profiles for the z/OSMF tasks and REST services that are associated with the service. IBM provides a set of IZUnnSEC jobs in SYS1.SAMPLIB with RACF commands to help with performing these changes. Each IZUnnSEC job is associated with a service, as described in “Security concepts in z/OSMF” on page 5.
- Perform the various z/OS system customization updates that are associated with each service, as described in the sections that follow.

Note: Some z/OSMF tasks use FTP to transmit data. If your network contains a firewall that blocks FTP traffic or does not allow authentication using FTP, you must perform an additional action to allow the traffic to pass. For considerations, see the online help for the Task Settings task.

Removing services

After a service is enabled, you might later decide to remove it. To do so:

1. Edit the IZUPRMxx parmlib member and remove the service identifier from the PLUGINS statement. Not all services have an associated identifier in IZUPRMxx.
2. Remove the security profiles and authorizations for the service from your external security manager (ESM).
3. Restart the z/OSMF server. This action removes the icons from the z/OSMF desktop. Any residual data that is associated with the service is saved in z/OSMF, in case you decide to enable it again later.

External applications

Besides the services that are supplied with z/OSMF, your installation can choose to add applications from other sources (IBM or other vendors) to your configuration. In such cases, a z/OSMF administrator can use the Import Manager task to import the applications into z/OSMF. For more information, see the online help for the Import Manager task.

As an example, z/OS System Display and Search Facility (SDSF) supplies a service for use with z/OSMF. For the installation and customization requirements for a particular application, see the documentation that is provided with the application. For example, the setup requirements for the SDSF service are described in the topic about z/OSMF considerations in *z/OS SDSF Operation and Customization*.

Further, your installation can create its own applications to use with z/OSMF. For more information, see *IBM z/OS Management Facility Programming Guide*.

Chapter 15. Configure the Security Configuration Assistant service

To use the z/OSMF Security Configuration Assistant, configure it as described in this topic.

Description

The Security Configuration Assistant provides a visual framework for examining the different elements of z/OSMF security. The Security Configuration Assistant layout consists of tabbed sections and tabular reports that can be expanded or compressed, as needed. This framework provides a comprehensive perspective on your z/OSMF security setup.

You can use the Security Configuration Assistant to check the authorizations for z/OSMF itself, including the nucleus, core and optional services, and advanced configuration options. You can also check the security setup for other products on your system for which you have installed the required security descriptor files.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUSASEC in SYS1.SAMPLIB. For a summary of the required profile authorizations, see [“Resource authorizations for the Security Configuration Assistant service”](#) on page 374.

Carefully review the contents of job IZUSASEC before you submit it. Observe the following considerations:

- Ensure that only the appropriate security administrators or system programmers are authorized to use the Security Configuration Assistant. As shipped from IBM, the IZUSASEC sample job grants authority to users in the IZUADMIN security group. If you do not want to enable all users in the IZUADMIN group to run the tool, edit the job and specify the permitted user ID or group. In the job, this authorization is created with the following PERMIT statement:

```
PERMIT IZUFLT.ZOSMF.CONFIGURATION.SECURITY_ASSISTANT +  
      CLASS(ZMFAPLA) ACCESS(READ) ID(IZUADMIN)
```

- The job includes JCL for authorizing a user ID to a number of BBG security profiles. Be aware that the BBG.SECCLASS.xx SERVER profiles should be permitted only to the z/OSMF started task user ID.
- Before you use the Security Configuration Assistant, verify that the z/OSMF server started task user ID:
 - Has READ access to the z/OSMF SAF prefix in the APPL resource class. By default, the resource is IZUFLT (APPL) and the z/OSMF server user ID is IZUSVR.
 - Is connected to the z/OSMF administrator security group, which is IZUADMIN by default.

To run the IZUSASEC job, do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUSASEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Optional extensions to this service

You can check the security configuration for external products on your z/OS system. This option requires that you obtain and install a security descriptor file from the product vendor. For more information, see [Appendix B, “Creating security descriptor files for the Security Configuration Assistant task,” on page 411.](#)

Chapter 16. Configure the Network Configuration Assistant service

To use the Network Configuration Assistant, you must configure it as described in this topic.

Description

You can use the Network Configuration Assistant task to configure TCP/IP policy-based networking functions.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUCASEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUCASEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Note: If your installation uses the Windows desktop version of Network Configuration Assistant, you can optionally transfer your existing configuration data into the z/OSMF environment. If you have a backing store that was exported from another version or instance of the Network Configuration Assistant, select **Transfer Backing Store File to z/OSMF** and provide the fully qualified path and file name for the exported backing store, and the z/OSMF backing store name to transfer the file.

Optional extensions to this service

You can further extend the function of this service by enabling other, corequisite services in z/OSMF. [Table 15 on page 85](#) describes these dependencies.

Table 15. Optional extensions to the Network Configuration Assistant service		
Functional extensions to this service	Co-requisite z/OSMF services	Reason for enabling
Networking configuration functions for IBM Cloud Provisioning and Management for z/OS	• Cloud Provisioning • Notifications • z/OSMF Settings • z/OSMF Workflows	If you plan to use the Cloud Provisioning service, the cloud networking functions of Network Configuration Assistant are a required service. The cloud networking functions of Network Configuration Assistant in turn, require the remainder of the services in the "Co-requisite z/OSMF services" column.

Table 15. Optional extensions to the Network Configuration Assistant service (continued)

Functional extensions to this service	Co-requisite z/OSMF services	Reason for enabling
z/OSMF workflows for policy-based networking functions	• z/OSMF Workflows	Network Configuration Assistant includes several workflows that can be useful for policy-based networking functions.

Chapter 17. Configure the Software Management service

To use the Software Management task, you must configure it as described in this topic.

Description

You can use the Software Management task to manage your z/OS software inventory, deploy SMP/E packaged and installed software, and generate reports about your software.

Dependencies on other z/OSMF services

The Software Management service requires the following services to be configured:

- TSO/E address space services, as described in [Chapter 13, “Configure the TSO/E address space services,” on page 77](#). In addition, to enable the Software Update task ensure the TSO/E region size is set to at least 2,000,000. Set this value on the COMMON_TSO REGION statement in the IZUPRMxx parmlib member. For more information, see [“IZUPRMxx reference information” on page 35](#).
- z/OS data set and file REST services, as described in [Chapter 12, “Configure the z/OS data set and file REST services,” on page 73](#)
- z/OS jobs REST services, as described in [Chapter 11, “Configure the z/OS jobs REST services,” on page 71](#)

You might also need to enable z/OSMF Workflows; see [“Optional extensions to this service” on page 87](#).

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUDMSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUDMSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

Your z/OS system requires further customization. Follow the instructions in the following topic:

- [“Updating z/OS for the Software Deployment service” on page 88](#)

Note: If your system uses JES3 as its primary subsystem, and you find that jobs are not running, verify that JES3 is configured to allow multiple jobs with the same name. The JES3 option DUPJOBNM option must be set to YES. For more information, see the note regarding JES3 in [“Host system customization” on page 77](#).

Optional extensions to this service

You can further extend the function of this service by enabling other, corequisite services in z/OSMF. [Table 16 on page 88](#) describes these dependencies.

Table 16. Optional extensions to the Software Management service

Functional extensions to this service	Co-requisite z/OSMF services	Reason for enabling
z/OSMF Workflows. For set-up instructions, see Chapter 7, “Configure the z/OSMF Workflows task,” on page 63.	z/OSMF Workflows	If you want to open a workflow from a software instance in Software Management, you require the z/OSMF Workflows service.

Updating z/OS for the Software Deployment service

If you selected to configure the Software Deployment service, you might have system customization to perform, as described in this topic.

The Software Deployment service contains the Software Management task, which becomes available to users in the z/OSMF desktop interface when you configure the service.

The Software Management task:

- Allows all users of the task to access deployment objects. Optionally, your installation can further restrict these authorizations, as described in the topic “Creating access controls for the Software Management task” on page 88.
- Works only with systems in the local sysplex. Optionally, your installation can allow the Software Management task to work with other sysplexes in your installation, as described in Chapter 37, “Configuring a primary z/OSMF for communicating with secondary instances,” on page 221.

Creating access controls for the Software Management task

The Software Management task allows users with proper authorization to manage global zones, software instances, deployments, and categories. For some actions, users must also have appropriate authorization to the physical resource these objects describe, such as a target zone or data set. This topic describes how to control user access to the objects in the Software Management task. Creating access controls for the actual physical resource is outside the scope of z/OSMF.

You can use a security manager, such as RACF, to control access to the task and to create more granular authorizations, such as restricting access to an object or an action. Access to the Software Management task and its objects are controlled through the following default resource profiles, which are defined in the ZMFAPLA class:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.**
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.PRODUCT_INFO_FILE.RETRIEVE
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.CATEGORIES.MODIFY
```

With the default access authorities in effect, z/OSMF users and administrators are allowed to perform all actions for all software instances, portable software instances, and deployments, and only z/OSMF administrators are allowed to retrieve information from product information files and add, modify, or remove categories.

Important: All users of the Software Management task should be permitted at least READ access to profile <SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.**. Otherwise, no actions can be performed because users will not have access to any objects.

To further restrict access to the objects and actions, define a SAF resource profile for each object and grant users the appropriate access authority. Regardless of where the physical resource that is described by an object resides, the SAF profiles for that object must be defined on the z/OS system that hosts the z/OSMF instance to which a user's web browser is connected. The Software Management task uses this z/OS system when it checks SAF authorizations.

Use the SAF resource names, which are generated by the Software Management task, to help you define profiles that control user access to an object or an action. The SAF resource names for each object are constructed using properties of the object. The casing that is used for each property value is preserved; therefore, SAF resource names are case-sensitive. The SAF resource name format that is used for each object type and supported actions are described in the sections that follow.

Authorizing users to software instances

A software instance describes a deployable unit of software, which is composed of data sets containing SMP/E installed software. To control access to a specific software instance, define a SAF resource profile for that resource. The SAF resource name for a software instance object has the following format:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.SWI.category.systemName.instanceName
```

Where:

- **SWI** indicates that the object that is associated with this SAF resource is a software instance.
- **category** is the name of the category that is assigned to the software instance. If multiple categories are assigned, a separate SAF resource name is created for each category. If no category is assigned, the category value is NOCATEGORY.

To perform an action, users must have the access authority that is required for that action for all the SAF resource names that are associated with the software instance.

- **systemName** is the name of the z/OSMF host system that has access to the volumes and data sets where the software instance resides. The system is inherited from the global zone that is associated with the software instance, and is defined in the Systems task.
- **instanceName** is the name of the software instance.

For example, if you have a software instance that is named z/OSV2R3_Test that can be accessed by system AQFT and is assigned to categories z/OS and Test, its SAF resource names would be:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.SWI.z/OS.AQFT.z/OSV2R3_Test
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.SWI.Test.AQFT.z/OSV2R3_Test
```

Table 17 on page 89 lists the access authorities that you can assign to software instance resources and the actions that are permitted for each access authority. The Software Management task does not perform authorization checks to determine which software instances to display in a list or table; therefore, all software instances are displayed regardless of access authority.

Table 17. Actions users can take against software instances by access authority	
Access Authority	Actions Allowed
READ	• View the properties of the software instance. • View information about the products, features, and FMIDs contained in a software instance. • View information about the data sets contained in a software instance. • Copy the properties of the software instance. • Deploy the software instance during a deployment. • Use the software instance as the model for priming a deployment configuration. • Generate reports for the software instance. • Export the software instance. • Perform workflows for the software instance.

Table 17. Actions users can take against software instances by access authority (continued)

Access Authority	Actions Allowed
UPDATE	In addition to the actions specified for READ access, users can perform the following actions: • Modify the software instance properties that are <i>not</i> used to create the SAF resource name for the software instance. This includes modifying the software instance explicitly using the Modify action or implicitly when completing a deployment where the objective is to replace the software instance. • Replace the software instance during a deployment. • Retrieve information from SMP/E about the products, features, and FMIDs contained in the software instance and make that information available to z/OSMF. • Update the software instance in the Software Update task. • Delete temporary catalog aliases for the software instance.
CONTROL	In addition to the actions specified for READ and UPDATE access, users can perform the following actions: • Create new software instances explicitly using the Add action or implicitly as part of the Copy action or when completing a deployment where the objective is to create a new software instance. • Modify the software instance properties that are used to create the SAF resource name for the software instance and control access to the software instance. This includes modifying the software instance explicitly using the Modify action or implicitly when completing a deployment where the objective is to replace the software instance. • Remove the software instance.

Authorizing users to portable software instances

Each software instance archive has a unique SAF resource name that can be used by your security manager to control access to the portable software instance. The SAF resource name for a portable software instance archive object has the following format:

```
<safPrefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.PSWI.category.systemName.portableSwiName
```

where:

PSWI

Indicates that the object that is associated with this SAF resource is a portable software instance.

category

Is the name of the category that is assigned to the portable software instance. If multiple categories are assigned, a separate SAF resource name is created for each category. If no category is assigned, the category value is NOCATEGORY. To perform an action, users must have the access authority that is required for that action for all the SAF resource names that are associated with the portable software instance.

systemName

Is the nickname of the z/OSMF host system that has access to the UNIX directory where the portable software instance resides. The system is defined in the z/OSMF Systems task.

portableSwiName

Is the name of the portable software instance.

The following describes the access authority levels that are used to control access to portable software instance objects and the actions that are permitted for each access authority. The Software Management task does not perform authorization checks to determine which portable software instances to display in a list or table; therefore, all portable software instances are displayed regardless of a user's allowed access authority.

<i>Table 18. Actions users can take against portable software instances by access authority</i>	
Access Authority	Actions Allowed
READ	• View the properties of the portable software instance. • Deploy the portable software instance during a deployment.
UPDATE	In addition to the actions specified for READ access, users can perform the following action: • Modify the portable software instance properties that are not used to create the SAF resource name for the portable software instance.
CONTROL	In addition to the actions specified for READ and UPDATE access, users can perform the following actions: • Create new portable software instances explicitly by using the Add action. • Modify the portable software instance properties that are used to create the SAF resource name for the portable software instance and control access to the portable software instance. • Remove the portable software instance.

Authorizing users to deployments

A deployment is a checklist that guides users through the process of cloning or deploying a software instance, and it is the object in which z/OSMF stores information about the clone, such as its data set names and locations, catalog structure, and SMP/E zone names. To control access to a specific deployment, define a SAF resource profile for that resource. The SAF resource name for a deployment object has the following format:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.DEP.category.deploymentName
```

where:

- **DEP** indicates that the object that is associated with this SAF resource is a deployment.
- **category** is the name of the category that is assigned to the deployment. If multiple categories are assigned, a separate SAF resource name is created for each category. If no category is assigned, the category value is NOCATEGORY.

To perform an action, users must have the access authority that is required for that action for all the SAF resource names that are associated with the deployment.

- **deploymentName** is the name of the deployment.

For example, if you have a deployment that is named z/OS_R21_Production that is not assigned to any category, its SAF resource name would be:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.DEP.NOCATEGORY.z/OS_R21_Production
```

Table 19 on page 92 lists the access authorities that you can assign to deployment resources and the actions that are permitted for each access authority. The Software Management task does not perform authorization checks to determine which deployments to display in a list or table; therefore, all deployments are displayed regardless of access authority.

Table 19. Actions users can take against deployments by access authority	
Access Authority	Actions Allowed
READ	• View the properties of the deployment. • Copy the properties of the deployment.
UPDATE	In addition to the actions specified for READ access, users can perform the following actions: • Modify the deployment properties that are <i>not</i> used to create the SAF resource name for the deployment. • Cancel the deployment. This action ends the deployment, unlocks the associated software instances, and limits all future actions for the deployment to View and Remove.
CONTROL	In addition to the actions specified for READ and UPDATE access, users can perform the following actions: • Create new deployments explicitly by using the New action or implicitly as part of the Copy action. • Modify the deployment properties that are used to create the SAF resource name for the deployment and control access to the deployment. • Remove the deployment.

Authorizing users to categories

A category is a string or label that is used to organize and group software instances and deployments. A category might denote a system, subsystem, software vendor, software life cycle state, business function, or geographic location. There are no predefined categories.

To control access to a specific category, define a SAF resource profile for that resource. The SAF resource name for a category object has the following format:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.CAT.categoryName
```

Where:

- **CAT** indicates that the object that is associated with this SAF resource is a category.
- **categoryName** is the name of the category.

For example, if you have a category that is named z/OS, its SAF resource name would be:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.CAT.z/OS
```

Table 20 on page 92 lists the access authorities that you can assign to category resources and the actions that are permitted for each access authority. Note that the Software Management task does not perform authorization checks to determine which categories to display in a list or table; therefore, all categories are displayed regardless of access authority.

Table 20. Actions users can take against categories by access authority	
Access Authority	Actions Allowed
READ	• View the properties of the category. • Copy the properties of the category. • Assign deployments and software instances to the category.

Table 20. Actions users can take against categories by access authority (continued)

Access Authority	Actions Allowed
UPDATE	In addition to the actions specified for READ access, users can perform the following action: Modify the category properties that are <i>not</i> used to create the SAF resource name for the category.
CONTROL	In addition to the actions specified for READ and UPDATE access, users can perform the following actions: - Create new categories explicitly by using the Add action or implicitly as part of the Copy action. - Modify the category properties that are used to create the SAF resource name for the category and control access to the category. - Remove the category.

Using categories to authorize users to groups of software instances and deployments

Because category names are part of the SAF resource name for software instances and deployments, you can use categories to control access to groups of software instances and deployments. For example, if you want to give Db2® system programmers CONTROL access to all software instances and deployments in the Db2 category and give other users READ access to these objects, define a SAF profile for the following resource:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.*.DB2.**
```

If your installation is using RACF and your Db2 system programmers are defined in a group that is called DB2PROG, you can create a profile like the following:

```
RDEFINE ZMFAPLA +
  (IZUDFLT.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.*.DB2.** ) UACC(NONE)
PERMIT +
  IZUDFLT.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.*.DB2.** +
  CLASS(ZMFAPLA) ID(DB2PROG) ACCESS(CONTROL)
PERMIT +
  IZUDFLT.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.*.DB2.** +
  CLASS(ZMFAPLA) ID(IZUUSER) ACCESS(READ)
```

Controlling who can manage categories

By default, z/OSMF users and administrators are authorized to add, copy, modify, and remove categories. However, if you plan to use categories to authorize users to groups of software instances and deployments, it is important to control who can perform these actions. Therefore, it is recommended that you permit READ access to the following resource to z/OSMF administrators or trusted users only:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.CATEGORIES.MODIFY
```

If your installation is using RACF and you want to allow only administrators to perform these actions, you can define a profile like the following:

```
RDEFINE ZMFAPLA +
```

```
(IZUDFLT.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.CATEGORIES.MODIFY) +  
UACC(NONE)  
PERMIT +  
IZUDFLT.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.CATEGORIES.MODIFY +  
CLASS(ZMFAPLA) ID(IZUADMIN) ACCESS(READ)
```

Users who are not permitted at least READ access to this profile can only view a list of the categories and assign categories to software instances and deployments. This is true even if other controls exist that would otherwise allow such a user to perform actions on a specific category.

Ensuring that all objects are assigned to a category

When using categories to control access to groups of software instances and deployments, it is also important to ensure that all software instances and deployments are assigned to a category. To do so, permit no users access to the following resource:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.*.NOCATEGORY.**
```

If your installation is using RACF and you want to force all objects to be assigned to at least one category, you can define a profile like the following and permit no users to the profile:

```
RDEFINE ZMFAPLA +  
(IZUDFLT.ZOSMF.SOFTWARE_DEPLOYMENT.DATA.*.NOCATEGORY.**) UACC(NONE)
```

Controlling who can retrieve product information files

A product information file is a file that contains information about one or more products, such as the product announce date and end of service date. Information that is extracted from these files are displayed in several views and reports in the Software Management task, such as in the *Products* view and in the End of Service report.

When you retrieve a product information file, z/OSMF reads the file and loads the extracted content into the database where data for the Software Management task is stored. The scope of this action is broad and spans all products in the database; therefore, this action should be carefully controlled.

To control who can retrieve product information files, permit users READ access to the following resource:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.PRODUCT_INFO_FILE.RETRIEVE
```

By default, only z/OSMF administrators are permitted READ access to this resource. That is, by default, only z/OSMF administrators can retrieve product information files.

Controlling who can delete software update processes

Software instances can be updated by using the Software Update Task. A completed or canceled software update process can be removed, and the associated output can be deleted from the file system if the user has UPDATE access to the associated software instance object.

To control who else can delete completed or canceled software update processes, permitted users need READ access to the following resource:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.SWUPDATE.COMPLETED.DELETE
```

By default, only z/OSMF administrators are permitted READ access to this resource. Hence, only z/OSMF administrators, or users with UPDATE access to the associated software instance object, can delete software update processes.

Controlling who can download output for software update processes

Software instances can be updated by using the Software Update Task. The saved SMP/E output for a completed or canceled software update process can be download if the user has UPDATE access to the associated software instance object. To control who else can download the saved SMP/E output for completed or canceled software update processes, permitted users need READ access to the following resource:

```
<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.SWUPDATE.COMPLETED.DOWNLOAD
```

By default, only z/OSMF administrators are permitted READ access to this resource. Hence, only z/OSMF administrators, or users with UPDATE access to the associated software instance object, can download saved SMP/E output for software update processes.

Creating product information files for the Software Management task

A *product information file* is a flat file, such as a text file, that contains information about one or more products. This information includes, for example, the product announce date, general availability date, and end of service date. You can create your own product information files or obtain them from a provider, such as IBM, another vendor, or a third party.

z/OSMF displays data from product information files in several views in the Software Management task. For example, this information is displayed in the Products page, the Products, Features, and FMIDs page, and the End of Service report.

Syntax for product information files

To be processed by z/OSMF, product information files must be formatted as JSON data and have the following syntax:

```
{
  "Version": "date-modified",
  "Products":
  [
    {
      "prodName": "product-name",
      "prodId": "product-identifier",
      "prodVRM": "version-release-modification",
      "GAAnnounceDate": "date-announced",
      "GADate": "general-availability-date",
      "URL": "URL",
      "EOSDate": "end-of-service-date",
      "country": "country"
    }
  ]
}
```

where,

date-modified

Date the file was created or last updated. The date must have the format YYYY-MM-DD. The date is required.

product-name

Name of the product. The name is optional, and is not used by z/OSMF. To omit the product name, exclude the field, type null as the value, or set the value equal to an empty string.

product-identifier

Identifier of the product. The product ID is required.

version-release-modification

Version, release, and modification level of the product. The value has the format *VV.RR.MM*, where *VV* is the two-digit version, *RR* is the two-digit release, and *MM* is the two-digit modification level. The version, release, and modification level are required.

date-announced

Date the vendor publicly announced the details of the product. The date must have the format YYYY-MM-DD. The date is optional. To omit the date, exclude the field or type null as the value.

general-availability-date

Date that a version or release of the product is available to all users. The date must have the format YYYY-MM-DD. The date is optional. To omit the date, exclude the field or type null as the value.

URL

URL that links to additional information about the product. This information can include, for example, product life cycle dates, product highlights, planning information, and technical descriptions. The URL is optional. To omit the URL, exclude the field, type null as the value, or set the value equal to an empty string.

end-of-service-date

Last date on which the vendor delivers standard support services for a particular version or release of the product. This date is the general end of service date. It does not account for lifecycle extensions. The date must have the format YYYY-MM-DD. The date is optional. To omit the date, exclude the field or type null as the value.

country

Country for which the end of service date is applicable. The country is optional. To omit the country, exclude the field, type null as the value, or set the value equal to an empty string.

The information for each product must be contained within separate braces ({ }) inside the brackets ([]), and each set of braces must be comma-separated. For a sample file that contains the information for two products, see [Figure 21 on page 96](#).

Sample product information file

```
{
  "Version": "2011-06-30",
  "Products":
  [
    {
      "prodName": "z/OS",
      "prodId": "5694-A01",
      "prodVRM": "01.10.00",
      "GAAnnounceDate": "2008-08-05",
      "GADate": "2008-09-26",
      "URL": "http://www-03.ibm.com/systems/z/os/zos/",
      "EOSDate": "2011-09-30",
      "country": "US"
    },
    {
      "prodName": "z/OS",
      "prodId": "5694-A01",
      "prodVRM": "01.13.00",
      "GAAnnounceDate": "2011-07-12",
      "GADate": null,
      "URL": "",
      "country": "US"
    }
  ]
}
```

Figure 21. Sample product information file for the Software Management task

Working with the IBM product information file

The product information file that IBM supplies for System z® software is available for download from [Product information file for IBM Z software products \(public.dhe.ibm.com/services/zosmf/JSONs/IBMProductEOS.txt\)](http://public.dhe.ibm.com/services/zosmf/JSONs/IBMProductEOS.txt)

To load the contents of the file into z/OSMF, do one of the following:

- Load directly from the URL.
- Manually download the file at the URL to your local workstation.

- Manually download the file at the URL to a z/OS data set or UNIX file that the primary z/OSMF host system can access.

When you transfer the file from a workstation to a z/OS data set or UNIX file, transfer the file in binary format. To avoid errors, do not convert the file to the EBCDIC character set.

After you store the file in your desired location to retrieve its contents, complete the steps that are provided in the *Retrieving product information from product information files* topic in the z/OSMF online help.

Chapter 18. Configure the Storage Management service

To use the Storage Management task, you must configure it as described in the following sections.

Description

The storage management services are an application programming interface (API) that are implemented through industry standard Representational State Transfer (REST) services. The storage management services provide a programming interface for system storage elements.

Dependencies on other z/OSMF services

SCDS activation of Storage Management Services requires the user to have the z/OSMF Console Service. For more information, see [Chapter 22, “Configure the Operator Console task,” on page 113](#).

Security setup

To assist you with the security setup, IBM provides the sample security job IZUSGSEC in SYS1.SAMPLIB. For more information about the required profile authorizations and the Storage Management Services user roles, see [“Resource authorizations for the Storage Management service” on page 408](#).

Follow these steps:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUSGSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Optional extensions to this service

None.

Enabling the service

To enable Storage Management, follow these steps:

1. Open the z/OSMF General Settings task.
2. "Enable" Storage Management listed under "Optional Services".
3. Restart the z/OSMF server.

To enable Storage Management when z/OSMF is not started ("General Settings" page is not available), follow these steps:

1. Go to the UNIX System Services directory `/global/zosmf/configuration/settings/zosmf/`.
2. Edit `zosmf.json`. Paste the following contents into the JSON if it is empty.

```
{
  "services": {
    "IZU_STORAGE_CONFIGURE": "Y"
  }
}
```

```
}  
}
```

If it is not empty, locate IZU_STORAGE_CONFIGURE.

- If IZU_STORAGE_CONFIGURE is found, change the value to "Y".
- If IZU_STORAGE_CONFIGURE is not found, it means a default value "N" is taken for IZU_STORAGE_CONFIGURE. To enable "Storage Management", add a new line in the "services" section with "IZU_STORAGE_CONFIGURE": "Y".

3. Save zosmf.json and exit.

4. Restart the z/OSMF server.

Note: Replace /global/zosmf with your data directory path if you customize the z/OSMF user directory. You must use a user who is connected to IZUADMIN group to access contents in directory /global/zosmf/configuration.

Removing the service

After Storage Management is enabled, you might later decide to remove it. Follow these steps to remove the service:

1. Open the z/OSMF General Settings task.
2. "Disable" Storage Management listed under "Optional Services".
3. Restart the z/OSMF server.

To remove Storage Management when z/OSMF is not started (when the "General Settings" panel is not available), follow these steps:

1. Go to z/OS UNIX System Services directory /global/zosmf/configuration/settings/zosmf/.
2. Edit zosmf.json. Locate IZU_STORAGE_CONFIGURE, and change the value of IZU_STORAGE_CONFIGURE to "N". If IZU_STORAGE_CONFIGURE is not found, it means a default value "N" is taken for IZU_STORAGE_CONFIGURE. No further action is required.
3. Save zosmf.json and exit.
4. Restart the z/OSMF server.

Note:

Replace /global/zosmf with your data directory path if you customize the z/OSMF user directory. You must use a user who is connected to IZUADMIN group to access contents in the directory /global/zosmf/configuration.

Chapter 19. Configure the ISPF service

To use the ISPF task, you must configure it as described in this topic.

Description

You can use the ISPF task to access traditional ISPF applications through a web browser UI.

Dependencies on other z/OSMF services

The ISPF service requires the following services to be configured:

- Common event adapter (CEA), as described in [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#)
- TSO/E address space services, as described in [Chapter 13, “Configure the TSO/E address space services,” on page 77](#)

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUISSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUISSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

Follow the instructions in:

- [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#)
- [“Updating z/OS for the ISPF service” on page 101](#)

Note: If your system uses JES3 as its primary subsystem, and you find that jobs are not running, verify that JES3 is configured to allow multiple jobs with the same name. The JES3 option DUPJOBNM option must be set to YES. For more information, see the note regarding JES3 in [“Host system customization” on page 77](#).

Optional extensions to this service

None.

Updating z/OS for the ISPF service

If you have selected to configure the ISPF service, you must ensure that each user of the ISPF task is an existing TSO/E user with a valid password.

Specifically, for each user of the ISPF task, ensure that the corresponding user ID:

- Is authorized to TSO/E on the z/OS host system and has a valid password.
- Is authorized to a valid logon procedure and TSO/E account number.

- Is authorized to the JES spool. This authorization allows the user to use various functions in TSO/E, such as the SUBMIT, STATUS, TRANSMIT, and RECEIVE commands, and to access the SYSOUT data sets through the command TSO/E OUTPUT command.
- Has an OMVS segment defined, which allows for access to z/OSMF.
- Has a home directory defined, which is required for z/OSMF.

By default, the ISPF task uses the logon procedure IKJACCNT, which is supplied by IBM in your ServerPac order, and an asterisk (*) for the account number. A user can select to use a different logon procedure or account number, as long as the user's logon procedure is properly configured for ISPF and the account number is valid.

Assigning the TRUSTED attribute to CEA

To allow the CEA TSO/E address space manager to access or create any resource it needs, the CEA started task requires the TRUSTED(YES) attribute to be set on the RDEFINE STARTED CEA.** definition.

For more information about the RACF TRUSTED attribute, see the topic on associating started procedures and jobs with user IDs in *z/OS Security Server RACF System Programmer's Guide*, and the topic on using started procedures in [z/OS Security Server RACF Security Administrator's Guide](#).

Customizing for reconnecting user sessions

For potentially faster logons for users of the ISPF task, you can customize your z/OS system to allow the use of reconnectable user sessions. Here, the user session is deactivated after log-off is requested, but the user is not logged off. Instead, the system maintains the session for a period of time so that the user can reconnect to it. Reconnecting to a session is faster and uses fewer resources than creating a new session because the session resources are retained and reused when the user reconnects to the session.

To set up this capability in z/OS, the common event adapter (CEA) component must have certain controls set. See the description of the CEA parmlib member, CEAPRMxx, in *z/OS MVS Initialization and Tuning Reference*, specifically, the descriptions of the RECONTIME and RECONSESSIONS statements. By default, reconnectable user sessions are not enabled.

Customizing for profile sharing

Some TSO/E users require the use of multiple ISPF sessions. For example, a user might need to:

- Log on simultaneously through a z/OSMF ISPF session and a telnet 3270 session, or
- Log on through multiple z/OSMF ISPF sessions (this is different than having split screens, which is also allowed).

If you plan to allow the use of multiple ISPF sessions, the user's logon procedure must be configured to allow profile sharing. This option avoids enqueue lock outs and loss of profile updates when the same profile data set is used for concurrent ISPF sessions. With profile sharing enabled, the user's logon procedure is required to allocate ISPF profile data sets with the disposition SHARED, rather than NEW, OLD, or MOD, and the data sets must already exist. Or, these data sets must be temporary data sets. For more information about ISPF customization, see [z/OS ISPF Planning and Customizing](#).

Profile sharing is only effective if enabled for each concurrent ISPF session. This includes running a 3270 z/OS ISPF session at the same time as a z/OSMF ISPF session. For a 3270 z/OS ISPF session, invoke ISPF with the SHRPROF option. For a z/OSMF ISPF session, select Profile Sharing "On" from the z/OSMF ISPF User Settings panel. If you intend to run ISPF by using a 3270 z/OS ISPF session and also with a z/OSMF ISPF session using the same user ID, specify the value of "YES" for the keyword PROFILE_SHARING in the ISPF Configuration Table. Here, SHRPROF becomes the default option for the ISPF or ISPSTART command.

Otherwise, the default for the 3270 ISPF command is EXCLPROF, which prevents profile sharing between a z/OSMF ISPF user and a 3270 instance of the same user.

Chapter 20. Configure the z/OS Management Services Catalog task

To use the z/OS Management Services Catalog (zMSC) task, you must configure it as described in the following sections. The security setup must be performed by Security Administrators.

Description

IBM® z/OS Management Services Catalog improves how z/OS system programmers manage their z/OS environment. Services can streamline repetitive and frequent tasks as well as complex, infrequent tasks. Associated institutional knowledge and processes are embedded in services, which help z/OS system programmers adhere to best practices and internal standards.

z/OS Management Services Catalog administrators create services. When the services are published to the Catalog, you can repeatedly run them to manage your z/OS.

Dependencies on other z/OSMF services

The z/OS Management Services Catalog task requires the following z/OSMF services to be configured:

- z/OSMF Workflow task. For more information, see [Chapter 7, “Configure the z/OSMF Workflows task,” on page 63.](#)
- z/OSMF Settings service. For more information, see [Chapter 9, “Configure the z/OSMF settings service,” on page 67.](#)

Security setup

To assist you with the security setup, IBM provides the sample security job IZUMSSEC in SYS1.SAMPLIB. For more information about the required profile authorizations and z/OS Management Services Catalog user roles, see [“Resource authorizations for the z/OS Management Services Catalog service” on page 404.](#)

Follow these steps:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUMSSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Optional extensions to this service

None.

Enabling the service

To enable z/OS Management Services Catalog, follow these steps:

1. Open the z/OSMF General Settings task.
2. "Enable" z/OS Management Services Catalog listed under "Optional Services".
3. Restart the z/OSMF server.

4. Open the z/OSMF "App Center" from the z/OSMF taskbar and drag the "Management Services Catalog" icon to the z/OSMF desktop.

Removing the service

After z/OS Management Services Catalog z/OS Management Services Catalog is enabled, you might later decide to remove it.

Follow these steps to remove the service:

1. Open the z/OSMF General Settings task.
2. "Disable" z/OS Management Services Catalog listed under "Optional Services".

Chapter 21. Configure the Resource Monitoring service

To use the z/OSMF Resource Monitoring service, you must configure it as described in this topic.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZURMSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZURMSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

Your z/OS system requires further customization. Follow the instructions in the following topics:

- [“Updating z/OS for the Resource Monitoring service” on page 105](#)

Optional extensions to this service

You can further extend the function of this service by enabling other, corequisite services in z/OSMF. [Table 21 on page 105](#) describes these dependencies.

Table 21. Optional extensions to the Resource Monitoring service		
Functional extensions to this service	Co-requisite z/OSMF services	Reason for enabling
Interactive functions with the Workload Management task.	• Workload Management • z/OSMF administrative tasks	Users of the Resource Monitoring task can interact with the Workload Management task. This function requires the Application Linking service of z/OSMF.

Updating z/OS for the Resource Monitoring service

If you selected to configure the Resource Monitoring service, you might have system customization to perform, as described in this topic.

This topic contains the following information:

- [“System customization for the Resource Monitoring and System Status tasks” on page 106](#)
- [“Enabling PassTicket creation for Resource Monitoring task users” on page 107](#)
- [“Establishing secure communications with the Distributed Data Server” on page 108](#)
- [“Browser consideration for the Resource Monitoring task” on page 111.](#)

System customization for the Resource Monitoring and System Status tasks

Table 22 on page 106 describes the z/OS system changes that are required or recommended. Some of this work might already be done on your system, or might not be applicable. If so, you can skip the particular setup action.

Table 22. z/OS setup actions for the Resource Monitoring and System Status tasks		
	z/OS setup action	Check when task is completed
1	Enable the optional priced feature, Resource Measurement Facility (RMF), on one of the systems in your enterprise. For information about enabling features, see z/OS Planning for Installation	
2	For data collection and monitoring of your systems, ensure that the RMF Distributed Data Server (DDS) is active on one of the systems in your sysplex. To monitor several sysplexes, ensure that a DDS is running on one system in each sysplex. You can use the following command to check for the existence of GPMSERVE address spaces in your sysplex: <pre>ROUTE *ALL,D A,GPMSERVE</pre> If your installation uses RMF Cross Platform Monitoring (RMF XP), the RACF profile name for the RMF XP DDS is GPM4CIM, rather than GPMSERVE. For information about setting up the DDS and RMF XP, see z/OS Resource Measurement Facility User's Guide.	
3	Determine whether the DDS on the target system is currently configured to require authentication. To check, use the following command to display the active DDS options: <pre>MODIFY GPMSERVE,OPTIONS</pre> If your installation uses RMF XP, the RACF profile name for the RMF XP DDS is GPM4CIM, rather than GPMSERVE. In the command output, check for the HTTP_NOAUTH setting, which indicates the scope of authentication for the DDS, as follows: HTTP_NOAUTH() All hosts must authenticate HTTP_NOAUTH(*) No authentication is required HTTP_NOAUTH(<i>specific_host_or_mask</i>) All hosts except those matching the mask must authenticate. If DDS authentication is not required in your enterprise, you are done. Otherwise, proceed to Step 4.	

Table 22. z/OS setup actions for the Resource Monitoring and System Status tasks (continued)

	z/OS setup action	Check when task is completed
4	Determine whether your installation security procedures require that the DDS should require authentication from the z/OSMF system and its users, and perform one of the following actions: • If DDS authentication is required from the z/OSMF system, you must ensure that the PassTicket is set up properly, and that the z/OSMF started task user ID is authorized to generate the PassTicket. See “Enabling PassTicket creation for Resource Monitoring task users” on page 107. • If DDS authentication is not required from the z/OSMF system, you can disable DDS authentication for the system on which z/OSMF is running. Doing so allows the Resource Monitoring and System Status tasks to access the DDS on behalf of z/OSMF users without potentially encountering authentication errors. To disable DDS authentication for the system on which z/OSMF is running (the server host name or IP address), modify the HTTP_NOAUTH statement in the GPMSRVxx parmlib member on the DDS system. In the following example, the HTTP_NOAUTH statement is updated to bypass DDS authentication for the host system represented by <i>host_system_IP_address</i>: <pre>HTTP_NOAUTH(host_system_IP_address)</pre> For more information about DDS authentication, see z/OS Resource Measurement Facility User's Guide, SC33-7990.	

Enabling PassTicket creation for Resource Monitoring task users

If the RMF Distributed Data Server (DDS) requires authentication from the z/OSMF system and its users, follow the steps in this procedure to set up the PassTicket support.

About this task

In this procedure, you ensure that the PassTicket is set up properly, and that the z/OSMF started task user ID is authorized to generate the PassTicket. The procedure shows how this setup can be done for a system that uses RACF as its security management product.

Note: If your installation uses RMF Cross Platform Monitoring (RMF XP), the RACF profile name for the RMF XP DDS is GPM4CIM. Use this profile name instead of GPMSERVE when you complete Steps 2 through 4 in the procedure.

Procedure

1. On the z/OSMF system, activate the security class PTKTDATA, if this class is not already active. If you plan to use generic profiles for the PTKTDATA class, include the GENERIC option on the **SETROPTS** command, for example:

```
SETROPTS CLASSACT(PTKTDATA)
SETROPTS RACLIST(PTKTDATA) GENERIC(PTKTDATA)
```

2. Define the profile GPMSERVE for the DDS in the PTKTDATA class and associate a secret secured signon key with the profile. The key must be the same on both the system on which the PassTicket is to be generated (the z/OSMF system) and the system on which the PassTicket is to be verified (the DDS system). For example:

```
RDEFINE PTKTDATA GPMSERVE SSIGNON(KEYMASKED(key))
SETROPTS RACLIST(PTKTDATA) REFRESH
```

where *key* is a user-supplied 16-digit value used to generate the PassTicket.

If a common cryptographic architecture (CCA) product is installed on the systems with the secured signon function, you can encrypt the secured signon key using a KEYENCRYPTED value. If not, you can

mask the secured signon key by using the SSIGNON option and a 64-bit KEYMASKED value, as shown in the preceding example.

If you plan to use a KEYENCRYPTED value, note that additional authorizations are required, such as access to security profiles in the CSFSERV class, and additional profiles for PassTicket creation and PassTicket validation. Be sure to review the RACF setup requirements for the CCA product.

3. To enable PassTicket creation for Resource Monitoring users, define the profile IRRPTAUTH.GPMSSERVE.* in the PTKTDATA class, and set the universal access authority to NONE. You can do enable PassTicket creation for either for all user IDs or for a specific user ID, as shown in the examples that follow.

- Example (for all user IDs):

```
RDEFINE PTKTDATA IRRPTAUTH.GPMSSERVE.* UACC(NONE)
```

- Example (for a specific user ID):

```
RDEFINE PTKTDATA IRRPTAUTH.GPMSSERVE.specific_dds_login_userid UACC(NONE)
```

4. Grant the z/OSMF started task user ID permission to generate PassTickets for users.

- Example (for all user IDs):

```
PERMIT IRRPTAUTH.GPMSSERVE.* CLASS(PTKTDATA) ID(passticket_creator_userid)  
ACCESS(UPDATE)
```

- Example (for a specific user ID):

```
PERMIT IRRPTAUTH.GPMSSERVE.specific_dds_login_userid CLASS(PTKTDATA)  
ID(passticket_creator_userid) ACCESS(UPDATE)
```

where *passticket_creator_userid* is the user ID of the z/OSMF started task user ID. By default, this is IZUSVR.

5. Activate the changes, for example: SETROPTS RACLIST(PTKTDATA) REFRESH

Establishing secure communications with the Distributed Data Server

You must ensure that communication between the Resource Monitoring tasks and the RMF Distributed Data Server (DDS) is protected. For secure network communications, it is recommended that you use Application Transparent Transport Layer Security (AT-TLS) and Transport Layer Security (TLS), as described in this topic.

Before you begin

- Ensure that the basic setup for the Policy Agent is done. For information about policy-based networking and data protection, in [z/OS Communications Server: IP Configuration Reference](#).
- Ensure that the basic certificate setup is done. For information about handling certificates for secure communications with RACF, see the topic on digital certificates in [z/OS Security Server RACF Security Administrator's Guide](#).
- To enable AT-TLS and encrypted communication with the DDS server, you require the following:
 - Valid server certificate and the associated server private key
 - Certificate from a trusted Certificate Authority (CA).

The example in [Figure 22 on page 109](#) uses a key ring that is named DDSServerKeyring to store these credentials. This key ring must be accessible by the DDS server user ID (for example, GPMSSERVE), and the server certificate must be the default certificate.

- To enable secure communication for the Resource Monitoring tasks, you require a certificate from a trusted Certificate Authority (CA). The example in [Figure 23 on page 110](#) uses a key ring that is named DDSClientKeyring to store the credentials. This key ring must be accessible to the z/OSMF server user ID, which is IZUSVR, by default.

For a sample setup that uses RACF, see "RACF and digital certificates" in *z/OS Security Server RACF Security Administrator's Guide*, specifically "Implementation Scenario 1" and "Implementation Scenario 2."

For other security management products, refer to your product documentation for information about handling certificates and key rings.

About this task

Use this procedure to establish secure communications between the Resource Monitoring tasks and the RMF DDS server.

Procedure

1. Configure the Policy Agent to allow secure communication with the RMF DDS server.

a) Enable the Policy Agent for AT-TLS.

For information about AT-TLS data protection, see *z/OS Communications Server: IP Configuration Reference*.

b) Configure the Policy Agent to specify secure communication for the DDS server.

For a sample policy, see Figure 22 on page 109.

```
# RMF Distributed Data Server Rule
TTLSRule
{
  LocalPortRange      8803
  Jobname              GPMSERVE
  Direction            Inbound
  Priority              1
  TTLSGroupActionRef   DDSServerGRP
  TTLSEnvironmentActionRef DDSServerENV
}
TTLSGroupAction        DDSServerGRP
{
  TTLSEnabled          On
  Trace                1
}
TTLSEnvironmentAction  DDSServerENV
{
  HandshakeRole         Server
  TTLSKeyringParms
  {
    Keyring              DDSServerKeyring
  }
  TTLSEnvironmentAdvancedParms
  {
    ServerCertificateLabel RMFDDS
    SSLv2                 Off
    SSLv3                 Off
    TLSv1                 Off
    TLSv1.1               Off
    TLSv1.2               On
  }
}
}
```

Figure 22. Sample Policy Agent policy for secure communication setup for the DDS

Where the AT-TLS policy properties are set, as follows:

TTLSRule: Jobname

The name value specifies the job name of the application. GPMSERVE is the job name of the DDS.

TTLSRule: LocalPortRange

The local port the application is bound to for this rule's action to be performed. 8803 is the default HTTP Port of the DDS.

TTLSRule: Direction

Specifies the direction the connection must be initiated from for this rule's action to be performed. In this example, Inbound is specified, which means that the rule applies to connection requests that arrive to the local host.

TTLRule: Priority

An integer value in the range 1 - 2000000000 that represents the priority associated with the rule. The highest priority value is 2000000000.

When using multiple rules for the DDS server, the more specific the rule, the higher its priority should be. Generic rules without detailed specifications of the incoming connections should have a low priority.

TTLSEnvironmentAction: HandshakeRole

Specifies the SSL handshake role to be taken for connections in this AT-TLS environment. In this example, Server is specified which means that the SSL handshake is performed as a sever.

TTLSEnvironmentAction: Keyring

Specifies the path and file name of the key database z/OS® UNIX file, the ring name of the SAF key ring, or the name of the z/OS PKCS #11 token. In this example, the RACF key ring DDSServerKeyring is specified.

TTLSEnvironmentAdvancedParms: ServerCertificateLabel

Specifies the label of the certificate for a server application to authenticate the server. In this example, the DDS server certificate with the label RMFDDS is used.

TTLSEnvironmentAdvancedParms: SSLv2

Specifies the state of the SSL Version 2 protocol. In this example, Off is specified, which means that the SSL Version 2 protocol is disabled.

TTLSEnvironmentAdvancedParms: SSLv3

Specifies the state of the SSL Version 3 protocol. In this example, Off is specified, which means that the SSL Version 3 protocol is disabled.

TTLSEnvironmentAdvancedParms: TLSv1

Specifies the state of the TLS Version 1 protocol. In this example, Off is specified, which means that the TLS Version 1 protocol is disabled.

TTLSEnvironmentAdvancedParms: TLSv1.1

Specifies the state of the TLS Version 1.1 protocol. In this example, Off is specified, which means that the TLS Version 1.1 protocol is disabled.

TTLSEnvironmentAdvancedParms: TLSv1.2

Specifies the state of the TLS Version 1.2 protocol. In this example, On is specified, which means that the TLS Version 1.2 protocol is enabled.

2. Configure the Policy Agent to require secure communication for the Resource Monitoring tasks.

For an example of a Policy Agent policy for setting up simple TLS protection for the Resource Monitoring tasks, see [Figure 23 on page 110](#).

```
# RMF Distributed Data Server Client Rule
TTLRule DDSClientRule
{
  RemotePortRange 8803
  RemoteAddr 1.0.0.0
  Direction Outbound
  TTLGroupActionRef DDSClientGRP
  TTLEnvironmentActionRef DDSClientENV
}
TTLGroupAction DDSClientGRP
{
  TTLEnabled On
  Trace 1
}
TTLEnvironmentAction DDSClientENV
{
  HandshakeRole Client
  TTLKeyringParms
  {
    Keyring DDSClientKeyring
  }
}
```

Figure 23. Sample Policy Agent rule for secure communication for DDS clients

Where the AT-TLS policy properties are set, as follows:

TTLRule: RemoteAddr

The remote IP address specification that must match for this rule's action to be performed—that is, the IP address of the DDS to which the client is connecting.

TTLRule: RemotePortRange

The remote port to which the application must be connecting for this rule's action to be performed. The default HTTP port of the DDS is 8803.

TTLRule: Direction

Specifies the direction from which the connection must be initiated to perform this rule's action. In this example, Outbound is specified, which means that the rule applies to connection requests that are initiated by the application.

TTLEnvironmentAction: HandshakeRole

Specifies the SSL handshake role to be taken for connections in this AT-TLS environment. In this example, Client is specified, which means that the SSL handshake is performed as a client.

TTLKeyringParms: Keyring

Specifies the z/OS UNIX path and file name of the key database file, the ring name of the SAF key ring, or the name of the z/OS PKCS #11 token. In this example, the RACF key ring DDSCClientKeyring is specified.

3. Refresh the Policy Agent to have your changes take effect.

You can use the following command to refresh the Policy Agent:

```
F PAGENT,REFRESH
```

Browser consideration for the Resource Monitoring task

Users who plan to use the Microsoft Edge browser with Resource Monitoring task, and who plan to export the data collected in a dashboard to a CSV file, should ensure that the browser is enabled for automatic prompting for file downloads. This setting prevents the file download blocker from being invoked when the user downloads service definitions to the workstation.

Otherwise, if automatic prompting is disabled (the default setting), the download blocker prompts the user to accept these file downloads, causing the browser session to be reloaded and the active tabs to be closed. Users can avoid this disruption by enabling automatic prompting for file downloads.

For more information, see [“Enabling automatic prompting for file downloads” on page 283](#).

Chapter 22. Configure the Operator Console task

To use the Operator Console task, you must perform the setup steps that are described in this topic.

Dependencies on other z/OSMF services

The console services require the following services to be configured:

- Common event adapter (CEA), as described in [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#)
- TSO/E address space services, as described in [Chapter 13, “Configure the TSO/E address space services,” on page 77](#)
- z/OSMF Settings, as described in [Chapter 9, “Configure the z/OSMF settings service,” on page 67](#)

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUGCSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUGCSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

The z/OS common event adapter (CEA) service must be configured and started in full-function mode to run this service, as described in [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#).

Also, customize the z/OS host system as described in [“Updating z/OS for the z/OS Operator Consoles task” on page 113](#).

Note: If your system uses JES3 as its primary subsystem, and you find that jobs are not running, verify that JES3 is configured to allow multiple jobs with the same name. The JES3 option DUPJOBNM option must be set to YES. For more information, see the note about JES3 in [“Host system customization” on page 77](#).

Optional extensions to this service

None.

Updating z/OS for the z/OS Operator Consoles task

Some setup is required before you can use a z/OS console with the z/OS Operator Consoles task, as described in this topic.

See the following topics:

- [“Authorizing users to the z/OS Operator Consoles task” on page 114](#)
- [“Host system customization for the z/OS Operator Consoles task” on page 114](#)
- [“Security setup for the z/OS Operator Consoles task” on page 115](#)

Authorizing users to the z/OS Operator Consoles task

To view and access the z/OS Operator Consoles task in the z/OSMF desktop interface, users require READ access to resource <SAF-prefix>.ZOSMF.CONSOLES.ZOSOPER in the ZMFAPLA class. By default, is<SAF-prefix>IZUDFLT.

Host system customization for the z/OS Operator Consoles task

To use a z/OS console with the z/OS Operator Consoles task, do the following:

- Ensure that the z/OS system is defined to z/OSMF. The systems that are displayed in the table in the **Overview tab** are retrieved from the Systems task. By default, z/OSMF provides a system definition for the z/OS system that hosts the z/OSMF instance to which your web browser is connected, and it provides a system definition for the systems that belong to the same JES2 multi-access spool (MAS) as the z/OSMF host system. If you want z/OSMF to interact with other systems, you must create the corresponding system definitions.
- Establish an extended MCS console for the system, then grant permission to a user to use that console. This work is performed outside of z/OSMF, typically by a security administrator. For instructions, see [“Security setup for the z/OS Operator Consoles task” on page 115](#).
- Ensure that message automation is defined for the message that you want to detect. In your active MPFLSTxx member, create an entry for the message ID and set the AUTO keyword to YES. By making the message eligible for automation processing, you ensure that the message is queued to the EMCS console if a message flood occurs and your installation uses message flood automation to suppress the display of messages. For more information, see the description of MPFLSTxx in [z/OS MVS Initialization and Tuning Reference](#).
- Verify that CONSPROF is defined as a TSO/E authorized command.
 1. Enter the command D IKJTS0. The command response should look like the following result:

```
RESPONSE=SY1
IKJ738I TSO/E PARMLIB SETTINGS : 505
  CIMSRE.R14ONLY.PARMLIB(IKJTS000) on volume ZOSTS3
  Activated by **IPL** on 2019-04-12 at 21:51:06 from system SY1
  Applies to : SY1
          THE FOLLOWING ARE THE OPTIONS FOR THE ALLOCATE STATEMENT:
```

2. Browse member CIMSRE.R14ONLY.PARMLIB(IKJTS000) and verify that CONSPROF is included in the AUTHCMD NAMES list. For example:

```
AUTHCMD NAMES(          /* AUTHORIZED COMMANDS */          +
      ADYOPCMD          /* DAE STOP/START   RAS */          +
      CONSPROF          /* TSO/E COMMANDS  @L3M*/          +
      LISTB  LISTBC     /*                  @L3M*/          +
      PARMLIB IKJPRMLB  /*                  @L3M*/          +
```

If CONSPROF is not in the AUTHCMD NAMES list, add it to the list and save the change. After you update IKJTS0xx, enter the following TSO/E command to make the change effective: parm1ib update (xx).

3. If your installation uses hardware cryptography with z/OS Integrated Cryptographic Service Facility (ICSF), the z/OSMF server requires access to the ICSF callable services. For more information, see [“Resource authorizations for hardware cryptography” on page 385](#).

Notes on the extended MCS console:

1. z/OSMF uses TSO/E address space services to create a TSO address space as the host for the extended MCS (EMCS) console. You can control the parameters that are used for creating the TSO address space by setting the appropriate parameters in parmlib member IZUPRMxx. For example:

```
COMMON_TSO ACCT(IZUACCT) REGION(50000) PROC(IZUFPROC)
```

Ensure that your settings are configured before z/OSMF Operator Console Service is used. Otherwise, the default values (shown in the example) are used.

2. The attributes of the EMCS console that is started by z/OSMF are controlled by the OPERPARM settings of the user profile <consolename>. Thus, for example, if a user wants the z/OS Operator Consoles task to create a console named console1, a user profile that is named console1 must exist and contain an OPERPARM segment with the appropriate settings.

Security setup for the z/OS Operator Consoles task

IBM provides job IZUGCSEC in SYS1.SAMPLIB to assist you with performing these updates. The job contains RACF commands for creating the required security authorizations. The job includes commands for setting up an extended MCS console and granting access to the console.

Chapter 23. Configure the Workload Management service

To use the z/OSMF Workload Management task, you must configure it as described in this topic.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUWMSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUWMSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

Your z/OS system requires further customization. Follow the instructions in the following topics:

- [“Updating z/OS for the Workload Management service” on page 117](#)

Optional extensions to this service

You can further extend the function of this service by enabling other, corequisite services in z/OSMF. [Table 23 on page 117](#) describes these dependencies.

Table 23. Optional extensions to the Resource Monitoring service		
Functional extensions to this service	Co-requisite z/OSMF services	Reason for enabling
Interactive functions with the Resource Management task.	• Resource Management • z/OSMF administrative tasks	Users of the Workload Management task can interact with the Resource Monitoring task. This function requires the Application Linking service of z/OSMF.

Updating z/OS for the Workload Management service

If you selected to configure the Workload Management service, you might have system customization to perform, as described in this topic.

This topic contains the following sections:

- [“Authorizing users to the MVSADMIN.WLM.POLICY profile” on page 118](#)
- [“Authorizing the z/OSMF started task user ID to the MVSADMIN.WLM.POLICY profile” on page 118](#)
- [“Required security group for Workload Management” on page 118](#)
- [“Using authorization levels for the Workload Management task” on page 119](#)

- “Using a browser with WLM service definitions” on page 120.

IBM provides job IZUWMSEC in SYS1.SAMPLIB to assist you with performing these updates. The job contains RACF commands for creating the required security authorizations.

The Workload Management task is used for managing WLM resources in the IBM Cloud Provisioning and Management for z/OS provisioning tasks. For more setup considerations, see [Chapter 27, “Configure the Cloud Provisioning services,”](#) on page 141.

Authorizing users to the MVSADMIN.WLM.POLICY profile

Users of the Workload Management task require UPDATE access to resources that are protected by the profile MVSADMIN.WLM.POLICY in class FACILITY.

The WLM security group WLMGRP is granted UPDATE access to profile MVSADMIN.WLM.POLICY with the IZUWMSEC job.

You can have your security administrator create a separate group for WLM users and grant this group UPDATE access to profile MVSADMIN.WLM.POLICY. If such a group exists in your environment, you can use the existing group instead of creating a new group. As an example, the following steps show sample RACF commands for creating a separate WLM group and authorizing it to profile MVSADMIN.WLM.POLICY:

1. Create the WLM group:

```
ADDGROUP "WLMGroupName" OMVS(GID("WLMGroupGID"))
```

2. Authorize the WLM group:

```
PERMIT MVSADMIN.WLM.POLICY CLASS(FACILITY) ID("WLMGroupName") ACCESS(UPDATE)
```

3. Have your changes take effect:

```
SETROPTS RACLIST(FACILITY) REFRESH
```

Authorizing the z/OSMF started task user ID to the MVSADMIN.WLM.POLICY profile

The Workload Management task performs periodic queries of WLM on the z/OS host system. To perform the queries, the Workload Management task uses the z/OSMF started task user ID. Therefore, you must ensure that the z/OSMF started task user ID has READ access to the profile MVSADMIN.WLM.POLICY . By default, z/OSMF started task user ID is IZUSVR.

With the IZUWMSEC job, the z/OSMF started task user ID IZUSVR is granted READ access to profile MVSADMIN.WLM.POLICY. In RACF, you can use the following command:

```
PERMIT MVSADMIN.WLM.POLICY CLASS(FACILITY) ID(IZUSVR) ACCESS(READ)
```

Required security group for Workload Management

With the IZUWMSEC job, your security administrator can supply the name of the WLM security group that your installation uses for authorizing users to the z/OS Workload Management component on your system. The IZUAUTH job contains commands for connecting users to the group. As supplied by IBM, the IZUAUTH job contains a RACF CONNECT command for a group called WLMGRP. You can substitute a different group name. See [Table 24 on page 118](#).

Table 24. Security group required for the Workload Management service		
Group	Purpose	Created by
WLMGRP	Security group for users of the Workload Management task.	ADDGROUP command or an equivalent security command for creating user groups.

Using authorization levels for the Workload Management task

Using predefined authorization levels, your installation can authorize users to specific functions within the Workload Management task.

The Workload Management task supports the following authorization levels:

View

This authorization level allows the user to invoke the Workload Management task, and view service definitions, service policies, and WLM status.

Install

This authorization level allows the user to install service definitions and activate service policies. A user who is authorized for this level also must be authorized for the View level to invoke the Workload Management task.

Modify

This authorization level allows a user to modify service definitions and to import service definitions from host data sets or local workstation files into z/OSMF. A user who is authorized for this level also must be authorized for the View level to invoke the Workload Management task. To install service definitions and activate service policies, the user must also be authorized for the Install level.

By default, the z/OSMF administrators security group is authorized for the View, Install, and Modify functions, which are equivalent to a WLM policy administrator. The z/OSMF users security group is authorized for the View function, which is equivalent to a WLM performance analyst.

Your installation can manage user authorizations through your security management product, such as RACF. Grant access authority to the users and groups, as described in [Table 25 on page 119](#).

Table 25. Workload Management task authorizations for z/OSMF	
Required authorization level of user or group	Required SAF access authority
View	READ access for profile <SAF-prefix>.ZOSMF.WORKLOAD_MANAGEMENT.WORKLOAD_MANAGEMENT.VIEW
Install	READ access for profile <SAF-prefix>.ZOSMF.WORKLOAD_MANAGEMENT.WORKLOAD_MANAGEMENT.INSTALL
Modify	READ access for profile <SAF-prefix>.ZOSMF.WORKLOAD_MANAGEMENT.WORKLOAD_MANAGEMENT.MODIFY

If these default settings do not meet your needs, you can change the SAF authority of these respective groups for the profiles that are shown in [Table 25 on page 119](#).

Alternatively, you can define new custom groups for the Workload Management task. For example, the following RACF commands can be used to define a custom group WLMPOLOP, which is authorized for the View and Install functions. This set of authorizations is equivalent to a WLM policy operator.

```
ADDGROUP WLMPOLOP
PERMIT <SAF-prefix>.ZOSMF.WORKLOAD_MANAGEMENT.WORKLOAD_MANAGEMENT.VIEW
      CLASS(ZMFAPLA) ID(WLMPOLOP) ACCESS(READ)
PERMIT <SAF-prefix>.ZOSMF.WORKLOAD_MANAGEMENT.WORKLOAD_MANAGEMENT.INSTALL
      CLASS(ZMFAPLA) ID(WLMPOLOP) ACCESS(READ)
SETROPTS RACLIST(ZMFAPLA) REFRESH
```

To authorize a user to this group in RACF, you can use a CONNECT command:

```
CONNECT "userid" GROUP(WLMPOLOP)
```

Using a browser with WLM service definitions

Users who plan to use the Microsoft Edge browser to work with WLM service definitions should ensure that the browser is enabled for automatic prompting for file downloads. This setting prevents the file download blocker from being invoked when the user downloads service definitions to the workstation. Otherwise, if automatic prompting is unavailable (the default setting), the download blocker prompts the user to accept these file downloads, causing the browser session to be reloaded and the active tabs to be closed. Users can avoid this disruption by enabling automatic prompting for file downloads. For more information, see [“Enabling automatic prompting for file downloads” on page 283](#).

Avoiding a potential synchronization error

After you begin using the Workload Management task, avoid using other applications, such as the z/OS WLM Administrative Application, to modify and install WLM service definitions. During a modify operation, the Workload Management task automatically extracts the installed service definition from the WLM couple data set and imports it into the service definition repository. This import fails if the Workload Management task finds that a service definition exists in both the repository and the couple data set with the same name and description, but different content. If this error occurs, you can resolve it by either changing the name or description of the service definition in the repository, or deleting it from the repository.

Chapter 24. Configure the Sysplex Management service

To use the Sysplex Management service, you must configure it as described in this topic.

Dependencies on other z/OSMF services

The Sysplex Management service requires the following services to be configured:

- Common event adapter (CEA); see [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#)
- z/OSMF Settings service; see [Chapter 9, “Configure the z/OSMF settings service,” on page 67](#)
- z/OS data set and file REST interface; see [Chapter 12, “Configure the z/OS data set and file REST services,” on page 73](#)
- TSO/E address space services; see [Chapter 13, “Configure the TSO/E address space services,” on page 77](#)
- Consoles services; see [Chapter 22, “Configure the Operator Console task,” on page 113.](#)

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUSPSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUSPSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

The z/OS common event adapter (CEA) service must be configured and started in full-function mode to run this service. See [“Ensure that common event adapter \(CEA\) is configured and active” on page 17.](#)

Note: If your system uses JES3 as its primary subsystem, and you find that jobs are not running, verify that JES3 is configured to allow multiple jobs with the same name. The JES3 option DUPJOBNM option must be set to YES. For more information, see the JES3 note in [“Host system customization” on page 77.](#)

Optional extensions to this service

You can further extend the function of this service by enabling other, corequisite services in z/OSMF. [Table 26 on page 122](#) describes these dependencies.

Table 26. Optional extensions to the Sysplex Management service

Functional extensions to this service	Co-requisite z/OSMF services	Reason for enabling
Query the topology of interconnected CPCs and LPARs in the sysplex. For setup instructions, see “Configure the Discover CPC function” on page 122.	Discover CPC function.	If you want to query the topology of interconnected CPCs and LPARs in the sysplex, you require the Discover CPC function to be configured.

Updating z/OS for the Sysplex Management service

If you selected to configure the Sysplex Management service, you have system customization to perform, as described in this topic.

To use the Sysplex Management task, your system requires the following updates:

- [“Configure the CPC information in Systems task”](#) on page 122
- [“Configure the Discover CPC function”](#) on page 122
- [“Authorize users to the z/OS console services REST API”](#) on page 123
- [“Create security structures for the Sysplex Management task”](#) on page 123
- [“Update z/OSMF settings for managing a remote sysplex”](#) on page 123

The examples in this topic use RACF commands. If your installation uses an external security manager other than RACF, your security administrator can refer to these examples when creating equivalent commands for your environment.

IBM provides job IZUSPSEC in SYS1.SAMPLIB to assist you with performing these updates. The job contains RACF commands for creating the required security authorizations.

Configure the CPC information in Systems task

Before you use the Sysplex Management task, it is recommended that you configure the CPC information in the z/OSMF Systems task. You can use either of the following methods to do so:

- Add CPC information manually.
- Use the Discovery CPC function in Systems task to discover the CPC topology of the currently interconnected CPCs and LPARs in the sysplex.

The Discovery CPC function is a long running action; it might take several minutes to complete.

Configure the Discover CPC function

In the z/OSMF Systems task, the Discover CPC function uses z/OS data set and file REST services and BCP internal interface (BCPii) services to query the topology of interconnected CPCs and LPARs in the sysplex. Therefore, you must ensure that both z/OS data set and file REST services and BCPii are configured in the sysplexes that are to be managed through the Systems task.

After BCPii is configured, have your security administrator ensure that the required authorizations are created for the BCPii services. In SYS1.SAMPLIB, the IZUDCSEC job includes sample RACF commands for the BCPii services.

For more information about BCPii, see [z/OS MVS Programming: Callable Services for High-Level Languages](#).

Authorize users to the z/OS console services REST API

Users of the Sysplex Management task require authorization to the z/OS console services REST API. For a system that uses RACF as the security manager, IBM provides job IZUGCSEC in SYS1.SAMPLIB to assist you with creating the authorizations. Ask your security administrator to edit the job for your environment and submit it.

Create security structures for the Sysplex Management task

To enable users to work with the Sysplex Management task, your external security manager, such as RACF, requires that a number of security structures are defined, as described in this topic, and that users are authorized to the appropriate system resources. If RACF or another security manager is installed, the security administrator can define profiles that control the use of these resources.

Before using the Sysplex Management task, have your security administrator verify that the following conditions exist:

- The security database, such as the RACF database, is shared across the sysplex.
- The SAFDFLT profile is defined in the REALM class. The SAFDFLT profile in the REALM class allows the security environment to be recognized.
- Each security database REALM has its own unique APPLDATA profile, which is associated with the SAFDFLT profile. The same SAFDFLT APPLDATA value is used across all systems in the sysplex. Define the name by using the SAFDFLT profile in the REALM class. Substitute an appropriate string for the *plexname*, such as the name of the sysplex or another unique string.

Example:

```
SETROPTS GENERIC(REALM)
RDEFINE REALM SAFDFLT APPLDATA('<plexname OR other unique string>')
SETROPTS RACLIST(REALM) CLASSACT(REALM)
SETROPTS RACLIST(REALM) REFRESH
```

- TRUSTED attribute must be assigned to the CEA started task.
- CEA address space is started in full function mode.
- Users are authorized to the appropriate resources. IBM provides job IZUSPSEC in SYS1.SAMPLIB to assist you with performing these updates. The job contains RACF commands for creating the required security authorizations.

To make the preceding updates effective, you must:

1. Refresh your security database. Example:

```
SETROPTS RACLIST(SERVAUTH) REFRESH
```

```
SETROPTS RACLIST(ZMFAPLA) REFRESH
```

2. Restart CEA.

The Sysplex Management service requires access to local resources on your z/OS system. To assist you with performing the security setup, IBM provides the sample security job IZUSPSEC in SYS1.SAMPLIB.

Update z/OSMF settings for managing a remote sysplex

If you plan to manage a remote sysplex in addition to the local sysplex in the primary z/OSMF instance, ask your z/OSMF administrator to perform the following updates:

1. The remote sysplex to be managed must have a z/OSMF instance running in one of its systems. Open **z/OSMF Settings > Systems table > Add system** on the primary z/OSMF instance and define the system on which the z/OSMF instance is running in the remote sysplex. Specify the URL of the z/OSMF instance when you update the Systems table.

2. Ensure that single sign-on is configured for the system that is running the primary z/OSMF instance and for the secondary z/OSMF instances in other sysplexes.
3. Open **z/OSMF Settings > Systems table** on the primary z/OSMF instance and define the CPC objects on the primary z/OSMF instance, either manually or by running the discovery function, which retrieves CPC information by calling BCPii services.

z/OSMF does not verify the accuracy of your input. Ensure that the information you provide is correct and complete. Incorrect or missing information can cause the major views of the Sysplex Management task to be disabled:

- **Physical View**
- **Connectivity View**
- **Connectivity Details View**

Chapter 25. Configure the Capacity Provisioning service

To use the Capacity Provisioning service, you must configure it as described in this topic.

Dependencies on other z/OSMF services

The Capacity Provisioning service requires the following services to be configured:

- Common Information Model (CIM) server, as described in [Chapter 47, “Configuring the CIM server for your system,”](#) on page 253.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUCPSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of the IZUCPSEC job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.
4. Ensure that your user ID has been connected to the correct groups. For a RACF installation, you can use the IBM supplied job, IZUAUTH, to connect your user ID to the following Capacity Provisioning groups: CFZUSRGP, CPOCTRL, and CPOQUERY.

Ensure that the IZUCPSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

Your z/OS system requires further customization: The z/OS Common Information Model (CIM) server and a capacity provisioning domain must be configured to run this service.

Follow the instructions in the following topics:

- [Chapter 47, “Configuring the CIM server for your system,”](#) on page 253
- [“Updating z/OS for the Capacity Provisioning service”](#) on page 125

Optional extensions to this service

None.

Updating z/OS for the Capacity Provisioning service

If you have selected to configure the Capacity Provisioning service, you might have system customization to perform, as described in this topic. These actions are needed to ensure that users of the Capacity Provisioning task have access to the capacity provisioning domain.

This topic contains the following information:

- [“System customization for the Capacity Provisioning task”](#) on page 126
- [“Enabling PassTicket creation for Capacity Provisioning task users”](#) on page 126
- [“Establishing secure communications with the CIM server”](#) on page 127.

IBM provides job IZUCPSEC in SYS1.SAMPLIB to assist you with performing these updates. The job contains RACF commands for creating the required security authorizations.

System customization for the Capacity Provisioning task

Table 27 on page 126 describes the z/OS system changes that are required or recommended. Some of this work might already be done on your system, or might not be applicable. If so, you can skip the particular setup action.

Table 27. z/OS setup actions for the Capacity Provisioning task		
	z/OS setup action	Check when task is completed
1	Ensure that a Capacity Provisioning Domain is implemented in your enterprise. For information about setting up and implementing Capacity Provisioning, see z/OS MVS Capacity Provisioning User's Guide .	
2	Ensure that potential users of the Capacity Provisioning task are defined to the Provisioning Manager query security group on the provisioning system (by default, the CPOQUERY group). On a system with RACF, you can query the users in a group through the LISTGRP command. For example: <pre>LISTGRP CPOQUERY</pre>	
3	Determine whether the CIM server on the provisioning system is configured to use PassTicket authentication. If so, proceed to Step 4. Otherwise, you must perform this set-up, following the steps described in z/OS MVS Capacity Provisioning User's Guide .	
4	Determine whether the Provisioning Manager is running in the same security domain as the z/OSMF system. If so, grant the z/OSMF started task user ID at least UPDATE access authority to the profile IRRPTAUTH.CFZAPPL.* in the PTKTDATA class. On a system with RACF, you can create this authorization through the PERMIT command. For example: <pre>PERMIT IRRPTAUTH.CFZAPPL.* CLASS(PTKTDATA) ID(passticket_creator_userid) ACCESS(UPDATE) SETROPTS RACLIST(PTKTDATA) REFRESH</pre> where passticket_creator_userid is the z/OSMF started task user ID. By default, this is IZUSVR. Otherwise, if the Provisioning Manager is running in a different security domain, follow the steps in “Enabling PassTicket creation for Capacity Provisioning task users” on page 126.	
5	Ensure that communication between the Capacity Provisioning task and the CIM server is protected. Follow the steps in “Establishing secure communications with the CIM server” on page 127.	

Enabling PassTicket creation for Capacity Provisioning task users

Use the following procedure to ensure that Capacity Provisioning task users on the z/OSMF system can access the CIM server on the provisioning system.

About this task

In this procedure, you do the following:

- Ensure that PassTickets are enabled for every user who might require access to the provisioning system
- Verify that the z/OSMF started task user ID is authorized to generate PassTickets.

The procedure shows how this setup can be done for a system that uses RACF as its security manager. Included are the definitions for using the secured signon function and generating PassTickets. This setup must be done on both systems, as follows:

- System on which the PassTicket is to be verified (the provisioning system). This work is assumed to be done; otherwise, you must set up authentication on the provisioning system, as described in [z/OS MVS Capacity Provisioning User's Guide](#).

- System on which the PassTicket is to be generated (the z/OSMF system), which is described here.

For more information about PassTickets, see [z/OS Security Server RACF Security Administrator's Guide](#).

Procedure

1. On the z/OSMF system, activate the security class PTKTDATA, if it is not already active. If you plan to use generic profiles for the PTKTDATA class, include the GENERIC option on the **SETROPTS** command, for example:

```
SETROPTS CLASSACT(PTKTDATA)
SETROPTS RACLIST(PTKTDATA) GENERIC(PTKTDATA)
```

2. Define the profile CFZAPPL in the PTKTDATA class and associate a secret secured signon key with the profile. The key must be the same on both the system on which the PassTicket is to be generated (the z/OSMF system) and the system on which the PassTicket is to be verified (the provisioning system). For example:

```
RDEFINE PTKTDATA CFZAPPL SSIGNON(KEYMASKED(key))
APPLDATA('NO REPLAY PROTECTION')
SETROPTS RACLIST(PTKTDATA) REFRESH
```

where *key* is a user-supplied 16-digit value that is used to generate the PassTicket.

If a common cryptographic architecture (CCA) product is installed on the systems with the secured signon function, you can encrypt the secured signon key by using a KEYENCRYPTED value. If not, you can mask the secured signon key by using the SSIGNON option and a 64-bit KEYMASKED value, as shown in the preceding example.

If you plan to use a KEYENCRYPTED value, more authorizations are required, including security profiles in the CSFSERV class and profiles for PassTicket creation and validation. Review the RACF setup requirements for the CCA product.

3. To enable PassTicket creation for Capacity Provisioning task users, define the profile IRRPTAUTH.CFZAPPL.* in the PTKTDATA class and set the universal access authority to NONE. For example:

```
RDEFINE PTKTDATA IRRPTAUTH.CFZAPPL.* UACC(NONE)
SETROPTS RACLIST(PTKTDATA) REFRESH
```

4. Grant the z/OSMF started task user ID permission to generate PassTickets for users. For example:

```
PERMIT IRRPTAUTH.CFZAPPL.* CLASS(PTKTDATA) ID(passticket_creator_userid)
ACCESS(UPDATE)
SETROPTS RACLIST(PTKTDATA) REFRESH
```

where *passticket_creator_userid* is the z/OSMF started task user ID. By default, this user ID is IZUSVR.

5. Activate the changes, for example: SETROPTS RACLIST(PTKTDATA) REFRESH

Establishing secure communications with the CIM server

You must ensure that communication between the Capacity Provisioning task and the CIM server is protected. For secure network communications, it is recommended that you use Application Transparent Transport Layer Security (AT-TLS) and Transport Layer Security (TLS), as described in this topic.

Before you begin

Ensure that the basic setup for the Policy Agent and the certificate is done. For information, see [z/OS Common Information Model User's Guide](#).

About this task

Use this procedure to establish secure communications between the Capacity Provisioning task and the CIM server.

This setup is not required for other z/OSMF tasks that use the CIM server, such as Incident Log.

Procedure

1. Enable the HTTPS connection port.

For information, see the topic *Configuring the CIM server HTTPS connection using AT-TLS* in *z/OS Common Information Model User's Guide* with attention to the example called *SSL protection only*.

2. Activate AT-TLS communication for the CIM server.

This means creating a policy in Policy Agent, creating client and server certificates, and then activating the policy. For information, see *z/OS Common Information Model User's Guide*.

3. Refresh the Policy Agent to have your changes take effect.

You can use the following command to refresh the Policy Agent:

```
F PAGENT,REFRESH
```

4. Select the communications protocol and port number.

Do the following:

- a) In the z/OSMF desktop interface, select Capacity Provisioning.
- b) Select the **Provisioning Manager** tab.
- c) In the Connections Table, select the following values for the **Host Address**:
 - For **Protocol**, select HTTPS
 - For **Port**, specify 5989.
- d) Click **OK** to confirm these settings.

Chapter 26. Configure IBM z/OS Encryption Readiness Technology (zERT) Network Analyzer

To use IBM z/OS Encryption Readiness Technology (zERT) Network Analyzer, you must configure it as described in this topic.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUNASEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of the IZUNASEC job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUNASEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

If you selected to configure IBM z/OS Encryption Readiness Technology (zERT), you have additional system customization to perform, as described in the following sections:

- [“Db2 for z/OS customization for the IBM zERT Network Analyzer task” on page 129](#)
- [“Connect IBM zERT Network Analyzer task with the Db2 for z/OS database” on page 130](#)

Db2 for z/OS customization for the IBM zERT Network Analyzer task

The IBM zERT Network Analyzer task stores and queries SMF data in a Db2 for z/OS database. Before you can use the task, this database must be created in a suitable Db2 for z/OS subsystem and the connectivity information for the database must be configured in the IBM zERT Network Analyzer.

Requirement: A given IBM zERT Network Analyzer database must only be accessed by one IBM zERT Network Analyzer plug-in at a time. Concurrent access to a single database by more than one zERT Network Analyzer plug-in will generate unpredictable results.

Procedure:

1. Determine the local resource requirements and location for the IBM zERT Network Analyzer database.
Your DBA decides which Db2 for z/OS subsystem contains the IBM zERT Network Analyzer database objects. The DBA also decides which specific Db2 for z/OS resources need to be allocated for these objects.
2. Create the IBM zERT Network Analyzer database.
Your DBA should use the sample database schema tooling (IZUZNADx members) in the SYS1.SAMPLIB dataset to create the database for your environment. See below for more information on this tooling.
3. Define the IBM zERT Network Analyzer database user ID (the z/OS user ID that is permitted to connect to, store data into, and query data in the IBM zERT Network Analyzer database).

4. Collect the connectivity information required to connect the IBM zERT Network Analyzer service with the Db2 database. This information is necessary for the steps below.

Connect IBM zERT Network Analyzer task with the Db2 for z/OS database

You must provide the IBM zERT Network Analyzer task with Db2 for z/OS database connectivity information before using the task for any additional functions.

Procedure:

1. Start the IBM zERT Network Analyzer task. The first time that you start the task, you are directed immediately to the **Database Settings** page.
2. Enter the database connectivity information that is provided to you by the DBA as part of completing the procedure in [“Db2 for z/OS customization for the IBM zERT Network Analyzer task”](#) on page 129.

Note: After you save the connectivity information, the IBM zERT Network Analyzer task restarts, using the configuration information.

Optional extensions to this service

None.

Updating z/OS for the IBM zERT Network Analyzer service

If you selected to configure the IBM zERT Network Analyzer service, you might have system customization to perform, as described in this topic.

This topic contains the following information:

- [Authorize users to the IBM zERT Network Analyzer task](#)
- [Db2 for z/OS customization for the IBM zERT Network Analyzer task](#)
- [Install Java Database Connectivity](#)
- [Connect IBM zERT Network Analyzer task with the Db2 for z/OS database](#)

IBM provides job IZUNASEC in SYS1.SAMPLIB to assist you with performing these updates. The job contains RACF commands for creating the required security authorizations.

The examples in this topic use RACF commands. If your installation uses an external security manager other than RACF, your security administrator can refer to these examples when creating equivalent commands for your environment.

Authorize users to the IBM zERT Network Analyzer task

About this task

Users of the IBM zERT Network Analyzer task require access to resources that are protected by the profile <SAF-prefix>.ZOSMF.ZERT_NETWORK_ANALYZER in class ZMAFPLA. Your z/OS security administrator must perform additional steps to create the necessary authorizations.

Procedure

1. During the z/OSMF configuration process, edit the IZUNASEC job before you run it.
2. Add the names of the users who should be authorized to access the IBM zERT Network Analyzer task.

```
/* Connect the users of the zERT Network Analyzer to the      */
/* zERT Network Analyzer group                                */
CONNECT USER1 GROUP(IZUZNA)
CONNECT USER2 GROUP(IZUZNA)
/* End connect the users to zERT Network Analyzer group      */
```

3. Save your changes and run the updated IZUNASEC job.

Db2 for z/OS customization for the IBM zERT Network Analyzer task

About this task

The IBM zERT Network Analyzer task stores and queries SMF data in a Db2 for z/OS database. Before you can use the task, this database must be created in a suitable Db2 for z/OS subsystem and the connectivity information for the database must be configured in the IBM zERT Network Analyzer.

Requirement: A given IBM zERT Network Analyzer database must only be accessed by one IBM zERT Network Analyzer plug-in at a time. Concurrent access to a single database by more than one IBM zERT Network Analyzer plug-in generates unpredictable results.

IBM provides tooling and templates in the SYS1.SAMPLIB data set to help your local Db2 for z/OS database administrator (DBA) create the IBM zERT Network Analyzer database in a manner that conforms to local naming and resource conventions. Collectively, this tooling is called the IBM zERT Network Analyzer database schema tooling and it consists of the following members in SYS1.SAMPLIB:

- The IZUZNADT and IZUZNADA members contain Data Definition Language (DDL) templates for creating the required database objects using two different approaches.
 - IZUZNADT contains a template for creating the required database objects using a fixed schema name (SYSIBM_EZB_ZNADB) and fixed table names that the IBM zERT Network Analyzer depends on. If your local naming conventions allow for these fixed names, then you should use this template.
 - IZUZNADA contains a template for creating the required database objects using your own customized schema and/or table names. It also creates aliases for those tables using the fixed schema name and table names that the IBM zERT Network Analyzer depends on. If your local naming conventions require you to change the schema name and/or the table names, you should use this template.

The templates contain all the appropriate DDL commands using variables for a wide variety of resource names and parameter values that the DBA may want to control. The DBA should set values for some or all these variables that will comply with the local Db2 for z/OS conventions and allocation strategies.

Note: The IZUZNADT and IZUZNADA prolog commentary explain each of the variables they include as well as the required variable syntax.

The IZUZNADT and IZUZNADA templates can also be used to apply service updates to the zERT Network Analyzer database schema if schema changes are required with a PTF. The APAR hold information will provide appropriate instructions when such updates are required.

Four other DDL templates are also provided to upgrade the zERT Network Analyzer database schema from a previously supported release and schema version to the most current version for the current release. These are:

- IZUZNAT1 - upgrades a V2R5 database that uses fixed table names (as created by the IZUZNADT template) to the most current 3.1 schema release and version
- IZUZNAT2 - upgrades a V2R4 database that uses fixed table names (as created by the IZUZNADT template) to the most current 3.1 schema release and version
- IZUZNAA1 - upgrades a V2R5 database that uses aliased table names (as created by the IZUZNADA template) to the most current 3.1 schema release and version
- IZUZNAA2 - upgrades a V2R4 database that uses aliased table names (as created by the IZUZNADA template) to the most current 3.1 schema release and version

The "1" and "2" suffix in the member names represent the number of releases prior to the current release. As such, the IZUZNAX1 templates operate on databases one release prior to the current (for 3.1, the prior release is V2R5) and the IZUZNAX2 templates operate on databases two releases prior to the current release (for 3.1, two releases prior is V2R4).

- The IZUZNADI member of the SYS1.SAMPLIB data set. IZUZNADI is a sample variable substitution data set in which the DBA specifies the desired values for each of the template variables that are defined in the templates. The IZUZNADI sample specifies the default values for each of the variables, but the DBA can override any of the variables as needed.

The best practice is to create a copy of the IZUZNADI sample and make your local customizations in that copy before you generate the DDL to create your IBM zERT Network Analyzer database. Then later, when you need to invoke the IZUZNADG exec again to generate DDL to update the database schema for a PTF or upgrade it to a newer release level, you can use the same customized copy. Before generating DDL for a PTF or a release upgrade, however, you should check the most current IZUZNADI sample to see if any new variables have been added. If so, add those variable definitions with appropriate values to your customized copy before invoking the IZUZNADG exec with the appropriate template.

- The IZUZNADG DDL generation REXX exec reads the specified template and a customized variable substitution data set and generates a data set that contains a complete set of customized DDL for creating a new IBM zERT Network Analyzer database, updating an existing database to the most current schema level (for applying service), or upgrading the schema release and version to the latest z/OS release and schema level.

Note: Complete instructions for using the IZUZNADG exec are available by running the exec with the --HELP parameter.

After running the IZUZNADG exec, your DBA can use the resulting DDL data set in SPUFI or as input to a local JCL stream used to create or update the IBM zERT Network Analyzer database in the Db2 for z/OS subsystem.

Here are a few examples for invoking the IZUZNADG exec under TSO to generate DDL for the IBM zERT Network Analyzer database schema. In these examples, the user ID executing the exec is DBAUSER and the customized variable substitution values are defined in the dataset DBAUSER.ZDBVARS.

1. To generate executable DDL commands that will create a brand new IBM zERT Network Analyzer database that uses aliased table names:

```
ex 'sys1.samplib(izuznadg)' ' 'dbauser.zdbvars' ' 'dbauser.newdbddl' '
  'sys1.samplib(izuznada)'
```

The generated DDL commands will be written to a dataset named DBAUSER.NEWDBDDL.

2. To generate executable DDL commands that will update an existing zERT Network Analyzer database that uses fixed table names that is at schema version 1.2.0 to the latest schema level of the same release (as instructed by the hold information in an APAR that updates the schema):

```
ex 'sys1.samplib(izuznadg)' ' 'dbauser.zdbvars' ' 'dbauser.ptfddl' '
  'sys1.samplib(izuznadt)' ' dbvers(1.2.0)'
```

The generated DDL commands will be written to a dataset named DBAUSER.PTFDDL.

3. To generate executable DDL commands that will upgrade a V2R4 zERT Network Analyzer database that uses fixed table names and is at schema version 1.3.0 to the latest z/OS 3.1 schema version:

```
ex 'sys1.samplib(izuznadg)' ' 'dbauser.zdbvars' ' 'dbauser.v24t130' '
  'sys1.samplib(izuznat2)' ' dbvers(1.3.0)'
```

The generated DDL commands will be written to a dataset named DBAUSER.v24t130.

Procedure

1. Determine the local resource requirements and location for the IBM zERT Network Analyzer database.

Your DBA decides which Db2 for z/OS subsystem contains the IBM zERT Network Analyzer database objects as well as the specific Db2 for z/OS resources (storage pools, buffer pools, and so forth) to be allocated to these objects.

Tip: You should initially deploy the IBM zERT Network Analyzer database and service on a test system. The test system should be a place where you can easily get familiar with the operation of the service and can better understand the Db2 and system resource requirements when running queries against your imported SMF record data. Depending on the number of imported SMF records and the complexity of your queries, you might consider initially limiting query execution to specific times of day or specific systems to minimize system impacts.

- **Determine the maximum number of query reports that will be open at any given time**

Your DBA needs to specify the number of partitions for a subset of tables in the IBM zERT Network Analyzer database called "Query Result Tables." These partitioned tables hold intermediate query results that are displayed through the network analyzer's **Report** tab. The IBM zERT Network Analyzer assigns one partition in each query result table to each active IBM zERT Network Analyzer query report for as long as that report is open in the web browser. Because of this, you must work with your DBA to determine an appropriate number of partitions to ensure that your database has enough partitions to support your community of IBM zERT Network Analyzer users.

The number of partitions is determined by the `<QRTParts>` variable in the DDL templates and is controlled by the DBA in the variable substitution data set used with the IZUZNADG exec (see the IZUZNADI member of SYS1.SAMPLIB).

To calculate the number of partitions you need, consider the number of users that will be using the IBM zERT Network Analyzer as well as the number of reports each user might have open at any given time (a single user can have multiple reports open at one time, with each one in its own web browser tab). Multiply those two numbers together to determine the maximum possible number of open reports. You should create at least that many partitions for each query result table. You might also want to increase this value by an appropriate percentage to ensure there is some room for growth over time.

The following equation summarizes the above calculation.

```
NumOpenReports = MaxNumberOfUsers * MaxNumberOfReportsPerUser  
<QRTParts> = NumOpenReports + ( NumOpenReports * ExtraSpace% )
```

- **Decide whether to use a separate database for the query result tables**

Two variables in the DDL templates control whether the query result tables are created in the same database as the persistent IBM zERT Network Analyzer tables, or in a different database.

<database>

specifies the name of the database that contains the IBM zERT Network Analyzer's persistent tables (all tables except the query result tables). When you are creating a brand new IBM zERT Network Analyzer database, set this variable to any valid database name. By default, the IZUZNADI sample sets this to 'ZNADB'.

<QRTDatabase>

specifies the name of the database that contains the query result tables. You may choose to store the query result tables in the same database as the persistent tables or in a different database. To use the same database, set the `<QRTDatabase>` and `<database>` variables to the same value. To use different databases, specify different values. By default, the IZUZNADI sample sets this to 'ZNAQRDB', placing the Query Result Tables in their own database.

- **Determine how much table space to allocate in the database**

The amount of Db2 for z/OS table storage required by IBM zERT Network Analyzer varies, but you can use the following guidelines to estimate the table storage required in your environment:

- Start with an allocation of 20 MB to hold the core security session data and operational data related to data management and user-built queries. If you are collecting zERT data from an unusually large number of unique security sessions across the z/OS systems, you might eventually need to add to this amount over time.
- Estimate the space that is required to store the maximum number of SMF Type 119 zERT Summary (subtype 12) records that will be represented in the database at one time and add that amount to the initial 20 MB. To do so, see details in [Table 28 on page 134](#).

Table 28. How to estimate the space for SMF Type 119 zERT Summary (subtype 12) records

Step	Task	Formula	Example
1	Estimate the number of unique security sessions that typically exist across the set of z/OS systems from which you are collecting zERT data.	Use local procedures to estimate this value.	Assume that 10,000 unique security sessions typically exist across all of the zERT-monitored systems. UniqueSessions = 10,000
2	Determine the maximum number of SMF intervals to be represented in the IBM zERT Network Analyzer database at a single time.	MaxIntervals = ((1440 / SMFInterval) * Days) where: 1440 = the number of minutes in a day; SMFInterval = the SMF interval in minutes as defined in your SMFPRMxx parmlib member. Note that if you use different SMF intervals across the zERT-monitored z/OS systems, use the average interval length across the different systems here; Days = the maximum number of days' worth of SMF data that is stored in the database at a single time.	Assume the average SMF interval is set to 20 minutes and you plan to store 30 days of SMF data in the zERT Network Analyzer database: MaxIntervals = (1440 / 20) * 30 = 72 * 30 = 2160
3	Determine how many SMF 119 subtype 12 records are collected over the Days value and imported into the database.	MaxRecords = UniqueSessions * MaxIntervals	MaxRecords = 10,000 * 2160 = 21,600,000
4	Determine how much DASD storage is required to store the data for the maximum number of records. Each SMF 119 subtype 12 record consumes about 500 bytes of DASD storage above and beyond the base space allocation of 20 MB.	DASDSpaceMB = 20 + ((MaxRecords * 500) / 1,048,576)	DASDSpaceMB = 20 + ((21,600,000 * 500) / 1,048,576) = 20 + 10,299 = 10,319MB or 10GB

According to the example shown in the table, you need to allocate a total of 10 GB of space in the storage group that is used for your IBM zERT Network Analyzer database's table spaces.

2. Create the IBM zERT Network Analyzer database.

Your DBA should use the database schema tooling described above to create the database for your environment.

3. Define the IBM zERT Network Analyzer database user ID.

This is the z/OS user ID that is permitted to connect to, store data into, and query data in the IBM zERT Network Analyzer database. The IBM zERT Network Analyzer uses this user ID to communicate with the Db2 for z/OS database and to perform all the operations in its database.

The database user ID must be given the INSERT, SELECT, UPDATE, DELETE privileges for the IBM zERT Network Analyzer database tables to ensure proper operation of the IBM zERT Network Analyzer's various functions as described below.

If you use the IZUZNADT template or if you use the IZUZNADA aliasing template with SQL GRANT access controls, then grant INSERT, SELECT, UPDATE, DELETE privileges to:

- SYSIBM_EZB_ZNADB.APPL
- SYSIBM_EZB_ZNADB.DATAMGMTHISTORY
- SYSIBM_EZB_ZNADB.DATASET
- SYSIBM_EZB_ZNADB.SECURITY_SESSION
- SYSIBM_EZB_ZNADB.SESSION_STATISTICS
- SYSIBM_EZB_ZNADB.IPSEC_INFO
- SYSIBM_EZB_ZNADB.SSH_INFO
- SYSIBM_EZB_ZNADB.TLS_INFO
- SYSIBM_EZB_ZNADB.TOPOLOGY
- SYSIBM_EZB_ZNADB.OPENJPA_SEQUENCE_TABLE
- SYSIBM_EZB_ZNADB.QUERY
- SYSIBM_EZB_ZNADB.SCOPE_FLTR
- SYSIBM_EZB_ZNADB.SCOPE_FLTR_ENDPT
- SYSIBM_EZB_ZNADB.SCOPE_FLTR_SYSSPEC
- SYSIBM_EZB_ZNADB.SEC_FLTR
- SYSIBM_EZB_ZNADB.SEC_IPSEC_FLTR
- SYSIBM_EZB_ZNADB.SEC_SSH_FLTR
- SYSIBM_EZB_ZNADB.SEC_TLS_FLTR
- SYSIBM_EZB_ZNADB.FILTEREDSECURITYSESSIONIDS
- SYSIBM_EZB_ZNADB.TCPSERVER_SUMMARIES
- SYSIBM_EZB_ZNADB.TCPCLIENT_SUMMARIES
- SYSIBM_EZB_ZNADB.EEPEER_SUMMARIES
- SYSIBM_EZB_ZNADB.TCPSERVER_CLIENTDETAILS
- SYSIBM_EZB_ZNADB.TCPCLIENT_CLIENTDETAILS
- SYSIBM_EZB_ZNADB.EEPEER_CLIENTDETAILS
- SYSIBM_EZB_ZNADB.TCPSERVER_CLEARSECURITYSESSIONDETAILS
- SYSIBM_EZB_ZNADB.TCPSERVER_IPSECSECURITYSESSIONDETAILS
- SYSIBM_EZB_ZNADB.TCPSERVER_SSHSECURITYSESSIONDETAILS
- SYSIBM_EZB_ZNADB.TCPSERVER_TLSSECURITYSESSIONDETAILS
- SYSIBM_EZB_ZNADB.TCPCLIENT_CLEARSECURITYSESSIONDETAILS
- SYSIBM_EZB_ZNADB.TCPCLIENT_IPSECSECURITYSESSIONDETAILS
- SYSIBM_EZB_ZNADB.TCPCLIENT_SSHSECURITYSESSIONDETAILS
- SYSIBM_EZB_ZNADB.TCPCLIENT_TLSSECURITYSESSIONDETAILS

- SYSIBM_EZB_ZNADB.EEPEER_CLEARSECURITYSESSIONDETAILS
- SYSIBM_EZB_ZNADB.EEPEER_IPSECSECURITYSESSIONDETAILS

If you use the IZUZNADA aliasing template with SAF-based access controls on Db2 for z/OS objects, the INSERT, SELECT, UPDATE, DELETE privileges must be granted for the base tables (not the aliases) as specified by the values of the following template variables:

- <schema>.<appTable>
- <schema>.<dmhistTable>
- <schema>.<dsTable>
- <schema>.<secsessTable>
- <schema>.<sessstatsTable>
- <schema>.<ipsecTable>
- <schema>.<sshTable>
- <schema>.<tlsTable>
- <schema>.<topoTable>
- <schema>.<queryTable>
- <schema>.<scopeFltrTable>
- <schema>.<scopeFltrEndptTable>
- <schema>.<scopeFltrSysspecTable>
- <schema>.<secFltrTable>
- <schema>.<secIpsecFltrTable>
- <schema>.<secSshFltrTable>
- <schema>.<secTlsFltrTable>
- <schema>.<openjpaTable>
- <schema>.<fssIdsTable>
- <schema>.<tsrvrsTable>
- <schema>.<tcIntsTable>
- <schema>.<teepTable>
- <schema>.<tsrvrcTable>
- <schema>.<tcIntcTable>
- <schema>.<teepcTable>
- <schema>.<tsrvrCSessTable>
- <schema>.<tsrvrISessTable>
- <schema>.<tsrvrSSessTable>
- <schema>.<tsrvrTSessTable>
- <schema>.<tcIntCSessTable>
- <schema>.<tcIntISessTable>
- <schema>.<tcIntSSessTable>
- <schema>.<tcIntTSessTable>
- <schema>.<teepCSessTable>
- <schema>.<teepISessTable>

4. Collect the connectivity information that is required to link the IBM zERT Network Analyzer service with the Db2 for z/OS database to be used by the service.

The DBA must provide database connectivity information to the person setting up the IBM zERT Network Analyzer service. This information includes:

- The hostname or IP address on which the Db2 for z/OS subsystem is running
- The TCP port number of the subsystem
- The database location name, which is the value of the LOCATION parameter of the DSNJU003 utility
- The database user ID
- The password for the database user ID
- The JDBC classpath for the Db2 for z/OS JDBC driver on the system where the IBM zERT Network Analyzer executes

See [Connect IBM zERT Network Analyzer task with the Db2 for z/OS database](#) for how to use this connectivity information.

5. **(Optional)** Modify the setting of the DSN6SYSP URLGWTH parameter.

Depending on the size of the SMF dump data sets that you plan to import into IBM zERT Network Analyzer and the setting of the DSN6SYSP URLGWTH parameter, you might see one or more DSNJ031I messages when importing the SMF dump data sets. You can modify the setting of the DSN6SYSP URLGWTH parameter to reduce the number of DSNJ031I messages. For more information, see *UR LOG WRITE CHECK field (URLGWTH subsystem parameter)* in *Installing and migrating Db2*.

6. **(Optional)** Define additional 4K and 32K work files to be used by IBM zERT Network Analyzer

By default, small 4K and 32K work files are defined for use by the Db2 for z/OS subsystem. You might need to increase the size of the 4K and 32K work files to allow IBM zERT Network Analyzer to operate more efficiently. For more information, see *DSNTIP9: Work file database panel* in *Installing and migrating Db2*.

Restriction:

- Db2 for z/OS packages associated with the NULLID collection will be used when IBM zERT Network Analyzer plug-in connects to the Db2 for z/OS subsystem. The collection-id is an optional parameter when binding Db2 for z/OS packages where NULLID is the default collection-id value for distributed applications such as IBM zERT Network Analyzer.

The IBM zERT Network Analyzer requires that the NULLID collection specify the RELEASE(COMMIT) and KEEP DYNAMIC(NO) options for proper operation.

For more information, review the DSNTIJLC and DSNTIJLR Db2 for z/OS jobs.

- APPLCOMPAT bind option for the DB2 for z/OS packages associated with dynamic SQL statements must be set to its respective Db2 for z/OS release level. This means that a Db2 12 for z/OS subsystem must use an APPLCOMPAT value of at least V12R1M500 and a Db2 11 for z/OS subsystem must use an APPLCOMPAT value of V11R1. This means that a Db2 12 for z/OS subsystem must use an APPLCOMPAT value of at least V12R1M500 and a Db2 11 for z/OS subsystem must use an APPLCOMPAT value of V11R1. APPLCOMPAT is an optional parameter when binding Db2 for z/OS packages where the default APPLCOMPAT value is the APPLCOMPAT subsystem parameter. For more information, review the DSNTIJLC and DSNTIJLR Db2 for z/OS jobs.

Install Java Database Connectivity

About this task

The IBM zERT Network Analyzer task uses the Java Persistence API (JPA) to access the contents of the Db2 database. JPA, in turn, uses Java Database Connectivity (JDBC). If you have not already installed JDBC, or you have not run the DB2Binder utility as part of that installation, you have additional customization steps to perform.

Procedure

Follow the instructions in *Installing the IBM Data Server Driver for JDBC and SQLJ as part of a Db2 installation* in *Programming for Db2 for z/OS*.

Connect the IBM zERT Network Analyzer task with the Db2 for z/OS database

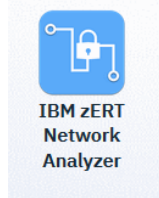
About this task

You must provide the IBM zERT Network Analyzer task with Db2 for z/OS database connectivity information before using the task for any additional functions.

Procedure

1. Click on the icon to launch the IBM zERT Network Analyzer task. The first time you launch the task, you are directed immediately to the **Database Settings** panel.

To access the IBM zERT Network Analyzer task, open the App folder on the z/OSMF desktop. Then,



click the **IBM zERT Network Analyzer** icon.

2. Connect the IBM zERT Network Analyzer task with the Db2 for z/OS database, enter the database connectivity information that is provided to you by the database administrator (DBA) as part of completing the Db2 for z/OS customization for the IBM zERT Network Analyzer task procedure. If you are upgrading the zERT Network Analyzer from an earlier supported release, you can use

Import database connection settings 

the **Import database connection settings** button. This opens a dialog through which you can import the database settings values (except the database user ID and password) from an earlier release. You must enter the database user ID and password or passphrase manually before saving the database settings. If the earlier release of the network analyzer plugin is installed under the same z/OSMF home directory as the new release, then this dialog will find the old settings and give you the option of importing them to the new release. If the earlier release was installed under a different z/OSMF home directory, you can use the **Select application settings**

Select application settings file 

button to locate and import the settings from that release. See the zERT Network Analyzer online help for importing settings for more details.

Note: Expect CWWKG0083W and CWWKE0701E messages in the z/OSMF joblog when the IBM zERT Network Analyzer plug-in initializes before any database connectivity information has been entered. WebSphere Liberty is attempting to process the server configuration that currently has an incomplete datasource definition.

What to do next

After you save the connectivity information, the IBM zERT Network Analyzer task restarts using the configuration information. You must stop and restart your browser session when the task restarts.

Tip: Use AT-TLS to cryptographically protect the IBM zERT Network Analyzer connections to the Db2 for z/OS database. The AT-TLS rule protecting the connection to the database must specify:

- Connection direction outbound
- The IP address of the Db2 for z/OS server (as the remote address)
- The secure port on which Db2 for z/OS is listening (as the remote port)
- Jobname of the main z/OSMF job (for example, IZUSVR1)
- Handshake role client

- The appropriate keyring and TLS settings that are compatible with the AT-TLS rule protecting the Db2 server's secure listening port

Tip: For more information on using AT-TLS to protect Db2 for z/OS connections, see [Application Transparent Transport Layer Security data protection in z/OS Communications Server: IP Configuration Guide](#).

Deployment guidelines for IBM zERT Network Analyzer

Here are some important guidelines to consider as you deploy the IBM zERT Network Analyzer.

There is no requirement to run the IBM zERT Network Analyzer service and database on a production system. You can deploy the IBM zERT Network Analyzer service and database on a non-production system as long as your local security requirements allow you to copy production SMF data to that system. Use a system where you can familiarize yourself with its operation and the Db2 for z/OS and system resource requirements. Depending on the number of imported SMF records and the complexity of your queries, you might also consider initially limiting query execution to specific times of day or specific systems to minimize system impacts.

Maximizing IBM zERT Network Analyzer performance

Database performance:

- Use a Db2 for z/OS subsystem that is co-located with IBM zERT Network Analyzer to reduce latency and elapsed times when you run operations such as SMF imports and queries.
- Configure PGSTEAL (LRU) for your Db2 for z/OS page stealing algorithm on the buffer pools that the zERT Network Analyzer database uses.
- Performance is also significantly enhanced when Db2 for z/OS performs its I/O operations over High Performance FICON (zHPF) or zHiperLink (zHL).

The IBM zERT Network Analyzer import and query processing for large amounts of data might take a long time and consume significant CPU cycles. Because IBM zERT Network Analyzer is a Java application, and Db2 for z/OS is used as its data store, much of the IBM zERT Network Analyzer processing is eligible to run on IBM z Integrated Information Processor (zIIP) specialty engines. You can run IBM zERT Network Analyzer on a system that has sufficient zIIP capacity available to minimize the general-purpose processor CPU costs that are associated with import and query operations. You can also use WLM policies to properly prioritize the DDF workload initiated by IBM zERT Network Analyzer, so it does not impact more important workloads on the system.

Specifying a long zERT aggregation recording interval in the TCPIP profile dataset on the GLOBALCONFIG ZERT AGGREGATION INTVAL parameter (for example, 12 or 24 hours) can greatly reduce the amount of SMF data the IBM zERT Network Analyzer has to import and query, speeding up both types of operations. For more information on specifying a zERT aggregation recording interval, see [Defining the zERT aggregation recording interval in z/OS Communications Server: IP Configuration Guide](#).

You can reduce the import time and processing costs by filtering out any of the SMF records that are not SMF type 119 subtype 12 before you run the import operation. These non-zERT records can be stripped out of your SMF dump data sets by using the IFASMFDP program. To do so, specify the SMF dump data set containing the SMF type 119 subtype 12 and other SMF records as the input data set (INDD) and specify OUTDD(<outDDname>, TYPE(119(12))). For more information about using the IFASMFDP SMF data set dump program, see [z/OS MVS System Management Facilities \(SMF\)](#).

Chapter 27. Configure the Cloud Provisioning services

This chapter describes how to quickly get started with IBM Cloud Provisioning and Management for z/OS, by using supplied jobs to set up a secure default domain and tenant. Included are descriptions of the key concepts and terms, and the resource profiles that must be defined. The examples in this chapter follow the default security setup for IBM Cloud Provisioning and Management for z/OS. Your installation can choose alternative values for the settings that are shown here.

IBM strongly recommends the use of groups, whenever possible, for ease of security administration. This chapter, the IZUPRSEC sample RACF setup job, and the automatic security management in IBM Cloud Provisioning and Management for z/OS all assume that you will use groups for security administration.

After you perform the initial security setup, you will not need to repeat the steps in this chapter. Instead, the Cloud Provisioning tasks will perform dynamic updates to your security environment, with one exception: The provisioning administrator group is maintained manually by your security administrator.

The instructions in this chapter assume that your installation shares its security database across the participating systems in the sysplex. If you use more than one security database, your security administrator must duplicate the Cloud Provisioning authorizations in each security database.

More information is provided in the sections that follow:

- [“What is Cloud Provisioning?” on page 141](#)
- [“Terms you should know” on page 142](#)
- [“Help with security setup” on page 145](#)
- [“Steps for setting up security” on page 146](#)
- [“Verify that security is set up for the domain administrator” on page 151](#)
- [“Automatic security management for Cloud Provisioning” on page 155](#)
- [“Summary of security requirements for the Cloud Provisioning tasks” on page 158](#)
- [“Cloud provisioning marketplace” on page 164](#)
- [“Considerations for a multiple sysplex domain” on page 167](#)
- [“Provisioning a z/OS software instance” on page 170.](#)

What is Cloud Provisioning?

With IBM Cloud Provisioning and Management for z/OS you can perform software provisioning for z/OS middleware. This work includes creating instances of IBM middleware, such as IBM Customer Information Control System (CICS, IBM Db2, IBM Information Management System (IMS), IBM MQ, and IBM WebSphere Application Server, and creating middleware resources, such as IBM MQ queues, CICS regions, and Db2 databases.

Using the Cloud Provisioning tasks, your system programmers and application programmers can perform the following actions:

- System programmers:
 - Define the cloud domain, administrators for the domain, and classes of users (tenants) for the domain.
 - Prepare software services templates that provision z/OS software. Service providers add templates, associate tenants with the templates, create resource pools for the templates, test the templates, then publish them to make them available for consumers.
- System programmers or application programmers:
 - Provision software from templates, creating software services instances.

- Manage software services instances.

For information about using the Cloud Provisioning tasks, see the online help that is included with z/OSMF. The z/OSMF online help is also available in IBM Documentation. For an overview of Cloud Provisioning, see [Cloud Provisioning in z/OSMF Cloud Provisioning](#).

The basic procedure for provisioning software is:

1. Define domains and tenants.
2. Create a template, specifying the workflow, action, and variables files that were provided by the software vendor.

The template is added to the software services catalog.

3. Modify the template as needed.
4. Add the template to a tenant.
5. Create the resource pool for the tenant and the template.
6. Approve any approval records. Approval records are created when a workflow or action definition file contains an element that identifies a user ID under which a workflow step or action is to be performed (a runAsUser ID). They can also be defined for the template in general, and for a domain.
7. Test the template and ensure that it successfully creates an instance, that is, that it provisions the software and that the actions that are defined for the instance perform as expected.
8. Publish the template to make it available to consumers.
9. Run the template to create a software instance.

Terms you should know

Security for Cloud Provisioning is based on resources and user groups. This topic describes the key concepts and terms that security administrators should know when creating authorizations for Cloud Provisioning.

Terms and concepts are described in the following topics:

- [“Resources” on page 142](#)
- [“User roles” on page 143](#)
- [“Objects” on page 144.](#)

For a summary of the required security setup, see [“Summary of security requirements for the Cloud Provisioning tasks” on page 158.](#)

Resources

The following are the key resources in the Cloud Provisioning tasks.

Table 29. Resources for Cloud Provisioning	
Resource	Description
<i>Domain</i>	Defines the management scope for tenants, services, and resource pools. A domain consists of one or more z/OS systems. A domain can include z/OS systems from more than one sysplex. A z/OS system can be in a single domain or in multiple domains that are managed by a single instance of z/OSMF. A cloud domain is defined by a z/OS system programmer who acts as the <i>provisioning administrator</i>. Each cloud domain is assigned one or more middleware system programmers who act as domain administrators. A base z/OSMF configuration includes one domain by default — the default domain.

Table 29. Resources for Cloud Provisioning (continued)

Resource	Description
<i>Resource pool</i>	Identifies the z/OS resources that are required by a z/OS software service. In a cloud domain with multiple tenants, the resource pool defines the scope of resource sharing and resource isolation. For example, a resource pool can define a range of dedicated IP addresses or ports for each tenant. A base z/OSMF configuration includes one resource pool by default — the default domain shared resource pool.
<i>Tenant</i>	Defines the group of users who have the authority to provision software instances. A tenant consists of a user or group of users that have contracted for the use of specified services and pooled z/OS resources that are associated with the services in a domain. A base z/OSMF configuration includes one tenant by default — the default tenant.

User roles

The following are the key roles in the Cloud Provisioning tasks.

Table 30. User roles for Cloud Provisioning

Role	Performer	Description	Group Name ¹
<i>Provisioning administrator</i>	System programmer	Defines the cloud domains and the associated system resources for the cloud. The provisioning administrator also designates one or more users as domain administrators.	IYU
<i>Domain administrator</i>	System programmer	Manages a domain. The domain administrator is responsible for defining services, tenants, and resource pools for the domain, and managing the relationship across tenants, services, and resource pools.	IYU0
<i>Resource pool networking administrator</i>	Network administrator	Manages the resource pool for the networking resources in the cloud, such as network configuration policies.	IYU0RPAN
<i>Resource pool WLM administrator</i>	Performance administrator	Manages the resource pool for the WLM resources in the cloud, such as WLM policies.	IYU0RPAW
<i>Security administrator</i>	Security administrator	Maintains the installation's external security manager. For example, in an installation that uses RACF as its security manager, the security administrator is responsible for creating the RACF profiles and classes that are required for Cloud Provisioning. The security administrator is a member of the z/OSMF security administrator group, which is named IZUSECAD by default. It is assumed that this user has RACF SPECIAL authority. If your installation plans to allow automatic security updates for Cloud Provisioning, you can specify this user ID for the CLOUD_SEC_ADMIN keyword in the active IZUPRMxx parmlib member for your system. For more information, see “IZUSVR reference information” on page 44.	IZUSECAD

Table 30. User roles for Cloud Provisioning (continued)

Role	Performer	Description	Group Name ¹
Template approver	System programmer or security administrator	Responsible for approving the pending approval records that are associated with the template.	N/A
Consumer	Application developer	Has access to the software services and resource pools for a tenant. This user can provision a software services instance by using a software services template, and can manage the lifecycle of a software services instance.	IYU000

The z/OSMF default group names are shown. Your installation can select different values for z/OSMF in the IZUPRMxx parmlib member.

Objects

The following are some basic objects that you work with in the Cloud Provisioning tasks.

Table 31. Objects for Cloud Provisioning

Object	Description
Instance, or software services instance	Represents software that is provisioned by using templates.
Template, or software services template	Represents z/OS or z/OS middleware, or a z/OS middleware resource service. A template consists of workflows and input variables that can be used to provision z/OS software, actions that can be used with the provisioned software (the instance), and documentation.

For a summary of Cloud Provisioning roles, see Figure 24 on page 144

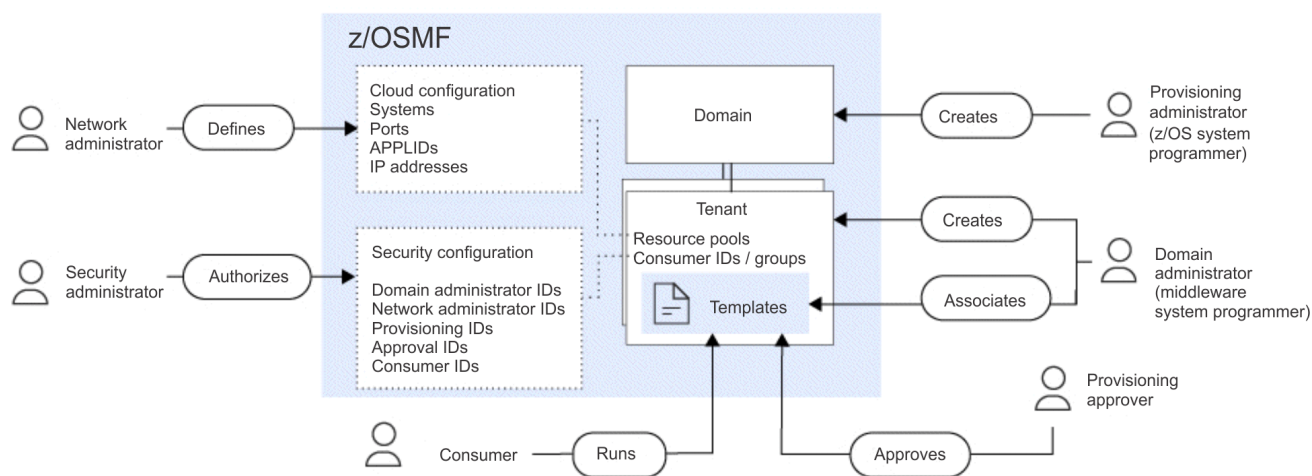


Figure 24. Roles in cloud provisioning

Help with security setup

In SYS1.SAMPLIB, the IZUPRSEC job represents the security definitions and authorizations that are needed for enabling the Cloud Provisioning functions. The job contains sample RACF commands for creating the required security authorizations.

Ask your security administrator to make a copy of this job and edit it for your environment.

Your security administrator can run the job to perform the following security setup actions:

- Define the required SAF resource profiles.
- Create the corresponding SAF security groups.
- Grant the appropriate authorizations.

As an alternative to running the IZUPRSEC job, your security administrator can perform the security setup manually. If so, see [“Steps for setting up security” on page 146](#) for instructions.

If your installation uses a security manager other than RACF, your security administrator can refer to the IZUPRSEC job for examples when creating equivalent commands for the security management product on your system.

Prerequisite services for Cloud Provisioning

Cloud Provisioning uses other z/OSMF services. Therefore, it is recommended that you enable these services prior to using Cloud Provisioning.

Cloud Provisioning uses the following z/OSMF services:

Network Configuration Assistant

Cloud Provisioning uses this service to define network resource pools. Resource pools are sets of z/OS resources that are required by a software service, such as port numbers.

Resource Monitoring

Cloud Provisioning uses this service for metering and capping. Metering helps you manage the use of resources by the tenant. Capping helps you limit the use of resources by the tenant.

Workload Management

Cloud Provisioning uses this service to enable metering and capping, and for defining workload management (WLM) resource pools. A WLM resource pool associates cloud information, such as a tenant name and domain ID, with WLM elements, such as report classes and classification rules. You define domains and tenants with the Resource Management task.

It is recommended that you enable the services prior to using Cloud Provisioning. To do so, you must do the following:

- In your active IZUPRMxx member, ensure that the PLUGINS statement is uncommented and includes at least the following options:

```
PLUGINS (COMMSERVER_CFG, RESOURCE_MON, WORKLOAD_MGMT)
```

- Create security profiles for the tasks that are associated with each z/OSMF service. IBM provides a set of IZUnnSEC jobs in SYS1.SAMPLIB with RACF commands to help with performing these changes. Each IZUnnSEC job is associated with a service, as follows:

IZUCASEC

Network Configuration Assistant

IZURMSEC

Resource Monitoring

IZUWMSEC

Workload Management

Modify the IZUxxSEC jobs for your environment. The IZUCASEC and IZUWMSEC jobs include commented sections for Cloud Provisioning, which you must uncomment.

Submit the IZUxxSEC jobs. Or, manually create the authorizations in your external security manager.

Steps for setting up security

In a z/OSMF base configuration, the initial IBM Cloud Provisioning and Management for z/OS environment includes a default domain and default tenant to help you quickly get started. This topic describes the steps for creating the security authorizations for the default domain and default tenant.

Before you begin

This procedure assumes that your installation has already created a base z/OSMF configuration.

This procedure is presented as an alternative for users who prefer to perform the security updates manually. The authorizations that it creates are equivalent to the security setup that is performed by running the IZUPRSEC job in SYS1.SAMPLIB. If you choose to run the IZUPRSEC job instead, locate the commented sections for Cloud Provisioning and uncomment them. Be sure to review and modify the job as necessary to ensure that its definitions work in your security environment. A summary of the IZUPRSEC authorizations is provided in [“Summary of security requirements for the Cloud Provisioning tasks” on page 158](#).

Regardless of whether you create authorizations manually or through IZUPRSEC, you need to connect one or more z/OS system programmer user IDs to the provisioning administrator group, as described in Step “2.d” on page 147 of the procedure. These users, called *provisioning administrators*, are responsible for managing the cloud environment.

Note: With the installation of the PTF for APAR PH29813, the default domain now supports manual security mode for creating templates and tenants. This option is intended for provisioning environments that cannot use automatic security mode. Previously, the default domain was required to run in automatic security mode. Now, when the default domain is created at z/OSMF startup time, it is placed in manual security mode if no security administrator is specified on the CLOUD_SEC_ADMIN statement in the IZUPRMxx parmlib member.

If you have incorrectly configured the security mode for Cloud Provisioning and Management, it is possible to change it. Doing so requires only that you edit the CLOUD_SEC_ADMIN statement in the IZUPRMxx parmlib member and restart the z/OSMF server. You can switch a domain from automatic security to manual security, and vice versa. Your changes to the CLOUD_SEC_ADMIN statement affect the security mode of all existing domains. The suggested practice is that you run Cloud Provisioning and Management in automatic security mode.

About this task

Use this procedure to define an initial set of security groups, user IDs, and resource profiles for your Cloud Provisioning environment.

This procedure involves the following changes to your security database:

- Activating the necessary RACF classes
- Creating the required SAF security groups
- Defining the required SAF resource profiles
- Granting the appropriate authorizations
- Refreshing the necessary RACF classes.

The examples in this section show the commands as they would be entered for a RACF installation. If your installation uses a security manager other than RACF, your security administrator can refer to the IZUPRSEC job for examples when creating equivalent authorizations for your system.

The instructions in this procedure assume that your installation shares its security database across the participating systems in the sysplex. If you use more than one security database, your security administrator must duplicate the Cloud Provisioning authorizations in each security database.

This procedure is intended only for your initial security set-up. Later, after you complete this procedure, you use the Software Services task and Resource Management task to maintain your security environment. However, managing the provisioning administrator IDs is a manual operation that you perform in your security database. This work involves connecting users to, or removing users from, the provisioning administrator group.

Procedure

1. Activate the ZMFCLLOUD resource class and enable the RACLIST and GENERIC profiles.

```
SETROPTS CLASSACT(ZMFCLLOUD) GENERIC(ZMFCLLOUD) RACLIST(ZMFCLLOUD)
```

2. Create the provisioning administrator identity.

- a) Define the provisioning administrator security group.

```
ADDGROUP IYU OWNER(some group)
```

Where IYU is the default SAF profile prefix for Cloud Provisioning. This prefix is used for the provisioning administrator group. User IDs with the provisioning administrator role have the authority to create domains, delete domains, and assign administrators within domains.

The IYU prefix is used in the examples in this procedure. Your installation can choose a different prefix by specifying it on the CLOUD_SAF_PREFIX keyword in the IZUPRMxx parmlib member. If so, substitute that value in the examples in this procedure.

- b) Define the SAF profile to be used for granting users access to the provisioning administrator role.

```
RDEFINE ZMFCLLOUD (IZUDFLT.ZOSMF.PROVISIONING.RESOURCE_MANAGEMENT.IYU) UACC(NONE)
```

Where IZUDFLT is the default SAF profile prefix for z/OSMF. This prefix is used for the z/OSMF resource profiles.

The IZUDFLT prefix is used in the examples in this procedure. Your installation can choose a different prefix by specifying it on the SAF_PREFIX keyword in the IZUPRMxx parmlib member. If so, substitute that value in the examples in this procedure.

- c) Grant the provisioning administrator group read access to the provisioning administrator profile.

```
PERMIT IZUDFLT.ZOSMF.PROVISIONING.RESOURCE_MANAGEMENT.IYU +  
CLASS(ZMFCLLOUD) ID(IYU IZUADMIN) +  
ACCESS(READ)
```

If you do not want all z/OSMF administrators to have the provisioning administrator role, remove the IZUADMIN group from the ID list.

- d) Select a user ID to be the provisioning administrator and connect it to the provisioning administrator group.

```
CONNECT <user-id> GROUP(IYU)
```

To authorize more provisioning administrator users, connect each user ID to the provisioning administrator group.

3. Set up security for the default domain.

- a) Define the domain administrator group for the default domain.

```
ADDGROUP IYU0 SUPGROUP(IYU)
```

Where IYU0 is the group name for domain administrators; it is defined under the Cloud Provisioning group (IYU), which is its RACF superior group.

- b) Define the SAF profile to be used for authorizing users to be domain administrators.

```
RDEFINE ZMFCLLOUD (IZUDFLT.ZOSMF.PROVISIONING.RESOURCE_MANAGEMENT.IYU0) UACC(NONE)
```

- c) Grant the provisioning administrator group (IYU), domain administrator group for the default domain (IYU0), and z/OSMF administrator group (IZUADMIN) read access to the domain administrator profile for the default domain.

```
PERMIT IZUDFLT.ZOSMF.PROVISIONING.RESOURCE_MANAGEMENT.IYU0 +  
CLASS(ZMFCLLOUD) ID(IYU IYU0 IZUADMIN) ACCESS(READ)
```

If you chose not to allow all z/OSMF administrators to be able to administer the default domain, remove the z/OSMF administrator group from the ID list. If you choose to later expand this authorization, you can use the Resource Management task in Cloud Provisioning to add individual users as domain administrators.

- d) Define the resource pool administrator group for networking for the default domain.

```
ADDGROUP IYU0RPAN SUPGROUP(IYU)
```

Where IYU0RPAN is the group name for networking administrators. It is defined as a subgroup of the Cloud Provisioning group.

- e) Define the resource pool administrator group for WLM for the default domain.

```
ADDGROUP IYU0RPAW SUPGROUP(IYU)
```

Where IYU0RPAW is the group name for WLM administrators. It is defined as a subgroup of the Cloud Provisioning group.

4. Set up security for the default tenant.

- a) Define the tenant consumer group for the default tenant.

```
ADDGROUP IYU000 SUPGROUP(IYU0)
```

Where IYU000 is the group name for tenant consumers. It is defined as a subgroup of the domain administrator group.

- b) Define the SAF profile to be used for authorizing users to be consumers in the default tenant.

```
RDEFINE ZMFCLLOUD (IZUDFLT.ZOSMF.PROVISIONING.RESOURCE_MANAGEMENT.IYU000) +  
UACC(NONE)
```

- c) Grant the tenant consumer group read access to the tenant consumer profile for the default tenant.

```
PERMIT IZUDFLT.ZOSMF.PROVISIONING.RESOURCE_MANAGEMENT.IYU000 +  
CLASS(ZMFCLLOUD) ID(IYU000) ACCESS(READ)
```

5. Define the SAF profile to be used for authorizing users to be template approvers for the default domain.

```
RDEFINE ZMFCLLOUD (IZUDFLT.ZOSMF.TEMPLATE.APPROVERS.IYU0) UACC(NONE)
```

6. Authorize users to be WLM administrators for the default domain.

- a) Define the SAF profile to be used for authorizing users to be resource pool administrators for WLM.

```
RDEFINE ZMFCLLOUD (IZUDFLT.ZOSMF.RESOURCE_POOL.WLM.IYU0) UACC(NONE)
```

- b) Grant the WLM administrator group and the z/OSMF administrator group (IZUADMIN) read access to the WLM administrator profile.

```
PERMIT IZUDFLT.ZOSMF.RESOURCE_POOL.WLM.IYU0 +  
CLASS(ZMFCLLOUD) ID(IYU0RPAW IZUADMIN) ACCESS(READ)
```

- c) Grant the z/OSMF server user ID access to the WLM administrator profile.

```
PERMIT IZUDFLT.ZOSMF.RESOURCE_POOL.WLM.IYU0 +  
CLASS(ZMFCLLOUD) ID(IZUSVR) ACCESS(READ)
```

Where IZUSVR is the default user ID for the z/OSMF server, which in turn has a default name of IZUSVR1. If you assigned a different user ID to the z/OSMF server started task, specify that user ID instead.

7. Authorize users to be network administrators for the default domain.

- a) Define the SAF profile to be used for authorizing users to be resource pool administrators for the network.

```
RDEFINE ZMFCLD (IZUDFLT.ZOSMF.RESOURCE_POOL.NETWORK.IYU0) UACC(NONE)
```

- b) Grant the network administrator group and the z/OSMF administrator group (IZUADMIN) read access to the network administrator profile.

```
PERMIT IZUDFLT.ZOSMF.RESOURCE_POOL.NETWORK.IYU0 +  
CLASS(ZMFCLD) ID(IYU0RPA IZUADMIN) ACCESS(READ)
```

- c) Grant the z/OSMF server user ID access to the network administrator profile.

```
PERMIT IZUDFLT.ZOSMF.RESOURCE_POOL.NETWORK.IYU0 +  
CLASS(ZMFCLD) ID(IZUSVR) ACCESS(READ)
```

Where IZUSVR is the default user ID for the z/OSMF server, which in turn has a default name of IZUSVR1. If you assigned a different user ID to the z/OSMF server started task, specify that user ID instead.

8. Define the ZMFAPLA profiles for the Cloud Provisioning, Workflows, Workflow Editor, and System Variables resources.

- a) Define the SAF profile to be used for authorizing users to the Software Services task.

```
RDEFINE ZMFAPLA (IZUDFLT.ZOSMF.PROVISIONING.SOFTWARE_SERVICES) UACC(NONE)
```

- b) Define the SAF profile to be used for authorizing users to the Resource Management task.

```
RDEFINE ZMFAPLA (IZUDFLT.ZOSMF.PROVISIONING.RESOURCE_MANAGEMENT) UACC(NONE)
```

- c) Define the SAF profile to be used for authorizing users to the Workflows task.

```
RDEFINE ZMFAPLA (IZUDFLT.ZOSMF.WORKFLOW.WORKFLOWS) UACC(NONE)
```

- d) Define the SAF profile to be used for authorizing users to the Workflow Editor task.

```
RDEFINE ZMFAPLA (IZUDFLT.ZOSMF.WORKFLOW.EDITOR) UACC(NONE)
```

- e) Define the SAF profile to be used for authorizing users to the System Variables resource.

```
RDEFINE ZMFAPLA (IZUDFLT.ZOSMF.VARIABLES.SYSTEM.ADMIN) UACC(NONE)
```

9. Grant z/OSMF access to the provisioning administrator, default domain administrator, and the default tenant consumer groups.

```
PERMIT IZUDFLT.ZOSMF CLASS(ZMFAPLA) ID(IYU IYU0 IYU000) ACC(READ)
```

10. Grant the resource administrator groups access to z/OSMF.

```
PERMIT IZUDFLT.ZOSMF CLASS(ZMFAPLA) ID(IYU0RPA IYU0RPAW) ACCESS(READ)
```

11. Grant the user groups access to the Software Services, Workflows, and Workflow Editor tasks.

```
PERMIT IZUDFLT.ZOSMF.PROVISIONING.SOFTWARE_SERVICES +
CLASS(ZMFAPLA) ID(IYU IYU0 IYU000) ACCESS(READ)
```

```
PERMIT IZUDFLT.ZOSMF.WORKFLOW.WORKFLOWS +
CLASS(ZMFAPLA) ID(IYU IYU0 IYU000) ACCESS(READ)
```

```
PERMIT IZUDFLT.ZOSMF.WORKFLOW.EDITOR +
CLASS(ZMFAPLA) ID(IYU IYU0 IYU000) ACCESS(READ)
```

12. Grant administrators access to the Resource Management task.

```
PERMIT IZUDFLT.ZOSMF.PROVISIONING.RESOURCE_MANAGEMENT +
CLASS(ZMFAPLA) ID(IYU IYU0) ACCESS(READ)
```

13. Grant the resource administrator groups access to the Workflows task and Software Services task.

```
PERMIT IZUDFLT.ZOSMF CLASS(ZMFAPLA) ID(IYU0RPAN IYU0RPAN) ACCESS(READ)
```

```
PERMIT IZUDFLT.ZOSMF.PROVISIONING.SOFTWARE_SERVICES +
CLASS(ZMFAPLA) ID(IYU0RPAN IYU0RPAN) ACCESS(READ)
```

```
PERMIT IZUDFLT.ZOSMF.WORKFLOW.WORKFLOWS +
CLASS(ZMFAPLA) ID(IYU0RPAN IYU0RPAN) ACCESS(READ)
```

14. Grant the z/OSMF administrator group the authority to modify or delete system variables by using the Systems task or a z/OSMF REST service.

```
PERMIT IZUDFLT.ZOSMF.VARIABLES.SYSTEM.ADMIN +
CLASS(ZMFAPLA) ID(IZUADMIN) ACCESS(READ)
```

15. Create the z/OSMF security administrator role if it does not exist already.

These users can perform automatic security updates in the Resource Management task.

- a) Define the z/OSMF security administrator group.

```
ADDGROUP IZUSECAD
```

Where IZUSECAD is the default group name.

- b) Define the SAF profile to be used for authorizing users to be z/OSMF security administrators.

```
RDEFINE ZMFCLD (IZUDFLT.ZOSMF.SECURITY.ADMIN) UACC(NONE)
```

Where IZUDFLT is the z/OSMF SAF profile prefix.

- c) Grant the security administrator group read access to the security administrator profile.

```
PERMIT IZUDFLT.ZOSMF.SECURITY.ADMIN CLASS(ZMFCLD) +
ID(IZUSECAD) ACCESS(READ)
```

Only users with read access to this profile can be selected as domain security administrators by the provisioning administrator.

16. Enable the z/OSMF server to perform authorization checks for ZMFCLD class resources.

- a) Create the SERVER class profile.

```
RDEFINE SERVER (BBG.SECCLASS.ZMFCLD) UACC(NONE)
```

- b) Grant the z/OSMF server user ID access to the SERVER class profile.

```
PERMIT BBG.SECCLASS.ZMFCLD CLASS(SERVER) ID(IZUSVR) +
ACCESS(READ)
```

Where IZUSVR is the default user ID for the z/OSMF server, which in turn has a default name of IZUSVR1. If you assigned a different user ID to the z/OSMF server started task, specify that user ID instead.

- c) Connect the z/OSMF started task user ID to the z/OSMF security administrator group (by default, IZUSECAD).

```
CONNECT IZUSVR GROUP (IZUSECAD)
```

17. Refresh the RACF classes to make the preceding changes effective.

```
SETROPTS RACLIST (ZMFAPLA ZMFCLOUD SERVER) REFRESH
```

What to do next

To verify that you configured IBM Cloud Provisioning and Management for z/OS correctly, you can use the supplied IVP template in the default domain. For the steps to follow, see [“Verify that security is set up for the domain administrator” on page 151](#).

Verify that security is set up for the domain administrator

You can verify that security is set up correctly for the domain administrator role in IBM Cloud Provisioning and Management for z/OS. To do so, you can add an IBM-supplied template to the Software Services catalog and test run the template. This verification is referred to as performing the *installation verification procedure* or IVP for IBM Cloud Provisioning and Management for z/OS.

Before you begin

The IVP is supplied by IBM in the following location on your system: `/usr/lpp/zosmf/samples/cpm-sample-ivp/`

The IVP contains the following parts:

cloud-provisioning-ivp-workflow.xml

Workflow definition file for the provisioning workflow.

cloud-provisioning-ivp.properties

Contains values for the console command and unsolicited message.

cloud-provisioning-ivp-actions.xml

Actions file that defines only a deprovision action.

cloud-provisioning-ivp-AdministratorDoc.pdf

Documentation file for the IVP.

cloud-provisioning-ivp.mf

Manifest file. This file provides a shortcut when you create the template. Rather than specifying each of the aforementioned files in the template individually, you can specify just the manifest file, then click **Load** to supply values for the other files.

About this task

The IVP contains a template that runs a provisioning workflow under your user ID.

The workflow consists of two steps:

- If Step 1 completes successfully, your user ID is set up correctly for issuing operator commands. This step issues the START command to start a non-existent job (IZUTEST), which results in an unsolicited message (IEFC452I) when the job is not found. To issue the command, the step uses a REST service.
- If Step 2 completes successfully, your user ID is set up correctly for reading messages that are written to the operations console. This step checks the result of the previous step for the presence of unsolicited message IEFC452I.

To perform the IVP, your user ID must be authorized as a domain administrator. If your installation defined security as described in “Steps for setting up security” on page 146 or by using the IZUPRSEC sample job, the user IDs in the IZUADMIN group are authorized as domain administrators.

Procedure

1. Add the sample template to the software services catalog.

- a) Log in to z/OSMF with a domain administrator user ID.
- b) In the z/OSMF desktop view, select **Software Services**.
- c) Select the **Templates** tab.
- d) In the *Templates* table, click **Add Template**, then select **Standard** to use a standard template.
If **Add Template** is not available, it might be because you are not a domain administrator. If so, contact your system programmer or security administrator for assistance.
- e) On the page that is displayed, supply the required values, as follows:
 - i) For *Template source file*, specify the absolute z/OS UNIX path of the template manifest file for the IVP: `/usr/lpp/zosmf/samples/cpm-sample-ivp/cloud-provisioning-ivp.mf`
 - ii) Click **Load** to supply values for other fields on the window.
 - iii) Specify a template name, for example, SampleIVP.
 - iv) Optionally, select the Workflows disposition and Jobs disposition to delete the workflow and job on completion. The default is *keep*, which means that the workflow and job are preserved. You can remove them later, if you prefer.
 - v) Click **OK**. The template is added to the software services catalog.

2. Associate the template with the default tenant and create a resource pool.

- a) In the *Templates* table, select the template by clicking the check box for the template that you created, then click **Actions > Associate Tenant**.
- b) On the **Associate Tenant** window, accept the defaults. For resource pool selection, ensure that **Create a dedicated resource pool** is selected.
A dedicated resource pool is allocated only to this template. In contrast, a *shared resource pool* can be used by multiple templates.
- c) Click **OK**.
The Resource Management task opens to the *Add Template and Resource Pool for Tenant* window.
- d) On the *Add Template and Resource Pool for Tenant* window, enter the following values:
 - For the software services instance name prefix, specify a meaningful value, such as IVP.
 - For the maximum number of software services instances, specify a low value, such as 10.
 - The instance runs under Job Class A, which is the IBM default. If this job class is defined and active at your installation, you can use it. Otherwise, you must include a JOB statement with a valid job class job in the *Add Template and Resource Pool for Tenant* window. You can optionally include other JCL values on the JOB statement, such as the accounting information.
- e) Click **OK**.
If message IYURP0013I is displayed, click **OK** to continue.
The resource pool for the template is created with no network or workload management resources.
- f) Having used the Resource Management task to add a template to the tenant, return now to the Software task. Click the **Software Services** tab.

3. Test run the template to provision a software instance.

- a) In the *Templates* table, select the template that you created.
Notice that the template is in *Draft* state, which means that the template is ready to be provisioned.
- b) Click **Actions**, then select **Test Run**.
- c) Click **OK**.

Message IYUSC0032I is displayed to indicate that the software services instance is started.

If you used the suggested values, the instance name is ConsoleCommand_IVP00.

4. Verify that the template is provisioned.

- a) Click the **Instances** tab.
- b) In the *Instances* table, check the state of your instance.
 - If the template state is **Being Provisioned**, click **Refresh** to refresh the table display. Provisioning might take several minutes to complete.
 - If the template state is **Provisioning-Failed**, your user ID needs an extra security authorization. Proceed to **Step 5** and **Step 6** for actions to take to resolve the problem.
 - If the template state is **Provisioned**, you started the instance successfully. Skip to **Step 7**.

5. Determine which step failed.

- a) In the *Instances* table, click the instance name.

The **Instance details** tab is shown, which includes the following details about the instance:

 - Domain name (*default*)
 - Tenant name (*default*)
 - Name of the provisioning workflow. The workflow name follows the convention `ConsoleCommand_<prefix><instance-count>provision<generated string>`.
- b) Click the workflow name to navigate to the workflow.
- c) In the workflow, check for the following results:
 - i) Step 1 is *Complete* or *Failed*.
 - ii) Step 2 is *Complete* or *Failed*.

6. Resolve the step failure.

- a) Work with your system programmer or security administrator to add the missing authorizations to your user ID.
 - If Step 1 failed, your user ID is not authorized to issue console commands.
 - If Step 2 failed, your user ID is not authorized to a console for viewing the unsolicited message.

For the required authorizations, see the sample security job for z/OSMF console services (IZUGCSEC) in SYS1.SAMPLIB.
- b) Repeat Steps 1-4 of this procedure.

7. Deprovision the instance.

- a) In **Software Services**, select the **Instances** tab.
- b) In the *Instances* table, select the instance that you created.
- c) Click **Actions > Perform > Deprovision**.
- d) In the Perform deprovision window, click **OK**.

What to do next

For a more advanced test of your security setup, you can create and test run a template that requires approval from a specified approver. In a production environment, the approver might be a middleware system programmer or a security administrator.

To perform this test, you create a new template based on the one you created previously. This time, you modify the workflow input variable file that was supplied with the IVP to add a performer (a *runAsUser*) and an approver for the template. You repeat some of the steps you performed in the previous procedure.

Follow these steps:

1. In the *Templates* table, select your template.
2. Create another template based on the one you created previously:

- a. Click **Actions > CreateBased on**.
 - i) For Template name, specify the name of a new template, for example SampleIVP2.
 - ii) For Target file path, specify the name of an empty or non-existent directory, for example: /tmp/xxx. If the directory does not exist, z/OSMF attempts to create it.
 - iii) For Domain, select default to use the default domain.
- b. Click **OK** to create the template. The template is created in a draft state.
3. Associate the template with the default tenant and create a resource pool, as you did in **Step 2** of the previous procedure. If message IYURP0013I is displayed, click **OK** to continue.
4. Specify a run-as-user and an approver for the template, as follows:
 - a. Select **Templates > Modify > Edit path**, which opens the Workflow Editor.
 - b. In the Workflow Editor, click the Input Properties tab, then specify your own user ID for the properties `CONSOLE_ADMIN` and `CONSOLE_APPROVER`.

Tip: In Cloud Provisioning, when a template specifies a user ID under which a step must be performed, an approval record is created. Here, the user ID is referred to as the *runAsUser ID* for the step. Approval records must be approved by the approvers before the template can be run or published.

In the example that follows, `IBMUSER` is specified for both properties.

```
# Licensed Materials - Property of IBM
# 5650-ZOS
# Copyright IBM Corp. 2018
#
# Status = HSMA230
#-----
#
# This is the command that will be issued
# via the z/OSMF REST Consoles API
#
# CONSOLE_CMD = S IZUTEST#
# This is the unsolicited keyword that
# z/OSMF REST Consoles API should expect
# in the response to the CONSOLE_CMD.
#
# UNSOL_KEY_TO_DETECT = IEFC452I#
# This is the console Administrator user ID
# that should be used to issue the
# z/OSMF REST Consoles API if the user ID
# running the template does not have appropriate
# authorization.
#
# CONSOLE_ADMIN = ibmuser
#
# This is the console Approval user ID used
# for approving the usage of the console
# Administrator user ID specified by
# the ADMIN_CONSOLE variable.
#
# CONSOLE_APPROVER = ibmuser
```

- c. Click **Save** to save the input properties file.
- d. Close the Workflow Editor window.
5. In the **Templates > Modify** page, click **OK**.
6. In the *Templates* table, check the state of the template:
 - If the template state is **Pending security update**, click **Refresh** to refresh the table display.
 - If the state is **Draft pending approval**, the template requires approval. Resolving this state requires the approver user ID that you specified earlier to approve the template.
7. Approve the template:
 - a. In the *Templates* table, select the template that is in **Draft pending approval** state, then click **Actions > Approvals**.
 - b. In the Approvals window, review the item to approve.
 - c. To approve the template, select the row, then click **Actions > Approve**.
 - d. Return to the *Templates* table. Notice that the template is now **Draft approved**.

8. Test run the template by clicking **Actions**, then **Test Run**.
9. In the *Instances* table, check the state of your instance:
 - If the template state is **Being Provisioned**, click **Refresh** to refresh the table display. Provisioning might take several minutes to complete.
 - If the template state is **Provisioning-Failed**, resolve the errors for any failed steps and test run the template again.
 - If the template state is **Provisioned**, you started the instance successfully.
10. Deprovision the instance.

You can remove the template from the software services catalog when you are done.

Exploring this function further: Try running the IVP with other user IDs specified for the `CONSOLE_ADMIN` and `CONSOLE_APPROVER` properties. When these user IDs do not match, Cloud Provisioning automatically generates an additional approval record for your security administrator to approve. This behavior helps to ensure that security is maintained when provisioning is performed under different user IDs.

In the Workflow Editor:

- For `CONSOLE_ADMIN`, specify the user ID under which the template is to run. This user ID requires the authority to enter commands from the z/OS operations console. Typically, this person is a middleware system programmer who provisions templates at your company.
- For `CONSOLE_APPROVER`, specify the user ID of the person who must approve the provisioning of the template.

Avoid using a functional user ID for the approver. The approver user ID must be able to log in to z/OSMF.

Automatic security management for Cloud Provisioning

During regular operations with Cloud Provisioning, your installation periodically adds or removes users for domains and tenants. Such changes require immediate updates to your security setup. If you select *automatic security* for Cloud Provisioning in the Resource Management task, or accept the default, these changes are performed for you automatically.

This topic describes the options that are available for enabling automatic security management for IBM Cloud Provisioning and Management for z/OS.

Automatic security management can be performed by using either of the following methods:

- XML security descriptor. Cloud Provisioning will generate an XML request that identifies the required security operations for your external security manager (ESM) to process.
- Security REXX exec that is provided by the vendor of the ESM. For example, IBM supplies the REXX exec **izu.provisioning.security.config.rexx** for use with RACF.

Automatic security is enabled by default. It uses the z/OS service **R_SecMgtOper** to perform security operations directly and synchronously. In contrast, the REXX exec is run by a Resource Management workflow.

Both of these methods require that a valid user ID be specified for the `CLOUD_SEC_ADMIN` keyword in the `IZUPRMxx` parmlib member.

Using the XML descriptor support for automatic security processing

This method of automatic security uses the **R_SecMgtOper** service (module `IRRSM000`) in z/OS to process an IBM-supplied XML document. The contents of the XML document are processed by the external security manager (ESM). For information about the **R_SecMgtOper** service, see *z/OS Security Server RACF Callable Services*.

Use of IRRSM000 by z/OSMF requires that the external security manager (ESM) is defined to your system. For a RACF installation, this means that the RACF subsystem is defined to your system, such as by using either of the following techniques:

- Add the following statement to IEFSSNxx:

```
SUBSYS SUBNAME(RACF) INITRTN(IRRSSI00) INITPARM('<')
```

- A temporary alternative is to enter the following command. This change does not persist across an IPL of the system.

```
SETSSI ADD,SUBNAME=RACF,INITRTN=IRRSSI00,INITPARM='<'
```

To enable automatic security processing based on XML security descriptors, do the following:

1. Ask your security administrator to do the following:

- a. Locate the security configuration properties file on your system:

```
/global/zosmf/configuration/workflow/izu.provisioning.security.config.properties
```

Locate the following property:

```
security-configuration-directsecurity-enabled=
```

If this property is not present, add it.

- b. To use XML security descriptors, ensure that the property is set to true:

```
security-configuration-directsecurity-enabled=true
```

- c. Save the properties file.

2. Restart the z/OSMF server. From the operator console, enter the START command for the z/OSMF server started task: START IZUSVR1

When the server initializes, the following message is written to the IZUG0.log to indicate that the **R_SecMgtOper** service is used for automatic security processing:

```
Cloud Provisioning and Management will use direct security processing via R_SecMgtOper
for automatic security domains.
```

Using a REXX exec for automatic security processing

This method of automatic security uses the security REXX exec from IBM or one that you have obtained from another vendor. When installed, the security REXX exec is owned by the z/OSMF server user ID (by default, IZUSVR) and is intended for use by security administrators only. The exec can be updated only by users in the z/OSMF security administrator group (by default, IZUSECAD).

If your installation uses a security manager other than RACF, you must obtain a REXX exec with equivalent security commands from your vendor and store it on your system.

Then, do the following:

1. Ensure that a security REXX exec is installed on your system. The IBM-supplied REXX exec for RACF is already included in the following directory on your system:

```
/global/zosmf/configuration/workflow/izu.provisioning.security.config.rexx
```

For other security managers, you must obtain an equivalent REXX exec from your vendor and install it on your system.

2. Recycle the z/OSMF server to ensure that the security configuration properties file is created with the default IBM content and the correct ownership and permission settings.

From the operator console, enter the operator commands in the following sequence: **STOP IZUSVR1 > START IZUSVR1 > STOP IZUSVR1**.

It is not necessary to stop or restart the z/OSMF angel process (IZUANG1).

3. With the z/OSMF server stopped, ask your security administrator to do the following:

a. Locate the security configuration properties file on your system:

```
/global/zosmf/configuration/workflow/izu.provisioning.security.config.properties
```

Locate the following property:

```
security-configuration-rexx-location=
```

By default, the property identifies the location of the IBM-supplied security REXX exec.

b. To use a different REXX exec, edit the property so that it refers to the location of the replacement REXX exec. The location can be a sequential data set, partitioned data set (PDS), or z/OS UNIX path and file name.

If the REXX exec resides in a data set, observe the following naming conventions:

- Enter the fully qualified data set name, including the member name if you are using a PDS.
- Do not enclose the data set name in quotation marks.

Example:

```
security-configuration-rexx-location=SYS1.REXX(ZOSMFSEC)
```

If the REXX exec resides in a z/OS UNIX file, observe the following naming conventions:

- Enter the full path name, beginning with the forward slash (/) and including the file name, or a relative path.
- The name cannot contain any path segments, such as /. / or /.. /

Example:

```
security-configuration-rexx-location=/u/cloud/zosmf/workflow/  
izu.provisioning.security.config.rexx
```

c. Save the properties file.

4. Restart the z/OSMF server. From the operator console, enter the START command for the z/OSMF server started task: **START IZUSVR1**

When the server initializes, the following message is written to the IZUG0.log to indicate that REXX processing is used for automatic security processing:

```
Cloud Provisioning and Management will use REXX processing for automatic security domains.
```

Applying service to the IBM-supplied REXX exec

IBM can ship service updates to Cloud Provisioning, which might include updates to the **izu.provisioning.security.config.rexx** exec. If you use the IBM exec, it is recommended that you apply the PTFs to stay current with the latest level of the exec.

If your installation uses a modified version of the IBM-supplied security REXX exec for RACF security:

- Ensure that the security configuration properties file identifies the location of the exec on your system. See the procedure for updating the properties file in [“Using a REXX exec for automatic security processing” on page 156](#).
- Work with your security administrator to reconcile any differences between your copy of the exec and a new version from IBM.

When you are working with service updates, always check the PTF ++HOLD action for specific instructions for deploying the updated code, such as the need to restart the z/OSMF server to have the updates take effect.

Summary of security requirements for the Cloud Provisioning tasks

This topic describes the resources that must be defined, and the groups that must be permitted to the resources.

The security configuration requirements for Cloud Provisioning are described in the sections that follow. Typically, these permissions are created by your security administrator.

- [“Select the Legacy Special user ID” on page 158](#)
- [“Group name prefix for Cloud Provisioning user groups” on page 158](#)
- [“Class activation for Cloud Provisioning” on page 158](#)
- [“Resource authorizations for security administrators” on page 159](#)
- [“Resource authorizations for network administrators” on page 159](#)
- [“Resource authorizations for WLM administrators” on page 159](#)
- [“Resource authorizations for application developers” on page 159](#)
- [“Resource authorizations for the Cloud Provisioning user roles” on page 160](#)
- [“Resource authorizations for the z/OSMF server user ID” on page 162.](#)

Select the Legacy Special user ID

During configuration, you select a user ID to use for authorizing groups to the domain. This user ID, which is referred to as the *Legacy Special* user ID, requires RACF SPECIAL authority. It must also be connected to the z/OSMF security group for z/OSMF security administrators (IZUSECAD, by default). Typically, this user is a security administrator.

The Legacy Special user is the first provisioning administrator to be defined for your configuration. After Cloud Provisioning is configured, remember the Legacy Special user ID and keep it active for future operations. For example, with the Legacy Special user ID, you can authorize other users to be provisioning administrators, or use the Resource Management task to create more domains and add default domain administrators.

Group name prefix for Cloud Provisioning user groups

Your installation must define a SAF group name to be used for Cloud Provisioning groups. The group name is prepended to the names of the groups that represent the various roles in Cloud Provisioning, such as provisioning administrators, domain administrators, and tenants. The group name prefix is used in the RACF commands for defining groups.

By default, the value IYU is the group name prefix for Cloud Provisioning groups. Your installation can select a different SAF group prefix. To do so, specify the value in the IZUPRMxx parmlib member. For more information, see the description of the CLOUD_SAF_PREFIX statement in [“IZUPRMxx reference information” on page 35.](#)

Your installation can select a different group name prefix for user groups. If so, substitute that value in the examples. If you plan to use a different value, ensure that it is 1-3 characters (alpha-numeric, uppercase, or the following special characters: \$, and @).

Class activation for Cloud Provisioning

For a RACF installation, the security class ZMFCLOUD must be active when you configure Cloud Provisioning. The RACF commands for activating the class (with generic profile checking activated) are

included in the IZUPRSEC job. If your installation uses an external security manager other than RACF, ask your security administrator to create equivalent commands for your environment.

The ZMFCLLOUD class requires the RACLIST option. If you change the profiles, you must refresh the ZMFCLLOUD class to have the changes take effect.

Table 32 on page 159 describes the class activation for Cloud Provisioning.

Table 32. Class activation for Cloud Provisioning		
Class	Purpose	RACF command for activating
ZMFCLLOUD	Allow the user to use the z/OSMF core functions and tasks that are related to Cloud Provisioning. z/OSMF defines a resource name for each core function and task that is related to Cloud Provisioning.	SETROPTS CLASSACT(ZMFCLLOUD) GENERIC(ZMFCLLOUD) + RACLIST(ZMFCLLOUD)

Resource authorizations for security administrators

Users who perform security administration tasks should be members of the z/OSMF security administrator group (IZUSECAD, by default). This group requires an OMVS group ID (GID).

Security administrators require access to the system resources that are used by the Cloud Provisioning tasks. For more information, see [Table 33 on page 160](#).

Resource authorizations for network administrators

Network administrators require access to the Network Configuration Assistant task, and to system resources that are used by the Network Configuration Assistant task. For more information, see [Table 33 on page 160](#).

Resource authorizations for WLM administrators

WLM administrators require access to resources, such as those that are protected by the profile MVSADMIN.WLM.POLICY. For more information, see [“Updating z/OS for the Workload Management service” on page 117](#) and [Table 33 on page 160](#).

Resource authorizations for application developers

z/OSMF includes the Swagger interface, which allows application developers and other users to display format descriptions of the Cloud Provisioning REST APIs. To enable the use of Swagger at your installation, define the Swagger resources in your external security manager, and grant READ access to the appropriate users and groups.

On a system with RACF as the security manager, you can use the following commands:

1. Define the allAuthenticatedUsers resource profile:

```
RDEFINE EJBROLE IZUDFLT.com.ibm.ws.management.security.resource.allAuthenticatedUsers UACC(NONE)
```

The profile includes the z/OSMF SAF profile prefix, which is IZUDFLT, by default. Your installation can select a different SAF profile prefix for z/OSMF in the IZUPRMxx parmlib member.

2. To give users and administrators access to Swagger, grant them READ access to the allAuthenticatedUsers resource profile:

```
PERMIT IZUDFLT.com.ibm.ws.management.security.resource.allAuthenticatedUsers CLASS(EJBROLE)  
ID(IZUUSER IZUADMIN) ACCESS(READ)
```

By default, the user and administrator groups for z/OSMF are IZUUSER and IZUADMIN.

3. Create an administrator role for Swagger by defining the Administrator resource profile:

```
RDEFINE EJBROLE IZUDFLT.com.ibm.ws.management.security.resource.Administrator UACC(NONE)
```

4. Assign the administrator role to the z/OSMF administrator group, which is IZUADMIN by default:

```
PERMIT IZUDFLT.com.ibm.ws.management.security.resource.Administrator CLASS(EJBROLE)  
ID(IZUADMIN) ACCESS(READ)
```

For more information about the Cloud Provisioning REST services, see [IBM z/OS Management Facility Programming Guide](#).

Resource authorizations for the Cloud Provisioning user roles

Table 33 on page 160 describes the authorization requirements for the common user roles in Cloud Provisioning. The IZUPRSEC job includes sample RACF commands for creating these authorizations on your system. A procedure for creating these authorizations manually is shown in [“Steps for setting up security”](#) on page 146.

Table 33. Security setup requirements for Cloud Provisioning user roles				
Resource class	Resource name	Who needs access?	Type of access required	Why
DATASET	<i>your_stack_include_dataset</i>	TCP/IP stack started task ID.	READ	Allows the TCP/IP stack to read the include data set when the TCP/IP stack is started. This definition is applicable only when your installation uses discrete or generic profiles to protect data set access.
DATASET	<i>your_stack_dynamic_update_dataset</i>	TCP/IP stack started task ID.	READ	Allows the TCP/IP stack to read the VARY OBEY data set that IBM Cloud Provisioning and Management uses to dynamically update the TCP/IP stack. This definition is applicable only when your installation uses discrete or generic profiles to protect data set access.
EJBROLE	<SAF-prefix>.IzuManagementFacilityProvisioning. izuUsers	- z/OSMF users group (IZUUSER) - z/OSMF administrators group (IZUADMIN)	READ	Allow the user to connect to the Software Services and Resource Management tasks.
EJBROLE	<SAF-prefix>.com.ibm.ws.management. security.resource.Administrator	- z/OSMF users group (IZUUSER) - z/OSMF administrators group (IZUADMIN)	READ	Allow the user to act as administrator for the Swagger function in z/OSMF.
EJBROLE	<SAF-prefix>.com.ibm.ws.management. security.resource.allAuthenticatedUsers	z/OSMF administrators group (IZUADMIN)	READ	Allow the user to use Swagger to display information about the z/OSMF REST APIs. For information about the REST services, see IBM z/OS Management Facility Programming Guide.
ZMFAPLA	<SAF-prefix>.ZOSMF.IBM_ CLOUDPORTAL.MARKETPLACE.CONSUMER	Consumers and domain administrators	READ	Allow the user to use the marketplace to provision and manage software services.

Table 33. Security setup requirements for Cloud Provisioning user roles (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF-prefix>.ZOSMF.IBM_ CLOUDPORTAL. MARKETPLACE. ADMIN	Domain administrators	READ	Allow the user to control which services are published to the marketplace, and manage the services to which consumers have subscribed.
ZMFAPLA	<SAF-prefix>.ZOSMF.PROVISIONING. RESOURCE_MANAGEMENT	- Provisioning administrator group - Domain group - Resource pool network administrator group - Resource pool WLM administration group - z/OSMF security administrators group (IZUSECAD)	READ	Allow the user to access the Resource Management task.
ZMFAPLA	<SAF-prefix>.ZOSMF.PROVISIONING. SOFTWARE_SERVICES	- Provisioning administrator group - Domain group - Tenant group - Resource pool network administrator group - Resource pool WLM administration group - z/OSMF security administrators group (IZUSECAD) - Consumers and domain administrators	READ	Allow the user to access the Software Services task.
ZMFAPLA	<SAF-prefix>.ZOSMF.VARIABLES. SYSTEM.ADMIN	z/OSMF administrators group (IZUADMIN)	READ	Allow the user to access the system variable definitions.
ZMFAPLA	<SAF-prefix>.ZOSMF.WORKFLOW. EDITOR	- Provisioning administrator group - Domain group - Tenant group - z/OSMF users group (IZUUSER) - z/OSMF administrators group (IZUADMIN)	READ	Allow the user to access the Workflow Editor task in z/OSMF.
ZMFAPLA	<SAF-prefix>.ZOSMF.WORKFLOW. WORKFLOWS	- Provisioning administrator group - Domain group - Tenant group - z/OSMF users group (IZUUSER) - z/OSMF administrators group (IZUADMIN)	READ	Allow the user to access the Workflows task in z/OSMF.

Table 33. Security setup requirements for Cloud Provisioning user roles (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF-prefix>.ZOSMF. WORKLOAD_MANAGEMENT. WORKLOAD_MANAGEMENT.ENWRP	- z/OSMF administrators group (IZUADMIN) - WLM resource pool administration group	READ	Allow the user to access the WLM Resource Pooling (WRP) functions of z/OSMF. Using a WRP definition, the user can associate cloud information (tenant name, domain ID, template type, service levels supported) with WLM elements (report classes and classification rules).
ZMFCLOUD	<SAF-prefix>.ZOSMF.PROVISIONING. RESOURCE_MANAGEMENT. <i>tenantGroupID</i>	Tenant group	READ	Allow the user to act as a tenant.
ZMFCLOUD	<SAF-prefix>.ZOSMF.PROVISIONING. RESOURCE_MANAGEMENT. <i>domainGroupID</i>	Domain group	READ	Allow the user to act as a domain administrator.
ZMFCLOUD	<SAF-prefix>.ZOSMF. RESOURCE_POOL.NETWORK. <i>domainGroupID</i>	Resource pool network administration group	READ	Allow the user to act as a network resource pool administrator.
ZMFCLOUD	<SAF-prefix>.ZOSMF. RESOURCE_POOL.WLM. <i>domainGroupID</i>	Resource pool WLM administration group	READ	Allow the user to act as a WLM resource pool administrator.
ZMFCLOUD	<SAF-prefix>.ZOSMF.SECURITY.ADMIN	z/OSMF security administrators group (IZUSECAD)	READ	Allow the user to access the security administration resource.
ZMFCLOUD	<SAF-prefix>.ZOSMF.TEMPLATE. APPROVERS. <i>domainGroupID</i>	Template approvers	READ	Allow the user to act as a cloud domain level template approver.
ZMFCLOUD	<SAF-prefix>.ZOSMF.TEMPLATE. APPROVERS. <i>domainGroupID. templateName</i>	Template approvers	READ	Allow the user to approve the specified template.
ZMFCLOUD	<SAF-prefix>.ZOSMF.TEMPLATE. INSTANCE. <i>domainGroupID. templateInstanceName</i>	Template instance owner	READ	Allow the user to access the specified template registry instance.

Resource authorizations for the z/OSMF server user ID

Table 34 on page 162 describes the Cloud authorizations that you must create for the z/OSMF server. By default, the server user ID is IZUSVR1. However, your installation might have selected a different user ID for the server during z/OSMF configuration. The IZUPRSEC job includes sample RACF commands for creating these authorizations on your system.

Table 34. Authorizations required for the z/OSMF server user ID

Resource class	Resource name	Type of access required	Why
DATASET	<i>your_stack_include_dataset</i>	ALTER	Allows the Network Configuration Assistant task to write to the configured include data sets when a network resource is provisioned or deprovisioned. There is one include data set for each stack that is defined for IBM Cloud Provisioning and Management for z/OS. This definition is applicable only when your installation uses discrete or generic profiles to protect data set access.

Table 34. Authorizations required for the z/OSMF server user ID (continued)

Resource class	Resource name	Type of access required	Why
DATASET	<i>your_stack_dynamic_update_dataset</i>	ALTER	Allows the Network Configuration Assistant task to write to the configured dynamic updates data sets when a network resource is provisioned or deprovisioned. There can be one dynamic update data set for each stack that is defined for IBM Cloud Provisioning and Management for z/OS. This definition is applicable only when your installation uses a discrete or generic profile to protect data set access.
OPERCMDs	MVS.VARY.TCPIP.OBEYFILE	CONTROL	Allows the Network Configuration Assistant task to issue the VARY TCPIP OBEYFILE command for IBM Cloud Provisioning and Management for z/OS. This definition is applicable only when your installation uses the OPERCMDS class to restrict access to the VARY TCPIP OBEYFILE command.
OPERCMDs	MVS.MCSOPER.ZCDPLM*	READ	Allows the Network Configuration Assistant task to issue various operator commands for IBM Cloud Provisioning and Management for z/OS. The console name for this extended MCS console is the text string ZCDPLM that is appended with the MVS sysclone value of the system of the z/OSMF instance.
OPERCMDs	MVS.DISPLAY.JOB	READ	Allows the Network Configuration Assistant task to issue the display A operator command for IBM Cloud Provisioning and Management for z/OS. This definition is applicable only when your installation uses the OPERCMDS class to restrict access to the DISPLAY A operator command.
OPERCMDs	MVS.DISPLAY.TCPIP	READ	Allows the Network Configuration Assistant task to issue the display TCPIP operator command for IBM Cloud Provisioning and Management for z/OS. This definition is applicable only when your installation uses the OPERCMDS class to restrict access to the DISPLAY TCPIP operator command.
OPERCMDs	MVS.DISPLAY.XCF	READ	Allows the Network Configuration Assistant task to issue the DISPLAY XCF operator command for IBM Cloud Provisioning and Management for z/OS. This definition is applicable only when your installation uses the OPERCMDS class to restrict access to the DISPLAY XCF operator command.
OPERCMDs	MVS.ROUTE.CMD.sysname	READ	Allows the Network Configuration Assistant task to issue the ROUTE operator command for IBM Cloud Provisioning and Management for z/OS. This definition is applicable only if the installation uses this profile to restrict the use of the ROUTE command.
SERVAUTH	EZB.NETWORKUTILS.CLOUD.mvsname	READ	Allows the Network Configuration Assistant task to issue operator commands for IBM Cloud Provisioning and Management for z/OS. <i>mvsname</i> is the name of the system where z/OSMF is running.

Table 34. Authorizations required for the z/OSMF server user ID (continued)

Resource class	Resource name	Type of access required	Why
SERVAUTH	EZB.NETSTAT.mvsname.tcprocname.CONFIG	READ	Allows the Network Configuration Assistant task to issue the command NETSTAT CONFIG . This definition is applicable only when your installation uses the SERVAUTH class to restrict usage of the NETSTAT command. When this definition is applicable, IZUSVR must be authorized for each stack defined for IBM Cloud Provisioning and Management for z/OS.
SERVAUTH	EZB.NETSTAT.mvsname.tcprocname.VIPADCFG	READ	Allows the Network Configuration Assistant task to issue the command NETSTAT VIPADCFG . This definition is applicable only when your installation uses the SERVAUTH class to restrict usage of the NETSTAT command. When this definition is applicable, IZUSVR must be authorized for each stack that is defined for IBM Cloud Provisioning and Management for z/OS.
SERVER	BBG.SECCLASS.ZMFCLOUD	READ	Allows the z/OSMF server to perform access checks in the ZMFCLOUD class.
ZMFCLOUD	<SAF-prefix>.ZOSMF.RESOURCE_POOL.NETWORK.domainGroupID	READ	Allows the z/OSMF server to access to the network administrator profile.
ZMFCLOUD	<SAF-prefix>.ZOSMF.RESOURCE_POOL.WLM.domainGroupID	READ	Allows the z/OSMF server to access the WLM administrator profile.

Cloud provisioning marketplace

Cloud Provisioning includes a sample marketplace, which makes software services available to marketplace consumers, and also includes functions for marketplace administrators. The sample marketplace is created when you import the Cloud Portal application into z/OSMF. Doing so adds the Marketplace and Marketplace Administration tasks to the z/OSMF desktop interface.

The marketplace is provided as-is, and is intended as a sample for learning purposes only.

If you plan to configure the marketplace, you have system customization to perform, as described in the following topics:

- [“Creating SAF authorizations for the marketplace tasks” on page 164](#)
- [“Creating role-based authorizations for the marketplace tasks” on page 165](#)
- [“Adding or removing the marketplace tasks” on page 165](#)
- [“Configuring the marketplace tasks” on page 166](#)
- [“Creating and managing subscriptions” on page 166](#)
- [“Modifying the Cloud Portal application” on page 166](#)

Creating SAF authorizations for the marketplace tasks

To enable the marketplace on your system, ask your security administrator to create the authorizations that are shown in [Table 35 on page 165](#).

Table 35. User authorization requirements for the marketplace tasks

Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF-prefix>.ZOSMF.IBM_CLOUDPORTAL.MARKETPLACE.CONSUMER	Consumers and domain administrators	READ	Allows the user to use the marketplace to provision and manage software services.
ZMFAPLA	<SAF-prefix>.ZOSMF.IBM_CLOUDPORTAL.MARKETPLACE.ADMIN	Domain administrators	READ	Allows the user to control which services are published to the marketplace, and manage the services to which marketplace consumers have subscribed.
ZMFAPLA	<SAF-prefix>.ZOSMF.PROVISIONING.SOFTWARE_SERVICES	Consumers and domain administrators	READ	Allows the user to access the Software Services task.

Creating role-based authorizations for the marketplace tasks

To perform tasks in the marketplace, users require the following authorizations:

- To associate a domain with the marketplace, the user must be defined to the domain as a domain administrator.
- To publish services to the marketplace, the user must be defined as either a domain administrator or a consumer in the domain that is associated with the marketplace.
- To subscribe to a published service, the user must be permitted to the template that is associated with the service.

Adding or removing the marketplace tasks

The Cloud Portal application is included with z/OSMF in the following location:

```
/usr/lpp/zosmf/samples/cloudportal
```

To add the marketplace tasks to z/OSMF, follow these steps:

1. Open the Import Manager task in z/OSMF.
2. Specify the following properties file as input:

```
/usr/lpp/zosmf/samples/cloudportal/cloudportal.properties
```

3. Click **Import**.

Later, if you want to remove the marketplace tasks from z/OSMF, you can do so by using a property file to remove the tasks.

Follow these steps:

1. Open the Import Manager task in z/OSMF.
2. Specify the following properties file as input:

```
/usr/lpp/zosmf/samples/cloudportal/cloudportaldelete.properties
```

3. Click **Import**.

If the removal is successful, the tasks are removed from z/OSMF. If an error occurs, resolve the error and import the property file again.

For more information, see the online help for the Import Manager task.

Configuring the marketplace tasks

When you access the marketplace for the first time, you are prompted as a marketplace administrator to supply information about the marketplace domain and its published services.

Specifically, you must provide the following information:

- On the **Settings** tab, specify the domain name for the marketplace. Specify one domain name only. Changing the domain name causes the deletion of any services that are published to the marketplace.

Also, on the **Settings** tab, you can indicate whether instances that are provisioned outside of the marketplace can be displayed in the **My Subscriptions** tab and **Manage Subscriptions** tab for marketplace consumers. By default, only entries that are provisioned in the marketplace can be displayed to marketplace consumers.

- On the **All Services** tab, select which services are to be published to the marketplace. You can add any of the templates that are listed in the Published Service Catalog to which you are permitted in the domain for the marketplace.

Creating and managing subscriptions

When a service is published, marketplace consumers can subscribe to it, which causes the service to be provisioned. In the Marketplace task, consumers can use the **All Services** tab to subscribe to any services to which they are permitted.

The marketplace provides the following functions for viewing and managing subscriptions:

- On the **My Subscriptions** tab, marketplace consumers can view their subscriptions. The tab shows which services are provisioned both within and outside of the marketplace, and allows consumers to take actions on the services.
- On the **Manage Subscriptions** tab, marketplace administrators can view all subscriptions in the marketplace domain to which they are permitted. The tab allows the administrator to manage the services to which marketplace consumers have subscribed.

The **All Services** tab has different functions, depending on whether you access the tab as a consumer (from the Marketplace task) or an administrator (from the Marketplace Administration task). In the Marketplace Administration task, the **All Services** tab allows the user (an administrator) to select which services are to be published to the marketplace.

Modifying the Cloud Portal application

The Cloud Portal application is provided as-is; you can modify it according to your needs. To modify the application, copy it to a local directory, and make changes to the copy.

To copy the application to another directory, you can use a command like the following, where /myuserdir is a local directory of your choice:

```
cp -R /usr/lpp/zosmf/samples/cloudportal /myuserdir/
```

To add or remove the modified Cloud Portal application from z/OSMF, you can use the Import Manager task, as described in [“Adding or removing the marketplace tasks”](#) on page 165. As input, specify the following properties file:

```
/myuserdir/cloudportal/cloudportal.properties
```

Considerations for a multiple sysplex domain

A domain can be defined to include systems from more than one sysplex. With a multiple sysplex domain, you can provision software instances across more than one sysplex in your enterprise, which allows your cloud provisioning environment to scale beyond the scope of a single sysplex.

In this configuration, you create the domain from a sysplex that you designate as the *primary z/OSMF system*. The objects that you create on the primary z/OSMF system are *managed domain objects* on the z/OSMF systems for the secondary sysplexes that are included in the domain.

Specifically:

- Tenants and templates that are created in the domain are *managed tenant and template objects* on the secondary z/OSMF systems.
- Resource pools that are created in the domain are *managed resource pool objects* on the secondary z/OSMF systems that are included in the resource pool system list.
- Registry instances that are created in the domain are *managed registry instances* on the secondary z/OSMF systems.

Managed domain objects can be viewed and used on any system in the domain. However, they are typically modified and removed only from the primary system.

Planning the systems for the domain

To participate in a multi-sysplex domain, the primary and secondary systems must be defined in the Systems table of the z/OSMF Systems task, configured to communicate with each other, and enabled for single sign-on.

For information about how to perform these setup actions, see the following:

- [Chapter 37, “Configuring a primary z/OSMF for communicating with secondary instances,” on page 221](#)
- [Chapter 38, “Enabling single sign-on between z/OSMF instances,” on page 225](#)
- The topic *“Defining your systems to z/OSMF”* in the online help for the z/OSMF Systems task.

For example, assume that your enterprise has three sysplexes and nine systems that are configured as shown in [Figure 25 on page 168](#). In this configuration, the z/OSMF instance in sysplex A is the primary z/OSMF instance. It manages sysplexes B and C by communicating (through HTTPS requests) with the secondary z/OSMF instances in sysplexes B and C.

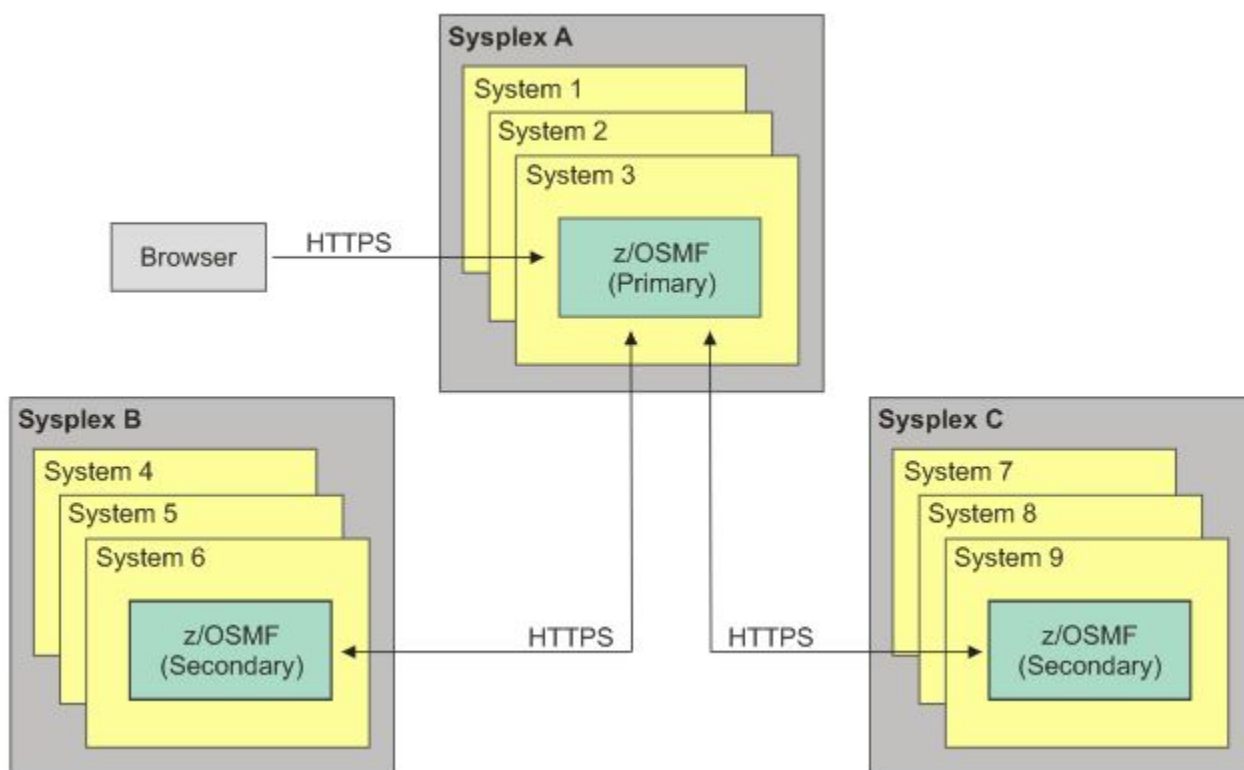


Figure 25. A multiple sysplex configuration includes one primary z/OSMF system and one or more secondary z/OSMF systems

The primary z/OSMF system is the one to which your web browser is connected, and it is the system that you use to create and modify objects in the domain. The other z/OSMF instances are referred to as secondary z/OSMF instances.

How provisioning is performed

In a multiple system configuration, creating and modifying templates and other objects is done from the sysplex that you designate as the primary z/OSMF system. Objects that are created on the secondary systems are managed by the primary z/OSMF system. To define the domain, templates and other objects, you use the Cloud Provisioning Resource Management and Software Services tasks. In the user interface, the objects that are created on the secondary sysplex are shown as *managed*. Managed objects are viewable and usable on the sysplex where they reside, but they should be modified and removed only from the primary system.

Table 36 on page 168 shows the types of created objects that are managed from the primary system.

Table 36. Managed object types in a secondary sysplex	
Object	Description
Domain	Domain for provisioning. When a domain is created in Cloud Provisioning the systems that are part of the domain are included in the definition. During domain creation, for each system in the domain that resides in a secondary sysplex, a managed domain is created in the Cloud Provisioning image in the secondary sysplex.
Tenant	Tenant for provisioning. When a tenant is created in Cloud Provisioning, for each system in the domain that resides in a secondary sysplex where a managed domain exists, a managed tenant is created in the managed domain in the secondary sysplex.

Table 36. Managed object types in a secondary sysplex (continued)

Object	Description
Resource pool	Resource pool for provisioning. A resource pool is created to define the resources in a template-to-tenant relationship. When a resource pool is created in a primary sysplex and the systems in the resource pool specify a secondary sysplex, the creation of the resource pool in the primary sysplex causes a managed resource pool to be created in the secondary sysplex. • If a template requires network resources, the network administrator must complete the network resource pool from each sysplex for the systems that are specified in the resource pool. • If a template requires WLM resources, the WLM administrator must complete the WLM resource pool from each sysplex for the systems that are specified in the resource pool.
Template	Template for provisioning. When a template is created in Cloud Provisioning, for each system in the domain that resides in a secondary sysplex where a managed domain exists, a managed template is created in the managed domain in the secondary sysplex. Templates can be run only from the primary z/OSMF system.
Registry instance	Registry instance for provisioning. When a template run or test run operation is performed on a template, if the target system is in a secondary sysplex, the provisioning workflow runs on the secondary sysplex. In this case, a managed registry instance is created on the secondary sysplex and the registry instance on the primary sysplex is updated to state of provisioning on the secondary sysplex. Registry instance actions must be performed on the primary z/OSMF system. Besides Deprovision, these actions can include Start, Stop, and Check Status.

Rules for a multiple sysplex environment

Observe the following rules for a multiple sysplex environment:

- Multiple sysplex domains, tenants, templates, and resource pools can be created and modified only from the primary sysplex. These objects should be removed from a secondary sysplex only in the event of an error, if they cannot be removed from the primary sysplex. Only the domain administrator can perform these actions.
- From z/OSMF on a secondary sysplex, do not create template, tenant, or resource pools in any "managed" domains. Templates, tenants, and resource pools for the managed domain must be created from the z/OSMF instance that is running in the primary sysplex.
- The primary sysplex and the secondary sysplexes must use the same cloud security mode: automatic or manual. A mix of automatic and manual cloud security modes between the primary and secondary sysplex is not supported.
- User IDs and group IDs that are used within the domain must exist in both the primary and the secondary sysplex. If the sysplexes have separate security databases, the user and group IDs must be defined in each security database. For example, consider consumer user IDs.
- Each sysplex has its own default domain. A primary sysplex cannot manage the default domain in a secondary sysplex. The multiple sysplex capability is not applicable to the default" domain. A default domain includes systems from the local sysplex only.
- Lower-level network resources to be used in the secondary sysplex must be configured by using the z/OSMF Network Configuration Assistant task in the secondary sysplex, not the primary sysplex.
- Lower-level WLM resources to be used in the secondary sysplex must be configured by using the z/OSMF Workload Management task in the secondary sysplex, not the primary sysplex.

- A multiple sysplex domain in a secondary sysplex includes only the z/OS systems in its local sysplex.
- The z/OSMF system settings in the primary sysplex must contain system definitions for all of the systems in the multiple sysplex domain. The z/OSMF system settings in the secondary sysplex must contain the system definitions for the systems in the secondary sysplex. The system definition for a system in the z/OSMF system settings in the secondary sysplex must match the system definition for a system in the z/OSMF system settings in the primary sysplex. That is, the system nicknames, systems, and sysplex names must be identical in the primary sysplex and the secondary sysplex.
- No more than one primary sysplex can be used to manage other secondary sysplexes.

Provisioning a z/OS software instance

IBM Cloud Provisioning and Management includes a set of templates that you can use to provision and deprovision z/OS systems. By selecting a z/OS provisioning template from the Cloud Provisioning software services catalog, you can provision a new instance of z/OS in a monoplex configuration in less than one hour.

The steps for provisioning a z/OS system are similar to the steps that you follow for provisioning other types of software instances. A key difference is that the z/OS system must be associated with a new type of dedicated resource pool that is called an *LPAR resource pool*.

During the provisioning process, an available LPAR entry is obtained from the LPAR resource pool automatically. The provisioning process uses properties that are associated with the selected LPAR entry, such as volume names, unit addresses, TCP/IP addresses, and OSA definitions to create and configure a new z/OS instance. Later, when you no longer require the instance, you can deprovision the instance. If so, the LPAR entry is returned to the pool, so that it can be reused when a new z/OS system is provisioned.

An LPAR resource pool can contain one or more LPAR entries. LPAR entries can be from the same CPC or a different CPC. If you have more than one LPAR resource pool, do not use duplicate LPAR entries.

Planning and setup

The z/OS system that you use to provision z/OS templates is known as the *provisioning system*. Some planning and setup is required to enable the provisioning system to drive the provisioning of z/OS templates. This work involves some host system customization and modification of the provisioning template's properties file, as described in the topic that follows.

You must plan for which logical partition (LPAR) to use for hosting the new z/OS image. You can use the Hardware Management Console (HMC) to locate an available LPAR, or use the HMC to create a new LPAR in your enterprise. If you use an existing LPAR, verify that the LPAR is not needed for any other systems at your installation.

This work might require the involvement of your system programmer, systems engineer, and network administrator.

IBM provides service for the z/OS provisioning templates in the form of program temporary fixes (PTFs). For a recommended approach to managing service updates, see [“Keep your z/OS provisioning templates up to date” on page 174](#).

Steps for provisioning a z/OS software instance

This topic describes the steps for provisioning a z/OS software instance.

Before you begin

This procedure assumes that you have a z/OS system available to use for provisioning the z/OS instance, with at least a default domain configured. If you prefer, you can use another domain for provisioning the z/OS instance.

This procedure refers to the following systems:

Provisioning system

An existing z/OS system that is used to provision (create) the new z/OS system. Also referred to as the *driving system*.

Source system

An existing z/OS system that is used for copying source libraries to the provisioned system. The source system must be identified in z/OSMF as a software instance.

Provisioned system

The new z/OS system that is created when the z/OS provisioning template is run successfully. Also referred to as the *target system*.

About this task

Use this procedure to provision a z/OS software instance in your environment.

Procedure

1. Define the source z/OS system as a z/OSMF software instance.

- a) From the z/OSMF desktop, select the Software Management task.
- b) In Software Management, select **Software Instances**.
- c) Select **Actions > Add** and follow the screen prompts for your z/OS system. For more information, see the online help for the Software Management task.
- d) Verify the z/OS product for your environment:

- View the software instance: **Actions > View > Products, Features, FMIDs**. Verify that the z/OS release is 2.5 or 2.4
- View the data sets and z/OS UNIX files: **Actions > View > Data sets**.

Notice that you can sort on the Volume column.

Determine the number of DASD volumes used for the z/OS software instance. Be sure to check the z/OS UNIX data sets, too.

- Work with your storage administrator to identify the type of DASD volumes that are associated with source z/OS software instance, such as Mod-9, Mod-18, or Mod-27. You will use the largest capacity of these volumes for the system residence (SYSRES) volume for the provisioned z/OS system (the target system). For example, if largest capacity SYSRES volume for the source z/OS software instance is Mod-18, all DASD volumes for the target system SYSRES must be Mod-18.

Make note of the following details:

Number of volumes used for z/OS
Largest capacity device type in use.

Later in this procedure, you will use this information when you define the LPAR pool entry for the target system.

2. Identify a logical partition (LPAR) to be used for provisioning the z/OS software instance.

- a) In the Hardware Management Console (HMC), locate a logical partition (LPAR) to be used for provisioning a z/OS instance. Or, define a new LPAR for this purpose.

Note: Consider the potential performance implications of sharing CPU resources between existing systems and the newly provisioned system. Though this consideration applies whenever you share CPU resources between systems, be aware that Cloud Provisioning makes it easier and faster to create z/OS systems. To ensure that production workloads are insulated from competition for CPU resources, work with your systems engineer to ensure that production LPAR weights are defined properly.

- b) In the **Sys Admin** page, make note of the LPAR characteristics, such as the number of I/O devices and IP addresses. Later in this procedure, you will specify the same number of devices for the system residence volume that you use for the z/OS software instance.

3. Ensure that Base Control Program internal interface (BCPii) is configured and active on the provisioning system (the driving system).

When you create a z/OS image, the provisioning system uses BCPii services to connect to system resources, such as the Support Element (SE) and the central processing complex (CPC). Therefore, the BCPii address space must be configured and active on the provisioning system.

To determine whether BCPii is active, enter the following command from the system console:

```
d a,hwibcpii
```

The BCPii address space (HWIBCPii) starts automatically during system IPL. If it is not active, you can start the address space manually by entering the START HWISTART command at the system console.

If the BCPii address space is unable to start, look for HWI* messages in the system log that indicate that BCPii failed to become active. Usually, the failures are caused by improper security configuration in the SE for BCPii communication with the local SE and the provisioning system. For more information, see the topic "BCPii address space does not start up at IPL" in *MVS Programming: Callable Services for High-Level Languages*.

If the provisioning system is not enabled for BCPii, you must configure the local Support Element (SE) to support BCPii and configure the BCPii address space. For the detailed steps, see the topic "BCPii setup and installation" in *MVS Programming: Callable Services for High-Level Languages*.

4. Ensure that System REXX is configured and active on the provisioning system.

For more information, see [“Ensuring that System REXX is set up and active” on page 192](#).

5. Obtain the provisioning properties file.

On the provisioning system, copy the `zosProvision.properties` file from the directory `/usr/lpp/zosmf/samples/cpm-sample-zos` to a user directory.

If you need to modify the provisioning workflow, you can copy the contents of the `/usr/lpp/zosmf/samples/cpm-sample-zos` directory to a user directory.

Future IBM service updates can include changes to the properties file. It is recommended that you use the most current version.

6. Modify the `zosProvision.properties` file with values for your environment.

For descriptions of the properties, refer to the properties file.

You can choose to provision the z/OS instance with a new RACF database with the base RACF definitions that are required for IPL and z/OSMF. The workflow variable `source_racf_db` is optional. By leaving this field empty, the provisioning process creates a new RACF database. Creating the new RACF database adds about 5 minutes to the provisioning time.

7. Create the z/OS provisioning template from the samples directory.

- In z/OS Cloud Provisioning, click **Software Services**.
- Select the **Templates** tab of Software Services to work with software services templates.
- Modify the template property file with values that are appropriate for your environment.
- Add the template by selecting **Actions > Add Template > Standard**.
- Specify the template name.

Rather than specifying each of the files individually, you can specify just the manifest file: `/usr/lpp/zosmf/samples/cpm-sample-zos/zos_provision.mf`. Then click **Load**.

- Modify the location of the properties file where you saved the updated properties file.
If you modified the workflow files or actions file, specify the location of the updated files.
- Click **OK** to create the template.

8. Associate the template with a tenant and create the LPAR resource pool.

The LPAR resource pool is required for the z/OS provisioning. It can be created in either a dedicated or shared resource pool.

- a) For the template you created, click **Actions > Associate Tenant** and select an existing tenant.
If you need to create a new tenant, use the Resource Management task to create a tenant in the domain in which the z/OS provisioning template is created. After a new tenant is created, you can perform **Actions > Associate Tenant**.
 - b) Select the option to either create a dedicated resource pool, use an existing tenant-shared resource pool, or use an existing domain-shared resource pool.
 - c) Click **OK** to associate the template with the tenant.
 - d) If you chose to create a dedicated resource pool, the resource management task will open with the **Add Template and Resource Pool** displayed. If you chose to use an existing tenant-shared resource pool or use an existing domain-shared resource pool, the template will be added to the shared pool, but you will need to open the Resource Management task and modify the shared resource pool that you selected.
 - To modify a tenant-shared resource pool, from the Resource Management task select the **Modify** action for the domain, select the **Modify** action for the tenant, and then select the **Modify** action for the shared resource pool.
 - To modify a domain-shared resource pool, from the Resource Management task select the **Modify** action for the domain, and then select the **Modify** action for the shared resource pool.
 - e) In the dialog **Add Template and Resource Pool** or the dialog **Modify Template and Resource Pool**, complete the fields in the **Instance Details** tab with appropriate values.
 - f) In the **Resource Pools** tab, click **Actions > Add Entry** to add the LPAR to the LPAR resource pool for the template. This LPAR is used to host the provisioned z/OS system.
 - g) In the dialog **Add LPAR Pool Entry**, complete each of the tabs to define the LPAR.
When you define the LPAR entry, ensure that the number of SYSRES volumes are equal or greater than number of SYSRES volumes identified in **Step 1**. Other operational volumes and network properties are set with the information you collected in **Step 2**.
 - h) When you have completed the input fields, click **Complete**.
 - i) You can define multiple LPARs for the LPAR resource pool. If you want to identify more LPARs for hosting provisioned z/OS systems, repeat steps **f** through **g**.
Note the following rules:
 - LPARs can be in same CPC or a different CPC.
 - The same LPAR/CPC cannot be specified in more than one LPAR resource pool.
 - j) Click **OK** to associate the template with the tenant and create the LPAR resource pool.
9. If the template is in *Pending Approvals* state, you can approve it by clicking **Actions > Approvals**.
10. **Test-run the template.**
To test run a template before publishing it, use the **Test Run** action that is provided in the Templates table.
If you choose to provision the z/OS instance with a new RACF database, you are required to add the values for `racf_user` and `racf_password`.
- a) Select the checkbox to Create a new RACF database.
 - b) Enter a value for the user ID for the new RACF database. This user ID is the only active user ID in the new RACF database and is used to log in into the provisioned system.
 - c) Enter a value for the password for the user ID. Be sure to save this value in a safe space, as it cannot be recovered or reset when the system IPLs. At initial login, this password expires and prompts you to create a new one.
- If the test run encounters a problem and provisioning fails, examine the workflow and resolve the issue that is associated with the failed step. It is advised to perform a complete deprovision after a failure.
11. **Publish the template.**

Publish makes a template available to consumers and prepares it for the **Run** action. It locks the template, allowing only limited modification, and puts it in the published state.

12. Run the template.

To run a template, use the **Run** action that is provided in the templates table. This action creates an instance. Repeat this step every time you want to provision a new instance of z/OS.

What to do next

To verify that the z/OS software instance is provisioned, try to log in to the system. You can, for example, use ssh or TN3270 to connect to the provisioned system host name, or open a web browser to z/OSMF to the provisioned system host name.

To determine the host name for the newly provisioned system, do the following:

1. In the Software Services task, select **Actions > View Instances**.
2. In the Private Variables tab, you can obtain the host name from the variable `fq_hostname`.

If you want to reIPL the provisioned z/OS system, do the following:

1. Select the **Instances** tab.
2. In the instances table, select the z/OS instance.
3. Click **Perform**, then select the action **Shutdown**.
4. Click **OK**.
5. In the instances table, select the z/OS instance.
6. Click **Perform**, then select the action **IPL**.
7. Click **OK**.

When you no longer need the z/OS software instance, you can deprovision it: **Actions > Perform > Deprovision**. The **Deprovision** action quiesces the partition, removes the operational data sets, deletes the master catalog, and initializes the volumes. The LPAR pool entry is returned back to the LPAR pool so that a new z/OS can be provisioned on that LPAR.

Keep your z/OS provisioning templates up to date

IBM periodically makes updates to the z/OS provisioning templates through service PTFs. It is recommended that you obtain these updates when they become available so that you can leverage them in your environment. For example, IBM might add properties to the properties file, `zosProvision.properties`, for a more complete z/OS configuration.

For the simplest approach to obtaining service updates, it is recommended that you make a practice of accessing the templates directly from the z/OSMF sample directory (`cpm-sample-zos`) whenever you create a provisioning template. Otherwise, if you choose to use customized versions of templates, you have a few more steps to perform to keep your templates up to date. Review the following scenarios and follow the one that matches your situation.

Scenario 1: You use the z/OS provisioning template directly from z/OSMF sample directory without modifications

This scenario is the easiest. If you are creating a new z/OS Provisioning template, simply create the z/OS Provisioning template by using the template manifest that is provided in z/OSMF samples directory (`cpm-sample-zos/zos-provision.mf`). As supplied by IBM, the updated template already references the new or changed properties in `zosProvision.properties`.

If you have already created a z/OS provisioning template but have not yet published it, refresh the template after applying updates to properties file, or modify the template to reference a new updated properties file. Select the template and perform **Refresh Template...** to import the template changes.

If your template is in “Published” state, it cannot be refreshed or modified. Select the template and choose **Create > New Version...** to create a new version of the published template.

Because the template references the z/OS provisioning template directly from z/OSMF sample directory, the new version picks up the workflow changes provided with the update. You need to provide a modified properties file that includes any new or updated properties that are provided with the update.

After you review and test the template, you can publish it.

Scenario 2: You use a copy of the z/OS provisioning template from your own directory, but you have not modified it

If your copy of the z/OS provisioning template is unchanged from the IBM-supplied template, it is recommended that you switch to using the template directly from the z/OSMF sample directory.

Then, do the following:

- If your current template is in draft state, select the template and perform **“Refresh Template...”** to pull in the template changes. Add the updated properties to your user directory, so that it replaces the previous version of the properties file.
- If your current template is in published state, create a new version of template that uses the z/OS provisioning workflow from z/OSMF samples directory and the updated properties file from your user directory. Select the template and choose **“Create”** then **“New Version...”** to create a new version of the published template. After you review and test the new template, you can publish it again.

Scenario 3: You use a modified copy of the z/OS provisioning template from your own directory

Besides updating the properties file, you must also apply your changes to your copy of the z/OS provisioning workflow, based on the new files from the PTF. Make a copy of z/OS provisioning workflow from z/OSMF sample directory and modify the copy as you require. Depending on whether the template is in "Draft" state or "Published" state, follow either of the preceding scenarios to update the provisioning template with the latest changes to the workflow file and properties file.

Chapter 28. Configure the Incident Log service

To use the Incident Log, you must configure it as described in this topic.

Dependencies on other z/OSMF services

The Incident Log service requires the following services to be configured:

- Common event adapter (CEA); see [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#).
- z/OSMF Settings service; see [Chapter 9, “Configure the z/OSMF settings service,” on page 67](#).
- Common Information Model (CIM) server; see [Chapter 47, “Configuring the CIM server for your system,” on page 253](#).

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUILSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUILSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

Follow the instructions in:

- [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#)
- [Chapter 47, “Configuring the CIM server for your system,” on page 253](#)
- [“Updating z/OS for the Incident Log service” on page 177](#)

Optional extensions to this service

None.

Updating z/OS for the Incident Log service

Enabling your z/OS system for the Incident Log service requires customization of the z/OS host system.

The Incident Log task requires that a number of z/OS components and facilities be enabled on your system. Much of this work might already be done on your system; for instructions, see the sections that follow.

System components used by the Incident Log task

As shown in [Figure 26 on page 178](#), a number of base z/OS functions are involved when the Incident Log task is used to manage diagnostic data for your system.

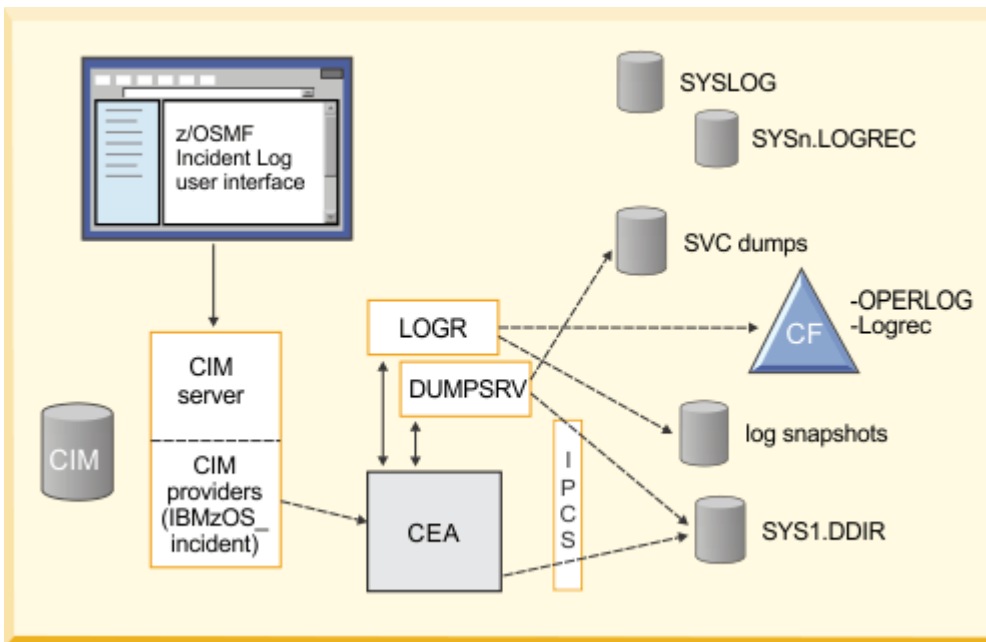


Figure 26. z/OS components that are used in Incident Log task processing

Specifically, z/OSMF and the Incident Log task interact with z/OS system functions in the following ways:

- Common Information Model (CIM) server for handling requests made by z/OSMF
- SDUMP component for managing the capture of OPERLOG, SYSLOG, and logrec snapshots
- IPCS dump directory services for managing the inventory of dumps related to incidents
- System Logger to capture log snapshots when sysplex-scope recording is requested through the OPERLOG or logrec system logger streams
- Dump analysis and elimination (DAE) for enabling the *Take Next Dump* function of the Incident Log task
- Environmental Record Editing and Printing (EREP) program for formatting the logrec data
- Common Event Adapter (CEA) for providing the data that is subsequently displayed in the Incident Log task user interface.

CEA helps to coordinate these system functions on behalf of z/OSMF incidents, in single system and sysplex environments.

Similar to other z/OS components, the CEA address space has the following attributes:

- Is started automatically during z/OS system initialization
- Supports a set of operator commands for interaction, such as `MODIFY CEA`
- Issues WTO messages (prefixed with CEA)
- Supports an abend code for handling incorrect actions (1D0)
- Requires security profile setup (through the CEA resource profile)
- Supports a variety of reason codes to indicate errors in CEA processing. Reason codes that might appear during z/OSMF operations are listed in [Appendix D, “Common event adapter \(CEA\) reason codes,” on page 419.](#)

The role of CEA in z/OSMF processing can be summarized, as follows:

- When CEA becomes active, it establishes an association with your installation's sysplex dump directory (typically SYS1.DDIR), which contains the inventory of SVC dumps taken in your sysplex, plus relevant information about each dump incident. This processing is done for SVC dumps taken on behalf of system abends, as well as those taken through the DUMP command and SLIP traps.

- Whenever an SVC dump is written to a data set, the DUMPSRV address space (on behalf of SVC dump processing) creates a new entry in the sysplex dump directory and informs CEA that the new incident has arrived. Then, CEA attempts to capture log snapshots, as follows:
 - If the system hardcopy log is recorded to the OPERLOG log stream, CEA directs the system logger component to create the log snapshot in a DASD log stream for the specified time duration. If the hardcopy is written to SYSLOG (that is, a single system scope), CEA uses spool allocation interfaces to access the SYSLOG data set and obtain the required snapshot, which is written to a DASD data set.
 - Similarly, if the logrec stream is written to a system logger log stream, CEA directs system logger to create a log snapshot of logrec data for the specified time period. If logrec is written to a data set, CEA invokes EREP to create the log snapshot.
 - Associates the snapshots with the corresponding incidents, based on snapshot data set name.
- When you use the Incident Log task to display incidents, CEA is invoked through the CIM server and uses IPCS functions to read the sysplex dump directory to obtain the inventory of SVC dumps taken on your system. CEA then extracts information from all relevant entries and returns it to z/OSMF for display. Similarly, when you use the Incident Log task to display details about an incident, z/OSMF receives those details from CEA, which obtains the information from the sysplex dump directory.
- When you request z/OSMF to send all or selected diagnostic materials to the specified URL, CEA is invoked to prepare the data, with different options, depending on whether you plan to use standard FTP or the z/OS Problem Documentation Upload Utility (PDUU). Here, all binary log data is formatted before being sent to the target system.
- In some instances, CEA performs its processing using System REXX execs, which are invoked through the AXREXX function.

As a result of this processing, your z/OS incidents are managed reliably on the system closest to the source of the information.

System customization needed for the Incident Log task

Table 37 on page 179 summarizes the z/OS system changes that are required or recommended for enabling the Incident Log task. Much of this work might already be done on your system, or might not be applicable. If so, you can skip the particular setup action. Other setup actions might require modifications to an existing setting, for example, if your installation has already defined a couple data set for the system logger component, you might need to increase the space allocation for system logger log stream records. For assistance with these setup actions, see the procedures referenced in the **Where described** column of Table 37 on page 179.

Table 37. z/OS setup actions for the Incident Log task			
	z/OS setup action	Where described	Check when task is completed
1	Ensure that the Common Information Model (CIM) server is configured on your system, including security authorizations and file system customization.	CIM includes jobs to help you perform these tasks (CFZSEC and CFZRCUST). See the chapter on CIM server quick setup and verification in <i>z/OS Common Information Model User's Guide</i> .	
2	Define a couple data set for the system logger component of z/OS.	See “Defining a couple data set for system logger” on page 181 .	

Table 37. z/OS setup actions for the Incident Log task (continued)

	z/OS setup action	Where described	Check when task is completed
3	Enable message log snapshots on the host system, or, optionally, on a sysplex-wide basis.	See the following topics: • “Setup considerations for log snapshots” on page 182 • “Enabling the operations log (OPERLOG)” on page 183 • “Defining and activating the LOGREC log stream” on page 185 • “Defining diagnostic snapshot log streams” on page 187 • “Enabling SYSLOG for diagnostic snapshots” on page 187.	
4	Enable error log snapshots on the host system, or, optionally, on a sysplex-wide basis.	See the following topics: • “Setup considerations for log snapshots” on page 182 • “Enabling the operations log (OPERLOG)” on page 183 • “Defining and activating the LOGREC log stream” on page 185 • “Defining diagnostic snapshot log streams” on page 187 • “Enabling SYSLOG for diagnostic snapshots” on page 187.	
5	Set up and configure automatic dump data set allocation (auto-dump).	See “Configuring automatic dump data set allocation” on page 188.	
6	Configure dump analysis and elimination (DAE) to suppress duplicate SVC dumps and use a sysplex-wide scope.	See “Configuring dump analysis and elimination” on page 189.	
7	Verify that a sysplex dump directory is defined for your system. If not, create a sysplex dump directory.	See “Creating the sysplex dump directory” on page 190.	
8	Ensure that the common event adapter (CEA) component is configured on your system, including security authorizations. Usually, the CEA address space is started automatically during z/OS initialization.	IBM provides the CEASEC job to help you create the security authorizations for CEA; see member CEASEC in SYS1.SAMPLIB. For information about running CEA, see “Ensure that common event adapter (CEA) is configured and active” on page 17.	
9	Ensure that System REXX (SYSREXX) is set up and active on your system.	See “Ensuring that System REXX is set up and active” on page 192.	

Table 37. z/OS setup actions for the Incident Log task (continued)

	z/OS setup action	Where described	Check when task is completed
10	If your installation has chosen to rename a dump data set, ensure that the data set name in the sysplex dump directory is correct.	See “Ensuring that dump data set names are correct” on page 193.	

Defining a couple data set for system logger

The Incident Log task requires that a couple data set be defined for the system logger component of z/OS to represent the diagnostic log snapshots. If your installation has not already defined the system logger data set, this topic describes the steps for doing so.

How to check if this step is done

To display LOGR couple data sets on a system, enter the following command:

```
D XCF,COUPLE,TYPE=LOGR
```

Figure 27 on page 181 shows the expected results:

```
IXC358I 15.15.26 DISPLAY XCF 038
LOGR COUPLE DATA SETS
PRIMARY  DSN: UTCXCF.SVPLEX6.LOGRR13.PRI
        VOLSER: X6CPLP      DEVN: 3D09
        FORMAT TOD          MAXSYSTEM
        10/21/2012 12:05:59      32
        ADDITIONAL INFORMATION:
          LOGR COUPLE DATA SET FORMAT LEVEL: HBB7705
          LSR(2000) LSTRR(1000) DSEXTENT(10)
          SMDUPLEX(1)
ALTERNATE DSN: UTCXCF.SVPLEX6.LOGRR13.ALT
        VOLSER: X6CPLA      DEVN: 3E08
        FORMAT TOD          MAXSYSTEM
        10/21/2012 12:17:05      32
        ADDITIONAL INFORMATION:
          LOGR COUPLE DATA SET FORMAT LEVEL: HBB7705
          LSR(2000) LSTRR(1000) DSEXTENT(10)
          SMDUPLEX(1)
LOGR IN USE BY ALL SYSTEMS
```

Figure 27. Expected results from the **D XCF,COUPLE,TYPE=LOGR** command

If this step is not already done

Define or update the system logger couple data set (LOGR CDS) with a large enough log stream records (LSR) value to allow sufficient space for managing the DASD-only log streams that will be created for capturing diagnostic log snapshots. The LSR value must be large enough to allow for two snapshot log streams for each dump recorded in z/OSMF, plus two model log streams, which are used as templates for defining the storage attributes for the snapshots. For information about modifying and reformatting a couple data set, see [z/OS MVS Setting Up a Sysplex](#).

System logger supports shared sysplex-scope (coupling facility resident) log streams and single-system DASD-only log streams, as follows:

- Coupling facility (CF) log streams are sysplex-wide in scope; any system in the sysplex can write to these log streams.

- DASD-only log streams can be written to by the local system only. When a DASD-only log stream is closed, it can be read from other systems in the sysplex if it resides on DASD that is shared by the other systems in the sysplex.

The system creates DASD-only log streams for the operations log (OPERLOG) and the sysplex logrec diagnostic snapshots. You do not need to predefine the DASD-only log streams. For the model used, see sample job CEASNPLG, which is supplied by IBM in SYS1.SAMPLIB(CEASNPLG).

Use shared DASD as the target for OPERLOG and logrec snapshots, so that the Incident Log task can access the log snapshots from any system in the sysplex.

In planning the space requirements for your system logger couple data set, plan for two DASD-only log streams per incident. To allow up to 100 incidents, for example, you must allow enough space for 200 log streams.

IBM recommends that you allow space for up to 1000 DASD-only log streams (or 500 incidents). To do so, use the IXCL1DSU format utility, for example:

```
//FMTLGCD S JOB MSGLEVEL=(1,1)
//          EXEC PGM=IXCL1DSU
//* S SUBMIT, JOB=LOGGER.ZOS17.JCL(FORMAT17)
//* SETXCF COUPLE, ACOUPLE=(LOGGER.OSR13.LARGE.INVNTRY, LOGR3), TYPE=LOGR
//* SETXCF COUPLE, PSWITCH, TYPE=LOGR
//SYSPRINT DD SYSOUT=*
//SYSIN DD *
DEFINEDS SYSPLEX( PLEX1) DSN(LOGGER.OSR13.LARGE.INVNTRY) VOLSER( LOGR3)
DATA TYPE( LOGR)
ITEM NAME( LSR)          NUMBER(2000)
ITEM NAME( LSTRR)        NUMBER(25)
ITEM NAME( DSEXTENT)     NUMBER(15)
ITEM NAME( SMDUPLEX)     NUMBER(1)
//
```

If the system logger couple data set lacks sufficient space to contain the diagnostic snapshots, the system issues message CEA0600I to indicate that the log streams could not be created.

To allow the Incident Log task to access diagnostic log snapshots on other systems in the sysplex, the log streams must reside on shared DASD. DASD-only log streams are expected to be written to SMS-managed DASD.

Related information

For more information, see [z/OS MVS Setting Up a Sysplex](#), which explains the following concepts:

- DASD-only log streams
- Setting up an SMS environment for DASD data sets
- Adding the data sets to the GRSRNL inclusion list
- Managing system logger log stream data sets
- Defining authorization.

Setup considerations for log snapshots

Enabling your z/OS system for the Incident Log service requires customization of the z/OS host system.

The Incident Log task can work with incident data from throughout your sysplex, or from just the system on which z/OSMF is installed. Your installation should determine the scope of incident related data collection, or *log snapshots*, to be used for the Incident Log task. To obtain the most benefit from the Incident Log task, it is recommended that your installation enable log snapshots on a sysplex-wide basis. If you cannot do so, however, z/OSMF is ready to work with incident data from a single system.

This section describes the system setup to be completed, based on the scope of data collection that you require.

- When message data is collected on a sysplex-wide basis, z/OSMF uses the operations log (OPERLOG) as the source for message data. This processing requires the following system setup:

- Enabling OPERLOG on each system for which message data is to be collected. See [“Enabling the operations log \(OPERLOG\)”](#) on page 183.
- Defining log streams for log snapshots to be obtained by the common event adapter (CEA) component of z/OS. See [“Defining diagnostic snapshot log streams”](#) on page 187.
- Defining a couple data set for sysplex-wide logging through system logger. See [“Defining a couple data set for system logger”](#) on page 181.

If you do not enable message data collection on a sysplex wide basis, z/OSMF collects message data for the z/OS host system only, using the system log (SYSLOG) as the source for creating diagnostic snapshots. See [“Enabling SYSLOG for diagnostic snapshots”](#) on page 187.

- When error log data is collected on a sysplex-wide basis, z/OSMF uses the logrec log stream as the source for error data. This processing requires that you set up system logger so that logrec data is written to a logger log stream. See [“Defining and activating the LOGREC log stream”](#) on page 185.

If you do not enable error log data collection on a sysplex wide basis, z/OSMF collects error log data for the z/OS host system only, using the logrec data set as the source for logrec data.

- These log stream snapshots are unique in that each incident creates two new log streams and are not deleted by Autodelete. The AUTODELETE function provides a method to automatically delete data from the log stream; it does not delete the log stream itself. In order for the RETPD value to be honored, the log streams must be connected and offload processing must be running regularly, which is not the case with these types of log streams.

You must manually delete snapshot log streams using either the **Delete Incident** action in the Incident Log task or the CEATool utility. For more information, see [Chapter 50, “Deleting incidents and diagnostic data,”](#) on page 261.

This action should be done frequently to mitigate the danger of running out of System Logger extents in the LOGR couple dataset. Consider scheduling a CEATool batch process to clean the snapshots.

Enabling the operations log (OPERLOG)

The operations log (OPERLOG) is a sysplex-wide log of system messages (WTOs) residing in a system logger log stream, comparable to SYSLOG, which is a single system message log residing on JES spool.

If OPERLOG is enabled on your system, z/OSMF can use OPERLOG to collect message data on a sysplex wide basis. Here, OPERLOG must be active in a system logger log stream. For the steps to follow, see [“Steps for setting up OPERLOG”](#) on page 184.

If you choose to defer this step, z/OSMF collects message data on a single system basis, using the system log (SYSLOG) as the source.

How to check if this step is done

To display the active medium where messages are recorded, enter the following command:

```
D C,HC
```

Figure 28 on page 183 shows the expected results:

```
CNZ4100I 15.19.16 CONSOLE DISPLAY 056
CONSOLES MATCHING COMMAND: D C,HC
MSG:CURR=0      LIM=9000 RPLY:CURR=0      LIM=9999  SYS=P02      PFK=00
HARDCOPY LOG=(SYSLOG,OPERLOG) CMDLEVEL=CMDS
ROUT=(ALL)
LOG BUFFERS IN USE: 0      LOG BUFFER LIMIT: 9999
```

Figure 28. Expected results from the **D C,HC** command

Steps for setting up OPERLOG

The following instructions are a summary of the details found in *IBM Redbook System Programmer's Guide to: z/OS System Logger*, which is available from <http://www.redbooks.ibm.com/>. For more information about setting up OPERLOG, see the topic on preparing to use system logger applications in [z/OS MVS Setting Up a Sysplex](#).

Before you begin

You must define the logger subsystem.

Procedure

1. Define the OPERLOG coupling facility structure in the CFRM policy. For example:

```
//OPERLOG JOB CLASS=A,MSGCLASS=A
//POLICY EXEC PGM=IXCMIAPU
//SYSPRINT DD SYSOUT=A
//SYSIN DD *
DATA TYPE(CFRM)
STRUCTURE NAME(OPERLOG)
SIZE(40448)
INITSIZE(40448)
PREFLIST(FACIL01,FACIL02)
```

2. Activate the CFRM policy through the **SETXCF START,POLICY,TYPE=CFRM,POLNAME=polname** command, or through the COUPLExx parmlib member.
3. Define the log stream to the LOGR policy. The following example is for illustrative purposes only; follow the recommendations in [z/OS MVS Setting Up a Sysplex](#) and [z/OS MVS Programming: Assembler Services Guide](#).

```
//OPERLOG JOB CLASS=A,MSGCLASS=A
//POLICY EXEC PGM=IXCMIAPU
//SYSPRINT DD SYSOUT=A
//SYSIN DD *
DATA TYPE(LOGR)
DEFINE STRUCTURE NAME(OPERLOG)
LOGSNUM(1)
MAXBUFSIZE(4092)
AVGBUFSIZE(512)
DEFINE LOGSTREAM NAME(SYSplex.OPERLOG)
STRUCTNAME(OPERLOG)
LS_DATACLAS(LOGR24K)
HLQ(IXGLOGR)
LS_SIZE(1024)
LOWOFFLOAD(0)
HIGHOFFLOAD(80)
STG_DUPLEX(NO)
RETPD(0)
AUTODELETE(No)
```

4. Create the security definitions for RACF (or an equivalent security manager). In the following example, the SYSplex.OPERLOG of the LOGSTRM resource CLASS is given READ permission, which allows all users to browse the operations log and *userid1* has UPDATE access level, which allows *userid1* to delete records from the log stream. That is, the user ID associated with the job running the IEAMDBLG program. For example:

```
RDEFINE LOGSTRM SYSPLEX.OPERLOG UACC(READ)
PERMIT SYSPLEX.OPERLOG CLASS(LOGSTRM) ID(userid1)
ACCESS(UPDATE) SETROPTS CLASSACT(LOGSTRM)
```

This example is for illustrative purposes only. Follow the guidelines for your installation.

5. Define the hardcopy device as OPERLOG in the HARDCOPY statement of the CONSOLxx parmlib member. You can change this setting using the **V OPERLOG, HARDCPY** command.

6. After you activate OPERLOG, you must manage the way in which records are handled.

SYS1.SAMPLIB contains a sample program, IEAMDBLG, to read log blocks from the OPERLOG log stream and convert them to SYSLOG format. The program is an example of how to use the services of the system logger component to retrieve and delete records from the OPERLOG log stream. It reads the records created in a given time span, converts them from message data block (MDB) format to hardcopy log format (HCL or JES2 SYSLOG), and writes the SYSLOG-format records to a file. It also has an option to delete from the log stream the records created before a given date.

When you use the delete option, you might want to first copy the records on alternate media and then conditionally delete the records in a separate JCL step to ensure that you have a copy of the data before deleting. If you do not run them on two separate conditional steps, deletion occurs simultaneously with copy without any guarantee that the copy process was successful.

For more information, see the topic on managing log data in [z/OS MVS Setting Up a Sysplex](#).

Results

To verify the completion of this work, enter the **DISPLAY CONSOLES, HARDCOPY** command to display the OPERLOG status.

What to do next

If you need to deactivate OPERLOG, you can use the **V OPERLOG, HARDCPY, OFF** command.

Defining and activating the LOGREC log stream

Logrec is the z/OS error log. It contains binary data describing error records that are written on behalf of system abends and other system recording requests. Logrec data is formatted through the batch utility EREP. The single-system version usually resides in a data set named SYS1.LOGREC or &SYSNAME.LOGREC. The sysplex version resides in a system logger log stream (the LOGREC log stream).

If the LOGREC log stream is active on your system, z/OSMF uses this log stream to collect logrec data on a sysplex wide basis. For information about defining and activating the LOGREC log stream, see [“Steps for setting up the LOGREC log stream”](#) on page 186.

If you choose to defer this step, z/OSMF collects logrec data on a single system basis, using the logrec data set as the source.

How to check if this step is done

To display the active medium for collecting logrec data, enter the following command:

```
D LOGREC
```

Figure 29 on page 185 shows the expected results:

```
IFB090I 15.22.12 LOGREC DISPLAY 062
CURRENT MEDIUM = DATASET
MEDIUM NAME = SYS1.P02.LOGREC
```

Figure 29. Expected results from the D LOGREC operator command

If the medium is DATASET, the logrec data is recorded using a data set. If the medium is LOGSTREAM, the logrec data is recorded in a LOGR logstream.

Steps for setting up the LOGREC log stream

The following instructions are a summary of the details found in *IBM Redbook System Programmer's Guide to: z/OS System Logger*, which is available from <http://www.redbooks.ibm.com/>. For more information about defining the log stream, see the topic on preparing to use system logger applications in [z/OS MVS Setting Up a Sysplex](#).

Before you begin

You must define the logger subsystem.

Procedure

1. IPL each system using its own logrec data set specified in the IEASYSxx parmlib member. Then, switch to using the log stream through the **SETLOGRC** command. This process allows your installation to fall back to using the data set if needed. To use the log stream immediately from the IPL, specify LOGREC=LOGSTREAM in IEASYSxx, as follows:

```
IEASYSxx with logrec data set:
LOGCLS=L,
LOGLMT=010000,
LOGREC=SYS1.&SYSNAME..LOGREC,    or  LOGREC=LOGSTREAM,
MAXUSER=128,
MLPA=00
```

2. Define the LOGREC log stream structure definition in the CFRM policy. For example:

```
//LOGREC JOB CLASS=A,MSGCLASS=A
//POLICY EXEC PGM=IXCMIAPU
//SYSPRINT DD SYSOUT=A
//SYSIN DD *
DATA TYPE(CFRM)
STRUCTURE NAME(LOGREC)
SIZE(2048)
INITSIZE(1024)
PREFLIST(FACIL01,FACIL02)
```

3. Define the system logger policy. For example:

```
//DEFINE EXEC PGM=IXCMIAPU
//SYSPRINT DD SYSOUT=A
//SYSIN DD *
DATA TYPE (LOGR)
DEFINE STRUCTURE NAME(LOGREC)
LOGSNUM(1)
AVGBUFSIZE(4068)
MAXBUFSIZE(4068)
DEFINE LOGSTREAM NAME(SYSPLEX.LOGREC.ALLRECS)
STRUCTNAME(LOGREC)
LS_DATACLAS(LOGR4K)
HLQ(IXGLOGR)
LS_SIZE(1024)
LOWOFFLOAD(0)
HIGHOFFLOAD(80)
STG_DUPLEX(NO)
RETPD(0)
AUTODELETE(NO)
```

4. Change the logrec recording medium:

```
SETLOGRC {LOGSTREAM|DATASET|IGNORE}
```

5. Create the required security definitions. For example:

```
RDEFINE LOGSTRM SYSPLEX.LOGREC.ALLRECS UACC(READ)
SETROPTS CLASSACT(LOGSTRM)
```


Results

To verify the completion of this work, enter the **DISPLAY LOGREC** command to display the current logrec error recording medium.

Defining diagnostic snapshot log streams

For optimal performance of the Incident Log task, it is recommended that your installation define operations log (OPERLOG) and logrec log streams for the CEA component of z/OS. Doing so allows the system logger component to determine the storage characteristics for storing diagnostic snapshots.

How to check if this step is done

To display the OPERLOG logstream, enter the following command:

```
D  LOGGER,L,LSN=SYSPLEX.OPERLOG
```

Figure 30 on page 187 shows the expected results:

```
IXG601I  15.26.03  LOGGER DISPLAY 070
INVENTORY INFORMATION BY LOGSTREAM
LOGSTREAM          STRUCTURE          #CONN  STATUS
-----
SYSPLEX.OPERLOG    LOGGER_STR1        000004  IN USE
SYSNAME: P00
DUPLEXING: LOCAL BUFFERS
SYSNAME: P01
DUPLEXING: LOCAL BUFFERS
SYSNAME: P02
DUPLEXING: LOCAL BUFFERS
SYSNAME: P03
DUPLEXING: LOCAL BUFFERS
GROUP: PRODUCTION

NUMBER OF LOGSTREAMS: 000001
```

Figure 30. Expected results from the **D LOGGER** command

If this step is not already done

To create the log streams, you can use a batch job like sample job CEASNPLG, which is supplied by IBM in SYS1.SAMPLIB(CEASNPLG). The CEASNPLG job deletes and redefines CEA diagnostic snapshot model log streams, using the IBM utility program, IXCMIAPU. For information about the IXCMIAPU utility, see [z/OS MVS Setting Up a Sysplex](#).

Enabling SYSLOG for diagnostic snapshots

If your installation collects messages about programs and system functions (the hardcopy message set) on a single system basis, the Incident Log task uses the system log (SYSLOG) as the source for diagnostic log snapshots.

Here, you must ensure that the proper security permissions exist, so that the JES subsystem can access SYSLOG on behalf of the common event adapter (CEA) component of z/OS. For example, in a system with RACF as the security management product, your security administrator can enter RACF commands like those shown in [Figure 31 on page 187](#). In the example, *node-id* is the NJE node ID of the JES2 or JES3 subsystem and *CEA_userid* is the user ID that you use to access CEA.

```
RDEFINE JESSPOOL node-id.MASTER+.SYSLOG.*.* UACC(NONE)
PERMIT node-id.MASTER+.SYSLOG.*.* CLASS(JESSPOOL) ID(CEA_userid) ACC(READ)
SETROPTS RACLIST(JESSPOOL)
```

Figure 31. RACF commands to enable CEA to access SYSLOG

Your installation might not have defined JESSPOOL under RACF authority. If so, your setting for the SETROPTS command will be different.

Configuring automatic dump data set allocation

For full functionality, the Incident Log task requires that automatic dump data set allocation (auto-dump) be active on the z/OS host system. If your installation has not already set up auto-dump, this topic describes the steps for doing so. If you choose to defer this step, the Incident Log task runs with limited functionality. If your installation uses automatic dump data set allocation, the Incident Log task uses the resulting dump data set names in the "Send Data" action, which allows your installation to transmit this data to a remote destination through FTP.

To set up automatic dump data set allocation, do the following:

1. Define the dump data set naming convention to be used by the system. Specify it using the "DUMPDS NAME=" command, for example:

```
$sysplex..DUMP.D&date..T&time..&SYSNAME..&S&seq
```

2. Determine where the dumps are to be stored. It is recommended that you use an SMS storage class or a shared DASD volume for dumps. Examples:

```
DUMPDS ADD,SMS=class  
DUMPDS ADD,VOL=(volser,volser,volser,..)
```

If you use a shared volume, ensure that the volume is managed through a shared catalog for the sysplex. Otherwise, for an incident with multi-system dumps, when deleting the incident, only the primary dump is deleted because the remote dumps are not accessible.

3. Start the function through the following command:

```
DUMPDS ALLOC=ACTIVE
```

For more details, see the following information:

- [DUMPDS command in z/OS MVS System Commands.](#)
- [SVC dump in z/OS MVS Diagnosis: Tools and Service Aids.](#)

If your installation does not use automatic dump data set allocation, it is likely that you have defined pre-allocated dump data sets (SYS1.DUMPxx) for the system to use. Typically, an installation archives an SVC dump to another data set as soon as the dump is complete, to avoid having the system overlay the data set with a subsequent dump. The archive data set name is defined by the installation and is not known to the system. If so, the following limitations result:

- Incident Log records identify the pre-allocated dumps. Thus, the same property information is shown for each incident.
- Send Data action does not locate the dump data set because the name is unknown to the Incident Log task. The system, however, continues to process the log snapshots.

To continue using pre-allocated dump data sets, your installation can use an IBM-supplied JCL step to rename the dump data set in the sysplex dump directory, to allow z/OSMF to locate the correct data set. For information, see ["Ensuring that dump data set names are correct" on page 193.](#)

Some installations use automatic dump data set allocation, but then, subsequently, copy the dump data sets to another volume (to preserve space in the SMS DASD set). If the copied data set has the same name as the original dump data set, and the data set is cataloged, the Incident Log "Send data" action will locate the copied dump data sets. However, if the copied dump data set has a different name, use the IBM-supplied JCL step to rename the dump data set in the sysplex dump directory, so that the Incident Log task will locate it.

Configuring dump analysis and elimination

To avoid capturing duplicate problems in the Incident Log task display, ensure that dump analysis and elimination (DAE) is running on the z/OS host system. If your installation has not already configured DAE, this topic summarizes the steps for doing so.

IBM recommends that you enable DAE to suppress SVC dumps with duplicate symptoms for all of the systems in the sysplex (or all systems that you want the Incident Log task to represent). Doing so ensures that the Incident Log task displays only the initial instance of a dump-related incident. If necessary, you can use the *Allow next dump* action on the Incident Log page to allow the system to take and report the next dump that occurs for the same symptoms. You might use this option, for example, after you apply a fix for the problem. The *Allow next dump* action allows you to collect diagnostic data for the next new occurrence of the same problem.

To configure DAE processing for Incident Log processing, create a pair of ADYSETxx parmlib members with the appropriate options specified. Use one member to start DAE processing and the other member to stop DAE processing.

Consider using the following steps:

1. Create an ADYSETxx member for starting DAE. To do so, copy the IBM-supplied ADYSET00 member in SYS1.PARMLIB to a new member, for example, ADYSETAA. Do not modify the IBM-supplied member itself.
2. Create an ADYSETxx member for stopping DAE. To do so, copy the IBM-supplied ADYSET01 member in SYS1.PARMLIB to a new member, for example, ADYSETBB. Again, do not modify the IBM-supplied member itself.
3. Edit the new members, as follows:
 - In the DAE start-up member, specify the option SUPPRESSALL on the SVCDUMP parameter to suppress duplicate SVC dumps. Also, include the options SHARE, DSN and GLOBAL to use DAE in a sysplex-wide scope. For example:

```
DAE=START,RECORDS(400),  
SVCDUMP(MATCH,SUPPRESSALL,UPDATE,NOTIFY(3,30)),  
SYSMDUMP(MATCH,UPDATE),  
SHARE(DSN,OPTIONS),DSN(SYS1.DAESH2) GLOBAL(DSN,OPTIONS)
```

In this example, DSN specifies a cataloged data set SYS1.DAESH2 that resides on a DASD volume with shared access to all of the systems in the sysplex.

- In the DAE shut-down member, include the option GLOBALSTOP on the DAE= parameter. For example:

```
DAE=STOP,GLOBALSTOP
```

4. Ensure that the active IKJTSOxx parmlib member includes the program name ADYOPCMD in the AUTHCMD NAMES section. For information, see the topic on accessing the DAE data set in [z/OS MVS Diagnosis: Tools and Service Aids](#).
5. To start DAE processing, enter the MVS command **SET DAE=xx** from the operator console, where xx is the suffix of the DAE start-up member. Enter the command for each system in the sysplex, for example, by using the **ROUTE** command to direct the **SET DAE=xx** command to the other systems:

```
R0 *ALL,SET DAE=xx
```

To ensure that DAE processing is started automatically at IPL-time, include this command in the COMMNDxx parmlib member for the affected systems. If you choose to defer this step, you will need to manually start DAE on each system after each IPL.

6. Thereafter, for the IPLed systems in the sysplex, starting or stopping DAE on any one system will result in the other participating systems automatically starting or stopping DAE processing with the same options.

For more information about how to set up DAE, see [z/OS MVS Diagnosis: Tools and Service Aids](#). For more information about the IBM-supplied ADYSETxx parmlib members, see [z/OS MVS Initialization and Tuning Reference](#).

Creating the sysplex dump directory

The sysplex dump directory is a shared VSAM data set that contains information about SVC dumps that are taken on each of the systems in the sysplex. As each SVC dump is written to a data set, an entry is added by the dumping services address space (DUMPSRV) to the sysplex dump directory to store information like dump data set name, dump title, and symptom string.

The Incident Log task uses the sysplex dump directory as the repository for information about incidents that occur in the sysplex. If your installation does not already have a sysplex dump directory, this topic describes the steps for creating one.

How to check if this step is done

A sysplex dump directory might already exist for your system. This data set is defined through the SYSDDIR statement, which is typically specified in the parmlib member BLSCUSER. An example of the SYSDDIR statement follows:

```
SYSDDIR SYS1.DDIR ENV(ESAME)
```

IBM recommends that you define the SYSDDIR statement in member BLSCUSER. Alternatively, your installation might have specified this statement in member BLSCECT or BLSCECTX, or another member.

If you locate the SYSDDIR statement, verify that the specified sysplex dump directory data set exists, and is accessible to all of the systems in the sysplex (or all of the systems that you want the Incident Log task to represent).

Otherwise, you must create the sysplex dump directory, as described in the section that follows.

Steps for creating the sysplex dump directory

To create the sysplex dump directory, follow these steps:

1. Run the BLSCDDIR CLIST, which resides in system data set SYS1.SBLSCLI0(BLSCDDIR). For example:

```
EXEC 'SYS1.SBLSCLI0(BLSCDDIR) '  
'DSNAME(SYS1.DDIR) VOLUME(volser) RECORDS(15000) '
```

Where:

- DSNAME specifies the data set name for the sysplex dump directory. As supplied by IBM, the CLIST specifies the name, SYS1.DDIR.
- VOLUME specifies the DASD volume. To allow the Incident Log task (running on one system in the sysplex) to deliver a sysplex view of SVC dumps that are taken, select a volume with shared access to all of the systems in the sysplex (or all systems that you want the Incident Log task to represent).
- RECORDS specifies the data set size in records. The Incident Log task requires a sysplex dump directory data set with at least 15,000 records, which is about 60 cylinders. Approximately 50 directory entries are used for each incident and more are used for multi-system dumps.

The CLIST creates SYS1.DDIR as a VSAM data set with SHAREOPTIONS(1,3).

This data set must be cataloged on the current system and any other backup systems that are running the CIM server to allow for access by the Incident Log task.

2. Specify the dump directory name on the SYSDDIR statement in member BLSCUSER. Alternatively, your installation might use another member, such as BLSCECT or BLSCECTX.
3. Recycle the DUMPSRV address space through the command **CANCEL DUMPSRV**. The DUMPSRV address space restarts automatically. This action registers the dump directory name with the DUMPSRV address space.
4. Start BLSJPRMI through the command **START BLSJPRMI**. This action registers the dump directory name to IPCS.

For more information about the BLSCDDIR CLIST, see [z/OS MVS IPCS User's Guide](#).

Considerations for using a sysplex dump directory

Observe the following considerations:

- The sysplex dump directory (SYS1.DDIR, by default) is a shared VSAM data set serialized with an exclusive ENQ on the data set. This ENQ is used only by:
 - DUMPSRV address space, when it writes an entry to the directory for a new SVC dump
 - CEA address space, when it reads or updates the dump directory for Incident Log requests.
- The sysplex dump directory is different from the IPCS user local dump directory. A local directory is created for each IPCS user to store detailed data that is related to the IPCS session. The sysplex dump directory is used only to save name and symptom data for all SVC dumps taken, and must not be used as an IPCS user local dump directory.
- **Do not access the sysplex dump directory from an IPCS user.** Instead, use a batch job to access the directory.
- If new entries are not being added to the Incident Log task, or if requests are not being satisfied, check for contention on the sysplex dump directory by using the command **D GRS,C**. Verify that no IPCS user is accessing the sysplex dump directory.

Establishing a larger sysplex dump directory

Over time, your sysplex dump directory might become full with the dumps that you save. To create more space for dumps, you can delete old dumps from the directory. However, if you must retain the saved dumps, you can instead migrate your existing dumps to a larger sysplex dump directory.

To establish a larger sysplex dump directory, follow these steps:

1. Create a sysplex dump directory data set through the BLSCDDIR CLIST, for example:

```
EXEC 'SYS1.SBLSCLI0(BLSCDDIR) '
'DSNAME(new.DDIR) VOLUME(volser) RECORDS(25000) '
```

If your existing dump directory was created with the default size of 15000 records, you might want to specify a larger size. Approximately 50 directory entries are used for each incident and more are used for multi-system dumps.

2. Use the command **IPCS COPYDDIR** to copy the old directory entries to the new directory data set, as follows:

```
COPYDDIR INDSNAME(SYS1.DDIR) DSNAME(new.DDIR)
```

3. Update BLSCUSER with the new dump directory name, but make note of the old dump directory name.
4. Recycle the DUMPSRV address space (CANCEL DUMPSRV; it restarts automatically). This action registers the new dump directory name to DUMPSRV.
5. Run BLSJPRMI (START BLSJPRMI). This action updates the in-storage copy of the dump directory name.

Your new sysplex dump directory now contains the old dumps and can be used to store new dumps.

Ensuring that System REXX is set up and active

For full functionality, the Incident Log task requires that the System REXX (SYSREXX) component be set-up and active on your z/OS system.

This topic contains the following information:

- [“Ensuring that System REXX is set-up properly” on page 192](#)
- [“Ensuring that System REXX is active” on page 192](#)
- [“Starting the SYSREXX address space” on page 192](#)

Ensuring that System REXX is set-up properly

Observe the following considerations regarding System REXX set-up:

- Ensure that you have an AXRnn JCL member in PROCLIB, similar to the AXRnn member in SYS1.IBM.PROCLIB.
- If you have an AXRnn member in SYS1.IBM.PROCLIB, ensure that SYS1.IBM.PROCLIB is in the MSTJCLxx IEFPSI DD concatenation.
- Ensure that the user ID specified for AXRUSER in AXRnn has the correct permissions.

For more information about setting up System REXX, see the following documents:

- [z/OS MVS Programming: Authorized Assembler Services Guide.](#)
- [z/OS MVS Initialization and Tuning Reference.](#)

Ensuring that System REXX is active

SYSREXX is started automatically during IPL. If your installation has stopped SYSREXX, it is recommended that you restart it.

If you choose to defer this step, the Incident Log task runs with limited functionality.

How to check if this step is done

If the AXR address space is active on the z/OS system, the System REXX component is active. To determine whether the AXR address space is active, enter the following command:

```
D A,AXR
```

Figure 32 on page 192 shows the expected result:

```
IEE115I 15.34.46 2010.132 ACTIVITY 111
  JOBS      M/S      TS USERS      SYSAS      INITS      ACTIVE/MAX VTAM      OAS
  00018      00040      00002      00043      00246      00002/03500      00043
  AXR        AXR        IEFPROC    NSWPR*    A=0019    PER=YES    SMC=000
                                PGN=N/A    DMN=N/A    AFF=NONE
                                CT=000.088S  ET=45.34.45
                                WKL=STC_WLD  SCL=STCLOW  P=1
                                RGP=N/A      SRVR=NO    QSC=NO
                                ADDR SPACE  ASTE=05A34640
                                DSPNAME=AXRTRDSP  ASTE=1002D880
                                DSPNAME=AXRRXENV  ASTE=06BED200
                                DSPNAME=AXRREQCP  ASTE=06029180
```

Figure 32. Expected result from the **D A,AXR** command

Starting the SYSREXX address space

To start the SYSREXX component, enter the following command from the operator console:

```
START AXRPSTRT
```

For information about configuring System REXX on your system, see https://www.ibm.com/docs/en/SSLTBW_3.1.0/pdf/e0zpdz00.pdf.

Ensuring that dump data set names are correct

If your installation has an automation program that copies an SVC dump data set to a different location using a different data set name, you must ensure that the dump data set name is changed accordingly in the sysplex dump directory. This action is necessary to allow the Incident Log task to locate the correct dump.

In your automation program, add a step to rename the dump data set in the sysplex dump directory; Figure 33 on page 193 provides an example of the JCL you can use.

```
//IPCS EXEC PGM=IKJEFT01,DYNAMNBR=20,REGION=1500K
//IPCSDDIR DD DSN=SYS1.DDIR,DISP=(SHR)
//SYSTSPRT DD SYSOUT=*
//SYSTSIN DD *
IPCS
ALTER DSNAME('OldDump') NEWNAME(DSNAME('NewDump'))
END
/*
```

Figure 33. Sample JCL to rename SVC dumps in the sysplex dump directory

In the example:

- Modify the keyword DSN=SYS1.DDIR to specify the name of your sysplex dump directory (the default name is SYS1.DDIR)
- Modify the values *OldDump* and *NewDump* to use the correct dump data set names.

Chapter 29. Configure z/OS Data Gatherer: SMF REST Services

z/OS Data Gatherer is the strategic IBM performance measurement tool in a z/OS host environment. This is the information that you need to set up the SMF REST Services for z/OS Data Gatherer.

For an overview of accessing SMF data by using REST services, see [Accessing data using REST services in z/OS Data Gatherer User's Guide](#) and [Accessing data using the z/OS Data Gatherer REST services in z/OS Data Gatherer Programmer's Guide](#).

For more information about setting up z/OS Data Gatherer SMF REST Services, see [Setting up z/OS Data Gatherer SMF REST services in z/OS Data Gatherer User's Guide](#).

z/OS Data Gatherer: SMF REST Services should be run on a separate z/OSMF server instance

Complex z/OSMF plugin applications like z/OS Data Gatherer SMF REST services should be deployed on separate z/OSMF server instances. If a separate z/OSMF server instance is already configured for any other z/OS Data Gatherer REST Services, then z/OS Data Gatherer SMF REST services should also be run on that z/OSMF server instance. This ensures that they do not impact or get impacted by other z/OSMF plugins. Plugins that have more computational complexity require more resources like CPU, memory, and network bandwidth.

Running z/OSMF plugins on separate z/OSMF servers also gives system programmers the flexibility to configure different z/OSMF plug ins on different z/OSMF servers. This enables system programmers to allocate more resources to the z/OSMF Servers that are running heavy workloads.

For more information about configuring a separate z/OSMF server instance, see [Chapter 35, “Configure a separate z/OSMF server instance,” on page 211](#).

Dependencies on other z/OSMF services

If you are configuring a separate z/OSMF server instance to run z/OS Data Gatherer SMF REST services, the following z/OSMF services should be configured in this new z/OSMF server instance:

- z/OSMF Settings service. For more information, see [Chapter 9, “Configure the z/OSMF settings service,” on page 67](#).

Enabling the service

After you configure a separate z/OSMF server instance and set up z/OS Data Gatherer SMF REST services, you can enable z/OS Data Gatherer SMF REST services. For more information, see [Chapter 35, “Configure a separate z/OSMF server instance,” on page 211](#) and [Setting up z/OS Data Gatherer SMF REST services in z/OS Data Gatherer User's Guide](#).

Enable the z/OS Data Gatherer SMF REST services by following these steps:

1. Start the separate z/OSMF server instance.
2. Open the z/OSMF General Settings task.
3. **Enable** z/OS Data Gatherer SMF REST services that are listed under **Optional Services**.
4. Restart the z/OSMF server.

Or follow these steps:

1. Copy the zosmf.json file from /usr/lpp/zosmf/samples/ to <USERDIR>/configuration/settings/zosmf/.
2. Start the separate z/OSMF server instance.

Note:

- 1.<USERDIR> should be the mount point that you mounted your new z/OSMF user file system. For more information, see [Chapter 35, “Configure a separate z/OSMF server instance,”](#) on page 211.
- 2.The directory <USERDIR>/configuration/settings/zosmf/ may not exist if this is your first time configuring the separate z/OSMF server. If so, create the directory, then copy the zosmf.json file. The z/OSMF server ID should have the READ/WRITE permission of the zosmf.json file.

Chapter 30. Configure z/OS Data Gatherer: Monitor II REST Services

z/OS Data Gatherer is the strategic IBM performance measurement tool in a z/OS host environment. This is the information that you need to set up the Monitor II REST Services for z/OS Data Gatherer.

For an overview of accessing Monitor II data by using REST services, see [Accessing data using REST services](#) in *z/OS Data Gatherer User's Guide* and [Accessing data using the z/OS Data Gatherer REST services](#) in *z/OS Data Gatherer Programmer's Guide*.

For more information about setting up z/OS Data Gatherer Monitor II REST Services, see [Setting up z/OS Data Gatherer SMF REST services](#) in *z/OS Data Gatherer User's Guide*.

z/OS Data Gatherer Monitor II REST Services should be run on a separate z/OSMF server instance. If a separate z/OSMF server instance is already configured for any other z/OS Data Gatherer REST Services, then z/OS Data Gatherer Monitor II REST Services should also be run on that z/OSMF server instance. This ensures that they do not impact or get impacted by other z/OSMF plugins. These plugins have more computational complexity and require more resources like CPU, memory, and network bandwidth.

Running z/OSMF plugins on separate z/OSMF servers also gives system programmers the flexibility to configure different z/OSMF plugins on different z/OSMF servers. This enables system programmers to allocate more resources to the z/OSMF Servers that are running heavy workloads.

For more information about configuring a separate z/OSMF server instance, see [Chapter 35, “Configure a separate z/OSMF server instance,”](#) on page 211.

Dependencies on other z/OSMF services

If you are configuring a separate z/OSMF server instance to run z/OS Data Gatherer: Monitor II REST Services, the following z/OSMF services should be configured in this new z/OSMF server instance:

- z/OSMF Settings service. For more information, see [Chapter 9, “Configure the z/OSMF settings service,”](#) on page 67.

Enabling the service

After you configure a separate z/OSMF server instance and set up z/OS Data Gatherer Monitor II REST services, you can enable z/OS Data Gatherer Monitor II REST services. For more information, see [Chapter 35, “Configure a separate z/OSMF server instance,”](#) on page 211 and [Setting up z/OS Data Gatherer SMF REST services](#) in *z/OS Data Gatherer User's Guide*.

Enable the z/OS Data Gatherer Monitor II REST services by following these steps:

1. Start the separate z/OSMF server instance.
2. Open the z/OSMF General Settings task.
3. **Enable** the z/OS Data Gatherer Monitor II REST services that are listed under **Optional Services**.
4. Restart the z/OSMF server.

Chapter 31. Configure z/OS Data Gatherer: Monitor III REST Services

z/OS Data Gatherer is the strategic IBM performance measurement tool in a z/OS host environment. This is the information that you need to set up the Monitor III REST Services for z/OS Data Gatherer.

For an overview of accessing Monitor III data by using REST services, see [Accessing data using REST services](#) in *z/OS Data Gatherer User's Guide* and [Accessing data using the z/OS Data Gatherer REST services](#) in *z/OS Data Gatherer Programmer's Guide*.

For more information about setting up z/OS Data Gatherer Monitor III REST Services, see [Setting up z/OS Data Gatherer SMF REST services](#) in *z/OS Data Gatherer User's Guide*.

z/OS Data Gatherer Monitor III REST Services should be run on a separate z/OSMF server instance. If a separate z/OSMF server instance is already configured for any other z/OS Data Gatherer REST Services, then z/OS Data Gatherer Monitor III REST Services should also be run on that z/OSMF server instance. This ensures that they do not impact or get impacted by other z/OSMF plugins. These plugins have more computational complexity and require more resources like CPU, memory, and network bandwidth.

Running z/OSMF plugins on separate z/OSMF servers also gives system programmers the flexibility to configure different z/OSMF plugins on different z/OSMF servers. This enables system programmers to allocate more resources to the z/OSMF Servers that are running heavy workloads.

For more information about configuring a separate z/OSMF server instance, see [Chapter 35, “Configure a separate z/OSMF server instance,”](#) on page 211.

Dependencies on other z/OSMF services

If you are configuring a separate z/OSMF server instance to run z/OS Data Gatherer: Monitor III REST Services, the following z/OSMF services should be configured in this new z/OSMF server instance:

- z/OSMF Settings service. For more information, see [Chapter 9, “Configure the z/OSMF settings service,”](#) on page 67.

Enabling the service

Before you enable z/OS Data Gatherer Monitor III REST services, you must configure a separate z/OSMF server instance. For more information, see [Chapter 35, “Configure a separate z/OSMF server instance,”](#) on page 211 and [Setting up z/OS Data Gatherer SMF REST services](#) in *z/OS Data Gatherer User's Guide*.

Enable the z/OS Data Gatherer Monitor III REST services by following these steps:

1. Start the separate z/OSMF server instance.
2. Open the z/OSMF General Settings task.
3. **Enable** the z/OS Data Gatherer Monitor III REST services that are listed under **Optional Services**.
4. Restart the z/OSMF server.

If you have already configured a separate z/OSMF server for any z/OS Data Gatherer REST services, you should enable z/OS Data Gatherer Monitor III Rest services on the same z/OSMF server instance by following these steps:

1. Stop the separate z/OSMF server instance which has z/OS Data Gatherer REST services configured.
2. Go to the z/OSMF user directory associated with the separate z/OSMF server, for example `/global/zosmf2`.
3. Switch to the `configuration/settings/zosmf` directory.
4. Edit the `zosmf.json` file.

Add a new line under “Services” to enable z/OS Data Gatherer Monitor III Rest services:

"IZU_ZM3_CONFIGURE": "Y"

5. Save the update to the `zosmf.json` file.
6. Restart the z/OSMF server.

Chapter 32. Configure the REST Hub service

To use the REST Hub service, you must configure it as described in this topic. This includes the Runtime Diagnostics REST interface service.

Description

The Runtime Diagnostics (RTD) services are an application programming interface (API) that are implemented through industry standard Representational State Transfer (REST) services. The RTD services provide a programming interface to get runtime diagnostic data about a specified set of address spaces on a defined system.

Dependencies on other z/OSMF services

None.

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZURHSEC in SYS1.SAMPLIB.

Do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZURHSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

None.

Optional extensions to this service

None.

Enabling the service

To enable Runtime Diagnostics, follow these steps:

1. Open the z/OSMF **General Settings** task.
2. Enable Rest Hub under **Optional Services**.
3. Restart the z/OSMF server.

To enable Runtime Diagnostics when z/OSMF is not started, or the **General Settings** page is not available, follow these steps:

1. Go to the UNIX Systems Services directory `/global/zosmf/configuration/settings/zosmf/`.
2. Edit `zosmf.json`.

If the JSON file is empty, paste in the following contents:

```
{
  "services": {
    "IZU_RH_CONFIGURE": "Y"
  }
}
```

```
}  
}
```

If the file is not empty, look for IZU_RH_CONFIGURE.

- If IZU_RH_CONFIGURE is found, change the value to "Y".
- If IZU_RH_CONFIGURE is not found, the default value of "N" is assigned. To enable REST Hub, add a new line in the services section with IZU_RH_CONFIGURE set to "Y" as shown in the previous example.

3. Save `zosmf.json` and exit.

4. Restart the z/OSMF server.

Note: Replace `/global/zosmf` with your data directory path if you customized your z/OSMF user directory. The user must be connected to the IZUADMIN group to access the contents in directory `/global/zosmf/configuration`.

Removing the service

After REST Hub is enabled, you might later decide to remove it. Follow these steps to remove the service:

1. Open the z/OSMF General Settings task.
2. Disable REST Hub listed under **Optional Services**.
3. Restart the z/OSMF server.

To remove REST Hub when z/OSMF is not started, or the **General Settings** page is not available, follow these steps:

1. Go to the UNIX Systems Services directory `/global/zosmf/configuration/settings/zosmf/`.
2. Edit `zosmf.json`. Locate IZU_RH_CONFIGURE and change the value of IZU_RH_CONFIGURE to "N". If IZU_RH_CONFIGURE is not found, the default value "N" is assigned. No further action is required.
3. Save `zosmf.json` and exit.
4. Restart the z/OSMF server.

Note: Replace `/global/zosmf` with your data directory path if you customized your z/OSMF user directory. The user must be connected to the IZUADMIN group to access the contents in directory `/global/zosmf/configuration`.

Chapter 33. Configure the AI Control Interface

To use the AI Control Interface, you must configure it as described in this topic.

Description

The AI Control Interface for IBM z/OS automates and enhances z/OS system management tasks with prebuilt AI models.

With AI Control Interface, you can perform management actions to train models, update model settings and view the status of AI enablement.

Dependencies on other z/OSMF services

None.

Security setup

IBM provides both a sample workflow definition file and the sample security job IZUAISEC to assist you with performing the security setup. You can use either method to proceed with the setup process.

The sample workflow definition file is located in:

```
<product_dir>/workflow/izu.aici.config.xml
```

where `<product_dir>` is the z/OSMF product directory. By default, this is `/usr/lpp/zosmf`.

Note: If you plan to use the AI Framework for z/OS Configuration Workflow to set up the AI Control Interface, you do not need to run `izu.aici.config.xml` separately. For more information, see [Configuring the AI Framework in *Infusing AI into IBM z/OS*](#).

The z/OSMF Workflows service is required to run the workflow. For more information, see [Chapter 7, “Configure the z/OSMF Workflows task,” on page 63](#).

To use the sample definition workflow for setup, do the following:

1. Import the workflow definition file into the Workflows task.
2. Tailor the variable values according to the setup of your system. This is used to populate the workflow.
3. Run the workflow to setup the plugin.
4. Verify the results by opening the AI Control Interface plugin.

Alternatively, to use the sample security job IZUAISEC in SYS1.SAMPLIB for setup, do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUAISEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

For the AI Control Interface to communicate with the AI provider, the AI Base Component for IBM z/OS must be configured and started in full-function mode. For more information, see [z/OS MVS System Commands](#).

To fully configure the AI Framework for z/OS, you must run the AI Framework Configuration Workflow as well as an AI-enhanced use case for z/OS. For more information, see [Configuring the AI Framework in *Infusing AI into IBM z/OS*](#).

Optional extensions to this service

None.

Chapter 34. Configure the Parmlib Management service

To use the z/OSMF Parmlib Management service, configure it as described in this topic.

Description

Parmlib is one of the most important configuration file formats on the z/OS platform. Parmlib management requires a deep learning curve and repetitive manual effort, which can lead to creating errors. z/OSMF Parmlib Management service provides a solution to simplify and modernize the user experience of managing Parmlib.

The following Parmlib types are currently supported:

1. ALLOCxx
2. AUTORxx
3. AXRxx
4. BPXPRMxx
5. CEAPRMxx
6. CEEPARM
7. CLOCKxx
8. COMMNDxx
9. CONSOLxx
10. COUPLExx
11. CUNUNIxx
12. DEVSUPxx
13. DIAGxx
14. EXITxx
15. FXEPRMxx
16. GRSCNFxx
17. GRSRNLxx
18. IEAFIXxx
19. IEALPAxx
20. IEAPAKxx
21. IEASVCxx
22. IEASYMxx
23. IEASYSxx
24. IEFOPZxx
25. IEFSSNxx
26. IFAPRDxx
27. IGDSMSxx
28. IGGCATxx
29. IKJTSoxx
30. IQPPRMxx
31. IRRPRMxx

- 32. IXGCNFxx
- 33. IZUPRMxx
- 34. LOADxx
- 35. LPALSTxx
- 36. PROGxx
- 37. SCHEDxx
- 38. VATLSTxx

Dependencies on other z/OSMF services

Common event adapter (CEA). For more information, see [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#).

TSO/E address space services. For more information, see [Chapter 13, “Configure the TSO/E address space services,” on page 77](#).

Security setup

To assist you with performing the security setup, IBM provides the sample security job IZUPMSEC in SYS1.SAMPLIB. For a summary of the required profile authorizations, see [“Resource authorizations for the Parmlib Management service” on page 408](#).

To run the IZUPMSEC job, do the following:

1. Make a copy of this job.
2. Review and edit the job, if necessary.
3. Submit the job as a batch job on your z/OS system.

Ensure that the IZUPMSEC job completes with return code 0000. To verify, check the results of the job execution in the job log, for example, by using SDSF.

Host system customization

[“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#).

Optional extensions to this service

None.

Enabling the service

Parmlib is disabled by default. To enable the Parmlib Management service, follow these steps:

1. Log in to z/OSMF and open the **General Settings** task.
2. Enable **Parmlib Management** under **Optional Services**.
3. Restart the z/OSMF server.

To enable Parmlib Management when z/OSMF is not started, or the **General Settings** task is not available, follow these steps:

1. Navigate to the UNIX System Services directory `/global/zosmf/configuration/settings/zosmf/`.
2. Edit `zosmf.json`.

If the file is empty, paste in the following contents:

```
{
  "services": {
    "IZU_PARMLIB_SERVICE": "Y"
  }
}
```

```
}  
}
```

If the file is not empty, locate IZU_PARMLIB_SERVICE.

- If IZU_PARMLIB_SERVICE is found, change the value to Y.
- If IZU_PARMLIB_SERVICE is not found, a default value of N is assigned. Add a new line in the services section with IZU_PARMLIB_SERVICE": "Y".

3. Save zosmf.json and exit.

4. Restart the z/OSMF server.

Note: Replace /global/zosmf with your data directory path if you customized the z/OSMF user directory. The user must be connected to the IZUADMIN group to access contents in the directory /global/zosmf/configuration.

Removing the service

Follow these steps if you decide to remove the service after it was enabled:

1. Log in to z/OSMF and open the **General Settings** task.
2. Disable **Parmlib Management** under **Optional Services**.
3. Restart the z/OSMF server.

To remove Parmlib Management when z/OSMF is not started, or the **General Settings** task is not available, follow these steps:

1. Navigate to the UNIX System Services directory /global/zosmf/configuration/settings/zosmf/.
2. Edit zosmf.json.

If the file is empty, paste in the following contents:

```
{  
  "services": {  
    "IZU_PARMLIB_SERVICE": "N"  
  }  
}
```

If the file is not empty, locate IZU_PARMLIB_SERVICE.

- If IZU_PARMLIB_SERVICE is found, change the value to N.
- If IZU_PARMLIB_SERVICE is not found, a default value of N is assigned. No further action is required.

3. Save zosmf.json and exit.

4. Restart the z/OSMF server.

Note: Replace /global/zosmf with your data directory path if you customized the z/OSMF user directory. The user must be connected to the IZUADMIN group to access contents in the directory /global/zosmf/configuration.

Part 5. Advanced configurations

You can optionally perform additional tasks to enhance your z/OSMF configuration. System programmers and administrators are the most likely IT personnel to participate in these activities.

Beyond creating the nucleus and adding core and optional services, you can extend your z/OSMF configuration in the following ways:

- [Chapter 36, “Autostart concepts in z/OSMF,” on page 213](#)
- [Chapter 37, “Configuring a primary z/OSMF for communicating with secondary instances,” on page 221](#)
- [Chapter 38, “Enabling single sign-on between z/OSMF instances,” on page 225](#)
- [Chapter 39, “z/OSMF in a DevOps context,” on page 227](#)
- [Chapter 40, “Configuring z/OSMF to provide redundancy for subset of z/OSMF function,” on page 231](#)
- [Chapter 41, “Configuring the z/OSMF server certificate and key ring,” on page 233](#)
- [Chapter 44, “Considerations for using ICSF services,” on page 247](#)
- [Chapter 45, “Considerations for using AT-TLS,” on page 249](#)
- [Chapter 46, “Linking z/OSMF tasks and external applications,” on page 251](#)
- [Chapter 47, “Configuring the CIM server for your system,” on page 253](#)
- [Chapter 48, “Configuring your system for asynchronous job notifications,” on page 255](#)
- [Chapter 49, “Adding links to z/OSMF,” on page 257](#)
- [Chapter 50, “Deleting incidents and diagnostic data,” on page 261](#)
- [Chapter 51, “Setting the KCINDEX parameter,” on page 265](#)

Chapter 35. Configure a separate z/OSMF server instance

To set up a separate z/OSMF server instance, you must configure it as described in this topic. You can then deploy an application, like a z/OSMF plugin, on this separate server so that it will not be impacted or impact other z/OSMF plugins. This is especially useful if your application is more complex and requires more resources such as CPU, memory, and network bandwidth.

Setup a separate z/OSMF server by reusing existing z/OSMF security setup

If you already have z/OSMF Server running on your system, this means your z/OSMF Configuration has already been set up. In this case, follow these steps to configure and start a separate z/OSMF server.

1. Run job IZUMKFS to create the z/OSMF user file system.
 - a. This step is the same as [“Step 2: Run job IZUMKFS to create the z/OSMF user file system”](#) on page 26 except you need to modify IZUMKFS to ensure that the new file system is mounted to the other mount point like /var/zosmf, not /global/zosmf. This is because /global/zosmf is supposed to be the mount point of your primary z/OSMF server’s user file system, which is already being used.
2. Make a copy of your current z/OSMF server’s PARMLIB and change the port number, and remove the unnecessary plugins in the PARMLIB.
 - a. For example, if the z/OSMF server you are currently running is using IZUPRM00, you can make a copy of IZUPRM00 and name it IZUPRM01. You can then change the HTTP_SSL_PORT statement in IZUPRM01 to ensure that the new separate z/OSMF server uses a port that is not yet being used.

For more information, see [“IZUPRMxx reference information”](#) on page 35.

3. Start this separate z/OSMF server.
 - a. This step is the same as [“Step 4: Start the z/OSMF server”](#) on page 28, except you don’t need to start the angel process. This is because the new separate z/OSMF server uses the same angel process that is running from your current running z/OSMF server.
 - b. Start this separate z/OSMF server with the following command: `S IZUSVR1 JOBNAME=<job name>,IZUPRM=<parmlib suffix>,USERDIR=<user directory>, SERVER=STANDALONE`
JOBNAME should be a different name than your current running z/OSMF server job name, for example, IZUSVR2.
IZUPRM should be the new PARMLIB created in Step 2, for example, 01.
USERDIR should be the mount point that you mounted your new z/OSMF user file system that is created in Step 1.
This z/OSMF server should be a stand-alone z/OSMF server. This means the separate server would not be a server of your autostart group. For more information about z/OSMF Autostart concepts, see [Chapter 36, “Autostart concepts in z/OSMF,”](#) on page 213.
For more information about these parameters, see [“IZUSVR reference information”](#) on page 44.
 - c. Here is an example command to start a separate server: `S IZUSVR1 jobname=izusvr2,izuprm=01,userdir='/var/zosmf', SERVER=STANDALONE`

4. Log in to z/OSMF.

- a. This is the same step as [“Step 5: Log in to z/OSMF”](#) on page 29, except you use the new port number.

When you log in to z/OSMF, the desktop interface is displayed. Only the options you are authorized to use are displayed.

5. Enable z/OSMF service with these steps:
 - a. Open the z/OSMF General Settings task.
 - b. "Enable" the services that you are interested in.
 - c. Restart the z/OSMF server.

Setup a separate z/OSMF server with new security configuration

In this case, it would be the same process as setting up your z/OSMF system for the first time. For more information about setting up a z/OSMF server for your specific needs, see [“Faster set-up with a z/OSMF Lite configuration” on page 9.](#)

Chapter 36. Autostart concepts in z/OSMF

You can configure z/OSMF so that it is started when you IPL your z/OS system. This behavior, which is referred to as *z/OSMF autostart*, means that z/OSMF is available for use as soon as the system is up. A z/OSMF realm refers to both the Angel and Server pair.

To make the best use of the z/OSMF autostart capability, plan to deploy one or more z/OSMF realms in your environment. Generally, having one z/OSMF realm active in a sysplex or monoplex is sufficient, but you might choose to have more, based on your workload requirements. The goal is to ensure that at least one z/OSMF realm is always active in your environment.

For a monoplex, little or no planning is needed. The z/OSMF realm is started when you IPL the system.

For a sysplex, more planning is required. You can choose to have one z/OSMF realm autostart on a particular system and be used by the other systems in the sysplex. Or, you can select a subset of systems, or several subsets, and associate each subset with a specific z/OSMF realm within an autostart group.

If you do not want to use the autostart capability, some planning is needed to either remove it or to disable autostart, even in a monoplex. For more information, see [“Scenario 4: The z/OSMF realm is not autostarted on any system”](#) on page 216.

The set of systems that is served by an autostarted server is known as the *autostart group*. z/OSMF includes one autostart group by default. To have more z/OSMF servers autostarted in a sysplex, you must associate each server—and the systems it serves—with a unique autostart group name.

In your planning, you must decide:

- What the autostart groups will be in your sysplex
- Which systems will autostart a z/OSMF server
- Which systems will share the use of the autostarted server; these systems must be defined to the same autostart group.

Only one z/OSMF server can be active per autostart group in the sysplex. An autostarted z/OSMF server holds an enqueue on the z/OSMF data directory file system, and handles the z/OSMF requests from other systems that are connected to the same autostart group. Based on your planning, you can enable the desired number of z/OSMF autostart groups for your sysplex.

During IPL, the IZU=xx parameter in the IEASYSxx configuration is used to select which IZUPRMxx options are used at IPL. To create one or more autostart groups in z/OSMF, use the following statements in parmlib member IZUPRMxx in combination:

AUTOSTART(LOCAL|CONNECT)

Specifies the capability for autostarting the z/OSMF server on this system.

- AUTOSTART(LOCAL) indicates that the system is capable of autostarting a z/OSMF server.
- AUTOSTART(CONNECT) indicates that the system cannot autostart a z/OSMF server. The system will, instead, use the z/OSMF server on another system in the same autostart group.

By default, AUTOSTART is set to LOCAL.

AUTOSTART_GROUP(IZUDFLT|'nnnnnnnn')

Assigns a name to the autostart group. z/OSMF includes one AUTOSTART_GROUP name by default (called IZUDFLT). To associate a group of systems with a different autostart group, ensure that each system specifies the same value for AUTOSTART_GROUP.

By default, AUTOSTART_GROUP is set to IZUDFLT.

If one autostart group is sufficient for your sysplex, it is recommended that you allow each system to use the IZUDFLT autostart group.

The following scenarios are valid in a multi-system environment:

- [“Scenario 1: One z/OSMF server is autostarted for the entire sysplex”](#) on page 214

- “Scenario 2: Multiple z/OSMF servers and autostart groups per sysplex” on page 214
- “Scenario 3: Some systems belong to an autostart group, and other systems do not” on page 215
- “Scenario 4: The z/OSMF realm is not autostarted on any system” on page 216.

Scenario 1: One z/OSMF server is autostarted for the entire sysplex

In this scenario, the z/OSMF server is autostarted on one system in the sysplex. All systems are associated with the default autostart group, which is named IZUDFLT.

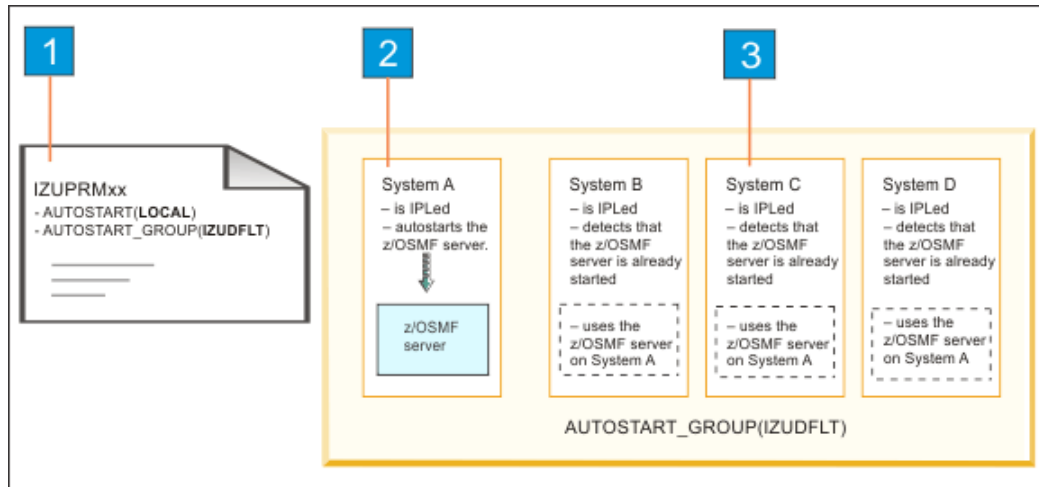


Figure 34. Scenario 1: One z/OSMF server is autostarted for the entire sysplex

In Figure 34 on page 214:

1. Each system uses the following default values for autostart:

```
AUTOSTART(LOCAL)
AUTOSTART_GROUP(IZUDFLT)
```

With these values set for all systems, the first one to complete IPL is the system on which the z/OSMF server is started.

2. System A is the first system to complete IPL in the sysplex. Its attempt to autostart the z/OSMF server is successful.
3. System B, C, and D complete IPL. These systems detect that an autostarted server is active on System A, so they do not attempt a server. Instead, they use the server on System A.

This scenario is enabled by default. If it is sufficient for your requirements, you can use the z/OSMF defaults. If you care *which* system in the sysplex autostarts the z/OSMF server, keep the default values for that system and change the AUTOSTART value to CONNECT for all other systems in the same autostart group.

Scenario 2: Multiple z/OSMF servers and autostart groups per sysplex

In this scenario, more than one z/OSMF server is to be autostarted in a sysplex. Suppose, for example, that you have a sysplex of four systems: A, B, C, and D. You plan to have System A autostart a server and share it with System B. Similarly, System C will autostart a server and share it with System D.

In this scenario, each server and the systems it serves are associated with an autostart group, as follows:

- System A and System B are associated with the autostart group IZUDFLT.
- System C and System D are associated with the autostart group ALTERNATE.

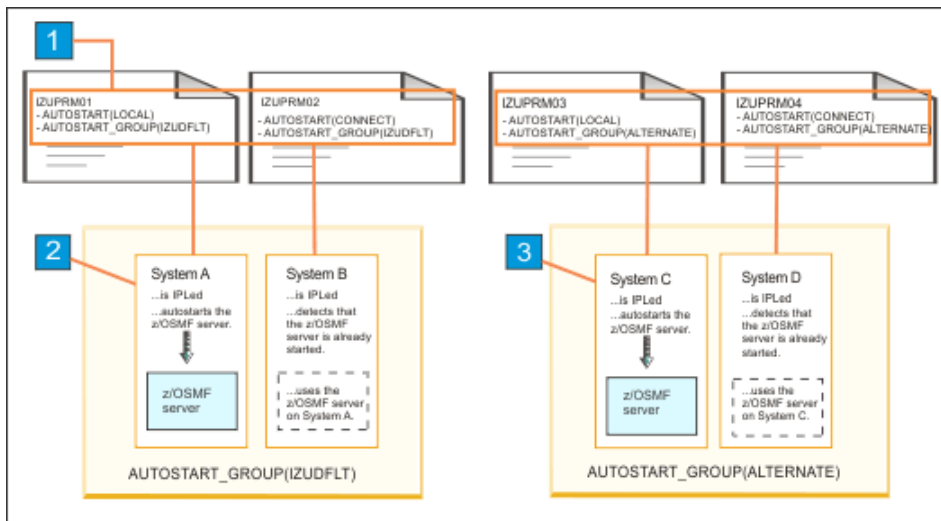


Figure 35. Scenario 2: Multiple z/OSMF servers and autostart groups per sysplex.

In Figure 35 on page 215:

1. Each system uses a different IZUPRMxx member with different settings for AUTOSTART and AUTOSTART_GROUP.
2. System A autostarts a z/OSMF server. System B uses the autostarted server on System A.
3. System C autostarts a z/OSMF server. System D uses the autostarted server on System C.

Scenario 3: Some systems belong to an autostart group, and other systems do not

In this scenario, some systems belong to an autostart group, and other systems do not. Suppose, for example, that you have a sysplex of four systems: System A, B, C, and D. In this sysplex, you plan to have System A autostart the z/OSMF server and share it with System B. System C and System D will not use an autostarted z/OSMF server. The z/OSMF server can be started on these systems manually, by using the **START** operator command with the name of the z/OSMF started procedure (IZUSVR1).

In this scenario:

- System A and System B are defined to autostart group IZUDFLT.
- System C and System D are not defined to an autostart group.

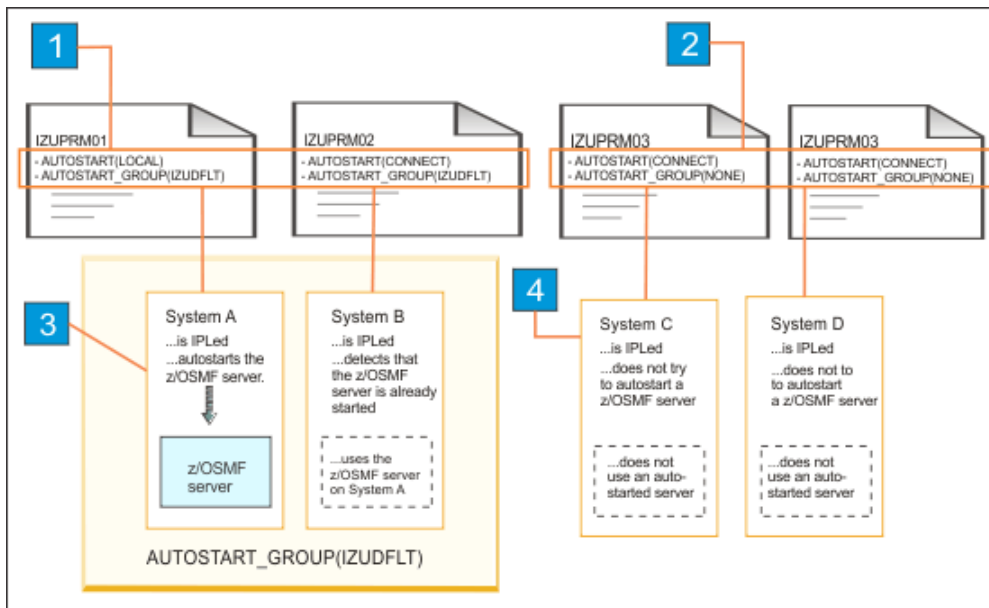


Figure 36. Scenario 3: One z/OSMF server is autostarted for a subset of systems in a sysplex.

In Figure 36 on page 216:

1. Systems A and B specify AUTOSTART_GROUP(IZUDFLT).
2. Systems C and D specify a non-functioning autostart group name, NONE.
3. System A autostarts a z/OSMF server. System B uses the autostarted server on System A.
4. Systems C and D do not use an autostarted z/OSMF.

Scenario 4: The z/OSMF realm is not autostarted on any system

In this scenario, no z/OSMF realms are started automatically during system IPL. That is, the autostart capability is disabled. Perhaps, you prefer to start the realm manually, with the **START** operator command, as done in previous releases.

To disable the autostarting of z/OSMF realms in a sysplex, do the following for each system in the sysplex:

- To prevent a z/OS system from autostarting the z/OSMF realm, ensure that the system uses a IZUPRMxx member that specifies AUTOSTART(CONNECT). This setting causes the system to connect to the autostart group that is specified on the AUTOSTART_GROUP statement, rather than autostarting its own realm.
- To prevent a z/OS system from connecting to an autostart group, specify the name of a group on the AUTOSTART_GROUP parameter that is not used by any autostart realm in the sysplex. For example, AUTOSTART_GROUP('NONE').
- Similarly, for each system for which you want to disable z/OSMF autostart, ensure that the AUTOSTART(CONNECT) and AUTOSTART_GROUP('NONE') settings are in effect.
- In your IZU= specifications, verify that the IZU= parameter identifies the suffixes of the IZUPRMxx members that contain the desired settings.

These actions must be taken if you want to disable the autostarting of z/OSMF realms. Otherwise, the default behavior for each system is to attempt to start the z/OSMF realm automatically during IPL.

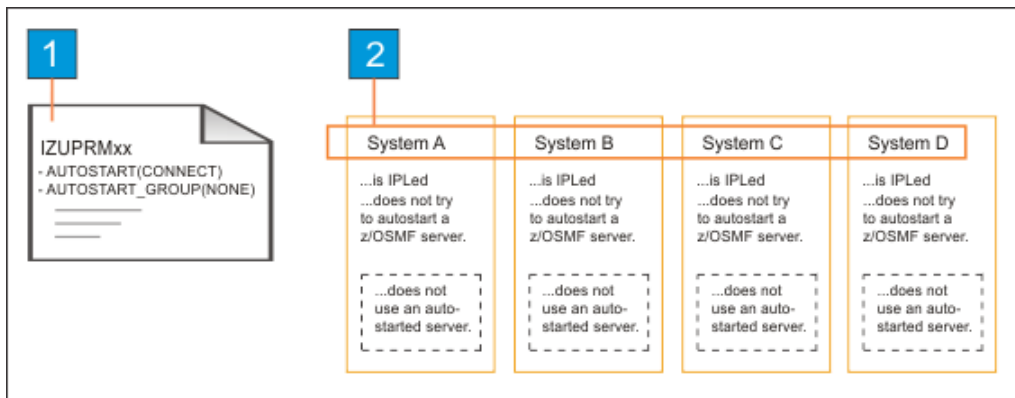


Figure 37. Scenario 4: No z/OSMF realms are started automatically.

In Figure 37 on page 217:

1. Each system uses the following values for autostart:

```
AUTOSTART(CONNECT)
AUTOSTART_GROUP(NONE)
```

With these values set for all systems, no system attempts to autostart an z/OSMF server.

2. Systems A, B, C, and D complete the IPL process. No z/OSMF servers are autostarted in the sysplex.
- The JES2 Email Delivery Services (EDS) function requires a z/OSMF server to be active in an AUTOSTART group that JES2 can access. Specifically, the z/OSMF server must be started with SERVER= 'AUTOSTART' in the IZUSVR1 started procedure, and JES2 must be running on a system that is included in the AUTOSTART_GROUP specification. Otherwise, if this setup is not done, JES2 cannot send e-mail messages to users who submit jobs.

The z/OSMF server is not required to be on the same system on which the JES2 EDS is used. However, you do need to ensure that the system from which you are using JES2 EDS is part of an z/OSMF AUTOSTART_GROUP in which there is an active server in that group. If so, JES2 automatically detects the presence of the z/OSMF server; you do not need to identify the location of the z/OSMF server to JES2.

For information about configuring JES2 EDS, see the topic [JES2 Email Delivery Services in z/OS JES2 Initialization and Tuning Guide](#).

- You can start the z/OSMF server manually on any system by using the **START** operator command with the name of the z/OSMF started procedure. By default, the procedure is IZUSVR1. For more information, see [“Stopping and starting z/OSMF manually”](#) on page 31.
- To change the AUTOSTART_GROUP name, issue the following command:

```
SETIZU AUTOSTART_GROUP=NEWVALUE
```

The new AUTOSTART_GROUP name is effective immediately. Make sure that the IZUPRMXX member is also updated so that z/OSMF will pick up the new value after you restart or reIPL.

- The z/OSMF autostart capability does not automatically restart a terminated server. If an autostarted server fails, you can resume z/OSMF operations by manually starting the server.
- Authorized programs can use the event notification facility (ENF) to determine whether the z/OSMF server is up or down. For more information, see [“Detecting whether the z/OSMF server is available”](#) on page 218.

Steps to enable or disable the autostart capability

The autostart capability requires that common event adapter (CEA) be configured on your system. For information, see [“Ensure that common event adapter \(CEA\) is configured and active”](#) on page 17 .

Plan your use of the autostart capability, based on the preceding scenarios.

- If you want to use the autostart capability, refer to [“Scenario 1: One z/OSMF server is autostarted for the entire sysplex”](#) on page 214, [“Scenario 2: Multiple z/OSMF servers and autostart groups per sysplex”](#) on page 214, and [“Scenario 3: Some systems belong to an autostart group, and other systems do not”](#) on page 215 to plan your z/OSMF environment. Then, do the following:
 1. Customize your parmlib IZUPRMxx to fit the scenario that you select. Add the IZUPRMxx member to your system's parmlib concatenation.
 2. Specify the suffix IZU=xx for IZUPRM in your IEASYSxx parmlib member.
 3. Review the job IZUASSEC in SYS1.SAMPLIB and run it to set up security for the AUTOSTART function.
 4. If your installation uses an external security manager other than RACF, ask your security administrator to create equivalent commands for your environment.
- If you do not want to use the autostart capability, refer to Scenario 4 to plan your z/OSMF environment. Then, do the following:
 1. Customize your IZUPRMxx parmlib member to fit "Scenario 4."
 2. Add the IZUPRMxx member to your system's parmlib concatenation.
 3. Specify the suffix IZU=xx for IZUPRM in your IEASYSxx parmlib member.
 4. Do not run job IZUASSEC.

Detecting whether the z/OSMF server is available

Your installation might choose to write a program that depends on the z/OSMF server being active.

A program can use one of the following methods to determine whether the z/OSMF server is up or down in the sysplex:

- An APF-authorized program can use the ENFREQ LISTEN service to specify a listen exit for ENF event code 83 that tells the program the z/OSMF server is up and running.
- An unauthorized program cannot use the ENFREQ LISTEN service. However, it can periodically check the global storage pointer, which is mapped by macro IZUGSP.

Using ENF event code 83 to listen for z/OSMF availability

An authorized program can use the ENFREQ LISTEN service to determine when the z/OSMF server is up and running. On the ENFREQ service, you specify the specific event for which you would like to listen (z/OSMF server availability) and the listener user exit routine that is to receive control after the specified event occurs. The listener user exit that is specified receives control when the z/OSMF server comes up and notifies your program.

To listen for ENF event code 83, you must specify the qualifying events on the QUAL parameter, which specifies a 4-byte field, a hexadecimal constant, or a register containing the address of a 4-byte field containing a bit-mapped qualifier that further defines the event. The qualifiers are mapped by mapping macro IZUENF83.

To use QMASK=BYTE1 to listen for server up and down events, the QUAL values are:

X'80'

z/OSMF server is available.

X'00'

z/OSMF server has ended and is not available.

To use QMASK=ALL, to listen for server up and down events, the QUAL values are:

X'80000000'

z/OSMF server is available.

X'00000000'

z/OSMF server has ended and is not available.

When your program no longer needs to know whether the z/OSMF server is up, it can issue the ENFREQ REQUEST=DELETE request to delete the listen request.

For coded examples that show how to query the status of the z/OSMF server, see [Appendix E, “ENF listener code examples,”](#) on page 423.

For more information about ENFREQ and listener exits, see:

- [*z/OS MVS Programming: Authorized Assembler Services Reference EDT-IXG*](#)
- Topic about listening for system events in [*z/OS MVS Programming: Authorized Assembler Services Guide*](#).

Chapter 37. Configuring a primary z/OSMF for communicating with secondary instances

z/OSMF can be configured to communicate with another instance of z/OSMF in a remote sysplex. This capability is important because it allows z/OSMF tasks to work with systems on other sysplexes in your enterprise. To enable z/OSMF-to-z/OSMF communication, you must configure a primary z/OSMF for communicating with secondary instances, as described in this topic. The key requirement is to enable the sharing of digital certificates between instances.

This information assumes the use of RACF. If you use another external security manager, consult the vendor for more information.

Each z/OSMF instance includes a server runtime and digital certificates

During the configuration process, z/OSMF creates a certificate authority (CA), optionally, and a server certificate, to be used for enabling Secure Sockets Layer (SSL) connections between z/OSMF instances. z/OSMF also creates a SAF key ring, and stores the CA and server certificate in the key ring.

These constructs are named, as follows:

- Key ring name is IZUKeyring.IZUDFLT
- CA name is:

```
CN('z/OSMF CertAuth for Security Domain')
OU('SAF_PREFIX')
WITHLABEL('zOSMFCA')
```

z/OSMF creates the CA and the server certificate if you uncomment the following commands for creating certificates in the IZUSEC job:

```
/* Create the CA certificate for the z/OSMF server          *
RACDCERT CERTAUTH GENCERT +
  SUBJECTSDN(CN('z/OSMF CertAuth for Security Domain') +
    OU('IZUDFLT')) WITHLABEL('zOSMFCA') +
  TRUST NOTAFTER(DATE(2028/05/17))
RACDCERT ADDRING(IZUKeyring.IZUDFLT) ID(IZUSVR)

/* Create the server certificate for the z/OSMF server      *
RACDCERT ID( IZUSVR ) GENCERT SUBJECTSDN(CN('PEV051.POK.IBM.COM') +
  O('IBM') OU('IZUDFLT')) ALTNAME(DOMAIN('PEV051.POK.IBM.COM')) WITHLABEL('DefaultzOSMFCert.IZUDFLT') , +
  SIGNWITH(CERTAUTH LABEL('zOSMFCA')) NOTAFTER(DATE(2028/05/17))
RACDCERT ALTER(LABEL('DefaultzOSMFCert.IZUDFLT')) ID(IZUSVR) TRUST
RACDCERT ID( IZUSVR ) CONNECT (LABEL('DefaultzOSMFCert.IZUDFLT') +
  RING(IZUKeyring.IZUDFLT) DEFAULT)
RACDCERT ID( IZUSVR ) CONNECT (LABEL('zOSMFCA') +
  RING(IZUKeyring.IZUDFLT) CERTAUTH)
```

Planning for secure communication between instances

In the sections that follow, the z/OSMF instance that initiates communication is considered to be the *primary* instance. It serves as the repository for the data that is generated by the z/OSMF instances running in your installation. When planning to enable communication between instances of z/OSMF, first determine which of the instances is to be the primary.

The primary instance communicates with other z/OSMF instances through Secure Sockets Layer (SSL) connections. Each SSL connection requires an exchange of digital certificates, which are used to authenticate the z/OSMF server identities. For the SSL connection to be successful, the primary instance must be configured to trust the server certificates from the secondary instances.

For signing the server certificates, each instance uses a certificate authority (CA) certificate. Establishing a trust relationship between instances will require knowing which CA certificate is used to sign each secondary instance server certificate.

Another consideration is whether the instances share the same security database or use separate security databases. Using a shared database can simplify the process of enabling secure communications if the same CA certificate is used by all participating systems. Sharing a RACF database is not feasible for every installation, however. If your installation uses separate security databases, you must ensure that the appropriate certificates are shared by the participating z/OSMF instances.

For more information about digital certificates, see [z/OS Security Server RACF Security Administrator's Guide](#).

Strategies for sharing CA certificates

This topic describes two scenarios for sharing CA certificates between multiple instances: You might choose to use one common CA certificate for all of the instances, or a different CA certificate for each instance. A third situation is also described, wherein the existence of identically named CA certificates can complicate certificate sharing.

If you have not yet created any secondary instances of z/OSMF, you might find it easier to create one CA certificate and use it to sign all of the server certificates in the primary and secondary instances. Using this approach, you export the CA certificate from the primary system and add it to each of the secondary system security databases. Then, you configure the additional instances of z/OSMF on each secondary system. Here, you should not run the IZUSEC job certificate commands on the secondary systems, as mentioned in [“Each z/OSMF instance includes a server runtime and digital certificates” on page 221](#). Instead, use the certificate that you have from the primary system. As a result, the same CA certificate is used to sign the server certificate for each instance. This approach is shown in [“Scenario 1: SSL connections using the same CA certificate” on page 222](#).

If you have already created one or more secondary instances of z/OSMF, and you want to enable them for communication with the primary, determine whether the secondary systems were configured to use identically-named CA certificates or uniquely-named CA certificates. If you created each of the secondary instances with unique SAF prefix values, each secondary instance uses a uniquely named CA certificate. To allow SSL connections in this case, you can make available the secondary system CA certificates on the primary system key ring (that is, export, add, and connect them). As a result, the primary system will trust the secondary system server certificates, and be able to establish SSL connections with those systems. This approach is shown in [“Scenario 2: SSL connections using different CA certificates” on page 223](#).

A third possibility exists. If you created the secondary instances using the default z/OSMF security execs, it is likely that you have identically named CA certificates on each secondary system— and a problem. The CA certificates have identical names (that is, label name and distinguished name), but different key ring material. The reason is that the default z/OSMF commands for creating the CA certificates all specify the same label name and distinguished name, but the resulting CA certificates contain system-specific key ring material.

The differences in key ring material prevent the primary system from trusting the server certificates from the secondary systems, unless the corresponding CA certificates can be added to the primary system key ring. However, you cannot add the secondary system CA certificates to the primary system key ring, because of naming conflicts; those different CA certificates are not "unique" enough to be added to the same database. Attempting to add a certificate into a database that already has a same-named certificate will result in an error and a message such as: IRRD109I The certificate cannot be added... already defined.

This potential problem can be avoided if the same CA certificate (from the primary system) is used by all of the instances (primary and multiple secondaries). Or, if the secondary instances are created with unique cell names, thus ensuring that each system's CA certificate can be added to the same security database.

Scenario 1: SSL connections using the same CA certificate

In this scenario, you use the primary system CA to generate a common CA certificate, and distribute this CA certificate to the secondary systems. This approach is recommended if the secondary instances do not already exist.

For example, in Figure 38 on page 223, both the primary z/OSMF and the secondary instances are identified by server certificates that were created using the same CA (Jupiter).

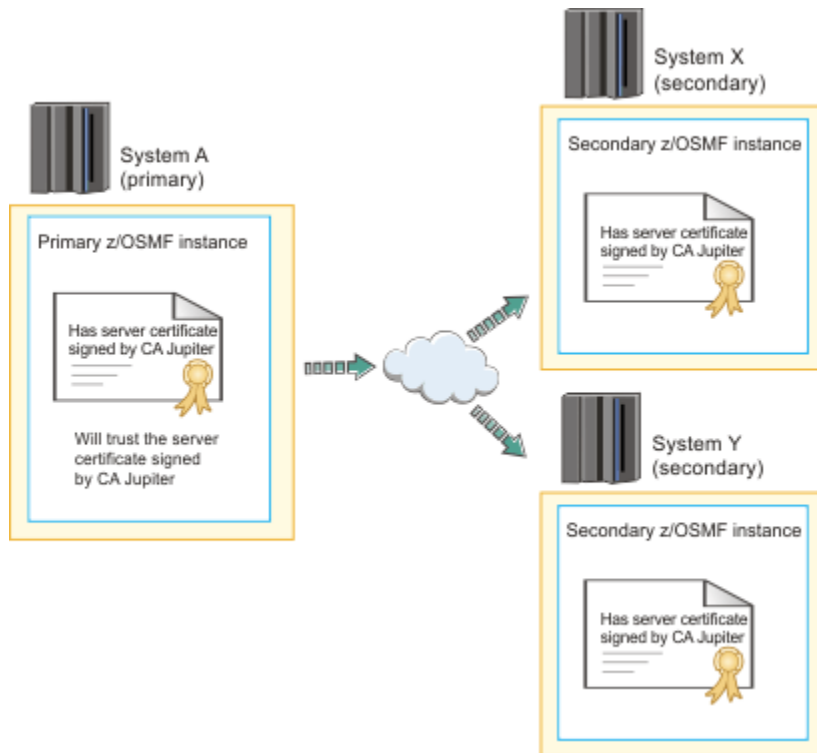


Figure 38. Trust relationship when server certificates are signed by the same CA certificate

Using the same CA to sign the server certificate for each system eliminates the need to import CA certificates from the secondary systems into the primary system security database.

Scenario 2: SSL connections using different CA certificates

In this scenario, each secondary instance of z/OSMF uses its own certificate authority and CA certificate to sign its server certificates. To enable SSL connections in this scenario, you must add each secondary system CA certificate to the primary system security database. This approach is recommended if the secondary instances already exist, and were created to use uniquely named CA certificates.

For example, in Figure 39 on page 224, the primary z/OSMF:

- Is identified by a server certificate created by the Jupiter CA
- Holds (in its security database) the CA certificates from CA Saturn and CA Mars, for the secondary instances, System X and System Y, respectively.

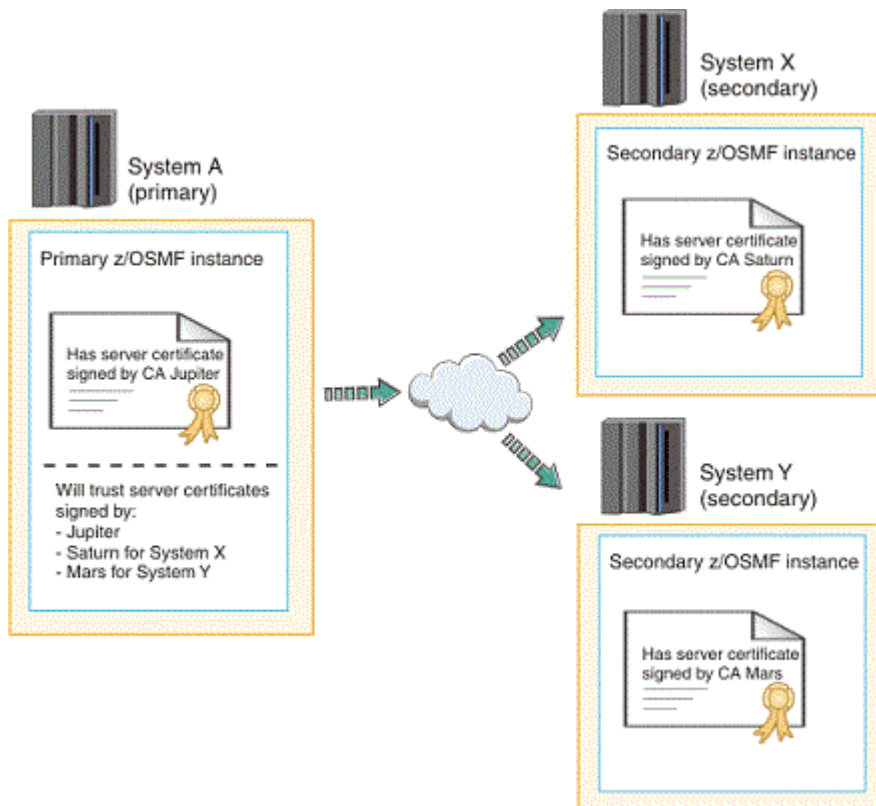


Figure 39. Trust relationship when the server certificates are signed by different CA certificates

To enable SSL connections between instances in this scenario, you would do the following:

1. Export the CA certificate from each secondary system
2. Import the CA certificates into the primary system security database
3. Connect the CA certificates to the primary system.

Chapter 38. Enabling single sign-on between z/OSMF instances

Single sign-on (SSO) enables users to log into one z/OSMF instance and to access other z/OSMF instances without getting prompted to log in again. z/OSMF uses the Lightweight Third Party Authentication (LTPA) security protocol to enable a secure single sign-on environment among z/OSMF instances.

The LTPA protocol uses an LTPA token to authenticate a user with the z/OSMF servers that are enabled for single sign-on. The LTPA token contains information about the user and is encrypted using a cryptographic key. The z/OSMF servers pass the LTPA token to other z/OSMF servers through cookies for web resources. If the receiving server uses the same key as the *primary z/OSMF server* -- the server that generated the key to be used for SSO, the receiving server decrypts the token to obtain the user information, verifies that the token has not expired, and confirms that the user ID exists in its user registry. After the receiving server validates the LTPA token, the server authenticates the user with that z/OSMF instance, and allows the user to access any resource to which the user is authorized.

To establish a single sign-on environment for z/OSMF, the following requirements must be satisfied:

- Enable the z/OSMF Settings service on both the primary and secondary instances. For more information, see [Chapter 9, “Configure the z/OSMF settings service,”](#) on page 67.
- Set up communications between the primary z/OSMF server and the secondary instances, as described in [Chapter 37, “Configuring a primary z/OSMF for communicating with secondary instances,”](#) on page 221.
- The z/OSMF servers participating in the single sign-on environment must reside in the same LTPA domain as the primary z/OSMF server. The LTPA domain name is the parent portion of the fully qualified hostname of the z/OSMF servers. For example, if the fully-qualified hostname is *server.yourco.com*, the LTPA domain is *yourco.com*. Due to browser restrictions, the hostname must be qualified with at least three levels (for example *server.yourco.com*). The domain name must have at least two levels (for example, *yourco.com*).
- The servers must share the same LTPA key. For z/OSMF, this is accomplished by invoking the **Enable Single Sign-on** action to synchronize the LTPA key on the primary and secondary z/OSMF servers. For instructions, see the z/OSMF online help.
- The user ID of the user must exist and be the same in all System Authorization Facility (SAF) user registries. It is recommended that you use the same user registry settings for all z/OSMF servers so that users and groups are the same, regardless of the server.
- The value specified for the SAF prefix during the z/OSMF configuration process must be the same for each z/OSMF server you want to enable for single sign-on. By default, the z/OSMF SAF prefix is IZUDFLT.

z/OSMF generates an LTPA keys file when you start the primary z/OSMF sever if an LTPA keys file does not exist. The file is encrypted with a randomly generated key, and a default password of *WebAS* is initially used to protect the file. When establishing a single sign-on environment, it is recommended that administrators change the default password on the primary z/OSMF server, restart the server to generate a new LTPA keys file, and then proceed with enabling single sign-on between one or more z/OSMF instances. For more information about changing the LTPA key password and enabling single sign-on, see the z/OSMF online help.

Chapter 39. z/OSMF in a DevOps context

The term *DevOps* (*development and operations*) describes a common approach to agile software development that application developers and operations teams can use to build, test, deploy, and monitor applications with speed, quality, and control.

Implementing z/OSMF in support of a DevOps initiative offers a number of benefits, including:

- Faster code delivery
- Feature-rich REST services that can be run locally or remotely
- Isolation of workloads
- More granular security, such as allowing access by DevOps user IDs, but not end users.

In general, a single z/OSMF server can easily handle requests from the system programming staff and from application developers or release engineers with their corresponding automation programs, such as Jenkins or other pipelines. However, if your business is implementing a DevOps strategy, z/OSMF offers a number of advantages in such an environment. It is possible to configure a second, dedicated server for handling workloads for specific applications or automation pipelines.

z/OSMF REST services extend beyond systems management

z/OSMF offers a variety of Representational State Transfer (REST) services that can be used for purposes beyond systems management by the various teams in your organization. By their nature, REST services are secure, can be run remotely, and are easy to include in programs. For example, consider the z/OS jobs REST interface services. With this API, any authorized user at your installation, if they are permitted to, can submit JCL jobs, retrieve job status, cancel or purge output, and retrieve job output. This API can also be used to handle code pages, provide regular expression searching, and so on. As with other z/OSMF REST services, the z/OS jobs REST interface services are not limited to use by system programmers.

REST services provide an ideal interface for creating DevOps style tooling for some z/OS operations. With automation programs, such as Jenkins, the pipelines can be run locally on z/OS and submitted through the z/OS jobs REST interface services directly or through the Zowe command-line interface (CLI). Or, the pipelines can be run remotely on another z/OS or non-z/OS platform. Ansible modules that can invoke z/OS jobs REST interface services are available in Ansible Galaxy.

As you consider how to deploy z/OSMF in your enterprise, consider the anticipated users. It can be convenient to divide them into three primary roles, as shown in the following table.

Role	Common use cases	z/OSMF advantage
System programmer	Software management, editing of system control files, system administration, such as RMF, SDSF, and workload management (WLM).	• Systems management task UIs • z/OSMF REST APIs for common programming tasks.
Application developer	Creating applications to drive your business.	z/OSMF REST APIs for common programming tasks: • z/OS jobs REST interface services • z/OS file and data set REST services • TSO/E address space services.
DevOps programmer	Automating processes, creating pipelines, and provisioning system resources for in-house development projects.	• z/OSMF REST APIs for common programming tasks (see above)

Role	Common use cases	z/OSMF advantage
		• z/OS Cloud Provisioning and Management services • Ansible includes modules that can invoke z/OS jobs REST interface services.

It is possible to serve all three areas (applications, systems programming, and DevOps) with one z/OSMF server instance per sysplex, if the performance and availability of the server remains at acceptable levels for each area of usage. However, for applications, the requirement for availability tends to be higher than for systems programming or DevOps. Because z/OSMF does not run as a clustered server for system management purposes, it is recommended that systems programming and DevOps workloads be isolated from the applications workload. This means having one z/OSMF server for system programmers and DevOps and another one for applications.

Further, if you limit a z/OSMF server to handling just a subset of the z/OSMF REST APIs, you can more easily run the server in a high availability configuration, as described in [Chapter 40, “Configuring z/OSMF to provide redundancy for subset of z/OSMF function,”](#) on page 231.

Defining a second z/OSMF server

It is recommended that you define z/OSMF servers so that they can run on any system in the sysplex:

- Define a home system and alternates.
- When you define a second server, give it a unique name, such as IZUSRV2, to allow you to easily distinguish one server from the other.
- The new server must be addressed separately; it is recommended that you use a virtual IP address (VIPA) with an independent hostname.
- The system programmers can use the hostname and port of the main z/OSMF management server. The DevOps and Applications teams can use the hostname and port of the second z/OSMF server.
- Define a unique SAF profile prefix for the new server so that you can define security for it separately, as described in [“Security”](#) on page 230.
- You cannot use the z/OSMF autostart capability with the second server. Instead, plan to start the server by operator command or through an automation script.

[Figure 40 on page 229](#) shows how workloads can be distributed between three z/OSMF servers. In the figure, requests from special workloads are handled by dedicated z/OSMF servers. Specifically, requests for z/OS containers (zCX address spaces) are handled by server IZUSVR2 and server IZUSVR3. All other z/OSMF requests are handled by server IZUSVR1.

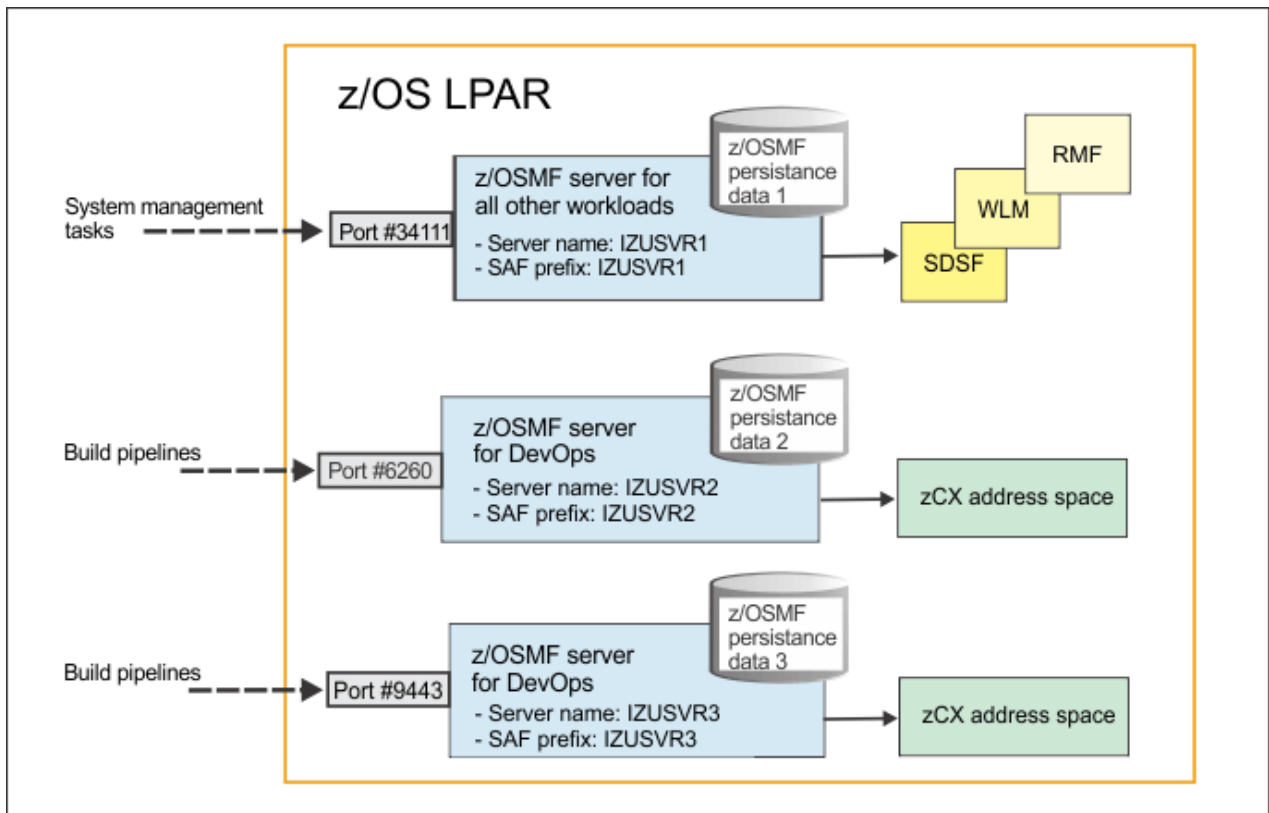


Figure 40. You can run multiple servers of z/OSMF, with each server processing requests from different groups in your organization.

As mentioned earlier, it is possible to handle diverse workloads with one z/OSMF server instance per sysplex. However, you might consider setting up additional servers as described here if you are concerned about workload constraints, RAS (reliability, availability, and serviceability) and isolation of workloads.

Scalability

Though performance measurement remains a system-specific activity, IBM has verified the z/OS jobs REST interface services in sample workloads, with many hundreds of requests per second, per CPU. For most z/OS installations, this activity would far exceed normal workloads.

Some activities in the z/OSMF server can require more time to complete than others. In those cases, you might occasionally find that the server is busy and rejects a request. If so, you must try the request again. This risk is the nature of REST-based programming.

Availability

It is possible to run a z/OSMF server in a high availability configuration. For more information, see [Chapter 40, “Configuring z/OSMF to provide redundancy for subset of z/OSMF function,”](#) on page 231.

As an additional consideration, evaluate your settings for workload manager (WLM) to ensure that z/OSMF for DevOps resource consumption is limited for cases of accidental or unintentional high usage. Guard against situations such as:

- Bad script in a Continuous Integration, Continuous Delivery (CICD) pipeline that drives repeated requests.
- Errors in an integrated development environment (IDE).

Security

From a security standpoint, using multiple z/OSMF servers means defining a unique SAF profile prefix for each new server so that you can control which users can access it. The SAF profile prefix is specified in the IZUPRMxx parmlib member. This setup also requires that you create separate security definitions for each z/OSMF server in your external security manager. To see sample RACF commands for creating the security definitions, refer to the IZUNUSEC or IZUSEC members in SYS1.SAMPLIB. For ease of security management, it is recommended that you define a separate user group for each new server, and permit the group to the new server's security definition.

Isolation

You might choose to run more than one z/OSMF server in a sysplex, for cases in which it is desirable to manage workloads in a more isolated manner. However, a potential downside to having multiple z/OSMF servers is that the z/OSMF persistence data is also isolated. This means having a separate copy of user preferences, saved column widths, and other UI control settings. Also, the z/OSMF Systems table, which is used for defining local to remote system relationships, is not shared and must be maintained on each z/OSMF instance. Therefore, you should try to use the same z/OSMF server instance for all activities that are related to the isolated workload.

Chapter 40. Configuring z/OSMF to provide redundancy for subset of z/OSMF function

In general, z/OSMF cannot be implemented in a high-availability cluster. Each z/OSMF server control access to its own persistence data store, which cannot be shared with other z/OSMF servers. However, it is possible to create a configuration of multiple servers to provide a level of redundancy for a subset of z/OSMF functions. This topic describes an example use case.

If a z/OSMF instance is not available on one LPAR, a manual process is required to start the instance on another LPAR.

For applications that rely z/OSMF REST services, it is possible to configure extra z/OSMF servers to ensure availability of those services. The goal of this configuration is to ensure that one z/OSMF server is always available to provide the REST services to your applications.

The following z/OSMF REST services are supported for high availability:

- z/OS jobs REST interface services
- z/OS data set and file REST service "Retrieve the contents of a z/OS data set or member."

These services are described in [IBM z/OS Management Facility Programming Guide](#).

To ensure that these services remain available to your applications, you can establish a pair of z/OSMF servers on different systems in your sysplex, with each server defined to its own autostart group. z/OSMF includes one autostart group by default. To have more z/OSMF servers autostarted in a sysplex, you must associate each server—and the systems it serves—with a unique autostart group name. Information about defining autostarted servers is provided in [Chapter 36, "Autostart concepts in z/OSMF,"](#) on page 213.

Use a workload router to switch between the z/OSMF servers if an outage occurs. [Figure 41 on page 231](#) shows how a z/OSMF workload can be distributed between multiple z/OSMF servers.

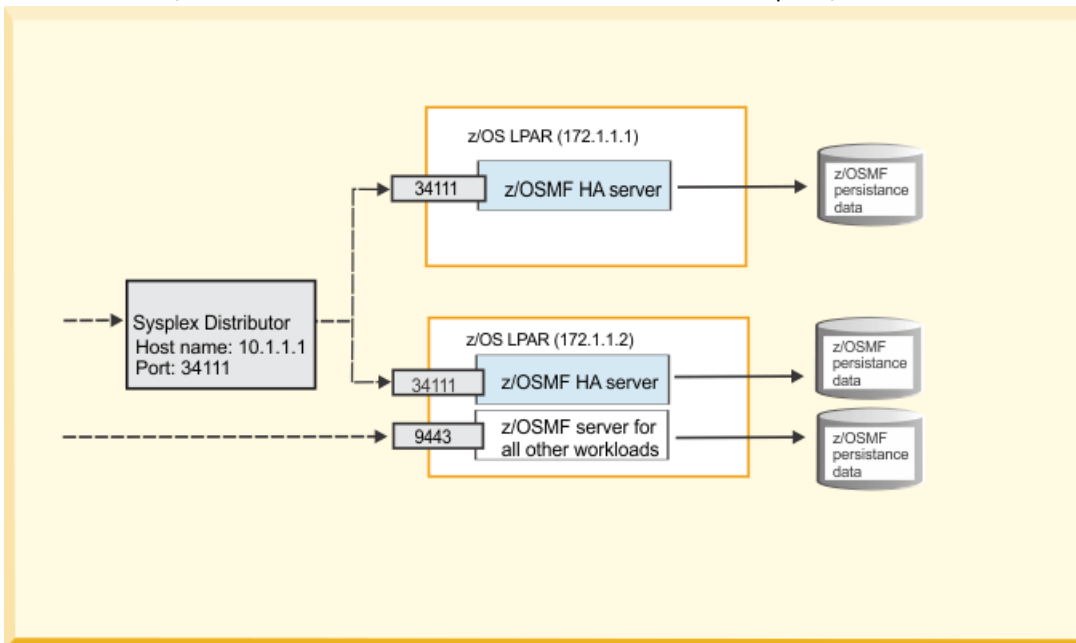


Figure 41. You can run multiple servers of z/OSMF, and use Sysplex Distributor to route requests to specific z/OSMF servers.

In this scenario, a Sysplex Distributor (or a similar workload router) is used to route application requests to a high availability (HA) server on a particular LPAR. If the HA server at system IP address 172.1.1.1 becomes unavailable, the Sysplex Distributor can redirect REST requests to the HA server at system IP

address 172.1.1.2. Each HA server maintains its own copy of the z/OSMF data file system, which contains persistence data. No persistence data is shared between the HA instances.

How to set up z/OSMF for high availability

Assume that two z/OS systems in a sysplex, SYS1 and SYS2, are updated, as follows:

1. Define an IP address for both systems:

- a. Add the following statement to the TCPIP profile for the SYS1 system:

```
IPCONFIG DYNAMICXCF 172.1.1.1 255.255.255.0 3
```

- b. Add the following statement to the TCPIP profile for the SYS2 system:

```
IPCONFIG DYNAMICXCF 172.1.1.2 255.255.255.0 3
```

2. Define a dynamic VIPA (DVIPA) for both SYS1 and SYS2:

```
VIPADYNAMIC
...
:-----
: Test HA for zOSMF
:
: VIPADEFINE 255.255.255.0 10.1.1.1
: VIPADISTRIBUTE DEFINE DISTM HOTSTANDBY 10.1.1.1 PORT 34111
:   DESTIP 172.1.1.1 PREFERRED
:         172.1.1.2 BACKUP
ENDVIPADYNAMIC
```

In this example, the VIPADEFINE statement is used to define the DVIPA 10.1.1.1. The VIPADISTRIBUTE statement with PREFERRED and BACKUP settings is used to enable automatic dynamic VIPA takeover to occur, if needed. The system SYS1 is defined as the preferred system and the system SYS2 is defined as the backup system.

These statements are added to the TCP profiles for both SYS1 and SYS2.

3. Register the DVIPA with one hostname so that z/OSMF can be bound to that hostname. Define the z/OSMF hostname in your name server, for example 10.1.1.1.
4. In the active IZUPRMxx parmlib member for SYS1 and SYS2, define the z/OSMF hostname and port, for example:

```
HOSTNAME('zosmfha.yourcompany.com')
HTTP_SSL_PORT(34111)
```

Now assume that both SYS1 and SYS2 are active. Each system has an active z/OSMF server with its own data directory (sometimes called the user directory). Both z/OSMF servers are bound to the DVIPA 10.1.1.1. With both z/OS systems active in the sysplex, the preferred z/OSMF server receives all new incoming requests. If the SYS1 system fails, new work requests for z/OSMF are routed to the server on SYS2. When SYS1 resumes normal operations, new work requests for z/OSMF are routed to SYS1 again. This behavior occurs because the IP parameter AUTOSWITCHBACK is in effect by default.

For more information about network configuration, see the following documents:

- [z/OS Communications Server: IP Configuration Guide](#)
- [z/OS Communications Server: IP Configuration Reference](#)

Chapter 41. Configuring the z/OSMF server certificate and key ring

If you plan to enable the z/OSMF server for Transport Layer Security (TLS) or Secure Sockets Layer (SSL), you must set up a key ring and certificates. This topic provides the steps for several common scenarios.

This topic describes the following scenarios for server authentication and client authentication:

- Server authentication:
 - [“1a. Create a server certificate and key ring for the z/OSMF server” on page 233](#)
 - [“1b. Share an existing server certificate and key ring with the z/OSMF server” on page 234](#)
- Client authentication:
 - [“2a. Use client and server certificates from the same CA” on page 235](#)
 - [“2b. Use a client certificate from a different CA” on page 236](#)

Additional information is provided in the following topics:

- [“Transfer the client certificate from z/OS to the client workstation” on page 237](#)
- [“Import the client certificate into the client web browser” on page 237](#)
- [“Grant the z/OSMF server access to the key ring and the certificate” on page 237](#)
- [“Tips for proper set-up” on page 238](#)

This information assumes the use of RACF. If you use another external security manager (ESM), contact the vendor for more information.

Note: The z/OSMF sample jobs IZUNUSEC and IZUSEC contain RACF commands for creating a CA and a server certificate. You might find that the commands in IZUNUSEC or IZUSEC are sufficient for your needs. If so, have your security administrator review these sample jobs carefully before you submit them.

1a. Create a server certificate and key ring for the z/OSMF server

In this scenario, you create a new certificate and key ring specifically for use by the z/OSMF server. The certificate and key ring are owned by the z/OSMF server. The certificate is signed by a local certificate authority (CA).

The z/OSMF sample jobs IZUNUSEC and IZUSEC contain RACF commands for creating a key ring and storing the CA and server certificate in the key ring. These constructs are named, as follows:

- Key ring name is IZUKeyring.IZUDFLT
- CA name is:

```
CN('z/OSMF CertAuth for Security Domain')
OU('SAF_PREFIX'))
WITHLABEL('zOSMFCA')
```

To configure a new certificate and key ring, follow these steps:

1. Create the root CA and the server certificate. In the following example, replace 'HOST NAME' with the local hostname.

```
/* Create the CA certificate for the z/OSMF server          *
RACDCERT CERTAUTH GENCERT SUBJECTSDN(CN('z/OSMF CertAuth for Security Domain') +
OU('IZUDFLT')) WITHLABEL('zOSMFCA') TRUST NOTAFTER(2028/04/12))

/* Create the server certificate for the z/OSMF server      */
/* Change HOST NAME in CN field to the real host name.      */
/* Usually the format of the host name is 'XXXX.XXX.XXX.XXX' */
RACDCERT ID(IZUSVR) GENCERT SUBJECTSDN(CN('HOST NAME') O('IBM') OU('IZUDFLT')) +
ALTNAME(DOMAIN('HOST NAME')) +
```

```
WITHLABEL('DefaultzOSMFCert.IZUDFLT') SIGNWITH(CERTAUTH LABEL('zOSMFCA')) +
NOTAFTER(DATE(2021/06/02))

/* RACDCERT ID(IZUSVR) ALTER(LABEL('DefaultzOSMFCert.IZUDFLT')) TRUST */
```

In the example, notice that the RACDCERT ALTER command is commented. This command is not needed if the server certificate validity range falls within the CA certificate validity range.

2. Create the key ring in your external security manager. For a RACF-protected system, use the following command:

```
RACDCERT ID(IZUSVR) ADDRING(IZUKeyring.IZUDFLT)
```

3. Connect the CA certificate and the server certificate to the key ring, as follows:

```
RACDCERT ID(IZUSVR)
CONNECT(CERTAUTH
LABEL('zOSMFCA')
RING(IZUKeyring.IZUDFLT)
)
```

4. Verify that the certificates are set up correctly, as follows:

- Server certificate indicates that it has a private key and is connected as DEFAULT.
- Both the CA and server certificates have TRUST status and are both connected to the key ring.

5. Authorize the z/OSMF server to use the key ring and the certificate, as described in [“Grant the z/OSMF server access to the key ring and the certificate” on page 237](#).

6. Export the CA certificate, as follows:

```
RACDCERT CERTAUTH
EXPORT(LABEL('zOSMFCA'))
DSN('TYQTYQ.ZOSMFCA.CRT')
FORMAT(CERTDER)
```

7. Distribute the CA certificate to users. In an installation with many z/OSMF users, you must decide on an appropriate way to distribute the certificate, such as:

- Sending the certificate to users in an email.
- Allowing users to download the certificate from the mainframe by FTP.
- Implementing a method for sending certificates to users automatically.

8. Set up the web browser with the CA certificate. For details, see [“Import the CA certificate into the client web browser” on page 237](#).

9. Update the KEYRING_NAME parameter in the active IZUPRMxx parmlib member for your system:

```
KEYRING_NAME('IZUKeyring.IZUDFLT')
```

For related considerations, see [“Tips for proper set-up” on page 238](#).

1b. Share an existing server certificate and key ring with the z/OSMF server

In this scenario, you share an existing certificate and key ring with the z/OSMF server. This procedure is shorter than [“1a. Create a server certificate and key ring for the z/OSMF server” on page 233](#) because the CA certificate and key ring exist already.

Follow these steps:

1. Authorize the z/OSMF server to use the key ring and the certificate, as described in [“Grant the z/OSMF server access to the key ring and the certificate” on page 237](#).
2. Export the CA certificate, as follows:


```
RACDCERT CERTAUTH
EXPORT(LABEL('<label of the existing CA>'))
DSN('TYQTYQ.SOMECA.CRT')
FORMAT(CERTDER)
```

3. Distribute the CA certificate to users. In an installation with many z/OSMF users, you must decide on an appropriate way to distribute the certificate, such as:
 - Sending the certificate to users in an email.
 - Allowing users to download the certificate from the mainframe by FTP.
 - Implementing a method for sending certificates to users automatically.
4. Set up the web browser with the CA certificate. For details, see [“Import the CA certificate into the client web browser” on page 237](#).
5. Update the KEYRING_NAME parameter in the active IZUPRMxx parmlib member for your system:

```
KEYRING_NAME('keyring.name')
```

6. Grant the z/OSMF started task user ID access to the key ring and the certificate, as described in [“Grant the z/OSMF server access to the key ring and the certificate” on page 237](#).
7. Update the parameter KEYRING_NAME in the IZUPRMxx parmlib member:

```
KEYRING_NAME('RING01')
```

8. Because the owner is not the z/OSMF started task user ID, you must create an override file that is named **local_override.cfg** in the user configuration directory, which is /global/zosmf/configuration/, by default. In the override file, add the following option, which indicates that the certificate and key ring are owned by BBGSRV, which is the WebSphere Liberty server user ID:

```
KEYRING_OWNER_USERID=BBGSRV
```

For related considerations, see [“Tips for proper set-up” on page 238](#).

2a. Use client and server certificates from the same CA

In this scenario, you use client and server certificates that were generated by the same CA to enable client authentication with the z/OSMF server. This procedure assumes that you completed the steps in [“1a. Create a server certificate and key ring for the z/OSMF server” on page 233](#).

Follow these steps:

1. Generate a client certificate from the same root CA. For example, assume that you want to create a client certificate with the label **Certificate for user DEBUG41**. This label is signed by the internal certificate authority (CA), which uses the label **zOSMFCA**. This certificate is created under the user ID DEBUG41.

To create the client certificate signed by the internal CA, enter the following RACF command:

```
RACDCERT ID(DEBUG41) GENCERT SUBJECTSDN(CN('User DEBUG41'))
O('Your Company') OU('Org A') C('US')) WITHLABEL('Certificate for DEBUG41')
SIGNWITH(CERTAUTH LABEL('zOSMFCA'))
```

Where the distinguished name consists of the following properties:

Common name (Domain Name)

User DEBUG41

Organization name

Your Company

Optional organizational unit

Org A

Country code

US

User ID under which the client certificate is to be added.

DEBUG41

Label of the client certificate

Certificate for DEBUG41

Label of the CA certificate that is used to sign the client certificate.

zOSMFCA

The client certificate is created with status TRUST, which indicates that the client certificate can be used to authenticate the user ID DEBUG41.

2. Export the client certificate and provide it to the client, as described in the following example:
 - a. To export the client certificate to a z/OS data set, enter the RACDCERT command, as shown in the following example:

```
RACDCERT ID(DEBUG41) EXPORT(LABEL('Certificate for DEBUG41'))  
DSN('TYQTYQ.CLIENTCR.DEBUG41.P12') FORMAT(PKCS12DER) PASSWORD('Test1234')
```

Where:

- DEBUG41 is the user ID associated with the client certificate to be exported.
 - Certificate for DEBUG41 is the label of the client certificate.
 - TYQTYQ.CLIENTCR.DEBUG41.P12 is the data set that will contain the client certificate.
 - PKCS12DER indicates that the client certificate and private key are DER encoded when saved to the data set.
 - Test1234 is the password that is associated with the encrypted certificate. You are required to provide this password when you import the client certificate into the browser. The password is case-sensitive.
- b. FTP the client certificate to the client's workstation. For details, see [“Transfer the client certificate from z/OS to the client workstation”](#) on page 237.
 - c. Import the client certificate into the user's web browser. For details, see [“Import the client certificate into the client web browser”](#) on page 237.
 3. Export the CA certificate and provide it to the client if you have not already done so, as described in [“1a. Create a server certificate and key ring for the z/OSMF server”](#) on page 233.

2b. Use a client certificate from a different CA

In this scenario, you use a client certificate that is generated by a different CA than was used for the server certificate. This procedure assumes that you completed the steps in [“1a. Create a server certificate and key ring for the z/OSMF server”](#) on page 233.

Follow these steps:

1. Obtain the client certificate from an external CA.
2. Add the external root CA to RACF and connect it to the server key ring. In the following example, the label zOSMFEXTCA is assigned to the external CA.

```
RACDCERT CERTAUTH  
ADD('TYQTYQ.ZOSMF.EXTCA.CRT')  
WITHLABEL('zOSMFEXTCA')  
  
RACDCERT ID(IZUSVR)  
CONNECT(CERTAUTH LABEL('zOSMFEXTCA'))  
RING(IZUKeyring.IZUFLT)  
)
```

- -
 3. Place the client certificate and CA certificate in the web browser, as described in [“2a. Use client and server certificates from the same CA”](#) on page 235.

Transfer the client certificate from z/OS to the client workstation

In the following example, the FTP command is used to transfer the PKCS12 data set that contains the signed client certificate to the client's workstation.

From the client workstation, do the following:

1. Enter the FTP command and the hostname or IP address of the server, for example, `ftp hostname.com`.
2. When prompted, enter your user ID and password.
3. Enter **bin** to transfer the file in binary format.
4. Transfer the file to the workstation by entering `get 'TYQTYQ.CLIENTCR.DEBUG41.P12' debug41.p12`
5. Enter **quit** to exit.

Import the client certificate into the client web browser

In the following example, the PKCS12 file is imported into the Mozilla Firefox browser.

From the client workstation, do the following:

1. Start the Firefox browser.
2. Access the Certificate Manager by selecting **Tools > Options > Privacy & Security > View Certificates**.
3. Under **Certificate Manager**, click **Your Certificates**.
4. To import the certificate, click **Import**.
5. Locate your PKCS12 certificate file and select it.
6. Click **Open** and enter the case-sensitive password to be used for protecting the file.
7. Click **OK**. The following message is displayed: "Successfully restored your security certificate(s) and private key(s)."
8. Click **OK**. Verify that the certificate label is shown in the window **These are Your Certificates**.
9. To make these changes effective, restart the browser.

Tip: For the Microsoft Edge browser, you can double-click the PKCS12 certificate file to import it.

Import the CA certificate into the client web browser

In the following example, the CA certificate file is imported into the Mozilla Firefox browser.

From the client workstation, do the following:

1. Start the Firefox browser.
2. Access the Certificate Manager by selecting **Tools > Options > Privacy & Security > View Certificates**.
3. Under **Certificate Manager**, click **Authorities**.
4. To import the certificate, click **Import**.
5. Locate the CA certificate file and select it.
6. Click **Open**.
7. Select **Trust this CA to identify websites** and click **OK**.
8. To make these changes effective, restart the browser.

Grant the z/OSMF server access to the key ring and the certificate

You can use the RDATA LIB class to grant the z/OSMF server access to the key ring, certificate, and private key, as described in the procedure that follows.

Before you begin:

1. The RDATA LIB class is case-insensitive.
2. The RACF commands in this procedure are provided in the sample job IZUSKSEC.

Follow these steps:

1. Define the RDATA LIB class, as follows:

```
RDEFINE RDATA LIB <keyRingOwner>.<keyRingName>.LST UACC(NONE)
```

Substitute the *keyRingOwner* and *keyRingName* with the actual key ring owner and key ring name.

- a. If the owner of the server certificate is the same as your z/OSMF started task ID (IZUSVR), enter the following command:

```
PERMIT <keyRingOwner>.<keyRingName>.LST CLASS(RDATA LIB) ID(IZUSVR) ACCESS(READ)
```

- b. If the owner of the server certificate is not the same as your z/OSMF started task ID (IZUSVR), enter the following command:

```
PERMIT <keyRingOwner>.<keyRingName>.LST CLASS(RDATA LIB) ID(IZUSVR) ACCESS(UPDATE)
```

- c. If the owner of the server certificate is SITE, enter the following command:

```
PERMIT <keyRingOwner>.<keyRingName>.LST CLASS(RDATA LIB) ID(IZUSVR) ACCESS(CONTROL)
```

2. If the RDATA LIB class is not already active, activate it and RACLIST it, as follows:

```
SETROPTS CLASSACT(RDATA LIB) RACLIST(RDATA LIB)
```

3. If the RDATA LIB class is already active and RACLISTed, refresh it, as follows:

```
SETROPTS RACLIST(RDATA LIB) REFRESH
```

Tips for proper set-up

Observe the following considerations:

- The z/OSMF server certificate that is created by the RACDCERT command does not contain any certificate revocation information. If a web browser attempts to check revocation when it validates the server certificate, it will fail. As an alternative, consider obtaining the server certificate from z/OS PKI Services.
- The IZUNUSEC and IZUSEC sample jobs provide commands for defining the z/OSMF server certificate and its local signing CERTAUTH certificate (a RACF defined CA). If you prefer, you can use an external CA instead of the local CA. If so, you must connect the external CA (which is used to sign the server certificate) into the key ring instead of the local CA.
- The certificate must be marked TRUST. A NOTRUST certificate is ignored and is not loaded on server start-up.
- The z/OSMF server key ring must contain a personal certificate that is owned by a personal ID. By default, the certificate is owned by the z/OSMF server started task user ID, which is IZUSVR. If the z/OSMF server started task user ID does not own the certificate, it must have authority to extract the private key. For more information, see [“Grant the z/OSMF server access to the key ring and the certificate” on page 237](#).
- If multiple personal certificates exist in the key ring that is owned by the z/OSMF server user ID, the most recently connected certificate is used.
- The certificate must have the appropriate access to <ringOwner>.<ringName>.LST, as follows:
 - READ access to <ringOwner>.<ringName>.LST, if it is also the certificate owner.
 - UPDATE access to <ringOwner>.<ringName>.LST, if it is not the certificate owner.

To avoid confusion, disable any certificates in the key ring that are not intended to be used, by doing either of the following actions:

- Remove the certificates from the key ring.
- Mark the certificates NOTRUST
- z/OSMF is based on WebSphere Liberty, which is in turn based on Java. You can find more information about key ring and certificate setup in the Java security website: <https://public.dhe.ibm.com/software/Java/Java11/IBMJCEHYBRID/HybridRefGuide.html>.

Chapter 42. Securing LTPA keys

Lightweight Third Party Authentication (LTPA) is the authentication method for the z/OSMF server. The z/OSMF server generates LTPA keys automatically on its initial start-up and retains the keys in storage until they are explicitly deleted by your installation. On subsequent starts, the z/OSMF server generates new LTPA keys to replace deleted keys.

Your company's security procedures might require you to provide extra security controls for the z/OSMF server LTPA keys. You can enhance security by enabling AES encryption on the LTPA password and restricting user access to the LTPA keys using a FSACCESS class profile.

Protecting the LTPA password using AES encryption

AES (Advanced Encryption Standard) encryption can be enabled to provide enhanced security for the LTPA password. Even if the encoded password is exposed, it still cannot be decoded unless the user has access to the private key in the certificate.

Pre-configuration - Setting up the AES encryption key

Before enabling the AES encryption function, you must set up the AES encryption key.

1. Define `<SAF-Prefix>.ZOSMF.SETTINGS.SYSTEMS.AES.MODIFY` in class ZMFAPLA. A user ID with READ access on this profile can enable or disable AES encryption, so grant this permission to only the user who enables this function. You can customize `<SAF-Prefix>`, the default value is `IZUDFLT`.
2. Copy `aeskey.xml` from `/usr/lpp/zosmf/defaults/servers/zosmfServer` to `/global/zosmf/configuration/servers/zosmfServer/resources/security`. The destination folder is created when z/OSMF starts for the first time. Rename `/global/zosmf` to your z/OSMF data directory path if you customized the z/OSMF data folder.
3. Review and edit the copied `aeskey.xml`. You might need to create a new certificate for the use of this function. The following is a sample of `aeskey.xml`:

```
<server>
  <variable name="sslCertLabel" value="DefaultzOSMFCert.IZUDFLT"/>
  <variable name="aesCertLabel" value="DefaultzOSMFCert.IZUDFLT"/>

  <ssl id="izuSSLConfig" serverKeyAlias="${sslCertLabel}"/>

  <featureManager>
    <feature>zosPasswordEncryptionKey-1.0</feature>
    <feature>transportSecurity-1.0</feature>
  </featureManager>

  <zosPasswordEncryptionKey keyring="safkeyring://${izu.ssl.key.store.owner.userid}/${
izu.ssl.key.store.saf.keyring}" type="JCERACFKS" label="${aesCertLabel}"/>
</server>
```

Review and edit the certificate names in the value field for variable `sslCertLabel` and `aesCertLabel`.

- Set the value of variable `sslCertLabel` to the label name of the certificate that is used by SSL. This certificate should be an existing certificate that is used by your z/OSMF for SSL.
- Set the value of variable `aesCertLabel` to the label name of the certificate that is used by AES encryption. This certificate can either be a new certificate, which is recommended, or an existing certificate such as your SSL certificate.

Notes:

- a. Both certificates reside in the same key ring that is configured in `IZUPRMXX`. It is highly recommended to set up a new certificate for AES encryption uses only, with the expiry date set far into the future. When the certificate for AES encryption expires or changes, all passwords that were previously encoded by AES in z/OSMF become undecodeable or corrupted. For more information

about the actions that are necessary before you change the AES certificate, see [“Post-configuration - Changing AES encryption key” on page 242](#).

- b. If IBM zERT Network Analyzer is enabled and you stored the database password in z/OSMF, that password is lost when the AES encryption key pre-configuration is completed. IBM zERT Network Analyzer uses a default AES key to encode the provided password. When the new AES encryption key is enabled, it applies globally across all z/OSMF plug-ins. IBM zERT Network Analyzer is not able to decode previously saved database passwords with the new AES key enabled. You must provide the database password again when you use the IBM zERT Network Analyzer plug-in. For more information, see [Chapter 26, “Configure IBM z/OS Encryption Readiness Technology \(zERT\) Network Analyzer,” on page 129](#).

The next time z/OSMF is started after pre-configuration is completed, message CWWKB0808I appears in the server job log. This message indicates that the AES encryption key is properly set to be read from a certificate for AES encryption.

Example:

```
CWWKB0808I: The DefaultzOSMFCert.IZUDFLT private key from the safkeyringhybrid:///
IZUKeyring.IZUDFLT key ring is set as the AES password encryption key.
```

Configuration – Setting a new LTPA password and enabling AES encryption of the LTPA password

After completing the pre-configuration process, perform the following to enable AES encryption for the LTPA password.

1. Log in to z/OSMF with the Administrator user ID. To enable the function, the user ID must have READ access on IZUDFLT.ZOSMF.SETTINGS.SYSTEMS.AES.MODIFY (ZMFAPLA).
2. Go to **Systems** task and click **Actions > Change LTPA Key**.
3. Input the new LTPA password and check option **Protect the LTPA Password with AES encryption**.
4. If z/OSMF SSO was enabled, you must send the newly set LTPA password to other z/OSMF systems that have SSO enabled. For more information about enabling z/OSMF SSO with AES encryption, see [Chapter 38, “Enabling single sign-on between z/OSMF instances,” on page 225](#).
5. Restart z/OSMF for the change to take effect.

Post-configuration - Changing AES encryption key

The password encoded by AES encryption is stored in the z/OSMF file system. When the AES key changes, z/OSMF also cannot decode all of the stored passwords that were encoded by a previous key.

If z/OSMF fails to decode the LTPA password, the whole LTPA feature fails. This failure can occur when your certificate expires or when you enable a new certificate for AES encryption. When the LTPA feature fails to initiate, users cannot log in to the z/OSMF UI. The following error message would appear in the job log:

```
ERROR CWWKS4000E: A configuration exception has occurred. The requested TokenService instance of
type Ltpa2 could not be found.
```

When a certificate expires, if the certificate is a self-signed certificate, you can use RACF to renew the certificate without changing the private key to correct error CWWKS400E. You must restart z/OSMF after extending the expiry date of the certificate if you already see the previous error message.

To prevent the potential outage caused by expired certificates, renew the certificate without changing the key before the certificate expires.

If you cannot renew the certificate due to the security policy, you must turn off the AES encryption for LTPA before the key expires. You can re-enable the AES encryption after you configure a new AES encryption key by either supplying a new certificate or rekeying the existing certificate.

To use a new certificate for AES encryption, make sure you do the following before the certificate expires.

1. Log in to z/OSMF with the user ID that enables the AES encryption for LTPA. Go to **Systems task > Change LTPA Key**.

2. Turn off the AES encryption for LTPA by clearing the AES encryption checkbox. You do not need to supply an LTPA password.
3. Stop the z/OSMF server.
4. Connect the new certificate to the key ring or rekey the certificate in the key ring. If the label name changes, make sure the variable *aesCertLabel* in *aeskey.xml* is also updated.
5. Start z/OSMF. Locate message CWWKB0808I in server job log.
6. Configure a new LTPA password and re-enable the AES encryption for LTPA on the **Change LTPA Key** window.
7. Restart z/OSMF for the change to take effect.

Post-configuration - Troubleshooting

The AES key and the encoded LTPA password value pair must match for the z/OSMF server to start. A mismatch causes initialization failures. The following error appears in the server job log when a mismatch occurs.

```
ERROR CWWKS4106E: LTPA configuration error. Unable to create or read LTPA key file: /var/zosmf/
configuration/servers/zosmfServer/resources/security/ltpa.keys
```

If for any reason you cannot make the AES key and the encoded LTPA password match each other, you must manually remove the AES encoded LTPA password. After you remove the password, z/OSMF will use the default XOR encoded LTPA password to start the LTPA service and start the z/OSMF server. You can re-enable the AES encoded LTPA password after the server is started.

Do the following to remove the AES encoded LTPA password.

1. Stop the z/OSMF server.
2. Locate *bootstrap.properties* in your z/OSMF file system. By default, the *bootstrap.properties* file resides in the directory */global/zosmf/configuration/servers/zosmfServer*.
3. Use a text editor to edit *bootstrap.properties* file, locate variable *izu.ltpa.key.password*. The value of this variable should start with *{aes}*.
4. Remove the line that contains the *izu.ltpa.key.password* variable. Save the change and exit the editor.
5. Remove the *ltpa.keys* file. By default, the *ltpa.keys* file resides in the directory */global/zosmf/configuration/servers/zosmfServer/resources/security*.
6. Start the z/OSMF server.

Restricting user access to the LTPA keys

This topic describes a procedure for using a FSACCESS class profile to restrict access to the LTPA keys file.

The following information assumes the use of RACF. If you use a different external security manager (ESM), check with the vendor for more information.

Steps for restricting user access to the LTPA keys

Perform the following steps to give selected users access to the LTPA keys file. IBM recommends that you limit access to the z/OSMF server user ID only. By default, this user ID is IZUSVR.

1. Activate the FSACCESS class profile, if it is not currently active at your installation. By default, it is not active.
2. Create a file system for storing the LTPA keys, for example:

```
ZOSMF.LTPA.FS
```

3. Ask your security administrator to restrict access to the file system. In the following example, RACF commands are used to define a profile in the FSACCESS class to protect the LTPA file system and grant access to the z/OSMF server user ID. For example:

```
RDEFINE FSACCESS ZOSMF.LTPA.FS UACC(NONE)
PERMIT ZOSMF.LTPA.FS CLASS(FSACCESS) +
  ID(IZUSVR) ACCESS(UPDATE)
SETROPTS RACLIST(FSACCESS) REFRESH
```

4. If the z/OSMF server is active on your system, stop the server. From the operator console, enter the following command: **STOP IZUSVR1**.

If you are setting up z/OSMF for the first time and never started the server, start it at least once to create the z/OSMF file systems. Then, stop the server before you continue with the next step: **START IZUSVR1 > STOP IZUSVR1**.

5. Mount the new file system at the z/OSMF global mount point:

```
/global/zosmf/configuration/servers/zosmfServer/resources/security
```

By default, the z/OSMF global mount point is `/global/zosmf/`. However, during z/OSMF configuration your installation might have selected a different path for the mount point. If so, specify that path instead.

6. Verify that users are blocked from accessing the LTPA file system. To do so, use a superuser ID to try to access the follow directory:

```
/global/zosmf/configuration/servers/zosmfServer/resources/security
```

7. Start the z/OSMF server. The server creates an `ltpa.keys` file automatically, which is now protected from all users, except the z/OSMF server user ID.
8. Ensure that the LTPA file system is mounted automatically for subsequent IPLs. To do so, add a MOUNT command for the LTPA file system to your currently active BPXPRMxx parmlib member. For example:

```
MOUNT FILESYSTEM('ZOSMF.LTPA.FS') TYPE(ZFS) MODE(RDWR)
MOUNTPPOINT('/global/zosmf') PARM('AGGRGROW') UNMOUNT
```

9. Optional: For added security, you can direct the z/OSMF server to generate new LTPA keys on each server restart. In the IZUSVR1 procedure, add a step to delete the existing `ltpa.keys` file, as shown in the following example. Insert this step before the step that starts the z/OSMF server.

```
//STEP1 EXEC PGM=BPXBATCH,
// PARM=('SH im -fr &USERDIR/configuration/servers/zosmfServer/resource
// s/security/ltpa.keys')
//BPXPRINT DD SYSOUT=&OUTCLS
//STDERR DD SYSOUT=&OUTCLS
//STDOUT DD SYSOUT=&OUTCLS
```

After you complete this procedure, the z/OSMF server LTPA keys can be accessed only by the z/OSMF server user ID.

Chapter 43. Restricting IP addresses from accessing the z/OSMF server

You can configure the z/OSMF server to limit inbound connections to specific IP addresses, and block requests from all other sites. This topic describes a procedure that you can follow to customize the z/OSMF server.

This work requires you to identify which sites (IP addresses) are to be allowed to access the z/OSMF server on the host system. You define this "allowlist" of exceptions (the trusted sites) in the `server_override.xml` file.

Important: Review the list of trusted sites with your security administrator to ensure that your installation security policies are followed.

How to restrict the IP addresses that can access the z/OSMF server

To restrict the IP addresses that can access the z/OSMF server, follow these steps:

1. Copy the file `server_override.xml` from `<product_dir>/defaults/servers/zosmfServer/` to `<user_dir>/configuration`

Where:

- `<product_dir>` is the z/OSMF product directory. By default, this is `/usr/lpp/zosmf`
- `<user_dir>` is the z/OSMF data directory. By default, this directory is `/global/zosmf`

2. Set the permissions to 755 for the file `server_override.xml` in `<user_dir>/configuration`. For example:

```
chmod 755 <user_dir>/configuration/server_override.xml
```

3. Copy the following statements to the `server_override.xml`:

```
<server>
  <httpEndpoint id="defaultHttpEndpoint">
    <tcpOptions addressIncludeList="a.aa.aa.aaa,b.bb.bb.bbb ... n.nn.nn.nnn"/>
  </httpEndpoint>
</server>
```

4. In the example, replace `a.aa.aa.aaa` with an IP address that is allowed to connect to the z/OSMF server on your system. To specify additional IP addresses, separate each address with a comma, as shown in the example. You can specify IPv4 or IPv6 addresses. All values in an IPv4 or IPv6 address must be represented by a number or by an asterisk wildcard character (*).

It is not necessary to specify the IP address of the z/OS host system.

5. Restart the z/OSMF server by entering the following command at the operations console:

```
START IZUSVR1
```

As a result, only the IP addresses that are specified in the `server_override.xml` file can connect to the z/OSMF server on your system. Inbound requests from all other IP addresses are blocked.

More information

You can further customize the endpoint behavior of the z/OSMF server by setting one or more of the following options in the `server_override.xml` file:

addressExcludeList

Addresses that are not allowed to make inbound connections on this endpoint.

hostNameExcludeList

Hostnames that are not allowed to make inbound connections on this endpoint.

hostNameIncludeList

Hostnames that are allowed to make inbound connections on this endpoint.

For descriptions of these options, see the following website: <https://www.ibm.com/docs/en/was-liberty/core?topic=configuration-httpendpoint>

Chapter 44. Considerations for using ICSF services

If your installation uses hardware cryptography with z/OS Integrated Cryptographic Service Facility (ICSF), the z/OSMF server uses a number of ICSF callable services, such as CSFRNGL, CSFDSV, CSFOWH, CSFIQF, and others. These services might be protected through profiles that are established in your external security manager, such as RACF. Because z/OSMF uses these services, you must permit the z/OSMF started task user ID to these profiles.

Follow these steps :

1. Determine whether your installation uses hardware cryptography with z/OS Integrated Cryptographic Service Facility (ICSF), and whether the ICSF services are protected by an external security manager.
2. If so, and your installation uses RACF as its security manager, review job IZUICSEC in SYS1.SAMPLIB and edit it for your environment. Then, run job IZUICSEC.
3. If your installation uses an external security manager other than RACF, ask your security administrator to create equivalent commands for your environment.

Table 38 on page 247 shows which permissions must be granted to the z/OSMF server user ID. Commands for the creating the permissions are included in commented sections in the IZUICSEC job.

Note: Table 38 on page 247 is not an exhaustive list of ICSF related authorizations. The z/OSMF server user ID might require additional authorizations, depending on which ICSF resources are used at your installation. For example, if your installation uses the CSFKEYS facility to control access to keys, the z/OSMF server user ID might require permission to the CSF-PKDS-DEFAULT profile. For more information about protecting ICSF resources, see *z/OS Cryptographic Services ICSF Administrator's Guide*.

Table 38. Security setup requirements for hardware cryptography with ICSF				
Resource class	Resource name	Who needs access?	Type of access required	Why
CSFSERV	CSFIQF	IZUSVR	READ	ICSF query facility callable service.
CSFSERV	CSFENC	IZUSVR	READ	Encipher callable service.
CSFSERV	CSFCVE	IZUSVR	READ	Cryptographic variable encipher callable service.
CSFSERV	CSFDEC	IZUSVR	READ	Decipher callable service.
CSFSERV	CSFSAE	IZUSVR	READ	Symmetric algorithm encipher callable service.
CSFSERV	CSFSAD	IZUSVR	READ	Symmetric algorithm decipher callable service.
CSFSERV	CSFOWH	IZUSVR	READ	One-way hash generate callable service.
CSFSERV	CSFRNG	IZUSVR	READ	Random number generate callable service.
CSFSERV	CSFRNGL	IZUSVR	READ	Random number generate long callable service.
CSFSERV	CSFPKG	IZUSVR	READ	PKA key generate callable service.
CSFSERV	CSFDSG	IZUSVR	READ	Digital signature generate service.
CSFSERV	CSFDSV	IZUSVR	READ	Digital signature verify callable service.
CSFSERV	CSFPKT	IZUSVR	READ	PKA key generate callable service.
CSFSERV	CSFRKL	IZUSVR	READ	Retained key list callable service.
CSFSERV	CSFPKX	IZUSVR	READ	PKA Public Key Extract callable service.
CSFSERV	CSFPKE	IZUSVR	READ	PKA encrypt callable service.
CSFSERV	CSFPKD	IZUSVR	READ	PKA decrypt callable service.
CSFSERV	CSFPKI	IZUSVR	READ	PKA key import callable service.
CSFSERV	CSFCKM	IZUSVR	READ	Multiple clear key import callable service.
CSFSERV	CSFKGN	IZUSVR	READ	Multiple clear key import callable service.

Table 38. Security setup requirements for hardware cryptography with ICSF (continued)				
Resource class	Resource name	Who needs access?	Type of access required	Why
CSFSERV	CSFEDH	IZUSVR	READ	ECC Diffie-Hellman callable service.

Chapter 45. Considerations for using AT-TLS

During z/OSMF server startup, if the z/OSMF server address space initializes before the Policy Agent address space, it is possible that the z/OSMF server might try to establish a listener socket and attempt to establish remote connections before the AT-TLS policy is available. In this situation, the connections fail.

The resource EZB.INITSTACK.sysname.tcpname in the SERVAUTH class controls the ability of applications to open a socket before the AT-TLS policy is loaded on the TCP/IP stack. If the z/OSMF server address space user ID does not have access to the EZB.INITSTACK.sysname.tcpname profile, and it attempts to establish a listener socket, an ICH408I message is issued to the console.

To allow the z/OSMF server to establish connections before the AT-TLS policy is available, follow these steps:

1. Determine whether you AT-TLS is configured in the initial PROFILE.TCPIP profile.
2. If you do not use AT-TLS, no other actions are needed.
3. If you have configured AT-TLS, and the Policy Agent address space is started, no other actions are needed.
4. If you have configured AT-TLS, but the Policy Agent address space is not started or has failed to start, do the following to allow the z/OSMF server to establish connections:
 - a. Review the security job IZUTLSEC in SYS1.SAMPLIB. Change the *sysname* and *tcpname* to appropriate values in your environment.
 - b. Submit the job to permit the z/OSMF server user ID to access to EZB.INITSTACK.sysname.tcpnameprofile.

Table 39 on page 249 shows which permissions must be granted to the z/OSMF server user ID. Commands for the creating the permissions are included in commented sections in the IZUTLSEC job.

Table 39. Security setup requirements for z/OSMF AT-TLS installations				
Resource class	Resource name	Who needs access?	Type of access required	Why
SERVAUTH	EZB.INITSTACK.sysname.tcpname	IZUSVR	READ	Allows the z/OSMF server to access the TCP/IP stack during TCP/IP initialization. This authorization is needed if the TCP/IP profile activates Application Transparent Transport Layer Security (AT-TLS) but before the Policy Agent address spaces are started.

Chapter 46. Linking z/OSMF tasks and external applications

To perform traditional system management tasks in z/OS, you might interact with several different interfaces, such as the TSO command line, graphical user interfaces, and web-style interfaces. In z/OSMF, it is possible to link or connect some of these tasks and external applications together for a smoother user experience. To help manage these connections, z/OSMF provides the Application Linking Manager task.

Key components

The key components of the Application Linking Manager task include the:

- **Event requestor.** z/OSMF task or external application that requests the launch of a specific function within another task or external application
- **Event.** Action requested by the event requestor. It includes the type of event and the event parameters.
- **Event type.** Object that connects an event requestor to an event handler. It identifies the handlers that can process an event and the possible parameters that can be supplied with an event.
- **Event handler.** z/OSMF task or external application that can process the event parameters and display the requested information.

Figure 42 on page 251 depicts the relationship of these components in the application linking process.

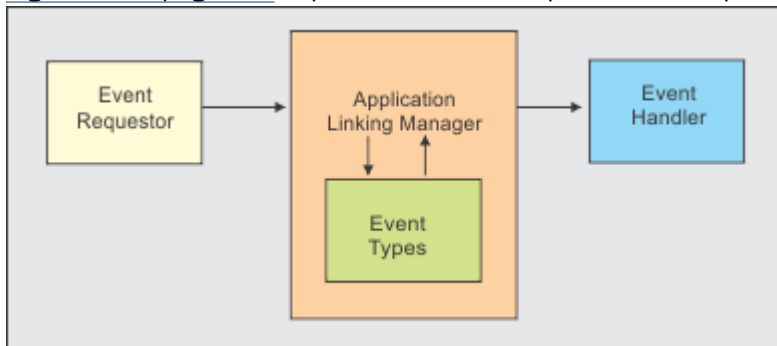


Figure 42. Key components in the application linking process

The process begins with a user action, such as clicking a link. In response to this action, the event requestor creates an event and sends it to the Application Linking Manager. The Application Linking Manager searches the set of known event types for the type identified by the event. If a match is found, the Application Linking Manager searches for event handlers that are registered for this event type. If only one handler is found, it is launched. Otherwise, the user is prompted to select the handler to launch. The Application Linking Manager provides the handler with the parameters that were supplied with the event. The event handler processes the parameters and displays the requested information.

z/OSMF includes a number of predefined event types, requestors, and handlers. For a list, see [IBM z/OS Management Facility Programming Guide](#).

Key features

To access the Application Linking Manager task, click on the App center to display the z/OSMF task icons. You can drag the icon for the Application Linking Manager task out of the App center to the desktop.

The Application Linking Manager task provides a web-based, user interface that you can use to:

- Define new event types, and view and delete existing event types.
- Define new handlers; view, enable, disable, and delete existing handlers; and make a handler the default handler.

For assistance with the Application Linking Manager task, see the online help.

Programming interface

The Application Linking Manager task also provides an application programming interface (API) that you can use to complete the aforementioned actions. For more details about the API, see [*IBM z/OS Management Facility Programming Guide*](#).

Chapter 47. Configuring the CIM server for your system

If your installation is using services that require the CIM server, see this section for additional considerations.

Some z/OSMF tasks require the Common Information Model (CIM) server to be running on the host z/OS system. Using these tasks requires that you ensure that the CIM server is configured on your system, including security authorizations and file system customization.

Ensure that the administrator role is authorized to the CIM server

If your z/OSMF configuration includes tasks that require the Common Information Model (CIM) server to be active, you must ensure that the z/OSMF administrator group has the proper level of access to CIM server resources. In effect, the z/OSMF administrator is also a CIM administrator. CIM includes the CFZSEC job to help you perform these authorization tasks.

Before you run the CFZSEC job, make the following changes.

1. Ensure that the first line in CFZSEC includes the appropriate information. Specify values for *jobname*, MEGCLASS, MSGLEVEL, and NOTIFY.

```
//CFZSEC JOB CLASS=A,MSGCLASS=H,MSGLEVEL=(1,1),NOTIFY=&SYSUID
```

2. Specify the CIM started procedure in the CFZSEC job. Find the following statement in CFZSEC, and verify that it uses CFZCIM as the started procedure name:

```
RDEFINE STARTED CFZCIM.* STDATA(USER(CFZSRV) GROUP(CFZSRVGP))
```

3. Submit the CFZSEC job.
4. After the job completes:
 - a. If necessary, start the CIM server, by using the following command: S CFZCIM
 - b. Your security administrator must connect the z/OSMF administrator user IDs to the CFZADMGP group.

If necessary, see the chapter on CIM server quick setup and verification in [z/OS Common Information Model User's Guide](#).

Note: By default, the CFZSEC job issues an ALTUSER command under OMVS(UID(0)). In some situations, the CFZSEC job might end with the error message: "Incorrect UID(0). This value is already in use by PDS." If so, edit the CFZSEC job and change UID(0) to UID(0) SHARED. For example:

```
ALTUSER CFZSRV DFLTGRP(CFZSRVGP) OMVS(UID(0) SHARED +  
PROGRAM('/bin/sh') +  
HOME('/u/cfzsrv')) NOPASSWORD NOIDCARD NOPHRASE
```

If your installation does not plan to run the CFZSEC job, your security administrator can perform these tasks manually, as follows:

1. Grant the z/OSMF administrator group UPDATE access to the CIMSERV profile in the WBEM class. This access can be granted through an explicit PERMIT command, or, if the CIM administrator group is already permitted with UPDATE access, you can connect the z/OSMF administrator user ID to the group. If necessary, refresh the WBEM class.
2. Ensure that the user ID under which the CIM server is running has SURROGAT access for the z/OSMF administrator group. If a generic BPX.SRV.** profile is already authorized in the SURROGAT class. No additional action is required. Otherwise, define a discrete profile for the z/OSMF administrator group and authorize it. If necessary, refresh the SURROGAT class.

These updates should be made before logging in to z/OSMF as the administrator.

Customizing the administrator role for running CIM commands

The CIM server commands are UNIX style programs that run in the z/OS UNIX shell. To ensure that the z/OSMF administrator can use the CIM commands, verify that the administrator role is properly set up for the z/OS UNIX shell environment, as described in this topic.

The file **profile.add**, which is included with the CIM server, provides the environment variables that you need to define for the administrator; see `/usr/lpp/wbem/install/profile.add`. If your installation used the job CFZRCUST from the installation SAMPLIB to customize the file systems and directories that are used by the CIM server, this setup is already done.

If your installation did not run the CFZRCUST job, you can perform this setup manually. Copy the contents of the **profile.add** file to the `.profile` file in the home directory of the z/OSMF administrator user ID. Modify the appropriate settings if you do not plan to use the defaults. The `.profile` file should be owned by the z/OSMF administrator; this person requires read-write-execute access to the file.

Or, you can use the following command to include the CIM profile settings during a shell session: `. /usr/lpp/wbem/install/profile.add`

Here, you must enter this command whenever the z/OSMF administrator logs in to the z/OS UNIX shell to run CIM command-line utilities.

Ensure that the CIM server is started

If your configuration includes a service that uses the CIM server, ensure that the CIM server is active on your system when using z/OSMF. You can verify that the CIM server is started by entering the following command from the operator console:

```
D A,CFZCIM
```

This example assumes that the CIM server runs as a started task that uses the default name CFZCIM.

If the CIM server is not already started, follow the steps that are described in [*z/OS Common Information Model User's Guide*](#) to start it. This book also includes information about customizing your CIM server start-up procedure and details on how to set environment variables for the CIM server.

It is recommended that you ensure that the CIM server is started automatically at IPL time. For information about customizing the CIM server startup, see [*z/OS Common Information Model User's Guide*](#).

Chapter 48. Configuring your system for asynchronous job notifications

To allow HTTP client applications on your z/OS system to receive asynchronous job notifications, your system must be configured as described in this topic.

The z/OS jobs REST interface provides a set of REST services that allow a client application to perform operations with batch jobs on a z/OS system. Through the z/OS jobs REST interface services, an application can:

- Obtain the status of a job
- List the jobs for an owner, prefix, or job ID
- List the spool files for a job
- Retrieve the contents of a job spool file
- Submit a job to run on z/OS
- Cancel a job
- Change the job class
- Delete a job (cancel a job and purge its output).

The z/OS jobs REST interface services can be invoked by any HTTP client application, running on the z/OS local system or a remote system, either z/OS or non-z/OS. The z/OS jobs REST interface services are described in the document [IBM z/OS Management Facility Programming Guide](#).

You can use the asynchronous job notifications function of z/OSMF to allow your programs to be notified of job events.

The asynchronous job notifications function is available for the JES2 subsystem only; it is not available for the JES3 subsystem.

You need to configure JES2 EDS for job notifications over HTTP to use asynchronous job notifications function. With this function, the program that submits the job through the z/OS jobs REST interface services PUT method specifies a URL when the job is submitted. You can define the events of a submitted job. If any of those events happen, z/OSMF returns an HTTP message to the URL location. The data returned is in the form of a JSON document.

JES2 EDS for job notifications over HTTP

The notification URL supplied to the z/OS jobs REST interface services is passed to JES2 via SYS_JOB_NOTIFY JES symbol when submitting a job through the internal reader (see description of SYS_JOB_NOTIFY in [JES Symbol Service \(IAZSYMBL\)](#) in *z/OS JES Application Programming*).

There is a new RACF check added for job notifications. Using RACF, you can control which users can request such job notifications by defining a profile in JESJOBS class. For additional information, see [Controlling job notifications in z/OS JES2 Initialization and Tuning Guide](#) and description of SYS_JOB_NOTIFY in [JES Symbol Service \(IAZSYMBL\)](#) in *z/OS JES Application Programming*.

Prerequisites

To use the asynchronous job notifications function of z/OSMF through JES2 EDS for job notifications over HTTP, you must enable the following z/OS elements:

- JES2 Email Delivery Services (EDS); see [Using JES2 EDS for job notification over HTTP](#) in *z/OS JES2 Initialization and Tuning Guide*.
- Enable JES2 support for job notifications over HTTP; see description of SYS_JOB_NOTIFY in [JES Symbol Service \(IAZSYMBL\)](#) in *z/OS JES Application Programming*.

Chapter 49. Adding links to z/OSMF

Generally, when you want to add a link to the z/OSMF user interface, you can do so through the Links task. In some situations, however, you might be asked at the direction of a vendor to add a link to z/OSMF through the link properties file. If so, you can follow the steps in this section.

After a link is added to z/OSMF, you can modify or remove the link through the Links task, as described in the online help.

Steps for adding a link to z/OSMF

A sample link properties file is supplied with z/OSMF:

```
<product_dir>/samples/sampleLink.properties
```

where *<product_dir>* is the z/OSMF product directory. By default, this is `/usr/lpp/zosmf`.

To add a link to z/OSMF, follow these steps:

1. **Make a copy of the sample link properties file.** Copy the sample link properties file to a read/write directory.
2. **Edit the new link properties file with your text.** As shown in [Figure 43 on page 257](#), the link properties file contains the following input fields for a link:

```
LinkName=  
LinkURL=  
LinkAuthorizedRoles=  
LinkSafSuffix=  
LinkLaunchWorkArea=
```

Figure 43. Content of the link properties file

In your link properties file, define the link using these input fields:

LinkName

Specify a name for the link, as it should be displayed in the z/OSMF desktop. Specify a value of up to 30 characters, including alphanumeric characters (A-Z a-z 0-9), blanks, mathematical symbols (+ - = | ~ () { } \), punctuation marks (? , . ! ; : ' " / []), and the following special characters: %, \$, #, @, ^, *, and _. Any leading or trailing white space is ignored.

Specify your input in the form of the ASCII, EBCDIC or Unicode character sets. To use Japanese language characters, enter the characters in Unicode. Each Unicode character (\uxxxx) is treated as one character.

The name you select must be unique among the existing links defined in z/OSMF. It is recommended that you choose a name that will be easily understood by users. Avoid names that might be confused with other links or tasks in z/OSMF.

LinkURL

Specify the location for the link (a URL), which is a valid Internet or intranet address, for example:

```
http://www.ibm.com
```

The URL can be up to 4000 characters, including alphanumeric characters (A-Z a-z 0-9), blanks, mathematical symbols (+ - = | ~ () { } \), punctuation marks (? , . ! ; : ' " / []), and the following special characters: %, \$, #, @, ^, *, and _. Any leading or trailing white space is ignored.

z/OSMF performs limited syntax checking of the specified URL. Ensure that the link location is a syntactically correct URL. Generally, a URL includes a protocol (such as `http://`), a host name (`www.hostname.com`), and, often, a resource such as a directory path and file.

To link to a file on the host system, ensure that the host name is included in the URL, for example:

```
file://localhost/C:/tmp/test.html
```

Note that the ability to connect to a particular location can depend on the user's browser settings.

LinkAuthorizedRoles

Specify the z/OSMF roles for which users are authorized to use the link. You can limit access to users with one or more of the following roles:

- z/OSMF Guest
- z/OSMF Authenticated Guest
- z/OSMF User
- z/OS Security Administrator
- z/OSMF Administrator

Enter the role name exactly as depicted here. To specify multiple roles names, separate each name with a comma. Any leading or trailing white space is ignored.

If you specify a role incorrectly, the role is ignored. If you specify no roles at all, or omit this property, the link is added to the table displayed in the Links task with no roles assigned to it.

LinkSafSuffix

Specify the system authorization facility (SAF) resource name suffix to be used for managing user authorizations to the link. To create a unique resource name for the link, z/OSMF appends this value to the z/OSMF SAF profile prefix (by default, IZUDFLT), followed by ZOSMF.LINK. Specify a unique resource name suffix, for example: IZUDFLT.ZOSMF.LINK.**mylink**

You can specify a suffix of up to 220 alphanumeric characters (A-Z a-z 0-9) and the following special characters: underscore (_), dash (-), period (.). The use of a period in a resource name is treated as a qualifier. As such, the first character after a period must be A-Z or a-z.

You must provide a unique SAF resource name suffix for each link. z/OSMF uses the resource name for locating and identifying links.

LinkLaunchWorkArea

Specify how the link is to open in the user's z/OSMF session, as follows:

- To have the link open in the user's session as a separate window or tab, set this value to FALSE. The link will open in the user's browser as a new window or tab, based on the user's browser settings.
- To have the link open as a tab in the z/OSMF desktop (like a z/OSMF task), set this value to TRUE.

Any other value is ignored and FALSE is used by default.

If you choose to have the link open as z/OSMF tab, verify that the link will work as intended in the z/OSMF desktop. You might find that some links display better in a separate browser window or tab. Also, some external web sites might cause the user's browser window to be re-sized or even redirect the browser to a new destination, rather than opening in the z/OSMF desktop. Therefore, it is strongly recommended that you verify the general usage of the link in z/OSMF before directing others to use the link.

Figure 44 on page 258 shows an example of a completed link definition.

```
LinkName=IBM
LinkURL=http://www.ibm.com
LinkAuthorizedRoles=z/OSMF Guest, z/OSMF User
LinkSafSuffix=IBM_COM
LinkLaunchWorkArea=false
```

Figure 44. Example of a link definition

3. **Restart the z/OSMF server to make your changes effective.** The new link does not appear in the z/OSMF desktop until after z/OSMF is started.

To start z/OSMF, enter the appropriate START command.

Managing security for links in z/OSMF

In z/OSMF, a link in the z/OSMF user interface is treated as a resource. Your installation should determine whether access to a particular link is to be limited to certain users or be unrestricted. This topic describes the security considerations for managing links in z/OSMF.

Managing a link in z/OSMF involves the following steps:

- Defining the link to z/OSMF through the Links task
- Controlling access to the link through the ZMFAPLA resource class profile.

The z/OSMF configuration process defines a generic resource profile for links and permits groups to it. Specifically, links in z/OSMF are protected under the generic resource profile: `<SAF-prefix>.ZOSMF.LINK.**` where `<SAF-prefix>` is the SAF profile prefix that was defined for your configuration (IZUDFLT by default). z/OSMF permits the groups for z/OSMF users (IZUUSER) and z/OSMF administrators (IZUADMIN) to this profile. As a result, these users will be able to see all of the links in the desktop interface. z/OSMF does not, by default, permit the z/OS security administrator role to the ZOSMF.LINK** profile.

For more information about the Links task, see the online help.

Defining a link as a protected resource

Depending on your installation's security procedures, a link might require further protection through a discrete profile. When planning for new links, it is recommended that the z/OSMF Administrator work with the security administrator to determine whether a new link requires protection through a discrete profile.

In the Links task, the z/OSMF Administrator defines a link by specifying a name for the link and its URL. The Links task also includes a text entry window that requires the z/OSMF Administrator to further qualify the link resource name with a SAF resource name, which can be used if a discrete profile is required for the link. If so, the z/OSMF Administrator can provide this fully-qualified resource name to the security administrator to use to create the user authorizations for the link.

As an example, [Figure 45 on page 259](#) shows the RACF commands that a security administrator can use to define a discrete profile for a new link (the z/OS Basics Information Center web site) and permit a group (IZUUSER) to that link.

```
RDEFINE ZMFAPLA (IZUDFLT.ZOSMF.LINK.Z_OS_BASICS_INFORMATION_CENTER) UACC(NONE)
PERMIT IZUDFLT.ZOSMF.LINK.Z_OS_BASICS_INFORMATION_CENTER CLASS(ZMFAPLA) ID(IZUUSER) ACC(READ)
```

Figure 45. Example: Defining a link resource name and permitting a group to it

If you change a link SAF resource name through the Links task, ensure that the new link resource name is adequately protected through a ZMFAPLA resource profile definition. You might need to create a new profile to properly secure the link.

Deleting an existing link will potentially require that your security administrator delete the discrete profile, if one is used to secure access to the link.

Chapter 50. Deleting incidents and diagnostic data

For installations that use the Incident Log task, the **ceatool** program provides a command line interface for deleting the incidents that you no longer want to retain.

When an incident occurs, the system typically creates an SVC dump and collects diagnostic log snapshots of the operations log, error log, and error log summary. This data can consume a large amount of system resources, such as DASD space and logstream slots, if incidents are not periodically deleted. To delete incidents, you can use the delete option provided in the **ceatool** command-line interface.

Tip: You can also use the **Delete Incident** action provided in the Incident Log task. For instructions, see the topic about *Deleting incidents* in the z/OSMF online help.

Overview

The **ceatool** command-line interface is a utility that you can use to send requests to the z/OS common event adapter (CEA) component. With this utility, you can manage the incidents that were created for the z/OSMF Incident Log task. Specifically, you can use a z/OS UNIX System Services shell, a JCL job, or a cron job to delete incidents and the associated diagnostic data. The diagnostic data to be deleted includes:

- Error log
- Error log summary
- Operations log
- Entry for the dump in the sysplex dump directory
- SVC dump (optional)

Note: The utility deletes only incidents that are not associated with a problem number or tracking ID. These incidents are referred to as *inactive incidents*. The utility ignores all active incidents. To delete active incidents, use the **Delete Incident** action provided in the Incident Log task.

Before invoking the utility

Before invoking the utility, complete the following steps:

1. Ensure that the common event adapter (CEA) component and the System REXX (SYSREXX) component are active on your z/OS system. For instructions, see [“Ensure that common event adapter \(CEA\) is configured and active” on page 17](#) and [“Ensuring that System REXX is set up and active” on page 192](#).
2. Ensure that the user ID you are using to invoke the utility is authorized to access SAF resource CEA.CEAPDWB.CEADELETEINCIDENT, which is defined in the SERVAUTH class.
3. Ensure that the PATH environment variable is set to the directory in which the utility is installed. By default, the utility is installed in the /bin directory.
4. Ensure that the NLSPATH environment variable contains /usr/lib/nls/msg/%L/%N, which is, by default, the directory in which the CEA message catalog, called *ceamsg.cat*, is installed.

If these requirements are not satisfied, errors will occur when you invoke the utility.

When you configure the Incident Log service for z/OSMF, you specify a high-level qualifier to use for naming log snapshot data sets. By default, this value is CEA. z/OS V2R1 increased the allowable length of this high-level qualifier from four- to eight-characters through the new HLQLONG statement in member CEAPRMxx. If your installation uses systems with a mix of shorter and longer high-level qualifiers, be sure to run the **ceatool** program from a system in your sysplex that specifies the HLQLONG value. Doing so ensures allows the **ceatool** program to locate all dump data sets, regardless of which style of high-level qualifier is used.

Invoking the utility

The **ceatool** command-line interface must be invoked from the z/OS UNIX System Services shell or a BPXBATCH environment. [Figure 46 on page 262](#) shows the format of the **ceatool** command, which invokes the utility.

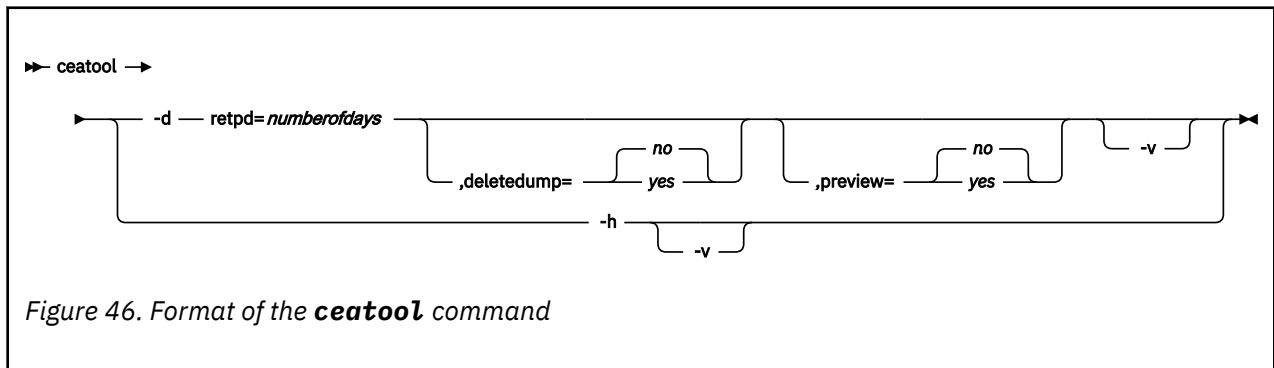


Figure 46. Format of the **ceatool** command

Where:

-d

Deletes incidents that satisfy the specified criteria. Use the following options to identify the incidents to be deleted:

retpd=numberofdays

This is a required parameter that indicates the number of days an incident must be kept before it can be deleted. All inactive incidents that are older than the retention period will be deleted. The value for *numberofdays* can be any whole number in the range of 0 - 9999.

The retention period is derived from the current time. For example, if the retention period is one (retpd=1) and the current time is 10:00 am, all incidents that occurred at or before 10:00 am yesterday will be deleted.

To delete all inactive incidents, use a retention period of zero (retpd=0).

deletedump

This is an optional parameter that indicates whether the SVC dumps associated with an incident will be deleted. The value can be:

yes

All diagnostic data associated with an incident, including the SVC dumps, will be deleted when the incident is deleted.

no

All diagnostic data associated with an incident, except the SVC dumps, will be deleted when the incident is deleted. This is the default.

Specify this value if your installation has procedures or policies for managing dump data sets. Doing so instructs the utility to ignore the dump data sets during delete processing.

preview

This is an optional parameter that indicates whether to activate preview mode. The value can be:

yes

Preview mode is enabled. In this case, the incidents that match the filter criteria will *not* be deleted. Instead, the tool will provide the number of incidents that are candidates for deletion.

no

Preview mode is disabled. In this case, the incidents that match the filter criteria will be deleted. This is the default.

-v

Activates verbose mode, which issues additional diagnostic messages while the **ceatool** command is processing.

-h

Displays usage help for the **ceatool** command.

You can use a JCL or cron job to invoke the utility, or you can enter the commands directly in the z/OS UNIX shell. To invoke the utility using a batch job, see sample job CEATool, which is supplied by IBM in SYS1.SAMPLIB(CEATool).

Important: Do not submit multiple, concurrent requests to delete incidents using the **ceatool** utility. Otherwise, errors might occur.

Examples

Table 40 on page 263 provides sample commands to invoke the **ceatool** utility and describes the expected result for each command.

Table 40. Sample ceatool commands	
Sample Command	Results
<code>ceatool -d retpd=7,deletedump=no</code>	Deletes inactive incidents and the corresponding diagnostic data, excluding SVC dumps, that are older than seven days.
<code>ceatool -d retpd=7,deletedump=yes</code>	Deletes inactive incidents and the corresponding diagnostic data, including SVC dumps, that are older than seven days.
<code>ceatool -d retpd=7,deletedump=yes -v</code>	Deletes inactive incidents and the corresponding diagnostic data, including SVC dumps, that are older than seven days. Because verbose mode is requested, additional diagnostic messages are displayed during processing.
<code>ceatool -d retpd=0</code>	Deletes all inactive incidents and the corresponding diagnostic data, excluding SVC dumps.
<code>ceatool -d retpd=7,preview=yes</code>	Displays the number of inactive incidents that are older than seven days. The incidents that satisfy the filter criteria are not deleted.
<code>ceatool -h</code>	Displays help for the ceatool command.
<code>ceatool -hv</code>	Displays help for the ceatool command, plus an additional message with the build date.

Verifying that the incidents were deleted

To verify that the incidents were deleted, complete one of the following steps:

- Display the list of incidents in the z/OSMF Incident Log task, and verify that the incidents in the specified retention period are not listed.
- Check the contents of the sysplex dump directory, and verify that the incidents in the specified retention period are not listed.

Note: If the utility encounters an error during delete processing, the processing will stop and any incidents that were *not* deleted before the error occurred will still be listed in the incident log and the sysplex dump directory.

Chapter 51. Setting the KCINDEX parameter

On a performance constrained z/OS system, or a simulation system such as zPDT, z/OSMF might take longer to start and might consume more CPU resource during start-up. In such cases, it might be possible to improve the performance of z/OSMF start-up by temporarily bypassing the rebuilding the IBM Documentation index for the z/OSMF online help system. Rebuilding the index allows for recent changes to the help system to be included in help system searches. By default, the index is rebuilt every time the z/OSMF server is restarted. However, you might want to disable the rebuilds for some or most of the z/OSMF server starts to improve performance.

To control whether the IBM Documentation index is rebuilt, set the parameter KCINDEX in the z/OSMF server started procedure (IZUSVR1 and IZUSVR2), as follows:

- If the KCINDEX parameter is specified as N, the index is not rebuilt when the z/OSMF server is started. This setting can shorten the server start-up time and lower z/OSMF CPU usage. However, any newly applied service updates to the z/OSMF help system are not available in searches until the index is rebuilt.
- If the KCINDEX parameter is set to Y, or KCINDEX parameter is omitted, the index is rebuilt every time the z/OSMF server is started. This is the default behavior, which is intended to keep the z/OSMF searches synchronized with the z/OSMF help system.

If you suspect that z/OSMF performance is slow because your system is constrained, consider restarting the z/OSMF server with index rebuilding disabled. Do this after the first time startup of z/OSMF, so that the indices needed for z/OSMF help system searches are created. Later, if new help files are introduced through service, it is recommended that you allow the index to be rebuilt, so that the new help files are included in searches.

You can disable index rebuilding dynamically by starting the z/OSMF server, as follows:

```
S IZUSVR1,KCINDEX=N
```

Chapter 52. Enabling JSON Web Token support

You can configure the z/OSMF server to build and use JSON Web Token (JWT) tokens.

z/OSMF supports the use of JWT tokens, as follows:

- The z/OSMF server returns a JWT token after the user authenticates with the z/OSMF server
- The z/OSMF-provided JWT token can be decrypted by a remote web application with or without requiring a connection to the z/OSMF server.
- The z/OSMF-provided JWT token can be used to access z/OSMF REST services, similar to the use of LTPA tokens.

By default, the JWT function is disabled on the z/OSMF server. For information about enabling the JWT function, see the sections that follow.

How to enable JWT function on z/OSMF server

To enable the JWT function on z/OSMF server, do the following:

1. Copy the file `server_override.xml` from `<product_dir>/defaults/servers/zosmfServer/` to `<user_dir>/configuration`

Where:

- `<product_dir>` is the z/OSMF product directory. By default, this is `/usr/lpp/zosmf`
 - `<user_dir>` is the z/OSMF data directory. By default, this is `/global/zosmf`
2. Set the permissions to 755 for the file `server_override.xml` in `<user_dir>/configuration`. For example:

```
chmod 755 <user_dir>/configuration/server_override.xml
```

3. Restart the z/OSMF server.

As a result, the JWT support is enabled with default values. Usually, the default values are sufficient for most installations.

How to obtain the JWT token from the z/OSMF server

On successful SAF authentication with the z/OSMF server, an application can receive both the JWT token and the LTPA token.

In the following example, assume that the POST request is issued with a valid user ID and password. If so, the JWT token is stored in the browser cookies:

```
POST /zosmf/services/authenticate
```

For more information about authenticating with z/OSMF, see [IBM z/OS Management Facility Programming Guide](#).

How to configure the JWT settings for the z/OSMF server

You can customize different aspects of the JWT token processing, as described in the following sections:

- [“JWT Single Sign On” on page 268](#)
- [“JWT builder” on page 268](#)
- [“MicroProfile JWT token” on page 269](#).

JWT Single Sign On

In the `server_override.xml` file, locate the following statement. Here, you specify the settings for configuring JWT Single Sign On.

```
<jwtSso cookieName="jwtToken" jwtBuilderRef="zOSMFBuilder" includeLtpaCookie="true"
useLtpaIfJwtAbsent="true" />
```

The parameters of this statement are described in [Table 41 on page 268](#).

Table 41. Statement for configuring JWT single sign-on			
Parameter	Type	Default value	Description
cookieName	String	jwtToken	Name of the cookie that is used to store the JWT token.
jwtBuilderRef	A reference to top-level jwtBuilder element (string).	zOSMFBuilder	A reference to the JWT Builder configuration element in <code>server.xml</code> that describes how to build the JWT token.
includeLtpaCookie	Boolean	true	After successful authentication with a JWT token, include an LTPA cookie in addition to the JWT cookie. z/OSMF requires this setting to be TRUE.
useLtpaIfJwtAbsent	Boolean	true	If the JWT cookie is missing, attempt to process an LTPA cookie if it is present. z/OSMF requires this setting to be TRUE.

For more information, see https://www.ibm.com/docs/en/was-liberty/zos?topic=SS7K4U_liberty/com.ibm.websphere.liberty.autogen.nd.doc/ae/rwlp_config_jwtSso.htm

JWT builder

In the `server_override.xml` file, locate the following statement. Here, you can specify the elements and attributes that are used to build the JWT token.

```
<jwtBuilder id="zOSMFBuilder" issuer="zOSMF" keyAlias="DefaultzOSMFCert.IZUDFLT"
expiresInSeconds="${izu.ltpa.expiration}"/>
```

The parameters of this statement are described in [Table 42 on page 268](#).

Table 42. Statement for configuring the JWT builder			
Parameter	Type	Default value	Description
id	String	zOSMFBuilder	This ID is used to identify the JWT builder.
issuer	String	zOSMF	The issuer information.

Table 42. Statement for configuring the JWT builder (continued)

Parameter	Type	Default value	Description
keyAlias	String	DefaultzOSMFCert.IZUDFLT	A key alias name that is used to locate the private key for signing the token with an asymmetric algorithm. This value is the certificate label value.
expiresInSeconds	A time period with second precision	<code>\${izu.ltpa.expiration}</code>	Indicates the token expiration time in seconds. z/OSMF requires JWT token expiration time be equal to LTPA token expiration time, so use one variable to set it. This value can be set by the statement SESSION_EXPIRE in parmlib.

For more information, see https://www.ibm.com/docs/en/was-liberty/zos?topic=SS7K4U_liberty/com.ibm.websphere.liberty.autogen.nd.doc/ae/rwlp_config_jwtBuilder.htm

MicroProfile JWT token

In the server_override.xml file, locate the following statement, which is used to configure the MicroProfile JWT token:

```
<mpJwt id="myMpJwt" issuer="zOSMF" jwksUri="https://${izu.jwks.hostname}:${izu.https.port}/jwt/ibm/api/zOSMFBuilder/jwk" />
```

The parameters of this statement are described in Table 43 on page 269.

Table 43. The statement to configure JWT builder

Parameter	Type	Default value	Description
id	String	myMpJwt	The unique ID.
issuer	String	zOSMF	The issuer information. It should match the issuer value in Builder.
jwksUri	String	<code>https://\${izu.jwks.hostname}:\${izu.https.port}/jwt/ibm/api/zOSMFBuilder/jwk</code>	Specifies a JSON Web Key service URL.

For more information, see https://www.ibm.com/docs/en/was-liberty/zos?topic=SS7K4U_liberty/com.ibm.websphere.liberty.autogen.nd.doc/ae/rwlp_config_mpJwt.htm

Chapter 53. Configuring the z/OSMF workflow signing certificate

If you plan to use the workflow signing function, you must create a workflow signing certificate and connect it to the z/OSMF server key ring. This topic provides the steps for the configuration.

Pre-configuration - Specifying the certificate label for SSL

To use the workflow signing function, you must specify the certificate label for SSL before you create a workflow signing certificate. This can be done by adding the file `server_override.xml`. If it is not specified, the certificate to be used for SSL might change when you connect the workflow signing certificate to your z/OSMF server key ring.

The following is an example `server_override.xml`. Replace the value `DefaultzOSMFCert.IZUDFLT` with your actual certificate label name. If you do not know the current SSL certificate settings, consult your security administrator for more information.

```
<server>
<variable name="sslCertLabel" value="DefaultzOSMFCert.IZUDFLT"/>
<ssl id="izuSSLConfig" serverKeyAlias="${sslCertLabel}"/>
</server>
```

Place `server_override.xml` in `/global/zosmf/configuration` if you use `/global/zosmf` as your z/OSMF user directory.

You must restart z/OSMF after this operation.

Configuration – Creating a workflow signing certificate and connect it to the z/OSMF server key ring

This information assumes the use of RACF. If you use another external security manager (ESM), contact the vendor for more information.

Note: The z/OSMF sample job `IZUWFSEC` contains RACF commands for creating a workflow signing certificate and connecting it to z/OSMF server key ring. You might find that the commands in `IZUWFSEC` are sufficient for your needs. If so, have your security administrator review these sample jobs carefully before you submit them.

Create a new certificate specifically for z/OSMF workflow signing. The certificate is owned by the z/OSMF server and is signed by a local certificate authority (CA).

The z/OSMF sample job `IZUWFSEC` contains RACF commands for storing the workflow signing certificate in the key ring. These constructs are named, as follows:

- The key ring name is `IZUKeyring.IZUDFLT`
- The CA name is:

```
('z/OSMF CertAuth for Security Domain')
OU('SAF_PREFIX')
WITHLABEL('zOSMFCA')
```

To configure a new certificate and connect it to the z/OSMF server key ring, follow these steps:

1. Create the workflow signing certificate.

```
/* Create the workflow signing certificate "WFSigningCert.IZUDFLT" *
/* Create the workflow signing certificate for the z/OSMF server */
/* Change HOST NAME in CN field to the real host name. */
/* Usually the format of the host name is 'XXXX.XXX.XXX.XXX' */
RACDCERT GENCERT ID(IZUSVR) +
SUBJECTSDN(CN('HOST NAME') +
```

```
O('IBM') +
OU('IZUDFLT')) +
RSA +
SIZE(2048) +
NOTAFTER( DATE(2030-07-31) ) +
WITHLABEL('WFSigningCert.IZUDFLT') +
SIGNWITH(CERTAUTH LABEL('zOSMFCA'))
```

2. Set the workflow signing certificate to TRUST state.

```
RACDCERT ALTER(LABEL('WFSigningCert.IZUDFLT')) ID(IZUSVR) TRUST
```

3. Connect the workflow signing certificate to the z/OSMF server key ring.

```
RACDCERT ID(IZUSVR) CONNECT (LABEL('WFSigningCert.IZUDFLT') +
RING(IZUKeyring.IZUDFLT))
```

4. Verify that the certificate is set up correctly, as follows:

- The workflow signing certificate indicates that it has a private key.
- The workflow signing certificate has TRUST status and is connected to the key ring.

After completing these steps, add the workflow signing certificate `WFSigningCert.IZUDFLT` to the Workflow Settings page as the Workflow signing keyLabel.

Post-configuration - Changing workflow signing certificate

The step signatures are generated using the workflow signing certificate and stored in the z/OSMF workflow definition file. z/OSMF Workflows and Workflow Editor verify the signatures using the workflow signing certificate.

If the step signing fails, the signature cannot be generated in the workflow definition. If the verification of signed step fails, the workflow definition cannot be used to create a workflow instance, and the existing workflow instance cannot run the signed `runAsUser` step under `runAsUser` identity. The failure can occur when your certificate expires or when you enable a new certificate for workflow signing.

When a certificate expires, if the certificate is a self-signed certificate, you can use RACF to renew the certificate without changing the private key. You must restart z/OSMF after extending the expiry date of the certificate.

To prevent a potential outage caused by expired certificates, renew the certificate without changing the key before the certificate expires.

If you cannot renew the certificate due to the security policy, you must create a new certificate. After do that, you must re-sign the `runAsUser` step that had been signed with the old certificate.

To use a new certificate for workflow signing, make sure you do the following before the certificate expires.

1. If the label name changes, put the new label name to workflow settings page.
2. Re-sign for the `runAsUser` steps which had been signed with the old certificate in Workflow Editor.
3. Create the workflow instance with the new workflow definition file that is signed by the new certificate.

Part 6. Troubleshooting problems

You can optionally perform additional tasks to enhance your z/OSMF configuration. z/OSMF administrators are the most likely IT personnel to participate in this activity.

Post-configuration in z/OSMF includes the following topics:

- [Chapter 54, “Troubleshooting problems,” on page 275](#)
- [Chapter 55, “Configuration messages,” on page 309.](#)

Chapter 54. Troubleshooting problems

This chapter provides tips and techniques for troubleshooting common problems. Included are procedures and methods for performing problem determination and for determining the status of the different components.

This chapter is organized into topics, as follows:

- “Resources for troubleshooting” on page 275
- “Tools and techniques for troubleshooting” on page 276
- “Common problems and scenarios” on page 289.

Resources for troubleshooting

z/OSMF is composed of a number of system "layers," each maintaining a different set of diagnostic information. Some errors that are intercepted at the lowest system levels can surface at the user interface layer. Some errors appear as messages in a CIM log, and others might be issued as standard z/OS messages to the system logs (SYSLOG or OPERLOG).

Table 44 on page 275 shows a summary of the diagnostic tools and data available for each of the layers in the z/OSMF stack and references for locating the information.

Table 44. Summary of tools and information for troubleshooting problems with z/OSMF			
Component or task	Tools to assist with troubleshooting	Where described	Associated messages
Workstation and web browser	Environment checker tool	“Verifying your workstation with the environment checker” on page 276.	N/A
z/OSMF core functions and system management tasks	• The About page • z/OSMF Diagnostic Assistant • z/OSMF log files and tracing.	• “Finding information about z/OSMF” on page 284 • “Using the z/OSMF Diagnostic Assistant” on page 284 • “z/OSMF log files” on page 285 • “Problems when using Network Configuration Assistant” on page 301.	Messages encountered while configuring z/OSMF; see Chapter 55, “Configuration messages,” on page 309. z/OSMF messages. For assistance, click on the message help link. For Network Configuration Assistant, messages and pop-ups are supplied with the task.
z/OSMF server	z/OSMF log files and tracing.	“IZUPRMxx reference information” on page 35.	• Chapter 55, “Configuration messages,” on page 309 • z/OSMF messages. For assistance, click on the message help link.
WebSphere Liberty profile	Troubleshooting information is provided in the WebSphere Application Server for z/OS information center.	See the topics at: http://www.ibm.com/software/webervers/appserv/was/library/v85/was-zos/index.html .	Messages prefixed by CW.
CIM server and CIM providers	• CIM server logging • CIM server trace • CIM provider trace.	These options are defined in the CIM server configuration properties and set through the cimconfig command; see z/OS Common Information Model User's Guide .	z/OS Common Information Model User's Guide .
Common event adapter (CEA)	System commands: • MODIFY CEA • MODIFY AXR • TRACE CT.	z/OS MVS System Commands	z/OS MVS System Messages for information about: • WTO messages • CTRACE • Reason codes.

Tools and techniques for troubleshooting

This section describes the tools and techniques available for troubleshooting problems with z/OSMF.

Verifying your workstation with the environment checker

To work with z/OSMF, your web browser and workstation require a number of settings for proper functioning. z/OSMF includes an environment checker tool to help you verify these settings. The environment checker tool inspects your web browser and workstation operating system for adherence to z/OSMF requirements and recommended settings.

Before you run the tool

Check to ensure that your workstation is set up correctly for z/OSMF. See [“Preparing your workstation for z/OSMF”](#) on page 10.

Your workstation requires a compatible operating system and web browser. For more information, see [“Software prerequisites for z/OSMF”](#) on page 7.

Ensure that your browser is enabled for JavaScript. For instructions, see [Table 46 on page 278](#) or [Table 47 on page 280](#).

Running the tool

To run the tool, do the following:

1. Open a web browser to the environment checker tool:

`https://hostname:port/zosmf/IzuUICommon/environment.jsp`

Where:

- *hostname* is the hostname or IP address of the system on which z/OSMF is installed
- *port* is the secure application port.

To find the hostname and port, see the link for z/OSMF in message IZUG349I. This message was written to the z/OSMF server job log, as described in [“Step 4: Start the z/OSMF server”](#) on page 28.

2. Follow the instructions for your particular browser in the online help for the tool.

Understanding the results of the tool

[Table 45 on page 276](#) describes the layout of the environment checker report.

Table 45. Columns in the environment checker tool results page

Column	Description
Environment Option	Browser setting that was examined by the environment checker tool.

Table 45. Columns in the environment checker tool results page (continued)

Column	Description
Settings as of <i>date-time</i>	Findings from the most recent invocation of the tool. This column indicates potential problems with your browser. In the column heading, the date and time (<i>date-time</i>) is represented in ISO 8601 format, a standard provided by the International Organization for Standardization (ISO). In this format: • Calendar date is represented in year-month-day format (<i>yyyy-mm-dd</i>). • Time of day (<i>T</i>) is based on the 24-hour clock: <i>hh:mm:ss:mmm</i>. • <i>Z</i> indicates zero offset from Coordinated Universal Time (UTC). In the report, the status of each setting is indicated, as follows: Items that are marked with a critical icon X Setting is not correct for z/OSMF. You must fix this problem before you continue with z/OSMF. Items that are marked with a warning symbol ! Setting is not optimal for z/OSMF. It is recommended that you update the setting before you continue with z/OSMF. No error indication Setting is correct for z/OSMF.
Requirements	Recommended setting for your environment.

For the steps to resolve a problem, see the appropriate entry in the tool's online help. After you update a setting, use the browser reload button to run the environment checker again. Repeat this process until you resolve all of the errors and warnings.

If you are using the Microsoft Edge browser:

- When you are working with WLM service definitions, ensure that automatic prompting for file downloads is enabled for the web link (a URL) to the active z/OSMF instance. See [“Enabling automatic prompting for file downloads” on page 283](#).
- When you are working with Resource Monitoring task, users who plan to export the data that is collected in a dashboard to a CSV file should ensure that automatic prompting for file downloads is enabled. See [“Enabling automatic prompting for file downloads” on page 283](#).

If you are using the Microsoft Edge browser on a Windows 10 system:

- When you are working with the Sysplex Management task or the System Status task, you might experience problems with the zoom function on graphical views. If so, upgrade the browser to a later version and try again. If the problem persists, use another tested browser.

For a list of the supported web browsers, see [“Software prerequisites for z/OSMF” on page 7](#).

Recommended settings for the Mozilla Firefox browser

[Table 46 on page 278](#) shows the recommended settings for the Mozilla Firefox browser.

Table 46. Recommended settings for Firefox

Environment Option	Response
JavaScript	To work with z/OSMF, your browser must have JavaScript enabled. To enable JavaScript, do the following: 1. From the <i>Tools</i> menu, click Options > Content tab. 2. Ensure that the JavaScript check box is selected. 3. Click OK.
Cookies	To work with z/OSMF, your browser must have cookies enabled—if not for all sites, then at least for the z/OSMF site at your installation. To enable cookies for use by any site, do the following: 1. From the <i>Tools</i> menu, click Options > Privacy tab. 2. Ensure that the Accept cookies from sites check box is selected. 3. Click OK. To enable cookies for only the z/OSMF site, clear the Accept cookies from sites check box. Then, do the following: 1. Click Exceptions. 2. Enter the URL for the z/OSMF site at your installation. 3. Click Enable > Close > OK.
Pop-up Windows	For proper functioning with z/OSMF, your browser must be enabled for pop-up windows. To enable your browser for pop-up windows, do the following: 1. From the <i>Tools</i> menu, click Options > Content tab. 2. Clear the Block pop-up windows check box. 3. Click OK. To enable pop-up windows for the z/OSMF site only, ensure that the Block pop-up windows check box is selected. Then, do the following: 1. Click Exceptions. 2. Enter the URL for the z/OSMF site at your installation. 3. Click Allow > Close > OK.
Frames	To work with z/OSMF, your browser must have frames enabled. By default, the Firefox browser is enabled for frames. If you need to enable your browser for frames, do the following: 1. In the browser input area, enter the following URL: <code>about:config</code>. 2. If a warranty warning message appears, click the I'll be careful, I promise! button to continue. 3. In the Filter field, enter <code>frames</code>. 4. Click <code>browser.frames.enabled</code> to set the Value field to <code>true</code>. 5. Close the browser to save the changes.

Table 46. Recommended settings for Firefox (continued)

Environment Option	Response
Screen Resolution	For optimal viewing with z/OSMF, your workstation requires a minimum screen resolution of 1024 by 768 pixels. To increase the screen resolution, do the following: 1. Right-click on the desktop and select Properties > Settings tab. 2. Move the slider to select a screen resolution of at least 1024 by 768 pixels. 3. Click OK.
Browser Content Dimensions	For optimal viewing with z/OSMF, your browser requires a usable content display area of at least 800 by 600 pixels. A number of factors can affect the size of your browser's usable content display area, such as Windows desktop appearance settings and the inclusion of toolbars for browser plug-ins. To check the desktop appearance settings, do the following: 1. Right-click on the desktop and select Properties to open the <i>Display Properties</i> dialog box. 2. Click the Appearance tab. 3. Click Advanced. 4. From the Item list, select Active Title Bar and verify that it is no larger than necessary (the default is 25 pixels). Similarly, check the setting for Scrollbar (the default is 17 pixels). 5. Click OK > OK. To remove unnecessary toolbars, do the following: 1. From the <i>View</i> menu in Firefox, click Toolbars. 2. For any unnecessary toolbars, clear the associated check box. As an alternative, you can maximize the browser window, thus eliminating the toolbars, by pressing the F11 function key. To restore the window to its previous size, press F11 again.
Add-ons	For optimal performance with z/OSMF, disable the Firebug add-on in your browser settings. To disable the Firebug add-on, do the following: 1. From the <i>Tools</i> menu, click Add-ons > Extensions tab. 2. Select the Firebug add-on and click the Disable option. 3. Restart the browser to have the changes take effect.

Table 46. Recommended settings for Firefox (continued)

Environment Option	Response
Plug-ins	Some plug-ins, such as JavaScript debuggers, can affect browser performance. For optimal performance with z/OSMF, include only required plug-ins with your browser. In the environment checker report, the Settings column shows the installed plug-ins for your browser. To verify this list, do the following: 1. In the browser input area, enter the following URL: <code>about:plugins</code>. 2. Compare the list of installed plug-ins to the list shown in the environment checker report to determine whether any add-ons should be disabled. To disable a plug-in, do the following: 1. From the <i>Tools</i> menu, click Add-ons > Plugins tab. 2. Scroll down the list to locate the plug-in. 3. Select the plug-in and click the Disable option. 4. Restart the browser to have the changes take effect.
Scrollbar	For optimal viewing with z/OSMF, enable the setting to always show scrollbars in your workstation. To enable this option, do the following: Mac: 1. Open Systems Settings, and select Appearance. 2. Under the Show scroll bars section, select the Always option. Windows: 1. From the Start menu , click Settings > Accessibility > Visual effects. 2. Toggle the Always show scrollbars setting to On.

Recommended settings for the Microsoft Edge browser

Table 47 on page 280 shows the recommended settings for Microsoft Windows web browsers.

For more considerations, see the following topics:

- “Considerations for Microsoft web browsers” on page 283
- “Enabling automatic prompting for file downloads” on page 283

Table 47. Recommended settings for Microsoft Edge web browser.

Environment Option	Response
JavaScript	To work with z/OSMF, your browser must have JavaScript enabled. To enable JavaScript, do the following: 1. From the <i>Tools</i> menu, click Internet Options > Security tab. 2. Click Custom Level. 3. Scroll down to <i>Scripting</i>, then <i>Active Scripting</i>. 4. Click Enable. 5. Click OK > OK.

Table 47. Recommended settings for Microsoft Edge web browser. (continued)

Environment Option	Response
Cookies	To work with z/OSMF, your browser must have cookies enabled—if not for all sites, then at least for the z/OSMF site at your installation. To enable cookies for use by any site, do the following: 1. From the <i>Tools</i> menu, click Internet Options > Privacy tab. 2. Click Advanced. 3. Select the Override automatic cookie handling check box. 4. Select Accept for <i>First-party Cookies</i> and <i>Third-party Cookies</i>. 5. Click OK > OK. To enable cookies for only the z/OSMF site, clear the Override automatic cookie handling check box and select Block for <i>First-party Cookies</i> and <i>Third-party Cookies</i>. Then, do the following: 1. From the <i>Tools</i> menu, click Internet Options > Privacy tab. 2. Click Sites. 3. Enter the URL for the z/OSMF site at your installation. 4. Click Allow. 5. Click OK > OK.
Pop-up Windows	For proper functioning with z/OSMF, your browser must be enabled for pop-up windows. To enable your browser for pop-up windows, do the following: 1. From the <i>Tools</i> menu, click Internet Options > Privacy tab. 2. Clear the Turn on Pop-up Blocker check box. 3. Click OK. To enable pop-up windows for the z/OSMF site only, ensure that the Turn on Pop-up Blocker check box is selected. Then, do the following: 1. Select Settings 2. Enter the URL for the z/OSMF site at your installation. 3. Click Add. 4. Click Close > OK.
Frames	To work with z/OSMF, your browser must have frames enabled. To enable your browser for frames, do the following: 1. From the <i>Tools</i> menu, click Internet Options > Security tab. 2. Click Custom Level. 3. Scroll down to <i>Miscellaneous</i>, then <i>Launching programs and files in an IFRAME</i>. 4. Click Enable. 5. Click OK.

Table 47. Recommended settings for Microsoft Edge web browser. (continued)

Environment Option	Response
Screen Resolution	For optimal viewing with z/OSMF, your workstation requires a minimum screen resolution of 1024 by 768 pixels. To increase the screen resolution, do the following: 1. Right-click on the desktop and select Properties > Settings tab. 2. Move the slider to select a screen resolution of at least 1024 by 768 pixels. 3. Click OK.
Browser Content Dimensions	For optimal viewing with z/OSMF, your browser requires a usable content display area of at least 800 by 600 pixels. A number of factors can affect the size of your browser's usable content display area, such as Windows desktop appearance settings and the inclusion of tool bars for browser plug-ins. To check the desktop appearance settings, do the following: 1. Right-click on the desktop and select Properties to open the <i>Display Properties</i> dialog box. 2. Click the Appearance tab. 3. Click Advanced. 4. From the Item list, select Active Title Bar and verify that it is no larger than necessary (the default is 25 pixels). Similarly, check the setting for Scrollbar (the default is 17 pixels). 5. Click OK > OK. To remove unnecessary tool bars, do the following: 1. From the <i>View</i> menu, click Toolbars. 2. For any unnecessary toolbars, clear the associated check box. As an alternative, you can maximize the browser window, thus eliminating the toolbars, by pressing the F11 function key. To restore the window to its previous size, press F11 again.
Add-ons	For optimal performance with z/OSMF, it is recommended that you include only required add-ons with your browser. To disable an add-on, do the following: 1. From the <i>Tools</i> menu, click Manage Add-ons > Enable or Disable Add-ons. 2. Scroll down the list to view the add-ons. 3. To disable an add-on, select it and click the Disable button. 4. Click OK. 5. Restart the browser to have the changes take effect.

Table 47. Recommended settings for Microsoft Edge web browser. (continued)

Environment Option	Response
Plug-ins	Some plug-ins, such as JavaScript debuggers, can affect browser performance. For optimal performance with z/OSMF, it is recommended that you include only required plug-ins with your browser. In the environment checker report, the Settings column shows the installed plug-ins for your browser. To verify this list, do the following: 1. From the <i>Tools</i> menu, click Manage Add-ons > Enable or Disable Add-ons. 2. Scroll down the list to view the add-ons. 3. To disable an add-on, select it and click the Disable button. 4. Click OK. 5. Restart the browser to have the changes take effect.

Considerations for Microsoft web browsers

If you are using a Microsoft web browser, such as Microsoft Edge:

- When you are working with WLM service definitions, ensure that automatic prompting for file downloads is enabled for the web link (a URL) to the active z/OSMF instance. See [“Enabling automatic prompting for file downloads”](#) on page 283.
- When you are working with Resource Monitoring task, users who plan to export the data that is collected in a dashboard to a CSV file should ensure that automatic prompting for file downloads is enabled. See [“Enabling automatic prompting for file downloads”](#) on page 283.

Enabling automatic prompting for file downloads

If you are using a Microsoft web browser, such as Microsoft Edge, to work with WLM service definitions or RMF exported data, ensure that automatic prompting for file downloads is enabled for the web link (a URL) to the active z/OSMF instance. If the feature is disabled, when you attempt to display the *File Download* dialog box, the browser window refreshes and all of your selections and unsaved changes are discarded. To enable automatic prompting for file downloads, use one of the procedures described in this section, depending on the version of the Internet Explorer browser.

Procedure

1. From the *Tools* menu, click **Internet Options > Security** tab.
2. Under *Select a zone*, click **Local intranet**.
3. Click **Sites**.
4. Click **Advanced**.
5. If the URL to the active z/OSMF instance is listed in the Add this web site to the zone field, click **Add**. Otherwise, enter the URL, and then click **Add**.
6. Click **Close**.
7. Click **OK**.
8. Click **OK**.

Finding information about z/OSMF

z/OSMF includes an **About z/OSMF** page to display the component version details that can be useful to IBM Support for diagnosing a problem.

About this task

To access the **About** page for z/OSMF, do the following:

Procedure

1. From the z/OSMF desktop, click the **menu** in the taskbar.
2. Click **About z/OSMF** in the menu.

Results

Details about the z/OSMF build level, and the SMP/E-installed z/OSMF services and their versions (FMIDs), are displayed in a new browser window. If no z/OSMF services are installed, this area is empty.

Even if you have configured only the z/OSMF nucleus, other z/OSMF services might be listed in the **About z/OSMF** page. When the z/OSMF server is initialized, it starts the following services automatically: z/OS Operator Consoles, Variables, Workflows, Import Manager, Security Configuration Assistant, and Cloud Provisioning. Though the code for these services is installed, the services are not displayed in the z/OSMF UI until you configure them according to the instructions in this book.

Using the z/OSMF Diagnostic Assistant

You can use the z/OSMF Diagnostic Assistant task to collect diagnostic data about z/OSMF and download it as a compressed file package. You can also set logging levels and manage the removal of old log files.

Before you begin

Ensure that the z/OSMF administrative tasks are enabled, as described in [Chapter 6, “Configure the z/OSMF administration tasks,”](#) on page 61.

About this task

You must be a system administrator to collect diagnostic data.

Complete the following steps to collect diagnostic data about z/OSMF.

Procedure

1. Select the Diagnostic Assistant icon in the z/OSMF desktop.
2. Ensure that the option **z/OSMF default diagnostics data** is selected.
3. If you want to limit data collection to **runtime and server side logs**, select this option.

If so, the diagnostic data is obtained from the following directory:

```
<USER_DIR>/data/logs
```

where <USER_DIR> is /global by default. Your installation might use another name for <USER_DIR>, such as /var/zosmf.

4. To include the **z/OSMF server job log** in the data collected, select this option.
5. Click **Export** to download your diagnostic data as a compressed (.zip) file.

What to do next

You can review the diagnostic data in a compressed (.zip) file.

For information about using the Diagnostic Assistant to set logging levels and manage the removal of old log files, see z/OSMF online help.

Types of messages in z/OSMF

z/OSMF records messages from the user interface, from tasks performed by z/OSMF users, and from programs that are running on the z/OS host system. Because of the various layers of functions involved in typical z/OSMF operations, locating a particular message might require you to check more than one location.

z/OSMF collects the following types of messages:

Operator console messages

z/OSMF writes some messages to the operator console with time stamps that are assigned by the console. These messages are also recorded in the z/OSMF server job log, with time stamps that are assigned by the JES subsystem. For example:

```
16.52.31 STC00049 IZUG400I: The z/OSMF Web application services are initialized.
```

Runtime data messages

z/OSMF collects its runtime data (log and trace messages) in the server logs directory. This directory contains one or more log files that are named IZUGn.log, where *n* is a numeral 0 - 9.

In a runtime log file, a message might appear like this:

```
[tx000000000000000008:*izubootstrap*]  
2013-09-06T20:52:31.937Z|0000001F|com.ibm.zosmf.navigation.listener.Boot  
strap|contextInitialized(ServletContextEvent)  
INFO:IZUG400I: The z/OSMF Web application services are initialized.
```

For more information about how runtime log files are processed, see [“z/OSMF log files” on page 285](#).

Messages from z/OSMF tasks

These messages are written to SYSOUT and the job log. In addition, some z/OSMF tasks might write messages to the standard UNIX streams (STDOUT and STDERR) or to z/OS data sets. Typically, messages that are written to the UNIX streams do not have time stamps, for example:

```
.AUDIT . CWWKZ0001I: Application IzuManagementFacilityWorkload...
```

Regardless of the message origin, z/OSMF records all of its messages and traces in the z/OSMF server logs directory. By default, the server logs directory is located in

```
<USER_DIR>/data/logs/zosmfServer/logs
```

where the default for <USER_DIR> is /global/zosmf.

z/OSMF log files

During normal operations, z/OSMF runtime data is created on the server (*server side*) or sent to the server by the client (*client side*). Both types of data are written to the z/OSMF log files.

Viewing the z/OSMF logs

The z/OSMF and WebSphere Liberty logs are available in the z/OSMF logs directory:

```
<USER_DIR>/data/logs/zosmfServer/logs
```

where the default for <USER_DIR> is /global/zosmf.

The z/OSMF runtime log files are written in English only, and are tagged as ASCII, using the ISO8859-1 code page. You can view the log files in ASCII format through ISPF option 3.17, using the VA action (View an ASCII file). Other viewing options, such as OBROWSE, or tools such as vi, emacs, or grep, might require that you first convert the files to EBCDIC. If you want ASCII files to be converted to EBCDIC automatically

when you browse them, set the z/OS UNIX System Services environment variable `_BPXK_AUTOCVT` to "ON".

To access the logs, you require a user ID with z/OSMF administrator authority (that is, a user ID defined to the z/OSMF administrator group).

There are three ways to change logging levels and activate tracing:

1. Use the **Set the z/OSMF logging level** tab in the z/OSMF Diagnostic Assistant. For more information, see z/OSMF online help.
2. Use the IZUPRMxx parmlib member. For more information, see [“IZUPRMxx reference information” on page 35](#).
3. Use the z/OS MVS **MODIFY** command. The change persists only until the z/OSMF server is restarted.

In the following example, the command is entered from the system console and enables the finest level of logging for z/OSMF console services:

```
F server_name,logging='com.ibm.zosmf.console.*=finest'
```

z/OSMF diagnostic data is stored in the following log files. By checking these log files, you can locate any of the messages that are written by z/OSMF.

IZUGx.log

Contains the runtime messages, including the standard output and standard error streams from the JVM process.

The IZUGx.log files are contained in the following directory:

```
<USER_DIR>/data/logs
```

z/OSMF names the log files IZUGn.log, where *n* is a numeral in the range 0 - 9. z/OSMF creates log files in a "cascading" manner. The most current log file is always named IZUG0.log. When this log file reaches its predefined limit, z/OSMF saves it as IZUG1.log and begins writing to a new IZUG0.log file. When the IZUG0.log file is again full, z/OSMF saves it as IZUG1.log after it renames the existing IZUG1.log file to IZUG2.log. z/OSMF continues this process, saving each log file under the next available name, up to a maximum of ten log files. Thereafter, z/OSMF discards the oldest log file (IZUG9.log) whenever a new log file is to be created.

If the current IZUG0.log file becomes unavailable, z/OSMF writes its runtime data to the z/OSMF server logs directory (trace.log and messages.log) until the problem is resolved.

For examples of z/OSMF runtime log data, and a description of the log file format, see [“Examples of working with z/OSMF runtime logs” on page 288](#).

FFDC log files

Contains the WebSphere Liberty first failure data capture (FFDC) log files. FFDC log files include the exception stack and optional additional data that is recorded when an unexpected exception occurs.

The FFDC log files are contained in the following directory:

```
<USER_DIR>/data/logs/zosmfServer/logs/ffdc
```

messages.log

Contains the WebSphere Liberty startup and runtime messages. Messages that are written to this file begin with CWW and include information such as the message timestamp and the ID of the thread that wrote the message. The messages.log does not contain messages that are written by the JVM process.

For example:

```
[9/6/13 20:52:21:569 GMT] 0000001f
com.ibm.ws.app.manager.internal.statemachine.StartAction A CWWKZ0001I:
Application IzuManagementFacilityWorkloadManagement started in 4.121
seconds.
```

The messages log is written to the following location:

```
<USER_DIR>/data/logs/zosmfServer/logs/messages.log
```

trace.log

Contains the same entries as found in messages.log. In addition, this file contains trace entries when additional tracing is enabled. This file does not contain messages that are written by the JVM process.

WebSphere Liberty defines this file as stderr. For example, JSSE tracing is enabled with the z/OSMF advanced setting `-Djavax.net.debug=all`.

The trace log is written to the following location:

```
<USER_DIR>/data/logs/zosmfServer/logs/trace.log
```

Periodic maintenance of log files is recommended

It is recommended that you periodically review the following directories and remove files that are no longer needed. You can manage the removal of old log files by using the **Space management** tab in the z/OSMF Diagnostic Assistant. For more information, see z/OSMF online help.

<USER_DIR>/data/logs/zosmfServer

Contains the JVM-generated diagnostic files for Java exceptions, such as Java core, heap dump, snap.trc, and jit dump files.

<USER_DIR>/configuration

Contains the backup_configuration files.

<USER_DIR>/data/logs/zosmfServer/logs/ffdc

Might contain many log entries due to log rotation; additional file names with time stamps might be generated. The file names in this directory are created with a date and time stamp. For example: `exception_summary_18.07.06_19.33.00.0.log`

Managing log lock files

When z/OSMF initializes, the log file handler creates a file that is named `IZUG0.log.lck`. This file represents a "lock" on the log data. Usually, lock files are cleaned up automatically as part of an application shutdown. However, if the z/OSMF server ends abnormally, the lock files might remain. If so, the log file handler appends numbers to the normal lock file name to find a file that is free.

If the z/OSMF server ends abnormally, inspect the log directory and delete the lock files. If more locks and log files were created, you can sort the files in the directory by timestamp to determine which files are the most recent. Back up these files if you want to preserve them, then clear the logs directory to conserve space.

If client data cannot be written to the server

If a communication problem prevents the client error log data from being written to the z/OSMF logs directory, the unlogged client data is displayed to the user in a separate browser window. This failover action allows for the client data to be retained until the communication with the z/OS system is restored. In some situations, IBM Support might request this data for diagnostic purposes. If the browser window is closed, the client data is not retained.

Other z/OSMF log files

Do not confuse the z/OSMF runtime log file with the job log files that are created during the configuration process. In contrast to runtime data, configuration log data is written to a file in the z/OSMF user file system. If a problem occurs with the configuration log file, the log data is written instead to the directory specified by the `/tmp parmlib` statement.

Examples of working with z/OSMF runtime logs

For your reference, this topic describes the attributes of the z/OSMF log files that are created at runtime.

Examining log data that originates from the server

Figure 47 on page 288 shows portions of an example of z/OSMF server side log data.

```
2009-04-29T18:38:51.285Z|00000012|com.ibm.zosmf.util.eis.cim.ccp.CimClientPool|getWBEMClient(Endpoint,
String,
Set<Locale>) INFO:IZUG911I: Connection to "http://null:5988" cannot be established, or was lost and
cannot be
re-established using protocol "CIM" .
com.ibm.zosmf.util.eis.EisConnectionException: IZUG911I: Connection to "http://null:5988" cannot be
established,
or was lost and cannot be re-established using protocol "CIM" .
com.ibm.zosmf.util.eis.EisException.getEisException(EisException.java:145)
com.ibm.zosmf.util.eis.EisException.diagnoseAndThrow(EisException.java:221)
com.ibm.zosmf.util.eis.cim.ccp.CimClientPool.getWBEMClient(CimClientPool.java:279)

o
o
o

+--> javax.wbem.WBEMException: JNI Exception type CannotConnectException:
Cannot connect to local CIM server. Connection failed.
org.sblim.cimclient.internal.jni.pegasus.CimReturnBuffer.getWBEMException(CimReturnBuffer.java:1244)
org.sblim.cimclient.internal.jni.pegasus.NativeCimClient.verifyResult(NativeCimClient.java:1834)

o
o
o

[tx000000000000000017:pegadm@IBM-FF0E8EC4FCB.xxx.yyy.com (GET) /zosmf/pdw/PdwServiceServlet/
Incidents?filters=IncidentTime(FROM1240704000000)&dojo.preventCache=1241030163470]
```

Figure 47. Portion of z/OSMF server side log data

As shown in Figure 47 on page 288, each log record begins with a line divided by 'pipe' (|) characters into the following components:

- Timestamp in ISO8601 format, set to UTC timezone. Example: 2009-03-10T18:04:08.051Z
- Thread ID as an 8 digit hex number. Example: 00000010
- Class name. Example: com.ibm.zosmf.util.eis.cim.ccp.CimClientPool
- Method name. Example: getClient(Endpoint, String).

The next line of a log record contains the logging level, followed by a colon, followed by the message text. Messages logged at level INFO, WARNING, or SEVERE begin with an eight character message ID at the start of the message text. Message IDs that begin with "IZU" are part of the z/OSMF product.

If the log record includes an exception, the exception is logged next. The exception class is logged, followed by a colon, followed by the message text of the exception. The lines following this make up the traceback information embedded in the exception, which is useful first-failure data capture. If the exception has attached causes, each cause is also logged with "+-->" indicating the start of an attached cause.

The final line in every log record is contained in brackets. If the log record is written during a specific user's context, information about that context is logged, as follows:

- "Transaction ID". An internal counter value that applies to all actions between a specific set and clear of a context. This identifier begins with "tx", followed by a sixteen digit hex ID, and ends with a colon ':'. Example: tx000000000000000017:
- Remote user name (null for a guest user). This value is followed by an 'at' symbol (@).
- Remote host name. This value is followed by a space.
- Servlet "verb" is next, contained in parenthesis. Examples include GET and POST.
- URL of the request and query string, ending with the closing bracket ']'.

If the log record is created during an initialization sequence, the transaction ID is printed and the user name is listed as "`*bootstrap*`". No other data are provided.

If the log record is created with no known context, only "`[tx:]`" appears on the final line.

Viewing client side log data

Included with the server statistics in the z/OSMF logs are client side data, which are used to monitor the JavaScript activity of each user login session. Client side log data differs in format from server side log data, as shown in Figure 48 on page 289.

```
[tx00000000000000ED5:debug2@9.10.83.13 (POST) /zosmf/IzuUICommon/UILoggerServlet?
preventCache=1243956783360]
2009-06-02T15:37:51.933Z|0000001A|com.ibm.zosmf.util.log.servlet.UILoggerServlet|
UILoggerServlet::doPost()
SEVERE: [2009-06-02T15:36:47.047Z] IZUG802E: An error occurred.
Error: "makeTree error: Error: timeout exceeded"
[tx00000000000000ED8:debug2@9.10.83.13 (POST) /zosmf/IzuUICommon/UILoggerServlet?
preventCache=1243956783360]
2009-06-02T15:37:52.020Z|0000001A|com.ibm.zosmf.util.log.servlet.UILoggerServlet|
UILoggerServlet::doPost()
SEVERE: [2009-06-02T15:36:47.203Z] IZUG802E: An error occurred.
Error: "makeTree error: Error: timeout exceeded"
[tx00000000000000ED9:debug2@9.10.83.13 (POST) /zosmf/IzuUICommon/UILoggerServlet?
preventCache=1243956783360]
```

Figure 48. Example of z/OSMF client side log data

Log records that originate from the client side are formatted using the same data as those that originate within the server. However, the "message text" itself is specially formatted to represent the state of the client when the message occurred. This is done to compensate for the fact that client side messages might not be immediately sent to the server.

The following fields are recorded on the client when the message occurs, and are formatted within the message text of a log record as such:

- Client timestamp in brackets []
- Browser name and level
- ENTRY or RETURN, to indicate the beginning or the end of a routine
- Package name, such as AuthorizationServices
- Module name, such as util.ui.messages.Message.js
- Method name, such as _getMessageType()
- Detailed message.

Common problems and scenarios

z/OSMF is based on a stack of components, starting with the application running in the user's workstation web browser and extending to the base z/OS functions and components that deliver much of the underlying function. This section discusses troubleshooting topics, procedures and tools for recovering from a set of known issues.

Troubleshooting topics are included for the following problems and scenarios:

- [“Problems during configuration” on page 290](#)
- [“Problems when accessing the user interface” on page 292](#)
- [“Problems when using Network Configuration Assistant” on page 301](#)
- [“Problems when using IBM zERT Network Analyzer” on page 302](#)
- [“Problems when using the Incident Log task” on page 305](#)

- [“Problems when using the ISPF task” on page 304](#)
- [“Problems when attempting to send data” on page 307.](#)

Problems during configuration

This topic provides troubleshooting tips for resolving problems that are related to the configuration and setup of z/OSMF.

Troubleshooting topics are included for the following problems and scenarios:

- [“z/OSMF server does not initialize or appears to hang” on page 290](#)
- [“IZUSEC job fails with an authorization failure for the issuer” on page 290](#)
- [“A z/OSMF script fails because no z/OS UNIX processes are available” on page 291](#)
- [“Initialization fails with messages IZUG401E and IZUG620E” on page 291](#)
- [“You receive message EDC5134I: Function not implemented” on page 292](#)
- [“RACDCERT or another RACF command abends during configuration” on page 292.](#)

A problem in the configuration of z/OSMF might be indicated by error messages from the common event adapter (CEA) component of z/OS. For a description of configuration-related CEA reason codes, which might be useful in diagnosing problems in your z/OSMF setup, see [Appendix D, “Common event adapter \(CEA\) reason codes,” on page 419.](#)

z/OSMF server does not initialize or appears to hang

Symptom: On start-up, the z/OSMF server (IZUSVR1) does not fully initialize. The following message or a similar message is written to the server job log:

```
ERROR CWWKZ0002E: An exception occurred while starting the application
IzuManagementFacilitySoftwareDeployment. The exception message was:
com.ibm.ws.container.service.state.StateChangeException: java.lang.IllegalStateException:
Unable to acquire TCCL store lock
```

Possible Cause: The thread context classloader (TCCL) store lock timeout threshold was exceeded. By default, this time limit is 5 seconds. In a resource constrained environment, this time limit can be exceeded.

Corrective Action: Increase the timeout threshold by specifying a larger value in an override file.

Follow these steps:

1. Create an override file in the z/OSMF data directory. By default, the data directory is `/global/zosmf`. For example:

```
/global/zosmf/configuration/local_override.cfg
```

2. In the override file, add the following statement on one line:

```
JVM_OPTIONS=-Dcom.ibm.ws.classloading.tcclLockWaitTimeMillis=300000
```

This statement sets the timeout to 300,000 milliseconds (5 minutes).

3. To have this change take effect, restart the z/OSMF server.

You can experiment with this value by reducing it to find the minimum possible timeout for your environment. The z/OSMF server is not negatively affected if the value is set higher than necessary.

IZUSEC job fails with an authorization failure for the issuer

Symptom: The job IZUSEC fails with an authorization failure message for the z/OSMF issuer’s user ID.

Possible Cause: Your installation uses the RACF PROTECT-ALL option to protect its data sets, but you did not define the CEA.* RACF profile.

Corrective Action: If your installation uses PROTECT-ALL, you must define a CEA.* data set profile to RACF and permit CEA and the z/OSMF installer user ID. For example:

```
ADDSD 'CEA.*' UACC(NONE)
PERMIT 'CEA.*' ID(CEA) ACCESS(ALTER)
PERMIT 'CEA.*' ID(USER-ID) ACCESS(ALTER)
```

A z/OSMF script fails because no z/OS UNIX processes are available

Symptom: A script fails with a message that indicates that no z/OS UNIX processes are available for the user ID that was used to run the script.

Possible Cause: The user ID exceeds the MAXPROCUSER setting for your system. MAXPROCUSER specifies the maximum number of z/OS UNIX processes that a single user can have active concurrently. Typically, an installation sets a system-wide limit through the MAXPROCUSER setting in the BPXPRMxx member of parmlib, and then sets higher limits for individual users and processes through PROCUSERMAX, a value in the OMVS segment. Though z/OSMF by itself does not add significantly to the number of z/OS UNIX processes for the user, the MAXPROCUSER setting can be reached when the user is also running a number of other processes on the system besides z/OSMF.

Corrective Action: Use the RACF ADDUSER or ALTUSER command (or an equivalent command for your external security manager) to specify a PROCUSERMAX value for the user ID that is higher than the MAXPROCUSER setting. Try adding 20 to the value that is specified through the MAXPROCUSER setting.

Suppose, for example, that your installation specified a MAXPROCUSER value of 80 in the BPXPRMxx member. Here, you would set the PROCUSERMAX value for this user ID to 100 to allow a greater number of processes for the user ID. For example:

```
ALTUSER USER-ID OMVS(PROCUSERMAX(100))
```

If the problem persists, repeat this process by increasing the PROCUSERMAX value by an extra 20, taking care not to exceed any limits that are appropriate for your installation; check with your security administrator.

Initialization fails with messages IZUG401E and IZUG620E

Symptom: During initialization, the z/OSMF server fails with the following error messages:

```
IZUG401E: Initialization has failed for the z/OSMF web application services.
IZUG620E: The required environment variable "IZU_DATA_DIR" is missing or blank.
```

Possible Cause: The TCP/IP resolver trace function is active on the system. This trace is used for debugging problems that are related to TCP/IP. The trace is enabled by including the TRACE RESOLVER statement in data set TCPIP.DATA.

When active, the resolver trace causes the z/OS UNIX command **hostname** to return diagnostic data with the host name. During z/OSMF initialization, the diagnostic data is erroneously supplied as input to the z/OSMF configuration file. The incorrect input data is indicated in message IZUG620E.

Corrective Action: Disable the TCP/IP resolver trace function on the z/OSMF system. If the trace is required, you can resume the trace after the z/OSMF server is initialized.

You can use the **MODIFY RESOLVER, REFRESH** command to change the TCPIP.DATA statements that are being used by the z/OSMF server. For more information about modifying statements in TCPIP.DATA, see [Abstract for IP System Administrator's Commands in z/OS Communications Server: IP System Administrator's Commands](#).

You receive message EDC5134I: Function not implemented

Symptom: You receive the following message and error code:

```
atoe_getcwd error: EDC5134I Function not implemented. (errno2=0x052C04DC)
```

Possible Cause: The error code indicates that the system root directory is not mounted. However, this message is also issued if the OMVS home settings for a user ID include a root directory (/) specification, but the user ID does not have access to the root directory.

Corrective Action: Verify that the system root directory is mounted and that the user ID OMVS home settings are correct.

RACDCERT or another RACF command abends during configuration

Symptom: A RACF command abends with code S684 or code 047 during the configuration process. On checking the script log, you find a message such as the following:

```
Script izutsoz.rexx returned with reason code -1668
```

Possible Cause: The RACF command is not defined in AUTHCMD section of your active IKJTSOxx parmlib member.

Corrective Action: Verify that the IKJTSOxx member defines the required RACF commands. See the list of IKJTSOxx parmlib updates in the *z/OS Program Directory*. The AUTHCMD section of member IKJTSOxx should list RACDCERT and a number of other RACF commands. You can update the IKJTSOxx member dynamically through the TSO command: PARMLIB UPDATE (xx) where xx is the correct suffix.

Problems when accessing the user interface

This topic provides troubleshooting tips for resolving problems related to the user interface of z/OSMF.

Troubleshooting topics are included for the following problems and scenarios:

- [“Browser cannot connect to z/OSMF” on page 292](#)
- [“Missing initialization message or JSP processing error when attempting to use z/OSMF” on page 293](#)
- [“Certificate error in the Mozilla Firefox browser” on page 294](#)
- [“Cannot log into z/OSMF” on page 296](#)
- [“Re-authenticating in z/OSMF” on page 297](#)
- [“User receives message ICH408I for insufficient authority to an EZB.STACKACCESS resource” on page 298](#)
- [“z/OSMF server appears to hang” on page 298](#)
- [“Help link does not work” on page 299](#)
- [“Help topics are missing or displayed out of sequence” on page 300](#)
- [“Action or link that was previously provided is not available” on page 300](#)
- [“A script takes too long to run or is not responding” on page 301.](#)

Browser cannot connect to z/OSMF

When you log in to z/OSMF for the first time, your browser either does not connect, or waits indefinitely. Verify that the browser has network connectivity to the host on which the z/OSMF instance is running. If your network connectivity is functioning properly, there might be an issue with the digital certificates that are used for SSL connections.

See the following topics:

- [“Check your network connection” on page 293](#)
- [“More information about certificates” on page 293](#)

Check your network connection

Try the following network diagnostic techniques:

- Entering the command NSLOOKUP to verify that the host name is resolvable
- Pinging the host system for a response
- Running the TRACEROUTE command.

More information about certificates

For information about using signed certificates, see [“2a. Use client and server certificates from the same CA” on page 235](#)

For an example that uses certificates in z/OSMF connections, see [Chapter 37, “Configuring a primary z/OSMF for communicating with secondary instances,” on page 221.](#)

Missing initialization message or JSP processing error when attempting to use z/OSMF

Symptoms: The following symptoms occur in this sequence:

1. You start z/OSMF, but see no message in the operator log about whether z/OSMF started successfully or failed.
2. You attempt to access the z/OSMF URL, but encounter a JSP processing error with HTTP code 500, along with text like the following with supporting messages:

JSPG0049E: /NavigationTree.jsp failed to compile
3. You examine the z/OSMF logs and find that they are empty or have no new messages since starting z/OSMF. No .lck file exists either, which suggests that the logs are not active.
4. You examine the z/OSMF logs and search for IZUG, looking for message codes. While none exist, you notice that the search reveals the following:

```
UTLS0002E: The shared library IzuSrvLibs contains a classpath entry
which does not resolve to a valid jar file, the library jar file is
expected to be found at /usr/lpp/zosmf/lib/izugjni.jar.
```

Possible Cause: A failure of the JSP to compile typically means that one or more required classes could not be found. Most likely, this is a problem with a referenced shared library. Failures with the shared libraries typically mean either of the following:

- Shared libraries class path entries are incorrect.
- Class path entries point to missing JAR files.

In this situation, the message shows which paths were not found.

Investigation: Use the following procedure to determine the cause of the error.

1. Examine the contents of the directory where the JARs are supposed to exist:

```
# ls /usr/lpp/zosmf/lib
ls: FSUM6785 File or directory "/usr/lpp/zosmf/lib" is not found
```

2. The directory does not exist, so determine which file systems are mounted.

Corrective Action: Mount the necessary file system in the correct location and restart z/OSMF.

Certificate error in the Mozilla Firefox browser

When logging into z/OSMF for the first time, you might notice that the Mozilla Firefox browser displays the error message: *Secure Connection Failed*.

If the error message indicates that the browser does not recognize the Certificate Authority (CA) certificate that is configured for z/OSMF, you can resolve the error by adding the certificate to the browser security exception list, or importing the certificate into the browser. For information, see the following sections:

- [“Adding the CA certificate to the security exceptions list” on page 294](#)
- [“Importing the CA certificate into the browser” on page 294](#).

If the error message indicates that the certificate contains the same serial number as another certificate issued by the CA, it is possible that your browser contains a CA certificate from a previous installation of z/OSMF. If so, you can remove the older certificate from the browser, as described in [“Removing the CA certificate from the browser” on page 295](#). Then, try again to access z/OSMF and allow the new certificate to be stored in the browser.

Adding the CA certificate to the security exceptions list

You can allow your browser to bypass the *Secure Connection Failed* message for z/OSMF.

Do the following:

1. On the error page, click **Or you can add an exception**.
2. Click **Add Exception**. The *Add Security Exception* dialog is displayed.
3. Click **Get Certificate**.
4. Click **View** to display a window that describes the problem with your z/OSMF site.

Examine the *Issued To* fields. Verify that the information identifies z/OSMF. The value for *Common Name (CN)* should match the host name for your installation of z/OSMF.

Examine the *Issued By* fields. Verify that the certificate was issued by the certificate authority (CA) that was used to generate the server certificate. By default, z/OSMF uses the certificate authority *zOSMFCA*.

To see the other fields of the certificate, select the *details* tab.

5. After you have verified the certificate, close the dialog. If you leave the **Permanently store this exception** check box selected, Firefox stores the certificate information to prevent the error from being displayed again for the z/OSMF site.
6. Click **Confirm Security Exception** to trust the z/OSMF site.

Your browser will now open to the z/OSMF interface.

Importing the CA certificate into the browser

You can import the CA certificate into your browser. Doing so involves exporting the z/OSMF certificate from RACF, transferring the CA certificate to your workstation, and importing the CA certificate into your browser.

The CA certificate is determined by your configuration setting for the variable `IZU_DEFAULT_CERTAUTH`. If this variable is set to `Y`, z/OSMF creates the CA during the configuration process. Otherwise, no CA is created, and z/OSMF uses `CERTAUTH LABEL('zOSMFCA')` to sign the certificate. z/OSMF uses the SAF key ring name `IZUKeyring.IZU_SAF_PROFILE_PREFIX`.

To import the CA certificate into your browser, do the following:

1. List the key rings for the z/OSMF server user ID, using the `RACDCERT` command, for example:

RACDCERT ID(IZUSVR1) LISTRING(*)

Figure 49 on page 295 shows an example of the output from the RACDCERT command.

```
Digital ring information for user IZUSVR1:

Ring:
>IZUKeyring.IZUDFLT<
Certificate Label Name      Cert Owner      USAGE      DEFAULT
-----
zOSMFCA                    CERTAUTH        CERTAUTH    NO
Verisign Class 3 Primary CA CERTAUTH        CERTAUTH    NO
Verisign Class 1 Primary CA CERTAUTH        CERTAUTH    NO
Thawte Server CA           CERTAUTH        CERTAUTH    NO
Thawte Premium Server CA   CERTAUTH        CERTAUTH    NO
Thawte Personal Basic CA   CERTAUTH        CERTAUTH    NO
Thawte Personal Freemail CA CERTAUTH        CERTAUTH    NO
Thawte Personal Premium CA CERTAUTH        CERTAUTH    NO
```

Figure 49. Digital ring information for the z/OSMF server user ID

Verify that the configured SAF key ring is shown for the z/OSMF server user ID. Note the key ring name and the certificate label (zOSMFCA, in this case).

2. Export the CA certificate using the RACDCERT command, for example:

```
RACDCERT EXPORT(LABEL(' zOSMFCA')) CERTAUTH
DSN('??????.CERT.AUTH.DER') FORMAT(CERTDER)
```

3. Transfer this file in binary format to your workstation. Keep the .der extension when you transfer the file.
4. To import the certificate into the Firefox browser, do the following:
 - a. From the *Tools* menu, click **Options > Advanced** tab.
 - b. Click **View Certificates**.
 - c. Select the *Authorities* tab.
 - d. Click **Import**.
 - e. From the *Select File* menu, navigate to the folder to which you transferred the CA certificate.
 - f. Select the certificate file and click **Open**.
 - g. In the dialog box, select the *Trust this CA to identify web sites* check box. You can also click **View** to examine the certificate.
 - h. To import the certificate to your browser, click **OK**.

Your browser will now open to the z/OSMF interface.

Removing the CA certificate from the browser

You can remove an older CA certificate from the browser to allow the CA certificate for the new release of z/OSMF to be added.

Do the following:

1. From the *Tools* menu, click **Options > Advanced** tab.
2. Click the **Encryption** tab.
3. Click *View Certificates*.
4. Click the **Servers** tab.
5. In the *Certificate Name* column, locate the *z/OSMF CertAuth* section.

6. Select the certificate files under z/OSMF and click **Delete**.

7. Click **OK**.

Try to access z/OSMF with your web browser. If prompted, allow the CA certificate to be stored in the browser. Your browser will now open to the z/OSMF user interface.

Cannot log into z/OSMF

If a user receives an error when attempting to log into z/OSMF, try troubleshooting with the following steps.

Procedure

1. Verify that the user ID is correct and try logging in. If the user is still not able to log in, continue to the next step.
2. Ensure that the password that is associated with the user ID is correct. If the user is still not able to log in, continue to the next step.
3. It is possible that the password for the user ID is expired. To check, try logging in to TSO through an emulator.
4. Ensure that your installation defined the z/OSMF unauthenticated guest user in your external security manager. This authorization is required so that users can access the z/OSMF Welcome page prior to login. In a system with RACF, for example, your security administrator can use the following commands to create the unauthenticated guest user:

```
/* Create the z/OSMF unauthenticated USERID */
ADDUSER IZUGUEST RESTRICTED DFLTGRP(IZUUNGRP) OMVS(UID(9011)) +
NAME('zOSMF Unauthenticated USERID') NOPASSWORD NOOIDCARD

/* Permit the z/OSMF unauthenticated USERID access */
PERMIT IZUDFLT CLASS(APPL) ID(IZUGUEST) ACCESS(READ)

/* Permit other users USERID access */
PERMIT IZUDFLT CLASS(APPL) ID(userid) ACCESS(READ)
```

5. If the user is attempting to log in with a password phrase (pass phrase), your installation's external security manager might need to be updated to allow mixed case passwords. In a system with RACF, for example, your security administrator can use the SETROPTS PASSWORD(MIXEDCASE) option to allow mixed-case passwords at your installation. After this change is made, you must restart the z/OSMF server.
6. Check the z/OSMF server job log for message BPXP014I with either of the following messages: ICH420I or BPXP015I. These message pairings indicate that the z/OSMF server did not connect to the z/OSMF angel process.

- For example:

```
ICH420I PROGRAM BPXBATSL FROM LIBRARY SYS1.LINKLIB CAUSED THE ENVIRONMENT TO BECOME
UNCONTROLLED.
BPXP014I ENVIRONMENT MUST BE CONTROLLED FOR DAEMON (BPX.DAEMON) PROCESSING.
```

- Or:

```
BPXP015I HFS PROGRAM /usr/lpp/zosmf/lib/libIzugJni64.so IS NOT MARKED PROGRAM CONTROLLED.
BPXP014I ENVIRONMENT MUST BE CONTROLLED FOR DAEMON (BPX.DAEMON) PROCESSING.
```

If these messages appear, check the Liberty log for message CWWKB0117W or CWWKB0118W:

```
CWWKB0117W: The angel-name angel process is not available. No authorized services
will be loaded. The reason code is 4.
```

```
CWWKB0118W: This server is not authorized to connect to the angel-name angel process.
No authorized services will be loaded.
```

- For message CWWKB0117W, you must start the address space for the angel that is identified in the message. Then, restart the z/OSMF server address space.
- For message CWWKB0118W, you must grant the z/OSMF server user ID read access to the profile BBG.ANGEL.*proc-name* in the SERVER resource class, where *angel-proc* is the name of the angel started procedure. Then, restart the z/OSMF server address space.

By default, the Liberty log is located in the following path `/global/zosmf/data/logs/zosmfServer/logs/messages.log`.

What to do next

If none of these steps resolves the problem, contact your system programmer for assistance. The system programmer should check the z/OSMF log files for messages that indicate that the user ID is not authorized.

User messages for authentication errors are often general by design to avoid providing malicious users with valuable information, such as whether a particular user ID is valid. More specific information about this error might be available to your system programmer in the form of messages that are written to the operator console or to the operator log. Typically, these problems are caused by incorrect passwords or user IDs that are revoked.

Re-authenticating in z/OSMF

When your z/OSMF session expires, you can re-authenticate using the re-authentication dialog box.

About this task

Your z/OSMF session expires after a period of time has elapsed. By default, this period is 495 minutes from the time you log into z/OSMF. Your installation can choose to modify this setting (SESSION_EXPIRE) using the IZUPRMxx parmlib member. z/OSMF. For details, see [“IZUPRMxx reference information” on page 35](#).

The re-authentication dialog box is displayed for 15 minutes. If you re-authenticate before the period ends, the tabs (in the work area) are unaffected by the re-authentication. If you do not respond before the re-authentication period ends, you are logged out, the tabs in the work area are closed, and any unsaved data is lost.

If you launched multiple instances of z/OSMF in the same browser (using new tabs or new windows) and your browser is configured to use the same browser session for new windows or tabs, the session for each instance will expire simultaneously; hence, a re-authentication dialog box is displayed in each tab or window. In this case, you can respond to one re-authentication dialog box and you are automatically re-logged into or logged out of each instance. If you launched multiple z/OSMF instances using different computers or different browsers or using multiple instances of a browser that is not configured to use the same browser session, the browser sessions are treated independently and each z/OSMF instance will require its own re-authentication.

While the re-authentication dialog box is displayed, you cannot interact with any tasks in that z/OSMF instance. You cannot explicitly close the dialog box. You can only close it by choosing to log in or log out.

Procedure

1. Verify the user ID.

You cannot modify the user ID. If it is incorrect, click **Log out**. Otherwise, proceed to Step 2. When you click **Log out**, z/OSMF closes all opened tabs and discards any unsaved changes.

2. Enter the password or pass phrase that corresponds with the z/OS user ID.
3. Click **Log in** to re-authenticate.

Results

If the password or pass phrase is valid, you are logged in again. If you selected to log out (by clicking **Log out**), the *Welcome* page is displayed. If the password or pass phrase is incorrect, an error message is displayed and the re-authentication dialog box is still displayed. In this case, try logging in again. If you are unable to authenticate before the re-authentication period expires, z/OSMF will automatically log you out.

User receives message ICH408I for insufficient authority to an EZB.STACKACCESS resource

When accessing z/OSMF as a guest user, the user receives message ICH408I for insufficient authority to an EZB.STACKACCESS resource.

About this task

In z/OSMF, a guest user is one who displays the z/OSMF home page, but has not yet logged in. Here, z/OSMF applies the guest user classification (IZUGUEST) to the user. By default, a guest user can access the z/OSMF home page and the default links.

A guest user might encounter an ICH408I message like the following:

```
ICH408I USER(IZUGUEST ) GROUP(GUESTGRP) NAME(ZOSMF UNAUTH USER)
EZB.STACKACCESS.nnn.nnn CL(SERVAUTH)
INSUFFICIENT ACCESS AUTHORITY
FROM EZB.STACKACCESS.*.* (G)
ACCESS INTENT(READ ) ACCESS ALLOWED(NONE )
```

This problem can occur if your installation created a NETACCESS or STACKACCESS rule to limit access to the TCP/IP stack. If so, a login attempt to the z/OSMF server is first performed under the IZUGUEST identity before the identity is switched to that of the user's TSO/E user ID. This behavior allows the z/OSMF Welcome page to be displayed to the user, prior to log-in.

Procedure

1. It is recommended that your installation define the IZUGUEST identity with the RESTRICTED attribute. Doing so means that the guest user cannot access any profiles unless explicitly granted. That is, the UACC specification is ignored for any NETACCESS or STACKACCESS rule.
2. If your installation uses a STACKACCESS or NETACCESS rule, grant the IZUGUEST identity READ access to the specific EZB.STACKACCESS.nnn.nnn or EZB.NETACCESS.nnn.nnn profile that is referenced in the ICH408I message. This authorization allows the user to access the TCP stack on which the z/OSMF server is running.

z/OSMF server appears to hang

During normal operations, the z/OSMF server becomes unresponsive.

About this task

The following message or a similar message is written to the server job log:


```
[err] Exception in thread "Scheduled Executor-thread-1"  
[err] java.lang.OutOfMemoryError  
[err] :  
[err] Failed to create a thread:
```

This message can result when the z/OSMF server exceeds the maximum number of connections (threads) that are allowed for a single address space. Set MAXTHREADS in SYS1.PARMLIB(BPXPRMxx) to 500 or higher. This is a system-wide limit, and thus active for all z/OS UNIX address spaces.

Procedure

1. Check the MAXTHREADS value for your system by entering the following command at the operations console: **D OMVS,0**.
Check the command response for the value of MAXTHREADS.
2. If the value is less than 500, reset it to 500 or more by entering the following command at the operations console: **SETOMVS MAXTHREADS=500**
3. Restart the z/OSMF server.

What to do next

If the problem persists, contact IBM for assistance.

Help link does not work

The online help information for the z/OSMF user interface (UI) pages or messages is not available.

Symptom

The user clicks a help link to open a new window with help information for a UI page or message, but the help is not displayed. Instead, the error message `file not found` is displayed in the user's web browser.

Possible cause

The help files are missing or are not readable. Or, new help files were installed and the z/OSMF server was not restarted. By default, the z/OSMF help files reside at the location `/global/zosmf/helps/eclipse/plugins`.

System programmer response

Use the following procedure to resolve this error:

1. Verify that symlinks exist in the `/global/zosmf/helps/eclipse/plugins` subdirectory. The symlinks must refer to the z/OSMF product directory, which, by default, is `/usr/lpp/zosmf`.
2. Verify that the EJBROLE resource class is defined properly; it is case-sensitive.
3. Restart the z/OSMF server, for example, through the MVS **START** command.

User response

No action is required.

Help topics are missing or displayed out of sequence

In the z/OSMF online help *Table of Contents*, one or more help topics is missing or displayed out of sequence.

Symptom

The user clicks a help link to open a new window with help information for a UI page or message. However, the online help *Table of Contents* is missing the topic or the topic is displayed out of sequence with the other help topics.

Possible cause

The online help *Table of Contents* is corrupted.

System programmer response

Use the following procedure to resolve this error:

1. Clear the z/OSMF server logs directory by deleting its contents. By default, the logs directory is located in

```
<USER_DIR>/data/logs/zosmfServer/logs
```

where the default for <USER_DIR> is /global/zosmf.

2. Restart the z/OSMF server, for example, through the MVS **START** command.

User response

No action is required.

Action or link that was previously provided is not available

Symptom: An action or link that was previously provided in the user interface is disabled, not listed, or no longer provided.

Possible Cause:

- No items have been selected against which to perform the action.
- Too many items have been selected.
- The action or link is not applicable for the selected items.
- The event type is not registered or no handlers are available to process the request.

Administrator Action:

1. Determine if the user-interface control invokes an event type. For IBM-supplied event requestors, see [IBM z/OS Management Facility Programming Guide](#).
2. If the user-interface control invokes an event type, do the following:
 - a. Verify that the event type is registered in the Application Linking Manager task.
 - b. If the event type is not registered, ensure that the z/OSMF service that registers the event type is configured in z/OSMF.
 - c. If the service is configured, you can use the Application Linking Manager task or the API to register the event type, or you can recycle the z/OSMF server to register it automatically. For IBM-supplied event types, to register them manually, specify the information included in the topic about the event types, requestors, and handlers that are shipped with z/OSMF in [IBM z/OS Management Facility Programming Guide](#).

- d. Verify that a handler is registered for the event type and that the user is authorized to access the handler.

User Action: Ensure that items are selected and that the correct number and type of items are selected.

A script takes too long to run or is not responding

When using z/OSMF, you might encounter the long-running script dialog, which means that a script is taking a long time to run or that a script has stopped responding. From the dialog, you can decide either to stop executing the script or to continue executing it. If you stop executing the script, the function on that web page that is dependent upon the script might not function properly. If you continue executing the script, the dialog will re-display each time the number of statements executed or the amount of time executing a script exceeds the browser's threshold.

To decrease the number of times the long-running script dialog is displayed, you can increase the maximum amount of time a script is allowed to execute or you can increase the maximum number of statements that can be executed. Whether you are modifying the amount of time or the number of statements is dependent upon the browser. For example, the Firefox threshold is based on time; while the Internet Explorer 11 threshold is based on the number of statements.

For more information about unresponsive or long-running scripts, see the appropriate support web site for your browser:

Firefox

- See the following Mozilla web site for information you might find useful: <http://support.mozilla.com/en-US/kb/Warning+Unresponsive+script>.

Internet Explorer 11

- See the following Microsoft web site for information you might find useful: <http://support.microsoft.com/kb/175500>.

Problems when using Network Configuration Assistant

This section provides a procedure you can use to send troubleshooting documentation to IBM Support.

Steps for sending information to IBM Support

For a failure in the Network Configuration Assistant task, use the following procedure to gather troubleshooting documentation. This information can be used by IBM Support to diagnose the problem.

Procedure

1. Transfer the z/OSMF runtime log files that contain Network Configuration Assistant logging data.

During normal operations, z/OSMF collects its runtime data (log messages and trace messages) in log files. z/OSMF runtime data is created on the server (server side) or sent to the server by the client (client side). Both types of messages are written to the z/OSMF runtime log files.

z/OSMF creates the log files in the product logs directory, which is, by default, `/global/zosmf/data/logs`. z/OSMF names the log files `IZUGn.log`, where `n` is a numeral in the range 0 - 9.

z/OSMF creates log files in a "cascading" manner. The most current log file is always named `IZUG0.log`. When this log file reaches its predefined limit, z/OSMF saves it as `IZUG1.log` and begins writing to a new `IZUG0.log` file. When the `IZUG0.log` file is again full, z/OSMF saves it as `IZUG1.log` after it renames the existing `IZUG1.log` file to `IZUG2.log`. z/OSMF continues this process, saving each log file under the next available name, up to a maximum of ten log files. Thereafter, z/OSMF discards the oldest log file (`IZUG9.log`) whenever a new log file must be created.

2. Transfer the currently active backing store file. From the **Tools** option, select **Manage Backing Stores** > **Actions** > **Transfer**.

3. For issues that are related to Cloud Provisioning, create a compressed copy (.zip) of the directory `/global/zosmf/data/datastore/NetworkResourceManager`.
4. Package the files and send them to IBM Support.

Problems when using IBM zERT Network Analyzer

This section provides a procedure you can use to send troubleshooting documentation to IBM Support.

Steps for sending information to IBM Support

For a failure in the IBM zERT Network Analyzer task, use the following procedure to gather troubleshooting documentation. This information can be used by IBM Support to diagnose the problem.

Procedure

1. Use the **Applications Settings** sub-panel on the IBM zERT Network Analyzer task to ensure that the proper level of logging is being used. By default, IBM zERT Network Analyzer writes Info level messages to the z/OSMF runtime logs, but writes no logging information to the IBM zERT Network Analyzer logs.

After the necessary level of logging is in place, re-create the problem that was reported, if needed.

2. Transfer the z/OSMF runtime log files that contain IBM zERT Network Analyzer logging data.

During normal operations, z/OSMF collects its runtime data (log messages and trace messages) in log files. z/OSMF runtime data is created on the server or sent to the server by the client. Both types of messages are written to the z/OSMF runtime log files.

In addition, if IBM zERT Network Analyzer logging is enabled, the IBM zERT Network Analyzer task writes some runtime data (log messages and trace messages) to the z/OSMF runtime log files.

z/OSMF creates the log files in the product logs directory, which is, by default, `/global/zosmf/data/logs`. z/OSMF names the log files `IZUGn.log`, where *n* is a numeral in the range 0 - 9.

z/OSMF creates log files in a "cascading" manner. The most current log file is always named `IZUG0.log`. When this log file reaches its predefined limit, z/OSMF saves it as `IZUG1.log` and begins writing to a new `IZUG0.log` file. When the `IZUG0.log` file is again full, z/OSMF saves it as `IZUG1.log` after it renames the existing `IZUG1.log` file to `IZUG2.log`. z/OSMF continues this process, saving each log file under the next available name, up to a maximum of ten log files. Thereafter, z/OSMF discards the oldest log file (`IZUG9.log`) whenever a new log file must be created.

3. Transfer the IBM zERT Network Analyzer log files that contain IBM zERT Network Analyzer logging data.

During normal operations, when logging is enabled, IBM zERT Network Analyzer collects runtime data (log messages and trace messages) specific to IBM zERT Network Analyzer processing in its own separate log files. Only server-side information is written to the IBM zERT Network Analyzer runtime log files.

By default, IBM zERT Network Analyzer creates its log files in the `/global/zosmf/data/app/<release>/debug` folder, where `<release>` is the current level of IBM zERT Network Analyzer. The format of `<release>` is `ZNAxxxx`, where `xxxx` represents the z/OSMF release level. For example, IBM zERT Network Analyzer running on z/OSMF V2R3 would use `ZNAV2R3` for `<release>`. The full name of the default folder would be `/global/zosmf/data/app/ZNAV2R3/debug`. IBM zERT Network Analyzer names the files `ZnaDbgn.log`, where *n* is a numeric value in the range 0 - 24.

IBM zERT Network Analyzer creates log files in a "cascading" manner. The most current log file is always named `ZnaDbg0.log`. When this log file reaches its predefined limit, IBM zERT Network Analyzer saves it as `ZnaDbg1.log` and begins writing to a new `ZnaDbg0.log` file. When the `ZnaDbg0.log` file is again full, IBM zERT Network Analyzer saves it as `ZnaDbg1.log` after it renames the existing `ZnaDbg1.log` file to `ZnaDbg2.log`. IBM zERT Network Analyzer continues this process, saving each log

file under the next available name, up to the current maximum of log files. Thereafter, IBM zERT Network Analyzer discards the oldest log file (ZnaDbgn.log) whenever a new log file must be created.

By default, the maximum number of log files maintained is 10. You can modify that setting on the IBM zERT Network Analyzer **Applications Settings** sub-panel, up to a maximum of 25 possible log files.

4. Package the files and send them to IBM Support.

Recovering unavailable partitions

In unusual and infrequent situations, an active partition of a query result table might become unavailable. When a partition is in unavailable state, the IBM zERT Network Analyzer is unable to use it to store query results or to free the partition. In order to make the partition available for use, database administrator must recover the partition.

Before you begin

To determine the partitions that are unavailable, the IBM zERT Network Analyzer logs the following message for each unavailable partition when the plugin starts up:

```
'IZUET0049E Error freeing up partition partition-id during IBM zERT Network Analyzer startup'
```

where *partition-id* is the partition identifier.

This message may be preceded by an IZUET0034 message:

```
IZUET0034E: Exception encountered: stack dump caused by "UNSUCCESSFUL EXECUTION CAUSED BY AN UNAVAILABLE RESOURCE, REASON reason-code, TYPE OF RESOURCE resource-type, AND RESOURCE NAME resource-name SQLCODE=-904, SQLSTATE=57011"
```

For more information on the SQL error code -904, see SQL error codes in the Db2 for z/OS *Codes Book*.

About this task

To recover an unavailable partition, see the following steps.

Procedure

1. Run the Db2 for z/OS RECOVER utility on all query result table spaces associated with the *partition-id* in the IZUET0049E message. The RECOVER utility recovers all data to the current state or to a previous point in time by restoring a copy and applying log records or by undoing all committed work. In order to use this utility, you must have one the following authorities:

- RECOVERDB privilege for the database
- DBADM or DBCTRL authority for the database
- System DBADM authority
- DATAACCESS authority
- SYSCTRL or SYSADM authority

Use the TABLESPACE option on the RECOVER utility control statement to recover all query result table spaces associated with the *partition-id*. To recover multiple table spaces, repeat the TABLESPACE option before each specified table space in the following statement:

```
RECOVER TABLESPACE database-name.tablespace-name DSNUM unavailable-partition-number
```

Where *database-name* is the value specified for the <QRTSDatabase> variable, *tablespace-name* is the value specified for each <*Space> variable, and *unavailable-partition-number* is the *partition-id* value in the IZUET0049E message.

You can find the variables mentioned above in the Query Result Tables database variables section of your customized version of the IZUZNADI dataset used to generate the DDL that created the IBM zERT Network Analyzer database.

For example:

```
RECOVER TABLESPACE QRTSDB.FSSIDSS DSNUM 1
        TABLESPACE QRTSDB.TCLNCDS DSNUM 1
        TABLESPACE QRTSDB.TSRVSSS DSNUM 1
        TABLESPACE QRTSDB.EPEERSS DSNUM 1
        ...
        TABLESPACE QRTSDB.TCLNSS DSNUM 1
```

For more information on syntax and examples for the recovery utility, see *Recover* and *Recovering a table space* in Db2 for z/OS *Utilities Book*.

2. Run the Db2 online utility. For detailed information on the different ways to invoke and steps to run the utility, see *Invoking Db2 online utilities* in the Db2 for z/OS book.

What to do next

After recovering the partitions, you must recycle z/OSMF for IBM zERT Network Analyzer to detect the availability of the partitions.

Problems when using the ISPF task

This topic provides troubleshooting tips for common problems that might occur while using the ISPF task.

Troubleshooting topics are included for the following problems and scenarios:

- [“Unexpected behavior occurs in the ISPF user session after the user logs on again” on page 304](#)
- [“Log-on or log-off through the ISPF task takes too long” on page 305](#)
- [“Log-on through the ISPF task takes too long, even though the system is enabled for reconnectable user sessions” on page 305.](#)

Unexpected behavior occurs in the ISPF user session after the user logs on again

Symptom: User logs off from an ISPF session. On logging on again, the user encounters an unexpected behavior, such as one of the following:

- z/OSMF ISPF environment is not reset
- Logon proc is not run
- Region size is not restored
- Session behaves unexpectedly in some other manner.

Probable cause: The user required a new session, but the ISPF task reconnected the user to an existing session. To save time and system resources, the ISPF task can reconnect a user to an existing session, rather than creating a new session. This reconnect capability requires that some aspects of the user session be preserved after logoff (the session is not completely ended). In some cases, this processing can pose a problem for users who require that their sessions be completely ended and cleaned up during logoff.

Corrective Action: The user can force z/OSMF to create a new session, rather than reconnect to an existing session, by changing one of the logon settings. For example, changing the screen size or region size slightly would result in a new session being created. If this problem occurs frequently or for multiple z/OSMF users, consider deactivating the reconnect capability for the ISPF task. You can do so through parmlib member, CEAPRMxx, which is used to specify options for the common event adapter (CEA) component of z/OS. In CEAPRMxx, the following statements control the reconnect capability for the ISPF task:

- RECONTIME limits the number of reconnectable sessions
- RECONSESSION limits the time that sessions can remain in a reconnectable state.

To deactivate the reconnect capability for the ISPF task, set one or both of these values to zero, as indicated in the commented section of IBM-supplied member, CEAPRM00. For more information about CEAPRM00, see [z/OS MVS Initialization and Tuning Reference](#).

Log-on or log-off through the ISPF task takes too long

Possible Cause: The extra time is used by the system during logon processing to perform a complete log-on for the user. Or, to log-off the user and clean-up the user address space.

Corrective Action: Enable the use of reconnectable sessions for ISPF task users. Doing so can allow for potentially faster logon processing when existing user sessions are eligible for re-use. Enabling reconnectable user sessions involves modifying the CEA component on your system through parmlib member CEAPRMxx. See the descriptions of statements TSOASMGR, RECONSESSIONS, and RECONTIME in [z/OS MVS Initialization and Tuning Reference](#). If reconnectable user sessions are already enabled, consider increasing either the RECONSESSIONS or RECONTIME values.

Log-on through the ISPF task takes too long, even though the system is enabled for reconnectable user sessions

Symptom: User selects the ISPF task, but the resultant log-on takes too long, even though the z/OS system is enabled for reconnectable user sessions.

Possible Cause: On a system enabled for reconnectable user sessions, the ISPF task checks for a session to which the user can reconnect. No eligible session was found, however, possibly because the session has expired, based on one or more system limits. Without an available reconnectable session, the ISPF task creates a new session for the user. The additional processing increases the time for the log-on request to complete. Another possibility is that the ISPF task has discarded its reconnectable user sessions as part of normal clean-up. This processing occurs when the ISPF task is idle (has no active users) for at least 15 minutes. After the clean-up is completed, a subsequent user of the ISPF task will always receive a new session.

Corrective Action: You can increase the number of reconnectable sessions allowed on your system and the time that sessions can remain connectable. See the descriptions of the RECONTIME and RECONSESSION statements of parmlib member CEAPRMxx in [z/OS MVS Initialization and Tuning Reference](#). Regardless of these settings, the ISPF task discards its reconnectable sessions when it is idle for 15 minutes.

Problems when using the Incident Log task

This topic provides troubleshooting tips for common problems that might occur while using the Incident Log task.

Troubleshooting topics are included for the following problems and scenarios:

- [“User cannot access the Incident Log task” on page 305](#)
- [“User encounters message ICH408I” on page 306](#)
- [“CEA address space is blocking the use of the sysplex dump directory” on page 306](#)
- [“CEA cannot allocate a data set for dump prepare or snapshot” on page 306](#)
- [“Diagnostic log streams and other incident data for deleted incidents are not being deleted over time” on page 306](#)
- [“Problems when attempting to send data” on page 307.](#)

User cannot access the Incident Log task

Symptom: On selecting the Incident Log task, the user receives an error message indicating a lack of authorization to CEA.

Probable cause: During the configuration of z/OSMF, the configuration script defines the resource CEA.CEAPDWB*. However, the resource CEA.* was already defined by your installation. Because CEA.CEAPDWB* takes priority over CEA.* no users are authorized to make CIM requests.

Corrective Action: Give z/OSMF users access to CEA.CEAPDWB*. If you have CEA security definitions configured, you might already have the CEA.* resource defined.

User encounters message ICH408I

```
ICH408I USER(user ) GROUP(group ) NAME(user ) 031
CATALOG.SYVPLEX.MASTER CL(DATASET ) VOL(volser)
INSUFFICIENT ACCESS AUTHORITY
FROM CATALOG.*.MASTER (G)
ACCESS INTENT(UPDATE ) ACCESS ALLOWED(NONE )
```

Possible Cause: A user with insufficient authority is attempting to update the master catalog while creating the data diagnostic files. As a result, an Incident Log task request to FTP materials cannot compress (terse) the diagnostic snapshot data set.

Corrective Action: Determine whether the user should be allowed to update the master catalog. If so, you can authorize the user to create entries in the master catalog through the appropriate security commands.

To authorize a user to create entries in a user catalog, use the following command:

```
DEFINE ALIAS(NAME(CEA) RELATE(<usercatalog name>))
```

CEA address space is blocking the use of the sysplex dump directory

Possible Cause: CEA holds an exclusive ENQ to serialize on the sysplex dump directory data set while processing a z/OSMF request. Usually, the ENQ is released in microseconds. But sometimes an I/O error could result in holding the ENQ for longer time periods, therefore blocking DUMPSRV from updating the dump directory with information about a new dump, or your installation from doing maintenance on the sysplex dump directory data set.

Corrective Action: Use the system console command **F CEA,DROIPCS** to disconnect CEA from the IPCS sysplex dump directory data set.

CEA cannot allocate a data set for dump prepare or snapshot

Possible Cause: CEA alias is not cataloged properly.

Corrective Action: If your installation has a user catalog setup instead of using the MASTER catalog, you might need to define the CEA alias to the user catalog. For example:

```
DEFINE ALIAS(NAME(CEA) RELATE(YOUR_CATALOG_NAME))
```

Diagnostic log streams and other incident data for deleted incidents are not being deleted over time

Possible Cause: If you modified the HLQ parameter value in the CEAPRMxx parmlib member, CEA no longer detects the previously-stored diagnostic data files stored under the old high level qualifier.

Corrective Action: Carefully remove the data manually. The data exists in both log stream and data set format. Use caution as to not remove any needed data. Remove data sets and log streams manually. To list the available log streams, enter the following system console command: **D LOGGR,L**.

Most log streams with the status of AVAILABLE are the result of diagnostic snapshots taken at the time of the dump. The old high level qualifier appears in the log streams that were created earlier by CEA. To delete log streams, enter the following command: **SETLOGR FORCE,DELETE,LSN=logstreamname**.

To remove data sets, do the following:

- List the data sets having the same HLQ as the available log streams.
- Delete the data sets.

Problems when attempting to send data

When you invoke the Send Diagnostic Data wizard from the Incident Log task, the information supplied in the page is used to produce one FTP job for each diagnostic data file being sent. Thus, if an incident has a dump data set and three log snapshot files, four FTP jobs are created (and the FTP Job Status table will have four entries). To debug the FTP jobs, you need access to the job output. Typically, this is done by using z/OS System Display and Search Facility (SDSF) to examine the spooled output from the job.

FTP job status codes and other information

The Incident Log task allows you to display the status of the FTP jobs. On the *FTP Job Status* page, you can display the status of all FTP jobs associated with a particular incident or the FTP jobs associated with diagnostic data.

For a description of each FTP job status condition and the actions you can take to resolve errors in the jobs, see the online help for the *FTP Job Status* page.

Chapter 55. Configuration messages

This chapter describes the z/OSMF messages that you might encounter during the configuration process. These messages have a message ID between IZUG000-IZUG399.

For each configuration message, this document provides a detailed explanation of the message; describes the reason codes (if any) listed in each message; and, suggests actions that you can take to resolve the issue. The messages are organized by message ID.

Information about other messages you might encounter while configuring z/OSMF is provided in the following documents:

- Messages for the common event adapter (CEA) component of z/OS are prefixed by CEA. See *z/OS MVS System Messages*. For more information, see <https://www.ibm.com/docs/en/zos/3.1.0?topic=do-cea-messages>.
- z/OS-specific messages for the CIM server are prefixed by CFZ. For information about CIM server logging and messages, see *z/OS Common Information Model User's Guide*.
- Messages for the WebSphere Liberty profile are prefixed by CW. For descriptions of the WebSphere messages, see *WebSphere Liberty message descriptions* (www.ibm.com/docs/en/was-liberty/zos?topic=liberty-messages).

All other messages for z/OSMF are documented in the z/OSMF node of IBM Documentation, which is available at <https://www.ibm.com/docs/en/zos/3.1.0?topic=help-zosmf-messages>.

Because of the various layers of function involved in typical z/OSMF operations, locating a particular message might require you to check more than one location. For more information, see “[Types of messages in z/OSMF](#)” on page 285.

IZUG000-IZUG399

This topic describes the z/OSMF messages that have a message ID between IZUG000-IZUG399.

IZUG001E **A system request for storage in the storage-area area ended in error. The return code is *return-code*.**

Explanation

While attempting to allocate storage for the z/OSMF global storage area control block (IZUGSP), the system encountered an error that prevented the storage request from completing. The return code from the STORAGE macro is included in the message for diagnostic purposes.

This message might indicate a storage constraint in your system.

In the message text:

storage-area

Storage area that was requested.

return-code

Return code from the STORAGE macro.

System programmer response

This problem can be caused by a shortage of common storage or private storage. Refer to the STORAGE macro return code for more information.

Most likely, the problem is that ECSA storage (subpool 241) is exhausted. When this happens, further storage requests are allocated from the much smaller CSA storage area. Determine whether ECSA is exhausted and, if so, why the shortage occurred. For example, it is possible that not enough common storage is currently allocated, or that an address space has allocated a large amount of ECSA storage.

For assistance with this problem, contact IBM Support.

User response:

Contact the system programmer for assistance.

IZUG002E **The address space *address-space-name* failed to start. The following error codes were returned: Return code *return-code*, reason code *reason-code*.**

Explanation

This message is issued when one of the following z/OSMF dependent address spaces cannot be started:

- IZUINSTP, which is used for detecting z/OS UNIX and TCP/IP.
- z/OSMF server, which is named IZUSVR1, by default.

- z/OSMF angel process, which is named IZUANG1, by default.

The return code from the ASCRE macro is included in the message for diagnostic purposes.

In the message text:

address-space-name

Address space that was not started.

return-code

Return code from the ASCRE macro.

reason-code

Reason code from the ASCRE macro.

System programmer response

Refer to the STORAGE macro return code for more information.

For assistance with this problem, contact IBM Support.

User response:

Contact the system programmer for assistance.

IZUG003E	The request to listen for ENF code 83 failed with return code <i>return-code</i>.
-----------------	--

Explanation

An error occurred that prevented the system from listening for events related to the z/OSMF server, such as server start-up or server shut down. The return code from the ENFREQ macro is included in the message for diagnostic purposes.

In the message text:

return-code

Return code from the ENFREQ macro.

System programmer response

Refer to the ENFREQ macro return code for more information.

For assistance with this problem, contact IBM Support.

User response:

Contact the system programmer for assistance.

IZUG004I	An autostarted z/OSMF server was detected on another system in the same AUTOSTART group. As a result, the local system will connect to the autostarted server, rather than attempting to start a server locally.
-----------------	---

Explanation

Only one z/OSMF server can be active per autostart group in the sysplex. An autostarted z/OSMF server holds an enqueue on the z/OSMF user directory file

system, and handles the z/OSMF requests from other systems that are connected to the same AUTOSTART group.

This message can be issued during IPL for either of the following situations:

- The initializing system determines that an autostarted z/OSMF server is already active in the sysplex for the local AUTOSTART group. Because only one autostarted server can be active for a particular AUTOSTART group, the system does not attempt to autostart a z/OSMF server on the local system. In IZUPRMxx, the default setting of AUTOSTART(LOCAL) is treated as AUTOSTART(CONNECT).
- Multiple z/OSMF servers are attempting to autostart at the same time. The first system to complete IPL will autostart the z/OSMF server. The other systems will be treated as AUTOSTART(CONNECT) systems.

System programmer response:

No action is required.

User response:

No action is required.

IZUG005E	The attempt to cancel the z/OSMF server, which is identified by job <i>job-name</i>, ended with an error. The return code is <i>return-code</i>.
-----------------	---

Explanation

Only one z/OSMF server can be active per AUTOSTART group in the sysplex. Because a z/OSMF server is already active on another system in the same AUTOSTART group, the local system attempted to cancel the z/OSMF server that was attempting to start locally. However, an error occurred that prevented the cancel request from completing. The return code from the IEFSSREQ macro is included in the message for diagnostic purposes.

In the message text:

job-name

Job name for the local z/OSMF server.

return-code

Return code from the IEFSSREQ macro.

System programmer response:

Contact IBM Support for assistance.

User response:

Contact the system programmer for assistance.

IZUG006E	The z/OSMF service <i>service-number</i> failed after detecting one or more errors in the IZUPRMxx parmlib member. The return code is <i>return-code</i>.
-----------------	--

Explanation

During IPL, the indicated z/OSMF service attempted to read the contents of the currently active IZUPRMxx parmlib member. However, the service failed when it detected errors in one or more of the parmlib member statements. The return code from the z/OSMF service is included in the message for diagnostic purposes.

In the message text:

service-number

Number that identifies the failing z/OSMF service.

return-code

Return code from the failing service.

System programmer response:

Contact IBM Support for assistance.

User response:

Contact the system programmer for assistance.

IZUG007I	A z/OSMF server is already running in AUTOSTART group autostart-group-name.
-----------------	--

Explanation

Only one z/OSMF server can be active per AUTOSTART group in the sysplex. Because a z/OSMF server is already active on another system in the same AUTOSTART group, the local system has canceled the z/OSMF server that was attempting to start locally.

In the message text:

autostart-group-name

AUTOSTART group name.

System programmer response:

No action is required.

User response:

No action is required.

IZUG008I	Initialization of the z/OSMF server is suspended until TCP/IP is active.
-----------------	---

Explanation:

The z/OSMF server initialization does not complete until TCP/IP is initialized on the system.

System programmer response:

No action is required.

User response:

No action is required.

IZUG009I	the z/OSMF AUTOSTART server cannot be started due to the needed ENQ has been hold by another server
-----------------	--

Explanation:

the z/OSMF AUTOSTART server cannot be started due to the needed ENQ hold by another system

System programmer response:

No action is required.

User response:

No action is required.

IZUG010E	The ENQ failed to be released. RC= return-code, RSN= reason-code.
-----------------	--

Explanation

Only one z/OSMF server can be active per AUTOSTART group in the sysplex in IPL. When the activated z/OSMF server starts, a ENQ (qname:zosmf/rname:AUTOSTART_GROUP plus FILE SYSTEM mounted on user directory) will be hold. When the z/OSMF server stops, the ENQ is intended to be released. The return code and reason code from the ISQENQ macro is included in the message for diagnostic purposes.

In the message text:

return-code

Return code from the ISQENQ macro.

reason-code

Reason code from the ISQENQ macro.

System programmer response:

Contact IBM Support for assistance.

User response:

Contact the system programmer for assistance.

IZUG011E	The server name followed by parameter SERVER is larger than 8 characters.
-----------------	--

Explanation:

The server name followed by parameter *SERVER* is larger than the maximum of 8 characters.

System programmer response:

Adjust the server name for parameter *SERVER*.

User response:

No action is required.

IZUG012I	The command syntax is DISPLAY IZU or DISPLAY IZU,SERVER=IZUSVR1. IZUSVR1 is the z/OSMF server process PROC name.
-----------------	---

Explanation:

The command syntax is DISPLAY IZU or DISPLAY IZU,SERVER=IZUSVR1.

System programmer response:

Follow the explanation to issue the correct command.

User response:

Follow the explanation to issue the correct command.

IZUG013E **The service *service-name* in subsystem creation failed. The following error codes were returned: Return code *return-code* reason code *reason-code*.**

Explanation

The service performing subsystem creation failed due to the RC/RSN.

In the message text:

service-name

The service name of subsystem creation.

return-code

The service return code. The return code provided by the subsystem service.

reason-code

The service reason code. The reason code provided by the subsystem service.

System programmer response:

Figure out the root cause through RC/RSN.

User response:

Contact the system programmer.

IZUG015I **The command DISPLAY IZU is performed in the MODIFY command against the server *server-name* on system *system-name*.**

Explanation

The command of DISPLAY IZU is performed in the MODIFY command against the z/OSMF server on a system.

In the message text:

server-name

The z/OSMF server job name.

system-name

The system name.

System programmer response:

No action is required.

User response:

No action is required.

IZUG016I **The server *server-name* on system *system-name* is not available.**

Explanation

The z/OSMF server in the specified system is not available.

In the message text:

server-name

The z/OSMF server job name.

system-name

The name of the system hosting the z/OSMF server.

System programmer response:

Check the status of the z/OSMF server in the system specified.

User response:

Contact the system programmer.

IZUG017I **The AUTOSTART server in AUTOSTART GROUP *autostart_group* is not available. The DISPLAY command in the CONNECT system cannot proceed.**

Explanation

When the DISPLAY command of z/OSMF is issued by the customer on the CONNECT system, if the AUTOSTART server the CONNECT system connects to is not available, this message is prompted.

In the message text:

autostart_group

AUTOSTART group name.

System programmer response:

Check the status of the AUTOSTART server in the autostart group specified.

User response:

Contact the system programmer.

IZUG018W **The property *the-property* is set to *the-value*. The value is incorrect. The default value of *the-default* will be used.**

Explanation

The specified timeout value is incorrect. The specified default value will be used.

In the message text:

the-property

The property that is set.

the-timeout

The value for the property.

the-default

The default value for the property.

System programmer response:

No action is required.

User response:

No action is required.

IZUG020I **The server needs to be restarted to make the change made by the SET command effective.**

Explanation:

The server needs to be restarted to make the change made by the SET command effective.

System programmer response:

No action is required.

User response:

No action is required.

IZUG021E **The argument *argument* is required.**

Explanation

The indicated argument is required and must be specified.

In the message text:

the-argument

Name of the required argument.

System programmer response:

Retry the operation and provide the specified argument.

User response:

No action is required.

IZUG021I **To make the changes from the SET command effective, restart the z/OSMF server with IZUPRM=PREV specified.**

Explanation:

The SET IZU or SETIZU command was used to change settings in the currently active IZUPRMxx parmlib member. One or more of the changes require the z/OSMF server to be restarted before the changes can take effect.

System programmer response

Restart the z/OSMF server, as follows: START IZUSVR1, IZUPRM=PREV.

Specifying IZUPRM=PREV, which is the default, ensures that you use the same set of IZUPRMxx parmlib values that were in effect in the previous instance of z/OSMF. Alternatively, IZUPRM=PREV can be specified in the server started procedure.

User response:

Ask the system programmer to restart the z/OSMF server.

IZUG022I **The SETIZU command is issued. The z/OSMF AUTOSTART server *server-name* home page with AUTOSTART_GROUP**

***autostart-group* in system *system-name* can be accessed at *URI*.**

Explanation

After the SETIZU command is issued, the z/OSMF server can be accessed at the new URI.

In the message text:

server-name

The z/OSMF server job name.

autostart-group

The autostart group name.

system-name

The name of the system.

URI

The URI of the z/OSMF server.

System programmer response:

No action is required.

User response:

No action is required.

IZUG023I **The SETIZU command was performed in the MODIFY command against server *server-name* on system *system-name*.**

Explanation

The SETIZU command was performed in the MODIFY command against the z/OSMF server running on the system.

In the message text:

server-name

The job name of the z/OSMF server.

system-name

The system name.

System programmer response:

No action is required.

User response:

No action is required.

IZUG024E **The keyword *keyword-name* is not valid. Enter a valid keyword and value.**

Explanation

The specified keyword is not recognized as a valid keyword of the SET command.

In the message text:

keyword

The keyword that was specified for the SET command.

System programmer response:

Enter a valid keyword and value. For information about the valid keywords and values, see *z/OS MVS System Commands*.

User response:

No action is required.

IZUG025E **The SETIZU command cannot be processed because it exceeds the maximum length.**

Explanation:

The specified SETIZU command exceeds 126 characters, which is longer than maximum length allowed.

System programmer response:

No action is required.

User response:

Enter a valid keyword and value. For information about the valid keywords and values, see *z/OS MVS System Commands*.

IZUG025I **The value *prop-value* for property *prop-name* was found in file *file-name*.**

Explanation

The indicated property was found in the specified file containing the indicated value.

In the message text:

prop-value

The value of the property.

prop-name

The name of the property.

file-name

The name of the file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG026E **The specified value for the parameter is too long.**

Explanation:

The specified value for parameter is too long for the SET command.

System programmer response:

Specify a valid value for the parameter.

User response:

No action is required.

IZUG027E **The specified value for the parameter is not valid.**

Explanation:

The specified value for parameter is not valid for SET command.

System programmer response:

Correct the error and retry. Ensure the value for the specified property is valid.

User response:

No action is required.

IZUG028I **The z/OSMF autostart group in system *system-name* was changed to *autostart-groupname* with the SETIZU command.**

Explanation

The z/OSMF autostart group for the indicated system was modified by using the SETIZU command. Specifically, the command was used to change the value of the AUTOSTART_GROUP parameter in the active IZUPRMxx parmlib member for your system.

In the message text:

system-name

System that was specified.

autostart-groupname

Autostart group name that was changed with the SETIZU command.

System programmer response:

No action is required.

User response:

No action is required.

IZUG029E **An autostart group with the same name is already associated with the primary z/OSMF server on another system. Select a unique autostart group name.**

Explanation

An attempt was made to change the autostart group for the primary z/OSMF server on this system. However, the specified autostart group name is already associated with the primary z/OSMF server on another system in the sysplex.

In a sysplex, the set of systems that is served by an autostarted z/OSMF server is known as the autostart group. z/OSMF includes one autostart group by default. To have more z/OSMF servers autostarted in a sysplex, you must associate each server and the systems it serves with a unique autostart group name.

System programmer response:

No action is required.

User response:

Specify a different autostart group name for the z/OSMF server.

IZUG030E **Script *script-name* requires the following input options: *input-options*.**

Explanation

The valid script options are displayed. For information about the script options, see [IBM z/OS Management Facility Configuration Guide](#).

In the message text:

script-name

Name of the script

input-options

Options required by the script.

System programmer response:

Correct the error and retry the operation.

User response:

No action is required.

IZUG031I **The *file-name* file will be used from the following location: *file-name-location***

Explanation

The specified file will be used from the specified location.

In the message text:

file-name

Name of the file.

file-name-location

Name of the file location.

System programmer response:

No action is required.

User response:

No action is required.

IZUG032W **The property *var-name* could not be found in *file-name*. Defaulting value to: *value-name***

Explanation

The specified variable could not be found in the specified file. The variable will default to the specified value. The value is obtained from the shipped default file.

In the message text:

var-name

Name of the variable.

file-name

Name of the file.

value-name

Value for the variable.

System programmer response:

No action is required.

User response:

No action is required.

IZUG033I **Examine each of the output execs and determine which exec is appropriate for your environment. Run one exec only. The output execs are: *rexex-exec-name1*, *rexex-exec-name2***

Explanation

The z/OSMF configuration process creates sample security execs to assist your security administrator in creating security authorizations for z/OSMF. The execs are tailored, based on the selections you made when running the script `izusetup.sh -config`, or specified in your override file.

z/OSMF creates several execs to accommodate a number of possible configuration paths, however, your installation should run only one of the execs. Your choice of which exec to run depends on whether:

- You are creating a new z/OSMF configuration or migrating from an earlier release of z/OSMF.
- The configuration process detected a change in the authorization mode for your installation.
- One or more of your selected services require additional security authorizations on your z/OS system.

In the message text:

rexex-exec-name1

Name of the first generated RACF exec.

rexex-exec-name2

Name of the second generated RACF exec.

Have your security administrator review the execs, and run the exec that is appropriate for your environment. Most likely, one of the following descriptions fits your environment.

- The exec named `configfilename.cfg.rexx` is the appropriate choice for new or first-time z/OSMF configurations. This exec contains the superset of required RACF commands, tailored for the service selections you specified when running the script `izusetup.sh -config`, or specified in your override file.
- The exec named `configfilename.cfg.convertFromSAftoREP.rexx` is the appropriate choice if your installation is migrating from an earlier release of z/OSMF for the new configuration. This exec contains the subset of RACF

commands that are needed to update an existing security setup to SAF based security.

z/OSMF creates the execs for any izusetup.sh invocation that updates your configuration file, even if you are just adding a service to an existing configuration (izusetup.sh -add). If the services to be added require no additional security setup, the created execs are "empty" and need not be run. It is recommended that your security administrator review each of the output execs to determine whether they require changes and should be run for your installation.

System programmer response:

Have your security administrator review the execs and determine which exec to run, based on the guidance information in this message. For more information about the security execs, see [IBM z/OS Management Facility Configuration Guide](#).

User response:

No action is required.

IZUG034I	The z/OSMF configuration process has created a set of sample security execs for your reference in directory <i>directory-name</i>.
-----------------	---

Explanation

The z/OSMF configuration process creates sample security execs to assist your security administrator in creating security authorizations for z/OSMF. The execs are tailored, based on the selections you made when running the script izusetup.sh -config, or specified in your override file. The execs are stored in the indicated directory.

In the message text:

directory-name

Directory in which the generated sample security execs reside.

System programmer response:

See the accompanying message for the names of the sample security execs.

User response:

No action is required.

IZUG035W	The default value <i>file-name</i> will be used because a fully-qualified path name was not provided for the file.
-----------------	---

Explanation

A fully-qualified path name was not provided for the file. The default value specified in the property IZU_CONFIG_DIR will be used.

In the message text:

file-name

Name of the file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG036W	The variable <i>var-name</i> could not be found in the configuration file <i>file-name</i>. Defaulting value to: <i>value-name</i>
-----------------	---

Explanation

The specified variable could not be found in the specified configuration file. The variable will default to the specified value.

In the message text:

var-name

Name of the variable.

file-name

Name of the configuration file.

value-name

Value for the variable.

System programmer response:

No action is required.

User response:

No action is required.

IZUG037E	The value <i>value</i> in file <i>file</i> is incorrect for property <i>property</i>.
-----------------	--

Explanation

The specified value is incorrect for the property.

In the message text:

value

The value for the property

file

File containing the value.

property

Property containing the value.

System programmer response:

Correct the error and retry. Ensure the value for the specified property is valid.

User response:

No action is required.

IZUG038E	The file system that is mounted on the user directory is not available.
-----------------	--

Explanation

The data directory cannot be accessed because its enqueue is held by another z/OSMF server.

When a z/OSMF server is started, it obtains an enqueue on the data directory (sometimes called the user directory) for the duration of the server start-up. This message is issued if the user attempts to start a z/OSMF server using a directory that is already enqueued by a z/OSMF server on another system.

System programmer response

Check the directory that is enqueued by another z/OSMF server. For the z/OSMF server that holds the enqueue, ensure that there is no in-progress work and then stop the server. When a z/OSMF server is stopped, it releases any enqueues it holds. Then, start the z/OSMF server on the local system.

To view the activity for a z/OSMF server, you can use the z/OSMF Usage Statistics task on the system on which the server is running.

For information about starting and stopping a z/OSMF server, see *z/OSMF Configuration Guide*.

User response:

Report this problem to your system programmer.

IZUG039I	The override file <i>config-file</i> has been migrated to the format: <i>release-level</i>.
-----------------	--

Explanation

The specified configuration file has been migrated to the specified release level.

In the message text:

config-file

Name of the configuration file.

release-level

Level of the release.

System programmer response:

No action is required.

User response:

No action is required.

IZUG040I	<i>system-time</i> DISPLAY IZU
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Explanation

The message displays 'DISPLAY IZU' to be part of the command output.

In the message text:

system-time

The time of performing the DISPLAY IZU command.

System programmer response:

Ensure that the role file or alias exists and retry the operation.

User response:

No action is required.

IZUG041I	The server started at <i>start-time</i> and has been running for <i>running-time</i>.
-----------------	--

Explanation

The time when the server started and how long the server has been running.

In the message text:

start-time

The time the server started.

running-time

The amount of time that the server has been running.

System programmer response:

No action is required.

User response:

No action is required.

IZUG042I	The override file <i>file-name</i> conforms to the expected format: <i>release-level</i>. No migration will be performed.
-----------------	--

Explanation

No migration is needed since the specified override file is at the correct version level.

In the message text:

file-name

Name of the override file.

release-level

Level of the release.

System programmer response:

No action is required.

User response:

No action is required.

IZUG043E	Unable to update override file <i>file-name</i>.
-----------------	---

Explanation

The specified override file could not be updated.

In the message text:

file-name

Name of the override file.

System programmer response:

Ensure that the caller is authorized to update the override file. For more information, review the log file that was created for the error.

User response:

No action is required.

IZUG044I **The input override file *over-file* was saved to a backup file *back-up-override-file*.**

Explanation

The data of the source override file has been saved to the specified override file.

In the message text:

over-file

Name of the override file.

back-up-override-file

Name of the backup override file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG045E **Unable to back up override file data.**

Explanation:

The data of the source override file could not be saved. Ensure that the permission settings are correct for the file and directory.

System programmer response:

Ensure that the permission settings are correct for the file and directory.

User response:

No action is required.

IZUG046I **Enter the existing *group-name* group name that is used to authorize users to the *service-name* resources. Enter *keyword-name* if no group exists.**

Explanation

The message prompts for the service group name. These group are expected to already exist. If a group does not exist or if the group has not yet been created enter the specified keyword. The RACF exec generated will have the required commands commented out. Once the group has been created, update and uncomment the commands in the RACF exec.

In the message text:

group-name

Name of the group

service-name

Name of the service

keyword-name

Name of the keyword

System programmer response:

Enter the information or enter the specified keyword if no group exists.

User response:

No action is required.

IZUG047I **Enter the existing *group-name* group name that is used to authorize users to the *service-name* resources. Press Enter to accept the default *default-value*, or enter *keyword-name* if no group exists.**

Explanation

The message prompts for the group name. These groups are expected to already exist. If a group does not exist or if the group has not yet been created enter the specified keyword. The RACF exec generated will have the required commands commented out. Once the group has been created, update and uncomment the commands in the RACF exec.

In the message text:

group-name

Name of the group

service-name

Name of the service

default-value

The default value

keyword-name

Name of the keyword

System programmer response:

Enter the information, press Enter to accept the default, or enter the keyword if no group exists.

User response:

No action is required.

IZUG048W **Group *group-name* does not exist.**

Explanation

The specified group does not exist.

In the message text:

group-name

Name of the group.

System programmer response:

Ensure that the specified group exists. If not create it and retry.

User response:

No action is required.

IZUG049I **z/OSMF configuration has detected a *current-auth-mode* to *new-auth-mode* authorization mode switch.**

Explanation

The current authorization mode will be changed to the new authorization mode specified.

In the message text:

current-auth-mode

The current authorization mode.

new-auth-mode

The new authorization mode.

System programmer response:

No action is required.

User response:

No action is required.

IZUG050I **z/OSMF configuration has detected a *current-auth-mode* to *new-auth-mode* authorization mode switch. The data file system *file-system* must be mounted.**

Explanation

The authorization mode switch indicated requires the data file system specified be mounted.

In the message text:

current-auth-mode

The current authorization mode.

new-auth-mode

The new authorization mode.

file-system

The data file system.

System programmer response:

No action is required.

User response:

No action is required.

IZUG051W **The permissions assigned to directory *directory-name* will be changed to *permissions*.**

Explanation

The current assigned permissions for the specified directory will be changed to the new permissions specified.

In the message text:

directory-name

The directory being checked.

permissions

The new permissions that will be assigned to the specified directory.

System programmer response:

No action is required.

User response:

No action is required.

IZUG052W **The group assigned to directory *directory-name* will be changed to *group-name*.**

Explanation

The current assigned group of the specified directory will be changed to the new group specified.

In the message text:

directory-name

The directory being checked.

group-name

The new group that will be assigned to the specified directory.

System programmer response:

No action is required.

User response:

No action is required.

IZUG053W **The owner assigned to directory *directory-name* will be changed to *new-owner*.**

Explanation

The current owner of the specified directory will be changed to the new owner specified.

In the message text:

directory-name

Directory being checked.

new-owner

User id of the new owner to be assigned to the specified directory.

System programmer response:

No action is required.

User response:

No action is required.

IZUG054I **To obtain the results of the *verification-type* verification, review report *report-name*.**

Explanation

Review the specified report file to obtain the results of the verification.

In the message text:

verification-type

The type of verification being performed.

report-name

Name of the verification report.

System programmer response:

Review the specified report.

User response:

No action is required.

IZUG055E	Group <i>group-name</i> not permitted to RACF class <i>class-name</i>.
-----------------	---

Explanation

The specified group name is not permitted to the specified RACF class.

In the message text:

group-name

Name of the group being evaluated.

class-name

Name of the RACF class.

System programmer response:

For more information, review the log file created for the error and the RACF report.

User response:

No action is required.

IZUG056I	The file <i>target-file</i> was saved to a backup file <i>back-up-file</i>.
-----------------	--

Explanation:

The data of the source file has been saved to the specified file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG057E	File <i>file-name</i> does not exist or is not accessible.
-----------------	---

Explanation

The specified file does not exist or is not accessible.

In the message text:

file-name

Name of the file.

System programmer response:

Ensure that the specified file exists and is accessible. Retry your request.

User response:

No action is required.

IZUG058E	File <i>file-name</i> is incomplete. The property <i>configuration-property</i> is missing.
-----------------	--

Explanation

This message indicates that the specified configuration property was not found. The script exits in error.

In the message text:

file-name

The configuration file.

configuration-property

The configuration property.

System programmer response:

Ensure that the specified property exists in the specified configuration file.

User response:

No action is required.

IZUG059I	The configuration operation already exists. The configuration parameters have not changed since the last time the operation was run.
-----------------	---

Explanation:

The configuration operation already exists. The configuration parameters have not changed since the last time the operation was run.

System programmer response:

No action is required.

User response:

No action is required.

IZUG060I	Specify the CEA high level qualifier (HLQ) to use for log snapshot data sets. The HLQ can be 1-4 characters. Or press Enter to accept the default <i>default-HLQ-mode</i>:
-----------------	---

Explanation:

The message prompts for the high level qualifier to use.

System programmer response:

Enter the high level qualifier value to use or press Enter to use the specified default value.

User response:

No action is required.

IZUG061I **What security mode do you want to use? To use SAF mode, enter S. To use Repository mode, enter R. Or press Enter to accept the current setting *current-mode*:**

Explanation

The message prompts for the security mode to use.

In the message text:

current-mode

Current security mode for the z/OSMF configuration.

System programmer response:

Enter S to use SAF security mode or R to use Repository mode or press Enter to use the current setting for security mode.

User response:

No action is required.

IZUG062I **What security mode do you want to use? To use SAF mode, enter S. To use Repository mode, enter R:**

Explanation:

The message prompts for the security mode to use.

System programmer response:

Enter S to use SAF security mode or R to use Repository mode.

User response:

No action is required.

IZUG063E **File *file-name* could not be found in *dataset-name*. This file is required for the configuration of Common Event Adapter (CEA) for Incident Log.**

Explanation

The specified file does not exist in specified data set. This file is used by the Incident Log verification to verify the Incident Log configuration. As part of the configuration of CEA for Incident Log, this file is copied to the specified target dataset where it will be used to create a test dump for the verification of Incident Log.

In the message text:

file-name

File name.

dataset-name

Data set name.

System programmer response:

Ensure that the specified file exists in the specified data set. Retry your request.

User response:

No action is required.

IZUG064I **Enter the name of the target data set to be used for saving the updated *member-name* parmlib member. Specify the fully qualified data set name, or press Enter to accept the default: *default-member-name*:**

Explanation

The message prompts you for the name of the data set to be used for saving the updated parmlib members, IEADMCnn and CEAPRMnn, which are used for Incident Log task processing. A fully qualified data set name is expected.

In the message text:

member-name

User-specified parmlib member

default-member-name

Default data set name.

System programmer response:

Specify the fully qualified data set name, or press Enter to accept the supplied default if it is correct for your environment. If you specify the data set name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG065I **Enter the name of the target data set to be used for saving the updated *member-name* parmlib member. Specify the fully qualified data set name, or press Enter to use SYS1.PARMLIB:**

Explanation

The message prompts you for the name of the data set to be used for saving the updated parmlib members, IEADMCnn and CEAPRMnn, which are used for Incident Log task processing. A fully qualified data set name is expected.

In the message text:

member-name

User-specified PARMLIB member.

System programmer response:

Specify the fully qualified data set name, or press Enter to save the updated member in SYS1.PARMLIB. If you specify the data set name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG066I **Enter the name of the source data set for the IEADMCZM parmlib member. Specify the fully qualified data set name, or press Enter to accept the default data-set-name:**

Explanation

The message prompts you for the name of the data set that contains the IEADMCZM parmlib member. This is shipped by default in SYS1.SAMPLIB. A fully qualified data set name is expected.

In the message text:

data-set-name

Default data set name.

System programmer response:

Specify the fully qualified data set name, or press Enter to accept the supplied default if it is correct for your environment. If you specify the data set name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG067I **Enter the name of the source data set for the IEADMCZM parmlib member. Specify the fully qualified data set name, or press Enter to use SYS1.SAMPLIB:**

Explanation:

The message prompts you for the name of the data set that contains the IEADMCZM parmlib member. This is shipped by default in SYS1.SAMPLIB. A fully qualified data set name is expected.

System programmer response:

Specify the fully qualified data set name, or press Enter to use SYS1.SAMPLIB as the source for the IEADMCZM member. If you specify the data set name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG068W **The configuration property *cfg-prop* was found in file *cfg-ovr-file*. This property will be ignored.**

Explanation

The indicated configuration property was found in either the configuration or override file. The indicated property will be ignored since the property can only be set by manually exporting it or through the use of

the environment file specified by environment variable IZU_ENV_FILE.

In the message text:

cfg-prop

Name of the property.

cfg-ovr-file

The configuration or override file.

System programmer response:

If the intent was to set the specified property, either update the file specified by IZU_ENV_FILE with the property and then export IZU_ENV_FILE OR manually export the property prior to calling the script. Otherwise, no action is required.

User response:

No action is required.

IZUG069I **The configuration property *cfg-prop* is set to the value *cfg-val*.**

Explanation

The indicated configuration property is set to the indicated value.

In the message text:

cfg-prop

Name of the property.

cfg-val

Value of the property.

System programmer response:

No action is required.

User response:

No action is required.

IZUG070I **If you have AUTOGID enabled, RACF can assign unused GIDs for your group ids. Do you want RACF to automatically assign GIDs to groups created by z/OSMF? For yes, enter Y. For no, enter N:**

Explanation:

RACF can automatically assign unused GIDs to your group ids if AUTOGID is enabled. If selected, all GID properties in the configuration file will be set to AUTOGID. This can also reduce the number of prompts for UIDs.

System programmer response:

Enter Y to have RACF automatically assign unused GIDs for your z/OSMF created group ids or enter N to assign your own.

User response:

No action is required.

IZUG071I **If you have AUTOGID enabled, RACF can assign unused GIDs for your group ids. Do you want RACF to automatically assign GIDs to groups created by z/OSMF? For yes, enter Y. For no, enter N. Or press Enter to accept the default default-value:**

Explanation

RACF can automatically assign unused GIDs to your group ids if AUTOGID is enabled. If selected, all GID properties in the configuration file will be set to AUTOGID. This can also reduce the number of prompts for UIDs.

In the message text:

default-value

The default value to use.

System programmer response:

Enter Y to have RACF automatically assign unused GIDs for your z/OSMF created group ids or enter N to assign your own.

User response:

No action is required.

IZUG072I **If you have AUTOUID enabled, RACF can assign unused UIDs for your user ids. Do you want RACF to automatically assign UIDs to user ids created by z/OSMF? For yes, enter Y. For no, enter N:**

Explanation:

RACF can automatically assign unused UIDs to your user ids if AUTOUID is enabled. If selected, all UID properties in the configuration file will be set to AUTOUID. This can also reduce the number of prompts for UIDs.

System programmer response:

Enter Y to have RACF automatically assign unused UIDs for your z/OSMF created user ids or enter N to assign your own.

User response:

No action is required.

IZUG073I **If you have AUTOUID enabled, RACF can assign unused UIDs for your user ids. Do you want RACF to automatically assign UIDs to user ids created by z/OSMF? For yes, enter Y. For no, enter N. Or press Enter to accept the default default-value:**

Explanation

RACF can automatically assign unused UIDs to your user ids if AUTOUID is enabled. If selected, all UID properties in the configuration file will be set to AUTOUID. This can also reduce the number of prompts for UIDs.

In the message text:

default-value

The default value to use.

System programmer response:

Enter Y to have RACF automatically assign unused UIDs for your z/OSMF created user ids or enter N to assign your own.

User response:

No action is required.

IZUG074I **Clearing cached content for z/OSMF online help at location: *help-dir*.**

Explanation

While processing your request, z/OSMF deployed or redeployed one or more services. This activity includes the deletion of the contents of the z/OSMF online help directory. This processing is normal.

In the message text:

help-dir

Name of the directory to be processed.

System programmer response:

No action is required.

User response:

No action is required.

IZUG075I **Environment file *env-file* has been sourced.**

Explanation

The indicated environment file has been sourced.

In the message text:

env-file

Name of the environment file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG076E **An unexpected error occurred.**

Explanation:

An error occurred, but the cause could not be determined.

System programmer response:

Check the job log for any other messages that might indicate a reason for this error. If the log messages do not explain the cause of the problem, contact IBM Support for assistance.

User response:

No action is required.

IZUG077E	The value specified for <i>attribute</i> is not valid. The value must start with an alpha character (A-Z, a-z) or a special character (# \$ @) and must contain <i>number</i> characters.
-----------------	--

Explanation

The value specified for the variable is not valid.

In the message text:

attribute

Attribute for the prompt.

number

Minimum and maximum number of characters the variable can contain.

System programmer response:

Enter a value that starts with an alpha character (A-Z, a-z) or a special character (# \$ @) and contains between the minimum and maximum number of characters specified.

User response:

No action is required.

IZUG078E	File <i>file-name</i> does not exist. This file is required for the configuration of Common Event Adapter (CEA) for Incident Log.
-----------------	--

Explanation

The specified file does not exist. This file is required for the configuration of the Incident Log service.

In the message text:

file-name

File name.

System programmer response:

Ensure that the specified file exists. Retry your request.

User response:

No action is required.

IZUG079E	File <i>file-name</i> could not be found in SYS1.SAMPLIB. This file is required for the configuration of Common Event Adapter (CEA) for Incident Log.
-----------------	--

Explanation

The specified file does not exist in SYS1.SAMPLIB. This file is used by the Incident Log verification to verify the Incident Log configuration. As part of the configuration of CEA for Incident Log, this file is copied to the specified target dataset where it will be used to create a test dump for the verification of Incident Log.

In the message text:

file-name

File name.

System programmer response:

Ensure that the specified file exists in SYS1.SAMPLIB. Retry your request.

User response:

No action is required.

IZUG080I	All of the available z/OSMF services have been configured already.
-----------------	---

Explanation:

All of the services that were shipped with z/OSMF have been configured with the product already. No other services remain to be configured.

System programmer response:

No action is required.

User response:

No action is required.

IZUG081E	The service -add request cannot be performed because the specified configuration file <i>file-name</i> was not found. This file is required for adding services.
-----------------	---

Explanation

The request to add one or more services could not be completed because the specified input configuration file was not found. This file is required for configuring services on your system.

In the message text:

file-name

Name of the configuration file.

System programmer response:

Ensure that the specified configuration file exists. If not, recreate the configuration file with the values for your existing z/OSMF configuration. Retry your request.

User response:

No action is required.

IZUG082E	File system <i>file-system-name</i> at mount point <i>file-system-mount-point</i> must be a ZFS or HFS file
-----------------	--

system and must be mounted in read-write mode.

Explanation

The specified file system at the specified mount point must be of type ZFS or HFS and must be mounted in read-write mode. This can be done by specifying rdwr for the mode when mounting the filesystem.

In the message text:

file-system-name

Name of the file system.

file-system-mount-point

The mount point of the file system.

System programmer response:

Ensure the file system is a ZFS or HFS. Also, ensure that the file system is mounted in read-write mode.

User response:

No action is required.

IZUG083I The verification of *verify-type* has completed successfully.

Explanation

The verification request completed.

In the message text:

verify-type

Type of verification that was requested.

System programmer response:

No action is required.

User response:

No action is required.

IZUG084W The IZU_DATA_DIR variable, which identifies the mount point of the z/OSMF data file system, has been reset to the default value *mount-point*.

Explanation

The z/OSMF configuration process has updated the IZU_DATA_DIR variable in your configuration file to the default value of /global/zosmf. In old releases of z/OSMF, the z/OSMF data file system was mounted at /var/zosmf by default.

In the message text:

mount-point

Default mount point for the z/OSMF data file system.

System programmer response:

Determine whether the z/OSMF data file system on your system is currently mounted at the old

default location /var/zosmf. If so, unmount it. You can remount the data file system manually at the new location /global/zosmf or you can allow z/OSMF processing to mount it at this location during the processing of the izusetup.sh -config script.

User response:

No action is required.

IZUG085I The IZU_IL_CONFIGURE variable must be set to Y before completing action *action*.

Explanation

The IZU_IL_CONFIGURE variable in the configuration file must be set to Y before the specified action can be completed.

In the message text:

action

The Incident Log action to be completed.

System programmer response:

Enter the izusetup.sh -config [filename.cfg] command, specifying as input the configuration file that you used previously for setting up z/OSMF. If you omit this file name, the IBM-supplied configuration file (izudflt.cfg) is used. Then, when prompted to configure the Incident Log, enter Y.

User response:

No action is required.

IZUG086E The Incident Log configuration request failed. The IZU_IL_CEA_CONFIGURE variable in the configuration file must be set to Y before the request can be processed.

Explanation:

The Incident Log configuration request failed because the IZU_IL_CEA_CONFIGURE variable is not set to Y.

System programmer response:

Enter the izusetup.sh -config [filename.cfg] command. The configuration file name is optional. If you omit this file name, the IBM-supplied configuration file (izudflt.cfg) is used. Then, when prompted to configure the Incident Log, enter Y.

User response:

No action is required.

IZUG087I The IZU_IL_CEA_CONFIGURE variable must be set to Y before completing action *action*.

Explanation

The IZU_IL_CEA_CONFIGURE environment variable in the configuration file must be set to Y before the specified action can be completed.

In the message text:

action

The Incident Log action to be completed.

System programmer response:

Enter the `izusetup.sh -config [ filename.cfg ]` command. Use the configuration file that you used previously for setting up z/OSMF. If you omit this file name, the IBM-supplied configuration file (`izudflt.cfg`) is used. Then, when prompted to configure the Incident Log, enter Y.

User response:

No action is required.

IZUG088E	The required environment variable <i>env-var</i> is not set.
-----------------	---

Explanation

For script processing, z/OSMF requires that the indicated environment variable be set to a valid value. However, no value was found for the variable.

In the message text:

env-var

Name of the variable that was not set.

System programmer response:

A serious error has occurred. Contact IBM Support.

User response:

No action is required.

IZUG089E	Directory <i>directory-name</i> must be writable.
-----------------	--

Explanation

Processing of the script has stopped. For processing to continue, the indicated directory must be writable.

In the message text:

directory-name

Name of the directory.

System programmer response:

Ensure that the user running the script has permission to write to the directory. After correcting the error, have the user run the script again.

User response:

No action is required.

IZUG090I	Environment variable <i>env-var</i> has been set to the default value <i>env-value</i>.
-----------------	--

Explanation

The indicated environment variable has been set to the specified default value.

In the message text:

env-var

Name of the variable.

env-value

Value of the variable.

System programmer response:

No action is required.

User response:

No action is required.

IZUG091I	Environment variable <i>env-var</i> is set to the value <i>env-value</i>.
-----------------	--

Explanation

The indicated environment variable is set to the indicated value.

In the message text:

env-var

Name of the variable.

env-value

Value of the variable.

System programmer response:

No action is required.

User response:

No action is required.

IZUG092E	Path <i>/usr/lib</i> was not found in LIBPATH variable in file <i>file-name</i>.
-----------------	---

Explanation

The path */usr/lib* was not found in the LIBPATH variable in the specified file.

In the message text:

file-name

Name of the file that was processed.

System programmer response:

Ensure that the path */usr/lib* in LIBPATH environment variable is set in the specified file.

User response:

No action is required.

IZUG093I	The directory <i>tmpdir-value</i> will be used for storing temporary files.
-----------------	--

Explanation

z/OSMF processing will use the indicated directory for storing temporary files.

In the message text:

tmpdir-value

Temporary directory value.

System programmer response:

No action is required.

User response:

No action is required.

IZUG094I In the previous configuration of z/OSMF, you allowed z/OSMF to configure the Common Information Model (CIM) server. In the current release of z/OSMF, the CIM configuration procedure is modified.

Explanation:

The procedure for configuring the CIM server has been modified in the current release of z/OSMF.

System programmer response:

No action is required.

User response:

No action is required.

IZUG095I The Common Information Model (CIM) server must be configured and started before proceeding with configuration.

Explanation:

After reviewing the RACF instructions for the CIM server, and running the exec, your installation must configure and start the CIM server before proceeding with the configuration of z/OSMF.

System programmer response:

Review the contents of the RACF exec that was created by the z/OSMF configuration process and run the exec, if appropriate. Then, configure and start the CIM server. For information about configuring the CIM server, see *z/OS Common Information Model User's Guide*, SC33-7998, which is available on-line in the [IBM z/OS Internet Library](#).

User response:

No action is required.

IZUG096I Do you need assistance in setting up security for the Common Information Model (CIM) server? To have z/OSMF create an exec with sample RACF commands, enter Y. Otherwise, enter N.

Explanation:

The z/OSMF configuration process includes the option of creating a REXX exec with sample RACF commands. Your security administrator can use these commands for authorizing z/OSMF users to the CIM server.

System programmer response:

To allow z/OSMF to create this exec, enter Y in response to this prompt. Otherwise, enter N.

User response:

No action is required.

IZUG097I Do you need assistance in setting up security for the Common Information Model (CIM) server? To have z/OSMF create an exec with sample RACF commands, enter Y. For no, enter N. Press Enter to accept the default value:

Explanation

The z/OSMF configuration process includes the option of creating a REXX exec with sample RACF commands. Your security administrator can use these commands for authorizing z/OSMF users to the CIM server.

In the message text:

value

Default for whether to set up RACF security for the CIM server.

System programmer response:

Enter Y or N, or accept the default value.

User response:

No action is required.

IZUG098E Unable to remove file *file-name*.

Explanation

z/OSMF processing of the `izusetup.sh -finish` request was unable to remove the indicated file. Possibly, the file is marked read-only or has permissions that do not allow for write access.

In the message text:

file-name

File that could not be removed.

System programmer response:

Ensure that the specified file exists. Ensure that the file and the file directory have permissions that allow for write access. Also, verify that the user ID for the request has update access to the file and its directory. Then, retry your request.

User response:

No action is required.

IZUG099W File *file-name* does not exist.

Explanation

In processing a `izusetup.sh -config` request, z/OSMF did not find the indicated file. If the file is needed, z/OSMF processing will create it using IBM defaults.

In the message text:

file-name

File that does not exist.

System programmer response:

No action is required.

User response:

If you are running the izusetup.sh script in interactive mode, the script will prompt you for a number of installation-specific values needed for configuration. In response to each prompt, you must either press Enter to use the default value, or type your installation specific value. Ensure that these values are appropriate for your setup. If you are running the script in fastpath mode, check the override file to ensure that the appropriate values have been specified for your installation.

IZUG100E Unable to register provider *name*.

Explanation

The specified provider could not be registered. Typically, this error occurs when the user is not authorized to write to the Common Information Model (CIM) server repository or when the providers are missing.

In the message text:

name

Name of the provider.

System programmer response:

Verify that the user is authorized to write to the Common Information Model (CIM) server repository. Ensure that the providers are available.

User response:

No action is required.

IZUG101W The file or parmlib member was not overwritten.

Explanation:

The specified file or parmlib member was not overwritten.

System programmer response:

No action is required.

User response:

No action is required.

IZUG102E The request to start the Common Information Model (CIM) server failed because the server is already running.

Explanation:

The Common Information Model (CIM) server could not be started because it is already running.

System programmer response:

Shutdown the CIM server by entering the cimserver -s command. Then, re-run the script.

User response:

No action is required.

IZUG104I Provider *namemodule* has already been registered with the Common Information Model (CIM) server.

Explanation

The specified provider module is already registered with the Common Information Model (CIM) server.

In the message text:

name

Name of the provider.

System programmer response:

No action is required.

User response:

No action is required.

IZUG105W Provider *namemodule* is not registered with the Common Information Model (CIM) server.

Explanation

The specified provider module is not registered with the Common Information Model (CIM) server. The script will register it.

In the message text:

name

Name of the provider.

System programmer response:

No action is required.

User response:

No action is required.

IZUG106I The provider *namemodule* is being registered with the Common Information Model (CIM) server.

Explanation

The provider module is not registered with the Common Information Model (CIM) server; therefore, the script is registering it.

In the message text:

name

Name of the provider.

System programmer response:

No action is required.

User response:

No action is required.

IZUG107E Unable to register provider *namemodule*.

Explanation

The specified provider module could not be registered. Typically, this error occurs when the user is not authorized to write to the Common Information Model (CIM) server repository or when the providers are missing.

In the message text:

name

Name of the provider.

System programmer response:

Verify that the z/OSMF administrator is authorized to write to the Common Information Model (CIM) server repository. Ensure that the providers are available.

User response:

No action is required.

IZUG108W The temporary directory *directory-namespecified* for environment variable **TMPDIR does not exist or cannot be accessed. The directory **/tmp** will be used.**

Explanation

The specified temporary directory either could not be found or is not writable. Thus, the directory **/tmp** will be used.

In the message text:

directory-name

Name of the directory specified for the **TMPDIR** environment variable.

System programmer response:

Verify that the directory exists. Ensure that the user running the script has permission to write to the directory.

User response:

No action is required.

IZUG109E Temporary directory *directory-namemust* exist and be writable: exiting script.

Explanation

For script processing, the named temporary directory must exist and be writable. If these requirements are not satisfied, processing of the script stops.

In the message text:

directory-name

Name of the temporary directory.

System programmer response:

Verify that the directory exists. Ensure that the user running the script has permission to write to the directory. After correcting the error, run the script again.

User response:

No action is required.

IZUG110I The **IZU_INCIDENT_LOG environment variable must be set to **Y** before completing action *action*.**

Explanation

The **IZU_INCIDENT_LOG** environment variable in the configuration file must be set to **Y** before the specified action can be completed.

In the message text:

action

The Incident Log action to be completed.

System programmer response:

Enter the **izusetup.sh -config [filename.cfg]** command. Use the configuration file that you used for setup. If the file name is omitted, the default configuration file is used. When prompted to configure the Incident Log, enter **Y**.

User response:

No action is required.

IZUG111E The value specified for variable *variable-name* is not valid. The variable must start with an alphanumeric character (A-Z**, **a-z**, and **0-9**) or a special character (**# \$ @**) and must contain *number* characters.**

Explanation

The value specified for the variable is not valid.

In the message text:

variable-name

Name of the input variable.

number

Minimum and maximum number of characters the variable can contain.

System programmer response:

Enter a value that starts with an alphanumeric character (**A-Z**, **a-z**, and **0-9**) or a special character (**# \$ @**) and contains between the minimum and maximum number of characters specified.

User response:

No action is required.

IZUG112I **Script *script-name* returned with reason code *code*.**

Explanation

The specified script returned with the specified reason code.

In the message text:

script-name

Name of the script.

code

Reason code.

System programmer response:

If the reason code is not 0, check the log for errors.

User response:

No action is required.

IZUG113I **The output of the command that was passed to script *script-name* is *output*.**

Explanation

The output of the command that was passed to the specified script is displayed.

In the message text:

script-name

Name of the script.

output

The output of the command.

System programmer response:

No action is required.

User response:

No action is required.

IZUG114I **Command *command-name* was passed to script *script-name*.**

Explanation

The specified command was passed to the specified script.

In the message text:

command-name

The command to execute.

script-name

Name of the script.

System programmer response:

No action is required.

User response:

No action is required.

IZUG115I **The RACF REXX executable was generated and saved in file *file-***

***name*. Review and execute the script before proceeding.**

Explanation

The RACF REXX executable has been created and saved in the specified file. The script sets up the RACF security for z/OSMF.

In the message text:

file-name

Name of the file in which the RACF REXX executable is stored.

System programmer response:

Review and execute the script. If you do not set up the security, you cannot proceed.

User response:

No action is required.

IZUG116E **User *user-name* does not exist.**

Explanation

The specified user does not exist.

In the message text:

user-name

User ID of the user.

System programmer response:

Provide a valid user name and try your request again.

User response:

No action is required.

IZUG117I **A *action* of the test incident for the Incident Log has occurred.**

Explanation

To verify that the Incident Log is configured properly, a test incident is created. Then, a series of tests are run against the incident. After verification is complete, the test incident is deleted. This message indicates that the test incident is either being created or that it is being deleted.

In the message text:

action

The action being performed as part of Incident Log verification.

System programmer response:

No action is required.

User response:

No action is required.

IZUG118I **Checking Incident Log dependencies.**

Explanation:

The PDW_IVP is being called to determine the status of Incident Log dependencies on the system.

System programmer response:

No action is required.

User response:

No action is required.

**IZUG119I Obtaining data for dependency
 dependency-name.**

Explanation

Dependency data is being collected for either the SysplexDumpDirectory provider or PDWLogstream provider.

In the message text:

dependency-name

Name of the Incident Log dependency.

System programmer response:

No action is required.

User response:

No action is required.

**IZUG120I Creating Incident Log report
 report-name.**

Explanation

The specified Incident Log report is being created.

In the message text:

report-name

Name of the Incident Log report.

System programmer response:

No action is required.

User response:

No action is required.

**IZUG121I To obtain the results of the
 Incident Log verification, review
 report *report-name*.**

Explanation

Review the Incident Log report to obtain the results of the verification.

In the message text:

report-name

Name of the Incident Log report.

System programmer response:

Review the specified report.

User response:

No action is required.

IZUG122E Verification failed for *item-name*.

Explanation

Verification failed because an error occurred while the specified item was being verified.

In the message text:

item-name

The item being verified.

System programmer response:

For more information, review the log file created for the error.

User response:

No action is required.

**IZUG123E An error occurred. The Common
 Event Adapter (CEA) parmlib
 member was not activated.**

Explanation:

The CEA parmlib member was not activated because an error occurred.

System programmer response:

For more information, review the log file created for the error.

User response:

No action is required.

**IZUG124I The Common Event Adapter (CEA)
 parmlib member *member-name* is
 being activated.**

Explanation

The specified CEA parmlib member is being activated on the system.

In the message text:

member-name

Name of the CEA parmlib member.

System programmer response:

No action is required.

User response:

No action is required.

**IZUG126E An error occurred. Variable
 variable-name is set to value
 actual-value. The expected value
 is *expected value*.**

Explanation

The specified variable is set to the specified value. The variable must be set to the expected value.

In the message text:

variable-name

Name of the variable.

actual-value

Actual value specified for the variable.

expected value

Value to which z/OSMF expects the variable to be set.

System programmer response:

For more information, review the log file created for the error and the RACF report.

User response:

No action is required.

IZUG127E **User *user-name* not connected to group *group-name*.**

Explanation

The specified user is not connected to the specified group.

In the message text:

user-name

User ID of the user.

group-name

Name of the group.

System programmer response:

For more information, review the log file created for the error and the RACF report.

User response:

No action is required.

IZUG128E **User *user-name* not permitted to RACF class *class-name*.**

Explanation

The specified user or group name is not permitted to the specified RACF class.

In the message text:

user-name

User ID of the user.

class-name

Name of the RACF class.

System programmer response:

For more information, review the log file created for the error and the RACF report.

User response:

No action is required.

IZUG129E **Unable to allocate the sysplex dump directory.**

Explanation:

The sysplex dump directory could not be allocated.

System programmer response:

For more information, review the log file created for the error.

User response:

No action is required.

IZUG130I **Allocating sysplex dump directory on volume *volume-name*.**

Explanation

The sysplex dump directory is being allocated on the specified volume.

In the message text:

volume-name

Name of the volume.

System programmer response:

No action is required.

User response:

No action is required.

IZUG131I **Activating sysplex dump directory.**

Explanation:

The sysplex dump directory is being activated.

System programmer response:

No action is required.

User response:

No action is required.

IZUG132E **Unable to activate sysplex dump directory.**

Explanation:

The sysplex dump directory could not be activated.

System programmer response:

For more information, review the log file created for the error.

User response:

No action is required.

IZUG133I **Enter the cluster transition name (case sensitive) for the server:**

Explanation:

Indicate the cluster transition name to be used. The name is case sensitive.

System programmer response:

Enter the cluster transition name.

User response:

No action is required.

IZUG134I **Enter the cluster transition name (case sensitive) for the server, or press Enter to accept the default *cluster-name*:**

Explanation

Indicate the cluster transition name to be used.

In the message text:

cluster-name

The default cluster transition name.

System programmer response:

To use the default cluster transition name, press Enter without entering a value. Otherwise, enter the name of the cluster transition.

User response:

No action is required.

IZUG135W	File <i>file-name</i> already exists. Ensure that the environment variables specified in the file have the same value as the corresponding variables in the configuration file.
-----------------	--

Explanation

The specified file already exists.

In the message text:

file-name

Name of the file.

System programmer response:

Ensure that the environment variables specified in the file have the same values as the corresponding variables in the configuration file. After you compare the variables and make any corrections, you can continue.

User response:

No action is required.

IZUG136I	The <i>item-type file-name</i> was created.
-----------------	--

Explanation

The specified file or directory has been created.

In the message text:

item-type

Type of item being created: file or directory.

file-name

Name of the file or directory.

System programmer response:

No action is required.

User response:

No action is required.

IZUG137E	File <i>file-name</i> already exists. The value specified in the file for the PEGASUS_HOME environment variable does not match the value
-----------------	---

specified in the configuration file for the IZU_WBEM_ROOT variable.

Explanation

The specified file already exists. An error occurred because the PEGASUS_HOME variable specified in the file does not have the same value as the IZU_WBEM_ROOT variable specified in the configuration file. The values for these two variables must be the same.

In the message text:

file-name

Name of the file.

System programmer response:

Update the specified file so that the PEGASUS_HOME variable has the same value as the IZU_WBEM_ROOT variable in the configuration file.

User response:

No action is required.

IZUG138E	Unable to read file <i>file-name</i>.
-----------------	--

Explanation

The permissions specified for the file does not allow read access.

In the message text:

file-name

Name of the file.

System programmer response:

Enable read access for the file.

User response:

No action is required.

IZUG139I	Has the Common Information Model (CIM) server been setup? [Y N]:
-----------------	---

Explanation:

The message prompts to determine if the Common Information Model (CIM) server has been set up.

System programmer response:

Enter Y or N.

User response:

No action is required.

IZUG140I	Has the Common Information Model (CIM) server been setup? [Y/N]. Or press Enter to accept the default value:
-----------------	---

Explanation

The message prompts to determine if the Common Information Model (CIM) server has been setup. A default value is provided.

In the message text:

value

The default response value for the CIM setup option.

System programmer response:

Enter Y or N, or accept the default. Default is NO

User response:

No action is required.

IZUG141W	No data directory specified. Using <i>directory-name</i> as the data directory.
-----------------	--

Explanation

The message indicates that no data directory was specified and that the default data directory will be used.

In the message text:

directory-name

The default data directory.

System programmer response:

Ensure the default data directory use is correct to the configuration.

User response:

No action is required.

IZUG142I	Enter the name of the target data set to be used for saving the updated parmlib members <i>ceaprm-parmlib-member</i> and <i>ieadmc-parmlib-member</i>. Specify the fully qualified data set name, or press Enter to accept the default: <i>parmlib-name</i>:
-----------------	---

Explanation

The message prompts you for the name of the data set to be used for saving the updated parmlib members, IEADMCnn and CEAPRMnn, which are used for Incident Log task processing. A fully qualified data set name is expected.

In the message text:

ceaprm-parmlib-member

User-specified CEAPRMxx member

ieadmc-parmlib-member

The user-specified IEADMCxx member

parmlib-name

Default data set name.

System programmer response:

Specify the fully qualified data set name, or press Enter to accept the supplied default if it is correct for your environment. If you specify the data set name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG143I	Enter the name of the target data set to be used for saving the updated parmlib members <i>ceaprm-parmlib-member</i> and <i>ieadmc-parmlib-member</i>. Specify the fully qualified data set name, or press Enter to use SYS1.PARMLIB:
-----------------	--

Explanation

The message prompts you for the name of the data set to be used for saving the updated parmlib members, IEADMCnn and CEAPRMnn, which are used for Incident Log task processing. A fully qualified data set name is expected.

In the message text:

ceaprm-parmlib-member

User-specified CEAPRMxx member.

ieadmc-parmlib-member

User-specified IEADMCxx member.

System programmer response:

Specify the fully qualified data set name, or press Enter to save the updated members in SYS1.PARMLIB. If you specify the data set name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG144I	Enter the mount point for the z/OSMF data file system:
-----------------	---

Explanation:

The message prompts for the mount point for where the z/OSMF data file system is to be mounted.

System programmer response:

Enter the mount point for where the z/OSMF data file system is to be mounted.

User response:

No action is required.

IZUG145I	Enter the mount point for the z/OSMF data file system, or press Enter to accept the default <i>mount-point</i>:
-----------------	--

Explanation

The message prompts for the mount point for where the z/OSMF data file system is to be mounted.

In the message text:

mount-point

The default mount point for the z/OSMF data file system.

System programmer response:

Enter the mount point for where the z/OSMF data file system is to be mounted.

User response:

No action is required.

IZUG146I	Invoking script <i>script-name-options</i>.
-----------------	--

Explanation

The message displays the script name and options that are being invoked.

In the message text:

script-name-options

The script name and options that are being invoked.

System programmer response:

No action is required.

User response:

No action is required.

IZUG147W	Path /usr/lib not found in LIBPATH variable.
-----------------	---

Explanation:

The message indicates the path /usr/lib was not found in the LIBPATH environment variable.

System programmer response:

Set the path /usr/lib in LIBPATH environment variable.

User response:

No action is required.

IZUG148I	Stopping Common Information Model (CIM) server.
-----------------	--

Explanation:

The message indicates that the CIM server is being stopped.

System programmer response:

No action is required.

User response:

No action is required.

IZUG149W	Path /usr/lib not found in LIBPATH variable in file <i>file-name</i>.
-----------------	--

Explanation

The message indicates the path /usr/lib was not found in the LIBPATH variable in the specified file.

In the message text:

file-name

The name of the file being checked.

System programmer response:

Ensure the path /usr/lib in LIBPATH environment variable is set in the specified file.

User response:

No action is required.

IZUG150E	Mount point <i>mount-point</i> must be a fully-qualified path name.
-----------------	--

Explanation

The message indicates the mount point provided is not a fully-qualified path.

In the message text:

mount-point

The mount point for the file system.

System programmer response:

Provide a fully-qualified path.

User response:

No action is required.

IZUG151I	z/OSMF data file system will be created using SMS managed storage.
-----------------	---

Explanation:

This message confirms your selection to use the z/OS storage management subsystem (SMS) to manage the storage of the z/OSMF data file system.

System programmer response:

No action is required.

User response:

No action is required.

IZUG157I	Enter the z/OSMF data file system type for the file system: <i>file-system-name</i>, or press Enter to accept the default <i>file-system-type</i>:
-----------------	---

Explanation

This message prompts for the type (zfs or hfs) of the specified file system. A default value is provided.

In the message text:

file-system-name

Name of the file system

file-system-type

Default file system type.

System programmer response:

No action is required.

User response:

No action is required.

IZUG158I	Enter the name of the volume to use for creating the z/OSMF data file system, enter an asterisk (*) to use SMS managed storage, or press Enter to accept the default volume-name:
-----------------	--

Explanation

The message prompts you for the name of the volume to create the z/OSMF data file system. To have the z/OS storage management subsystem (SMS) manage the storage, enter an asterisk (*). A default value is provided.

In the message text:

volume-name

Default volume name.

System programmer response:

Perform the requested action. If you specify a volume, the volume must be on-line. If you specify SMS managed storage, ensure that you have an automatic class selection (ACS) routine in place to assign the appropriate SMS construct, based on the name of the data set to be used for the z/OSMF file system.

User response:

No action is required.

IZUG159I	Enter the size (in cylinders) to allocate for the data file system, or press Enter to accept the default file-system-size:
-----------------	---

Explanation

Enter the initial space allocation, in cylinders, for the z/OSMF data file system. z/OSMF uses 90 percent of this value for the primary allocation and 10 percent for the secondary allocation. The minimum suggested size is 100 cylinders, which causes the script to use 90 cylinders for the primary allocation and 10 cylinders for the secondary allocation. A default value is provided.

In the message text:

file-system-size

Default size for the file system.

System programmer response:

Perform the requested action.

User response:

No action is required.

IZUG160E	The file extension specified for the override file is incorrect. The file must have a .ovr extension.
-----------------	--

Explanation:

An error occurred because the specified override file does not have a .ovr extension.

System programmer response:

Modify the override file name so that it has the .ovr extension.

User response:

No action is required.

IZUG161E	Directory <i>directory-name</i> must be a fully-qualified path name.
-----------------	---

Explanation

The message indicates that the directory provided is not a fully-qualified path.

In the message text:

directory-name

Name of the directory.

System programmer response:

Provide a fully-qualified path.

User response:

No action is required.

IZUG162I	Select the services to be configured. Multiple services can be selected by separating services with a comma.
-----------------	---

Explanation:

The message indicates that multiple services can be selected by separating plug-in IDs with a comma.

System programmer response:

No action is required.

User response:

No action is required.

IZUG163I	Select <i>plug-in-id</i> to configure <i>service-name</i>.
-----------------	---

Explanation

The message indicates the plug-in ID and service name for selection.

In the message text:

plug-in-id

Identifier of the service

service-name

Name of the service.

System programmer response:

No action is required.

User response:

No action is required.

IZUG164I **Which services do you want to configure?**

Explanation:

Enter the plug-in IDs for selection. For multiple selections, separate plug-in IDs with a comma.

System programmer response:

Select the service IDs for configuration.

User response:

No action is required.

IZUG165I **You have selected to configure service-name.**

Explanation

The message indicates the specified service was selected for configuration.

In the message text:

service-name

Name of the service.

System programmer response:

No action is required.

User response:

No action is required.

IZUG166I **No configuration prompts are required for the service service-name.**

Explanation

The message indicates there are no prompts to be displayed for the selected service.

In the message text:

service-name

Name of the service.

System programmer response:

No action is required.

User response:

No action is required.

IZUG167E **Value plug-in-id is ignored. Service was already selected.**

Explanation

The plug-in ID is ignored because the service has already been selected for configuration.

In the message text:

plug-in-id

Plug-in ID.

System programmer response:

No action is required.

User response:

No action is required.

IZUG168E **Expecting number arguments.**

Explanation

The message indicates the value that represents the number of services is incorrect.

In the message text:

number

Number of services.

System programmer response:

No action is required.

User response:

No action is required.

IZUG169E **Configuration file variable variable-name is not valid.**

Explanation

The message indicates the configuration file variable is not valid.

In the message text:

variable-name

The configuration file variable.

System programmer response:

No action is required.

User response:

No action is required.

IZUG170E **Log file variable variable-name is not valid.**

Explanation

The message indicates the log file is not valid.

In the message text:

variable-name

Log file variable.

System programmer response:

No action is required.

User response:

No action is required.

IZUG171I **Do you want to configure the Common Information Model (CIM) server as part of z/OSMF**

customization? If so, enter Y. To skip this step, enter N:

Explanation:

Specify whether the z/OS Common Information Model (CIM) server is to be configured as part of the z/OSMF configuration process. z/OSMF requires that the CIM server be operational on your system. To have z/OSMF configure the CIM server for you, enter Y. Otherwise, if you have already configured the CIM server or plan to do this step yourself, specify N.

System programmer response:

Enter Y or N.

User response:

No action is required.

IZUG172I **Do you want to configure the Common Information Model (CIM) server as part of z/OSMF customization? If so, enter Y. To skip this step, enter N. To accept the default, press Enter: value:**

Explanation

Specify whether the z/OS Common Information Model (CIM) server is to be configured as part of the z/OSMF configuration process. z/OSMF requires that the CIM server be operational on your system. To have z/OSMF configure the CIM server for you, enter Y. Otherwise, if you have already configured the CIM server or plan to do this step manually, specify N. To accept the default value displayed in the message, press Enter.

In the message text:

value

Default selection for setting up the CIM server.

System programmer response:

Enter Y or N, or accept the default value.

User response:

No action is required.

IZUG173I **Enter "N" to select none of these services.**

Explanation:

The value N indicates that no services are selected.

System programmer response:

No action is required.

User response:

No action is required.

IZUG174E **The value *value* is incorrect for property.**

Explanation

The specified value is incorrect for the indicated property. During the configuration process, the izusetup.sh script collects installation-specific data that is used in the configuration of the product. The script starts with the variable settings that are contained in the configuration file (izudflt.cfg), and substitutes any installation-specific changes that you supply (through interactive prompting or an optional override file) to tailor the configuration for your environment.

In the message text:

value

Value that was specified for the property

property

Property containing the value.

System programmer response:

Specify a valid value for the indicated property and retry the operation. Depending on how you choose to configure z/OSMF, you might need to respecify this value interactively or as a setting in the optional override file. Some values are case sensitive. For more information, see [IBM z/OS Management Facility Configuration Guide](#) . Do not edit the izudflt.cfg file directly.

User response:

No action is required.

IZUG175I **The configuration file *config-file* will be migrated to the format: *release-level*. Enter the *release-level* z/OSMF product file system mount point, or press Enter to accept the default path *default-code-root*:**

Explanation:

The specified configuration file will be migrated to the specified release level. This message prompts for the default code root directory.

System programmer response:

Enter the root code directory path or press Enter to accept the default.

User response:

No action is required.

IZUG176I **The configuration file *config-file* conforms to the expected format: *release-level*. No migration will be performed.**

Explanation:

No migration is needed since the specified configuration file is at the correct version level.

System programmer response:

No action is required.

User response:

No action is required.

IZUG177I **The configuration file *config-file* has been migrated to the format: *release-level*.**

Explanation:

The specified configuration file has been migrated to the specified release level.

System programmer response:

No action is required.

User response:

No action is required.

IZUG178I **The input configuration file *config-file* was saved to a backup file *back-up-config-file*.**

Explanation:

The data of the source configuration file has been saved to the specified configuration file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG179E **Unable to back up configuration data.**

Explanation:

The data of the source configuration file could not be saved. Ensure that the permission settings are correct for the file and directory.

System programmer response:

Ensure that the permission settings are correct for the file and directory.

User response:

No action is required.

IZUG180E **The configuration file *config-file* does not conform to the expected format: *release-level*. Migrate the configuration file to the correct format and retry the operation.**

Explanation:

The configuration file is not at the correct release level.

System programmer response:

Migrate the configuration file to the correct release level and retry the request.

User response:

No action is required.

IZUG181E **The value for the property *plugin-property* is set inconsistently in the configuration file and the override**

file. In the configuration file, *plugin-property* is set to *plugin-property-value*. In the override file, *plugin-property* is set to *plugin-property-value*.

Explanation

In processing a `izusetup.sh -add request, z/OSMF` detected that the indicated property is specified inconsistently in the configuration file and the override file.

In the message text:

plugin-property

The property name

plugin-property

The property for the service.

plugin-property-value

The value for the property for the service.

plugin-property

The property for the service.

plugin-property-value

The value for the property for the service.

System programmer response:

Update the property with the correct value in the configuration file and in the override file. Then, retry the request.

User response:

No action is required.

IZUG182I **The property *plugin-property* is set inconsistently in the configuration file and the override file. The property *plugin-property* will be set to *plugin-property-value*.**

Explanation

In processing a `izusetup.sh -add request, z/OSMF` detected that the indicated property is specified inconsistently in the configuration file and the override file. Z/OSMF processing will set the property as indicated in the resulting configuration file.

In the message text:

plugin-property

Property for the service

plugin-property

Property for the service

plugin-property-value

The value for the property for the service.

System programmer response:

No action is required.

User response:

IZUG183I **The property *plugin-property* in the override file contains the value *plugin-property-value*. The value for the property *plugin-property* will be set to *plugin-property-value*.**

Explanation

The indicated property was set incorrectly in the override file. z/OSMF processing uses a reset value as indicated and ignores the value specified in the override file.

In the message text:

plugin-property

Property for the service

plugin-property-value

Value of the property

plugin-property

The property for the service

plugin-property-value

The new value for the property

System programmer response:

No action is required.

User response:

No action is required.

IZUG184E **The property *plugin-property* in the specified configuration file is set to an incorrect value *plugin-property-value*.**

Explanation

In processing the izusetup.sh -add request, z/OSMF processing detected that the indicated variable was set incorrectly in the specified configuration file.

In the message text:

plugin-property

Property for the service

plugin-property-value

Value that is incorrect

System programmer response:

Check the override file for errors. Some variables are initially set to the following value, which is not a valid setting: NO.DEFAULT.VALUE. Correct the errors and try the request again.

User response:

No action is required.

IZUG185I **Enter the value for the Common Information Model (CIM) server attribute *server-attribute*, or press**

Enter to accept the default *server-attribute-value*:

Explanation:

The message prompts for CIM server attribute values.

System programmer response:

Provide the value for the server attribute.

User response:

No action is required.

IZUG186I **You have selected to add the following services.**

Explanation:

This message precedes the list of one or more services that have been selected for configuration.

System programmer response:

No action is required.

User response:

No action is required.

IZUG187I **Service: *service-name*.**

Explanation

The specified service has been selected for configuration.

In the message text:

service-name

Name of the service to be added.

System programmer response:

No action is required.

User response:

No action is required.

IZUG188I **To accept these service selections, press Enter. To edit these selections, enter E.**

Explanation:

The message prompts you to confirm your selection of which services are to be configured. You can change your selection.

System programmer response:

Enter E to modify the selection. Press enter with no value to accept the current selection.

User response:

No action is required.

IZUG189I **No services were selected for configuration.**

Explanation:

The izusetup.sh -add request identified no services to be added to the z/OSMF configuration.

System programmer response:

No action is required.

User response:

No action is required.

IZUG190I **The service *service-name* is set to the value *service-value*, this indicates that it is already configured. The request to add this service is ignored.**

Explanation

The service is already configured. Your request is ignored.

In the message text:

service-name

Name of the service

service-value

Value of the service

System programmer response:

No action is required.

User response:

No action is required.

IZUG191I **No security setup procedure is required for the specified services.**

Explanation:

The RACF setup procedure is not required for the specified services.

System programmer response:

No action is required.

User response:

No action is required.

IZUG192I **Enter the Common Information Model (CIM) Server attribute *server-attribute*:**

Explanation:

You requested that z/OSMF set this CIM server attribute, but no value was supplied for the attribute in the configuration file or override file. Therefore, the script prompts you for the value.

System programmer response:

Enter the appropriate value for your installation.

User response:

No action is required.

IZUG193E **Group *group-name* does not exist.**

Explanation

In processing the izusetup.sh -verify racf request, z/OSMF detected that the specified group is not defined.

In the message text:

group-name

Name of the group.

System programmer response:

For more information, check the log file created for the error and the RACF report. Also, examine the generated RACF exec to ensure that the indicated group was created.

User response:

No action is required.

IZUG194E **The value for variable *property-name* contains an incorrect character *char-value*.**

Explanation

The specified value is incorrect because it contains an incorrect character.

In the message text:

property-name

The incorrect property.

char-value

The incorrect character within the input value.

System programmer response:

Correct the value.

User response:

No action is required.

IZUG195E **The value for variable *property-name* contains one or more spaces. Enter the value without spaces.**

Explanation

The value specified for the variable is not valid because it contains one or more spaces, which is not allowed.

In the message text:

property-name

Name of the incorrect property.

System programmer response:

Specify a value that does not contain spaces.

User response:

No action is required.

IZUG196E **The variable *property-name* contains an incorrect value *property-value*.**

Explanation

The specified value is incorrect.

In the message text:

property-name
Name of the property.

property-value
Value of the property.

System programmer response:
Correct the value.

User response:
No action is required.

IZUG197E **The file system name *file-system-name* is incorrect. The maximum allowable length is 44 characters.**

Explanation
The specified value is incorrect.

In the message text:

file-system-name
The incorrect value.

System programmer response:
Correct the value.

User response:
No action is required.

IZUG198E **Parmlib data set *parmlib-name* does not exist.**

Explanation
The specified parmlib data set does not exist.

In the message text:

parmlib-name
Parmlib name.

System programmer response:
Ensure that the specified parmlib exists. Retry your request.

User response:
No action is required.

IZUG199W **File *file-name* already exists.**

Explanation
The specified file already exists. Later during the configuration of CEAPRM parmlib member you will be given the option to overwrite the file.

In the message text:

file-name
File name.

System programmer response:
No action is required.

User response:
No action is required.

IZUG200I **The home page of the z/OSMF server *server-name* in AUTOSTART_GROUP *autostart-group* can be accessed at *URI* in SYSTEM *system-name*.**

Explanation
The message displays the information related to the z/OSMF server as the output of DISPLAY command.

The information includes the job name, autostart group and URI of the z/OSMF server, as well as the system name.

In the message text:

server-name
The z/OSMF server job name.

autostart-group
The autostart group that was specified for the z/OSMF server.

URI
The URI of the z/OSMF server.

system-name
The system name.

System programmer response:
No action is required.

User response:
No action is required.

IZUG201E **User *user-id* could not be primed for z/OSMF. The action failed with return code *return-code*.**

Explanation
The -prime request failed for the specified user ID. A return code is provided to indicate the cause of the error.

In the message text:

user-id
User ID that could not be processed by the -prime request

return-code
Return code indicating the result of the process.

The following return codes are valid:

- 1**
Usage error.
- 2**
Problem with the log directory.
- 3**
Error writing to the log file.

- 4 Script encountered an error when running a z/OS UNIX shell command, such as mkdir or cp.
- 5 A repository already exists.
- 6 Specified user ID is not defined to the z/OS system.
- 7 The data directory specified by IZU_DATA_DIR does not exist or is not accessible.

This message is accompanied by one or more related messages with more information about the error.

System programmer response:

For more information, check for related messages. For return code 6, see the z/OSMF log file. After correcting the error, run the script again.

User response:

No action is required.

IZUG202E z/OSMF could not make user *user-name* owner of directory-file name.

Explanation

z/OSMF could not make the specified user owner of the specified file or directory.

In the message text:

user-name

User name

directory-file

Indication of directory or file

name

Name of the directory or file.

System programmer response:

Ensure that the caller has permission to set ownership. For more information, review the log file created for the error.

User response:

No action is required.

IZUG203E The request to set permissions for the files in directory *directory-name* failed.

Explanation

z/OSMF could not set permissions for the files in the specified directory.

In the message text:

directory-name

Name of the directory.

System programmer response:

Ensure that the caller has permission to set ownership. For more information, review the log file created for the error.

User response:

No action is required.

IZUG204E The request to set permissions for file *file-name* failed.

Explanation

z/OSMF could not set permissions for the specified file.

In the message text:

file-name

File name.

System programmer response:

Ensure that the caller has permission to set ownership. For more information, review the log file created for the error.

User response:

No action is required.

IZUG205E The file extension specified for the configuration file is incorrect. The file must have a .cfg extension.

Explanation:

An error occurred because the specified configuration file does not have a .cfg extension.

System programmer response:

Modify the configuration file name so that it has the .cfg extension.

User response:

No action is required.

IZUG206E The variables specified in configuration file *file-name* could not be exported.

Explanation

The variables included in the specified configuration file were not exported because an error occurred.

In the message text:

file-name

Name of the configuration file.

System programmer response:

For more information, review the log file created for the error.

User response:

No action is required.

IZUG207E File *file-name* does not exist.

Explanation

The specified file does not exist.

In the message text:

file-name

File name.

System programmer response:

Ensure that the specified file exists. Retry your request.

User response:

No action is required.

IZUG208E	The configuration file is incomplete. The value for variable <i>variable-name</i> is missing.
-----------------	--

Explanation

The request could not be completed because an error occurred. The configuration file is missing the specified information.

In the message text:

variable-name

Name of the variable that is missing from the configuration file.

System programmer response:

Enter the `izusetup.sh -config [ filename.cfg ]` command. *filename.cfg* is the name of the configuration file that is missing the specified data. When prompted, provide a value for the specified variable.

User response:

No action is required.

IZUG209I	Script <i>script-name</i> supports one or more of the following input options: <i>input-options</i>.
-----------------	---

Explanation

The valid script options are displayed. For information about the script options, see [IBM z/OS Management Facility Configuration Guide](#).

In the message text:

script-name

Name of the script

input-options

Options supported by the script.

System programmer response:

No action is required.

User response:

No action is required.

IZUG210I	The script <i>script-name</i> has completed.
-----------------	---

Explanation

The specified script completed.

In the message text:

script-name

Name of the script.

System programmer response:

No action is required.

User response:

No action is required.

IZUG211E	Script <i>script-name</i> encountered errors: exiting script.
-----------------	--

Explanation

Processing of the script stopped because one or more errors occurred.

In the message text:

script-name

Name of the script.

System programmer response:

For more information, review the log file created for the error. Correct any errors and re-run the script.

User response:

No action is required.

IZUG212E	Directory <i>directory-name</i> does not exist or is not accessible.
-----------------	---

Explanation

The specified directory does not exist or is not accessible.

In the message text:

directory-name

Name of the directory.

System programmer response:

Ensure that the specified directory exists and is accessible. Retry your request.

User response:

No action is required.

IZUG213I	Log information will be written to file <i>file-name</i>.
-----------------	--

Explanation

Log information will be saved to the specified file.

In the message text:

file-name

Name of the file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG214E **Failed to create *directory-file* *directory-file-name*.**

Explanation

The specified file or directory could not be created.

In the message text:

directory-file

Directory or file

directory-file-name

Name of the directory or file.

System programmer response:

Ensure that the caller is authorized to create files or directories. For more information, review the log file created for the error.

User response:

No action is required.

IZUG215I **Starting *z/OSMF procedure-name* procedure.**

Explanation

The specified procedure is being processed.

In the message text:

procedure-name

Name of the procedure.

System programmer response:

No action is required.

User response:

No action is required.

IZUG216E **The command is missing one of the required arguments: *argument-name*.**

Explanation

The command could not be completed because the specified argument was not found.

In the message text:

argument-name

Name of the argument.

System programmer response:

Re-enter the command and include the missing argument.

User response:

No action is required.

IZUG217E **The command could not be completed because it contains an incorrect argument.**

Explanation:

An incorrect argument was provided with the command. Typically, this error occurs when an argument that is not supported by the command is used or when the argument is misspelled.

System programmer response:

Verify that the correct argument is being used. Ensure that it is spelled correctly. Correct any errors and re-enter the command.

User response:

No action is required.

IZUG218E **The command could not be completed because it contains an incorrect argument *argument-name*.**

Explanation

An incorrect argument was provided with the command. The name of the incorrect argument is provided. Typically, this error occurs when an argument that is not supported by the command is used or when the argument is misspelled.

In the message text:

argument-name

Name of the incorrect argument.

System programmer response:

Verify that the correct argument is being used. Ensure that it is spelled correctly. Correct any errors and enter the command again.

User response:

No action is required.

IZUG220E **The Incident Log configuration request failed. The *IZU_INCIDENT_LOG* variable in the configuration file must be set to Y before the request can be processed.**

Explanation:

The Incident Log configuration request failed because the *IZU_INCIDENT_LOG* variable is not set to Y.

System programmer response:

Enter the `izusetup.sh -config [ filename.cfg ]` command. The configuration file name is optional. If the file name is omitted, the default configuration file is used. When prompted to configure the Incident Log, enter Y.

User response:

No action is required.

IZUG221E **A value must be provided for argument *argument-name*.**

Explanation

An error occurred because no value was found for the specified argument.

In the message text:

argument-name

Name of the required argument.

System programmer response:

Correct the input to the request.

User response:

No action is required.

IZUG222E	Unable to update configuration file <i>file-name</i>.
-----------------	--

Explanation

The specified configuration file could not be updated.

In the message text:

file-name

Name of the configuration file.

System programmer response:

Ensure that the caller is authorized to update the configuration file. For more information, review the log file created for the error.

User response:

No action is required.

IZUG223I	For more information, review log file <i>file-name</i>.
-----------------	--

Explanation

For more information, review the log file created for the error.

In the message text:

file-name

Name of the log file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG224I	The configuration data was saved in file <i>file-name</i>.
-----------------	---

Explanation

The configuration data was saved in the specified file.

In the message text:

file-name

Name of the configuration file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG225E	Unable to mount file system <i>file-system-name</i>.
-----------------	---

Explanation

The specified file system could not be mounted.

In the message text:

file-system-name

Name of the file system.

System programmer response:

For more information, review the log file created for the error.

User response:

No action is required.

IZUG226E	Unable to allocate file system <i>file-system-name</i>.
-----------------	--

Explanation

The specified file system could not be allocated.

In the message text:

file-system-name

Name of the file system.

System programmer response:

For more information, review the log file created for the error.

User response:

No action is required.

IZUG227I	Creating directory-file <i>directory-file-name</i>.
-----------------	--

Explanation

The specified file or directory is being created.

In the message text:

directory-file

Directory or file

directory-file-name

Name of the directory or file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG228I	Enter the fully qualified name of the z/OSMF <i>file-system-type</i> file system:
-----------------	--

Explanation

The message prompts you for the name to be used for the z/OSMF data file system. A fully qualified name is expected.

In the message text:

file-system-type

File system type.

System programmer response:

Specify the fully qualified name of the z/OSMF data file system. If you specify the file system name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG229I	Enter the fully qualified name of the z/OSMF <i>file-system-type</i> file system, or press Enter to accept the default <i>value</i> file system name :
-----------------	---

Explanation

The message prompts you for the name to be used for the z/OSMF data file system. A fully qualified name is expected.

In the message text:

file-system-type

File system type.

value

Default file system name.

System programmer response:

Specify the fully qualified name of the z/OSMF data file system, or press Enter to accept the supplied default if it is correct for your environment. If you specify the file system name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG230E	The value <i>value</i> is incorrect.
-----------------	---

Explanation

The specified value is incorrect.

In the message text:

value

Name of the input field.

System programmer response:

Correct the value.

User response:

No action is required.

IZUG231W	A file system with the name <i>file-system-name</i> already exists. Do you want to use the existing file system as the z/OSMF <i>file-system-type</i> file system (Y N)?
-----------------	---

Explanation

The specified file system already exists. Indicate whether you want to use the existing file system.

In the message text:

file-system-name

Name of the file system

file-system-type

File system type.

System programmer response:

To use the existing file system, enter Y. Otherwise, enter N. Prior to mounting a new file system, you must unmount the existing file system.

User response:

No action is required.

IZUG232I	The specified z/OSMF <i>file-system-type</i> file system with name-<i>type</i> <i>file-system-name-type</i> was accepted.
-----------------	--

Explanation

The value specified for the file system name or type was accepted.

In the message text:

file-system-type

File system type

name-type

The word name or type

file-system-name-type

File system name or file system type.

System programmer response:

No action is required.

User response:

No action is required.

IZUG233E	File system <i>file-system-name</i> could not be mounted. A file system with the same name is already mounted at <i>mount-point</i>.
-----------------	---

Explanation

The file system could not be mounted at the specified mount point because a file system with the same name is already mounted at another mount point.

In the message text:

file-system-name

Name of the file system

mount-point

Mount point of the file system.

System programmer response:

To mount a new file system at that mount point, you must unmount the existing file system and then mount the new file system.

User response:

No action is required.

IZUG234I **File system *file-system-name* is already mounted at mount point *mount-point*. Do you want to use the existing file system as the z/OSMF *file-system-type* file system (Y|N)?**

Explanation

The specified file system is already mounted at the mount point. Indicate whether you want to use the existing file system.

In the message text:

file-system-name

Name of the file system

mount-point

Mount point of the file system

file-system-type

File system type.

System programmer response:

To use the existing file system, enter Y. Otherwise, enter N. Prior to mounting a new file system, you must unmount the existing file system.

User response:

No action is required.

IZUG235E **The file system could not be mounted at mount point *mount-point*. File system *file-system-name* is already mounted at that mount point.**

Explanation

The file system could not be mounted at the specified mount point because another file system is already mounted at that mount point.

In the message text:

mount-point

Name of the mount point

file-system-name

Name of the file system.

System programmer response:

To mount a new file system at that mount point, you must unmount the existing file system and then mount the new file system.

User response:

No action is required.

IZUG236I **Enter zFS or HFS as the z/OSMF data file system type for the file system: *file-system-name*:**

Explanation

This message prompts for the type (zfs or hfs) of the specified file system.

In the message text:

file-system-name

Name of the file system.

System programmer response:

No action is required.

User response:

No action is required.

IZUG237I **Enter the name of the file to save the configuration data (must be .cfg extension), or press Enter to save as file *default-cfg-file*:**

Explanation

This message prompts the user to provide the name of the configuration file where the configuration data is to be saved. A default name is provided.

In the message text:

default-cfg-file

Configuration file name.

System programmer response:

No action is required.

User response:

No action is required.

IZUG238E **File name must be specified with the path.**

Explanation:

A value was provided but did not contain a file name.

System programmer response:

Provide a valid value and retry.

User response:

No action is required.

IZUG239W **File name *file-name* already exists: Overwrite (Y|N)?**

Explanation

The specified file name already exists. The message prompts the user to overwrite it.

In the message text:

file-name

File name.

System programmer response:

Try the action again.

User response:

No action is required.

IZUG240E **Overwrite reply was not (Y). Try again.**

Explanation:

A value of Y was not received to overwrite the file. The message prompts the caller to try again.

System programmer response:

Try the action again.

User response:

No action is required.

IZUG241E **File *file-name* cannot be saved to a read-only file system.**

Explanation

The file cannot be saved to a read-only file system.

In the message text:

file-name

File name.

System programmer response:

Review the location of where to save the file and try again.

User response:

No action is required.

IZUG242I **Do one of the following: Enter the system name, enter NONE not to set the name, or press Enter to accept the default *system-name*:**

Explanation

The message prompts the caller for the system name value to use. A default value is provided. Enter a value of NONE if you do not want to set the system name.

In the message text:

system-name

Default system name.

System programmer response:

No action is required.

User response:

No action is required.

IZUG243E **z/OSMF is not registered as a feature in z/OS.**

Explanation:

z/OSMF is not registered as a feature in z/OS.

System programmer response:

No action is required.

User response:

No action is required.

IZUG244I **z/OSMF is unregistered as a feature from z/OS.**

Explanation:

z/OSMF is unregistered as a feature from z/OS.

System programmer response:

No action is required.

User response:

No action is required.

IZUG245I **z/OSMF is registered as a feature in z/OS.**

Explanation:

z/OSMF is registered as a feature in z/OS.

System programmer response:

No action is required.

User response:

No action is required.

IZUG246I **Enter the name of the volume to use for creating the z/OSMF data file system, or enter an asterisk (*) to use SMS managed storage:**

Explanation:

The message prompts you for the name of the volume to create the z/OSMF data file system. If you enter an asterisk (*), it indicates that you want the z/OS storage management subsystem (SMS) to manage the storage.

System programmer response:

Perform the requested action. If you specify a volume, the volume must be on-line. If you specify SMS managed storage, ensure that you have an automatic class selection (ACS) routine in place to assign the appropriate SMS construct, based on the name of the data set to be used for the z/OSMF file system.

User response:

No action is required.

IZUG247I **z/OSMF data file system will be created on volume: *volume-name***

Explanation

The file system will be created on the specified volume.

In the message text:

volume-name

Name of the volume to create the file system.

System programmer response:

No action is required.

User response:

No action is required.

IZUG248I

Enter the size (in cylinders) to allocate for the data file system:

Explanation:

Enter the initial space allocation, in cylinders, for the z/OSMF data file system. z/OSMF uses 90 percent of this value for the primary allocation and 10 percent for the secondary allocation. The minimum suggested size is 100 cylinders, which causes the script to use 90 cylinders for the primary allocation and 10 cylinders for the secondary allocation.

System programmer response:

Perform the requested action.

User response:

No action is required.

IZUG249E

Volume size must be greater than 10 cylinders.

Explanation:

The specified volume is too small (less than 10 cylinders).

System programmer response:

Specify a volume that is at least 10 cylinders in size.

User response:

No action is required.

IZUG250I

The z/OSMF data file system *file-system-name* has a *primary-secondary* allocation size of *cylinder-size*cylinders.

Explanation

The specified file system was allocated with the specified number of cylinders for the primary or secondary extent.

In the message text:

file-system-name

Name of the file system

primary-secondary

Primary or secondary allocation for the file system.

cylinder-size

Size in cylinders of the allocation.

System programmer response:

No action is required.

User response:

No action is required.

IZUG251I

Allocating z/OSMF data file system *file-system-name*.

Explanation

The procedure to allocate the specified file system has started.

In the message text:

file-system-name

Name of the file system.

System programmer response:

No action is required.

User response:

No action is required.

IZUG252I

Mounting *file-system-name* at *mount-point*.

Explanation

The procedure to mount the specified file system at the specified mount point has started.

In the message text:

file-system-name

Name of the file system

mount-point

Mount point of the file system.

System programmer response:

No action is required.

User response:

No action is required.

IZUG253I

Enter the Common Information Model (CIM) administrator user ID, or press Enter to accept the default *default-value*:

Explanation

The message prompts for the Common Information Model (CIM) administrator user ID. A default attribute value is provided.

In the message text:

default-value

Default value for the CIM administrator user ID.

System programmer response:

Perform the requested action, or accept the default.

User response:

No action is required.

IZUG254E

Unable to copy *source-file-name* to *target-file-name*.

Explanation

Attempt to copy the specified file failed.

In the message text:

file-name

Name of the file source

target-file-name

Name of the file target

System programmer response:

Ensure that the caller is authorized to perform the copy.

User response:

No action is required.

IZUG255E	The instance of IzuConfigProperties cannot be retrieved. The SETIZU command might not function properly.
-----------------	---

Explanation:

When the SETIZU command of z/OSMF is issued, if the configuration properties service depend is not available, this message is prompted in DD STDOUT of the z/OSMF job log.

System programmer response:

Review the job log of the z/OSMF server to contact IBM support for assistance.

User response:

No action is required.

IZUG256I	Enter the z/OSMF administrator attribute-name-keyword, or press Enter to accept the default value:
-----------------	---

Explanation

The message is used to prompt for the z/OSMF administrator attributes. The message individually prompts for the following attributes:

- User ID
- Home directory
- Shell program name
- Logon Procedure Name
- Account number
- Region size

These attributes are used to create the z/OSMF administrator user ID. A default attribute value is provided.

In the message text:

attribute-name-keyword

Name of the attribute

value

Default value of the attribute.

System programmer response:

Enter the requested information, or accept the default.

User response:

No action is required.

IZUG257W	User user-id already exists.
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Explanation

The user ID provided already exists.

In the message text:

user-id

User name.

System programmer response:

No action is required.

User response:

No action is required.

IZUG258I	Enter the Common Information Model (CIM) administrator user ID:
-----------------	--

Explanation:

The message prompts for the Common Information Model (CIM) administrator user ID.

System programmer response:

No action is required.

User response:

No action is required.

IZUG259I	Enter the default RACF-defined group for the z/OSMF administrator:
-----------------	---

Explanation:

The message prompts for the default group for the z/OSMF administrator.

System programmer response:

No action is required.

User response:

No action is required.

IZUG260I	Enter the default RACF-defined group for the z/OSMF administrator, or press Enter to accept the default group-id:
-----------------	--

Explanation

The message prompts for the default group for the z/OSMF administrator. A default value is provided.

In the message text:

group-id

Name of the default group.

System programmer response:

No action is required.

User response:
No action is required.

IZUG261E **Attribute *attribute-name* must be *attribute-size*.**

Explanation
The value provided for the attribute does not conform to the expected range or size in the number of characters.

In the message text:

attribute-name
Name of the attribute

attribute-size
Expected attribute size.

System programmer response:
Specify the value within the correct range or size.

User response:
No action is required.

IZUG262I **Enter the server attribute *attribute-name*:**

Explanation
The message prompts for the name of the z/OSMF server attributes.

In the message text:

attribute-name
Name of the attribute for the server.

System programmer response:
Enter the server attribute name.

User response:
No action is required.

IZUG263I **Enter the server attribute *attribute-name*, or press Enter to accept the default value *value*:**

Explanation
The message prompts for the z/OSMF server attributes. A default value is provided.

In the message text:

attribute-name
Name of the attribute

value
Name of the attribute to which the default applies.

System programmer response:
Enter the requested information, or accept the default.

User response:
No action is required.

IZUG264E **Value *attribute-name* must be alphanumeric and must be *attribute-size* characters.**

Explanation
The value provided for the z/OSMF server is incorrect or outside the expected range or size for that attribute.

In the message text:

attribute-name
Name of the attribute for the z/OSMF server.

attribute-size
Size or range for the attribute for the z/OSMF server.

System programmer response:
Specify with the correct range or size.

User response:
No action is required.

IZUG265I **Enter the root directory path of the z/OSMF server:**

Explanation:
The message prompts for the root directory path for the z/OSMF server.

System programmer response:
Enter the root directory path.

User response:
No action is required.

IZUG266I **Enter the root directory path of the z/OSMF server, or press Enter to accept the default *server-root-directory*:**

Explanation
The message prompts for the root directory path for the z/OSMF server. A default value is provided.

In the message text:

server-root-directory
Default root directory path of the z/OSMF server.

System programmer response:
Enter the root directory path or accept the default.

User response:
No action is required.

IZUG267I **Enter the SAF profile prefix (case sensitive) for z/OSMF resources:**

Explanation:
The message prompts for the SAF profile prefix.

System programmer response:
Enter the SAF profile prefix.

User response:

No action is required.

IZUG268I **Enter the SAF profile prefix (case sensitive) for z/OSMF resources, or press Enter to accept the default *saf-profile*:**

Explanation

The message prompts for the SAF profile prefix. A default value is provided.

In the message text:

saf-profile

Default SAF profile prefix.

System programmer response:

Enter the SAF profile prefix, or accept the default.

User response:

No action is required.

IZUG271I **Do you want to enable the common event adapter (CEA) component and update related parmlib options for using the Incident Log task? For yes, enter Y. For no, enter N:**

Explanation

The message prompts you to determine whether the Incident Log task is to be configured. When you select to configure the Incident Log task, z/OSMF verifies that the Common Information Model (CIM) server and the common event adapter (CEA) are properly configured. If you have already configured CIM and have set up the CEA parmlib, you still must enter Y. z/OSMF provides additional prompts allowing you to indicate whether the CIM server and the CEA parmlib need to be configured.

If you do not configure the Incident Log task, you cannot complete any other Incident Log set up steps, such as setting up RACF permissions for the Incident Log. In this case, the Incident Log task stills displays in z/OSMF; however, it will not be functional. To remove it from the z/OSMF desktop, do not authorize any roles to access the Incident Log task.

System programmer response:

Enter Y or N.

User response:

No action is required.

IZUG272I **Do you want to enable the common event adapter (CEA) component and update related parmlib options for using the Incident Log task? For yes, enter**

Y. For no, enter N. Or press Enter to accept the default *value*:

Explanation

The message prompts you to determine whether the Incident Log task should be configured. When you select to configure the Incident Log task, the Common Information Model (CIM) server and the common event adapter (CEA) are configured so that they can support the Incident Log task. If you have already configured CIM and have set up the CEA parmlib, you still need to enter Y. When you are asked whether CIM needs to be configured, you can say no. In this case, confirming that you want to set up the Incident Log task gives z/OSMF permission to verify that all of the settings are correct.

If you do not configure the Incident Log task, you cannot complete any other Incident Log set up steps, such as setting up RACF permissions for the Incident Log. The Incident Log task still displays in z/OSMF; however, it will not be functional. To remove the Incident Log task from the z/OSMF desktop, do not authorize any roles to access this task.

In the message text:

value

Default value to specify setup of the Incident Log task.

System programmer response:

Enter Y or N, or accept the default, which is Y.

User response:

No action is required.

IZUG273I **Enter the *dependency-name* *dependency-attribute*:**

Explanation

The message prompts for the Common Information Model (CIM) or common event adapter (CEA) attributes. The *attribute-name-keyword* can be a group user ID or the keyword AUTOID, the user ID, or the keyword AUTOUID, or the group name. The *attribute-name* can be a group user ID, user ID, or group name.

In the message text:

dependency-name

Name of the Incident Log dependency

dependency-attribute

Name of the Incident Log attribute.

System programmer response:

Enter the incident dependency name and log attribute names.

User response:

No action is required.

IZUG274I **Enter the *component-name*
attribute-name-keyword, or press
Enter to accept value:**

Explanation

The message prompts for the Common Information Model (CIM) or common event adapter (CEA) attributes. The *attribute-name-keyword* can be a group user ID or the keyword AUTOID, the user ID, or the keyword AUTOUID, or the group name. The *attribute-name* can be a group user ID, user ID, or group name. A default value is provided.

In the message text:

component-name

Name of the component

attribute-name-keyword

Name of the attribute keyword

value

Default value.

System programmer response:

Enter the information, or accept the default.

User response:

No action is required.

IZUG275I **Enter the member name suffix to
use for the *parmlib-member-name*
parmlib member, or press Enter to
accept the default *suffix-value*:**

Explanation

The message prompts for the suffix to use for IEADMC and CEAPRM members. A default value is provided.

In the message text:

parmlib-member-name

Name of the parmlib member

suffix-value

Default suffix of the parmlib member.

System programmer response:

No action is required.

User response:

No action is required.

IZUG276I **Enter the member name suffix
to use for the *parmlib-member-
name*parmlib member:**

Explanation:

The message prompts for the suffix to use for IEADMC and CEAPRM members.

System programmer response:

Enter the parmlib suffix.

User response:

No action is required.

IZUG277I **Enter the *branch-country-name*
code, or press Enter to accept the
default *attribute-value*:**

Explanation

The message prompts for the country code or branch code value. A default is provided.

In the message text:

branch-country-name

Name of the branch or country

attribute-value

Default value for the branch or country.

System programmer response:

Enter the country or branch code, or accept the default.

User response:

No action is required.

IZUG278I **Enter the *branch-country-
name*code:**

Explanation

The message prompts for the country code or branch code value.

In the message text:

branch-country-name

Name of the branch or country.

System programmer response:

enter the country or branch code.

User response:

No action is required.

IZUG279E **The *branch-country-name*
code must be *branch-country-
range*alphanumeric characters (A-
Z, 0-9).**

Explanation

The value specified for the branch or country code does not conform to guidelines.

In the message text:

branch-country-name

Name of the branch or country

branch-country-range

Range for the branch or country attribute.

System programmer response:

Specify the correct value.

User response:

No action is required.

IZUG280I **Do you want to accept storage value *storage-name*? (Y|N)?**

Explanation:

The message prompts whether you want to use the existing specified storage option.

System programmer response:

Enter Y or N.

User response:

No action is required.

IZUG281I **What storage option do you want to use? Enter V for VOLSER or S for STORCLAS.**

Explanation:

The message prompts for the storage option to use.

System programmer response:

Enter a value.

User response:

No action is required.

IZUG282I **Enter the name of the SMS-storage-class:**

Explanation

The message prompts for the name of the specified SMS storage class.

In the message text:

SMS-storage-class

Type of storage option.

System programmer response:

Enter a storage class name.

User response:

No action is required.

IZUG283I **Specify one or more of the non-SMS direct access volumes to use. When you are finished entering the values, press Enter again without a value to complete:**

Explanation:

The message prompts for the volumes to use for the storage option.

System programmer response:

Enter the volume information. When you have entered all of the information for volume, to complete the input press Enter without specifying a value.

User response:

No action is required.

IZUG284I **Enter the name of the source data set for your existing CEAPRM00 parmlib member. Specify the**

fully qualified data set name, or press Enter to accept the default *parmlib-name*:

Explanation

The message prompts you for the name of the data set that contains your existing CEAPRM00 parmlib member. A fully qualified data set name is expected.

In the message text:

parmlib-name

Default data set name.

System programmer response:

Specify the fully qualified data set name, or press Enter to accept the supplied default if it is correct for your environment. If you specify the data set name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG285I **Enter the name of the source data set for your existing CEAPRM00 parmlib member. Specify the fully qualified data set name, or press Enter to use SYS1.PARMLIB:**

Explanation:

The message prompts you for the name of the data set that contains your existing CEAPRM00 parmlib member. A fully qualified data set name is expected.

System programmer response:

Specify the fully qualified data set name, or press Enter to use SYS1.PARMLIB as the source for the CEAPRM00 member. If you specify the data set name in quotes, the quotes are ignored. Your input is stored without quotes in the configuration file.

User response:

No action is required.

IZUG286W **Arguments are ignored.**

Explanation:

The additional unknown arguments that have been supplied in the call will be ignored.

System programmer response:

No action is required.

User response:

No action is required.

IZUG287I **z/OSMF RACF *racf-procedure* processing complete. Review and run *racf-rexx-file* before proceeding with configuration.**

Explanation

RACF processing has completed for the specified procedure.

In the message text:

racf-procedure

Name of the RACF procedure being performed

racf-rexx-file

Name of the RACF REXX exec.

System programmer response:

Review and run the REXX script before proceeding.

User response:

No action is required.

IZUG288I	The .profile is being created for the user.
-----------------	--

Explanation:

User .profile was not found. Attempting to create a .profile for the user.

System programmer response:

No action is required.

User response:

No action is required.

IZUG289I	The .profile is being updated with Common Information Model (CIM) environment variables.
-----------------	---

Explanation:

User .profile does not contain Common Information model (CIM) environment variables. Attempting to update .profile with CIM environment variables.

System programmer response:

No action is required.

User response:

No action is required.

IZUG290E	An attempt to update <i>file-name</i> has failed.
-----------------	--

Explanation

Attempt to update the specified file failed.

In the message text:

file-name

File name.

System programmer response:

Review log file for details.

User response:

No action is required.

IZUG291I	The .profile update is complete.
-----------------	---

Explanation:

The .profile has been updated.

System programmer response:

No action is required.

User response:

No action is required.

IZUG292W	Common Information Model (CIM) environment variables already set up in .profile: <i>wbem-root-value</i>
-----------------	--

Explanation

The .profile already contains Common Information model (CIM) environment variables.

In the message text:

wbem-root-value

Home directory of WBEM in the .profile.

System programmer response:

Ensure that the value in .profile matches the value specified in the configuration.

User response:

No action is required.

IZUG293I	Procedure <i>procedure</i> is being started.
-----------------	---

Explanation

An attempt to start the specified procedure has been made.

In the message text:

procedure

Procedure being started.

System programmer response:

No action is required.

User response:

No action is required.

IZUG294E	Common Information Model (CIM) server failed to start.
-----------------	---

Explanation:

Attempt to start the Common Information Model (CIM) server failed.

System programmer response:

Review log file for details.

User response:

No action is required.

IZUG295E	Verification process <i>ivp-name</i> has failed.
-----------------	---

Explanation

The verification process has failed.

In the message text:

ivp-name

Name of the IVP task.

System programmer response:

Review the log file for details.

User response:

No action is required.

IZUG296I **Verification process *ivp-name* has completed.**

Explanation

The specified verification process has completed.

In the message text:

ivp-name

Name of the IVP task.

System programmer response:

No action is required.

User response:

No action is required.

IZUG297I **Provider *provider-name* is already registered with Common Information Model (CIM).**

Explanation

The specified provider was found to have been already registered with Common Information Model (CIM).

In the message text:

provider-name

Name of the provider.

System programmer response:

No action is required.

User response:

No action is required.

IZUG298E **Provider *provider-name* is not registered with Common Information Model (CIM).**

Explanation

The specified provider is not registered with Common Information Model (CIM).

In the message text:

provider-name

Name of the provider.

System programmer response:

No action is required.

User response:

No action is required.

IZUG299I

The provider *provider-name* is being registered with Common Information Model (CIM).

Explanation

An attempt has been made to register the provider with Common Information Model (CIM).

In the message text:

provider-name

Name of the provider.

System programmer response:

No action is required.

User response:

No action is required.

IZUG300I **Processing of script *script-name* has started at *date-and-time*.**

Explanation

Script processing has started. The script name, data, and time are included.

In the message text:

script-name

Name of the script

date-and-time

Date and time that script processing started.

System programmer response:

No action is required.

User response:

No action is required.

IZUG301I **Log directory *log-directory* does not exist or is not writable: using temporary directory for log file.**

Explanation

For script processing, the named log directory (**logs**) within the z/OSMF data directory does not exist or the user who is executing the script does not have permission to write to this directory. The log file for processing of the script will be created in the temporary directory.

In the message text:

log-directory

Name of directory for the log files.

System programmer response:

No action is required.

User response:

No action is required.

IZUG302I **Log will be written to file *log-file-path-and-name*.**

Explanation

The path name of the log file for script processing is provided.

In the message text:

log-file-path-and-name

Directory and file name of the log.

System programmer response:

No action is required.

User response:

No action is required.

IZUG303I **Environment name and value being used are *env-var*.**

Explanation

The name and value for an environment setting is provided.

In the message text:

env-var

Name and value of an environment setting.

System programmer response:

No action is required.

User response:

No action is required.

IZUG304E **An error occurred writing to log file *log-file-path-and-name*: exiting script.**

Explanation

An error was encountered while attempting to write to the log file.

In the message text:

log-file-path-and-name

Directory and file name of the log.

System programmer response:

Check for additional error messages on the screen that describe the error. Rerun after correcting the error.

User response:

No action is required.

IZUG305E **The script *script-name* failed with reason code *reason-code*; see log file *log-file-path-and-name*.**

Explanation

The indicated script failed. A return code is provided to help indicate the cause of the error.

In the message text:

script-name

Script that failed

reason-code

Reason code for the error

log-file-path-and-name

Directory and file name of the log file.

For the **izuadmin.sh** script, the following reason codes are valid:

- 1** Script was called with incorrect arguments.
- 2** Problem with the log directory.
- 3** Error writing to the log file, or the log file is not accessible.
- 4** Required environment variable is missing or set incorrectly. Or, the izuadmin.env file does not exist.
- 5** Required environment setting is missing or incorrect. This error can occur if an expected configuration property or properties file, such as izuapps.properties, is not set, cannot be found, or is not readable.
- 6** Problem found during verification processing.
- 7** Installed z/OS level is incorrect for z/OSMF.
- 105** Exception encountered by an internal script.

For the **izuprime.sh** script, the following reason codes are valid:

- 1** Usage error.
- 2** Problem with the log directory.
- 3** Error writing to the log file.
- 4** Script encountered an error when running a z/OS UNIX shell command, such as mkdir or cp.
- 5** A repository already exists.
- 6** Specified user ID is not defined to the z/OS system.

System programmer response:

For more information, see the z/OSMF log file for related messages. After correcting the error, run

the script again. For reason code 105, contact IBM Support for assistance.

User response:

No action is required.

IZUG306I **Script *script-name* was invoked with options *input-options*.**

Explanation

The options specified as input to the named script are provided.

In the message text:

script-name

Name of the script

input-options

Options passed to the script.

System programmer response:

No action is required.

User response:

No action is required.

IZUG311E **IZU_APPSERVER_ROOT *server-root-directory* is not valid: exiting script.**

Explanation

The z/OSMF server root directory is not valid. The processing for the script stops.

In the message text:

server-root-directory

Root directory of the z/OSMF server.

System programmer response:

Set IZU_APPSERVER_ROOT to the valid root directory and run again.

User response:

No action is required.

IZUG312I **The administration request is being processed.**

Explanation:

Processing of the administration request has started.

System programmer response:

No action is required.

User response:

No action is required.

IZUG313E **A usage error has occurred: *error*.**

Explanation

A problem with the usage has occurred. Context of the error is provided.

In the message text:

error

Explanation for the incorrect usage.

System programmer response:

Correct the problem indicated by the explanation of the error and run again.

User response:

No action is required.

IZUG314E **IZU_CODE_ROOT *product-root-directory* is not valid: exiting script.**

Explanation

The z/OSMF product root directory is not valid.

In the message text:

product-root-directory

Root directory of the z/OSMF product.

System programmer response:

Set IZU_CODE_ROOT to the valid z/OSMF product root directory and run again.

User response:

No action is required.

IZUG315E **An incorrect environment setting has been detected: *env-var*.**

Explanation

A problem exists with a setting in the environment file. Context of the error is provided.

In the message text:

env-var

Environment setting and associated problem.

System programmer response:

Review the included environment setting and the associated problem. Correct the error and run again.

User response:

No action is required.

IZUG316E **PEGASUS_HOME directory *CIM-server-root-directory* is not valid: exiting script.**

Explanation

The Common Information Model (CIM) server WBEM root directory is not valid. Processing for the script stops.

In the message text:

CIM-server-root-directory

WBEM root directory of the CIM server.

System programmer response:

Set PEGASUS_HOME to the Common Information Model (CIM) server WBEM root directory and run the script again.

User response:

No action is required.

IZUG317E **IZU_CONFIG_DIR configuration-directory is not valid: exiting script.**

Explanation

The z/OSMF configuration directory is not valid. Processing for the script stops.

In the message text:

configuration-directory

Configuration directory of the z/OSMF product.

System programmer response:

Set IZU_CONFIG_DIR to the valid z/OSMF configuration directory and run again.

User response:

No action is required.

IZUG318E **Path path-setting member member-name must exist: exiting script.**

Explanation

A directory or path that is a member of the specified path setting does not exist. Processing of the script stops.

In the message text:

path-setting

Name of the path setting

member-name

Directory or file specified in the path that does not exist.

System programmer response:

Determine why the file or directory does not exist. Correct the problem and run again.

User response:

No action is required.

IZUG319E **Data directory data-directory must exist and be writable: exiting script.**

Explanation

For script processing the z/OSMF data directory must exist and be capable of being written to. Processing of the script stops.

In the message text:

data-directory

Name of the data directory.

System programmer response:

Ensure the z/OSMF data directory exists. Ensure that the user running the script has permission to write to the directory. After correcting the error run again.

User response:

No action is required.

IZUG320E **Users will not be able to launch z/OSMF. The installed z/OS level installed-z/OS-level is earlier than the minimum z/OS level minimum-z/OS-level that is required by z/OSMF.**

Explanation

z/OSMF cannot be launched because it is installed on a system that is earlier than the minimum supported level of z/OS.

In the message text:

installed-z/OS-level

Installed operating system level

minimum-z/OS-level

Minimum operating system level that z/OSMF requires.

In the message text, the software level for the product (z/OS or z/OSMF) is indicated through a standard convention: *aa.bb.cc*, where:

- *aa* is the version
- *bb* is the release
- *cc* is the modification level.

You can correlate the returned value as follows:

- 04.05.00 indicates V2R5 of z/OS
- 04.04.00 indicates V2R4 of z/OS
- 04.03.00 indicates V2R3 of z/OS

Thus, for example, the value 04.05.00 indicates z/OS V2R5.

System programmer response:

Upgrade to a z/OS level that is supported by z/OSMF.

User response:

No action is required.

IZUG321W **The installed z/OSMF level product-level is earlier than the z/OS level os-level.**

Explanation

Your system is running z/OSMF level *product-level*, but a newer z/OSMF level might be available from IBM. Most likely, your installation has migrated to a new release of z/OS without upgrading the z/OSMF product. To allow z/OSMF to use the latest functions

in z/OS level *os-level*, it is recommended that you upgrade z/OSMF to the latest level. Until you do so, z/OSMF will continue to operate at its current level of functionality.

In the message text:

product-level

Installed level of z/OSMF.

os-level

Operating system level.

In the message text, the software level for the product (z/OS or z/OSMF) is indicated through a standard convention: *aa.bb.cc*, where:

- *aa* is the version
- *bb* is the release
- *cc* is the modification level.

You can correlate the returned value as follows:

- 04.05.00 indicates V2R5 of z/OS
- 04.04.00 indicates V2R4 of z/OS
- 04.03.00 indicates V2R3 of z/OS

Thus, for example, the value 04.05.00 indicates z/OS V2R5.

System programmer response:

Upgrade z/OSMF to the latest level that is supported on your z/OS system.

User response:

No action is required.

IZUG333I	Enter the z/OSMF Unauthenticated <i>unauthenticated-UID</i>, or enter the keyword AUTOUID:
-----------------	---

Explanation

The message prompts you to input unauthenticated guest user UID in z/OSMF.

In the message text:

unauthenticated-UID

unauthenticated user UID.

System programmer response:

Enter a valid value.

User response:

No action is required.

IZUG334I	Enter the z/OSMF Unauthenticated <i>unauthenticated-UID</i>, or enter the keyword AUTOUID, or press Enter to accept the default <i>default-unauthenticated-UID</i>:
-----------------	--

Explanation

The message prompts you to input unauthenticated guest user UID in z/OSMF. To accept the default, press Enter.

In the message text:

unauthenticated-UID

unauthenticated guest user UID.

default-unauthenticated-UID

Default unauthenticated user UID.

System programmer response:

Enter a valid value.

User response:

No action is required.

IZUG335E	A symbolic link is required for the directory: <i>/etc/zosmf</i>. The link could not be created, however, because the directory already exists or <i>/etc/zosmf</i> is already defined as the symbolic link for another directory.
-----------------	---

Explanation

While processing the `izusetup.sh -finish` script, z/OSMF detected that the z/OSMF configuration directory is set to use a directory name other than the product default: `/etc/zosmf`. This directory name is specified through the variable `IZU_CONFIG_DIR`. Most likely, your installation chose another name for this directory when configuring z/OSMF on your system.

Because the z/OSMF online help system requires `/etc/zosmf` as its mount point, z/OSMF attempts to create a symbolic link "etc/zosmf" that resolves to the path name of your specified directory. The link could not be created, however, either because directory `/etc/zosmf` already exists on your system, or "etc/zosmf" is already defined as a symbolic link for another directory.

System programmer response

To resolve this error, take one of the following actions, as appropriate:

- If the directory `/etc/zosmf` already exists on your system, examine the directory and its contents. Determine whether the directory can be deleted safely, or its contents moved to another directory. If so, take these steps to remove the directory. Then, run the configuration request again.
- Change your installation's specification for the `IZU_CONFIG_DIR` variable to the default value `/etc/zosmf`, and re-run the z/OSMF configuration process, starting with the `izusetup.sh -config` invocation. You can specify this directory name in the override file

for variable IZU_CONFIG_DIR, or interactively, in response to the script prompt for the name of the z/OSMF configuration directory.

User response:

Contact your z/OSMF administrator or system programmer.

IZUG336I	Work manager <i>work-manager-name</i> is being created.
-----------------	--

Explanation

The work manager is being created.

In the message text:

work-manager-name

Name of the work manager.

System programmer response:

No action is required.

User response:

No action is required.

IZUG337I	Work manager <i>work-manager-name</i> property <i>property-name</i> is being set to value <i>value</i>.
-----------------	--

Explanation

The work manager property is being set to the indicated value.

In the message text:

work-manager-name

Name of the work manager

property-name

Name of the property

value

Value for the property.

System programmer response:

No action is required.

User response:

No action is required.

IZUG340I	Variable substitution entry <i>variable-name</i> is being updated with value <i>value</i>.
-----------------	---

Explanation

The variable substitution entry is being updated with the specified value.

In the message text:

variable-name

Name of the variable

value

Value of the variable.

System programmer response:

No action is required.

User response:

No action is required.

IZUG341I	Variable substitution entry <i>variable-name</i> is being created with value <i>value</i>.
-----------------	---

Explanation

The variable substitution entry is being created with the specified value.

In the message text:

variable-name

Name of the variable

value

Value of the variable.

System programmer response:

No action is required.

User response:

No action is required.

IZUG343I	Shared library <i>shared-library-name</i> with class path <i>class-path</i> and native path <i>native-path</i> is being deleted.
-----------------	---

Explanation

The specified shared library with the specified class path and native path is being removed.

In the message text:

shared-library-name

Name of the shared library

class-path

classpath value

native-path

Native path value.

System programmer response:

No action is required.

User response:

No action is required.

IZUG344I	Shared library <i>shared-library-name</i> with class path <i>class-path</i> and native path <i>native-path</i> is being created.
-----------------	---

Explanation

The specified shared library with the specified class path and native path is being created.

In the message text:

shared-library-name

Name of the shared library

class-path

classpath value

native-path

Native path value.

System programmer response:

No action is required.

User response:

No action is required.

IZUG345I **Service *plugin-name* is being removed.**

Explanation

The specified service is being removed from z/OSMF.

In the message text:

plugin-name

Name of the service.

System programmer response:

No action is required.

User response:

No action is required.

IZUG346I **Service *service-name* from location *file-location* is being installed.**

Explanation

The service is being installed into z/OSMF from the specified location.

In the message text:

plugin-name

Name of the service.

file-location

Location of the Enterprise Archive (EAR) file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG347I **Reference to shared library *shared-library-name* with scope *scope* is being added.**

Explanation

A reference to the shared library is being added with the specified scope.

In the message text:

shared-library-name

Name of the shared library

scope

Scope of the shared library reference.

System programmer response:

No action is required.

User response:

No action is required.

IZUG348I **Processing of your request has started. This process might require several minutes or more to complete.**

Explanation:

The requested script processing is running, but might take some time to complete. As it runs, the script writes messages to the script log file.

System programmer response:

No action is required.

User response:

No action is required.

IZUG349I **The function *function-name* can be accessed at link *link-name* after the z/OSMF server is started on your system.**

Explanation

The requested configuration process completed. z/OSMF will be available to users at the indicated URL after the z/OSMF server is restarted on this system.

In the message text:

function-name

The z/OSMF function that is available.

link-name

The link for accessing z/OSMF.

System programmer response:

No action is required.

User response:

No action is required.

IZUG350I **The z/OSMF server on this system will attempt to connect to the angel *angel-name*.**

Explanation

On start-up, the z/OSMF server attempts to connect to a WebSphere Liberty angel process. Your installation specifies the angel name on the ANGEL_PROC parameter in the IZUPRMxx member for this system. By default, the angel name is IZUANG1.

In the message text:

angel-name

Angel to be used by the z/OSMF server.

System programmer response:

No action is required.

User response:

No action is required.

IZUG352W **During the processing of the z/OSMF server started procedure, the PARM parameter was found to be missing one or more expected values: *missing-values***

Explanation

In the z/OSMF server started procedure, the PARM statement is missing the indicated values.

In the message text:

missing-values

Values that are missing from the PARM statement.

System programmer response:

No action is required.

User response:

This message indicates a potential problem with the z/OSMF server started procedure. In the started procedure, verify that the PARM= parameter list is complete and correct for your installation. For reference, see the sample procedures IZUSVR1 in SYS1.PROCLIB and IZUSVR2 in SYS1.SAMPLIB.

IZUG354I **Security option *option-name* with value *option-value* is being set.**

Explanation

A security setting in the z/OSMF server is being updated to the specified value.

In the message text:

option-name

Name of the option being set

option-value

Value of the option being set.

System programmer response:

No action is required.

User response:

No action is required.

IZUG356I **Service *plugin-name* is being stopped.**

Explanation

The specified service is being stopped.

In the message text:

plugin-name

Name of the service.

System programmer response:

No action is required.

User response:

No action is required.

IZUG357I **Arguments in effect: *CODE_ROOT* = *code-root* *USER_DIR* = *user-dir* *TRACE* = *trace* *KCINDEX* = *kcindex***

Explanation

The message displays the arguments that were passed from the z/OSMF started task to the z/OSMF configuration step.

In the message text:

code-root

The code root of z/OSMF.

user-dir

The user directory of z/OSMF.

trace

The trace level of z/OSMF configuration.

kcindex

Flag indicating if the KC index is rebuilt.

System programmer response:

No action is required.

User response:

No action is required.

IZUG358W **The property *IZU_TRACE* is set to *parameter-value*. The value is incorrect.**

Explanation

The value specified to the parameter *IZU_TRACE* is incorrect.

In the message text:

parameter-value

The value of the parameter *IZU_TRACE*.

System programmer response:

Specify a valid value for *IZU_TRACE*.

User response:

No action is required.

IZUG360I **Script option *option-name* is deprecated. The z/OSMF configuration process ignores this option.**

Explanation

The specified script option is deprecated. The z/OSMF process ignores the option and continues processing as normal. If you received this message when running the *izusetup.sh* script with the *-service*

option, understand that the -service option is no longer required when you apply z/OSMF service to your system.

In the message text:

option-name

Option that was specified.

System programmer response:

To avoid receiving this message in the future, do not specify the indicated option. If you received this message when applying z/OSMF service, you are using an obsolete option. Review the HOLDDATA section of the PTF for instructions on applying service to your system.

User response:

No action is required.

IZUG361I **Do you want to create a Certificate Authority? For yes, enter Y. For no, enter N:.**

Explanation

The message prompts you to indicate whether (Y or N) the z/OSMF security setup should include the creation of a Certificate Authority (CA). The CA is used to sign server certificates that are used for secure (SSL) communication between the user's web browser and the z/OSMF server. Y is the default.

If you specify N, you must provide your own CA for enabling secure communications.

System programmer response:

Enter a valid value.

User response:

No action is required.

IZUG362I **Do you want to create a Certificate Authority? For yes, enter Y. For no, enter N. Or press Enter to accept the default value *default-value*.**

Explanation

The message prompts you to indicate whether (Y or N) the z/OSMF security setup should include the creation of a Certificate Authority (CA). The CA is used to sign server certificates that are used for secure (SSL) communication between the user's web browser and the z/OSMF server. The default value is provided.

In the message text:

default-value

Default value for creating Certificate Authority (Y or N).

System programmer response:

Enter the a valid value (Y or N) or press Enter to select the default value.

User response:

No action is required.

IZUG363E **User *user-name* is not permitted to access the digital certificate *certificate-label*.**

Explanation

The specified user lacks sufficient authorization to the indicated digital certificate.

In the message text:

user-name

Name of the user

certificate-label

Label of digital certificate.

System programmer response:

Determine whether the user requires access to the digital certificate. If so, grant access to the user.

User response:

No action is required.

IZUG364E **User *user-name* did not connect label *certificate-label* to keyring *certificate-keyring*.**

Explanation

The specified user lacks sufficient authorization to the indicated keyring.

In the message text:

user-name

Name of the user.

certificate-label

Label of the digital certificate.

certificate-keyring

Keyring of the digital certificate.

System programmer response:

Determine whether the user requires access to the keyring. If so, grant access to the user.

User response:

No action is required.

IZUG365I **Process *process-name* with start command arguments is being updated to include value *value-1*. The value of the arguments is now *value-2*.**

Explanation

The specified argument is being added to the start command arguments for the specified process.

In the message text:

process-name

Name of the server process

value-1

Value of the new argument being added

value-2

New value of the start command arguments.

System programmer response:

No action is required.

User response:

No action is required.

IZUG366I **The locale is switched from
current-locale to designated-locale.**

Explanation

The message will be prompted if the locale in use is not standard ASCII locale.

In the message text:

current-locale

The locale the user specifies.

designated-locale

C (standard ASCII locale).

System programmer response:

No action is required.

User response:

No action is required.

IZUG367W **Script *setup_knownPlugins*
encountered errors: startup
continues.**

Explanation:

A problem occurs while setting up specified plug-ins. Configuration procedure continues instead of being terminated.

System programmer response:

Search the configuration job log to check out why performing *setup_knownPlugins* runs into a problem.

User response:

No action is required.

IZUG368I **Enter the z/OSMF unauthenticated
user name *unauthenticated-name*.**

Explanation

The message prompts you to input unauthenticated guest user name in z/OSMF.

unauthenticated-name

unauthenticated user name.

System programmer response:

Enter a valid value.

User response:

No action is required.

IZUG369I **Enter the z/OSMF unauthenticated
user name *unauthenticated-name*,
or press Enter to accept
the default value *default-
unauthenticated-name*.**

Explanation

The message prompts you for the unauthenticated guest user name in z/OSMF. To accept the default, press Enter.

unauthenticated-name

unauthenticated guest user name.

default-unauthenticated-name

Default unauthenticated user name.

System programmer response:

Enter a valid value, or press Enter to accept the default value.

User response:

No action is required.

IZUG370I **User registry is being initialized
with user ID *user-id*.**

Explanation

The z/OSMF user registry is being initialized with the specified user ID.

In the message text:

user-id

User ID with which the user registry is being initialized.

System programmer response:

No action is required.

User response:

No action is required.

IZUG371I **Role repository is being initialized
for user ID *user-id*.**

Explanation

The z/OSMF role repository is being initialized for the specified user ID.

In the message text:

user-id

User ID for which the role repository is being initialized.

System programmer response:

No action is required.

User response:

No action is required.

IZUG372E **Command *command-name* returned an error. Command return code is *return-code*.**

Explanation

An error was received from a command invocation.

In the message text:

command-name

Command that returned the error

return-code

Return code from the command.

System programmer response:

Search the log for other error messages that indicate the problem. Correct the problem indicated by the messages and run again.

User response:

No action is required.

IZUG373E **Repository *repository-name* was not initialized because it already exists: exiting script.**

Explanation

A z/OSMF repository was not initialized because it already exists. A z/OSMF repository can only be initialized if it does not exist. Processing of the script stops.

In the message text:

repository-name

Name of the existing repository.

System programmer response:

Do not attempt to initialize the existing repository.

User response:

No action is required.

IZUG374E **User ID *user-id* for the z/OSMF administrator must exist: exiting script.**

Explanation

The z/OSMF repositories were not initialized because the administrator user ID does not exist. Processing of the script stops.

In the message text:

user-id

User ID that does not exist.

System programmer response:

Search the log for other error messages that might indicate the problem. Correct the problem indicated by the messages and run again.

User response:

No action is required.

IZUG375I **Verification has completed for *item-name*.**

Explanation

Verification has completed for the specified item.

In the message text:

item-name

Item that was verified.

System programmer response:

No action is required.

User response:

No action is required.

IZUG376E **Verification failed for *item-name* because of the following reason: *reason***

Explanation

Verification failed for the item because of the specified reason. Context of the error is provided.

In the message text:

item-name

Item that failed verification

reason

Reason verification failed.

System programmer response:

Perform action to correct the problem based on the indicated reason.

User response:

No action is required.

IZUG377E **Unable to write to *directory-name*: exiting script.**

Explanation

Attempt to write to the specified directory failed.

In the message text:

directory-name

Name of the directory being written to.

System programmer response:

Ensure user has access to write to the directory.

User response:

No action is required.

IZUG378I **Process *process-name* JVM custom property *property-name* that has a value of *value* is being deleted.**

Explanation

The specified property for the named process is being removed.

In the message text:

process-name

Name of the server process

property-name

Name of the property

value

Value of the property.

System programmer response:

No action is required.

User response:

No action is required.

IZUG379I	Process <i>process-name</i> JVM custom property <i>property-name</i> that has a value of <i>value</i> is being created.
-----------------	--

Explanation

The specified property for the named process is being added.

In the message text:

process-name

Name of the server process

property-name

Name of the property

value

Value of the property.

System programmer response:

No action is required.

User response:

No action is required.

IZUG380E	Unable to unmount file system <i>file-system-name</i>.
-----------------	---

Explanation

Attempt to unmount the indicated file system failed.

In the message text:

file-system-name

Name of the file system.

System programmer response:

For more information, see the log file.

User response:

No action is required.

IZUG381I	Unmounting <i>file-system-name</i>.
-----------------	--

Explanation

The procedure to unmount the specified file system has started.

In the message text:

file-system-name

Name of the file system.

System programmer response:

No action is required.

User response:

No action is required.

IZUG382E	File system <i>file-system-name</i> does not exist.
-----------------	--

Explanation

The specified file system does not exist.

In the message text:

file-system-name

Name of the file system.

System programmer response:

Specify a file system that does exist.

User response:

No action is required.

IZUG383I	File system <i>file-system-name</i> is mounted at mount point <i>mount-point</i>.
-----------------	--

Explanation

The indicated file system is mounted at that mount point.

In the message text:

file-system-name

Name of the file system

mount-point

Name of the mount point.

System programmer response:

No action is required.

User response:

No action is required.

IZUG384I	Object <i>object-name</i> property <i>property-name</i>, which has a value of <i>value</i>, is being deleted.
-----------------	--

Explanation

The indicated property for this object is being deleted. The current setting for the property is shown.

You have either selected to change the current setting of a property, or you are deleting the property altogether. When you change the value of a property,

the property is first deleted and then created again with the new value. When you delete a property, z/OSMF uses the property default instead.

In the message text:

object-name

Name of the object

property-name

Name of the property

value

Value of the property.

System programmer response:

No action is required.

User response:

No action is required.

**IZUG385I The z/OSMF server is not started.
To allow the -addlink request
to complete, restart the z/OSMF
server.**

Explanation:

The -addlink request cannot complete until you start the z/OSMF server.

System programmer response

Start the z/OSMF server.

After the server is started, see the z/OSMF log file for an indication of the success or failure of this request. The z/OSMF log file is named IZUGn.log, where *n* is a number from 0 to 9. The z/OSMF log file resides in the /logs subdirectory directory of the z/OSMF data file system. Your installation specified the z/OSMF data file system on the IZU_DATA_DIR variable when configuring z/OSMF. By default, this is directory /var/zosmf/data.

User response:

No action is required.

**IZUG386E The command is missing a
required argument: *object-name*.**

Explanation

The command is missing the indicated argument and thus, cannot be performed.

In the message text:

argument-name

Name of the missing argument.

System programmer response:

Enter the command again with all of its required arguments.

User response:

No action is required.

**IZUG387I Setting *setting-name* has a value of
value.**

Explanation

The setting will be set to the indicated value. The current value of the setting in the z/OSMF configuration is shown.

In the message text:

setting-name

Name of the setting

value

Value for the setting.

System programmer response:

No action is required.

User response:

No action is required.

IZUG388I Setting *setting-name* is not set.

Explanation

The indicated setting is not currently set in the z/OSMF configuration. z/OSMF will use the setting default.

In the message text:

setting-name

Name of the setting

value

Value for the setting.

System programmer response:

No action is required.

User response:

No action is required.

**IZUG397I The -addlink request was
processed. To verify that the link
was added, check the z/OSMF log
file.**

Explanation:

To add a link to z/OSMF, you invoked the izusetup.sh script with the -addlink option. For an indication of the success or failure of this request, see the z/OSMF log file.

System programmer response:

No action is required.

User response

To verify that the link was added, check the z/OSMF log file. This file is named IZUGn.log, where *n* is a number from 0 to 9. The z/OSMF log file resides in the /logs subdirectory directory of the z/OSMF data file system. Your installation specified the z/OSMF data file system on the IZU_DATA_DIR variable when

configuring z/OSMF. By default, this is directory /var/zosmf/data.

To modify or remove a link after it is added, you must use the Links task in z/OSMF.

**IZUG398I The z/OSMF server is not started.
To allow the -addlink request to
complete, start the server.**

Explanation:

The -addlink request cannot complete until you start the z/OSMF server.

System programmer response

Start the z/OSMF server.

After the server is started, see the z/OSMF log file for an indication of the success or failure of this request. The z/OSMF log file is named IZUGn.log, where *n* is a number from 0 to 9. The z/OSMF log file resides in the /logs subdirectory directory of the z/OSMF data file system. Your installation specified the z/OSMF data file system on the IZU_DATA_DIR variable when configuring z/OSMF. By default, this is directory /var/zosmf/data.

User response:

No action is required.

**IZUG399I Successfully copied source-file-
name to target-file-name.**

Explanation

The input file was successfully copied to the destination.

In the message text:

source-file-name

Name of the source file

target-file-name

Name of the destination file.

System programmer response:

No action is required.

User response:

No action is required.

Appendix A. Security structures for z/OSMF

Using z/OSMF requires sufficient authority in z/OS. Specifically, on the z/OS system to be managed, the resources to be accessed on behalf of users (data sets, operator commands, and so on) are secured through the external security manager at your installation, such as RACF. Your installation's security administrator must create the authorizations in your external security manager. To assist your security administrator, z/OSMF provides sample jobs in SYS1.SAMPLIB and the information in this document. Your security administrator can use the sample jobs to create the groups, user IDs, and resource profiles for your z/OSMF configuration. Later, these z/OSMF constructs require more permissions to a number of existing groups, user IDs, and resources on your system.

This appendix describes the security configuration requirements for z/OSMF. Included are the resource authorizations that are created when your installation runs one or more of the following sample security jobs:

- IZUNUSEC, to help you set up basic security for a z/OSMF nucleus configuration.
- Individual IZUxxSEC jobs for the core services
- IZUSEC job that consolidates the security set-up for both the z/OSMF nucleus and the core services
- Individual IZUxxSEC jobs for the optional services.

Also listed are the resource authorizations that your installation must define outside of the configuration process.

The security configuration requirements for z/OSMF are described in the sections that follow. Creating these permissions requires the assistance of your security administrator.

- [“Class activations that z/OSMF requires” on page 372](#)
- [“SAF profile prefix for z/OSMF resources” on page 374](#)
- [“User IDs that z/OSMF creates during configuration” on page 374](#)
- [“Security groups that z/OSMF creates during configuration” on page 374](#)
- [“Resource authorizations for the Security Configuration Assistant service” on page 374](#)
- [“Resource authorizations for the z/OSMF core functions” on page 375](#)
- [“Resource authorizations for hardware compression” on page 384](#)
- [“Resource authorizations for hardware cryptography” on page 385](#)
- [“Resource authorizations for Common Information Model” on page 385](#)
- [“Resource authorizations for Capacity Provisioning Manager” on page 386](#)
- [“Resource authorizations for common event adapter \(CEA\)” on page 387](#)
- [“Resource authorizations for the z/OS compliance REST interface” on page 387](#)
- [“Resource authorizations for the z/OS console REST interface” on page 387](#)
- [“Resource authorizations for the z/OS data set and file REST interface” on page 390](#)
- [“Resource authorizations for the z/OS jobs REST interface” on page 390](#)
- [“Resource authorizations for the Capacity Provisioning service” on page 391](#)
- [“Resource authorizations for the Network Configuration Assistant service” on page 392](#)
- [“Resource authorizations for the Incident Log service” on page 393](#)
- [“Resource authorizations for the ISPF service” on page 396](#)
- [“Resource authorizations for the Resource Monitoring service” on page 397](#)
- [“Resource authorizations for the Software Deployment service” on page 397](#)
- [“Resource authorizations for the Sysplex Management service” on page 401](#)
- [“Resource authorizations for the Workload Management service” on page 402](#)

- “Resource authorizations for the IBM zERT Network Analyzer service” on page 403
- “Resource authorizations for the z/OS Management Services Catalog service” on page 404
- “Resource authorizations for the Storage Management service” on page 408
- “Resource authorizations for the Parmlib Management service” on page 408

Class activations that z/OSMF requires

For a RACF installation, the security classes that are shown in [Table 48 on page 372](#) must be active when you configure z/OSMF. Commands for activating the classes (with generic profile checking activated) are included in commented sections in the IZUxxSEC jobs. To allow the commands to be issued when the jobs run, uncomment the sections. Or, ask your security administrator to enter the commands directly, as shown in [Table 48 on page 372](#).

Table 48. Class activations that z/OSMF requires		
Class	Purpose	RACF commands for activating
ACCTNUM	Controls access to the account number used for the procedure for the z/OSMF REST interfaces.	SETROPTS CLASSACT(ACCTNUM)
APPL	Controls access to the z/OSMF application domain. This access is required by: • Security group for z/OSMF administrators (IZUADMIN, by default) • Security group for z/OSMF unauthenticated guest users (IZUGUEST, by default) • Security group for the z/OSMF users (IZUUSER, by default) • Security group for the z/OS security administrator (IZUSECAD, by default). If there is no matching profile in the APPL class, RACF allows the user to access the application.	SETROPTS CLASSACT(APPL) SETROPTS RACLIST(APPL) GENERIC(APPL)
EJBROLE	Controls the user’s ability to connect to the z/OSMF core functions and tasks. z/OSMF defines a resource name for each core function and task.	SETROPTS CLASSACT(EJBROLE) SETROPTS RACLIST(EJBROLE) GENERIC(EJBROLE)
FACILITY	Controls the user’s access to profiles when the user performs an action. This access is required by the z/OSMF started task user ID (IZUSVR, by default). Examples include the profiles that are used to control privileges in the z/OS UNIX environment.	SETROPTS CLASSACT(FACILITY) SETROPTS RACLIST(FACILITY) GENERIC(FACILITY)
JESSPOOL	Allows the user to retrieve messages from the system log (SYSLOG).	SETROPTS CLASSACT(JESSPOOL) SETROPTS RACLIST(JESSPOOL)
LOGSTRM	Allows the user to retrieve messages from the operations log (OPERLOG).	SETROPTS CLASSACT(LOGSTRM) SETROPTS RACLIST(LOGSTRM)

Table 48. Class activations that z/OSMF requires (continued)		
Class	Purpose	RACF commands for activating
OPERCMD S	Allows the user to create an EMCS console by using the z/OS Operator Consoles task.	SETROPTS CLASSACT(OPERCMDS) SETROPTS RACLIST(OPERCMDS)
SERVAUTH	Controls the user's ability to use CEA TSO/E address space services. In z/OSMF, this access is required by: • z/OSMF started task user ID (IZUSVR, by default) • Callers of the z/OS data set and file REST interface services • Users of the ISPF task.	SETROPTS CLASSACT(SERVAUTH) SETROPTS RACLIST(SERVAUTH) GENERIC(SERVAUTH)
SERVER	Allows the z/OSMF started task user ID to request services from z/OS system components, such as the System Authorization Facility (SAF), workload management (WLM), and SVCDUMP services.	SETROPTS CLASSACT(SERVER) SETROPTS RACLIST(SERVER) GENERIC(SERVER)
STARTED	Assigns an identity to the z/OSMF started task during the processing of an MVS START command. By default, the started task runs under the IZUSVR user ID.	SETROPTS CLASSACT(STARTED) SETROPTS RACLIST(STARTED) GENERIC(STARTED)
TSOAUTH	Allows the user to create an EMCS console by using the z/OS Operator Consoles task.	SETROPTS CLASSACT(TSOAUTH) SETROPTS RACLIST(TSOAUTH)
TSOPROC	Controls access to the procedure for the z/OSMF REST interfaces.	SETROPTS CLASSACT(TSOPROC)
ZMFAPLA	Controls the user's ability to use the z/OSMF core functions and tasks. z/OSMF defines a resource name for each core function and task. • Profile names in this class are case-sensitive. • The ZMFAPLA class requires the RACLIST option.	SETROPTS CLASSACT(ZMFAPLA) SETROPTS RACLIST(ZMFAPLA) GENERIC(ZMFAPLA)
ZMFCLLOUD	Allows the user to use the z/OSMF core functions and tasks that are related to IBM Cloud Provisioning. z/OSMF defines a resource name for each core function and task for IBM Cloud Provisioning. For more information, see Chapter 27, "Configure the Cloud Provisioning services," on page 141. The ZMFCLLOUD class requires the RACLIST option.	SETROPTS CLASSACT(ZMFCLLOUD) GENERIC(ZMFCLLOUD) RACLIST(ZMFCLLOUD)

If your installation uses an external security manager other than RACF, ask your security administrator to create equivalent commands for your environment.

SAF profile prefix for z/OSMF resources

During the configuration process, your security administrator runs the IZUxxSEC jobs to secure z/OSMF resources. In these jobs, your installation specifies a System Authorization Facility (SAF) profile prefix to be used for naming z/OSMF resources. The SAF prefix is prepended to the names of z/OSMF resource profiles, and is used in some of the RACF commands that are contained in the IZUxxSEC jobs.

In the examples in this document, the SAF prefix is shown as *<SAF-prefix>*. By default, the SAF prefix is IZUDFLT. If your installation selects to use a different value, substitute the value in the examples.

User IDs that z/OSMF creates during configuration

The IZUSEC job creates the user IDs that are described in [Table 49 on page 374](#).

Table 49. User IDs that z/OSMF creates during the configuration process			
User ID	Purpose	Default UID	Created by
IZUGUEST	User ID for performing unauthenticated work, such as guest user access to the Welcome page.	9011	IZUSEC job
IZUSVR	User ID for the z/OSMF started tasks, which are named IZUANG1 and IZUSVR1, by default.	9010	IZUSEC job

Table 49 on page 374 shows the IBM default values. Your security administrator can specify different user IDs in place of the default user IDs in the IZUSEC job.

Security groups that z/OSMF creates during configuration

The IZUSEC job creates a base set of security groups for your z/OSMF configuration. These groups are necessary for giving users the proper level of access to z/OSMF and z/OS system resources.

Your security team might determine that the existing group names would be preferred. If so, you can use your existing group names in place of the supplied z/OSMF default group names. For example, you might already have a group that is aligned with administrators; if so, you can use that group, instead of the z/OSMF default group for administrators, IZUADMIN.

Table 50 on page 374 lists the groups that the IZUSEC job creates. The group names can change, based on the values you provide during the configuration process. [Table 50 on page 374](#) shows the IBM default values.

Table 50. Security groups that z/OSMF creates during the configuration process		
Group	Purpose	Created by
IZUADMIN	Security group for the z/OSMF administrator role. Any user IDs connected to this group are considered to be z/OSMF administrators.	IZUSEC job
IZUUSER	Security group for the z/OSMF user role.	IZUSEC job
IZUSECAD	Security group for the z/OS security administrator role in z/OSMF.	IZUSEC job
IZUUNGRP	Security group for the z/OSMF unauthenticated user ID.	IZUSEC job

Resource authorizations for the Security Configuration Assistant service

Table 51 on page 375 describes the access requirements for the Security Configuration Assistant service. The IZUSASEC job includes sample RACF commands for creating these authorizations on your system. These values can vary, based on the values you use at your installation. [Table 51 on page 375](#) shows the IBM default values.

Table 51. Security setup requirements for the Security Configuration Assistant service

Resource class	Resource name	Who needs access?	Type of access required	Why
SERVER	BBG.SECCLASS.ACCTNUM	IZUSVR	READ	Grant the server permission to perform authorization checks against the ACCTNUM profile in the SERVER class.
SERVER	BBG.SECCLASS.APPL	IZUSVR	READ	Grant the server permission to perform authorization checks against the APPL profile in the SERVER class.
SERVER	BBG.SECCLASS.CSFSESV	IZUSVR	READ	Grant the server permission to perform authorization checks against the CSFSESV profile in the SERVER class.
SERVER	BBG.SECCLASS.EJBROLE	IZUSVR	READ	Grant the server permission to perform authorization checks against the EJBROLE profile in the SERVER class.
SERVER	BBG.SECCLASS.FACILITY	IZUSVR	READ	Grant the server permission to perform authorization checks against the FACILITY profile in the SERVER class.
SERVER	BBG.SECCLASS.JESSPOOL	IZUSVR	READ	Grant the server permission to perform authorization checks against the JESSPOOL profile in the SERVER class.
SERVER	BBG.SECCLASS.LOGSTRM	IZUSVR	READ	Grant the server permission to perform authorization checks against the LOGSTRM profile in the SERVER class.
SERVER	BBG.SECCLASS.OPERCMDS	IZUSVR	READ	Grant the server permission to perform authorization checks against the OPERCMDS profile in the SERVER class.
SERVER	BBG.SECCLASS.RDATALIB	IZUSVR	READ	Grant the server permission to perform authorization checks against the RDATALIB profile in the SERVER class.
SERVER	BBG.SECCLASS.SERVAUTH	IZUSVR	READ	Grant the server permission to perform authorization checks against the SERVAUTH profile in the SERVER class.
SERVER	BBG.SECCLASS.SERVER	IZUSVR	READ	Grant the server permission to perform authorization checks against the SERVER profile in the SERVER class.
SERVER	BBG.SECCLASS.STARTED	IZUSVR	READ	Grant the server permission to perform authorization checks against the STARTED profile in the SERVER class.
SERVER	BBG.SECCLASS.TSOAUTH	IZUSVR	READ	Grant the server permission to perform authorization checks against the TSOAUTH profile in the SERVER class.
SERVER	BBG.SECCLASS.TSOPROC	IZUSVR	READ	Grant the server permission to perform authorization checks against the TSOPROC profile in the SERVER class.
SERVER	BBG.SECCLASS.UNIXPRIV	IZUSVR	READ	Grant the server permission to perform authorization checks against the UNIXPRIV profile in the SERVER class.
SERVER	BBG.SECCLASS.ZMFAPLA	IZUSVR	READ	Grant the server permission to perform authorization checks against the ZMFAPLA profile in the SERVER class.
SERVER	BBG.SECCLASS.ZMFCLD	IZUSVR	READ	Grant the server permission to perform authorization checks against the ZMFCLD profile in the SERVER class.
ZMFAPLA	<SAF-prefix>.ZOSMF.CONFIGURATION.SECURITY_ASSISTANT	IZUADMIN	READ	Allow the user to access the Security Configuration Assistant task. See Table Notes 1 and 2.

1. User authorizations to functions, tasks, and links are controlled through the system authorization facility (SAF) profile prefix. By default, the SAF prefix is IZUDFLT.
2. Users require READ access to at least the profile <SAF-prefix>.ZOSMF to do work in z/OSMF. Without this authorization, the user is treated as an authenticated guest. That is, the user can log in to z/OSMF and display the **Welcome** page, but cannot access the z/OSMF functions and tasks.

Resource authorizations for the z/OSMF core functions

Table 52 on page 376 describes the access requirements for the z/OSMF core functions. The IZUSEC job includes sample RACF commands for creating these authorizations on your system. These values can

change, based on the values you provide during the configuration process. [Table 52 on page 376](#) shows the IBM default values.

Table 52. Security setup requirements for z/OSMF core functions				
Resource class	Resource name	Who needs access?	Type of access required	Why
ACCTNUM	IZUACCT	IZUADMIN IZUUSER	READ	Allows callers to access the account number that is used for the procedure for the z/OSMF REST interfaces.
APPL	<SAF-prefix>	IZUADMIN IZUGUEST IZUUSER IZUSECAD	READ	Allow access to the z/OSMF application domain. If there is no matching profile in the APPL class, RACF allows the user to access the application.
CERT	DefaultzOSMFCert.<SAF-prefix>	Owned by the IZUSVR user ID	N/A	Needed for secure communications between the browser and the z/OSMF server.
CERT	zOSMFCA	N/A	N/A	Certificate authority that is needed for secure communications between the browser and the z/OSMF server.
CSFSERV	CSF* profiles	IZUSVR	READ	z/OS Integrated Cryptographic Service Facility (ICSF) callable services. If your installation uses hardware cryptography with ICSF, you must permit the z/OSMF server user ID to these services, as described in “Resource authorizations for hardware cryptography” on page 385 .
DATASET	your_stack_include_dataset	IZUSVR	ALTER	Allows the z/OSMF server to write to the configured include data sets when a network resource is provisioned or de-provisioned. There is one include data set per stack defined for IBM Cloud Provisioning. This definition is applicable only when your installation uses discrete or generic profiles to protect data set access.

Table 52. Security setup requirements for z/OSMF core functions (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
DATASET	<i>your_stack_dynamic_update_dataset</i>	IZUSVR	ALTER	Allows the z/OSMF server to write to the configured dynamic updates data sets when a network resource is provisioned or de-provisioned. One dynamic update data set per stack can be defined for IBM Cloud Provisioning. This definition is applicable only when your installation uses a discrete or generic profiles to protect data set access.
EJBROLE	<SAF-prefix>.IzuManagementFacility.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to log on to z/OSMF and view the Welcome page.
EJBROLE	<SAF-prefix>.IzuManagementFacilityHelpApp.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the z/OSMF online help system.
EJBROLE	<SAF-prefix>.IzuManagementFacilityImportUtility.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to use the Import Manager task to import services, event types, event handlers, and links into z/OSMF.
EJBROLE	<SAF-prefix>.IzuManagementFacilityRestConsoles.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the z/OS console REST interface.
EJBROLE	<SAF-prefix>.IzuManagementFacilityRestFiles.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the z/OS data set and file REST interface.
EJBROLE	<SAF-prefix>.IzuManagementFacilityRestJobs.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the z/OS jobs REST interface.
EJBROLE	<SAF-prefix>.IzuManagementFacilityTsoServices.izuUsers	IZUADMIN	READ	Allow the user of the Operator Consoles task to start or reconnect to address spaces on other systems in the sysplex.
EJBROLE	<SAF-prefix>.IzuManagementFacilityWorkflow.izuUsers	IZUADMIN IZUUSER IZUSECAD	READ	Allow the user to connect to the Workflows task.
EJBROLE	<SAF-prefix>.com.ibm.ws.management.security.resource.allAuthenticatedUsers	IZUADMIN IZUUSER	READ	Allow the user to display information about the IBM Cloud Provisioning and Management for z/OS REST APIs. For more information about the REST services, see IBM z/OS Management Facility Programming Guide.IBM z/OS Management Facility Programming Guide.
FACILITY	BBG.SYNC.<SAF-prefix>	IZUSVR	CONTROL	Allow the z/OSMF server to synchronize any RunAs identity with the OS identity.

Table 52. Security setup requirements for z/OSMF core functions (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
FACILITY	BPX.CONSOLE	IZUSVR	READ	Allow the user to filter z/OS UNIX messages. Specifically, this setting suppresses the BPXM023I message prefix from any write-to-operator (WTO) messages that z/OSMF writes to the console.
FACILITY	BPX.WLMSEVER	IZUSVR	READ	Allows the z/OSMF server to use WLM functions to create and manage work requests.
FACILITY	HWI.APPLNAME.HWISERV	IZUADMIN	READ	Grant the administrator groups access to BCPii services.
FACILITY	HWI.TARGET.<netid.nau>	IZUADMIN	READ	Allow the administrator to access the BCPii request type of CPC.
FACILITY	HWI.TARGET.<netid.nau>.<imagename>	IZUADMIN	READ	Allow the administrator to access the BCPii request type of LPAR.
FACILITY	IRR.DIGTCERT.LISTRING	IZUSVR	READ	Allow the started task user ID to list and get the certificate keyring.
FACILITY	IRR.RUSERMAP	IZUSVR	READ	Allow the started task user ID to use the R_usermap service. This authorization is required for the z/OSMF notification function. The z/OSMF server uses the R_usermap service to determine the application user identity associated with a RACF user ID, or to determine the RACF user ID associated with an application user identity or digital certificate.
KEYRING	IZUKeyring.<SAF-prefix>	IZUSVR	N/A	Needed for secure communications.
OPERCMDS	MVS.MCSOPER.IZU@*	IZUADMIN IZUUSER	READ	Allow the user to operate an extended MCS console.
OPERCMDS	MVS.VARY.TCPIP.OBEYFILE	IZUSVR	CONTROL	Allows the z/OSMF server to issue the VARY TCPIP OBEYFILE command for IBM Cloud Provisioning. This definition is applicable only when your installation utilizes the OPERCMDs class to restrict access to the VARY TCPIP OBEYFILE command.

Table 52. Security setup requirements for z/OSMF core functions (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
OPERCMDS	MVS.MCSOPER.ZCDPLM*	IZUSVR	READ	Allows the z/OSMF server to issue various operator commands for IBM Cloud Provisioning. The console name for this extended MCS console is the text string ZCDPLM, which is appended with the MVS sysclone value of the system of the z/OSMF instance.
OPERCMDS	MVS.DISPLAY.XCF	IZUSVR	READ	Allows the z/OSMF server to issue the DISPLAY XCF operator command for IBM Cloud Provisioning. This definition is applicable only when your installation utilizes the OPERCMDs class to restrict access to the DISPLAY XCF operator command.
OPERCMDS	MVS.ROUTE.CMD<sysname>	IZUSVR	READ	Allows the z/OSMF server to issue the ROUTE operator command for IBM Cloud Provisioning and Management for z/OS. This definition is applicable only if the installation uses this profile to restrict the use of the ROUTE command.
SERVAUTH	CEA.CEATSO.TSOREQUEST	IZUADMIN IZUUSER	READ	Allow the HTTP client applications on your z/OS system to start and manage TSO/E address spaces.
SERVAUTH	CEA.CEATSO.TSOREQUEST	IZUSVR	READ	Allow the z/OSMF server to start and manage TSO/E address space services.
SERVAUTH	CEA.SIGNAL.ENF86	IZUSVR (z/OSMF started task ID)	READ	Allow callers to access the CEA service responsible for signal event 86 across sysplex.
SERVAUTH	CEA.SIGNAL.ENF83	IZUSVR	READ	Allow the z/OSMF server to use ENF83 to indicate its status to other systems in the sysplex.
SERVAUTH	EZB.INITSTACK.sysname.tcpname	IZUSVR	READ	Allows the z/OSMF server to access the TCP/IP stack during TCP/IP initialization. This authorization is needed if the TCP/IP profile activates Application Transparent Transport Layer Security (AT-TLS).
SERVAUTH	EZB.NETWORKUTILS.CLOUD.mvsname	IZUSVR	READ	Allows the z/OSMF started task user ID issue operator commands for IBM Cloud Provisioning. <i>mvsname</i> is the name of the system on which the z/OSMF server is running.

Table 52. Security setup requirements for z/OSMF core functions (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
SERVAUTH	EZB.NETSTAT.<mvsnam>.<tcpnam>	IZUSVR	READ	Allows the z/OSMF started task user ID to issue the NETSTAT command. Otherwise, the z/OSMF server fails on initialization. This definition is applicable only when your installation has configured an AT-TLS policy.
SERVAUTH	EZB.NETSTAT.<mvsnam>.<tcpprocnam>.CONFIG	IZUSVR		Allows the Network Configuration Assistant task to issue the command NETSTAT CONFIG . This definition is applicable only when your installation uses the SERVAUTH class to restrict usage of the NETSTAT command. When this definition is applicable, IZUSVR must be authorized for each stack defined for IBM Cloud Provisioning and Management for z/OS.
SERVAUTH	EZB.NETSTAT.<mvsnam>.<tcpprocnam>.VIPADCFG	IZUSVR	READ	Allows the z/OSMF started task user ID to issue the NETSTAT VIPADCFG command. This definition is applicable only when your installation uses the SERVAUTH class to restrict usage of the NETSTAT command. When this definition is applicable, the z/OSMF started task user ID must be authorized for each stack that is defined for IBM Cloud Provisioning.
SERVER	BBG.ANGEL	IZUSVR	READ	Allow the z/OSMF server to access the angel process.
SERVER	BBG.ANGEL.IZUANG1	IZUSVR	READ	Allow the z/OSMF server to access the z/OSMF named angel process.
SERVER	BBG.ANGEL.proc-name	IZUSVR	READ	Allows the z/OSMF server to use z/OS authorized services.
SERVER	BBG.AUTHMOD.BBGZSAFM	IZUSVR	READ	Allow the z/OSMF server to access the SAF authorized registry.
SERVER	BBG.AUTHMOD.BBGZSAFM.SAFCRED	IZUSVR	READ	Allow the z/OSMF server to access the SAF authorization services.
SERVER	BBG.AUTHMOD.BBGZSAFM.TXRRS	IZUSVR	READ	Allow the z/OSMF server to access the transaction services.
SERVER	BBG.AUTHMOD.BBGZSAFM.ZOSDUMP	IZUSVR	READ	Allow the z/OSMF server to access the SVC dump services.

Table 52. Security setup requirements for z/OSMF core functions (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
SERVER	BBG.AUTHMOD.BBGZSAFM.ZOSWLM	IZUSVR	READ	Allow the z/OSMF server to access the WLM services.
SERVER	BBG.SECCLASS.ZMFAPLA	IZUSVR	READ	Allow the z/OSMF server to authorize checks for the ZMFAPLA class.
SERVER	BBG.SECPFX.<SAF-prefix>	IZUSVR	READ	Allow the z/OSMF server to make authentication calls against the APPL-ID.
STARTED	IZUINSTP.IZUINSTP	IZUADMIN	N/A	Defines the started task for the z/OSMF dependent address space, which is used to determine whether z/OS UNIX and TCP/IP are available. The job name must be IZUINSTP. Otherwise, the z/OSMF dependent address space is not initialized during z/OSMF autostart processing.
STARTED	IZUSVR1.jobname	IZUADMIN	N/A	Define the started task for the z/OSMF server process.
STARTED	IZUANG1.jobname	IZUADMIN	N/A	Define the started task for the z/OSMF angel process.
TSOAUTH	CONSOLE	IZUADMIN IZUUSER	READ	Allow the user to issue the TSO/E CONSOLE command to activate the extended MCS console.
TSOPROC	IZUFPROC	IZUADMIN IZUUSER	READ	Allows callers to access the procedure for the z/OSMF REST interfaces.
UNIXPRIV	SUPERUSER.FILESYS.CHOWN	IZUSRV	SUPERUSER	Allows file access to server controlled directories.
UNIXPRIV	SUPERUSER.FILESYS.CHOW	IZUSRV	SUPERUSER	Allows file access to server controlled directories.
ZMFAPLA	<SAF-prefix>.ZOSMF	IZUADMIN IZUGUEST IZUUSER IZUSECAD	READ	Designates the user as a z/OSMF user, rather than an unauthenticated guest user. This authorization is the minimum requirement for allowing a user to do more than log in to z/OSMF and view the Welcome page. Without this authorization, the logged-in user is treated as an authenticated guest. Use the other ZMFAPLA resource names that follow in this table to create specific controls for each core function and task. See Table Notes 1 and 2.
ZMFAPLA	<SAF-prefix>.ZOSMF.GENERAL.SETTINGS	IZUADMIN	READ	Allow the user to access the Task Settings task.

Table 52. Security setup requirements for z/OSMF core functions (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF-prefix>.ZOSMF.ADMINTASKS.APPLINKING	IZUADMIN	READ	Allow the user to access the Application Linking Manager task.
ZMFAPLA	<SAF-prefix>.ZOSMF.ADMINTASKS.DIAGNOSTIC_ASSISTANT	IZUADMIN	READ	Allow the user to access the z/OSMF Diagnostic Assistant task.
ZMFAPLA	<SAF-prefix>.ZOSMF.ADMINTASKS.IMPORTMANAGER	IZUADMIN	READ	Allow the user to access the Import Manager task.
ZMFAPLA	<SAF-prefix>.ZOSMF.ADMINTASKS.LINKSTASK	IZUADMIN	READ	Allow the user to access the Links task.
ZMFAPLA	<SAF-prefix>.ZOSMF.ADMINTASKS.LOGGER	IZUADMIN	READ	Allow the user to manage the settings that control the behavior and content of the z/OSMF logs. This capability is used only in service situations.
ZMFAPLA	<SAF-prefix>.ZOSMF.ADMINTASKS.UI_LOG_MANAGEMENT	IZUADMIN	READ	Allow the user to manage the settings that control the behavior of the user interface (UI) portion of z/OSMF logging. This capability is used only in service situations.
ZMFAPLA	<SAF-prefix>.ZOSMF.ADMINTASKS.USAGESTATISTICS	IZUADMIN	READ	Allow the user to collect usage statistics about z/OSMF.
ZMFAPLA	<SAF-prefix>.ZOSMF.LINK.linkName	IZUADMIN IZUUSER	READ	Allow the user to view an installation-specified link. See Table Notes 3 and 4.
ZMFAPLA	<SAF-prefix>.ZOSMF.LINK.SHOPZSERIES	IZUADMIN IZUUSER	READ	Allow the user to view the ShopzSeries web site link.
ZMFAPLA	<SAF-prefix>.ZOSMF.LINK.SUPPORT_FOR_Z_OS	IZUADMIN IZUUSER	READ	Allow the user to view the Support for z/OS web site link.
ZMFAPLA	<SAF-prefix>.ZOSMF.LINK.SYSTEM_Z_REDBOOKS	IZUADMIN IZUUSER	READ	Allow the user to view the IBM Redbooks® web site link.
ZMFAPLA	<SAF-prefix>.ZOSMF.LINK.WSC_FLASHES_TCHDOCS	IZUADMIN IZUUSER	READ	Allow the user to view the WSC Flashes and Techdocs web site link.
ZMFAPLA	<SAF-prefix>.ZOSMF.LINK.Z_OS_BASICS_INFORMATION_CENTER	IZUADMIN IZUUSER	READ	Allow the user to view the z/OS Basic Skills Information Center web site link.
ZMFAPLA	<SAF-prefix>.ZOSMF.LINK.Z_OS_HOME_PAGE	IZUADMIN IZUUSER	READ	Allow the user to view the z/OS Home Page web site link.
ZMFAPLA	<SAF-prefix>.ZOSMF.LINK.Z_OS_INTERNET_LIBRARY	IZUADMIN IZUUSER	READ	Allow the user to view the z/OS Library web site link.
ZMFAPLA	<SAF-prefix>.ZOSMF.NOTIFICATION.MODIFY	IZUADMIN IZUUSER	READ	Allow the user to compose a notification.

Table 52. Security setup requirements for z/OSMF core functions (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF-prefix>.ZOSMF.NOTIFICATION.SETTINGS	IZUADMIN IZUUSER	READ	Allow the user to define an mail account for receiving notifications from z/OSMF. This action is performed through the Notification Settings task of z/OSMF.
ZMFAPLA	<SAF-prefix>.ZOSMF.NOTIFICATION.SETTINGS.ADMIN	IZUADMIN	READ	Allow the user to access the Notification Settings task of z/OSMF
ZMFAPLA	<SAF-prefix>.ZOSMF.SEND.IBM.FEEDBACK	IZUADMIN IZUUSER	READ	Allow the user to send feedback data to IBM by using the Provide IBM Feedback option in the z/OSMF desktop.
ZMFAPLA	<SAF-prefix>.ZOSMF.SETTINGS.FTP_SERVERS	IZUADMIN IZUUSER	READ	Allow the user to access the FTP Servers task.
ZMFAPLA	<SAF-prefix>.ZOSMF.SETTINGS.FTP_SERVERS.VIEW	IZUADMIN IZUUSER	READ	Allow the user to access the FTP Servers task <i>View</i> function.
ZMFAPLA	<SAF-prefix>.ZOSMF.SETTINGS.FTP_SERVERS.MODIFY	IZUADMIN	READ	Allow the user to access the z/OSMF Task Settings task <i>Modify</i> function.
ZMFAPLA	<SAF-prefix>.ZOSMF.SETTINGS.SYSTEMS	IZUADMIN IZUUSER	READ	Allow the user to access the Systems task.
ZMFAPLA	<SAF-prefix>.ZOSMF.SETTINGS.SYSTEMS.AES.MODIFY	IZUADMIN	READ	Allow the user to enable or disable AES encryption for the LTPA password.
ZMFAPLA	<SAF-prefix>.ZOSMF.SETTINGS.SYSTEMS.VIEW	IZUADMIN IZUUSER	READ	Allow the user to access the Systems task <i>View</i> function.
ZMFAPLA	<SAF-prefix>.ZOSMF.SETTINGS.SYSTEMS.MODIFY	IZUADMIN	READ	Allow the user to access the z/OSMF Task Settings task <i>Modify</i> function.
ZMFAPLA	<SAF-prefix>.ZOSMF.VARIABLES.SYSTEM.ADMIN	IZUADMIN	READ	Allows the user to access the system variables in the Systems task.
ZMFAPLA	<SAF-prefix>.ZOSMF.WORKFLOW.ADMIN	IZUADMIN	READ	Allow the user to change the assigned owner of a workflow.
ZMFAPLA	<SAF-prefix>.ZOSMF.WORKFLOW.EDITOR	IZUADMIN IZUUSER	READ	Allow the user to access the Workflow Editor task.
ZMFAPLA	<SAF-prefix>.ZOSMF.WORKFLOW.RUNASUSER	IZUUSER	READ	Allow the user to be defined as the runAsUser ID in a workflow instance that does not originate from z/OS Management Services Catalog or IBM Cloud Provisioning and Management for z/OS.
ZMFAPLA	<SAF-prefix>.ZOSMF.WORKFLOW.SIGNER	IZUADMIN	READ	Allow the user to be granted the runAsUser step signer role.

Table 52. Security setup requirements for z/OSMF core functions (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF-prefix>.ZOSMF.WORKFLOW.WORKFLOWS	IZUADMIN IZUSECAD IZUUSER	READ	Allow the user to access the z/OSMF Workflows task. See Table Note 5.

1. User authorizations to functions, tasks, and links are controlled through the system authorization facility (SAF) profile prefix. By default, the SAF prefix is IZUDFLT.
2. Users require READ access to at least the profile <SAF-prefix>.ZOSMF to do work in z/OSMF. Without this authorization, the user is treated as an authenticated guest. That is, the user can log in to z/OSMF and display the Welcome page, but cannot access the z/OSMFz/OSMF functions and tasks.
3. In a default z/OSMF configuration, all users are granted authority to all links through a wildcarded profile: <SAF-prefix>.ZOSMF.LINK.* *
4. You must provide a SAF resource name prefix for any links that you add to z/OSMF. You can control access to specific links by specifying a unique resource name for the link, for example, by including the link name as part of the resource name. For example: IZUDFLT.ZOSMF.LINK.mylink
For more information about defining links to z/OSMF, see [Chapter 49, “Adding links to z/OSMF,”](#) on page 257.
5. A user with access to the Workflows task can access any of the workflows that are displayed in the Workflows task. By default, the z/OSMF defined security groups IZUADMIN, IZUSECAD, and IZUUSER have access to the Workflows task.
6. If your installation uses hardware cryptography with z/OS Integrated Cryptographic Service Facility (ICSF), be aware that services such as CSFRNGL, CSFDSV, CSFOWH, CSFIQF, and others, might be protected through profiles that are established in your external security manager, such as RACF. In some cases, z/OSMF uses these services; therefore, you must permit the z/OSMF started task user ID to these profiles. For more information, see [“Resource authorizations for hardware cryptography”](#) on page 385.
7. All z/OSMF users must have a TSO segment that is defined in your installation’s security database. Failure to have a TSO segment causes some z/OSMF functions not to work.

Resource authorizations for hardware compression

If your installation uses IBM zEnterprise® Data Compression (zEDC), the z/OSMF server requires READ access to the FPZ.ACCELERATOR.COMPRESSION resource in the FACILITY class. Otherwise, if this authorization is not in place, the z/OSMF server runs without the use of zEDC. The system issues an error message, such as the following:

```
XAT1 IZUSVRU IZUSVR1 RACF ACCESS violation for IZUSVRU:
(READ,NONE) on FACILITY FPZ.ACCELERATOR.COMPRESSION
```

You can ignore the message.

Table 53 on page 384 shows which permissions must be granted to the z/OSMF server user ID. Commands for the creating the permissions are included in commented sections in the IZUSEC job. To issue the commands when the job runs, uncomment the sections.

Table 53. Security setup requirements for IBM zEnterprise Data Compression (zEDC)

Resource class	Resource name	Who needs access?	Type of access required	Why
FACILITY	FPZ.ACCELERATOR.COMPRESSION	IZUSVR	READ	Enable the z/OSMF server to run with IBM zEnterprise Data Compression (zEDC).

Resource authorizations for hardware cryptography

If your installation uses hardware cryptography with z/OS Integrated Cryptographic Service Facility (ICSF), the z/OSMF server requires access to the ICSF callable services. Table 54 on page 385 shows which permissions must be granted to the z/OSMF server user ID. Commands for the creating the permissions are included in commented sections in the IZUSEC job. To issue the commands when the job runs, uncomment the sections.

Table 54. Security setup requirements for hardware cryptography with ICSF				
Resource class	Resource name	Who needs access?	Type of access required	Why
CSFSERV	CSFIQF	IZUSVR	READ	ICSF query facility callable service.
CSFSERV	CSFENC	IZUSVR	READ	Encipher callable service.
CSFSERV	CSFCVE	IZUSVR	READ	Cryptographic variable encipher callable service.
CSFSERV	CSFDEC	IZUSVR	READ	Decipher callable service.
CSFSERV	CSFSAE	IZUSVR	READ	Symmetric algorithm encipher callable service.
CSFSERV	CSFSAD	IZUSVR	READ	Symmetric algorithm decipher callable service.
CSFSERV	CSFOWH	IZUSVR	READ	One-way hash generate callable service.
CSFSERV	CSFRNG	IZUSVR	READ	Random number generate callable service.
CSFSERV	CSFRNGL	IZUSVR	READ	Random number generate long callable service.
CSFSERV	CSFPKG	IZUSVR	READ	PKA key generate callable service.
CSFSERV	CSFDSG	IZUSVR	READ	Digital signature generate service.
CSFSERV	CSFDSV	IZUSVR	READ	Digital signature verify callable service.
CSFSERV	CSFPKT	IZUSVR	READ	PKA key generate callable service.
CSFSERV	CSFRKL	IZUSVR	READ	Retained key list callable service.
CSFSERV	CSFPKX	IZUSVR	READ	PKA Public Key Extract callable service.
CSFSERV	CSFPKE	IZUSVR	READ	PKA encrypt callable service.
CSFSERV	CSFPKD	IZUSVR	READ	PKA decrypt callable service.
CSFSERV	CSFPKI	IZUSVR	READ	PKA key import callable service.
CSFSERV	CSFCKM	IZUSVR	READ	Multiple clear key import callable service.
CSFSERV	CSFKGN	IZUSVR	READ	Multiple clear key import callable service.
CSFSERV	CSFEDH	IZUSVR	READ	ECC Diffie-Hellman callable service.

Resource authorizations for Common Information Model

If your z/OSMF configuration includes tasks that use the Common Information Model (CIM) server on the host z/OS system, users of the services require the proper level of access to CIM server resources.

These authorizations are required for using any of the following optional services or core functions:

- Capacity Provisioning
- Incident Log

CIM includes the CFZSEC job to help you create these authorizations. See the topic on CIM server quick setup and verification in *z/OS Common Information Model User's Guide*. IBM supplies the CFZSEC job in SYS1.SAMPLIB. If your installation does not plan to run the CFZSEC job, ensure that z/OSMF users have UPDATE access to the CIMSERV profile in the WBEM class. If necessary, refresh the WBEM class.

For more information about CIM authorization requirements, see [Chapter 47, “Configuring the CIM server for your system,”](#) on page 253.

Table 55 on page 386 lists the CIM security groups that are required for the optional services.

Table 55. CIM groups that might be required for the optional services			
Group	Purpose	Default group ID (GID)	Created by
CFZADMGP	Security group for the CIM administrator role.	9502	Member CFZSEC in SYS1.SAMPLIB.
CFZUSRGP	Security group for the CIM user role. This group grants a user access to all resources that are managed through CIM. Depending on how granular you want to control user access to CIM, your installation might have created more groups to allow access to only a subset of resources that are managed through CIM.	9503	Member CFZSEC in SYS1.SAMPLIB.

With the IZUAUTH job, your security administrator can supply the names of the CIM groups, based on your selection of optional services. These values include the names of the CIM administrators group (by default, CFZADMGP) and the CIM users group (by default, CFZUSRGP). The IZUAUTH job contains commands for connecting users to the groups and thus, depend on the groups to exist.

Resource authorizations for Capacity Provisioning Manager

If your z/OSMF configuration includes the Capacity Provisioning service, users of the service must be defined and authorized for all resources that are accessed by the Provisioning Manager. IBM provides the CPOSEC1 and CPOSEC2 jobs in SYS1.SAMPLIB to help you create these authorizations when you set up a Capacity Provisioning domain. For more information, see the topic on setting up a Capacity Provisioning domain in [z/OS MVS Capacity Provisioning User's Guide](#).

Table 56 on page 386 lists the default values for the Provisioning Manager. Your installation might have selected different values for these settings.

Table 56. Name information for a Capacity Provisioning domain	
Provisioning Manager setting	Default value
Domain name	DOMAIN1
Started task procedure name	CPOSERV
High-level qualifier for runtime data set	CPO
Provisioning Manager user	CPOSRV

With the IZUCPSEC job, your security administrator can supply the names of the security groups that your installation created for authorizing users to the Provisioning Manager on your system. The IZUAUTH job contains commands for connecting users to the groups and thus, depend on the groups to exist.

Table 57 on page 386 lists the security groups that are required for the Capacity Provisioning service.

Table 57. Security groups required for the Capacity Provisioning service			
Group	Purpose	Default group ID (GID)	Created by
CPOCTRL	Security group for users of the Capacity Provisioning task <i>Edit</i> function.	None; your installation must specify a GID for this group.	Member CPOSEC1 in SYS1.SAMPLIB.

Table 57. Security groups required for the Capacity Provisioning service (continued)

Group	Purpose	Default group ID (GID)	Created by
CPOQUERY	Security group for users of the Capacity Provisioning task <i>View</i> function.	None; your installation must specify a GID for this group.	Member CPOSEC1 in SYS1.SAMPLIB.

Resource authorizations for common event adapter (CEA)

If your z/OSMF configuration includes tasks that use the common event adapter (CEA) component on the z/OS host system, users of the services require the proper level of access to CEA resources. IBM provides the CEASEC job in SYS1.SAMPLIB to help you create these authorizations.

These authorizations are needed if you plan to use one or more of the following z/OSMF tasks:

- Incident Log
- ISPF
- Sysplex Management

CEA has security profiles in the SERVAUTH class for protecting different portions of its processing. When you run the IZUILSEC job, you permit the z/OSMF groups to the CEA resources.

For more information, see the topic on customizing for CEA in [z/OS Planning for Installation](#).

Resource authorizations for the z/OS compliance REST interface

In z/OS V2R4 and later, the process of collecting compliance data is assisted with the introduction of SMF type 1154. This record type is used to collect system settings and other forms of compliance data. On receiving an event notification facility (ENF) code 86 signal from the z/OS compliance REST interface, selected z/OS components and products collect and write compliance data to their associated SMF 1154 subtype records.

For more information about SMF record type 1154 and its associated mapping macros and subtypes, see [z/OS MVS System Management Facilities \(SMF\)](#).

Table 58. Security setup requirements for the z/OS compliance REST interface

Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF-prefix>. IzuManagementFacilityRestCompliance. izuUsers	IZUADMIN IZUUSER	READ	Allows callers to connect to the z/OS compliance REST interface.
SERVAUTH	CEA.SIGNAL.ENF86	IZUSVR started task ID	READ	Allow callers to access the CEA service responsible for the signal event 86 across the sysplex.

Resource authorizations for the z/OS console REST interface

In z/OSMF, users require access to z/OS console services when they use the following functions:

- z/OS console REST interface
- z/OS Operator Consoles

- IBM Cloud Provisioning and Management for z/OS, when using templates that issue operator commands or check for unsolicited command responses.
- Storage management services, when activating an SCDS or getting an SCDS activation result.

Users of the z/OS console REST interface require access to an extended MCS (EMCS) console for issuing commands and receiving console messages. Specifically, users require the following authorizations:

- READ access to the MVS.MCSOPER.*consolename* resource in the OPERCMDS class, where *consolename* is the name of the EMCS console that is used to issue the command.
- READ access to the CONSOLE resource in the TSOAUTH class.
- READ access to the <SAF_PREFIX>.IzuManagementFacilityRestConsoles.izuUsers resource in the EJBROLE class. Or, READ access to the <SAF_PREFIX>.*.izuUsers profile in the EJBROLE class.

z/OSMF uses TSO/E address space services to create a TSO address space as the host for the EMCS console. Therefore, users of the z/OS console REST interface require the following authorizations:

- READ access to the resource *account* in the ACCTNUM class, where *account* is the value that is specified in the COMMON_TSO ACCT option in parmlib member IZUPRMxx.
- READ access to the resource CEA.CEATSO.TSOREQUEST in the SERVAUTH class.
- READ access to the resource *proc* in the TSOPROC class, where *proc* is the value that is specified with the COMMON_TSO PROC option in parmlib member IZUPRMxx.

Also, the z/OSMF started task user ID, which is IZUSVR by default, requires READ access to the resource CEA.CEATSO.TSOREQUEST in the SERVAUTH class.

You can control which parameters are used for creating the TSO address space by setting the appropriate parameters in parmlib member IZUPRMxx. For example:

```
COMMON_TSO ACCT(IZUACCT) REGION(500000) PROC(IZUPPROC)
```

Ensure that your settings are configured before the z/OS console REST interface is used. Otherwise, the default values (shown here) are used.

The attributes of the EMCS console that is started by z/OSMF are controlled by the OPERPARM settings of the user profile <*consolename*>. Thus, for example, if a user wants the z/OS Operator Consoles task to create a console named console1, a user profile named console1 must exist and contain an OPERPARM segment with the appropriate settings.

Most IBM Cloud Provisioning and Management for z/OS templates use the *defcn* Console REST API endpoint, which expects a predefined console name. The convention is to use *userid* plus "CN", where the value for *userid* is truncated to the first six characters. For example, if the user ID is IBMUSER, the *defcn* value is expected to be IBMUSECN.

Typically, z/OSMF uses the following console attributes from the user's OPERPARM segment:

AUTH

Specifies the command authority for the console.

ROUTE

Specifies the routing codes for the console, which affects which messages can be received by the console. The default value is NONE, which prevents the console from receiving any messages.

MSCOPE

Specifies the system message scope in the sysplex.

For more information about setting these attributes, see the commented sections in SAMPLIB jobs IZUGCSEC and IZUPRSEC. For information about creating OPERPARM segments for users, see [z/OS MVS Planning: Operations](#).

In addition to the local system (the system on which z/OSMF is installed), users can enter system commands on other systems in the sysplex. To do so, users require READ access to the resource MVS.ROUTE.CMD.<*sysname*> in the OPERCMDS class.

Users can retrieve messages from OPERLOG or SYSLOG. To do so, users require the following authorizations:

- To retrieve messages from OPERLOG, users require READ access to the resource SYSPLEX.OPERLOG in the LOGSTRM class.
- To retrieve messages from SYSLOG, users require READ access to the resource *node-id*.+MASTER+.SYSLOG.*.* in the JESSPOOL class, where *node-id* is the NJE node ID of the JES2 or JES3 subsystem.

Table 59 on page 389 summarizes the security requirements for users of the z/OS console REST interface. IBM provides job IZUGCSEC in SYS1.SAMPLIB to assist you with performing these updates. The job contains RACF commands for creating the required security authorizations.

Table 59. Security setup requirements for the z/OS console REST interface				
Resource class	Resource name	Who needs access?	Type of access required	Why
N/A	User profile <consolename> with the appropriate OPERPARM segment.	N/A	N/A	The attributes of the EMCS console that is started by the z/OS Operator Consoles task are controlled by the OPERPARM setting of user profile <consolename>. The setting of OPERPARM can restrict which messages are received by the EMCS console and limit the commands that the EMCS console can issue.
ACCTNUM	IZUACCT	Users of the z/OS console services REST interface.	READ	Allow the user to access the account number for the procedure for the z/OS console services.
EJBROLE	<SAF-prefix>.IzuManagementFacilityRestConsoles.i zuUsers	Users of: • z/OS console services • z/OS Operator Consoles task.	READ	Allow the user to use the z/OS console services to issue operator commands.
EJBROLE	<SAF-prefix>.IzuManagementFacilityTsoServices.i zuUsers	IZUADMIN	READ	Allow the user of the Operator Consoles task to start or reconnect to address spaces on other systems in the sysplex.
JESSPOOL	<i>node-id</i> .+MASTER+.SYSLOG.*.*	Users of the z/OS Operator Consoles task.	READ	Allows the user to retrieve messages from SYSLOG by using the z/OS Operator Consoles task. <i>node-id</i> is the NJE node ID of the JES2 or JES3 subsystem.
LOGSTRM	SYSPLEX.OPERLOG	Users of the z/OS Operator Consoles task.	READ	Allows the user to retrieve messages from OPERLOG by using the z/OS Operator Consoles task.
OPERCMD5	MVS.MCSOPER.<consolename>	Users of the z/OS console services REST interface.	READ	Allow the user to operate the specified extended MCS console.
OPERCMD5	MVS.ROUTE.CMD.<sysname>	Users of the z/OS Operator Consoles task.	READ	Allows the user to use the ROUTE command to route commands to another system in sysplex, which is indicated by <i>sysname</i> . Otherwise, the user is limited to entering commands on the local system (the system on which z/OSMF is installed).
SERVAUTH	CEA.CEATSO.TSOREQUEST	Users of the z/OS console services REST interface.	READ	Allow the user to access the CEA TSO/E address space services. This setting allows HTTP client applications on your z/OS system to start and manage TSO/E address spaces.

Table 59. Security setup requirements for the z/OS console REST interface (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
SERVAUTH	CEA.CEATSO.TSOREQUEST	IZUSVR	READ	Allows the z/OSMF server to access the CEA TSO/E address space services. This setting allows the z/OSMF server to start and manage TSO/E address space services.
TSOAUTH	CONSOLE	Users of the z/OS console services REST interface.	READ	Allow the user to issue the TSO/E CONSOLE command to activate the extended MCS console.
TSOPROC	IZUFPROC	IZUADMIN IZUUSER	READ	Allow the user to access the procedure for the z/OS console services.
ZMFAPLA	<SAF-prefix>.ZOSMF.CONSOLES. ZOSOPER	Users of the z/OS Operator Consoles task.	READ	Allows the user to view and access the z/OS Operator Consoles task in the z/OSMF desktop interface.

Resource authorizations for the z/OS data set and file REST interface

The z/OS data set and file REST interface requires access to local resources on your z/OS system. [Table 60 on page 390](#) describes the security requirements for the z/OS data set and file REST interface.

For more information about the z/OS data set and file REST interface services, see [IBM z/OS Management Facility Programming Guide](#).

Table 60. Security setup requirements for the z/OS data set and file REST interface

Resource class	Resource name	Who needs access?	Type of access required	Why
ACCTNUM	IZUACCT	IZUADMIN IZUUSER	READ	Allows callers to access the account number that is used for the procedure for the z/OS data set and file REST interface services.
EJBROLE	<SAF-prefix>.IzuManagementFacilityRestFiles.izuUsers	IZUADMIN IZUUSER	READ	Allows callers to connect to the z/OS data set and file REST interface.
SERVAUTH	CEA.CEATSO.TSOREQUEST	IZUADMIN IZUUSER	READ	Allows callers to access the CEA TSO/E address space services. This setting allows HTTP client applications on your z/OS system to start and manage TSO/E address spaces.
SERVAUTH	CEA.CEATSO.TSOREQUEST	IZUSVR	READ	Allows the z/OSMF server to access the CEA TSO/E address space services. This setting allows the z/OSMF server to start and manage TSO/E address space services.
TSOPROC	IZUFPROC	IZUADMIN IZUUSER	READ	Allows callers to access the procedure for the z/OS data set and file REST interface services.

Resource authorizations for the z/OS jobs REST interface

For programs that use the z/OS jobs REST interface services to perform job modify operations, the caller's user ID must be authorized to the appropriate resources in the JESJOBS class, as shown in [Table 61 on page 391](#).

Table 61. JESJOBS class authorizations needed for performing job modify operations

Operation	JESJOBS resource	Access required
Hold a job	HOLD.nodename.userid.jobname	UPDATE
Release a job	RELEASE.nodename.userid.jobname	UPDATE
Change the job class	MODIFY.nodename.userid.jobname	UPDATE
Cancel a job	CANCEL.nodename.userid.jobname	ALTER
Delete a job (cancel a job and purge its output)	CANCEL.nodename.userid.jobname	ALTER

For more information about the z/OS jobs REST interface services, see [IBM z/OS Management Facility Programming Guide](#).

If run asynchronously, the z/OS jobs REST interface services also require that the caller's user ID is authorized to the CIM server and permitted to the JES2-JES3Jobs CIM provider. CIM includes jobs (CFZSEC and CFZRCUST) to help you configure the CIM server, including security authorizations and file system customization. For more information, see the topic on CIM server quick setup and verification in [z/OS Common Information Model User's Guide](#). IBM supplies the CFZSEC job in SYS1.SAMPLIB.

Resource authorizations for the Capacity Provisioning service

The Capacity Provisioning service requires access to local resources on your z/OS system. [Table 62 on page 391](#) describes the security requirements for the Capacity Provisioning service. The IZUCPSEC job includes sample RACF commands for creating these authorizations.

Table 62. Security setup requirements for the Capacity Provisioning service				
Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF-prefix>.IzuManagementFacilityCapacityProvisioning.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the Capacity Provisioning task.
ZMFAPLA	<SAF-prefix>.ZOSMF.CAPACITY_PROVISIONING.CAPACITY_PROVISIONING.EDIT	IZUADMIN	READ	Allow the user to display and access the Capacity Provisioning task <i>Edit</i> function.
ZMFAPLA	<SAF-prefix>.ZOSMF.CAPACITY_PROVISIONING.CAPACITY_PROVISIONING.EDIT.DOMAIN	IZUADMIN	READ	Allow the user to use the Capacity Provisioning task <i>Edit</i> function to edit a Capacity Provisioning domain.

Table 62. Security setup requirements for the Capacity Provisioning service (continued)				
Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF-prefix>.ZOSMF.CAPACITY_PROVISIONING.CAPACITY_PROVISIONING.EDIT.POLICY	IZUADMIN	READ	Allow the user to use the Capacity Provisioning task <i>Edit</i> function to edit a Capacity Provisioning policy.
ZMFAPLA	<SAF-prefix>.ZOSMF.CAPACITY_PROVISIONING.CAPACITY_PROVISIONING.VIEW	IZUADMIN IZUUSER	READ	Allow the user to access the Capacity Provisioning task <i>View</i> function.

More authorizations are required as follows:

- The Capacity Provisioning service requires the CIM server; thus, you must also create the authorizations that are described in [“Resource authorizations for Common Information Model” on page 385](#).
- Users of the Capacity Provisioning service must be authorized for resources that are accessed by the Provisioning Manager. IBM provides the CPOSEC1 and CPOSEC2 jobs in SYS1.SAMPLIB to help you create these authorizations. For more information, see the topic on setting up a Capacity Provisioning domain in [z/OS MVS Capacity Provisioning User's Guide](#).

Resource authorizations for the Network Configuration Assistant service

The Network Configuration Assistant service requires access to local resources on your z/OS system. [Table 63 on page 392](#) describes the security requirements for the Network Configuration Assistant service. The IZUCASEC job includes sample RACF commands for creating these authorizations.

Table 63. Security setup requirements for the Network Configuration Assistant service				
Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF-prefix>.IzuConfigurationAssistant.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the Network Configuration Assistant task.
ZMFAPLA	<SAF-prefix>.ZOSMF.CONFIGURATION_ASSISTANT.CONFIGURATION_ASSISTANT	IZUADMIN IZUUSER	READ	Allow the user to access the Network Configuration Assistant task.

Resource authorizations for the Incident Log service

The Incident Log service requires access to local resources on your z/OS system. Table 64 on page 393 describes the security requirements for the Incident Log service. The IZUILSEC job includes sample RACF commands for creating these authorizations.

Table 64. Security setup requirements for the Incident Log service

Resource class	Resource name	Who needs access?	Type of access required	Why
ALIAS	CEA	N/A	N/A	If your installation has a user catalog set-up instead of using the master catalog, you might need to define CEA alias to the user catalog.
DATASET	CEA.*	IZUADMIN IZUUSER	ALTER	Allow the user to create data sets with the CEA high-level qualifier (HLQ).
DATASET	<i>your_master_catalog</i>	IZUADMIN IZUUSER	UPDATE	If your installation has master catalog setup, you might need to permit a user to the master catalog data set class.
EJBROLE	<SAF- <i>prefix</i> >.IzuManagementFacilityIncidentLog.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the Incident Log task.

Table 64. Security setup requirements for the Incident Log service (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
JESSPOOL	<i>node-id</i> .+MASTER+.SYSLOG.**	CEA	READ	If your installation is using the system log (SYSLOG) as the source for diagnostic log snapshots, the CEA user ID requires READ access to the JESSPOOL class. This authorization allows the JES subsystem to access SYSLOG on behalf of the common event adapter (CEA) component. <i>node-id</i> is the NJE node ID of the JES2 or JES3 subsystem.
OPERCMDS	MVS.SETLOGR.LOGR	IZUADMIN IZUUSER	READ	Allow the user to issue the SETLOGR command which Incident log will issue via CEA to delete an Incident.
SERVAUTH	CEA.CEADOCONSOLECMD	IZUADMIN IZUUSER	READ	Allow the calling program to issue operator commands to accomplish its function.
SERVAUTH	CEA.CEADOCMD	IZUADMIN IZUUSER	READ	Allow the user to cancel the FTP job.

Table 64. Security setup requirements for the Incident Log service (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
SERVAUTH	CEA.CEAGETPS	IZUADMIN IZUUSER	READ	Allow the user to obtain information about the FTP job.
SERVAUTH	CEA.CEAPDWB.CEACHECKSTATUS	IZUADMIN IZUUSER	READ	Allow the user to check status and return incident information.
SERVAUTH	CEA.CEAPDWB.CEDELETEINCIDENT	IZUADMIN IZUUSER	READ	Allow the user to delete selected incidents, including the dumps, all diagnostic snapshot files, and the corresponding sysplex dump directory entry.
SERVAUTH	CEA.CEAPDWB.CEGETINCIDENT	IZUADMIN IZUUSER	READ	Allow the user to obtain data that is associated with a specific incident.
SERVAUTH	CEA.CEAPDWB.CEGETINCIDENTCOLLECTION	IZUADMIN IZUUSER	READ	Allow the user to obtain collection of incident data for all incidents that match a filter.
SERVAUTH	CEA.CEAPDWB.CEAPREPAREINCIDENT	IZUADMIN IZUUSER	READ	Allow the user to prepare data for FTP (locate and compress/terse).
SERVAUTH	CEA.CEAPDWB.CESETINCIDENTINFO	IZUADMIN IZUUSER	READ	Allow the user to set information that is associated with the incident, such as the Notes field.

Table 64. Security setup requirements for the Incident Log service (continued)				
Resource class	Resource name	Who needs access?	Type of access required	Why
SERVAUTH	CEA.CEAPDWB.CEASETPROBLEMTRACKINGNUMBER	IZUADMIN IZUUSER	READ	Allow the user to set a problem ID, such as a PMR number, or problem management tracking ID.
SERVAUTH	CEA.CEAPDWB.CEAUNSUPPRESSDUMP	IZUADMIN IZUUSER	READ	Allow user to allow a dump that is marked for suppression through DAE to be taken.
ZMFAPLA	<SAF-prefix>.ZOSMF.INCIDENT_LOG.INCIDENT_LOG	IZUADMIN IZUUSER	READ	Allow the user to access the Incident Log task.

Additional authorizations are required as follows:

- The Incident Log service requires the CIM server; thus, you must also create the authorizations that are described in [“Resource authorizations for Common Information Model”](#) on page 385.
- Users of the Incident Log service must be authorized for resources that are accessed by the common event adapter (CEA) component of z/OS. IBM provides the CEASEC job in SYS1.SAMPLIB to help you create these authorizations. For more information, see [“Resource authorizations for common event adapter \(CEA\)”](#) on page 387.

Resource authorizations for the ISPF service

The ISPF service requires access to local resources on your z/OS system. Table 65 on page 396 describes the security requirements for the ISPF service. The IZUISSEC job includes sample RACF commands for creating these authorizations.

Note that users of this service must also be authorized for resources that are accessed by the common event adapter (CEA) component of z/OS. IBM provides the CEASEC job in SYS1.SAMPLIB to help you create these authorizations. See [“Resource authorizations for common event adapter \(CEA\)”](#) on page 387.

Table 65. Security setup requirements for the ISPF service				
Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF-prefix>.IzuManagementFacilityISPF.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the ISPF task.

Table 65. Security setup requirements for the ISPF service (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
SERVAUTH	CEA.CEATSO.TSOREQUEST	IZUADMIN IZUUSER	READ	Allow the user to access the CEATSOREQUEST API so that the user's session can be managed through the ISPF task.
SERVAUTH	CEA.CEATSO.TSOREQUEST	IZUSVR	READ	Allow the z/OSMF server to access the CEATSOREQUEST API.
ZMFAPLA	<SAF-prefix>.ZOSMF.ISPF.ISPF	IZUADMIN IZUUSER	READ	Allow the user to access the ISPF task.

Resource authorizations for the Resource Monitoring service

The Resource Monitoring service requires access to local resources on your z/OS system. Table 66 on page 397 describes the security requirements for the Resource Monitoring service. The IZURMSEC job includes sample RACF commands for creating these authorizations.

Table 66. Security setup requirements for the Resource Monitoring service

Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF-prefix>.IzuManagementFacilityResourceMonitoring.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the Resource Monitoring and System Status tasks.
ZMFAPLA	<SAF-prefix>.ZOSMF.RESOURCE_MONITORING.PERFDESKS	IZUADMIN IZUUSER	READ	Allow the user to access the Resource Monitoring task.
ZMFAPLA	<SAF-prefix>.ZOSMF.RESOURCE_MONITORING.OVERVIEW	IZUADMIN IZUUSER	READ	Allow the user to access the System Status task.

Resource authorizations for the Software Deployment service

The Software Deployment service requires access to local resources on your z/OS system. Table 67 on page 398 describes the security requirements for the service. The IZUDMSEC job includes sample RACF commands for creating these authorizations.

Table 67. Security setup requirements for the Software Deployment service				
Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF- <i>prefix</i> >.IzuManagementFacilitySoftwareDeployment.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the Software Management task.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT	IZUADMIN IZUUSER	READ	Allow the user to access the Software Management task.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.SOFTWARE_DEPLOYMENT.DATA. <i>objectType</i> . <i>objectSuffix</i> For more information about the possible values for <i>objectType</i> and <i>objectSuffix</i> , see “Creating access controls for the Software Management task” on page 88.	IZUADMIN IZUUSER	CONTROL	Allow the user to access the Software Management task objects.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.PRODUCT_INFO_FILE.RETRIEVE	IZUADMIN	READ	Allow the user to access the Software Management task <i>Product Information File Retrieve</i> function.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.CATEGORIES.MODIFY	IZUADMIN	READ	Allow the user to add, copy, modify, or remove Software Management categories.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.SWUPDATE	IZUADMIN IZUUSER	READ	Allow the user to access the Software Update task.

Table 67. Security setup requirements for the Software Deployment service (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
UNIXPRIV	SUPERUSER.FILESYS.MOUNT	IZUADMIN IZUUSER	UPDATE	Allow the user to create workflow instances from workflow definition files that reside in UNIX file systems that are not currently mounted.
UNIXPRIV	SUPERUSER.FILESYS.USERMOUNT	IZUADMIN, IZUUSER	READ	Allow the user to mount a temporary work space UNIX file system data set created and used by the Deployment Unzip job and the Export job. Note: You can read about the SUPERUSER.FILESYS.USERMOUNT (and SUPERUSER.FILESYS.MOUNT) resource here: https://www.ibm.com/docs/en/zos/2.4.0?topic=security-using-unixpriv-class-profiles

Table 67. Security setup requirements for the Software Deployment service (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
FACILITY	STGADMIN.ADR.COPY.INCAT STGADMIN.ADR.DUMP.INCAT	IZUADMIN IZUUSER	READ	Allow the user access to the COPY and DUMP commands for program ADRDSSU. The COPY and DUMP commands are used by Deployment and Export JCL that is generated by Software Management. Note: INCAT is used only when source data sets are not cataloged in the current active catalog environment. This is an unlikely scenario. See Table Note 1.
ZMFAPLA	<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.SWUPDATE.COMPLETED.DELETE	IZUADMIN	READ	Allow an administrator to remove a completed or cancelled software update process and to delete the saved SMP/E output associated with that process.
ZMFAPLA	<SAF-prefix>.ZOSMF.SOFTWARE_DEPLOYMENT.SOFTWARE_MANAGEMENT.SWUPDATE.COMPLETED.DOWNLOAD	IZUADMIN	READ	Allow an administrator to download the saved SMP/E output associated with a completed or cancelled software update process.

1. If a resource profile is defined, then READ access is required. If a resource profile is not defined, then all users have access to that resource. More specifically:
 - If a profile for a resource is not defined, then the user can use the resource.
 - If a profile for a resource is defined and the user has at least READ access, then the user can use the resource.
 - If a profile for a resource is defined and the user does not have at least READ access, then the user cannot use the resource.

Resource authorizations for the Sysplex Management service

The Sysplex Management service requires access to local resources on your z/OS system.

Table 68 on page 401 describes the security requirements for the Sysplex Management service. The IZUSPSEC job includes sample RACF commands for creating these authorizations.

Table 68. Security setup requirements for the Sysplex Management service				
Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF-prefix>.IzuManagementFacilitySysplexManagement.izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the Sysplex Management task.
FACILITY	MVSADMIN.XCF.CFRM	IZUADMIN	READ or UPDATE	Allow the user to use the CFRM Policy Editor to edit CFRM policies. Assign UPDATE access authority to users who must alter or maintain the policy. Assign READ access authority to users who can view the policy, but not change it.
SERVAUTH	CEA.XCF.CDS	IZUADMIN IZUUSER	READ	Allow the user to access the couple data set for the Sysplex Management task.
SERVAUTH	CEA.XCF.CF	IZUADMIN IZUUSER	READ	Allow the user to access the coupling facility for the Sysplex Management task.
SERVAUTH	CEA.XCF.FLOW.<sysname>	IZUADMIN IZUUSER	READ	Allow the user to access the sysplex resources on remote systems for the Sysplex Management task. Replace <sysname> with the 8 character name of the system in the sysplex.
SERVAUTH	CEA.XCF.STRUCTURE	IZUADMIN IZUUSER	READ	Allow the user to access the coupling facility structures for the Sysplex Management task.

Table 68. Security setup requirements for the Sysplex Management service (continued)				
Resource class	Resource name	Who needs access?	Type of access required	Why
SERVAUTH	CEA.XCF.SYSPLEX	IZUADMIN IZUUSER	READ	Allow the user to access the sysplex general information and systems for the Sysplex Management task.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.SYSPLEX	IZUADMIN IZUUSER	READ	Allow the user to access the Sysplex Management task.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.SYSPLEX.LOG	IZUADMIN or a particular z/OS user ID.	READ	Allow the user to use the Sysplex Management task to clean up the command log table and specify clean-up settings.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.SYSPLEX.MODIFY	IZUADMIN	READ	Allow the user to use the Sysplex Management task to modify sysplex resources. This authorization also allows the user to use the CFRM Policy Editor to update Sysplex CFRM administrative policy information.

Resource authorizations for the Workload Management service

The Workload Management service requires access to local resources on your z/OS system. [Table 69 on page 402](#) describes the security requirements for the service. The IZUWMSEC job includes sample RACF commands for creating these authorizations.

Table 69. Security setup requirements for the Workload Management service				
Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF- <i>prefix</i> >.IzuManagementFacilityWorkloadManagement. izuUsers	IZUADMIN IZUUSER	READ	Allow the user to connect to the Workload Management task.
FACILITY	MVSADMIN.WLM.POLICY	IZUSVR	READ	Allow the z/OSMF server to access the WLM policies.

Table 69. Security setup requirements for the Workload Management service (continued)				
Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.WORKLOAD_MANAGEMENT.WORKLOAD_MANAGEMENT.ENWRP	IZUADMIN WLM resource pool administration group	READ	For z/OS Cloud Provisioning, allow the user to access the WLM Resource Pooling (WRP) functions of z/OSMF. Using a WRP definition, the user can associate cloud information (tenant name, domain ID, template type, service levels supported) with WLM elements (report classes and classification rules).
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.WORKLOAD_MANAGEMENT.WORKLOAD_MANAGEMENT.VIEW	IZUADMIN IZUUSER	READ	Allow the user to access the Workload Management View function.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.WORKLOAD_MANAGEMENT.WORKLOAD_MANAGEMENT.MODIFY	IZUADMIN	READ	Allow the user to access the Workload Management Modify function.
ZMFAPLA	<SAF- <i>prefix</i> >.ZOSMF.WORKLOAD_MANAGEMENT.WORKLOAD_MANAGEMENT.INSTALL	IZUADMIN	READ	Allow the user to access the Workload Management Install function.

Resource authorizations for the IBM zERT Network Analyzer service

The IBM zERT Network Analyzer service provides access to sensitive network security information. Only users authorized to manage this data should be allowed to access the IBM zERT Network Analyzer service. The IZUNASEC job includes sample RACF commands to create the group IZUZNA. The IZUZNA group should be used to control access to the IBM zERT Network Analyzer service.

Your security team might determine that existing group names would be preferred. If so, you can use your existing group names in place of the supplied z/OSMF default group names. For example, you might already have a group that is aligned with network security administrators; if so, you could use that group instead of the default group enabling access to the IBM zERT Network Analyzer service, IZUZNA.

The IBM zERT Network Analyzer service requires access to local resources on your z/OS system. Table 70 on page 404 describes the security requirements for the IBM zERT Network Analyzer service. The IZUNASEC job includes sample RACF commands for creating these authorizations.

Table 70. Security setup requirements for the IBM zERT Network Analyzer service				
Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF-prefix>.IzuZertNetworkAnalyzer.izuUsers	IZUZNA	READ	Allow the user to connect to the IBM zERT Network Analyzer task.
EJBROLE	<SAF-prefix>.com.ibm.ws.management.security.resource.Administrator	IZUSVR	READ	Allow the IBM zERT Network Analyzer to run the required WebSphere Liberty administrative actions.
ZMFAPLA	<SAF-prefix>.ZOSMF	IZUZNA	READ	Designates the user as a z/OSMF user
ZMFAPLA	<SAF-prefix>.ZOSMF.ZERT_NETWORK_ANALYZER	IZUZNA	READ	Allow a user to access the IBM zERT Network Analyzer task.

Resource authorizations for the z/OS Management Services Catalog service

The security configuration requirements for z/OS Management Services Catalog are described in the sections that follow. These sections describe the resources that must be defined, and the groups that must be permitted to the resources. Typically, these permissions are created by your security administrator.

z/OS Management Services Catalog uses a software defined role-based authorization model. A user ID's role determines what the user ID can do in the product.

Table 71. z/OS Management Services Catalog User Roles

Role	Recommended group	Capabilities
User	IZUUSER	Users have access to the Catalog, Activity, and History pages. Users can submit services from the Catalog, manage queued and active service submissions in Activity, and access completed and terminated service submissions in History. Users can see service submissions from other users and administrators. A service submission that has been started but is not yet submitted is saved in My drafts and cannot be seen by other users or administrators. Users can control their notification preferences using Settings.
Administrator	IZUADMIN	Administrators have additional authority that grants them access to the Administration page and the plug-in Global settings page. Administrators can manage existing services, create new services, and request approval to publish new services to the Catalog. Administrators can manage the plug-in's Global settings and take actions on service submissions that are created by other users.

Table 71. z/OS Management Services Catalog User Roles (continued)

Role	Recommended group	Capabilities
Publishing approver	IZUMSPAP	The publishing approver role authorizes a user ID to be assigned as an approver of services that are requested to be published in the Catalog. User IDs given this role must be that of a real user that can review and approve requests to publish services so that they are available on the Catalog page. Approvers are assigned by an administrator in the Publishing approvals table of Global Settings. Approvers have access to the Administration page and all services for which they are an approver.
RunAsUser step approver	IZUMSRAP	A user ID must have this role if a service's underlying workflow definition file requires the user ID to approve the use of a runAsUser step. User IDs given this role must be that of a real user that can review and approve the use of the runAsUser step. Do not use functional or application IDs as approvers. A runAsUser step is a step in the workflow that is performed by a specific user ID that might not be the user that runs the workflow. The user ID that the step is performed as is not necessarily the same as the user ID that approves the step. Every runAsUser step has an assigned user ID to approve it. This user ID's approval is required to publish the service in the Catalog. RunAsUser step approvers have access to the Administration page and all services for which they are an approver.

Table 71. z/OS Management Services Catalog User Roles (continued)

Role	Recommended group	Capabilities
RunAsUser user ID	IZUMSRAU	This role is required for any user ID assigned as the runAsUser for a runAsUser step in a workflow definition. Authorization to this role is checked when you create a new service from a workflow definition that contains runAsUser steps. It is also checked when the Workflows task runs a runAsUser step for workflow instances that are created by service submissions. This role does not grant any access to z/OS Management Services Catalog.

Note: At least one publish approver is required when publish approval is enabled in **Global settings**.

Table 72. Resource Authorization requirements for the z/OS Management Services Catalog service

Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAF-prefix> .IzuManagementFacilityManagementServicesCatalog.izuUsers	- z/OSMF users (IZUUSER) - z/OSMF administrators (IZUADMIN)	READ	Allow the user to open the Management Services Catalog desktop app.
ZMFAPLA	<SAF-prefix> .ZOSMF.MGMT_SERVICES.MGMT_SERVICES	- z/OSMF users (IZUUSER) - z/OSMF administrators (IZUADMIN)	READ	Allow the user to open the Management Services Catalog desktop app.
ZMFAPLA	<SAF-prefix>.ZOSMF.MGMT_SERVICES.ADMIN	z/OSMF administrators (IZUADMIN)	READ	Grants the administrator role to the user.
ZMFAPLA	<SAF-prefix>.ZOSMF.MGMT_SERVICES.USER	z/OSMF users (IZUUSER)	READ	Grants the user role to the user.
ZMFAPLA	<SAF-prefix>.ZOSMF.MGMT_SERVICES.PUBLISH.APPROVER	IZUMSPAP	READ	Grants the publishing approver role to the user.

Table 72. Resource Authorization requirements for the z/OS Management Services Catalog service (continued)

Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF-prefix>.ZOSMF.MGMT_SERVICES.RUNASUSER.APPROVER	IZUMSRAP	READ	Grants the runAsUser step approver role to the user.
ZMFAPLA	<SAF-prefix>.IZUDFLT.ZOSMF.MGMT_SERVICES.RUNASUSER	IZUMSRAU	READ	Grants authority to the user ID to be used as the runAsUser user ID in any workflow instance created by a service.

Resource authorizations for the Storage Management service

The Storage Management services require update access to the Source Control Data Set (SCDS) on your z/OS system to perform SCDS modification, validation and activation.

Table 73. Resource Authorization requirements for the Storage Management Services service

Resource class	Resource name	Who needs access?	Type of access required	Why
ZMFAPLA	<SAF-prefix>.ZOSMF.STORAGE.SG.VOLUME	z/OSMF administrators (IZUADMIN)	UPDATE	Allow the user to add volumes to storage group.
ZMFAPLA	<SAF-prefix>.ZOSMF.STORAGE.SCDS	z/OSMF administrators (IZUADMIN)	UPDATE	Allow the user to validate or activate the SCDS specified.
OPERCMDS	MVS.SETSMS.SMS	z/OSMF administrators (IZUADMIN)	UPDATE	Allow the user to activate the SCDS by using the SETSMS SCDS("scds-name") command.

Resource authorizations for the Parmlib Management service

The Parmlib Management service requires access to local resources on your z/OS system. Table 74 on page 409 describes the security requirements for the Parmlib Management service. The IZUPMSEC job includes sample RACF commands for creating these authorizations.

Table 74. Resource Authorization requirements for the Parmlib Management service

Resource class	Resource name	Who needs access?	Type of access required	Why
EJBROLE	<SAFprefix>.IzuManagementFacilityParmlibManagement.izuUsers	• IZUADMIN • IZUUSER	READ	Allow the user to connect to the Parmlib Management task.
ZMFAPLA	<SAFprefix>.ZOSMF.PARMLIB_MGMT	• IZUADMIN • IZUUSER	READ	Allow the user to use the Parmlib Management task.

Appendix B. Creating security descriptor files for the Security Configuration Assistant task

In the Security Configuration Assistant task, you can check the security configuration for external products on your z/OS system. This option requires a security descriptor file, which is typically provided by the product vendor. A *security descriptor file* is a flat file, such as a text file, that contains security information about the product.

You can create your own security descriptor files or obtain them from a provider, such as IBM, another vendor, or a third party. IBM supplies security descriptor files for the z/OSMF functions and services at the following location: `/usr/lpp/zosmf/configuration`. You can refer to the IBM-supplied files as examples for creating your own security descriptor files.

Syntax for security descriptor files

The security descriptor file format is JSON. This file must comply with the following syntax:

```
{
  "ServiceId": "service-identifier",
  "ServiceName": "service-name",
  "MetaValidationItemVersion": "validation-version",
  "Vendor": "vendor-name",
  "SecurityValidationItems": [
    {
      "ItemID": "item-identifier",
      "ItemType": "item-type",
      "ItemCategory": "item-category",
      "ResourceProfile": "resource-profile",
      "ResourceClass": "resource-class",
      "WhoNeedsAccess": "user name or group name",
      "LevelOfAccessRequired": "access-level",
      "ItemDescription": "item-description"
    }
  ]
}
```

Where:

ServiceId

An identifier that represents the specific product to be evaluated by the Security Configuration Assistant task. This value must be unique.

For reference, each z/OSMF core service is associated with a security descriptor file and is assigned a unique service identifier as follows:

- 5655S28SM00 - z/OSMF Nucleus
- 5655S28SM01 - z/OSMF Security Configuration Assistant
- 5655S28SM02 - z/OSMF AUTOSTART function
- 5655S28SM03 - z/OSMF Notification function
- 5655S28SM04 - z/OSMF Workflow Editor
- 5655S28SM05 - z/OSMF Configuration workflow

ServiceName

Service name. This value is displayed for the product in the Security Configuration Assistant task user interface (UI).

MetaValidationItemVersion

Version identifier for the security descriptor file, for example 1.0. The provider of the security descriptor file can increment this value to indicate a new version of the file.

Vendor

The provider of the security descriptor file, for example, IBM.

SecurityValidationItems

An array in which each element contains the following key value pairs:

ItemID

A unique identifier to represent the resource to be protected (the *security item*). This value can be used to control the order in which items are listed in the Security Configuration Assistant task UI. It is recommended that *nnnnnnnn* is a discontinuous number so that new items can be inserted in the future.

As an example, the item ID for a z/OSMF service consists of *ServiceID+I+nnnnnnnn*.

ItemType

One of the following values:

- PROGRAMMABLE indicates that the security item can be verified automatically by the Security Configuration Assistant task.
- MANUAL indicates that the security administrator must verify the security item manually.
- SEMI-PROGRAMMABLE indicates that the security item depends on configurable settings, which require input from the security administrator. For example, a RACF resource profile name with one or more variables. In the following example, the security administrator must provide the values for *sysname* and *tcpname*:

```
Resource EZB.INITSTACK.<sysname>.<tcpname>
```

When you add values to a resource profile, you are replacing the variable portion with an actual value. You must ensure that the resource profile is correct for your system.

ItemCategory

An optional value that can be used to group related security items in the Security Configuration Assistant task UI. For example, you might assign an item category to security items that are used to protect the same function.

ResourceProfile

SAF resource profile name. You can specify a generic profile.

ResourceClass

SAF resource class.

WhoNeedsAccess

Users (security groups) who require access to this resource. The Security Configuration Assistant task does not verify that security groups are defined for the external product. The security administrator must verify that the groups exist.

LevelOfAccessRequired

Level of access that is required, such as READ, UPDATE, ALTER, or CONTROL.

ItemDescription

Descriptive text. For example, an explanation of why the authorization is needed.

The information for each product must be contained within separate braces ({}) inside the brackets ([]), and each set of braces must be comma-separated. For an example file that contains the information for the z/OSMF ISPF plug-in, see [“Sample security descriptor file” on page 413](#).

Sample security descriptor file

Figure 50 on page 413 shows the contents of the security descriptor file for the z/OSMF ISPF service.

```
{
  "ServiceId": "5655S280100",
  "ServiceName": "z/OSMF ISPF",
  "MetaValidationItemVersion": 1.01,
  "Vendor": "IBM",
  "SecurityValidationItems": [
    {
      "ItemID": "5655S280100I0001000",
      "ItemType": "PROGRAMMABLE",
      "ItemCategory": "z/OSMF ISPF functions",
      "ResourceProfile": "IZUDLFT.ZOSMF.ISPF.ISPF",
      "ResourceClass": "ZMFAPLA",
      "WhoNeedsAccess": "<user or your group name>",
      "LevelOfAccessRequired": "READ",
      "ItemDescription": "Allow the user to access the ISPF task."
    }
  ]
}
```

Figure 50. Security descriptor file for the z/OSMF ISPF service

Working with a security descriptor file

Do the following:

1. Obtain the security descriptor file from the product vendor.
2. Install the security descriptor file in the following z/OSMF directory: <IZU_CONFIG_DIR>/configuration/security. By default, this directory is named /global/zosmf/configuration/security.

The Security Configuration Assistant task can access and display the security descriptor files in this directory.

Ensure that the z/OSMF server ID (by default, IZUSVR) has at least READ permission to the security descriptor file. On start-up, the z/OSMF server checks the files in the z/OSMF configuration directory. You can avoid a warning message or error message from the server if you set the file permissions to read/write: 660 (rw- -rw- - -).

Also, to avoid errors, if you transfer the file from a workstation to the z/OSMF directory, be sure to convert the file to the EBCDIC character set on the host system.

3. In the Security Configuration Assistant task, in the **Imported Products** view, click **Import**. This action displays a list of the available security descriptor files.
4. Select the security descriptor file for the product that you want to verify and click **OK**. If the file contains an error that prevents it from being loaded, an error message is displayed. For more details, such as the line number of the error, click the information icon for the file name.

In general, configuring the security for a product involves the following activities:

- Creating security profiles for the product.
- Performing the various z/OS system customization updates, if any, that are required by the product.
- Creating a security descriptor file for the product and using the Security Configuration Assistant to validate its security configuration.

For more information, see the online help for the Security Configuration Assistant task.

Appendix C. z/OSMF Configuration Workflow

This topic describes how to use the z/OSMF Configuration Workflow to perform the system customization for the z/OSMF optional services.

Part 4, “z/OSMF optional services,” on page 81 describes the manual steps that are required for customizing your system for the z/OSMF optional services. As an alternative, you can use the z/OSMF Configuration Workflow to perform the system customization for each service. If you use the workflow, you are guided through the system customization steps.

About the z/OSMF Configuration Workflow

For each service to be added, the z/OSMF Configuration Workflow performs the following actions:

- Creates and updates parmlib members as needed for the services to be configured. For example, if you configure the Incident Log service, the workflow creates members in the target parmlib data set.
- Prepares your z/OS system for running the tasks that are associated with the services.
- Verifies the setup for the z/OSMF tasks. If you configure the Incident Log service, the workflow verifies the setup of the following z/OS system components:
 - Sysplex dump directory
 - System logger
 - Common event adapter (CEA)
 - System REXX.

The workflow identifies any areas that might require further action on your part.

- Adds the names of the optional services to the PLUGINS statement in your IZUPRMxx member.
- Creates authorizations for the z/OSMF tasks. The workflow includes steps that create RACF commands for connecting users and groups to the appropriate SAF profiles. If your installation uses a security management product other than RACF, your security administrator can refer to the RACF commands as a reference.
- Completes the deployment of the services by restarting the z/OSMF server to make these changes effective.

To run the z/OSMF Configuration Workflow, you require a user ID that is connected to the z/OSMF Administrator security group, which is IZUADMIN, by default. Your user ID also requires:

- RACF SPECIAL attribute, which gives the user full control over the RACF profiles in the RACF database.
- Authorizations that are described in “Grant the user access to the IRRXUTIL program” on page 415 and “Grant the user access to the OPERCMDS resources” on page 416.

If you prefer, you can manually perform the system customization for each service. For descriptions of the customization that must be performed for each service, see Part 4, “z/OSMF optional services,” on page 81.

Grant the user access to the IRRXUTIL program

The z/OSMF Configuration Workflow uses the IRRXUTIL program to retrieve profile information about users, groups, general resources, and general RACF settings administered by the SETROPTS command. Therefore, your user ID requires READ authorization to the resource names listed in Table 75 on page 415.

Table 75. IRRXUTIL program authorizations required for using the z/OSMF Configuration Workflow			
Resource name	Class	Access	Purpose
IRR.RADMIN.LISTUSER	FACILITY	READ	Read USER profiles.

Table 75. IRRXUTIL program authorizations required for using the z/OSMF Configuration Workflow (continued)

Resource name	Class	Access	Purpose
IRR.RADMIN.LISTGRP	FACILITY	READ	Read group profiles.
IRR.RADMIN.RLIST	FACILITY	READ	Read profiles of general resources.
IRR.RADMIN.SETROPTS.LIST	FACILITY	READ	Read RACF SETROPTS settings.

The IZUSEC job contains sample RACF commands for creating these authorizations. [Figure 51 on page 416](#) shows the commands that are provided in the job.

```

/* Allow users of the z/OSMF Configuration Workflow to extract profile
information */
RDEFINE FACILITY IRR.RADMIN.LISTUSER
RDEFINE FACILITY IRR.RADMIN.LISTGRP
RDEFINE FACILITY IRR.RADMIN.RLIST
RDEFINE FACILITY IRR.RADMIN.SETROPTS.LIST

/* Permit the z/OSMF administrator access */
PERMIT IRR.RADMIN.LISTUSER CLASS(FACILITY) ID(IZUADMIN) ACCESS(READ)
PERMIT IRR.RADMIN.LISTGRP CLASS(FACILITY) ID(IZUADMIN) ACCESS(READ)
PERMIT IRR.RADMIN.RLIST CLASS(FACILITY) ID(IZUADMIN) ACCESS(READ)
PERMIT IRR.RADMIN.SETROPTS.LIST CLASS(FACILITY) ID(IZUADMIN) ACCESS(READ)

SETROPTS RACLIST(FACILITY) REFRESH

```

Figure 51. RACF commands for authorizing the users of the z/OSMF Configuration Workflow

Grant the user access to the OPERCMDS resources

The z/OSMF Configuration Workflow uses the extended MCS console to issue operator commands. Therefore, your user ID requires READ authorization to the CONSOLE profile of the TSOAUTH class. Also, if the OPERCMDS class is active on your system, your user ID requires READ authorization to the generic profile MVS.MCSOPER.IZU@*. These authorization requirements are summarized in [Table 76 on page 416](#).

Table 76. OPERCMDS authorizations required for using the z/OSMF Configuration Workflow

Resource name	Class	Access	Purpose
MVS.MCSOPER.IZU@*	OPERCMD	READ	Allow the user to operate an extended MCS console.
CONSOLE	TSOAUTH	READ	Allow the user to issue the TSO/E CONSOLE command to activate the extended MCS console.

The IZUSEC job contains sample RACF commands for creating these authorizations. [Figure 52 on page 417](#) shows the commands that are provided in the job.

```

/* Allow workflow users to issue MVS commands from EMCS consoles */
SETROPTS CLASSACT(TSOAUTH)
SETROPTS RACLIST(TSOAUTH)
RDEFINE TSOAUTH CONSOLE UACC(NONE)
PERMIT CONSOLE CLASS(TSOAUTH) ID(IZUADMIN) ACCESS(READ)
PERMIT CONSOLE CLASS(TSOAUTH) ID(IZUUSER) ACCESS(READ)

SETROPTS RACLIST(TSOAUTH) REFRESH

/* Allow workflow users to access EMCS consoles. The console names are */
/* prefixed with the value "IZU@" */
SETROPTS CLASSACT(OPERCMDS)
SETROPTS RACLIST(OPERCMDS)
RDEFINE OPERCMDS MVS.MCSOPER.IZU@* UACC(NONE)
PERMIT MVS.MCSOPER.IZU@* CLASS(OPERCMDS) ID(IZUADMIN) ACCESS(READ)
PERMIT MVS.MCSOPER.IZU@* CLASS(OPERCMDS) ID(IZUUSER) ACCESS(READ)

SETROPTS RACLIST(OPERCMDS) REFRESH

```

Figure 52. OPERCMDS authorizations that are required for using the z/OSMF Configuration Workflow

Getting started

To create the z/OSMF Configuration Workflow, import the following workflow definition file into the Workflows task:

```
<product_dir>/workflow/izu.config.setup.xml
```

where `<product_dir>` is the z/OSMF product directory. By default, this directory is `/usr/lpp/zosmf`.

When you create the workflow, specify the accompanying variable input file, which was generated when you created the base z/OSMF configuration. This file, which is used to populate the workflow with your configuration values, resides in the following directory path:

```
<user_dir>/configuration/workflow/izu.config.workflow.cfg
```

where `<user_dir>` is the data directory. By default, this directory is `/global/zosmf`.

More information about the Workflows task is provided in the online help.

Steps for adding services to z/OSMF

To add services to z/OSMF, follow these steps:

1. Run the z/OSMF Configuration Workflow to customize your system for the services to be added.
2. Verify the results of your work by opening a web browser to the **Welcome** page. For more information, see [“Step 5: Log in to z/OSMF” on page 29](#).

Appendix D. Common event adapter (CEA) reason codes

A problem in the configuration of z/OSMF might be indicated by reason codes from the common event adapter (CEA) component of z/OS.

This section describes the configuration-related CEA reason codes and includes a cross-reference of reason codes to CIM messages and z/OSMF messages. Where an associated z/OSMF message is indicated, check the z/OSMF message for more information about the error.

“CEA reason codes for the Incident Log task” on page 419 describes the CEA reason codes you might encounter during the configuration of the task. “CEA reason codes for the z/OS jobs REST interface services” on page 422 describes the CEA reason codes that an HTTP client application might encounter when using the z/OS jobs REST interface services. For other CEA reason codes, see the topic on using CEA TSO/E address space services in [z/OS MVS Programming: Callable Services for High-Level Languages](#).

CEA reason codes for the Incident Log task

Table 77 on page 419 describes the CEA reason codes you might encounter when setting up or using the Incident Log task. By default, CEA reason codes without an associated z/OSMF message are accompanied by z/OSMF message IZUP631E.

Table 77. CEA reason codes related to Incident Log task processing						
Reason code (decimal)	Reason code (hex)	Description	System programmer action	CIM message	z/OSMF message	IBM Support information
256	100	The CEA address space is not running.	Follow the steps in “ Ensure that common event adapter (CEA) is configured and active ” on page 17.	CEZ05002E	IZUP634E	CEAUNAVAIL
289	121	CIM indication processing is not available because the CEA address space is running in minimum (MIN) mode. To support Incident Log processing, CEA must be operated in full mode.	Use the MODIFY CEA, MODE command to change the CEA mode of operation to full mode. To do so, enter the command, as follows, from the operator console: F CEA, MODE=FULL Running CEA in full mode requires that z/OS UNIX system services is available.	CEZ05013E		CEAFORCEMINMODE
813	32D	The user is not authorized for this request.	Define the appropriate authority for the user.	CEZ05003E	IZUP635E	CEANOINSTRAUTH
830	33E	An abend occurred in the CEA task that interacts with the IPCS environment.	Report the problem to IBM Support.	CEZ05001E	IZUP639E	CEAIPRQSERVER ABENDED
834	342	The sysplex dump directory is empty.	Ensure that the sysplex dump directory is not empty.			CEASDDIREMPTY
835	343	A dump incident was not found. Most likely, the incident was deleted by another user.	No action is required.	CEZ05004E	IZUP636E	CEAADDFAILED

Table 77. CEA reason codes related to Incident Log task processing (continued)

Reason code (decimal)	Reason code (hex)	Description	System programmer action	CIM message	z/OSMF message	IBM Support information
850	352	The dump analysis and elimination (DAE) data set name (typically SYS1.DAE) could not be determined. Most likely, DAE is not configured or is not running. Or, the user attempted to unsuppress a dump without having write access to the DAE data set.	Ensure that: • DAE is active. • DAE is configured, as described in z/OS MVS Diagnosis: Tools and Service Aids. • User has write access to the active DAE data set. For more information, see “Configuring dump analysis and elimination” on page 189 .		IZUP637E	CEADAEDSN NOTAVAILABLE
855	357	The called function could not generate a prepared data set name (DSN).	Verify that the compiled REXX exec CEACDMPP exists and can be run by System REXX.			CEAGENPREPARED DSNFAIL
857	359	An internal CEA error occurred when attempting to invoke a SYSREXX exec.	If this reason code is accompanied by the following codes (in decimal), check the SYSREXX concatenation for a missing exec: • DIAG=8 • DIAG2=851. Also, check message CEZ05000E in SYSLOG. CEAERRO_Msg contains the name of the SYSREXX exec.	CEZ05000E		CEAAXREXXERROR
866	362	The source description for a requested dump incident was not found in the sysplex dump directory.	Determine why the dump incident was not identified in the sysplex dump directory. Possible reasons include: • Dump has not yet been taken • Dump has not yet been written out • Dump is being entered into a different sysplex dump directory than the one that is used by the Incident Log task.	CEZ05001E	IZUP631E	CEADMPINCIDENT NOTFOUND
869	365	The System REXX address space or the functions it provides are not available.	Follow the steps in “Ensuring that System REXX is set up and active” on page 192 .	CEZ05005E	IZUP640E	CEASYSREXX NOTACTIVE
870	366	System REXX cannot process an exec.	This problem usually indicates that the run time support for compiled REXX has not been set up. See “Ensuring that System REXX is set up and active” on page 192 .	CEZ05006E	IZUP643E	CEASYSREXXBAD ENVIRONMENT
871	367	System REXX cannot process the exec at this time.	Try the request again later.	CEZ05007W	IZUP644E	CEAEXEETIMEOUT
872	368	System REXX cannot schedule the exec to run at this time.	Try the request again later.	CEZ05008W	IZUP645E	CEASYSREXX OVERLOADED
879	36F	The user is not authorized to view the operations log (OPERLOG) snapshot information.	Ask the security administrator to authorize the user to the data set, which is specified in the CEAPRMxx parmlib member.	CEZ05010E		CEANOSAF OPERLOGSNAP

Table 77. CEA reason codes related to Incident Log task processing (continued)

Reason code (decimal)	Reason code (hex)	Description	System programmer action	CIM message	z/OSMF message	IBM Support information
880	370	The system logger component is not available.	For an explanation of the logger reason code in CEAERRO_DIAG4, see mapping macro IXGCON. If the system is not running with a logger couple data set, this is a permanent condition for the IPL. Otherwise restart system logger and enter the request again. For more information, see “Defining a couple data set for system logger” on page 181 . For information about the IXGCON macro, see z/OS MVS Programming: Authorized Assembler Services Reference EDT-IXG .	CEZ05011E		CEALOGGER NOTAVAIL
881	371	The function that prepares incident materials to be sent through FTP could not allocate a new data set for the tersed diagnostic snapshot.	Check the CIM trace file for system messages associated with the return code indicating the reason for the failure. For assistance, contact IBM Support.			CEABADALLOCNEW
882	372	The function that prepares an incident to be sent through FTP could not allocate the data set to be tersed.	Check the CIM trace file for system messages associated with the return code indicating the reason for the failure. For assistance, contact IBM Support.			CEATERSE BADALLOC1
886	376	The operations log (OPERLOG) snapshot was not created. When attempting to access the OPERLOG snapshot, the system logger service IXGCONN received a bad return or reason code indicating that the OPERLOG snapshot does not exist.	Check SYSLOG for message CEA0600I, which contains the return and reason codes.			CEANOSNAPSHOT
888	378	No log data was accumulated in diagnostic snapshot.	If this problem occurs frequently, adjust the DUMPCAPTURETIME setting in the CEAPRMxx parmlib member.			CEAPDWB DIAGDATAEMPTY
889	379	An incorrect format or value was supplied for the IBM PMR number.	Correct the IBM PMR number and try again. The format of the IBM PMR number should be <i>nnnnn.ccc.bbb</i> where <i>nnnnn</i> is the PMR number, <i>bbb</i> is the branch code, and <i>ccc</i> is the country code.			CEAWRONG IBMPMRFORMAT
893	37D	An attempt to obtain the enqueue on the sysplex dump directory failed; another program already holds the enqueue.	Ensure that only one user is attempting to access the dump information at one time. To check for enqueue contention, enter the command D GRS,C at the operator console. Wait for the enqueue to be released and try again.	CEZ05017E	IZUP641E	CEAIPCSENQ ERROR
894	37E	The requested function failed to open the sysplex dump directory.	Verify that the sysplex dump directory (default name SYS1.DDIR) is set up and usable. For more information, see “Creating the sysplex dump directory” on page 190 .	CEZ05016E	IZUP642E	CEASDDIR OPENERORR
898	382	The component table is corrupted.	Report the problem to IBM Support.			CEAXMLTAGS TOODEEP
901	385	The diagnostic data to be sent is currently in use.	Try the request again later.			CEAPREPARE OBJINUSE
902	386	The diagnostic data to be sent is currently in use.	Try the request again later.			CEAPREPAREENQERR
908	38C	The sysplex dump directory has no space available to record new SVC dumps.	See “Establishing a larger sysplex dump directory” on page 191 .			CEACKST INVALIDALLOC VALUE

Table 77. CEA reason codes related to Incident Log task processing (continued)

Reason code (decimal)	Reason code (hex)	Description	System programmer action	CIM message	z/OSMF message	IBM Support information
913	391	The JES subsystem is not available.	Determine why the JES subsystem is not accessible. Perhaps, it has not been started.			CEAJESNOT AVAILABLE
919	397	The Set Incident field data was truncated at 256 characters.	Specify a smaller amount of data for the user comment field to prevent truncation. Retry the request.			CEASETINCIFVAL DATATRUNC
920	398	The request failed because one or more of the affected dump data sets are migrated.	If the data set is migrated and automatic recall is enabled for the hierarchical storage manager (HSM), the system issues a recall request for the data set. Wait for the recall request to complete and then retry the request.			CEAMIGRATED DATASETS
921	399	The request failed because one or more of the requested dump data sets are migrated and the hierarchical storage manager (HSM) encountered an error occurred when attempting to recall the data sets.	Determine why HSM is not functioning properly. The problem might be that HSM is inactive or unresponsive. Correct the problem and retry the request.			CEAMIGRATED DATASETSWHSMERR
922	39A	The request failed because CEA could not allocate an internal buffer to satisfy the request.	Try the request again. If the problem persists, determine why there is insufficient storage on the system. Consider reducing the number of inactive incidents on your system through the ceatool program, which is described in Chapter 50, “Deleting incidents and diagnostic data,” on page 261 . Correct the problem and retry the request. If the problem persists, contact IBM Support.			CEAUNABLETO ALLOCATE3

CEA reason codes for the z/OS jobs REST interface services

Table 78 on page 422 describes the CEA reason codes that an HTTP client application might encounter when using the z/OS jobs REST interface services.

Table 78. CEA reason codes related to z/OS jobs REST interface processing

Reason code (decimal)	Reason code (hex)	Description	System programmer action	IBM Support information
923	39B	The request failed because the caller is not authorized to modify the job.	Check with your security administrator to ensure that the caller’s user ID is authorized to the appropriate resources in the JESJOBS class.	CEANOJESAUTHORITY
925	39D	An internal CEA error occurred.	Report the problem to IBM Support.	CEANOENTITY POSSIBLE
926	39E	The request failed because the specified job was not found on the system.	Examine the request to determine whether the job was identified correctly, either through the job name and job ID (jobname/jobid), or the job correlator.	CEASSIJOBNOTFOUND

Appendix E. ENF listener code examples

A program can use one of the following methods to determine whether the z/OSMF server is up or down in the sysplex:

- An APF-authorized program can use the ENFREQ LISTEN service to specify a listen exit for ENF event code 83 that tells the program the z/OSMF server is up and running. For an example of this technique, see the coded samples in [“Examples for an authorized program” on page 424](#).
- An unauthorized program cannot use the ENFREQ LISTEN service. However, it can periodically check the global storage pointer, which is mapped by macro IZUGSP. For an example of this technique, see the coded sample in [“Example for an unauthorized program” on page 427](#).

Examples for an authorized program

Example program IZULSTEN shows how an APF-authorized program can listen for z/OSMF server status. IZULSTEN invokes the sample exit routine, IZULST00, which must reside in the link pack area (LPA). Both sample programs are written in assembler language.

- [Figure 53 on page 424](#)
- [Figure 54 on page 425](#)

```

IZULSTEN  CSECT
IZULSTEN  AMODE 31
IZULSTEN  RMODE ANY

          STM 14,12,12(13)      Save caller's registers
          BALR 12,0             Establish module base

@ESTART  EQU *
          SAM31                 Ensure 31 bit mode
          USING @ESTART,12      Establish addressability
          MODID                 Eyecatcher and date
          SR 15,15             Set return code to 0

* Set mode to Supervisor State
          WTO 'ENTERING PGM...'
          MODESET MODE=SUP
          MODESET EXTKEY=ZERO,SAVEKEY=(2),WORKREG=7
          LR 8,2                Save user key

* Load ENF IZULST00 Listen Exit from LPA area
          LOAD EP=IZULST00,LOADPT=IZULS00@

          SR 15,15             Set return code to 0
          L 2,IZULS00@
          L 4,A31MASK
          OR 2,4                Must be 31-bit addressing

          ST 13,SAVEA+4
          LA 13,SAVEA           Provide save area

          WTO 'Now registering as a listener for ENF code 83.'
* Issue LISTEN Request for z/OSMF event code (all functions)
          ENFREQ ACTION=LISTEN, -- Function
          CODE=ENFC83,          -- Event code
          EXIT=(2),             -- Exit address
          QUAL=ENF83CUP,        -- z/OSMF is up?
          QMASK=ALL,           -- Set mask of all 4 bytes
          ESTBNME=THISMOD,      -- Establisher name
          EXITNME=IZULST00,     -- Exit name
          DTOKEN=IZULTOKN       -- Returned token field
          LTR 15,15
          JNZ ERRGO
          B @LEXIT

* Error exit, to print the return code
ERRGO    DS 0H
          ST 15,RETC
          WTO 'ENFREQ request error!'

          MVC DataToConvert,RETC
          UNPK ZonedArea,DataToUnpack
          TR CharData,TRTBL
          MVC ERRCODE,CharData

          MVC WTOAREA,WTOLIST
          LA 5,ERRLEN
          STH 5,ERRMSG
          MVC ERRMSG,ERRMSGC

          LA 2,ERRMSG
          WTO TEXT=(2),MF=(E,WTOAREA) Point to message in storage
                                          Write the message

@LEXIT   DS 0H
          LR 2,8                Restore user key
          MODESET KEYADDR=(2),WORKREG=7
          MODESET MODE=PROB
          WTO 'End of zOSMF listener routine.'

```

Figure 53. Example of an authorized program that listens for z/OSMF server events (Part 1 of 2)

```

* Return control
L      13,SAVEA+4
L      14,12(13)
LM     0,12,20(13)
BR     14
                                RESTORE CALLERS REGS
                                PRESERVE REG 15 - RC
                                RETURN TO CALLER

ENFC83 DS      0F
EQU     83
ENF83CUP DC     X'80000000'
A31MASK DC     X'80000000'
                                z/OSMF is up
                                High bit for 31-bit addressing

IZULTOKN DS      F
ENFPTR  DS      A
SAVEA   DS      18F
RETC    DS      F
IZULST00 DC     CL8'IZULST00'
IZULS000 DS      A
THISMOD DC     CL8'IZULSTEN'

* Work area for conversion of addr to char string
DataToUnpack DS      CL5
                                ORG      DataToUnpack
DataToConvert DS      CL4
DataToCvtXtra DS      CL1
ZonedArea    DS      CL9
                                ORG      ZonedArea
CharData     DS      CL8
ExtraChar    DS      CL1

* Table for converting
TRTBL      DC     XL256'00'
                                ORG      TRTBL+240
                                DC     C'0123456789ABCDEF'
                                DS      0F
                                Advance to offset F0 in table
                                Get ch 0 (F0x) at off F0

WTOLIST    WTO     TEXT=,MF=L      List form
WTOLEN     EQU     *-WTOLIST      Length to move
WTOAREA    DS      CL(WTOLEN)     WTO in dyn area for modifiable msgc
ERRMSGC    DC      C'The return code of ENQREQ is: '

* Error message format
DS      0F
ERRMSG    DC      AL2(ERRLEN)
ERRMSG    DS      CL(L'ERRMSGC)
ERRCODE   DS      CL8
ERRLEN    EQU     *-ERRMSG

*
* - External control blocks
CMT      DSECT=YES
* ENF facility vector table create by system. Pointed to from CVT.
IEFENFCT
* ENFREQ Macro area mapping
DATAAREA  DSECT
IEFENFPM
LENODATA  EQU     *-DATAAREA
END

```

Figure 54. Example of an authorized program that listens for z/OSMF server events (Part 2 of 2)

Figure 55 on page 426 shows IZULST00, which is an example of an ENF listener exit routine for the listener program that is shown in Figure 53 on page 424 and Figure 54 on page 425. To be used by an authorized program, the exit routine must reside in the link pack area (LPA).

```

IZULST00 CSECT
IZULST00 AMODE 31
IZULST00 RMODE ANY

STM 14,12,12(13)      Save caller's registers
BALR 12,0              Establish module base

@ESTART EQU *
SAM31 *               Ensure 31 bit mode
USING @ESTART,12      Establish addressability
MODID *               Eyecatcher and date
SR 15,15             Set return code to 0
LR 2,1               Save Register 1
WTO 'z/OSMF listen exit (IZULST00) receives control.'
L 3,0(2)             Address of IZUENF83 data area
USING IZUENF83,3      Establish addressability

* For reentrancy, obtain storage for the local modifiable variables.
USING WORKAREA,8      WORKAREA mapped at addr in Reg 8
GETMAIN RU,LV=WORKAREL,LOC=31 Allocated save/work area
LR 8,1               Save the addr of obtained storage

* Clear WORKAREA
MVI WORKAREA,X'00'    Clear the first byte
MVC WORKAREA+1((L'WORKAREA)-1),WORKAREA
ST 13,SAVEA+4
LA 13,SAVEA          Provide save area

* Add your process logic here, for example ...
* Check if the event is for UP/DOWN
L 7,IZUENF83_Status
L 9,TESTBIT
NR 7,9
LTR 7,7
BNZ IZUON             When High bit is on
WTO 'z/OSMF is down.'
B NEXTDO
IZUON DS 0H
WTO 'z/OSMF is up.'

NEXTDO DS 0H
* Print host and uri information
MVC WTOMSG(PARTURIL),IZUENF83_URI
MVC WTOAREA,WTOLIST
LA 9,WTOMSG
STH 9,WTOMSG
LA 5,WTOMSG
WTO TEXT=(5),MF=(E,WTOAREA) Point to msg in storage
                                Write the message

* Return control
L 13,SAVEA+4
FREEMAIN RU,A=(8),LV=WORKAREL Free save/workarea
L 14,12(13)           RESTORE CALLERS REGS
LM 0,12,20(13)        PRESERVE REG 15 - RC
BR 14                RETURN TO CALLER

WTOLIST WTO TEXT=WTOMSG,MF=L List Form
WTOLIST EQU *-WTOLIST Length to move
TESTBIT DC X'80000000'

WORKAREA DSECT
SAVEA DS 18F
WTOAREA DS CL(WTOLEN) WTO in dyn area for modifiable msgs

WTOMSGA DC AL2(WTOMSG)
WTOMSG DS CL(PARTURIL)
WTOMSG EQU *-WTOMSG
PARTURIL EQU 80
WORKAREL EQU *-WORKAREA WORKAREA DSECT length
IZUENF83 DSECT=YES
END

```

Figure 55. Example of an ENF listener exit routine for ENF code 83

Example for an unauthorized program

Figure 56 on page 427 shows a sample program in REXX that can be used to query the z/OSMF global storage area, which is mapped by the macro IZUGSP.

```
/* rexx */

/* Locate the system ECVT area */
CVT@ = C2d(Storage(10,4))
ECVT@ = C2d(Storage(D2x(CVT@ + 140),4))

/* Locate Address of z/OSMF Global Storage */
/* ECVTIZUGSP offset is 976 */
IZUGSP@ = C2d(Storage(D2x(ECVT@ + 976),4))
if IZUGSP@ = 0 then exit

/* Format the address to HEX string for visible */
IZUGSP$HEX = C2x(Storage(D2x(ECVT@ + 976),4))

/* Print values from area mapped by IZUGSP */
IZUGSP_ID = Storage(D2x(IZUGSP@),8)
IZUGSP_STATUS$HEX = C2x(Storage(D2x(IZUGSP@+8),4))
IZUGSP_URI = storage(D2x(IZUGSP@+12),274)

Say "Address of z/OSMF Global Storage is : "IZUGSP$HEX
Say "Eyecatcher field is : "IZUGSP_ID

/* High Bit is on when z/OSMF is active,
otherwise z/OSMF is inactive. */
Say "Status and URI length field in HEX format is : "IZUGSP_STATUS$HEX

Say "URI field is : "IZUGSP_URI

exit
```

Figure 56. Example of a REXX routine that allows an unauthorized program to listen for z/OSMF server events

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